

Health Economics



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Peter Zweifel
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PREFACE

In health economics, texts do not abound yet to the extent that authors must justify the appearance of another one. Nevertheless, it may be worthwhile to state the ways in which this book differs from others, at least in the authors' view. First, this text is truly international. Being nationals of two neighboring European countries, its authors are keenly aware of the institutional differences characterizing the health care sectors of industrialized countries. In spite of all the similarities between Germany and Switzerland, we have found many peculiarities hiding below the surface! However, we did not stop at just mentioning institutional differences but have made efforts to reflect them in our microeconomic modeling.

This points to the second distinguishing feature of this book. Rather than simplifying the issues facing today's health care systems to the point where they can be discussed using conventional introductory microeconomic analysis, this text seeks to model existing complexities as they are. Following a stylized sickness episode, economic models are presented to analyze the behavior of the (insured) patient, the treatment choices of the physician, the way hospital management deals with organizational slack, right down to the decision of a pharmaceutical company regarding how much to spend on research and development. It is designed to show the advanced undergraduate and beginning graduate student how economics can be adapted to reflect to a considerable extent the views of the "person on the street" with regard to health rather than "bending" them so they fit into the conventional toolbox.

Yet this emphasis on a consistent (mainly microeconomic) theoretical foundation does not constitute an aim for its own sake. Great care is taken to work out the implications of the models and to confront them with empirical evidence whenever possible. This amounts to what we believe to be the third special feature of this text. It may serve to render the policy suggestions emanating from the analysis more credible (if not palatable) to our international readers.

Many people have generously contributed their time and energy (but one hopes not their health) to this work, some of them mainly to the earlier German version, others to the present English version, and some to both. This list includes A. Bohm of the University of Constance (Germany), P.F. Clever of the University of Hagen (Germany), W.S. Comanor of the University of California at Santa Barbara and Los Angeles (U.S.), A. Culyer of the University of York (Great Britain), P. Feldstein of the University of California at Irvine (U.S.), M. Ferrari of the Dipartimento Opere Sociali of Bellinzona (Switzerland), M. Filippini of the University of Zürich (Switzerland),

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Finally, we beg our female readers' pardon for using the male form throughout. This should be understood as a tribute to simplicity rather than discrimination.

Zürich and Constance, December 1995

P.Z.

F.B.

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1

INTRODUCTION

1.1 HEALTH—A PRICELESS COMMODITY?

In trying to connect “health” and “economics” one usually thinks of

1. Health being the most precious good. In order to remain in good health just about anything should be done;
2. *Health care* being in a crisis. If costs increase at the present rate, then health might be a good that could not be afforded by most people.

Although the two statements seem to be contradictory, there is some accord in that both assume that health is “priceless” either in an ethical sense (“invaluable”) or in a more economic sense (“very expensive”).

So why should economists, presumably emphasizing the latter, deal with health-related subjects? In the eyes of most people, the answer lies in the development of expenditure. In all Western industrialized countries, *outlays on public health* have strongly increased, not only in absolute figures but also as a share of GNP (gross national product) during the past two decades (that is the 1970s and 1980s; see Table 1.1). Given continuing growth, the entire GNP of many an industrialized country would be consumed by health care expenditure before the end of the twenty-first century.

This so-called “cost explosion” has led to a number of legislative measures in many industrialized countries. They all had the objective of reducing the rate of increase in public health care expenditure.¹

Obviously health care costs are not the only costs that have increased greatly in the past. So have the costs of shoes and haircuts. Nonetheless no one has ever heard of a

1. In countries having social health insurance such as Germany, policy-makers seek to stabilize the contribution rate in terms of labor income that workers have to pay to their sick funds. In other countries, such as the United States, the federal government tries to limit its ever-rising subsidies paid to Medicaid (a program for the poor) and Medicare (for the aged). In Great Britain, where the budget of the National Health Service directly derives from the public purse, the government has sought to free resources from health in favor of other public activities. Thus there seems to be a general sense of crisis in health care, apparently calling for still more public intervention.

Table 1.1. Health Care Expenditure as a Share of Gross National Product (in Percent)

<i>Year</i>	<i>1960</i>	<i>1970</i>	<i>1980</i>	<i>1990</i>
Canada	5.5	7.1	7.4	9.0
Federal Republic of Germany	4.7	5.9	8.4	8.1
France	4.2	5.8	7.6	8.9
Italy	3.3	5.2	6.8	7.6
Japan	2.9	4.4	6.4	6.5
Netherlands	3.9	6.0	8.0	8.1
Sweden	4.7	7.2	9.4	8.7
Switzerland	3.3	5.2	7.3	7.4
United Kingdom	3.9	4.5	5.6	6.1
U.S.	5.3	7.4	9.3	12.4

Source: OECD (1990).

“crisis in hair care” and the media has not drawn our attention as much to the risk of walking around barefoot one day, as it has to the nightmare of not being able to afford medical treatment.

In looking for *peculiarities of health care* that have prevented such a comparison (so far), one comes across, above all, the three following characteristics.

1. Size of health care sector. Health care services constitute an industry of considerable significance. Its share in GNP today is close to ten percent in most Western industrialized countries (depending on the method of measurement and the delimitation of the industry; see Table 1.1 and Figure 1–1). This implies, on the other hand, that a great number of people owe their income to this industry (see Table 1.2). For economists, this is reason enough to gain knowledge of the subject and to analyze the industry, following the example of agricultural, energy, transportation, and recently tourism economics.

2. Public regulation of health services. Even more important than its sheer size is the fact that health services are regulated by the state to a considerable extent. Just consider compulsory health insurance in many countries, the fixing of fee schedules for medical services, or even the National Health Services of the UK and Italy, where almost all services are carried out by civil servants. These departures from market forces raise the question of whether an optimal allocation of scarce resources can be achieved.

3. Emotional significance. Finally, health and everything related to it creates many more emotions than other wants, which brings us again to the above-mentioned peculiarity of the good “health.” This leaves it to economists to find out which rules of allocation do justice to the specific character of this group of wants. In particular, conflict arises between the economic and the ethical way of looking at the allocation of scarce resources, both within health care and between health and other wants.

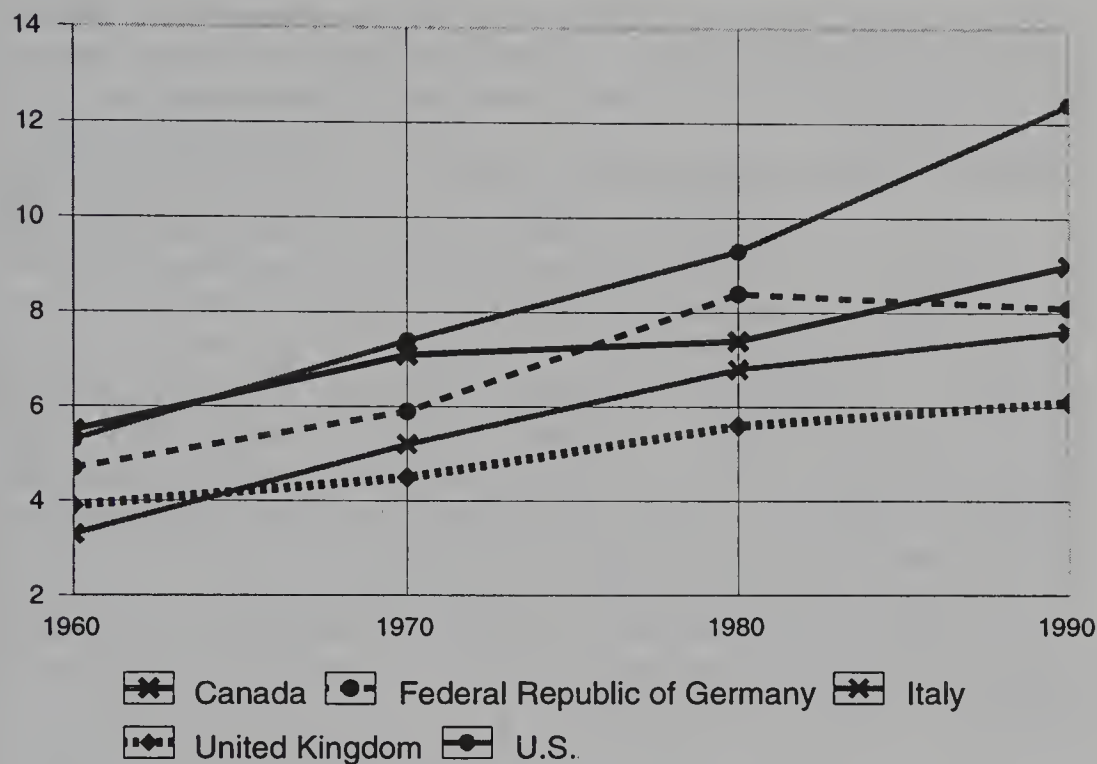


Figure 1-1. The development of the share of health in GNP in selected OECD countries (in percent). *Source:* OECD (1990).

Nevertheless, the development of cost in the health care sector should not become a fixation. First, the development of expenditure is the more appropriate term anyway, for we are dealing with a changing group of goods over time. Second, from a welfare point of view, the problem is not so much the *development* of expenditures of health care, but rather their *structure*. The economic principle generally requires that a given degree of satisfaction of one's needs be achieved with the minimum amount of scarce resources. Thus the demand for a group of goods or services may have grown—as has

Table 1.2. Employment in Health Care as a Share of Total Employment (in Percent)

Year	1970	1980	1989
Federal Republic of Germany	2.9	4.5	5.5
UK	3.1	4.7	4.7
Italy	1.6	3.9	4.3
Netherlands	4.0	6.4	7.1
Switzerland	2.8	4.6	n.a.
U.S.	3.7	5.3	6.3

Source: OECD (1991), Health Data File.

been the case in the services sector during recent decades—causing total expenditure on this group to increase in spite of perfect efficiency in the use of resources. Conversely, the shrinking of an industry does not imply efficient production.

CONCLUSION 1.1: From an economic perspective, the rules that determine the allocation of resources rather than the amount of expenditure on health are of interest. They indicate if the participants involved—both those supplying and those demanding health services—have incentives to use scarce resources economically.

One of the reasons for focusing on the *expenditure development* has to do with the wide spread *system of social security* in industrialized countries. A compulsory contribution is levied on an employee's salary, serving to finance health insurance, among other things. The concern is that increasing charges tend to turn wage-earners against the system of social security and to encourage moonlighting.

This argument, although reasonable at first sight, must be qualified, however. What counts is what one gets for one's money. As long as the contribution to social security is deemed equivalent to services received in the eyes of the individual, there is no incentive to withdraw from the system.

1.2 THE MICRO- AND MACROECONOMIC VIEWS OF HEALTH

In this textbook, chief importance will be given to the microeconomic approach. The behavior of an insured individual or a physician, the management of a hospital or a pharmaceutical company, is the focus of interest. Within the public debate, in contrast, the macroeconomic approach dominates, with the share of health expenditure in the social product usually serving as the reference quantity. Therefore, it seems appropriate to connect these two approaches and to clarify the above-mentioned criticism of trying to fix this share.

1.2.1 A Simplified Microeconomic View of Health

For an individual, health has a double function. On the one hand, perfect health represents a value of its own, a *target* that is to be reached as closely as possible. On the other hand, there are other aims in life. The gourmet who prefers a (fat) foie gras to a wholesome salad lets us presume that health is traded off for other objectives. And who has never crossed a busy street instead of using the underpass, only to save a little time? These modes of behavior contradict the assertion of health being an invaluable good (see statement No. 1 at the beginning of Section 1.1)

This contradiction rarely becomes obvious because no one sacrifices his health in an immediate sense. People only allow the probability of living healthy in the future to be a little smaller than it could be. This distinctive feature of health behavior will be brought up in Chapter 3. For a beginning, it is sufficient to record the fact that

individuals, in the end, trade off “health” (H) against all other aims, which for simplicity will be aggregated into “consumption” (C). This weighing up of objectives is, as usual in microeconomics, symbolized by a set of indifference curves (see quadrant I of Figure 1–2).

However, there are two characteristics that make health particularly important to most people.

I. The *consumptive benefit* that one can derive from one’s income depends on one’s state of health. Depression, for instance, the most widespread mental illness, makes it just about impossible for the person affected to enjoy the nice things in life. Similarly, in the event of a disease of the digestive tract, a foie gras is not a pleasure even to the gourmet.

This characteristic can be reproduced in quadrant I of Figure 1–2 by giving the indifference curves a particular form. As soon as the ratio of health to consumption becomes small, additional consumption does not have positive marginal utility any

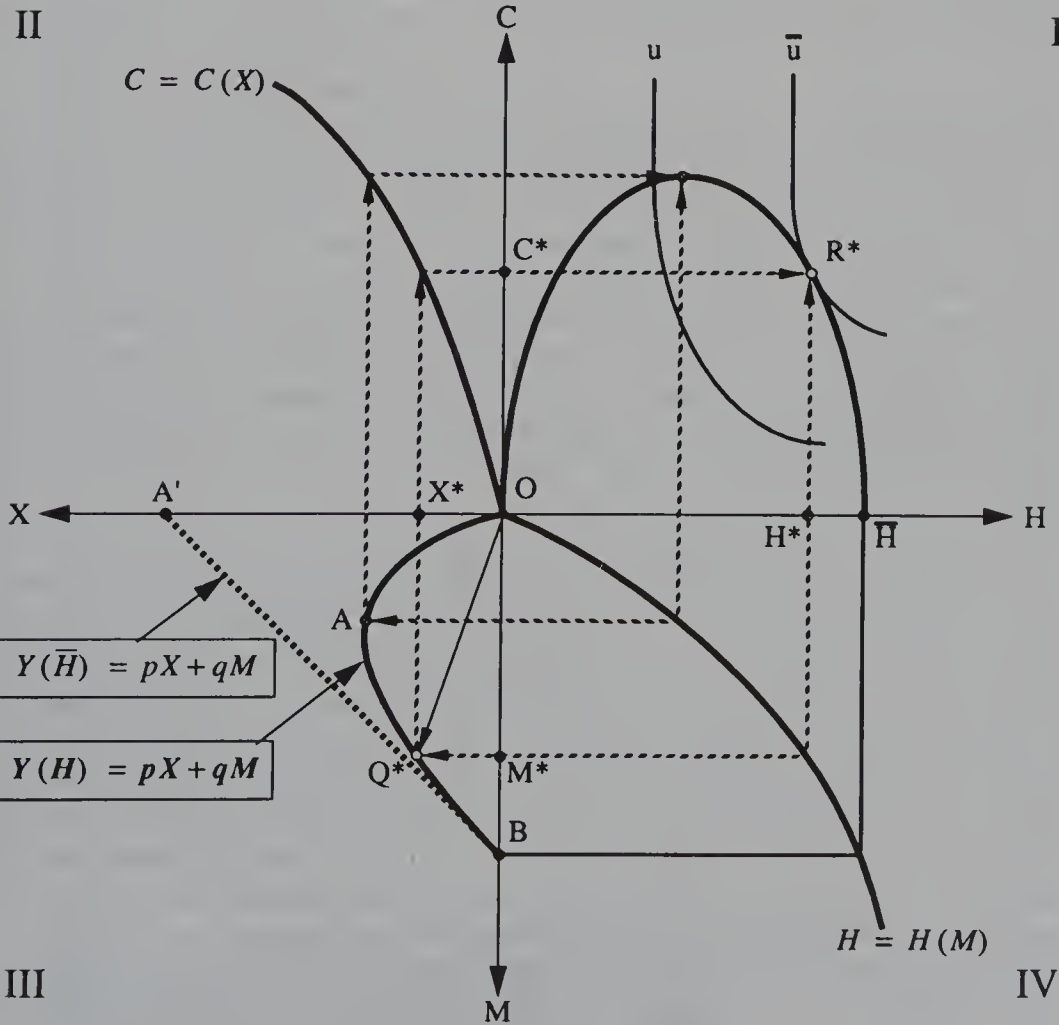


Figure 1-2. Health, consumption, and optimum share of income devoted to health.

longer, causing the indifference curve in the northwest area of quadrant I to run vertical rather than negatively sloped.

2. Only someone who is in good health can earn an *income* on the labor market.

This feature can be represented in quadrant III of Figure 1–2. This quadrant shows how the individual can divide his *budget* (disposable income, Y) between medical care (M) and consumer goods (X). The prices for medical care (net price q after deduction of insurance benefits) and consumer goods (p) are exogenous. The peculiarity of the *budget constraint*,

$$Y(H) = pX + qM \quad (1.1)$$

lies in the fact that disposable income Y depends on the state of health H , due, e.g., to inability to do overtime work. If income did not depend on the state of health (i.e., if H were fixed to $\bar{H}$), the budget constraint would be linear, as shown by the dashed line $A'B$. Point B is the point where the entire income is spent on medical care.

If, starting at point B , the expenditure on medical care is reduced, the state of health H deteriorates according to the function $H = H(M)$ shown in quadrant IV (see below). This decrease results in a reduction of income $Y(H)$ in the third quadrant. Consequently, the quantity of consumer goods rises at a decreasing rate up to point A . At this turning point, a further reduction of medical care lets the physical condition deteriorate so much that the associated drop in income neutralizes the savings in health expenditure. Beyond point A , a further reduction of M would result in a loss of income to such an extent as to require a reduction of consumption. And finally, at $M = 0$, income would drop to $Y = 0$ and with it purchases of consumer goods ($X = 0$).

Quadrant IV shows the amount of medical care M required to guarantee a given state of health. The more medical care (of a curative nature), the better the state of health. The individual's own efforts at preserving his good health (prevention) are not taken into consideration here. In Chapter 3 these efforts will be discussed in detail, though only taking into account the expense in terms of time, not in terms of money.

Finally, quadrant II shows the positive relationship between consumer goods (X) and consumable services (consumption, C) according to the equation $C = C(X)$. This equation and the one pictured in the fourth quadrant $H = H(M)$ allows one to assign to each point on the budget curve in quadrant III a point in the (C, H) space of quadrant I (see dashed arrows in Figure 1–2). The set of all these points indicates the individual's *production frontier*. It differs from the usual production frontiers known in microeconomics by running through the origin. That is because, as explained above, a very poor physical constitution ($H = 0$) is associated with an income of zero and consequently also with a consumption of $C = 0$. At first, it slopes upward with an increasing value of H . This means that, initially, better health enables one to consume more, not less (detailed reasons for this will be given in Chapter 3).

The tangent point of the highest attainable indifference curve and the production frontier indicates the combination of consumption and health (C^*, H^*) yielding maximum utility. Quadrant IV shows the corresponding *optimal expenditure on*

medical care M^* and quadrant II the optimal purchase of consumption goods X^* . Point Q^* on the budget curve in quadrant III finally indicates the individual's optimal budget allocation. Connecting Q^* with the origin, one can interpret the slope of this line as the "optimal health share." The steeper the line, the bigger the optimal share of income spent on medical care. These reflections can be summed up as follows.

CONCLUSION 1.2: Both the role of "consumption" and "health" as objectives and the tradeoff in their production, with "health" also serving as an input, can be represented as parts of a conventional microeconomic optimization problem.

1.2.2 Health Care at the Macroeconomic Level

The functions shown in Figure 1–2 can be aggregated over all individuals in principle. They are therefore valid at a macroeconomic level as well. The problem is their limited observability. The two quantities of utmost importance to the individual, state of health H and consumption C , are registered only rudimentarily by official statistics. On the other hand, the quantity of medical treatment (M^* or its monetary value qM^*) and purchases of consumer goods (X^* and pX^* , respectively) are well known. As a consequence, attempts at controlling health care usually start from these quantities.

Why should there be a need for *controlling health care expenditure*? Chapter 5 will give an answer to this question. It will be shown that subsidies for health insurance, possibly even compulsory insurance for individuals combined with compulsory open enrolment by insurers, can be justified. However, more far-reaching interventions such as stabilizing the share of health expenditure in the Gross National Product (GNP), limiting the increase in costs to a specified percentage, or stabilizing contributions as a quota of wage income can only be justified by postulating a collective welfare function.² A "benevolent dictator" would then have the duty of maximizing this function, possibly against the interests of certain individuals [Frey (1983), Chapter 3]. We will in the following dispense with such an assumption because it is incompatible with the principal hypothesis of economic theory, according to which all individuals (therefore also political decision-makers) pursue their own interests. In contrast, we would like to prove that *fixing the share* of health care expenditure in the GNP or in wage income can result in *efficiency losses* that accumulate over time.

As soon as the state is involved in the financing of health care, its *budget is affected* by individual decisions with regard to health. The more an individual relies on medical treatment, the bigger are the expenditures for the state at least in the long run: additional residencies for medical students, higher subsidies for planned

2. The United States instead has publicly financed programs for the medical supply of senior citizens ("Medicare") and those depending on welfare ("Medicaid"). In the Federal Republic of Germany the premiums for statutory health insurance are tied to income from salaried employment, in contrast to, e.g., Swiss social health insurance where the premium is not linked to income. An important objective of the German "Health Reform Law," in force since 1989, and the "Health Structure Law," in force since 1993, is to stabilize the contribution rate at around 13 percent.

investments of hospitals, higher subsidies to hospitals for covering their operating deficits in many countries, higher payments for the medical treatment of senior citizens in countries such as the United States and Germany, to name just a few of the delayed influences on the budget. These influences also exist when individuals decide to buy and use consumer goods (trash collection, road construction), but they are not as important as in health care.

On the other hand, public expenditure brings in *votes*, helping to secure re-election to politicians. Chapter 11 will focus on this issue; here we would simply like to point out that politicians prefer a specific division of the budget to any other division, just as the individual of Figure 1–2 prefers the division of his budget according to Q^* to other solutions. But by pushing through its preferences regarding the budget, the government more or less fixes the share of health in the GNP. The consequences of this fixation will be highlighted in the following section.

1.2.3 The Problem of Global Budgeting

In their struggle against the “cost explosion” in health care, governments refer to the figures of Table 1.1 to show that the share of health in the Gross National Product is “too high.” Assuming that one could succeed in stabilizing this share at a certain (lower) percentage of the GNP, what could be gained?

To answer this question, the transition from the micro- to the macroeconomic level is made in the simplest possible way by applying the features of Figure 1–2 to all individuals of a society, who are assumed to be identical. Accordingly H becomes the number of person-years spent in good health, qM the national health expenditure, and pX the aggregate consumption expenditure. The allocation of the GNP between these two categories is given by point Q^* in the third quadrant, symbolizing the aggregate outcome of individual decisions. For the sake of simplicity, the health share strived for by the government is assumed to coincide with the optimal Q^* in the initial situation.

This equality may now be disturbed by an exogenous shock, for example, an *improvement in medical technology*. The function $H(M)$ is then shifted to $H^+(M)$ in the fourth quadrant of Figure 1–3, implying that a given supply of medical services now guarantees a better health status than before. This leads to an outward shift of the budget curve in quadrant III. In all, the production possibility frontier shown in the first quadrant shifts outward. Society evidently can reach new (C, H) combinations, lying to the right and above the old optimum R^* .

Let us now assume that the government learns about the improved medical technology but wants the *share of the GNP devoted to health to remain unchanged*. This means that it is trying to keep the global allocation between consumption and health expenditure the same, which requires the new point Q^+ to lie on the same radial line as Q^* . Thereby a new (C, H) combination would be reached, symbolized by R^+ in the first quadrant (see the solid straight lines in Figure 1–3).

If the (identical) members of this society value their health as highly as the relative steepness of their indifference curves in the first quadrant indicates, they would opt for R^{**} instead, implying the following changes:

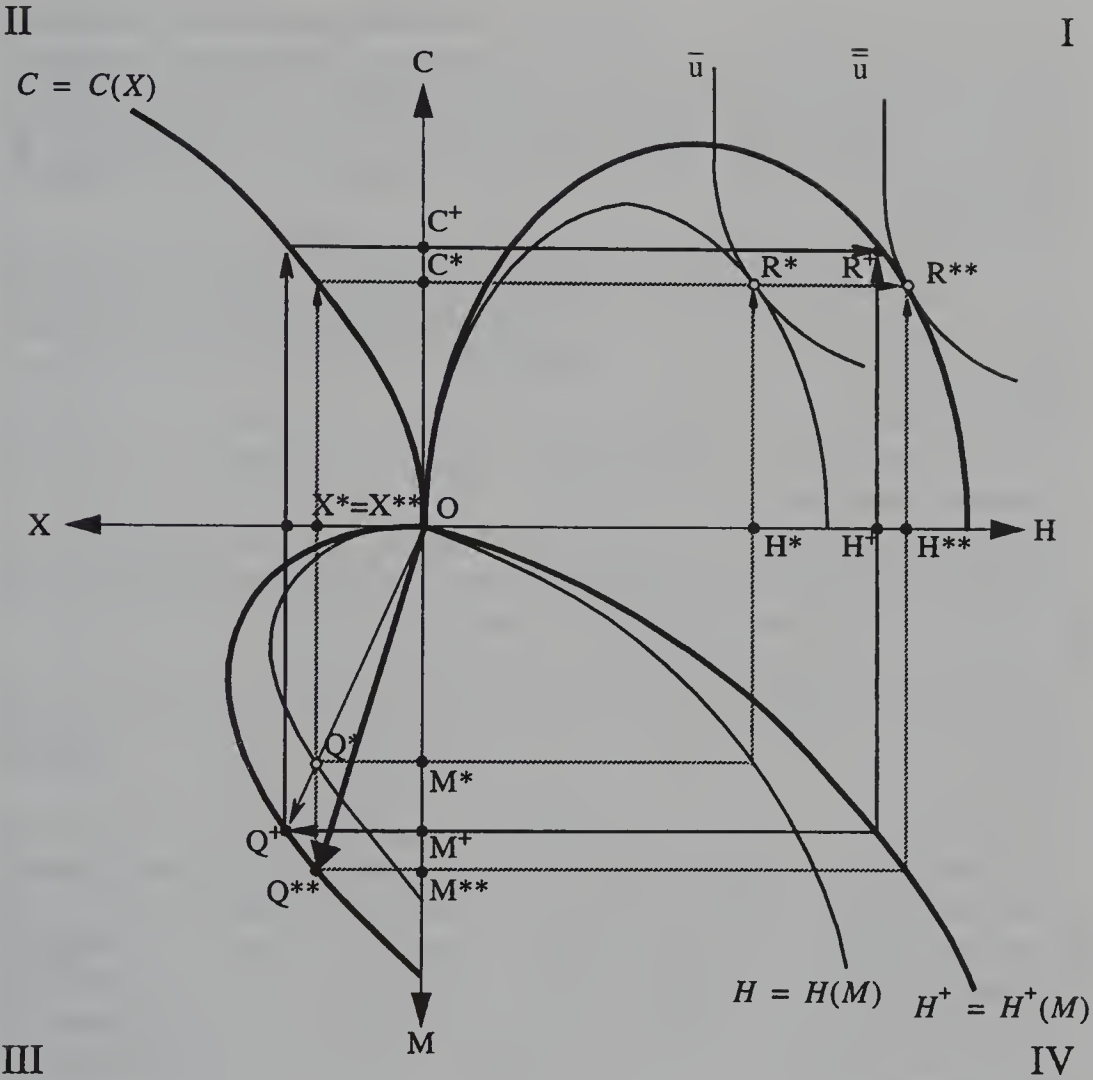


Figure 1-3. Change in medical technology and adjustment of optimal health share.

1. By assumption, the enlargement of the choice set will be used exclusively for an *improvement of health status* (see the transition from R^* to R^{**}).³

2. This is achieved by an *increasing use* of medical care (new value: M^{**}) not only in comparison to the initial optimum M^* , but also in comparison to M^+ fixed by the government.

3. Due to the improved state of health, there is a rise in *labor income* and therefore in available financial means, so that this increase of M is possible without changing consumption expenditure ($X^{**} = X^*$).

4. The new optimal budget allocation Q^{**} (in quadrant III) of society calls for an *increase of the optimal health share*. This is shown by the rotation of the ray from the origin to OQ^{**} , which is steeper than the original ray OQ^*Q^+ .

3. This case is presupposed in order not to complicate the figure. The same conclusions can be drawn from a predominant tendency toward improved health.

Now if politicians hold on to the existing allocation of the public budget, they force individuals as a group to settle with Q^+ or R^+ , respectively. This means, for example, that medical services are not available because of restricted access to medical schools or that hospital treatments are not available due to a lack of investment in equipment. Thus individuals reach a lower level of utility in quadrant I than the highest possible value at point R^{**} .

The example of technological change in medicine stands for just *one of many possible changes* that can lead to discrepancies between the optimal health share in income from the perspective of the individual and a politically established health share in the GNP. Going through the four quadrants of Figure 1–3, one comes across the following changes.

I. The *preferences* between health and consumption may change, for example, in the course of a fitness wave.

II. The relationship between consumer goods and consumption services does not remain unchanged either. The better the individual's education, for example, the more productive he is in nonmarket work as well, and the more *consumption services he is able to extract* from a given quantity of consumer goods.

III. The budget constraint is subject to continuous change. *Increasing incomes* shift it away from the origin, whereas its slope becomes steeper if prices of consumer goods increase.

IV. The connection between the use of medical care and the state of health is not only modified by technological developments in medicine but also by *environmental factors* or an increase or decrease of health-enhancing efforts on the individuals' part, the latter especially as a result of insurance coverage (see Chapter 6 for a detailed discussion). Besides, the performance of physicians and hospitals generally *falls short of their efficient levels* as a consequence of the remuneration system. Therefore, the location of the function $H = H(M)$ also reflects the system of remuneration (see Chapters 7 and 9).

This clearly shows that there are a lot of reasons for the politically determined health share to differ from its optimal value. In sum, one has

CONCLUSION 1.3: The political fixing of a share of health expenditure in the GNP would have to be founded on a social welfare function, which does not exist. Rather, it entails a risk of causing losses in efficiency that tend to accumulate over time.

1.3 "ECONOMICS OF HEALTH" VERSUS "ECONOMICS OF HEALTH CARE": SOME GENERALITIES

Up to this point, reasons for the existence of "health economics" as a discipline were given and the relationship between macroeconomic and microeconomic ways of looking at health and health care examined. In order to classify issues and to discuss them later in this book, we propose the distinction between economics of health and economics of health care.

1.3.1 Economics of Health

In their titles, some publications in the field (textbooks and anthologies) talk of "economics of health," while others talk of "economics of health care" or "economics of medical care."⁴ This differentiation suggests that *health itself* is an interesting economic issue. There are a great many questions connected with health, which have nothing to do with what we call "health care," that is, the provision of medical services by physicians and other professional suppliers. There are both positive and normative ways of looking at the problem of health.

One of the most important questions raised by the economics of health concerns the valuation of health in money terms, that is, the weighing of good health against other objectives, in particular consumption of other goods. This *normative* issue ("How much consumption should society sacrifice to increase the average life expectancy by one year? How much should an improvement of the state of health measured by suitable indicators be worth?") arises primarily in connection with public projects, financed by levies, such as taxes or social security contributions, whenever they have an effect on the life expectancy or health of citizens. The typical feature is that the individual does not weigh health against consumption himself. Rather government or parliament *decide on behalf of the citizens*, and it is the task of health economics to provide these authorities with useful, that is, welfare theoretically, sound rules for making such decisions. This will be the subject of Chapter 2.

The *positive* branch of health economics, in contrast, deals with individual health behavior, using the instruments of modern microeconomic theory. The basic paradigm of rational choice is applied to health; that is, the individual is considered to maximize utility, which depends, among other things, on material consumption as well as on health. With the help of such a model (see Chapter 3), one can examine the effect on the individual's "demand" for health (i.e., his health behavior) of changes of his budget constraint, that is, of his income, or of changes in relative prices, caused, for example, by changes in health insurance coverage.

1.3.2 Health and Consumption of Health Services

The connection between the health status of the individual (or the population on the whole) and consumption of medical services builds the link between "economics of health" and "economics of health care." There are two relationships to be distinguished. They are represented by the headings "production function of health" and "demand function for medical services."

I. In the first case, the point at issue is the question, how much do *medical services contribute* to the preservation or improvement of health status? This question sounds rather trivial, but becomes legitimate when the enormous increase in health care expenditure over the past decades is compared with the relatively modest growth of life

4. Examples are Feldstein (1993), Folland, Goodman, and Stano (1993), Fuchs (1986), Jacobs (1991), Klarman, ed. (1970), McGuire, Henderson, and Mooney (1988), Mooney (1992), Newhouse (1978), Perlman, ed. (1974), Phelps (1992), Van der Gaag and Perlman, eds. (1981), Zweifel and Frech, eds. (1992).

expectancy during the same period. It is therefore not surprising to see that more and more people, having become sceptical about classical medicine, turn to alternatives such as nonmedical methods and lay medicine.⁵ The empirical quantification of the influence of health care on health (see Chapter 4) thus also belongs to the domain of economists, statisticians, and econometricians.

In spite of the objections mentioned, this influence can be assumed:

- a. To be *positive*: Higher consumption of medical services results in better health.

For logical reasons, this effect is supposed:

- b. To set in with a *timelag*. It is to be expected in empirical investigations that consumption of medical services in a period t does not give rise to better health before some later period (say $t + 1$).

2. The second relationship runs from the state of health to the consumption of medical services, the so-called *demand function for medical services*. In this function, the state of health represents the explanatory variable. Here, the plausible assumptions are that:

- a. The theoretically expected effect is *negative* (worse health goes along with increased consumption of medical services) and
- b. This relationship applies to *simultaneously* measured values of the two quantities.⁶

1.3.3 Economics of Health Care

The demand function for medical services already belongs to the domain of the economies of health care. Taking the productivity of medical services as a given fact, the economics of health care—in its positive form—raises the issues of what determines the quantity and quality of medical services produced in a society. In its normative form, it sets out to investigate the mechanisms of *production of these services* and its distribution among demanders, identifying those mechanisms that are particularly *appropriate from an economic point of view*, taking the scarcity of resources used in their production into account.

This leads us to the issues of organizing and financing medical services. Microeconomic analysis, with its concept of “incentives,” is well suited to investigate the effects of alternative regulations.

The terms *financing* and *incentives* in fact amount to two sides of a coin.

1. On the one hand, they describe how the *recipients of medical services* (the patients) *pay* for these services. Here, the point at issue is health insurance with its incentive effects on the insured, which is dealt with in Chapters 5 and 6.

5. Classical medicine has been criticized with particular vehemence by Ivan Illich (1975).

6. If it were possible to achieve equality between actual and desired state of health, the demand function would be simply the inverse of the production function of health, comparable to a factor demand function derived from a production function (see Section 4.4).

2. On the other hand, they describe the way the money is *channeled to the providers of services*. Here, the analysis deals with the incentive effects of alternative payment systems for physicians (Chapters 7 and 8), hospitals (Chapter 9), and suppliers of drugs (Chapter 10).

1.4 A SYSTEMS ANALYSIS OF HEALTH CARE

To conclude this introduction, an attempt is made to show the issues dealt with in this book in a diagram in the guise of a strongly simplified systems analysis (see Figure 1–4). The diagram begins with the individual and his objectives, among them to live as long and healthy as possible while consuming as much as possible. In his health-related behavior, he will be guided by a variety of incentives, which are shaped especially by the form of his health insurance (see Chapter 5 and 6).

However, the state of health is not fully under the individual's control. While chance has an important impact, the individual can still opt for more or less health by his behavior that exerts a systematic influence (see Chapter 3). This also means that an additional day spent in good health has its "price" even if there is no market where it can be traded. This price is the *opportunity foregone*, summarized in additional consumption the individual could have afforded.

Being able to opt for a certain state of health also calls for a decision on how to reach this state. Very often people are willing to go through a phase of illness with a minimal use of pharmaceuticals and no additional medical treatment. Instead they use their time as an input. Presumably, material incentives influence the choice between one's own and other inputs (see Chapter 3; for empirical evidence see Chapter 4). If this choice increasingly tends towards inputs provided by a third party (i.e., health care services), then the result can be a cost explosion in health care.

The extent to which changes in individual health-related behavior affects health care expenditure is very much dependent on the *physician*. He, positioned just below the patient in Figure 1–4, performs as a "*gatekeeper*" to health care. His choice between outpatient or inpatient treatment has grave consequences in terms of costs incurred. This is due to the fact that a hospital stay typically goes together with the use of a good deal of nursing services as well as expensive technology. Again we will look for material incentives that may govern a physician's decisions regarding his own services (Chapter 7) and those of other suppliers (Chapter 8). This view, of course, is totally *contrary* to traditional medical ethics claiming that a decision to use a treatment is only based on medical need.

By referring a patient to the *hospital*, the physician passes on part of the demand for medical treatment. Closer inspection reveals *material incentives* at work in hospitals as well, which can influence the conduct of chief physicians (see Chapter 9). Traditional economic theory has a lot to say about the way profit-oriented enterprises respond to and manage changes in the environment. We will see to what extent this body of knowledge is applicable to nonprofit organizations such as hospitals.

Finally, doctors as well as hospitals determine the inpatients' demand for *pharmaceuticals*, another input to the treatment process. In this context we will be interested to see (in Chapter 10) how prices are determined in the pharmaceutical

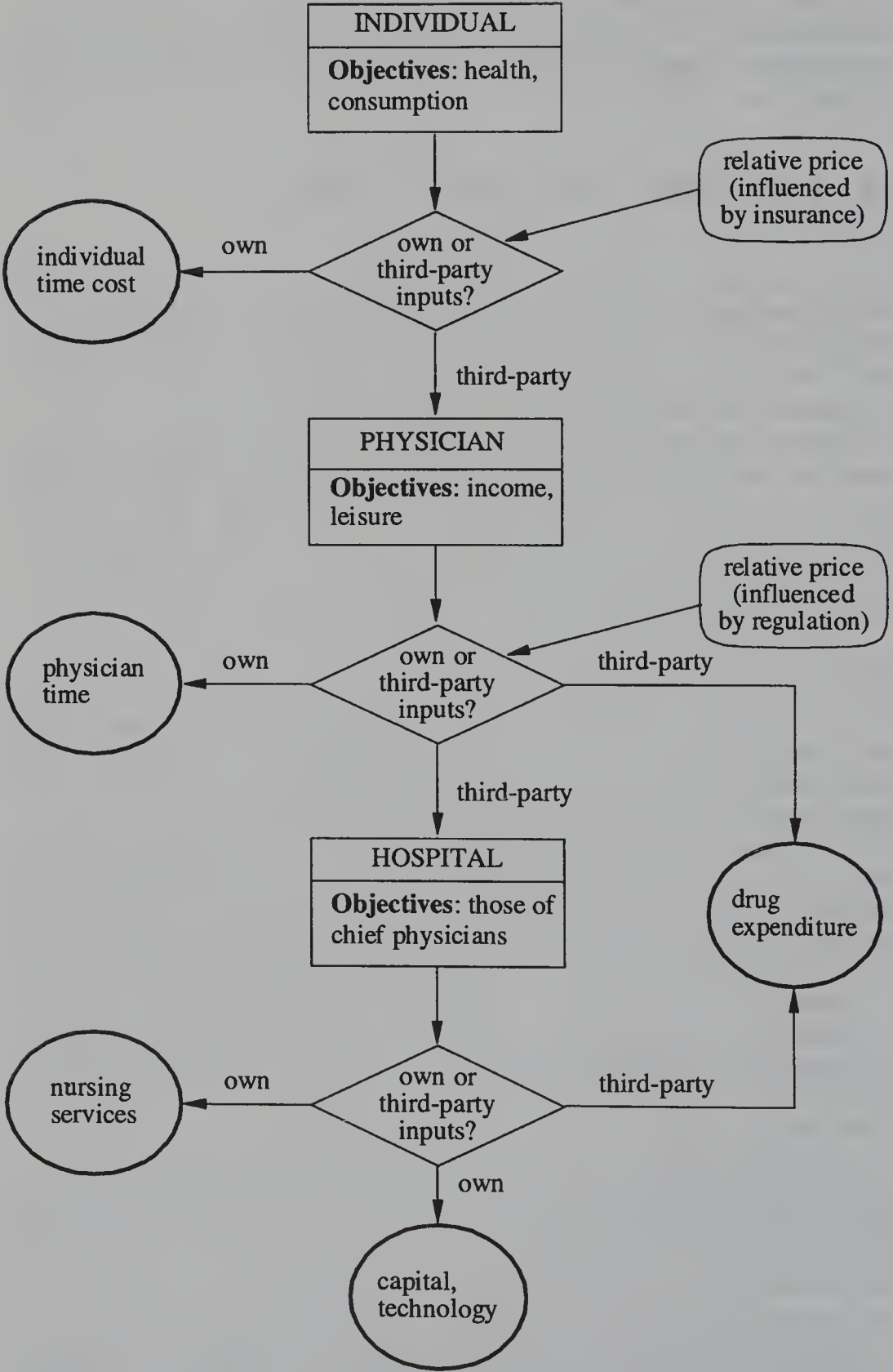


Figure 1-4. A systems analysis of health care.

market and how attempts to promote price competition may affect advances in technology, that is, the widening of therapeutic choices. Besides the market for pharmaceuticals, *other derived markets* could be examined, as, for example, those for health aids, mental treatment, and medical appliances but also the labor market in health care. In order to keep this book within manageable proportions, we will not deal with these derived markets.

The circles of Figure 1–4 stand for *resource inputs*, which are triggered by health care. If incentives are such as to induce individuals increasingly to substitute physician time, pharmaceuticals, and hospital services for their own time input in favor of health, this cannot happen at constant costs. As in any other industry an increasing volume of services entails increasing costs. The peculiarity in health care is that a single medical treatment and a single day in hospital have become so much more expensive. It is this cost increase that became what we called the “cost explosion.” This aspect of the problem also goes back to the *particular incentives governing decisions in health care* discussed in this book. Future challenges of health policy, such as aging of the population and faster technological development in medicine, will be focused on in Chapter 11. Finally, we will give some rather basic recommendations regarding health policy (Chapter 12).

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2 ECONOMIC VALUATION OF LIFE AND HEALTH

2.1 THE VALUATION OF HEALTH AND ITS APPLICATIONS

“Life is the most precious good and cannot be measured in terms of money.” Most people would probably agree with this statement. However, decisions affecting lives are not only made by individuals, but (necessarily) also by parliaments and public authorities on a regular basis. This implies weighing the preservation and lengthening of human life against the input of scarce resources having alternative uses. Thus prolonging human lives requires sacrificing the consumption of goods and services that could be produced using these resources. Examples for such decisions in the public sector can be found not only in health care but also in many other sectors, especially those related to transportation and the environment.

Many countries that have a national health service or a national health insurance let political authorities decide on new pharmaceuticals, new therapies, and new medical devices to be covered by the plan. As a rule such new products involve *additional financial expenditure*—so we are not dealing with cost-reducing process innovations. Instead, they bring about therapeutic advantages, often by reducing the risk of early death in a certain risk population.

Thus the provision of a mobile coronary unit with total costs of several million dollars may help treat heart attack patients on the spot, serving to reduce significantly the number of those dying prior to arriving at the hospital. Long-term drug therapy of hypertonic patients using antihypertensives may prevent a heart attack as well. Again, a reduced mortality risk is accompanied by considerable costs to the economy for research and development of pharmaceuticals. Another well-known example is the installation of dialysis equipment for patients with chronic kidney insufficiency.

Beyond health care, there are numerous other examples where “life” and “costs” have to be pitted against each other. Communities and countries have to decide whether notorious scenes of accidents, for example, narrow blind curves, should be eliminated by widening and straightening. In residential areas, the opposite can be appropriate; for

example, planting trees and installing bumps may reduce risks for playing children by lowering driving speed. All those measures of course involve additional expenditure from the public purse.

Environmental policy provides us with further examples. Costly “redundant” safety systems in nuclear power plants do not only reduce the likelihood of catastrophes with thousands of deaths (as in the case of Chernobyl in 1986) but also the emission of radioactivity exposing the population to an increased risk of suffering from leukemia. Costly filters retaining sulfur dioxide and other harmful substances from the combustion of coal serve to improve the quality of air and make suffering from diseases of the respiratory tracts less likely.

In all the areas of application mentioned so far, rational decisions can only be made by authorities if there is a comprehensive and precise *valuation* of future advantages (and possibly disadvantages) resulting from a specific measure taken, permitting comparison with the present value of the cost stream associated with the project. It is obvious that such an analysis of costs and benefits is most helpful to the decision-makers if the measures are commensurable, that is, that they have a common unit of measurement. As the project costs are usually measured in monetary units, it seems desirable to measure all advantages in the same way. Of course, this implies that the prolongation of human life or the improvement of the state of health due to the realization of a project must be valued in monetary terms as well.

As there are severe objections to a monetary evaluation of health, let alone human life as such, economists have developed alternative methods of evaluation that work without such a monetary evaluation. In the following we shall also examine the possibilities and limitations of these approaches.

This chapter is organized as follows. In Section 2.2 we give a general overview of the different approaches to evaluation in the health sector. Section 2.3 is devoted to the *monetary* evaluation of the *length* of human life. First (in Section 2.3.1) we discuss moral objections to the idea of comparing human life and health with financial outcomes, and second (in Section 2.3.2) we present two basic concepts of monetary evaluation, the human-capital and the willingness-to-pay approach. The measurement problems involved in the second approach are treated in Section 2.3.3. In Section 2.4 the scope of the question is broadened to the monetary evaluation of the *quality* of life, that is, of health status.

2.2 METHODS OF EVALUATION IN HEALTH CARE

The existing methods of evaluation in health care can be classified according to the units in which positive and negative effects of a project (medical or other) are measured. For the sake of clarity of exposition, we shall discuss the alternative methods using the example of a medical intervention whose only “negative effect” consists in the use of scarce resources (labor, raw material etc.). We shall further assume that these resources have a market price so that the costs of the intervention are already measured in monetary terms.

As regards the positive effects, we shall distinguish the following three possibilities of measurement,

- a. In natural units on a unidimensional scale;
 - b. In units of a cardinal utility function, which is used to transform the multidimensional concept of “health” into a scalar index;
 - c. In monetary units.
- (a) The “natural” scale mentioned above may be (1) a clinical parameter, for example, the lowering of blood pressure by x mm Hg, or (2) the length of life in years. The application of this method of measurement makes sense only if the alternatives to be compared have one specific effect and no side effects. In the case (1), the alternatives may be two different drugs for lowering blood pressure, both without side effects, while in case (2) they may be two safety measures designed to avoid fatal traffic accidents.

The method of evaluation applicable under these circumstances is called cost-effectiveness analysis (CEA). It can be used only for the comparison of mutually exclusive projects. In the two examples cited above, the CEA criterion is given by

$$t_{CEA}^1 = \frac{\text{costs in units of money}}{\text{benefits in mm Hg}}$$

and

$$t_{CEA}^2 = \frac{\text{costs in units of money}}{\text{benefits in additional life years}}$$

respectively.

Of two alternative projects, the one with the lower value of t_{CEA} is to be preferred.

The limitations of the CEA approach are obvious.

1. Its application implies that who obtains the additional life years is of no relevance. However, this criticism applies to all methods of evaluation discussed here and in the following.

2. It does not lend itself to the evaluation of projects with several different (positive) effects. If, for example, the safety measure mentioned above does not only avoid fatal accidents but also those with injuries to persons, the CEA is silent because it has no device to aggregate multidimensional effects.

3. While CEA provides a rank order of preference among mutually exclusive projects, it does not answer the question which of these projects should be realized and which should not. In particular, it is not suited for the evaluation of individual projects. The only practical case in which the CEA provides a sensible answer is the one of a fixed budget, which is to be divided among a fixed set of possible projects. In this case the allocation of funds should be started at the project with the lowest t_{CEA} value, and it is to be continued until the budget is used up completely. The problem with this procedure is that it leaves open the question of how the size of the budget itself can be determined in a rational way.

- (b) The second alternative takes account of the multidimensionality of the concept of “health” by trying to encompass all effects of an intervention—prolonging life and

changing health status—in the evaluation, using a suitable weighing scheme. Its most important instrument is a normalized cardinal utility function, which assigns to each conceivable health status a value on a scale on which “death” defines the origin, while the state of perfect health has the value of 1. The remaining values of this utility function are determined such that a representative individual is indifferent, for example, between the following two scenarios: “Living for one more year in the state of health with the utility value $1/2$ ” and “Living for another six months in perfect health” (for the details of this procedure, see Section 2.4). Through this utility function, all health consequences of a project are made comparable so that they can be aggregated into an index value. This index value may be interpreted as “quality-adjusted life years (briefly QALYs) gained.”

The corresponding method of evaluation is called *cost-utility analysis* (CUA). As cost-effectiveness analysis (CEA), it is suitable only for comparing mutually exclusive projects. Its criterion for comparison is defined by

$$t_{\text{CUA}} = \frac{\text{costs in units of money}}{\text{benefits in quality-adjusted life years (QALYs)}}$$

Out of two projects, the one that has the lower t_{CUA} value is to be preferred. Unlike CEA, the CUA is suitable for comparing medical interventions of heterogeneous kind and purpose. The effects in different (e.g., clinical) dimensions are made commensurable through evaluation in utility terms. By the same token, CUA may also be used for comparing projects in health care with projects in the remainder of the economy (investment in education, for example).

The results of cost-utility analyses may be arranged into “league tables” of medical interventions. These league tables show the observer—and the political decision-maker who is responsible for the allocation of resources in the health care sector—the cost at which a comparable gain in quality of life may be accomplished, depending on the medical intervention selected. An example of such a league table is given in Table 2.1.

This approach has its own limitations, however, which must be cited in addition to the objection mentioned above, viz. that the distribution of the QALYs gained among individuals may be of relevance.

1. It must be decided whose utility function is to be used when evaluating the different states of health.
2. Precisely as CEA, CUA provides only a ranking of mutually exclusive projects, but it does not say how far up on the scale of t_{CUA} values projects should be realized. This question can only be answered when the budget is given. However, the optimal size of the budget is not determined (again as in CEA).

(c) Cost-benefit analysis (CBA) goes one step further than CUA in that monetary equivalents are assigned to prolongations of life and change of health status (for more details concerning problems involved and evaluation method, see Section 2.3).

In CBA, both negative and positive effects of a project are expressed in units of money. This is the only one of the methods of evaluation discussed here that is suited

Table 2.1. League Table of Medical Interventions: QALYs Gained per £1,000

Therapy	QALYs per 1000 £ ^a
Stop-smoking campaign	5.88
Implantation of pacemaker	1.43
Hip replacement	1.33
Bypass operation in case of severe angina pectoris	0.96
Control of serum cholesterol	0.59
Kidney transplant	0.33
Breast cancer mass screening program	0.29
Heart transplant	0.20
Bypass operation in case of light angina pectoris	0.079
Hospital dialysis	0.068

^a Present value of all future costs, in prices of 1983/1984.
Source: Williams (1985).

to evaluate any conceivable project in its own right. The criterion for decision making in CBA is given by

$$t_{CBA} = \frac{\text{cost in units of money}}{\text{benefits in units of money}}$$

with realization of the project recommended if $t_{CBA} < 1$. Equivalently, one can recommend a project if $T_{CBA} > 0$, where T_{CBA} is defined by

$$T_{CBA} = \text{benefit in money} - \text{costs}$$

The application of this latter rule is justified in welfare economic terms with reference to the criterion of “potential Pareto improvement” (Kaldor–Hicks–Criterion).

It is easy to see that CBA is suitable to answer some of the questions that are left open by the two other instruments, in particular the question of the total amount of money that should be spent for prolonging life and enhancing its quality. As it is not possible to dodge this question forever if rational decisions in the public sphere are desired, a cost-benefit analysis that measures negative and positive consequences of each project on the same scale is eventually required. In view of this necessity, one may ask whether cost-utility analysis with its enormous effort of evaluation based on a utility function is justified.

CONCLUSION 2.1: Unlike cost-benefit analysis (CBA), cost-effectiveness analysis (CEA) and cost-utility analysis (CUA) circumvent the problem of a monetary evaluation of life and health. However, they provide only a relative evaluation of mutually exclusive

projects, whereas CBA permits evaluation of each project on its own.

The obvious problem of cost-benefit analysis is assigning a monetary value to changes in health status and even in length of life. The remainder of this chapter is devoted to attempts to solve this problem.

2.3 MONETARY EVALUATION OF LENGTH OF LIFE

2.3.1 Ethical Objections and Justifications

Economists are said to be people “who know the price of everything but the value of nothing.” The idea of assigning a value measured in monetary units to human life may therefore meet with widespread disapproval. The objections raised are of two different kinds. A principal attitude is that any attempt in trying to compare life and money is morally reprehensible. On a more pragmatic level, such a comparison would be accepted, but any other result than that of an infinite value of life would be called into question. These two arguments will be discussed in turn.

2.3.1.1 *Objections against Weighing Life against Money*

The weighing of life and freedom from bodily harm on one side and money on the other is considered profane by moral rigorists, whether inspired by Christian belief, the oath of Hippocrates, or humanistic philosophy of life. Sometimes economic approaches to these valuations are even put on a par with the euthanasia programs of the Third Reich. Does such a valuation not imply that it is acceptable to kill those whose “value” does not cover the cost of living, such as food and medical treatment?

First, this argument fails to take into consideration the *morally relevant difference* between actively intervening and letting nature run its course, that is, between the act of killing incurables and making do without life-support devices. This distinction plays a major role in the discussion on “dying in dignity.” Of course it can be said that refusing to give a person essential food for free is morally equivalent to killing that person. However, the greater the efforts required to save a human life, the less valid this argument is. For example, let us assume that rescuing a group of miners buried in a pit would cost thousands of billions of dollars, causing the rest of the country to survive in misery. Can a refusal to rescue the miners then be treated as equivalent to killing them?

Second, we have to keep in mind that a lot of public decisions do not affect “identified” but “*statistical lives*.” If life-threatening dangers of personally identified individuals are involved, politicians usually are expected to do everything possible to save the lives of the victims, probably because the public is aware that even a maximum effort would only consume a very small percentage of the Gross National Product. Let us assume it would take 10 million US\$ for every miner buried in the pit. This does not imply that the public would consent to the government spending the same amount on averting a risk of one in a thousand for one thousand miners of becoming buried. “Statistical” lives tend to cause less emotions than “identified” lives.

Realizing that the issue is not so much weighing life against money but rather *length of the life span* against money could help to de-emotionalize the subject. When thinking of medical treatment or public security measures, we usually think of lives actually saved. In reality, however, life is prolonged only (often with some heroic performances of high-tech medicine resulting in a gain of a few months), for in the long run everybody has to die. So actually a weighing of an extended life span against a better quality of life is what this is all about, since “more money” means more of consumption possibilities, permitting a higher quality of life. Therefore the moral argument of an inadmissible valuation of life in terms of money is not as convincing as it appears at first sight, but see Mishan (1971) and Williams (1992).

However, the *main justification* for elaborating and applying an economic approach is based on the simple observation, mentioned in Section 2.1, that political decisions involving such valuations have to be made regularly. The mere fact that some action is undertaken (or refrained from) implicitly means that a weighing of (prolonged statistical) lives against money has occurred. The economic calculus facilitates awareness of this fact, helping to make policies more consistent.

Conversely, doing without an economic valuation of “life” entails risks of its own. A country may take a measure (such as introducing new medical technology) resulting in costs of 1 million US\$ for avoiding premature deaths failing to take into account alternative measures (for example, road works) that could do the same for only 50,000 US\$. Society as a whole will then be deprived of a longer life expectation *and* of higher consumption.¹

For decades now, parliaments in many countries have usually approved public projects only after a thorough valuation of all of their costs and benefits. But when it comes to human lives, there has been a lack of an approved procedure to evaluate them in monetary units. As a substitute, it has become customary to mention such costs and benefits known as “intangibles,” without, however, considering them as net benefits in the actual evaluation process. As a consequence, projects with a high risk in terms of human lives are too easily approved, while those with a lower risk tend to be rejected. Ironically, those who are against a valuation of life in monetary units thus obtain the exact opposite of what they intend.

We believe that public policies in a *democracy* should not only be consistent in the above-mentioned sense but also capable of reflecting the preferences of the citizens involved. Since permitting affected citizens to express their preferences is a prerequisite of democracy, moral condemnation of weighing money against life is at odds with the democratic principle.² By determining the preferences of citizens regarding life span and quality of life, economic analysis thus also serves to enhance the process of democratic decision-making.

1. This reference to society as a whole presupposes that life years won or premature deaths avoided do not depend on *who* is obtaining them (see Subsection 2.3.2.2.3).

2. The persons affected are not only those having a certain disease, but all accountable people with a positive probability of suffering from the disease (in the future). For those who cannot be held accountable (young or mentally ill persons), the preferences of suitable agents should be considered. According to this delimitation, the preferences of men regarding the treatment of gynaecological disorders, for example, would be irrelevant; however, see also the discussion of external effects in Subsection 2.3.2.2.4.

These considerations make clear that expressing the value of a human life in monetary terms is not the same as estimating its market or financial value. It rather amounts to describing the preferences of the individual concerned or of society at large among mutually exclusive alternatives.

CONCLUSION 2.2: Since many public decisions inevitably imply a weighing of prolonging statistical lives against other goods, it is advantageous for society to make such a valuation explicitly. However, the preferences of the citizens should be reflected in this valuation.

2.3.1.2 *Arguments against a Finite Value of Life*

The reliance on the preferences of the individuals stated at the end of the previous section gives rise to another point of criticism. While the idea of an economic valuation of life is not questioned as such, any resulting value less than infinity is deemed unacceptable. It is argued that there are in principle only two possible approaches to determining the value an individual places at his life.³ One either has to find

1. The amount an individual would be willing to pay to avoid certain (and immediate) death; or
2. The compensation that would have to be paid to the individual to make him accept his (immediate) death.

The amount determined according to formulation (1) is not very informative, as most people would be willing to give up their entire wealth including their future income stream (except maybe for a small reserve to maintain a subsistence level) when facing death. Therefore, this tells us more about an individual's wealth and ability to run up debts than about his or her preferences.

On the other hand, question (2) will fail to call forth a limited amount—abstracting from a bequest motive—for the simple reason that money is of no use to the dead. The two approaches to measuring the willingness to pay also differ in their *allocation of "property rights."* For only alternative (2) says that an individual has the right to live, which he can give up voluntarily. Given such a right to live, the correct "value of a personalized life" should indeed be infinite.

This train of thought goes back to John Broome (1982a, b), who turns against a distinction of personalized and statistical lives, as in his view the latter concept is based on incomplete information about who is going to lose his life. If statistics tell us that in the course of a construction project one worker, who is not identified yet, is going to be killed, this statistical life will in fact be infinitely precious. For as soon as the veil of ignorance is lifted and the name of the victim known, approach (2) would call for an infinite compensation for the loss of a personalized life.

The flaw in Broome's argument is revealed when one takes a closer look at the way he constructs his case.

3. Both formulations can be accommodated within the framework of the expected utility rule. See Subsection 2.3.2.2 for further details.

It is very hard to imagine a risk where the number of victims is known in advance with certainty. In most cases, it cannot even be said with certainty that there will be any victims at all. For example, a road bend may have taken an average toll of one human life a year in the past. However, this does *not* mean that *exactly one* person will die during the next year. To the contrary, observing exactly one death during a given year is a rather unlikely event. To see this, let there be 100,000 users of the road per year, each of whom facing the risk of 1 in 100,000 of being killed in an accident. Adding their individual probabilities over time would imply a *perfectly negative correlation* of the individual risks.⁴ Assuming stochastic independence or even a positive correlation—which is far more plausible when speaking of accidents—no one will be killed with a certain, strictly positive probability, while with a positive though very small probability all 100,000 persons will be killed. The total number of casualties thus is unknown, causing the “statistical life” to be the relevant concept for an economic valuation of security measures.

In sum, most relevant decisions with regard to life and death seem to involve *small risks* that can be avoided (or must be taken). In these situations, we can expect limited amounts of money to be sufficient to compensate an individual for taking a risk. There are numerous examples showing that people are willing to risk their lives for the sake of pleasure, comfort, or thrill. Activities such as smoking, driving without seat belts, traveling by car or plane rather than by train, and riding on a roller coaster demonstrate that avoiding small risks is not infinitely valuable to people.

Since individuals clearly act as if their lives have only a finite value to them, the government should not (implicitly nor explicitly) attribute an infinite value to life when taking decisions in the public domain. Otherwise, inefficiencies due to the discrepancy between the costs of lives saved in the private and public sector would be the result. Such a discrepancy could only be justified if external effects were present.

2.3.2 Basic Conceptions of Valuing Lives

After arriving at a fundamental agreement on the necessity of attributing a monetary value to human life, the next step is to clarify how this value should be determined. For this purpose, two completely different conceptions have been developed: the human-capital approach and the willingness-to-pay approach. In this section, both of them are portrayed and discussed in terms of their definitions, their ideological underpinnings, and their problems. Measurement issues and empirical results are the subject of Section 2.3.3.

2.3.2.1 The Human–Capital Approach

The value of an object can be determined by measuring the owner’s financial loss from losing it. This loss, in turn, is measured by the revenue stream that he could have

4. Perfect positive correlation between two single risks (exposed persons) *A* and *B* means that person *A* dies if and only if *B* dies. Perfect negative correlation would mean that *A*’s death would preclude *B*’s death (like in Russian roulette). In our example it would follow that if one person were already killed in an accident, everybody else would be safe for the rest of the year.

obtained through careful management of the object. The application of this simple accounting rule (productive value principle) to human beings leads to a definition of the value of life based on the loss of human capital resulting from the death of that human being. The value of life is therefore equal to the discounted sum of the individual's future (marginal) *contributions to the social product*, which corresponds to his future labor income, provided his or her wage is equal to the value marginal product.

When applied to the situation of a human being that falls victim to an accident at work, this definition seems a reasonable way to ascertain the claims for compensation of his relatives. The summation of foregone earnings is called "gross human capital." In deducting the deceased's future consumption from his earnings, "net human capital" can be estimated. This quantity corresponds to the material loss that *others* suffer due to an individual's death.

The human capital approach is based on two implicit postulates.

1. An individual's value depends on the contribution that he or she makes to the welfare of its fellow citizens.
2. The appropriate measure of society's welfare is the Gross National Product (GNP).

Postulate 1 is more appropriate to a *slave-holder society* than to a liberal democracy of the twentieth century. It makes no essential difference between a person and a machine. Moreover, reliance on the "net human-capital" approach implies that the individual himself does not even count as a member of society, because his own loss (of future consumption) is not included in the calculation of the value of his life.

The main advantage of the human-capital approach is that it can be made operational rather easily.⁵ For this reason, it has often been applied in cost-benefit analyses in the past. But it is completely foreign to microeconomic theory, which focuses on individual valuation. In addition, its ethical underpinnings are very much open to attack. One criticism sets in at the result of such a human-capital calculation, which many people consider unacceptable. According to it, the value of the life of pensioners and others who are unable to work is always zero (even negative according to the net human-capital approach). Of even more fundamental nature is the objection to the second postulate on the grounds that the GNP measure entirely ignores the *pleasure of living as such*. According to the opinion of most students of the field, this objection has served to discredit the human-capital approach despite its advantages of application. To sum up:

CONCLUSION 2.3: According to the human-capital approach, the value of life is determined by the contribution the individual could make to the social product. Its relatively easy application is outweighed by serious economic and ethical shortcomings.

5. However, there remain the well-known problems of estimating the contribution of housewives (and housemen) to the national product and possible discrepancies between wage and marginal productivity of labor owing to imperfect labor markets.

2.3.2.2 The Willingness-To-Pay Approach

2.3.2.2.1 Definition

In contrast to the human-capital approach, the concept of willingness to pay is firmly rooted in the axiomatics of subjective valuation. The preferences of the persons affected and possibly their relatives are in the focus of interest. As mentioned briefly in Section 2.3.1, asking for the amount of money that is worth just as much to an individual as his own life leads to ambiguities, because the answer depends on whether losing or saving one's life is at issue. These difficulties can be avoided in considering alternatively

1. Marginal changes in the *probability* π to die within a given period of time; or
2. Marginal changes of the expected *life span*.

In case (1), one seeks to determine the amount of money Z that the individual is willing to pay for a small *decrease* in the risk of dying $\Delta\pi$ (e.g., 1 percentage point). In full accord with microeconomic theory, Z is called “equivalent variation.” By way of contrast, the amount K that would be required to compensate the individual for an *increase* in π by $\Delta\pi$ is called “compensating variation” for a change in risk. When $\Delta\pi$ goes towards zero, the two quotients $Z/\Delta\pi$ and $K/\Delta\pi$ converge. Their common limit, that is, the marginal rate of substitution between wealth and the risk of dying, can be interpreted as the individual's marginal willingness to pay for infinitesimal changes in risk.

In case (2), the amount of money that the individual would be prepared to sacrifice in order to prolong his life by one year can be interpreted as his marginal willingness to pay for a year of life gained.

Both of the two formulations have their specific legitimation. From the ex-post point of view, the concept of years of life gained makes more sense than the one of changes in risk. On the other hand, for an individual faced with the ex-ante uncertainty of the time of his death, the concept of life expectancy is less concrete than the one of a change in risk for the subsequent period of time because the life expectancy is determined by the entire probability distribution of death risks over the remaining lifetime. That is why we confine ourselves to case (1) in the following.

2.3.2.2.2 Decision-Theoretic Foundation

The objective of the theoretical analysis presented below is to derive an individual's willingness to pay from his maximization of expected utility and to provide answers to the following questions. How does his willingness to pay depend on

- The initial level of his risk of death,
- The size of his wealth,
- His age?

Consider a simple model, in which the individual's utility depends on the first two factors only⁶: on whether he survives the subsequent period or dies right away and on the size of his wealth (or his estate in case of death) y . Survival can be expressed by

6. This presentation follows Weinstein et al. (1980).

a dummy variable L , which assumes the values 1 (“life”) and 0 (“death”). The individual’s preference relation is hence defined over all vectors (L, y) with $L \in \{0, 1\}$ and $y \in R$ and can be represented by a cardinal utility function $u(\cdot)$, about which the following assumptions are made.

ASSUMPTION 2.1: $u(L, y)$ is twice continuously differentiable, strictly increasing, and strictly concave, which means $u_y[L, y] > 0$, $u_{yy}[L, y] < 0$ for all (L, y) .⁷

ASSUMPTION 2.2: For each y we have: $u[1, y] > u[0, y]$.

ASSUMPTION 2.3: For each y we have: $u_y[1, y] > u_y[0, y]$.

These assumptions are illustrated by Figure 2–1. Whereas the first part of Assumption 2.1 is of a wholly technical nature, its second part expresses increasing utility of wealth and its third part, *risk aversion with regard to variation in wealth*. This is shown for the case $L = 1$ as follows. Let there be a fifty-fifty chance of wealth taking on the low value of y^- and the high value of y^+ , yielding utility $u[1, y^-]$ and $u[1, y^+]$, respectively. Expected utility, being a linear, probability-weighted combination of these utilities, can then be read off from the straight line connecting these two points. In the fifty-fifty case, it is given by the point A with the ordinate value $EU[1, y]$. A risk-averse individual, however, prefers the alternative of certain wealth even if this alternative is financially equivalent, such as $Ey = (1/2)y^+ + (1/2)y^-$ in the fifty-fifty case. This preference is shown by a certain utility $u[1, Ey]$ being located above the expected utility $EU[1, y]$ in Figure 2–1. For this result to obtain, the function $u(1, y)$ must be concave from below [$u_{yy}(1, y) < 0$], thus symbolizing risk aversion.

Assumption 2.2 states that the individual—at any level of wealth—prefers life to death: the $u(1, y)$ curve lies everywhere above the $u(0, y)$ curve. Assumption 2.3, on the other hand, says that the *marginal utility* of wealth in case of survival is higher than the marginal utility of the estate in case of death: at any value of y , the $u(1, y)$ curve is steeper than the $u(0, y)$ curve. Assumption 2.2 is plausible; the whole analysis of willingness to pay would be paradoxical without it. Assumption 2.3, however, is more restrictive than appears at first sight. If there were perfect markets for term life insurance as well as for life annuities, a rational individual with a concave utility function in y would buy insurance to the point where his marginal utilities of wealth (or estate) are equal in both states of nature. Thus Assumption 2.3 implicitly presupposes imperfection of insurance markets.

The utility function $u(L, y)$ can also be expressed by the two *conditional* utility functions $u_1(y)$ and $u_0(y)$, with

$$u_1(y) = u[1, y], \quad u_0(y) = u[0, y] \quad (2.1)$$

and thus

$$u(L, y) = Lu_1(y) + (1 - L)u_0(y) \quad (2.2)$$

7. The expression $u_y[L, y]$ stands for $\partial u[L, y]/\partial y$, the partial derivative of function u with respect to the argument y evaluated at (L, y) . Analogously, u_{yy} stands for $\partial^2 u[L, y]/\partial y^2$. The brackets mean “evaluated at.”

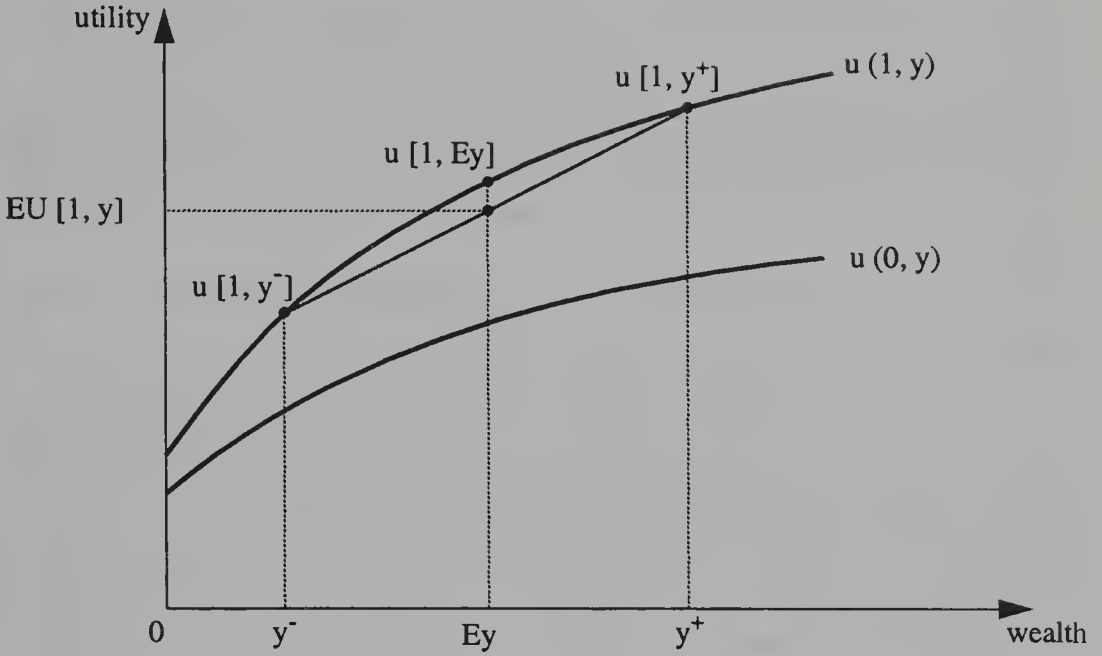


Figure 2-1. State-dependent utility of wealth.

This function may be generalized by replacing the dummy variable $(1 - L)$ by the probability of death π , which can take on any value in the $[0,1]$ interval. Accordingly, the individual's *expected utility* can be obtained by weighing the utility payoffs obtained in the states of nature “life” and “death” by their respective probabilities and adding them up. Using the symbol EU for expected utility, one has

$$EU(\pi, y) = (1 - \pi)u_1(y) + \pi u_0(y) \quad (2.3)$$

In order to get the marginal rate of substitution between changes of risk and wealth, an indifference curve is constructed by varying the expected utility by dEU but setting this change equal to zero. Evaluating the total derivative at some point (π, y) gives

$$\begin{aligned} 0 = dEU[\pi, y] &= \frac{\partial EU}{\partial y} dy + \frac{\partial EU}{\partial \pi} d\pi \\ &= \{ (1 - \pi) u'_1[y] + \pi u'_0[y] \} dy + (u_0[y] - u_1[y]) d\pi \end{aligned} \quad (2.4)$$

By solving this equation with respect to the ratio $dy/d\pi$, one obtains the slope of the indifference curve and hence the desired marginal rate of substitution,

$$m[\pi, y] = \frac{dy}{d\pi} \Big|_{dEU=0} = \frac{u_1[y] - u_0[y]}{(1 - \pi)u'_1[y] + \pi u'_0[y]} > 0 \quad (2.5)$$

due to Assumptions 2.1. and 2.2.

The marginal rate of substitution, $m[\pi, y]$, reveals how much the individual's wealth must increase in order to compensate him for an infinitesimal increase in the risk of death (or decrease in probability of survival) starting from an initial value π . The numerator shows the utility difference between "life" and "death" at a given level of wealth, which is positive owing to Assumption 2.2. The denominator shows the expected value of the marginal utility of wealth, which is positive as well, due to Assumption 2.1.

We summarize the results from the decision theoretical model (see Table 2.2) by

CONCLUSION 2.4: The willingness-to-pay approach is based on the concept of subjective utility. According to it, the monetary value of a marginal reduction of the death risk corresponds to the

Table 2.2. The Decision-Theoretic Model of Willingness to Pay

$$u(L, y) = Lu_1(y) + (1 - L)u_0(y) \quad (2.2)$$

$$EU(\pi, y) = (1 - \pi)u_1(y) + \pi u_0(y) \quad (2.3)$$

$$m[\pi, y] = \frac{dy}{d\pi} \Big|_{dEU=0} = \frac{u_1[y] - u_0[y]}{(1 - \pi)u'_1[y] + \pi u'_0[y]} > 0 \quad (2.5)$$

$$\frac{\partial m}{\partial \pi} = \frac{d^2y}{d\pi^2} = \frac{-(u_1[y] - u_0[y])(u'_0[y] - u'_1[y])}{\{(1 - \pi)u'_1[y] + \pi u'_0[y]\}^2} > 0 \quad (2.6)$$

$$\frac{\partial K}{\partial \pi} = \frac{(u_1[y + K] - u_1[y]) - (u_0[y + K] - u_0[y])}{(1 - \pi - \delta)u'_1[y + K] + (\pi + \delta)u'_0[y + K]} \quad (2.9)$$

$$\begin{aligned} \lim_{\substack{\delta \rightarrow 0 \\ K \rightarrow 0}} \frac{K}{\delta} &= \lim_{\substack{\delta \rightarrow 0 \\ K \rightarrow 0}} \frac{\partial K}{\partial \delta} \\ &= \lim_{\substack{\delta \rightarrow 0 \\ K \rightarrow 0}} \frac{u_1[y + K] - u_0[y + K]}{(1 - \pi - \delta)u'_1[y + K] + (\pi + \delta)u'_0[y + K]} = m(\pi, y) \end{aligned} \quad (2.12)$$

$$\frac{\partial m(\pi, y)}{\partial y} = \frac{EU_y(u'_1[y] - u'_0[y]) - (u[y] - u_0[y])EU_{yy}}{(EU_y)^2} > 0 \quad (2.13)$$

$$B = \sum_{i=1}^n \left(\delta_i \frac{Z_i}{\delta} \right) \approx \sum_{i=1}^n \delta_i m_i \quad (2.14)$$

y :	Wealth of the individual	δ :	Finite variation of the risk of death
L :	Dummy variable for "death" ($L = 0$) and "survival" ($L = 1$)	Z :	Willingness to pay for a (finite) reduction of the risk of death
π :	Risk of death	K :	Compensation demanded for a (finite) increase in the risk of death
$u(L, y)$:	Utility of the individual	C :	Cost of a project
EU :	Expected utility	B :	Benefit of a project
$m(\pi, y)$:	Marginal rate of substitution between wealth and risk of death		

individual's marginal rate of substitution between his wealth and his probability of survival.

How does this marginal rate of substitution m change in response to an increase in the initial level of the death risk π ? This question can be answered by differentiating $dy/d\pi$ with respect to π to obtain

$$\frac{\partial m}{\partial \pi} = \frac{d^2 y}{d\pi^2} = \frac{-(u_1[y] - u_0[y]) (u'_0[y] - u'_1[y])}{\{(1 - \pi) u'_1[y] + \pi u'_0[y]\}^2} > 0 \quad (2.6)$$

This expression is positive because the first term in parentheses in the numerator is positive due to Assumption 2.2 and the second is negative due to Assumption 2.3. The marginal rate of substitution and hence marginal willingness to pay for *infinitesimally small changes* of death risk thus rises with an increase from its initial level. The economic explanation for this is simple. The higher the risk of death, the higher the probability of the compensation adding to the estate in the case of death rather than to wealth in the case of survival. However, the marginal utility of an increased estate is comparatively low. This means that the compensation, being valued at a lower expected marginal utility of money, has to increase.

Up to this point, we considered willingness to pay for infinitesimally small changes of the death risk. But in practice, we are usually faced with changes in death risk of *finite magnitude*. How does one go about to determine demands for compensation in this case, for example, for an increase of exposure to risk from π to $\pi + \delta$ (with $\pi + \delta \leq 1$)? To define the compensation demanded K as a function of y , π , and δ , one equates expected utilities *before* and *after* the change in risk,

$$(1 - \pi)u_1[y] + \pi u_0[y] = (1 - \pi - \delta)u_1[y + K(y, \pi, \delta)] + (\pi + \delta)u_0[y + K(y, \pi, \delta)] \quad (2.7)$$

Analogously, we define the willingness to pay for a reduction of risk from δ to $\pi - \delta$, viz. $Z(y, \pi, \delta)$, by the equation

$$(1 - \pi)u_1[y] + \pi u_0[y] = (1 - \pi + \delta)u_1[y - Z(y, \pi, \delta)] + (\pi - \delta)u_0[y - Z(y, \pi, \delta)] \quad (2.7a)$$

To determine how the compensation K responds to a change in the risk π , Eq. (2.7) is considered as an implicit function of π . Its differentiation with respect to π , keeping in mind that $\partial u_i / \partial K = \partial u_i / \partial y := u'_i$, yields

$$-u_1[y] + u_0[y] = -u_1[y + K] + u_0[y + K] + \{(1 - \pi - \delta)u'_1[y + K] + (\pi + \delta)u'_0[y + K]\} \frac{\partial K}{\partial \pi} \quad (2.8)$$

Solving Eq. (2.8) for $\partial K / \partial \pi$, one obtains

$$\frac{\partial K}{\partial \pi} = \frac{(u_1[y + K] - u_1[y]) - (u_0[y + K] - u_0[y])}{(1 - \pi - \delta)u'_1[y + K] + (\pi + \delta)u'_0[y + K]} \quad (2.9)$$

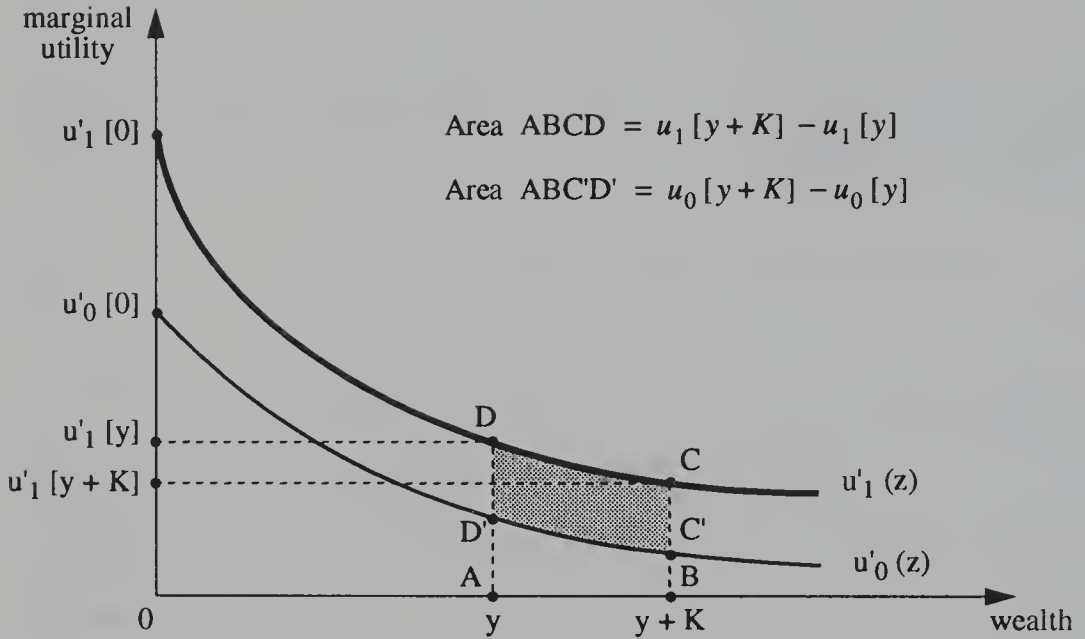


Figure 2-2. Utility differences as areas below the marginal utility curves.

The denominator of Eq. (2.9) is positive due to Assumption 2.1. In order to determine the sign of the numerator, note that the utility differences in the parentheses can be represented as the difference of areas below the corresponding marginal utility curves between wealth y and wealth $y + K$ (see Figure 2-2). This gives

$$(u_1[y+K] - u_1[y]) - (u_0[y+K] - u_0[y]) = \int_y^{y+K} u'_1(z) dz - \int_y^{y+K} u'_0(z) dz > 0 \quad (2.10)$$

With reference to Figure 2-2, the first term of Eq. (2.10) corresponds to the area ABCD, while the second term corresponds to ABC'D'. The difference between the two terms is positive owing to Assumption 2.3. Therefore, $\partial K/\partial \pi$ of Eq. (2.9) is positive; that is, the amount of compensation demanded for *finite* changes in the risk of death increases with a rising initial level of risk. By analogy, it can also be said that willingness to pay $Z(y, \pi, \delta)$ for a diminution of the death risk is higher, the higher the initial risk level.

What is the relationship between the average willingness to pay Z or the demand for compensation K “per risk unit” for a finite change of death risk $\delta = \Delta \pi$ on the one hand and the marginal willingness to pay for infinitely small changes of π on the other hand? The following proposition provides an answer to this question.

PROPOSITION 2.1: If δ goes to zero, both $K(y, \pi, \delta)/\delta$ and $Z(y, \pi, \delta)/\delta$ converge to the marginal rate of substitution $m(\pi, y)$.

PROOF: Only the first part of the proposition will be proved here. For $\delta = 0$, it obviously follows from Eq. (2.7) that $K(\delta) = 0$. Thus it is not possible to

calculate the limit K/δ with $\delta \rightarrow 0$ directly. Rather, the *rule of l'Hôpital* must be invoked,

$$\lim_{\substack{\delta \rightarrow 0 \\ K \rightarrow 0}} \frac{K}{\delta} = \lim_{\substack{\delta \rightarrow 0 \\ K \rightarrow 0}} \frac{\partial K}{\partial \delta}$$

Equation (2.7) is now viewed as an implicit function of δ . Its differentiation with respect to δ allows to derive the following expression for $\partial K/\partial \delta$,

$$0 = u_0[y + K] - u_1[y + K] + \{(1 - \pi - \delta)u'_1[y + K] + (\pi + \delta)u'_0[y + K]\} \frac{\partial K}{\partial \delta} \quad (2.11)$$

Solving this equation for $\partial K/\partial \delta$ and applying the rule of l'Hôpital, one obtains for K/δ

$$\begin{aligned} \lim_{\substack{\delta \rightarrow 0 \\ K \rightarrow 0}} \frac{K}{\delta} &= \lim_{\substack{\delta \rightarrow 0 \\ K \rightarrow 0}} \frac{\partial K}{\partial \delta} \\ &= \lim_{\substack{\delta \rightarrow 0 \\ K \rightarrow 0}} \frac{u_1[y + K] - u_0[y + K]}{(1 - \pi - \delta)u'_1[y + K] + (\pi + \delta)u'_0[y + K]} \\ &= \frac{u_1[y] - u_0[y]}{(1 - \pi)u'_1[y] + \pi u'_0[y]} = m(\pi, y) \end{aligned} \quad (2.12)$$

by reason of Eq. (2.5). QED.

Next we examine how the marginal rate of substitution $m(\pi, y)$ changes with a variation of the *initial level of wealth* y . For this purpose we differentiate (2.5) with respect to y to obtain

$$\frac{\partial m(\pi, y)}{\partial y} = \frac{EU_y(u'_1[y] - u'_0[y]) - (u_1[y] - u_0[y])EU_{yy}}{(EU_y)^2} > 0 \quad (2.13)$$

with

$$EU_y = \frac{\partial EU}{\partial y} = (1 - \pi)u'_1[y] + \pi u'_0[y] > 0$$

and

$$EU_{yy} = \frac{\partial^2 EU}{\partial y^2} = (1 - \pi)u''_1[y] + \pi u''_0[y] < 0$$

In view of Assumptions 2.2 and 2.3, the numerator on the right-hand side of Eq. (2.13) is clearly positive and so is the denominator. Hence the marginal willingness to pay for a reduction of the death risk *increases* with wealth. This result is obvious because the

marginal utility of money drops with an increase in wealth. The marginal rate of substitution between wealth and the probability of survival therefore must change in favor of the latter with increasing wealth.

It goes without saying that wealth includes tangibles and financial assets as well as future labor income (i.e., human capital). In this respect, there is indeed a connection between the “value of life” and human capital, even if it does not amount—in contrast to the human-capital approach—to a definitional identity.

CONCLUSION 2.5: The individual’s marginal willingness to pay for a decrease in his risk of death increases *ceteris paribus* if the initial level of his risk or his wealth increases.

Finally it should be remembered that the entire analysis refers only to the risk of death during an imminent period of time. The distribution of survival probabilities over the whole remaining (potential) lifetime must therefore be assumed as given. This distribution of course has a direct impact on willingness to pay. An individual presuming that he must most probably die, if not during the present then during the following period (e.g., because he suffers from a *second* life-shortening illness), will probably exhibit a smaller willingness to pay than another individual whose remaining life expectancy is quite high due to the elimination of the immediate risk. Thus age and additional life-threatening illnesses are examples of other factors explaining differences in willingness to pay between different groups of persons.

2.3.2.2.3 Implications for the Value of a Statistical Life

Most of the issues calling for a valuation of life mentioned in Section 2.1 involve projects having the property of a collective good. This means that they change the death risk of many people at the same time (and probably by differing amounts). Suppose the number of persons affected is n , while the death risk of individual i is being lowered by δ_i . For a fixed value of δ (e.g., $\delta = 10^{-6}$), let Z_i be individual i ’s willingness to pay for a decrease in π by δ . Thus the willingness to pay for a reduction of his individual death risk by δ_i equals $Z_i \delta_i / \delta$.

In keeping with the theory of collective goods, the “value” B of the scheme can now be defined as the *aggregate willingness to pay* of the n individuals,

$$B = \sum_{i=1}^n \left(\delta_i \frac{Z_i}{\delta} \right) \approx \sum_{i=1}^n \delta_i m_i \quad (2.14)$$

The approximation is founded on the fact that Z_i / δ converges towards the marginal rate of substitution m_i if δ goes towards zero (see Proposition 2.1).

The calculation of the sum on the right-hand side of Eq. (2.14) presupposes a great deal of information because the project planners would have to know for every single potentially affected person (i.e. from every $i = 1, \dots, n$):

1. The amount δ_i by which his individual risk of death is lowered; and
2. His individual value of the marginal rate of substitution m_i .

This information can be procured with extreme difficulty only. On the other hand, one is usually able

- 1'. To estimate the number of statistical lives saved, that is, the *sum* of the δ_i ; and
- 2'. To determine the *average* marginal willingness to pay $(1/n)\sum_i m_i = \bar{m}$ (see Section 2.3.3 about its measurement).

There is the following statistical connection between the desired quantity on the right-hand side of Eq. (2.14) and the two quantities just mentioned.⁸

$$B \approx \sum_{i=1}^n \delta_i m_i = \bar{m} \sum_{i=1}^n \delta_i + n \text{Cov}(\delta_i, m_i) \quad (2.15)$$

The product of the expected number of lives saved and the average marginal willingness to pay thus differs from the desired quantity exactly by n times the covariance between the individual quantities “risk reduction” and “willingness to pay.” Supposing that these two quantities are uncorrelated, the covariance drops out, and, according to Eq. (2.15), the benefit of the project B simply emerges as the product of the average marginal willingness to pay, $\bar{m}$, and the number of “statistical lives” saved, $\sum \delta_i$.

CONCLUSION 2.6: An approximate value for the benefit of a project designed to reduce risk of death is provided by the product of the number of statistical lives saved and the average marginal willingness to pay of the persons affected.

For the sake of simplicity, let us ignore that the project at issue may have additional desirable aspects. The calculated benefit B can then be compared with the costs of the project C . If the benefit exceeds the costs, realization of the project results in a potential Pareto improvement; this is the so-called Kaldor–Hicks criterion [Kaldor (1939)]. The improvement is potential, not real, as long as the calculated compensating payments for a reduction of the death risk are not actually collected and used to finance the project. Thus the same objections can be put forward against the willingness-to-pay approach that are raised against the criterion of potential Pareto improvement in general.⁹

According to Conclusion 2.5 and the ensuing discussion, an individual’s willingness to pay for a decrease of risk depends on various factors, implying that the benefit of a project does not only depend on the *number* of statistical lives “saved” ($\sum_i \delta_i$).

8. We have $\text{Cov}(\delta, m) = E(\delta m) - E(\delta)E(m)$, so that $E(\delta m) = E(m)E(\delta) + \text{Cov}(\delta, m)$ [see, e.g., Pindyck and Rubinfeld (1991, p. 21)]. Replacing population moments by sample analogs, one obtains

$\frac{1}{n} \sum_i \delta_i m_i = \bar{m} \frac{1}{n} \sum_i \delta_i + \text{Cov}(s_i, m_i)$. Multiplying this equation by n , one obtains Eq. (2.15).

9. This criterion says that an improvement of social welfare is already achieved if compensating losers by winners would result in a Pareto improvement, even though the compensation payments are not actually made.

First, it was shown that the willingness to pay Z or K rises with the *initial risk level*. A given reduction of risk among people (initially) exposed to a great deal of risk should therefore be *more* valuable than among a group of less exposed people. To illustrate this, let us consider two measures. Alternative X lowers the death risk of 1,500 people from 1:1,000 to zero, while alternative Y lowers it for 100 particularly exposed people from 50 percent to 49 percent. The first measure thus saves 1.5, the second only 1 statistical life. Nonetheless, it is conceivable that the m_i values for Y are high enough to yield an aggregate willingness to pay in excess of that pertaining to X , as given by Eq (2.15).

Second, provided that willingness to pay diminishes with decreasing residual life expectancy, should the average *age* of the persons affected be taken into account, or should age be considered “ethically irrelevant information” for public decisions? Opinions about this issue are divided. To clarify it, consider the example of two cancer patients, both with a probability of 50 percent of dying if denied tumor surgery. Suppose that the operation reduces their immediate risk of death to zero. One of the patients is 20 years old, and surgery would give him a 90 percent chance of being perfectly cured and having his life expectancy restored to the customary value of his age. The other patient is 80 years old, and owing to metastases, he will survive another three months at best. Due to limited capacity, only one of them may receive surgery. Is it ethically legitimate to argue that a decrease in the risk to life is more valuable for the first patient than for the second? By answering this question in the affirmative, one agrees to the statement that the remaining life expectancy, with age possibly serving as an indicator, constitutes relevant information for the “value” of life.

Third, should the value of life depend on the *wealth* of those affected? Here it seems much easier to reach agreement. As shown in Subsection 2.3.2.2.2, individual willingness to pay increases with wealth, but the compensation in question is never paid in reality. Therefore, society gains *ceteris paribus* just as much by the rescue of a rich as of a poor person. A decision on a public project having risk effects should hence be based on the willingness to pay of persons with average wealth.

This argument appears fairly clear but has to be modified when a public safety measure is taken either by a central authority or—on a smaller scale—by a local authority (e.g., province, district, or municipality). Suppose that the district consists only of two municipalities with very different per capita incomes. Furthermore, suppose that the average willingness to pay of the district is too small to make up for the costs of the project, which are presumed to be proportional to the number of “statistical” lives saved. In the high-income municipality, however, it is large enough to carry out the project on a reduced scale, whereas this is not possible for the “poor” municipality. The transfer of authority to the municipalities will cause the “value of life” to depend on the wealth of the persons affected. This may be considered an ethically undesirable effect, but it is the price that has to be paid for federalism.

2.3.2.2.4 Inclusion of the Willingness to Pay of Other Persons

So far, the value of a risk reduction has been equated with the willingness to pay of the person affected. The death of a human being, however, also gives rise to *external effects* in causing his relatives and friends grief and pain. These would often be

prepared to raise considerable amounts of money in order to lower the probability of death of the person affected. In the literature it is therefore suggested to add these amounts to the willingness to pay of the person affected, i , in order to determine the value of his life.

The following argument has been advanced against this addition. Clearly, j 's willingness to pay for the increase of i 's probability of survival is an expression of his altruism. If person j is really altruistic, then he is happy for i 's life and his wealth y_i . In that ideal case (absence of paternalism), it is the utility of i that enters the utility function of j ,

$$u_j = u_j\{L_j, y_j, u_i(L_i, y_i), \dots\} \quad (2.16)$$

However, this implies that the marginal rate of substitution of j with respect to life and wealth of i is given by

$$-\frac{dL_i}{dy_i} \bigg|_{du_j=0} = \frac{(\partial u_j / \partial u_i) (\partial u_i / \partial y_i)}{(\partial u_j / \partial u_i) (\partial u_i / \partial L_i)} = \frac{\partial u_i / \partial y_i}{\partial u_i / \partial L_i} = -\frac{dL_i}{dy_i} \bigg|_{du_i=0} \quad (2.17)$$

Thus the marginal rate of substitution of j corresponds with i 's own marginal rate of substitution. Hence, i uses his resources exactly in the way the unselfish person j desires. A specific subsidization of the reduction of risk affecting i would therefore fail to lead to a Pareto improvement.

Things look different if i 's willingness to pay does not reflect his altruism towards the "victim" but the loss he would suffer due to j 's death. However, it can still be argued that these feelings of others (e.g., of his own children) are contained in the willingness to pay of j , so that Eq. (2.17) remains valid.

In addition to these fundamental objections, *adding* the willingness to pay of other persons would face practical problems. The question of whether it is only appropriate if the other person stays alive ought to be settled first. The argument that a dead person is unable to feel grief speaks in favor of such a restriction. However, accounting for affected survivors makes the evaluation of a project that modifies risk even more difficult. In addition to the number of statistical lives saved, it has to be taken into consideration whether the persons affected are related on familiar terms with one another or not. Owing to external effects, a grave danger that simultaneously threatens several single persons ought to be fought against more than a grave danger threatening a family or a circle of friends (containing the same number of persons).

2.3.3 Measuring Willingness to Pay

As is true of all collective goods, public projects serving to increase safety (or to decrease it) are confronted with the problem of measuring preferences, here the willingness to pay or the demand for compensation, respectively. In general there are two alternative approaches to the problem. Either individuals can be interviewed or—in the spirit of revealed preferences—their behavior can be observed. Both methods have their specific pros and cons, which will be discussed in the following.

2.3.3.1 *The Direct Method: Questionnaire Studies*

The questionnaire is not only the most direct but also the most transparent method to find out preferences. There have been extensive and careful studies, some of them comprising more than 1,000 interviews, on the subject of “willingness to pay for changing risk” (for their results, see Subsection 2.3.3.3). Some of the difficulties involved in those questionnaire studies may be that those sampled do not understand the questions, fail to take them seriously enough—the situations being only hypothetical—or even hide their real preferences. Specifically, the following problems usually occur when working with questionnaires in this context.

1. *Dealing with small probabilities.* If the scenarios described in the questionnaire are to reflect reality, it is essential to consider very small probabilities and probability differences. Small probabilities, however, have little meaning to most people, who often fail to distinguish between probabilities that differ by powers of ten.¹⁰ So answers will not be very reliable, and by asking similar questions involving different probabilities, it is easy to obtain results that contradict the transitivity of preferences or axioms of the expected utility maximization. In this case the theoretical framework of Subsection 2.3.2.2.2 is not applicable anymore.

2. *Emotional rejection of questions.* Another problem is getting people to answer delicate questions involving the tradeoff between life and wealth. Refusal of some sampled persons (usually selected at random) to answer such questions is likely to bias the results if it occurs predominantly among people with a high esteem for the value of life. Emotional rejection of the questionnaire could also result in a conscious or unconscious distortion of revealed preferences.

3. *Lacking motivation of the interviewed.* Even if there is a basic willingness to answer the questions, there is no motivation—as usual when dealing with hypothetical situations—to think seriously about one’s preferences and one’s reactions in a real rather than hypothetical situation. Respondents could be tempted to express what they think the interviewer wants to hear or what they feel will help their personal image.

2.3.3.2 *The Indirect Method: Evaluation of Market Data*

The problem of lacking motivation to act according to true preferences can be solved by dealing with real rather than hypothetical situations. In principle it is possible to infer the risk preference of an individual from his actions to avoid risks. A well-known example is the choice (or avoidance) of a job that is known for its risks to life and health (for example, racing car driver, truck driver, stuntman, miner, or electrician). But also everyday situations like putting on a safety belt can serve as an indicator of risk preference derived from observed behavior.

However, this method of *measuring “revealed preferences”* has its problems as well. Let us take the example of occupational choice. The idea is to infer the compensation required for accepting a higher personal risk from comparing the wage rate for occupations with and without a threat to life. The problems are the following.

10. This has been found, e.g., by Kahneman and Tversky (1979).

1. Separating risk from other influences. In contrast to the hypothetical situations described in questionnaires, real-life situations have many aspects that make it difficult to isolate a single influencing factor. One hardly finds two occupations that are identical except for their risk to life and health. Differences in the rates of pay also reflect differences regarding educational requirements, mental and physical demands, and many other characteristics of occupations. As long as these characteristics cannot be held constant, it is quite daring to interpret a wage differential unambiguously as a risk premium.

2. Discrepancy between subjective probability and relative frequency. Even if the wage differential were a plain risk premium, the marginal rate of substitution between risk and wealth of an employee can only be determined with the help of his subjective estimates of relevant probabilities. According to expected utility theory, decisions are based on subjective probabilities. In contrast, only relative frequencies of occupational deaths (for example, industrial accidents) are observable. As these deaths are relatively rare, it is questionable whether affected employees know the frequencies, let alone use them as a basis for their own probability estimates. Surveys among car drivers indicate that almost everybody underestimates his personal risk of being involved in an accident, compared to the relative frequency derived from the total population.

Another question is whether observed behavior actually can be interpreted as a maximization of the expected utility as required by theory. Empirical observations indicate [as pointed out already by Eisner and Strotz (1961)] that when dealing with relatively small risks, individuals systematically violate this rule. This holds true of answers to hypothetical questions as well as real-life decisions.

3. Representativeness of persons having risky occupations. Finally it can be called into question whether persons having risky occupations are representative of the total population. The fact that such an occupation and no other was chosen by these people implies that (disregarding objections No. 1 and 2 for the moment), the wage differential constitutes the *upper limit* of their marginal rate of substitution between life and wealth (i.e. the willingness to pay for a risk reduction). Simultaneously, it marks the *lower limit* for the rest of the population (assuming that these individuals basically are suited for the job as well). However, considering the very small or even negative compensating wage differential reported below, this statement is not very helpful. We have to conclude that members of such a profession may have a special preference for risky situations (perhaps for the thrill that comes with them), which is not shared by the rest of the population.¹¹

We can summarize these considerations in

CONCLUSION 2.7: Individuals' willingness to pay for reducing their risk of death can either be measured by questionnaires or by observing their behavior. Both methods have their specific problems. In the first case, a hypothetical question may fail to reveal one's true

11. It is debatable, however, whether people who take high risks professionally or privately for the sake of "thrill" (stuntmen, hang gliders) feel the same way about a less spectacular risk of equal magnitude (for example, poisoning due to harmful substances).

preferences; in the second case, it would have to be made sure that the individual knows exactly what the relevant risks are and that there are no other motives for his observed behavior.

2.3.3.3 Empirical Results

2.3.3.3.1 Results from Questionnaire Studies

During the past two decades, there have been well over a dozen studies in the literature using questionnaires with differing sample sizes (from 30 to 1,000 respondents) conducted mainly in the United States and Great Britain. In some cases, respondents were students attending classes given by the authors. Other studies used representative samples of the country's population. All questionnaires had realistic "scenarios"; that is, hypothetical risk changes were given by plausible reasons such as additional safety measures in road transportation, in nuclear power plants, or in toxic waste disposal. This should have motivated respondents to take the questions seriously.

The results show a *marked variation* of the calculated "values of life." This is due to the fact that considerable risk reductions of about $1:10^3$ do not call forth a substantially larger willingness to pay than reductions of around $1:10^5$ to $1:10^6$. Studies of the first group thus lead to a value of "statistical life" that is lower by several orders of magnitude. For example, Acton (1973) found values ranging between 59,000 and 91,000 US\$, Frankel (1979), one of 57,000 US\$. When working with smaller risk reductions, Frankel arrived at a value of 3.37 million US\$ in the same study. Jones-Lee, Hammerton, and Phillips (1985) found a value of 1.5 million British pounds in their study, which also belongs to the second group.

The expectation formulated in Subsection 2.3.3.1, namely that most respondents have difficulties in dealing with very small probabilities, is confirmed by these results. Indeed, hypothetical probabilities below a certain level (which could be around $1:1,000$) are "mentally adjusted" to a higher value.

It is noteworthy that Jones-Lee et al. (1985) report their (arithmetic) mean to be strongly influenced by some outliers on the positive side. Using the median instead of the mean, the result would be a value of human life of only 800,000 rather than 1.5 million pounds. However, the criterion of potential Pareto improvement (see Subsection 2.3.2.2.3) would have to be based on the arithmetic mean, while using the median can be justified by the principle of majority rule, which applies when risk-reducing measures proposed by the government are voted upon in a poll.

There are some considerations suggesting that the results presented may fail to mirror "true" willingness to pay.

- A considerable percentage of respondents indicate a *constant* willingness to pay for risk reductions, although the risk had been repeatedly *changed* from its initial level;
- Some respondents even indicate *smaller* amounts for *higher risk* reductions [see Jones-Lee et al. (1985)];
- A study by Smith and Desvouges (1987) finds that for the majority of the respondents a given risk reduction evokes a *decreased* willingness to pay if the

initial risk level is raised, a result that is at variance with expected utility theory (see the decision-theoretical model of Subsection 2.3.2.2.2);

- Furthermore the same study shows considerable differences between willingness to pay for a small decrease in risk (Z) and compensation demanded for small increases in risk (K);
- Finally, the type of death described in the scenarios (for example, dying of cancer versus dying in a car accident) affects the expressed willingness to pay. As willingness to pay for avoiding a long suffering may be relevant in the case of cancer, the interpretation of the results of questionnaire studies involving different types of death calls for even more caution.

CONCLUSION 2.8: The considerable variation of the results and the widespread inconsistencies of answers to hypothetical questions confirms the doubts expressed in Conclusion 2.6, challenging the reliability of the “direct method” to the measurement of willingness to pay for reductions of risk.

2.3.3.3.2 Results from Observed Behavior

The first extensive empirical survey of wage differentials between occupations entailing differing threats to life was conducted by Thaler and Rosen (1975). The underlying data refer to 900 workers in 37 *risky occupations*. The authors try to trace the workers’ wage incomes to various causes and to isolate occupational risk using multiple regression analysis. Probabilities of death are inferred from life insurance statistics. The resulting regression coefficients yield—depending on the specification of the estimated equation—a “value of a statistical life” between 136,000 and 260,000 US\$ (referring to 1967).

More recent surveys using data from the United States and Great Britain indicate larger wage differentials in response to comparable risk increments. The value of a statistical life often exceeds one million US\$ in these studies, which, however, continue to exhibit variations of up to two powers of ten.

Another group of empirical surveys aim at deriving “the value of life” from the observed *behavior of consumers*. Among other things, the data used refer to

- The higher price of houses in neighborhoods with better air quality;
- The purchase and installation of smoke detectors in wooden houses;
- The use of safety belts and the choice of speed when driving a car;
- The use of underpasses to cross streets with heavy traffic.

Surprisingly, the results of these completely different studies bunch more closely than those comparing wage rates. They imply a value of life between 200,000 and 600,000 US\$ at 1983 prices [see Jones-Lee et al. (1985)]. Due to the reservations expressed both against the direct and the indirect method of measuring the willingness to pay for changes in the risk of death (see Subsection 2.3.3.2), it cannot be concluded with certainty that the “value of a statistical life” lies within this interval mentioned.

2.4 VALUATION OF THE QUALITY OF LIFE

The methods of valuation hitherto discussed refer solely to the risk of premature death. The question has been left open so far as to how a valuation should be undertaken if a public health program results in a change of health status rather than a change of survival probability. In other words: How should one proceed if not changes in mortality but changes in *morbidity* are at issue? The latter can either occur isolated or in connection with changes of the rate of mortality, and they may affect identified or unidentified individuals.

Many medical programs, such as campaigns against tooth decay or the development and implantation of artificial joints, aim at improving the *physical well-being* or ability to perform a social function of the people affected. Numerous other public projects have an impact on the health status of the population as a side effect. The building of noise barriers along freeways, for example, can avoid physical and psychological stress (e.g., evoked by sleeplessness), while measures to prevent air pollution may check respiratory diseases.

Finally, changes of health status may also be the consequence of measures designed to prolong life. Suppose an infarction patient can be kept alive thanks to newly developed surgery, which, however, causes irreparable damage to his health. Is it in this case permissible to put a value only on the length of life, leaving the quality of life completely out of consideration?

These examples purport to show that measurement and valuation of the “output” of health care and of other public projects not only call for valuing changes of the length of life (or the probability of survival) but also for changes of its quality. For this purpose, all relevant states of health have to be operationalized and methods for their valuation be given.

2.4.1 The QALY Concept

To solve this task, we make use of the monetary valuation of a prolongation of life. In Subsection 2.3.2.2.1 it was shown how to determine the “value” of an increase in the probability of survival using the willingness-to-pay approach. Moreover, mention was made of the equivalence between an increase in the probability of survival and the prolongation of the life span. Thus the value of extending life expectancy by one year (say) for a representative individual can be calculated as well, and in order to calculate the utility of a project, this value (US\$ per gained life year) can be multiplied by the number of life years gained.

In this analysis, the *state of health* in which the gained years are spent is assumed to be known. If interviews were used to determine individual willingness to pay of potentially affected persons, these persons are supposed to spend their additional life years in their present state of health (possibly adjusted for the usual changes in the course of age). But in order to compare “life years gained” of different projects, the valuation must be based on one and the same health status. The ideal state of health (complete physical, psychological, and social well-being) serves as the preferred benchmark in this normalization.

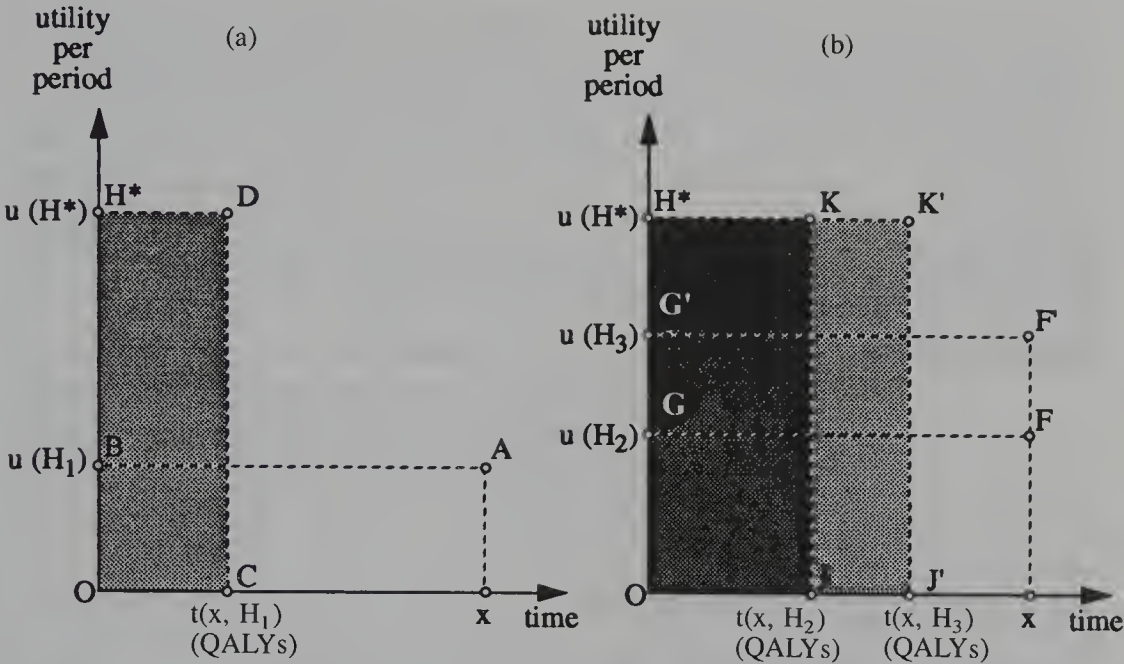


Figure 2-3. QALYs as the connecting link between length of life and health status with (a) converting years into QALYs and (b) converting a health status difference into a QALY difference.

Let $\{H_h; h = 1, 2, \dots\}$ be the (not necessarily ordered) set of possible states of health, with H^* denoting the ideal condition. Moreover, let Z be the monetary value of a prolongation of life by one year in perfect health H^* , obtained by using the method described in Subsection 2.3.2.2.2. But actually we face the problem of valuing either

- 1. A prolongation of life by x years that have to be spent in a *worse* health status, for example, H_1 ; or
- 2. A *change of health status*, for example, from H_2 to H_3 , lasting for x years.

To deal with these problems, Zeckhauser and Shepard (1976) have developed the concept of “quality-adjusted life years” (QALYs). Referring to (1), this concept converts the number x of years spent in a given health status H_1 into a (smaller) number $t(x, H_1)$ of years in perfect health H^* , which, in the individuals’ opinion, are worth just as much. This conversion is illustrated in Figure 2–3a. The utility of spending x years in health status H_1 is given by the area $OxAB$. The same utility can be attained by spending $t(x, H_1)$ years only, but in perfect health ($OCDH^* = OxAB$). The number of QALYs corresponding to x years spent in a less than perfect health status H_1 is thus given by $t(x, H_1) < x$. More generally, QALYs allow one to weigh the life years gained in different states of health using the valuation by persons affected so that they can be compared with each other.

Now it is only a short step to the solution of a problem of type (2). Let $t(x, H_2)$ be the number of QALYs corresponding to x years in state H_2 , while $t(x, H_3)$ denotes the number of QALYs corresponding to x years in a better state H_3 . The difference $t(x, H_3) - t(x, H_2)$ thus represents the number of life years gained in perfect health, which is

equal in value to the underlying change of health status from H_2 to H_3 for x years. Figure 2–3b illustrates this. First, x years lived in state H_2 are converted into $t(x, H_2)$ QALYs using the equality of areas $OxFG$ and $OJKH^*$. Next, x years lived in state H_3 are turned into $t(x, H_3)$ QALYs, due to the equality of areas $OxF'G'$ and $OJ'K'H^*$. The QALY difference then reflects the health status difference as evaluated by the affected individual. Therefore, the instruments available for valuing prolongations of life can also be applied to changes of health status, provided that the utility of different states of health was determined using the function $t(\cdot)$. We come to

CONCLUSION 2.9: The concept of “quality-adjusted life years” (QALYs) allows one to make changes in the quality of life and changes of the length of life comparable. Thereby it becomes possible to derive the value of changes of health status from the “value of a statistical life.”

2.4.2 Evaluating Health States

By nature, health is a multidimensional concept. It includes aspects of organ functioning, the capability of the human being to perform physical activities and to play a “normal” social role as well as a subjective feeling of well-being. As a consequence, the term *health status* is problematic because its complete description would require measurements for every single health component.

In the 1960s and early 1970s, medical statisticians thought that they could reduce the complexity of the problem of output measurement in health care by trying to develop a *one-dimensional scale*, the so-called “health status index.”¹² It purports to designate a value $I(H_h)$ varying between 0 (immediate death) and 1 (ideal state according to the WHO definition¹³), to every health state H_h defined by the components mentioned above. Next, a function would have to be determined that maps index I into either a utility $u(I)$ or better still into a monetary value $Y(I)$ such that the difference between the monetary values of the two states H_h and $H_{h'}$ reflects the willingness to pay for a transition from H_h to $H_{h'}$.

$$Z(H_h \rightarrow H_{h'}) = Y[I(H_{h'})] - Y[I(H_h)] \quad (2.18)$$

Constructing such an index I from a multidimensional description of health status calls for a clarification of many details.

1. The *aspects* of health to be considered must be specified.
2. For every aspect, the states distinguished need to be *scaled* (e.g., limitations on performing one’s social role, such as getting dressed without any help, with little help, etc.; other aspects considered might be ability to work, to prepare a meal, etc.).

12. A standard work in this field is by Fanshel and Bush (1970). A more recent index is due to Kind, Rosser, and Williams (1982).

13. The WHO defines health ‘as the state of complete physical, mental and social well-being and not just the absence of disease or handicap’; see World Health Organization (1948).

3. For each of these scales, it has to be determined *who* is actually scaling the different levels (doctors, other experts, or those potentially affected?).
4. Finally, the *weights* used for an aggregating of the partial indices to a comprehensive index have to be determined.

These considerations lead to

CONCLUSION 2.10: The valuation of health status in terms of monetary units might be carried out in two steps. Step one consists of the construction of a cardinal “health index”; step two, of the conversion of this index into monetary units. However, this procedure involves considerable effort.

In the meantime, many have come to realize that the construction of such a dimensionless health status index may amount to a detour on the way to a monetary evaluation of changes in morbidity.¹⁴ Instead, it is possible to achieve a valuation of different health states *directly in terms of utility*. This will also help to find the function $t(\cdot)$ permitting the calculation of QALYs. Specifically, $t(1, H_h)$ is sought for every state H_h , the amount of time spent in the ideal health status that is equivalent to spending x years in state H_h . In the following, we will describe a procedure that is analogous to the determination of willingness to pay for risk reductions [for the generality of this analogy, see Torrance (1986)].

In a preliminary step, the relevant health states have to be described fully in terms of their essential aspects. With regard to the validity and reliability of the results, care should be taken to ensure that no important components are left out of the description while respondents should not be burdened with excessive detail. Respondents are then confronted with a specific hypothetical choice problem for every health state distinguished. This problem will be either

- a. Of the “time tradeoff” type; or
- b. Of the “standard gamble” type.

Time tradeoff method. The question here is: “Suppose you had a disease that would leave you in a state of H_h for T years if left without treatment. The only possible treatment is free and would cure you perfectly; however, it shortens the rest of your life span to t years.” Now *life span* t is varied until the person interviewed is *indifferent* between the alternatives “treatment” and “no treatment.” As shown in Figure 2–4, this amounts to the same problem as the one of transforming years spent in a less than perfect health status into QALYs (see Figure 2–3a). There are only two modifications here. The reference duration x is set equal to the remaining life expectancy T , and the utilities are normalized to $u[H^*] = 1$ for perfect health and $u[\text{death}] = 0$, respectively. Thus t^* of Figure 2–4 is determined such that area $OTAB$ equals area Or^*DH^* . The

14. That does not mean that in situations where the available budget is already fixed, comparing the cost per additional point on the health status index would not be useful. Such an index appears as a latent variable in Chapter 4 (Sections 4.2.2 and 4.4.2); for a comparison of several indices see Leu, Gerfin, and Spycher (1992).

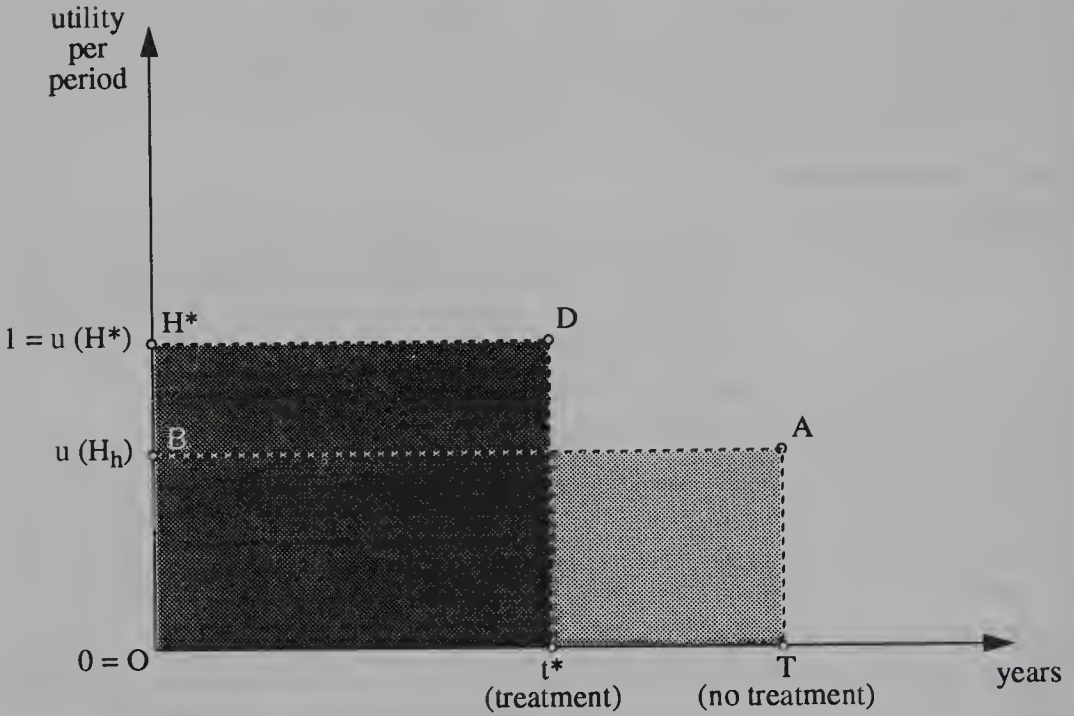


Figure 2-4. Time tradeoff method to evaluate health states.

resulting ratio $t^*(T, H_h)/T$ is then interpreted as the weight $w_a(1, H_h)$ with which to weigh health status H_h ¹⁵

$$w_a(1, H_h) := \frac{t^*(T, H_h)}{T} \tag{2.19}$$

Standard gamble method. Here we have the scenario, "Assuming you had a disease that would leave you permanently in state H_h without treatment. The only possible treatment is free and would cure you perfectly with probability π but lead, with probability $1-\pi$, to your immediate death" (see Figure 2–5). *Probability π* is now varied until the respondent is indifferent between the alternatives "treatment" and "no treatment." The resulting $\pi^*(H_h)$ is interpreted as the relative weight $w_b(1, H_h)$ with which to weigh health status H_h ¹⁶

$$w_b(1, H_h) := \pi^*(H_h) \tag{2.20}$$

At first sight, the two scenarios seem to be so different that the question arises whether the *same thing or something different* is being measured. In the latter case, one would

15. The subscript a in $w_a(1, H_h)$ reminds us that method (a) has been applied.
16. This way of putting the tradeoff presupposes that the respondent prefers health state H_h to immediate death. For situations in which the individual would opt for death, a slightly modified lottery can be constructed. See Torrance (1986, p. 21 ff).

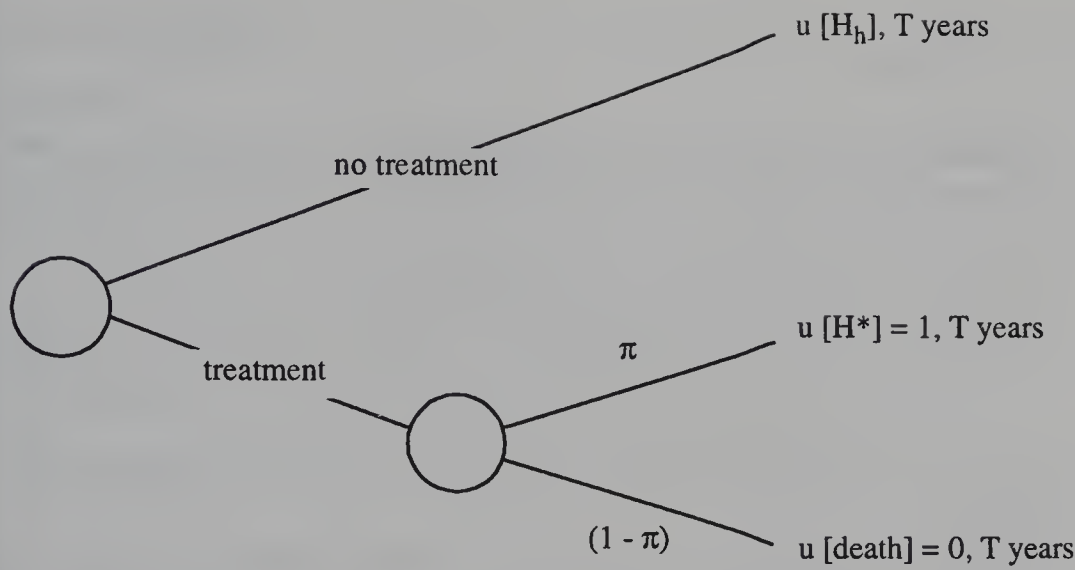


Figure 2-5. Standard gamble method to evaluate health states.

have to decide which measure is the “adequate” one. However, it can be shown that there is a plausible theoretical model *encompassing both scenarios* such that for any state H_h , the resulting $w_a(1, H_h)$ and $w_b(1, H_h)$ are equivalent.

This model considers an individual having T years to live at most. There are G possible health states (including death), and state H_h occurs with probability π_h and lasts t_h years ($h = 1, \dots, G$).

Furthermore, the assumption is that the respondent maximizes his expected utility in the form of

$$EU = \sum_{h=1}^G \pi_h u(H_h) t_h \tag{2.21}$$

using the normalization

$$u[H^*] = 1 \text{ and } u[\text{death}] = 0 \tag{2.22}$$

This expected utility function is restrictive in that it views total utility as a linear function of duration and utility weights of health states irrespective of their chronological order. Note that the utility weights $u(H_h)$ are originally only known to the individual and have to be investigated by the interviewer through hypothetical variation of t_h and π_h , respectively.

The *time tradeoff method* (scenario a) considers the two *certain* options, (A) to live in state H_h for T years, and (B) to live in state H^* for t^* years. These two options are equivalent in terms of (expected) utility. In both of them, probabilities $\pi_h = 1$ because health states are certain. The time weights assume the values

$$\left. \begin{array}{ll} T, & \text{for } H_h \\ 0, & \text{for "death"} \end{array} \right\} \text{ in Option A}$$

$$\left. \begin{array}{l} t^*, \quad \text{for } H^* \\ T-t^*, \quad \text{for "death"} \end{array} \right\} \quad \text{In Option B}$$

Inserting these values in Eq. (2.21) and simultaneously considering Eq. (2.22), equality of expected utilities holds if

$$1 \cdot u(H_h) + T \cdot 0 \cdot 0 := EU^A = EU^B := 1 \cdot 1 \cdot t^* + 1 \cdot 0 \cdot (T - t^*)$$

$$\Leftrightarrow u(H_h) = \frac{1}{T} t^* (T, H_h) \quad (2.23)$$

The $w_a(1, H_h)$ derived from scenario (a) in Eq. (2.19) thus coincides with the utility weight $u(H_h)$ for the health state H_h in the expected utility function (2.21) of the individual.

In the *standard gamble method* (scenario b), every state of health is considered to last permanently; that is, the factor t_h in Eq. (2.21) generally equals T . Equality of expected utilities of alternatives A “treatment” and B “no treatment” implies, in analogy to Eq. (2.23)

$$\pi^* \cdot 1 + (1 - \pi^*) \cdot 0 \cdot 1 = EU^A = EU^B = 1 \cdot u(H_h)$$

$$\Leftrightarrow u(H_h) = \pi^* (H_h) \quad (2.24)$$

and it follows from Eqs. (2.23) and (2.24) that

$$w_a(1, H_h) = u(H_h) = w_b(1, H_h) \quad (2.25)$$

This line of reasoning gives rise to

CONCLUSION 2.11: If the respondents behave according to the expected utility model of Eqs. (2.21) and (2.22), the time tradeoff method and the standard gamble method lead to the same result, in that they measure the utility weights of the affected health state on a scale ranging from 0 (death) to 1 (perfect health).

However, actual applications of the time tradeoff method and of the standard gamble method lead to *differing* results. On the one hand, one may not always succeed in defining health states H_h in such a way that they are interpreted uniformly by all respondents. On the other hand, there are a lot of experimental results questioning the validity of the expected utility rule [Kahneman and Tversky (1979), Pommerehne, Schneider, and Zweifel (1982)]. The difficulties associated with applying the two methods should therefore not be underestimated. Nevertheless, directly focusing on the utility weights of health states is the most convincing alternative, from a theoretical point of view.

2.5 EXERCISES

Exercise 2.1:

Consider an individual whose utility depends on whether he survives the subsequent period or dies immediately. His utility also depends on his wealth y ($y > 1$). The probability of death is 0.2. In case of survival, utility is given by $u_1 = 100y^{3/4}$, and in case of death, it is $u_0 = 100y^{1/4}$. (a) Why must the time horizon extend beyond the life span of an individual in this model? (b) Determine and interpret the marginal rate of substitution (MRS) between a change of risk and a change of wealth. (c) Show how the MRS changes with a marginal change of the death risk and discuss the result. (d) Show how the MRS changes with the initial level of wealth and explain the result.

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3 THE INDIVIDUAL AS THE PRODUCER OF HIS HEALTH

3.1 STATEMENT OF THE PROBLEM

As a point of departure, consider the following proverb: “Health is not everything in life, but without health, life is nothing.” This proverb points to the dual property of health.

- ***Health is a highly valued asset.*** Sometimes it is even claimed that health is the only thing that counts in life. The first part of the proverb reminds us that other values and goals do exist in life, yet, compared to health, they rank lower on the preference scale of most people. While this *priority of health* is accepted as fact in the following section, it does not rule out that health risks (i.e., an increased probability of poor health) are taken in order to achieve other goals.

- ***Health is a prerequisite for other activities.*** The second part of the proverb points to health as a precondition for success in other activities. Poor health limits the *production capabilities* of the affected person, including his or her ability to enjoy the good things of life (apart from health). In particular, consumption services do not flow automatically from consumer goods but require a great deal of time, knowledge, and skills, all of which can be impaired by a poor state of health.

An aspect not included in the proverb cited above is that health can be produced. Health economists, however, have adopted this idea, which is mirrored in the title of this chapter, the individual as the “producer of his health.” Not so long ago, good health was considered to be a gift from God and poor health to be simply bad luck. The successes of modern medicine have fostered the conviction that almost everyone can reach a good state of health, provided he uses the appropriate means. However, if health is perceived as being produced, the immediate question of the *producer* arises. Even though much of healing continues to be attributed to the

physician, it cannot be denied that every healing process ultimately begins in the psyche and body of the individual concerned. The fact that many illnesses heal spontaneously also lends support to the view that the individual is the ultimate producer of his health.

The expression “producer of health” or even “health production” must not mask the fact that the individual (with or without the help of a physician) can only influence his state of health but not effectively determine it. Apart from hereditary and environmental factors, chance may always annihilate one’s efforts to preserve or improve health. Thus large, unexpected changes in health can occur at any time.

The second part of the proverb cited at the beginning of this chapter, “... but without health, life is nothing” refers to the *second property* of health. Good health is a precondition for success in other activities. Evidently, good health contributes to the production of consumable services because the better the state of health, the more time available for income-generating or health-enhancing activities. Health is considered here as a capital good: the greater the quantity of capital goods, the greater the output of goods and services, which can be plowed back into investment, for still more goods and services.

The aim of this chapter is to define the conditions under which the individual acts as the producer of his health. In particular we are concerned with deriving tradeoffs, that is, with finding out what sacrifices the individual must make for a given improvement of his health. Answers to the following questions need to be found.

1. From the individual’s point of view, what are the conditions that determine the *optimal resource allocation* between health and other goods?
2. Can the often volatile pattern of behavior (“sinning against health as long as health is good, sacrificing everything for health when health is bad”) be brought into line with *rationality*? Or are individual preferences indeed inconsistent, implying that medical experts should perhaps determine the appropriate amount of preventive measures?
3. Can the economic concept of *substitution* be applied to health production, even though situations can easily be envisaged in which only medical services (rather than one’s own health-enhancing efforts) offer the prospect of a cure?

With these questions in the background, an attempt will be made to develop a model of health production in this chapter. This will then serve as a basis for interpreting empirical studies in Chapter 4. The next section is devoted to critical objections to the application of the economic production concept to health. It also introduces the reader to the capital approval to health pioneered by Grossman (1992). However, in an attempt to alleviate the concerns mentioned, a formulation is proposed in which the individual (and his physician) can do no more than marginally *influence the probability* of transition from one health state to another. Willingness to pay for improved chances of health in this short-run context is derived in Section 3.3. In Section 3.4, a model of state-dependent production is presented. If healthy in the current period, the individual may himself enhance health through his activities; if, on the other hand, he is ill, his only chance for recovery is to rely on medical care. From this it can be seen that the opportunity costs of “better health” are state-dependent, which could also explain the volatile nature of health behavior noted above.

3.2 THE CONCEPT OF HEALTH PRODUCTION

3.2.1 Basic Considerations

For the economic layman it is strange, perhaps even offensive, to talk of health production. Associating the concept of production with methods of cultivation in agriculture and in particular with production processes in industry, he may immediately see two difficulties for the application of this concept to health care.

- **Lack of control.** Production processes in agriculture and industry can be controlled to a large extent. If the producer expects a future increase in demand, he can produce on inventory; if he expects a drop in demand, he will limit production or reduce stock. The process of producing health does not seem to permit this type of control.

- **Lack of tradability.** The output of production is usually sold to a third party. In the case of health, such a sale would be perceivable only in a slave economy, in that a higher price would be obtained for a healthy slave than for a sick one. In today's society a higher wage can be obtained by a healthy working person [see Bartel and Taubman (1979)]. However, other characteristics of the worker, in particular his skills, determine his wage as well. These skills are not transferable either, posing an additional challenge to noneconomists who are reluctant to view education as the "production" of skills.

While these objections are not without merit, they are not as strong as they appear. In the first place, lack of complete control over the production process does not preclude the existence of a systematic relationship between inputs and outputs. The fact that the harvest yield in agriculture in a given year importantly depends on the weather does not rule out that increased use of fertilizer generally increases the yield, and farmers behave in accordance with this. In the same way, efforts taken to maintain health can be considered as inputs in a production process. While they do not always help to avoid illness, on the whole they do contribute to better health.

Part of the lack of control is the apparent perishability of the finished product. However, both health and education may be viewed as *invisible capital stocks*, which are augmented by investment, provide services, and are subject to depreciation (see the following subsection). For instance, the QALYs of Section 2.4.1 can be considered as the service flowing from health capital. For education capital, the service would be an increase in wage income. The annual holiday, and more generally a lifestyle that avoids excessive demands on one's health, corresponds to investment in health. If a loss in health capital (depreciation) occurs in the wake of illness, then the health stock of an individual who has previously invested in his health is still higher than that of another. It is a known fact that rested people withstand most illnesses better than over-tired people, nonsmokers better than smokers, those of normal weight better than overweight people. Thus, while "producing on inventory" cannot be directly observed, the concept of production need not be dismissed out of hand.

Lack of tradability of the produced good seems to present another obstacle for adopting the concept of a production process or a production function. However, the individual *trades with himself* as it were. Indeed, he must trade better health against

something else of value. Admittedly, he does not have to give up money as a rule because medical services in most industrialized countries are almost free, thanks to comprehensive health insurance. From the economic point of view, however, the sacrifice in favor of health may involve other inputs, in particular time, which could be used for other activities.

On the other hand, applying the concept of production, or more precisely, of a production function, to health has considerable advantages.

1. ***Optimal conditions as a point of reference.*** Production has a cost, and the costs resulting from securing better health can be compared to the marginal costs of other outputs, in particular, consumption services. In an optimal state, the marginal cost of health relative to the marginal cost of consumption should be equal to the individual's marginal willingness to pay for health relative to consumption. The relative (marginal) willingness to pay for health can be determined in principle using the model of Section 2.3.3, and it can be compared to the relative marginal costs. There is reason to expect a discrepancy between the two ratios, since the marginal willingness to pay in the health care sector is distorted both by insurance protection and the physician, while the negotiated fees for medical services do not have much in common with their true marginal costs.
2. ***Relative marginal productivity of inputs.*** The concept of the production function comprises the notion that different inputs make different contributions to the resulting output. One such input are medical services, whose true marginal costs are extremely high in today's health sectors. The huge expenditure on medical care can only be justified, from the economist's viewpoint, if it is characterized by a very high marginal productivity compared with other inputs in health production. In this context it should be noted that, for example, an improvement of environmental quality could perhaps have a greater marginal productivity in terms of health (measured, e.g., in QALYs; see Section 2.4.1), than a continued increase in health care expenditure.

These advantages appear on the whole to justify the application of the economic concept of the production function to health.

CONCLUSION 3.1: The fact that the health status of a person is largely determined by chance, that health is perishable and lacks tradability, does not rule out considering health as being the result of a production process. Moreover, the concept of a production function is the key to an assessment of the optimality of the individual's behavior and the efficient use of scarce resources.

3.2.2 Health as Part of Human Capital

If we are to accept the economist's view of health as an asset capable of being produced, then we can broaden the concept of capital as follows. On the one hand, there are the assets that are tradable in the marketplace. Included in this are financial assets as well as

tangible components such as real estate. On the other hand, there is human capital consisting of knowledge and skills, and health. Health production can therefore be viewed as an investment that makes up for the capital consumption connected with illness, possibly even resulting in a net increase of the capital stock “health” [see Grossman (1972), Muurinen (1982), and Wagstaff (1986)]. Investment is achieved by the input of curative medical services, whose productivity generally depends on the age of the recipient. The stock of health capital is subject to *depreciation*, the rate of which again varies with age but also depends on the type of consumer goods used (such as alcohol and tobacco).

3.2.2.1 Outline of the Grossman Model

In the following, the Grossman model is presented in the version of Wagstaff (1986). Thus let there be an individual having a planning horizon ending at period T , the time of his death. During each period τ , there is a nonnegative amount of sick time t^s , which is the lower the larger the health stock H . Put the other way around, healthy time constitutes the nontradable service flowing from the unobserved stock of health. The individual derives disutility from sick time $t^s(\tau)$ while deriving utility from consumption goods X . The utility function U defined over these two arguments is assumed to be time independent; thus the marginal rate of substitution between sick time and consumption (indicating the relative importance of the two arguments) does not change with aging. Continuous discounting of future utility flows using the subjective rate of time preference ρ results in exponentially declining weights. In all, the individual's maximand is given by (for the equations of the whole model, see also Table 3.1)

$$W = \int_0^T e^{-\rho\tau} U\{t^s(\tau), X(\tau)\} d\tau, \quad \frac{\partial U}{\partial t^s(\tau)} \leq 0, \quad \frac{\partial U}{\partial X(\tau)} \geq 0, \quad \frac{\partial t^s(\tau)}{\partial H(\tau)} < 0, \quad \text{for all } \tau \quad (3.1)$$

The crucial component of the Grossman model is the next equation, which defines the law of *motion of the health stock over time*. On the one hand, the individual may invest in health by purchasing medical care. If there were only his gross investment I , then the rate of change of the health stock $\dot{H}$ would be positive, causing the individual to become healthier all the time. However, health stock also depreciates at a rate δ , and it is the task of medical care services (which make up investment I) to keep health stock H on the trajectory deemed optimal by the individual. The rate of depreciation increases with time τ to reflect aging, while it may be reduced or increased by consumption, depending on the lifestyle of the individual considered. In all, the law of motion governing the development of health stock constitutes a constraint that will enter the Lagrangean through the multiplier $\mu(\tau)$, the value of which generally depends on time.

$$\dot{H}(\tau) = I(\tau) - \delta\{\tau, X(\tau)\}H(\tau), \quad \{\mu(\tau)\}$$

$$\frac{\partial \delta(\tau)}{\partial \tau} > 0, \quad \frac{\partial \delta(\tau)}{\partial X(\tau)} \geq 0 \quad (3.2)$$

Table 3.1: The Grossman Model

$$W = \int_0^T e^{-\rho \tau} U\{t^s(\tau), X(\tau)\} d\tau, \quad \frac{\partial U(\tau)}{\partial t^s(\tau)} \leq 0, \quad \frac{\partial U}{\partial X(\tau)} \geq 0, \quad \frac{\partial t^s(\tau)}{\partial H(\tau)} < 0, \quad \text{for all } \tau \quad (3.1)$$

$$\dot{H}(\tau) = I(\tau) - \delta\{\tau, X(\tau)\}H(\tau), \quad \{\mu(\tau)\}$$
$$\frac{\partial \delta(\tau)}{\partial \tau} > 0, \quad \frac{\partial \delta(\tau)}{\partial X(\tau)} \geq 0 \quad (3.2)$$

$$\dot{A}(\tau) = rA(\tau) + Y\{t^s(\tau)\} - q(\tau)I(\tau) - p(\tau)X(\tau), \quad \{\lambda(\tau)\}$$
$$\frac{\partial Y(\tau)}{\partial t^s(\tau)} < 0 \quad (3.3)$$

$$H[T] \geq H', \quad A[T] \geq 0 \quad (3.4)$$

$$\left(\frac{\partial U / \partial t^s(\tau)}{\lambda[0]} e^{-(\rho-r)\tau} + \frac{\partial Y(\tau)}{\partial t^s(\tau)} \right) \frac{\partial t^s(\tau)}{\partial H(\tau)} = \left(r + \delta(\tau) - \frac{\dot{q}(\tau)}{q(\tau)} \right) q(\tau) \quad (3.5)$$

- Change of a variable over time, e.g., $\dot{H} = \partial H / \partial \tau$
- W: Welfare of the individual
- U: Per period utility
- H: Health stock
- δ : Rate of depreciation of the health stock
- A: Stock of financial assets
- r: Interest rate
- Y: Labor income
- X: Consumption goods
- p: Unit price of consumption goods
- I: Investment in health
- q: Net price of medical care, per unit
- τ : Time
- t^s : Sick time
- ρ : Time preference rate
- μ, λ : Lagrangean multiplier

In spite of health insurance, medical care generally carries a positive net price q , while consumption goods cost p per unit. These expenditures can be offset by labor and property income. Labor income Y is the higher the less sick time is lost on the job, while property income flows from the stock of financial assets A at a rate r , which denotes the real rate of interest, the general price level being kept constant in the model. In all, the law of motion of financial assets is given by [Lagrangean multiplier $\lambda(\tau)$]

$$\dot{A}(\tau) = rA(\tau) + Y\{t^s(\tau)\} - q(\tau)I(\tau) - p(\tau)X(\tau), \quad \{\lambda(\tau)\}$$
$$\frac{\partial Y(\tau)}{\partial t^s(\tau)} < 0 \quad (3.3)$$

Dynamic optimization problems of this type require boundary conditions both for the initial and the final period [for more details, see Intriligator (1971, Chap. 14) or Léonard and Van Long (1992, Chaps. 2 and 3)]. In the present context, the stock of health at the end of the planning period should be at least as large as the minimum H' , which in fact defines death, while financial assets should be nonnegative

$$H[T] \geq H', \quad A[T] \geq 0 \quad (3.4)$$

Methods of optimal control theory can be applied to Eqs. (3.1) through (3.4). Basically, the two laws of motion (3.2) and (3.3) are added to the objective function (3.1) using Lagrangean multipliers $\mu(\tau)$ and $\lambda(\tau)$, respectively. Thus, contrary to static optimization, the value of these multipliers changes over time, reflecting the fact that *current period decisions may serve to relax or tighten* the constraints. For example, current investment in health increases current and future health stock *ceteris paribus*, which means that additional increments of H would not add very much to the (optimal) value of the target function anymore, a fact that in turn is reflected by a reduced value of the Lagrangean multiplier $\mu(\tau)$. The Lagrangean function is then maximized at a particular point in time τ with respect to I , H , X , and A , the logic being that optimization of one element of the utility flows defined in Eq. (3.1) serves to optimize the entire stream of such flows. This stream is evaluated at starting time $\tau = 0$, as indicated by the lower bound of the integral of Eq. (3.1). Focusing on the necessary optimality conditions derivable from the two health-related decision variables H and $\dot{H}$, one obtains [see Wagstaff (1986), Appendix]

$$\left(\frac{\partial U / \partial \dot{r}^s(\tau)}{\lambda[0]} e^{-(\rho-r)\tau} + \frac{\partial Y(\tau)}{\partial \dot{r}^s(\tau)} \right) \frac{\partial \dot{r}^s(\tau)}{\partial H(\tau)} = \left(r + \delta(\tau) - \frac{\dot{q}(\tau)}{q(\tau)} \right) q(\tau) \quad (3.5)$$

This condition requires the marginal utility of an investment in health to be equal to its marginal cost. The marginal utility side of the equation may be interpreted as follows.

- **Effectiveness as a precondition.** For the investment in health to have a positive payoff, it must bring about a reduction of sick time. A negative value of $\partial \dot{r}^s(\tau) / \partial H(\tau)$, in combination with the negative values of the terms in the large parentheses, results in a positive value of the right-hand side of Eq. (3.5), indicating positive marginal utility.
- **Valuation of health as a consumption good.** The reduction of lost sick time (and hence gain of healthy time) increases utility directly since $\partial U / \partial \dot{r}^s(\tau) < 0$. Subjectively discounted to decision time $\tau = 0$, this benefit amounts to $[\partial U / \partial \dot{r}^s(\tau)] e^{-(\rho-r)\tau}$. However, the gain in terms of healthy time could also be used for relaxing the constraint on the development of financial assets as specified in Eq. (3.3). The present value of this relaxation in period $\tau = 0$ is given by $\lambda[0] e^{(\rho-r)\tau}$, with $\lambda[0]$ indicating the marginal impact of a relaxation of the wealth constraint (3.3) on the Lagrangean function in the initial period. The factor $e^{-(\rho-r)\tau}$ reflects the force of these two discounting processes, one health-

related and subjective, the other asset-related and objective. In spite of these asset-related influences, versions of condition (3.5) retaining only this first component of the marginal utility caused by an investment in health are commonly called “*pure consumption*” models.

- **Valuation of health as an investment good.** The reduction of sick time has an immediate impact on labor income through $\partial Y(\tau)/\partial t^s(\tau)$. Thus, even if sick time were not disliked for its own sake, investment in health has a return in terms of added labor income.¹ Since health then becomes a commodity that is not valued per se but only for its impact on labor income and ultimately on the individual’s stock of financial assets, versions of condition (3.5) retaining only this component are known as “*pure investment*” models.

The right-hand side of condition (3.5) symbolizes the marginal cost associated with holding an additional unit of health capital. Since health stock is viewed as durable, this marginal cost generally falls short of the marginal outlay occasioned by the necessary investment in health [its unit price being $q(\tau)$]. Rather, marginal cost amounts to interest forgone [r per monetary unit of $[\dot{q}(\tau)]$] and depreciation [$\delta(\tau)$ per unit of $q(\tau)$]. Finally, in a dynamic context, the cost of holding any capital good decreases if its purchasing price rises over time [$\dot{q}(\tau) > 0$]. In this case, the purchaser is happy to have bought early in time. Conversely, the cost of holding an asset can become very high if its value decreases over time [$\dot{q}(\tau) < 0$]. These considerations hold true of the capital good “health” as well, with the price of a marginal unit given by the price of the investment good, which primarily depends on the *net price of medical care* in this context. It may also depend on the price of the individual’s own time (i.e., his wage rate) as soon as personal preventive efforts are admitted as an input into the production of investment I .

CONCLUSION 3.2: In the Grossman model, health and wealth constitute two interrelated assets the values of which are optimally controlled over time by the individual. In the case of health, the marginal utility of holding a marginal unit of stock has a consumption and an investment component, which together must always be equal to its marginal user cost consisting of interest, depreciation, and a possible change in the value of the capital good “health” over time.

3.2.2.2 Critical Assessment of the Grossman Model

This section is devoted to the derivation of testable hypotheses from the Grossman model and to their comparison with available empirical evidence. The first type of such hypotheses concerns the joint development of the endogenous variables over time. For example, the model may predict the path of health capital with increasing age. The second type relates to behavioral adjustment to a change in some exogeneous variable.

1. This financial return can be transformed into marginal utilities as well by multiplying Eq. (3.5) through by $\lambda[0]$, the increment in the value of the target function (which is expressed in utility units) due to a relaxation of the wealth constraint in the initial period.

Table 3.2: Main Predictions of the Grossman Model

Partial correlation with	Demand for health $H(\tau)$				Demand for medical care $I(\tau)$			
	Pure consumption (1)		Pure investment (2)		Pure consumption (3)		Pure investment (4)	
Health	..		..		$+^a$	[NC]	?(+)	[NT]
Age	$-^b$	[C]	$-^b$	[C]	$+^b$	[NC]	$+^b$	[NT]
Education	$+^a$	[NT]	$+^a$	[C]	$-^a$	[NC]	$-^a$	[NT]
Lifetime wage	$-^b$	[NC]	$+^b$	[NT]	?(-)	[NT]	$+^b$	[NT]

+: positive partial effect predicted; -: negative partial effect predicted; ..: no theoretical prediction
C: confirmed; NC: not confirmed; NT: not tested
Sources: ^aMuurinen (1982); ^bWagstaff (1986).

To this end, the optimality condition Eq. (3.5) is totally differentiated with respect to the exogenous variable of interest [like, for example, Eqs. (2.5) and (2.6) of Subsection 2.3.2.2.2], to find out the adjustments required to restore dynamic optimality. Theoretical predictions of this type were derived by Muurinen (1982) and Wagstaff (1986) for both the pure consumption and pure investment variants of the Grossman model. These predictions appear in Table 3.2 and will be discussed in the light of existing empirical evidence.

- **Health.** It should be noted that better health status cannot be defined separately from the demand for health because desired health stock is always equal to actual health stock in the Grossman model. Better health status may be defined as a separate entity in the determination of the demand for medical care, however. Still, $dH(\tau) > 0$ should not be regarded as an exogenous shock like aging [$d\tau > 0$] because such a change is caused by an investment in health, which brings $H(\tau)$ back up to its desired level during the period considered. Therefore, health stock $H(\tau)$ is an endogenous quantity. The entries in columns (3) and (4) of Table 3.2 thus reflect a partial correlation between two endogenous variables, $I(\tau)$ and $H(\tau)$. Clearly, *this partial correlation must be positive*. This has been rigorously demonstrated in the literature for the case of the pure consumption model only; however, the logic of the argument that investment in health adds to health stock carries over to the pure investment model. Unfortunately, empirical evidence consistently fails to confirm this crucial prediction [Wagstaff (1986), Leu and Gerfin (1992)]: When health status is introduced as a latent variable through the use of simultaneous indicators,² all components of medical care distinguished exhibit a very definite and highly significant *negative* partial relationship with health.

2. See Section 4.4.2 for a description of latent variable modeling.

- **Age.** In both variants of the model, the demand for health is predicted to decrease in the course of aging. As stated above, payoffs from health stock in the guise of healthy time may decrease in the course of aging. In addition, the rate of depreciation $\delta(\tau)$ increases, thus pushing up the marginal cost of holding an additional unit of health capital [see the right-hand side of condition (3.5)]. Demand for medical care typically should increase because it is not optimal for the individual to let health stock decline in step with depreciation, calling for increasing amounts of investment. Turning again to the empirical evidence, the prediction with regard to health stock is confirmed [Leu and Doppmann (1986), Leu and Gerfin (1992)]. The prediction with regard to the *demand for medical care, however, is consistently contradicted* by empirical evidence in the case of ambulatory medical care [Wagstaff (1986), Leu and Gerfin (1992)], especially if the probability of seeing a physician at all is distinguished as an indicator of patient-initiated demand not yet confounded by medical advice provided by a self-interested physician [Duan et al. (1984), Newhouse and Phelps (1974), Zweifel (1985)].
- **Education.** In both variants of the model, education should be positively related to health stock. On the one hand, improved education may enable the individual to select a consumption basket $X(\tau)$ that is more favorable to his health, with the effect of lowering the rate of depreciation $\delta(\tau)$. On the other hand, it may also make him a more productive contributor to investment in health once the model is generalized to admit not only medical care but also personal health-enhancing efforts as determinants of $I(\tau)$. With regard to the demand for medical care, better education should lower the demand for investment in health because there is less depreciation to be made up for. The first prediction is empirically *confirmed* in Wagstaff's (1986) test of the pure investment model, while it is rather consistently *rejected* in his test of the pure consumption model. As a determinant of the demand for medical care, education either fails to have a statistically recognizable effect [Leu and Gerfin (1992)] or is even found to bolster demand, contrary to the entries of columns (3) and (4) of Table 3.2 [Wagstaff (1986)].
- **Lifetime wage.** Since the current wage of the observation period presumably varies with health status, it is not fully exogenous. This problem may be relieved by estimating a lifetime or permanent value of the wage. The predicted relationship with health stock is negative in the pure consumption model for the following reason. The lifetime wage approximates $\lambda[0]$, the utility value of a relaxation of the wealth constraint (made possible by the additional healthy time available for work). However, as shown in the optimality condition (3.5), $\lambda[0]$ serves to "deflate" the utility gain $\partial U/\partial t^s(\tau)$ that comes from holding an additional unit of health stock. Therefore, the higher the lifetime wage, the lower the marginal incentive to hold health as an asset for consumptive use. By way of contrast, in the pure investment model a high lifetime wage serves to increase the term $\partial Y(\tau)/\partial t^s(\tau)$ in Eq. (3.5) and hence to reinforce the incentive to hold health as an asset. The prediction with regard to the demand for medical care is ambiguous in the pure consumption variant of the model as soon as the

individual may contribute his own time as an input to investment, the opportunity cost of which increases with the wage rate. In the pure investment model, the direct effect on labor income from reduced sick time is thought to be sufficiently strong as to permit the purchase of additional medical care. Again, the *empirical evidence is mixed*. Wagstaff (1986) finds a positive wage effect on health stock in the pure consumption model, contrary to predictions [see column (1) of Table 3.2], while it comes out insignificant in the work of Leu and Gerfin (1992). Lifetime wage is positively related to health in the pure investment variant of the Grossman model, in accord with column (2) of Table 3.2. However, it does not belong to the significant explanatory variables in any of the equations relating to the demand for medical care [see columns (3) and (4) of Table 3.2 and Wagstaff (1986)].

CONCLUSION 3.3: Many of the implications of the Grossman model are contradicted by available empirical evidence. Most important, the notion that expenditure on medical care constitutes a demand derived from an underlying demand for health cannot be upheld because health status and demand for medical care are negatively rather than positively related.

Admittedly, the final verdict on the Grossman model is not in yet. More conclusive evidence may be expected from panel data tracing individuals over time in their efforts to keep their health stock on the optimal trajectory prescribed by dynamic optimization. It seems safe to say at this point, however, that emphasis on long-run optimization generally tends to *detract from the uncertainty* surrounding health status and survival itself. Thus it runs the danger of overstating the degree of control an individual may have on his state of health. In particular, neglect uncertainty appears to give rise to two shortcomings in the Grossman model.

1. It should be noted that the rate of depreciation $\delta(\tau)$ is *defined deterministically* in Eq. (3.2). Thus there are no stochastic shocks such as accidents or major illnesses that may impart very large values to $\delta(\tau)$. Indeed, $\delta(\tau)$ may take on a value so large as to prevent the individual from returning to the optimal value of health stock during the current period because medical care simply is not effective enough to make up for such a large health loss. Thus current health status may *deviate from optimal* health status, an eventuality that is excluded from the Grossman model.

2. The rate of depreciation and hence loss of health stock may become so life threatening as to let the *planning horizon* T shrink to a few days or even hours. In that event, rules of optimal long-term intertemporal allocation of the type described by condition (3.5) lose much of their relevance.

For these reasons, an alternative approach will be proposed in the remainder of this chapter. Contrary to the Grossman model, it will emphasize the individual's *lack of control over health status*. The starting point is provided by health status as being governed by a stochastic process. All the individual can influence in this process is the probabilities of transition from one state to another. This view opens the door to making willingness to pay for health state-dependent (see Section 3.3). Even more important,

the individual's production possibilities can be made state-dependent as well, with one technology (based on preventive efforts) prevailing in the "healthy" state but another technology (relying on medical care) in the "sick" state. A corresponding model will be presented in Sections 3.4 and 3.5. A survey on empirical research into the production of health in the light of both the Grossman model and the alternative approach viewing the development of health as a stochastic process will be given in Chapter 4.

3.3 SHORT-RUN OPTIMIZATION AND WILLINGNESS TO PAY FOR HEALTH

3.3.1 The Production of Health as the Modification of a Stochastic Process

The health status of a given period (e.g., a particular day) can be viewed as a sample taken from a sequence of states. In principle, the individual is subject to chance, deprived of a choice between two sequences such as

hhhhssssshhhhhhhhhshhhhhhhhh...

and

hhhshhhssssssssssshhsssss...

where h is healthy and s is sick.

All the individual can do is to influence the transition probabilities and thus to increase the likelihood of the preferred sequence among the two to be realized. Considering only two consecutive periods, there are four possible transitions: hh , hs , sh , and ss (see Table 3.3). Accordingly, ϕ_{hs} symbolizes the probability of being sick in period 2 after having been healthy in period 1. If the probabilities entered in Table 3.3 remain constant over time, the sequence has the properties of a Markov process.

In the following, the probability of being healthy in period 2, $\pi_{h,2}$, is of particular interest. This probability is given by (see Table 3.3)

$$\pi_{h,2} = \pi_{h,1}(1 - \phi_{hs}) + \pi_{s,1}(1 - \phi_{ss}) \quad (3.6)$$

The first term reflects the possibility of remaining in good health in period 2, after having been in good health during period 1, with $1 - \phi_{hs}$ symbolizing the relevant transition probability. The second term reflects the case of a person who is sick in period 1 but turns healthy at the transition to period 2, with $1 - \phi_{ss}$ denoting the transition probability.

Now let the state of health in period 1 be known to the individual. In the case of good initial health, $\pi_{s,1} = 0$ in Table 3.3, and the only way to influence $\pi_{h,2}$ is through ϕ_{hs} . Being healthy, the individual has a wide choice of preventive measures at his disposal, ranging from nutrition to recreation. However, all of these measures cost *time spent in favor of health*, symbolized by t^H . Conversely, given sickness in period 1, $\pi_{h,1} = 0$, and the assumption is that *only medical services* M can contribute to reducing

Table 3.3: Transition and State Probabilities

Period 1	Period 2	
	Healthy (<i>h</i>)	Sick (<i>s</i>)
Healthy (<i>h</i>)	$1 - \phi_{hs}$	ϕ_{hs}
Sick (<i>s</i>)	$1 - \phi_{ss}$	ϕ_{ss}
$\pi_{h,2} (= 1 - \pi)$	$\pi_{h,1} (1 - \phi_{hs}) + \pi_{s,1} (1 - \phi_{ss})$	
$\pi_{s,2} (= \pi)$	$\pi_{h,1} \phi_{hs} + \pi_{s,1} \phi_{ss}$	

ϕ_{hs} : Probability of transition from healthy to sick
 ϕ_{ss} : Probability of transition from sick to sick
 $\pi_{h,2}$: State probability of being healthy in period 2, “chance of health”
 $\pi_{s,2}$: State probability of being sick in period 2, “risk of sickness” (denoted π below)

the probability ϕ_{ss} and hence increasing the probability of being healthy again in period $\pi_{h,2}$. Hence, the chance of health in period 2 depends on different factors, conditional upon the state of health prevailing in period 1,

$$\pi_{h,2} = \begin{cases} \pi_{h,2}[\phi_{hs}(t^H, \dots)], & \text{if healthy in period 1} \\ \pi_{h,2}[\phi_{ss}(M, \dots)], & \text{if sick in period 1} \end{cases} \tag{3.7}$$

CONCLUSION 3.4: In a formulation that seems to be in accordance with everyday experience, the production function for health is dependent on the state of health prevailing during the decision period. Production of health amounts to a modification of transition probabilities in a sequence of states of health that is essentially governed by chance.

In contradistinction to the Grossman model, production of health thus is not defined in terms of medical care (possibly combined with prevention) once and for all. Rather, it involves only one of the two inputs, depending on the prevailing state of health. In addition, the output of the process is not a stock of health but merely an increased probability to be in a better state of health during the following period.

3.3.2 Are Supply and Demand Independent of One Another?

In health policy, the question of whether demand for health care can be separated from the (self-produced) supply is of fundamental importance. An example illustrating this is the debate about the “rationing of health” [see Callahan (1987)]. Such a rationing amounts to excluding some of the more expensive medical interventions from the benefits of social health insurance if the patient is beyond a certain age. This serves to limit the production possibilities of the aged when in bad health, a state of considerable

probability. Thus rationing may cause an important loss of welfare. This loss can be mitigated by providing cheaper substitutes (for example, homecare). This works well if the individual's own productivity in terms of health does not decrease too fast with age. However, should preference for health care increase with age, then rationing would still burden the individuals concerned with a high loss of welfare.

From the economist's point of view, the *desirable separation of demand from supply* is made difficult by the patient's reliance on information provided by his physician. The physician could be tempted to exploit his superior knowledge. By overstating the risk of illness $\pi_{s,2}$, he could bolster patients' marginal willingness to pay for health care and therefore for his own services. From Eq. (2.6) it follows that an increase in the risk of death increases the marginal willingness to pay for a reduction of this risk. The services offered by the physician are part of such a risk reduction. By exaggerating the probability of illness to his patient, the physician can increase the demand for his services. Inasmuch as this supplier-induced demand also strengthens the preference for health, *demand and production of health can no longer be separated* [see Evans (1974) and Reinhardt (1985)].

Another interpretation of supplier-induced demand is that the physician may exaggerate his contribution to recovery. Whether this occurs in practice will be examined further in Section 7.2. Here the underlying preferences of the patient are assumed unchanged. Rather, a discrepancy between perceived and actual form of the transformation curves (as shown in Figure 3–3) occurs. In this chapter only this second variety of supplier-induced demand will be discussed, so that, at least in principle, an analysis of the production of health separate from preferences is still possible.

3.3.3 Deriving the Short-Run Willingness to Pay for Health Chances

In Chapter 2, methods to estimate the marginal willingness to pay for improved health were discussed. Willingness to pay means nothing else but a willingness to go without other desirable things in favor of health. If these other things are understood to be consumer goods or, in short, “consumption,” then this willingness to pay is reflected by the ratio of marginal utilities between consumption C and health H , and therefore by the slope of the indifference curves in the (H, C) space (as introduced in Chapter 1).

According to the considerations of Section 3.1, health itself cannot be controlled, but the probability of reaching a particular state of health in the following period can be modified. In full analogy with Eq. (3.7), which makes the production function state-dependent, the utility derived from consumption can be made *state-dependent* as well. In this case, the objective function of an individual with a long-term planning horizon may be written

$$EU = \sum_{t=0}^T (1 - \rho)^t \{ (1 - \pi_t) u[C_{h,t}, h] + \pi_t u[C_{s,t}, s] \} \quad (3.8)$$

According to this equation, future streams of utility are summed up after being discounted by the individual using the subjective rate of time preference ρ . In each period, the expected utility depends on consumption ($C_{h,t}$ and $C_{s,t}$, respectively) and

the state of health in the following way: $u[C_{h,t}, h] > u[C_{s,t}, s]$ for values of C and all t . A given quantity of consumption services always gives rise to more utility in the healthy state than in the sick state. Therefore, the utility function changes form according to health status. This fact is symbolized by including H as an argument in the function as a variable that can take on the values h and s , precisely as in Subsection 2.3.2.2.2. Finally, π_t symbolizes the probability of being sick in period t .

In the following, the planning horizon is shortened to comprise periods 1 and 2 only. Since the period is defined as a very short time span such that the transition probabilities of Table 3.3 can be changed marginally, time preference may be discarded ($\rho = 0$) and Eq. (3.8) simplified to

$$EU = (1 - \pi_1)u[C_{h,1}, h] + \pi_1 u[C_{s,1}, s] + (1 - \pi_2)u[C_{h,2}, h] + \pi_2 u[C_{s,2}, s] \quad (3.9)$$

As in the derivation of Eq. (3.7), let the state of health in period 1 be known, for example, $\pi_1 = 0$, (healthy in period 1). Therefore, first period consumption $C_{h,1}$ is left as the one decision variable relating to the current period. As will be shown in Section 3.4, making do with less consumption in period 1 serves to reduce the probability π_2 of being sick in the following period. Therefore, it is appropriate to determine the reduction of π_2 that would compensate for the sacrifice of one unit of $C_{h,1}$ or $C_{s,1}$ (consumption in period 1). In this calculation, consumption in period 2 is kept constant at $C_{h,2} = C_{s,2} = C_2$. The following relationship defines the marginal willingness to pay for improved health in the sense of reducing the risk of illness π_2 ,

$$dEU = 0 = \frac{\partial u[C_{h,1}, h]}{\partial C_{h,1}} dC_{h,1} - (u[C_2, h] - u[C_2, s])d\pi_2 \quad (3.10)$$

In order to represent the subjective tradeoff in preference (i.e., the indifference curve) between consumption and improved chances of health as two goods, Eq. (3.10) is solved for $dC_{h,1}/d(1 - \pi_2)$ rather than $dC_{h,1}/d\pi_2$, resulting in

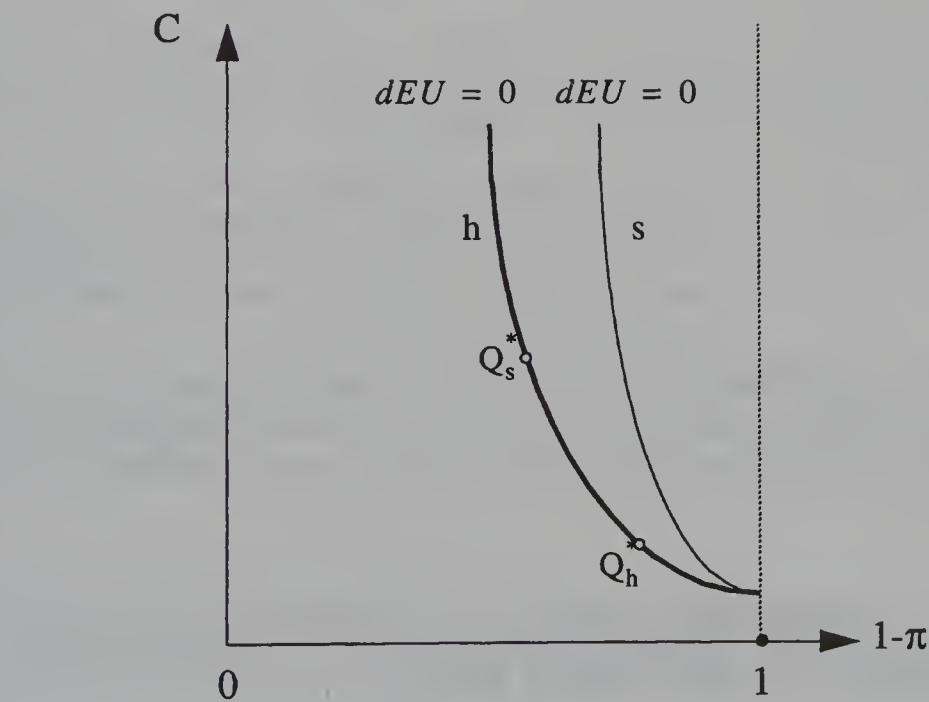
$$\frac{dC_{h,1}}{d(1 - \pi_2)} = - \frac{dC_{h,1}}{d\pi_2} = - \frac{u[C_2, h] - u[C_2, s]}{\frac{\partial u[C_{h,1}, h]}{\partial C_{h,1}}} \quad (3.11)$$

However, the individual could have been in a state of sickness in period 1. Giving up a unit of consumption $dC_{s,1}$ would again have to be compensated by an improvement in the chance of health $d(1 - \pi_2)$. Setting $\pi_1 = 1$ in Eq. (3.9) this time, one obtains another slope of the indifference curve

$$\frac{dC_{s,1}}{d(1 - \pi_2)} = - \frac{dC_{s,1}}{d\pi_2} = - \frac{u[C_2, h] - u[C_2, s]}{\frac{\partial u[C_{s,1}, s]}{\partial C_{s,1}}} \quad (3.12)$$

Equations (3.11) and (3.12) can be interpreted as follows.

- Both equations indicate a *marginal rate of substitution* (MRS) between consumption and the probability of being healthy. The individual is willing to pay for a marginal improvement of his chances of being healthy in the following period by sacrificing some consumption during the current period.
- The MRS is given by the ratio of two differences of utility (which become marginal utilities in the limit) in the usual way. The greater the *difference between “healthy” and “sick”* in terms of utilities in the numerator of Eqs. (3.11) and (3.12), the greater the absolute value of MRS and hence marginal willingness to pay for health. The greater the *loss in utility* occasioned by a sacrifice of consumption in the current period (denominator of the equations), the smaller the willingness to pay for an improvement of one’s chance of health.
- The MRS may well be *state-dependent*. The reason for this is that the marginal utility of consumption takes on a different value according to health state. It could be small in times of good health [large denominator of Eq. (3.11)] while large in times of bad health [small denominator of Eq. (3.12)].



C: Consumption in period 1 ($= C_{h,1} , C_{s,1}$)
1 - π : Probability of being healthy in period 2 ($= 1 - \pi_2$)

Figure 3-1. Marginal willingness to pay for health (short term).

This last point is illustrated in Figure 3–1. For simplicity, no differentiation with regard to periods is made and C used as the common symbol for a unit of consumption services, regardless of state of health. Accordingly, indifference curves are shown in a $[C, (1 - \pi)]$ space, one having index h , the other s . Both have high absolute slopes, mirroring considerable marginal willingness to pay for health throughout. However, the indifference curve with index s (initial state sick) has an even more marked (negative) slope, reflecting an even *higher marginal willingness to pay for health*. Again, the reason is that additional consumption has a lower marginal utility in the denominator of Eq. (3.12) compared to Eq. (3.11). In this way, a behavior can be modeled that patients are often criticized for by their physicians, “Do not care much about health as long as it is good, sacrifice everything for health as soon as it is bad.”

In the remainder of this section, this alleged instability of preferences with regard to health will not be pursued any further, for three reasons.

1. The argument that marginal utility of consumption is *lower in the state of sickness* than in the state of health is *not fully convincing*. Quite possibly, a sick person has to rely to a particularly high degree on high-quality accommodation and nutrition, implying that additional consumption services are of very high utility to him (see Section 6.3.2).

2. *Health insurance* enables an individual to redistribute his income between health states in a way as to make the marginal utilities of additional income or consumption converge; under certain conditions, they will even become equal (see again Section 6.3.2).

3. MRS also varies along a *stable indifference curve*. In Figure 3–1, the realization of a point like Q^*_h in good health and a point like Q^*_s in bad health is sufficient for creating the impression of a volatile valuation of health.

CONCLUSION 3.5: Short-run marginal willingness to pay for health can be represented by a subjective tradeoff between “consumption in the current period” against “probability of being healthy in the subsequent period.” It may be regarded as state-dependent, but there is no necessity to do so.

Since this chapter deals primarily with the individual as a producer and not as a demander of health, it will be shown in the following section that the seemingly volatile valuation of health may be due to the stochastic nature of individual production possibilities.

3.4 A MODEL WITH STATE-DEPENDENT PRODUCTION POSSIBILITIES: THE SHORT RUN

3.4.1 State-Dependent Optimization in the Stochastic Process

In Section 3.2.3 the production of health was defined as the influence exerted on probabilities characterizing a stochastic process. The subjective valuation of improved chances of health was introduced, using terms such as transition probabilities, state

probabilities, and expected utilities. These terms will now be used to describe the short-term decision situation of the individual, which will give rise to state-dependent production possibilities and hence tradeoffs.

A sample taken from a sequence of health states is shown in Figure 3-2. In a previous period 0, the health status of the individual was fixed. By assumption, this also determines his possibilities of influencing transition probabilities during period 1 in his attempt to control the state probabilities of period 2. If the individual is initially healthy (upper branch of Figure 3-2), he can spend time in favor of health (t^H), lowering the probability of falling ill ϕ_{hs} and hence the probability π of being sick in period 2 (see Table 3.2). He can thus realize a utility payoff $u(C_h, h)$ with probability $[1 - \pi(t^H)]$. With probability $\pi(t^H)$, he must settle for the lower payoff $u(C_s, s)$. His *short-run objective* is to select a value t^H such that expected utility EU , calculated over the conditional payoffs of period 2, is maximized.

If the initial state in period 1 is “sickness,” then efforts will be made to reduce the probability ϕ_{ss} in order to avoid spending period 2 in the state of sickness as well. By assumption, this is only possible *through the utilization of medical services M* but not using one’s own means t^H . Accordingly, there is the probability $[1 - \pi(M)]$ of attaining a utility payoff amounting to $u(C_h, h)$ in period 2. With probability $\pi(M)$, however, only the utility payoff from consumption in the state of sickness $u(C_s, s)$ can be

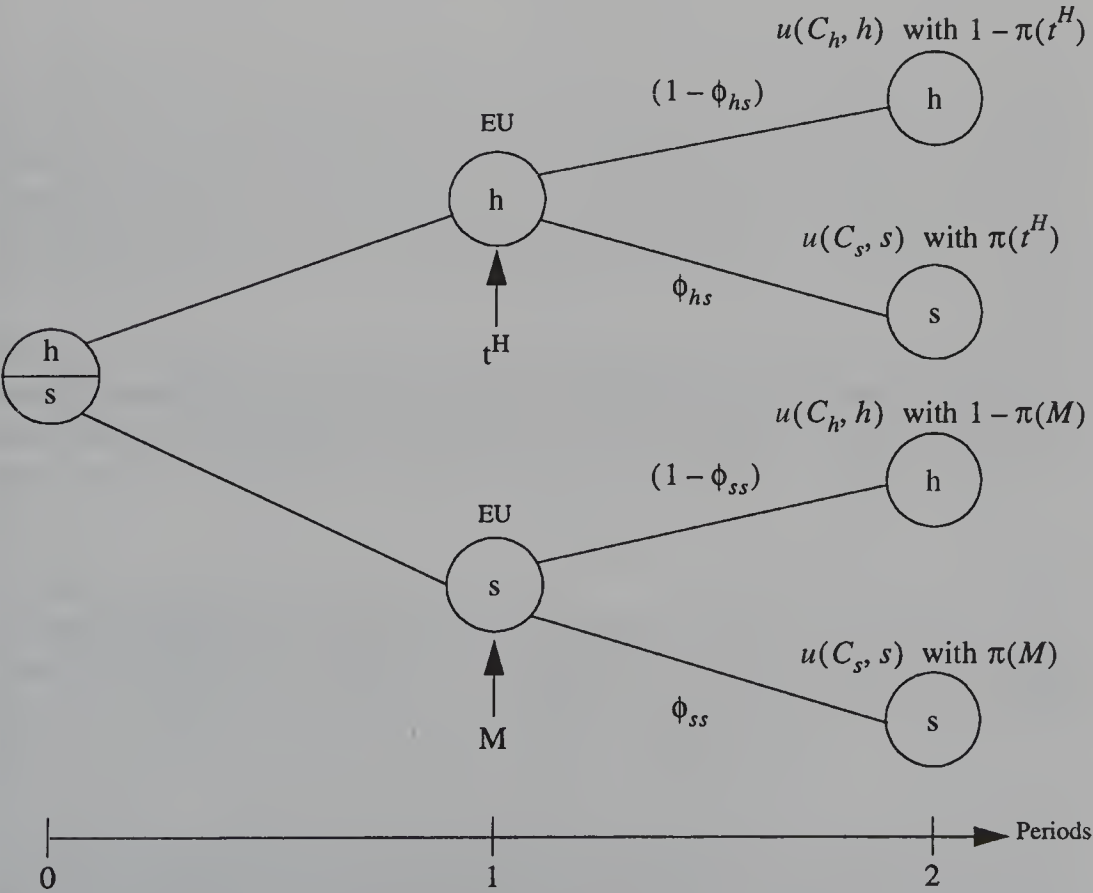


Figure 3-2. Sequence of health states as stochastic process.

attained. The individual will try to maximize expected utility again, this time by optimizing his use of medical services.

The peculiarities of the production process as shown in Figure 3–2 can be seen in the following three points.

- The role and responsibility of the individual as a producer of his own health is limited to an influence on *probabilities*.
- This influence can take the form of the deployment of one's own efforts only in a *state of good health*, whereas inputs provided by a third party are indispensable in a state of sickness.
- Health status is not only the result of a production process but also has the effect of a *stochastic input factor*, causing the production function to be conditional on health.

This specification constitutes one way of expressing in economic terms why “health is different.” In sum, one has

CONCLUSION 3.6: From an economic point of view, the peculiarity of health can be seen in its dual role as an output of a production process and as a determinant of the individual's possibilities to contribute to this production process.

3.4.2 The Equations of the Model

In the model presented in this subsection, the individual is presumed to distinguish two states of health, “healthy” and “sick,” and to limit his planning horizon to the current and one subsequent period. Whereas the production possibilities are different between the two states, a joint objective function is postulated as given by Eq. (3.9). Therefore, the model described below (and shown in Table 3.4) is in the tradition of “contingent claims,” the analysis of which was pioneered by Arrow (1951) and Debreu (1959, Chap. 7).

As long as the individual is *healthy*, he can himself have an influence on his chance of health, while medical services have no effect. In particular, this means that additional input of time in favor of health t^H can serve to reduce the probability of being sick in the following period π , with π remaining strictly positive, however

$$\pi = \pi(t^H), \quad \text{with } \partial\pi/\partial t^H < 0 \text{ and } \pi > 0 \quad (3.13)$$

Consumption services produced depend not only on the input of consumption goods X but also on time available for consumption t^C , in the line of thought of Becker (1965)

$$C_h = C_h(X, t^C), \quad \text{with } \partial C_h/\partial X > 0, \partial C_h/\partial t^C > 0 \quad (3.14)$$

If healthy (and active in the labor force), the individual earns a *labor income* given by his working time t^W and the wage rate w . By assumption, this income is just sufficient for financing the purchases of consumption goods at price p

Table 3.4: A Model of State-Dependent Production of Health (Short Term)

<i>Initial situation</i> healthy ($1 - \pi$)		<i>Initial situation</i> sick (π)	
$\pi = \pi(t^H)$, with	(3.13)	$\pi = \pi(M)$, with	(3.17)
$\partial\pi/\partial t^H < 0$, $\pi > 0$		$\partial\pi/\partial M < 0$, $\pi > 0$	
$C_h = C_h(X, t^C)$, with	(3.14)	$C_s = C_s(X, t^C)$, with	(3.18)
$\partial C_h/\partial X > 0$, $\partial C_h/\partial t^C > 0$		$\partial C_s/\partial X > 0$, $\partial C_s/\partial t^C > 0$	
$wt^W = pX_h$	(3.15)	$\bar{Y} = pX_s + qM$	(3.19)
$1 = t_h^C + t^H + t^W$	(3.16)	$1 = t_s^C + \mu M$	(3.20)

C: Consumption services

h: State of good health (subscript)

s: State of sickness (subscript)

M: Medical services

 μ : Time cost per unit of medical services

p: Price of consumption goods

 π : Probability of being sick in the subsequent period

q: Net price of medical services

 t^C : Time spent on consumption t^H : Time spent in favor of health t^W : Working time

w: Wage rate

X: Consumption goods

 $\bar{Y}$: Social security income in case of sickness

$$wt^W = pX_h \quad (3.15)$$

Finally, *time* amounting to one period is available for consumption t^C , health maintenance t^H , and work t^W

$$1 = t_h^C + t^H + t^W \quad (3.16)$$

Turning to the *production possibilities in the state of sickness*, the equations listed on the right-hand side of Table 3.4 reveal several differences. First, the assumption is that the probability of spending the subsequent period in sickness can only be lowered by using medical services M . It is still, however, not possible to reduce this probability to zero

$$\pi = \pi(M), \text{ with } \partial\pi/\partial M < 0 \text{ and } \pi > 0 \quad (3.17)$$

The *production of consumption services* requires both time and consumption goods, exactly as in the healthy state. However, these two inputs take on a lower value, since illness restricts the productive activities of the household

$$C_s = C_s(X, t^C), \text{ with } \partial C_s/\partial X > 0, \partial C_s/\partial t^C > 0 \quad (3.18)$$

Of crucial importance is the fact that the *income in the event of sickness* does not depend anymore on working time in today's industrial countries but is equal to a

replacement income $\bar{Y}$ guaranteed by social security. This income is used to finance the purchase of consumption goods and of medical services (at net price q)

$$\bar{Y} = pX_s + qM \quad (3.19)$$

Since working time is zero in the state of sickness, the *time constraint* comprises only time for consumption and for utilization of medical services. The individual, as a patient, views each physician hour or each day in hospital as costing μ hours or days of his own time. By assumption, working time t^W as well as time spent in favor of health t^H are also zero, because of social security and due to lack of effectiveness, respectively

$$1 = t_s^C + \mu M \quad (3.20)$$

The two sets of Eqs. (3.13)–(3.16) and (3.17)–(3.20) admittedly overstate the differences between the two health states. For example, healthy individuals also visit a physician, mostly in the hope of detecting a potential threat to their health early enough. Clearly, they perceive the probability of falling ill ϕ_{hs} and hence π as dependent on medical services M in the healthy state as well. Conversely, health-enhancing efforts by the individual (reducing ϕ_{ss} and hence π) may also be worthwhile in the state of sickness, contrary to assumptions. Finally, especially independent workers frequently continue working even though they are ill, causing t^W to appear in their time constraint regardless of their state of health. These qualifications will be neglected, however, in the aim of sharpening

CONCLUSION 3.7: The productive possibilities of an individual depend on the health status inherited from the previous period in many ways. If in good health, he can contribute himself to an increase of his chance of health while earning a labor income. If in bad health, he must rely on medical services, does not work, and receives a transfer income that must cover expenditures on consumption goods as well as the net cost of medical care.

3.4.3 Conditional Short-Term Production Frontiers

This section is devoted to the analysis of two production frontiers (tradeoffs).

1. Tradeoff between consumption and the probability of being healthy in the subsequent period; *good* health in the current period.
2. Tradeoff between consumption and the probability of being healthy in the subsequent period; *bad* health in the current period.

3.4.3.1 The Short-Term Tradeoff Given Good Health (No. 1)

The tradeoff between health and consumption can be illustrated by the case of a pedestrian who crosses the street against a red light. Such a person does not directly opt

for a worse state of health but for a probability distribution containing *unfavorable states with increased probability*. In terms of Table 3.3, he accepts an increased probability of becoming ill (due to an accident) ϕ_{hs} , which implies a smaller probability $(1 - \pi)$ of being in good health during the next period. On the other hand, the reduced waiting time in front of the light (amounting to a reduction of t^H) is now available for additional consumption t^C . Therefore, the tradeoff confronting our pedestrian can be depicted as a transformation curve in $(C_h, 1 - \pi)$ space (see Figure 3–3a). The shape of this curve is given by its slope, the marginal rate of transformation (MRT). Total differentiation of Eqs. (3.13)–(3.16) yields [see Appendix 3.A.1, Eq. (3.A.6)]

$$\frac{dC_H}{d(1 - \pi)} = \frac{\partial C_h}{\partial t^C} \bigg/ \frac{\partial \pi}{\partial t^H} < 0 \tag{3.21}$$

The concavity of this frontier is derived in the appendix [Eq. (3.A.7)].

The indifference curves shown in Figures 3–3a and b could in principle be members of different families [see Eqs. (3.11) and (3.12)]. In order to focus on the impact of conditional production possibilities, however, they are assumed to be of the same family and to be *homothetic*, that is, exhibiting the same slope (MRS) along a radial line from the origin. If the individual is healthy, he will attain Q_h^* , where the marginal rates of transformation MRT and substitution MRS are equal.

The following parameters influencing Eq. (3.21) are discussed here.

- **Increase in the real wage rate (w/p).** As a first approximation, this change does not have an immediate influence on the transformation curve [see the explanation of Eq. (3.A.6) in the Appendix for a justification]. Basically, an increase in the real wage serves to boost both labor income and the time cost of consumption in this simplified model, resulting in a cancellation of the two

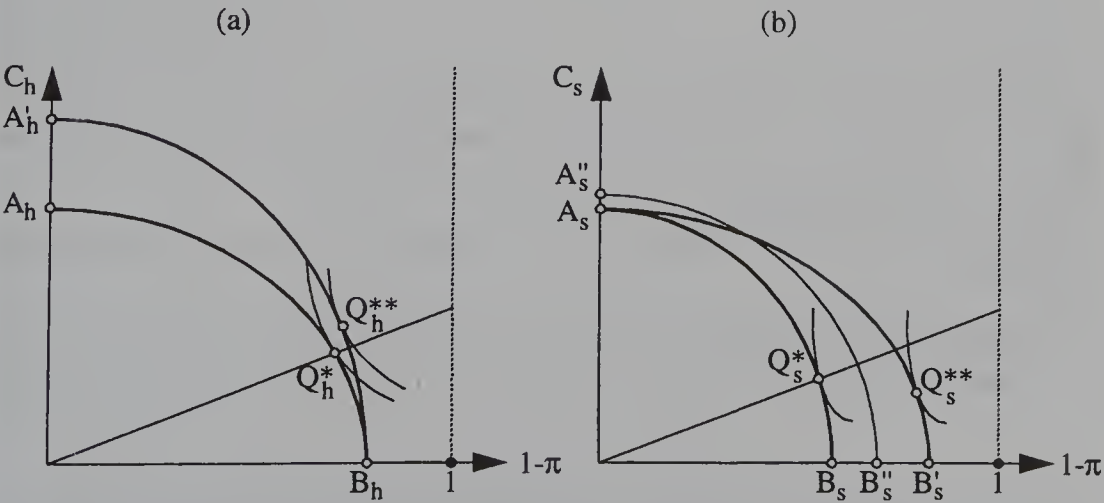


Figure 3-3. Tradeoffs between consumption and health, influenced by exogenous changes with (a) healthy in current period and (b) sick in current period.

effects.³ Therefore, the tradeoff between health chances and consumption has approximately the same slope for poor and rich individuals.

- **Technological change in the household.** Being mainly labor-saving, this type of technological change results in an increase of $\partial C_h / \partial t^C$. Under the presumption that the marginal productivity $\partial \pi / \partial t^H$ remains unaffected, the transformation curve becomes steeper, running from A'_h to B_h in Figure 3–3a. Given a high marginal willingness to pay for an improved chance of health, the initial optimum would have to be close to point B_h , such as Q_h^* . Therefore, the production possibility set will increase comparatively little in the horizontal direction, resulting in a new optimum such as Q_h^{**} . Accordingly, the diffusion of labor-saving consumption technology should have little impact on the health behavior of individuals—at least as long as they are healthy.

3.4.3.2 The Short-Term Tradeoff Given Bad Health (No. 2)

On the basis of Eq. (3.17)–(3.20), one obtains for the production possibility frontier [see Eq. (3.A.13)]

$$\frac{dC_s}{d(1 - \pi)} = \frac{\frac{\partial C_s}{\partial t^C} \mu}{\frac{\partial \pi}{\partial M}} + \frac{\frac{\partial C_s}{\partial X} \frac{q}{p}}{\frac{\partial \pi}{\partial M}} < 0 \quad (3.22)$$

(+)
(+)

(–)
(–)

This equation shows that an improved chance of health can only be achieved at the price of a sacrifice of consumption services, since both terms of the sum are unambiguously negative. On the one hand, utilization of medical services requires time to be spent by the patient (first term); on the other hand, medical services and consumption goods compete for the fixed income (second term). The opportunity cost of an improved chance of health depends significantly on $\partial C_s / \partial t^C$ and $\partial C_s / \partial X$. The first quantity (the individual's productivity in producing consumption services) is low in the state of sickness, whereas the second quantity (productivity of purchased goods and services) may be higher than in good health.

The transformation curve appears in Figure 3–3b. It is influenced by the following parameters.

- **Technological change in the household.** As in the healthy state, this serves to increase the (absolute value of the) slope of the transformation curve. (This change is not shown in Figure 3–3b).

3. This is in contrast to the Grossman model, which distinguishes a negative wage effect on the demand for health stock in its pure consumption version from a positive wage effect in its “pure investment” version (see Table 3.1).

- **Technological change in medicine.** This can be symbolized by an increased absolute value of $\partial\pi/\partial M$. Both terms of Eq. (3.22) therefore decrease in absolute value, resulting in a flatter frontier, symbolized by the shift from $A_s B_s$ to $A_s B'_s$ in Figure 3–3b. The extreme point on the C axis remains unchanged because the reproduction possibilities of the current period are not affected by technological change in medicine. However, the extreme point on the $(1 - \pi)$ axis *moves away from the origin* because the maximum attainable probability of being healthy in the subsequent period increases. A comparison between the optima Q_s^* and Q_s^{**} leads to the prediction that individuals profit from technological change in medicine by opting for a greatly increased chance of health.
- **Increased density of supply.** This can be modeled as a reduction of μ , implying that the patient has to spend less time when using medical services. In view of the time constraint (3.22), t^C and/or M may take on higher values, which means that both extreme points of Figure 3–3b move away from the origin. To the extent that medical services are subject to *decreasing marginal returns*, the new extreme point B''_s will remain to the left of B'_s , obtained for the case of technological change in medicine discussed in the previous paragraph. And according to Eq. (3.22), a lowered value of μ serves to reduce the absolute value of $dC_s/d(1 - \pi)$, imparting a flatter slope to the new transformation curve, corresponding to the curve $A''_s B''_s$ in Figure 3–3b. Apparently, optimization by the sick individual will lead to a limited improvement of his chance of health, in conjunction with additional outlay on medical services.
- **Extended coverage by health insurance.** This serves to decrease q/p because the insured has to bear a smaller share of the true cost of a physician hour, a drug, or a hospital day. Therefore, his income can be spent on additional consumption goods and/or medical services. In terms of Figure 3–3b, the transformation curve again shifts away from the origin, to a position such as $A''_s B''_s$ because decreasing marginal returns continue to set in as early as before given an unchanged state of technology in medicine.

These considerations give rise to

CONCLUSION 3.8: In both states “health” and “sickness,” the short-term transformation curve is decreasing and concave from the origin. Technological change in the household serves to increase its steepness, leading only to a slight increase in chances of staying healthy. Technological change in medicine, increased density of supply, and extended coverage by health insurance serve to make it flatter and to move it away from the origin, resulting in improved chances of health in the state of sickness.

3.4.3.3 The Alleged Instability of Health Behavior

In Section 3.3, the claim was made that the observed instability of health behavior need not reflect a volatile valuation of health but can be traced to the stochastic nature of

individual production possibilities. The truth of this claim shall be proved now, subject to two assumptions.

ASSUMPTION 3.1: The indifference curves have identical MRS in the two states (see the discussion following Conclusion 3.3) and are homothetic.

ASSUMPTION 3.2: The difference between the states “healthy” and “sick” has a stronger impact on the maximum attainable chance of health ($1 - \pi$) than the maximum attainable amount of consumption (C).

Let the individual opt for Q_h^* when healthy (see Figure 3–4). MRS_h and MRT_h have the same value in this optimum. In the state of sickness, the individual must settle for point Q_s^* ; at that point, MRS_s and MRT_s are again equal. In the majority of cases, this new optimum must lie *above* the ray $0Q_h^*$ (as marked in Figure 3–4) because the transformation curve A_sB_s runs more steeply on average than A_hB_h , in keeping with Assumption 3.2. In particular, point $\bar{Q}_s$, having the same MRS as in point Q_h^* by assumption 3.1, cannot qualify as an optimum. MRS must take on a *higher value* in order to satisfy the condition $MRS_s = MRT_s$.

This comparison among state-dependent production possibilities therefore leads to the prediction that in the *state of sickness a higher absolute value* of the marginal rates of substitution MRS as well as of transformation MRT will be realized. The observable health behavior of individuals accordingly suggests that an improvement of their

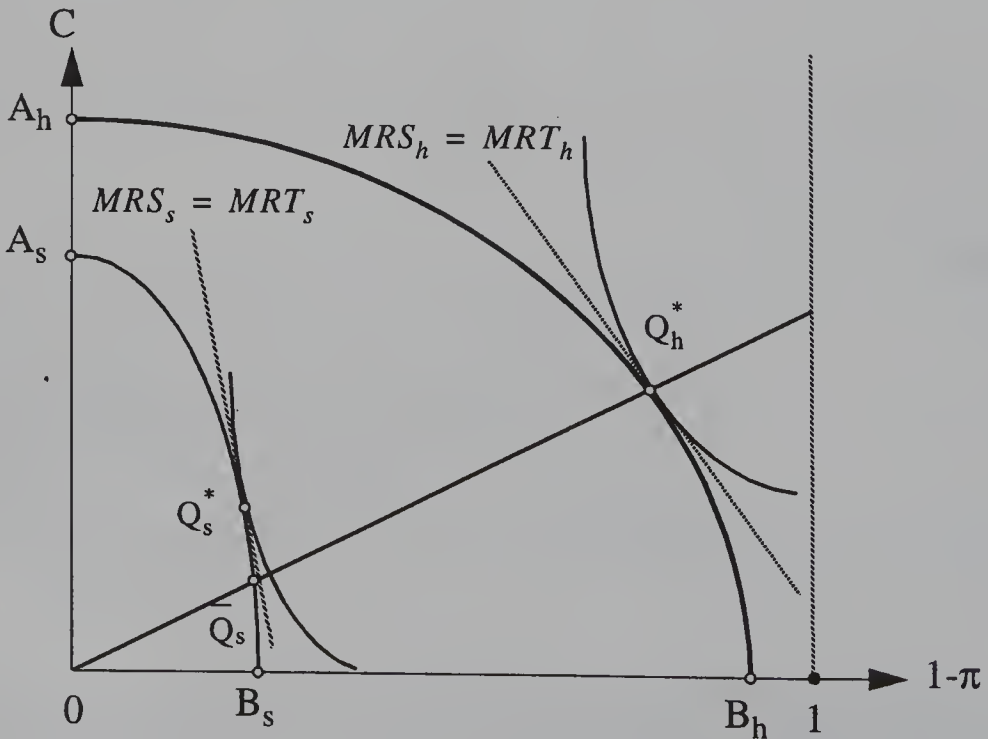


Figure 3-4. The alleged instability of health behavior.

chance of health is valued more highly in the event of illness than in good health. This inconsistency was already explained in Section 3.3, using economic terms. The usual explanation by physicians (shared by many economists), however, points to the aforementioned *inconsistency of preferences*, that people care too little about their health as long as they are healthy while being prepared to sacrifice almost everything as soon as they fall ill.

By way of contrast, the observable increase of willingness to pay for an improved chance of health in the state of sickness is *not related to a volatility of preferences* at all in Figure 3–4. Rather it can be traced to objective production possibilities and their stochastic changes, which are occasioned by the instability of health itself. Moreover, physicians only observe the revealed willingness to pay for medical services. To the extent that the assumption made above holds true, viz. that medical services do not contribute to an improvement of chances of health as long as the healthy state prevails, willingness to pay for such services should be small. Therefore, the demand for *medical services* becomes state-dependent even if the demand for health is stable. This fact is of crucial importance to empirical research dealing with the demand for health services, as will be shown in Section 4.4. These considerations lead us to

CONCLUSION 3.9: The alleged instability of individual health behavior does not permit one to conclude that there is a lack of rationality but may simply reflect state-dependent production possibilities of the individual.

This conclusion is important because health is a nontradable commodity, which makes the separation of subjective preferences from objective production possibilities especially difficult. If chances of health could be traded between buyers and sellers, one would not only *observe* optima such as Q_h^* and Q_s^* of Figure 3–4, but also the associated *terms of trade*. Therefore, relative prices could be compared with MRT, that is, the relative marginal costs in the production of consumption services and health, for finding out whether individuals are at their optimum indeed and what factors are responsible for any systematic deviation from that optimum. However, the lack of tradability of health renders relative prices nonobservable, which would represent objective information and could contribute to an evaluation of health-related policies.

3.5 A MODEL WITH STATE-DEPENDENT PRODUCTION POSSIBILITIES: THE LONGER RUN

If an individual succeeds in improving his chances of health in a permanent way, he modifies his longer-run time constraint. An increased probability of spending subsequent periods in good health results in an accumulation of healthy periods. The *time thus gained* can in turn be used to produce more consumption services and/or invest in health, a notion that is also prominent in the Grossman model [see Eq. (3.2)]. Conversely, an increase in the probability of being sick in subsequent periods results in an increased number of periods spent in bad health.

For simplicity, the individual is assumed to extrapolate his current state of health and to anticipate the resulting production possibilities. His optimization thus remains *myopic* in that he does not formulate a strategy for the event that the state of health should change again at the end of the current phase of good or bad health. Thus a sick person would not make commitments to step up his preventive efforts if he should ever become healthy again. Accordingly, the two short-run tradeoffs are simply extrapolated as follows.

1. Tradeoff between average duration of a future phase of good health and consumption, current state *healthy*.
2. Tradeoff between average duration of a future phase of sickness and consumption, current period *sick*.

3.5.1 The Longer-Run Tradeoff Given Good Health (No. 1')

In order to analyze the longer run, the Markov process described in Table 3.3 is now extended to more than two periods. If the state probability (the risk of sickness) remains constant after a once-and-for all modification by the individual, then the Markov process becomes a binomial process [sick with probability π , healthy with probability $(1 - \pi)$]. A binomial process can be transformed into a geometric distribution of waiting times. The mean number of periods that elapse before the event "sickness" materializes is given by [see Bhattacharyya and Johnson (1977), p. 154]

$$T_h = \frac{1}{\pi} \quad (3.23)$$

where T_h is the mean number of periods lived in good health.

Accordingly, the time constraint is changed in that t_h^C , t^H , and t^W are not limited by the current time period as in Eq. (3.16) of Table 3.4, but by T_h , which is an implicit function of t^H

$$T_h = t_h^C + t^H + t^W \quad (3.24)$$

By means of total differentiation of Eq. (3.13)–(3.15) and (3.24), the slope of the transformation curve can be determined, which is now defined in (T_h, C_h) space rather than in $(1 - \pi, C_h)$ space [see Appendix 3.A.3, Eq. (3.A.19)]

$$\frac{dC_h}{dT_h} = \frac{\begin{matrix} (+) & (+/-) \\ \frac{\partial C_h}{\partial t^C} \left(\frac{\partial T_h}{\partial \pi} \frac{\partial \pi}{\partial t^H} - 1 \right) \end{matrix}}{\begin{matrix} \frac{\partial T_h}{\partial \pi} \frac{\partial \pi}{\partial t^H} \\ (-) \quad (-) \end{matrix}} \geq 0 \quad (3.25)$$

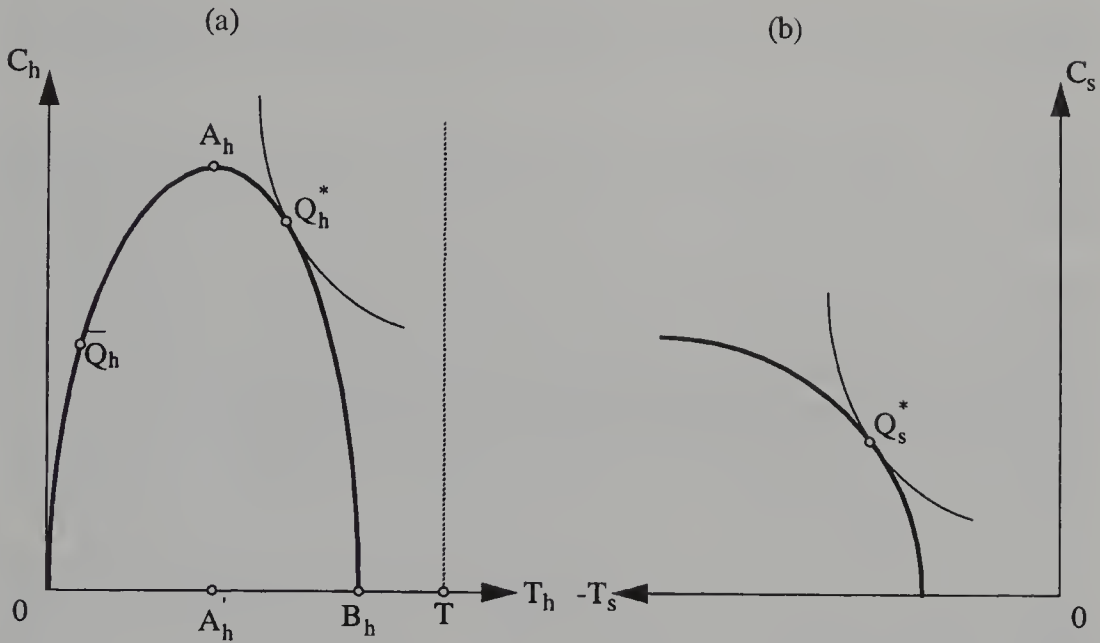


Figure 3-5. Tradeoffs between consumption and time spent in good health (longer term) with (a) healthy in current period and (b) sick in current period.

For small values of T_h , this transformation curve has a *positive slope*. In this situation, time spent on health is still small, while its marginal productivity $\partial \pi / \partial t^H$ is large (in absolute value). The product of $\partial T_h / \partial \pi$ and $\partial \pi / \partial t^H$ in the parentheses of Eq. (3.25) indicates whether an additional hour spent returns more or less than an hour of time in good health. As long as t^H returns more than one hour of healthy time, the numerator of Eq. (3.25) is positive, resulting in a positive slope of the transformation curve. Thus efforts in favor of maintaining or enhancing health constitute an investment, making more health and more consumption possible at the same time. This view corresponds to an early strand of thought in health economics, especially as applied to developing countries [see Mushkin (1962)].

The shape of the longer-run transformation curve is shown in Figure 3–5a. It is valid also at the *aggregate level* because the majority of a given population is healthy, capable of extending the duration of their current phase of good health by spending time on preventive efforts. This serves to increase aggregate productivity, and therefore health as well as consumption can be increased. For this reason, the tradeoff between consumption and health presented in Figure 1–2 is similar to the one shown in Figure 3–5a.

Sooner or later, however, the law of diminishing marginal returns will set in, causing marginal productivity of time spent in favor of health to fall. The location of the maximum point A_h in Figure 3–5a can be determined by setting the numerator of Eq. (3.25) equal to zero. From Eq. (3.23) one obtains for $\partial T_h / \partial \pi$

$$\frac{\partial T_h}{\partial \pi} = \frac{\partial}{\partial \pi} \left(\frac{1}{\pi} \right) = - \frac{1}{\pi^2}, \quad \text{or} \quad \frac{\partial T_h}{\partial (1 - \pi)} = \frac{1}{\pi^2} \tag{3.26}$$

Upon substitution into Eq. (3.25), the *critical value* for marginal productivity $\partial\pi/\partial t^H$ is given by

$$\frac{dC_h}{dT_h} \geq 0, \quad \text{if } \left| \frac{\partial\pi}{\partial t^H} \right| \geq |-\pi^2| \quad (3.27)$$

This critical value can be interpreted in terms of Figure 3–5a. Beyond point A_h , the influence of health-enhancing time on the risk of sickness, while still existing, is too small (too close to zero) to be compensated by a longer duration of the phase of good health. Concavity from the origin is derived in the Appendix [Eq. (3.A.7)].

On the preference side, the indifference curves of Figures 3–1 and 3–3, respectively, can be transformed into the (T_h, C_h) space of Figure 3–5a because T_h increases monotonically with $(1 - \pi)$, as can be seen from Eq. (3.26). This implies that the optimal point Q_h^* must lie beyond the maximum point A_h , where health is a *consumer good*. A situation in which health still amounts to an investment cannot be optimal (point $\bar{Q}_h$ of Figure 3–5a is not admissible). Nobody will purposefully opt for a short phase of health with $T_h < A'_h$.

Finally, the case $|\partial\pi/\partial t^H| < |-\pi^2|$ of Eq. (3.27) characterizes the *short-term tradeoff No. 1* of Subsection 3.4.3.1. This implies that the marginal productivity of preventive efforts usually take on a rather low value in the healthy state. Accordingly, the opportunity costs for additional improvement of health chances appears to be high [see Eq. (3.21)]. Therefore, the short-term optimum Q_h^* of Figure 3–3a has been put where the absolute value of the MRT and thus the opportunity cost of improved chances of health is high.

CONCLUSION 3.10: At the individual level, time spent in good health has the property of a consumption good rather than an investment good. This property of time as an input in the production of consumption services and health results in long phases of good health being opted for.

The fact that the transformation curve of Figure 3–5a shows a maximum at point A_h has another consequence. Differences in preferences should have a *smaller influence* on observable health-related behavior than differences in production possibilities. For differences in preferences can only lead to a shift in optimal points between A_h and B_h (and thus between A'_h and B_h on the T_h axis), whereas different individual production possibilities change the actual position of points A'_h and B_h themselves. These points can move on the T_h axis up to a maximum given by the number of years to death (T in Figure 3–5a). Therefore, notable differences between individuals may well arise, basically due to their age.

3.5.2 The Longer-Run Tradeoff Given Bad Health (No. 2')

If the individual is sick in the current period, he must settle for a phase of sickness, the mean duration of which is given in analogy to Eq. (3.23) by

$$T_s = \frac{1}{1 - \pi} \quad (3.28)$$

where T_s is the mean number of periods lived in bad health.

The time constraint of Eq. (3.20) in Table 3.4, valid for one period, must be replaced by

$$T_s = t_s^C + \mu M \quad (3.29)$$

It will be in the interest of the individual to keep this phase of sickness as short as possible; therefore the tradeoff derived below is presented in $(C_s, -T_s)$ space. The slope of the production possibility frontier is given by [see Appendix 3.A.4, Eq. (3.A.24)]

$$\frac{dC_s}{-dT_s} = - \frac{\frac{\partial C_s}{\partial t^C} \left(\frac{\partial T_s}{\partial \pi} \frac{\partial \pi}{\partial M} - \mu \right)}{\frac{\partial T_s}{\partial \pi} \frac{\partial \pi}{\partial M}} + \frac{\frac{\partial C_s}{\partial X} \frac{q}{p}}{\frac{\partial T_s}{\partial \pi} \frac{\partial \pi}{\partial M}} < 0 \quad (3.30)$$

This slope is negative throughout, implying that the transformation curve of Figure 3–5b is *strictly decreasing*. In bad health, an improvement of the chance of health therefore will never have the property of an investment because it can only be brought about by the use of medical services. While medical care contributes to a shortening of the phase of sickness, it does cost time that would have been available for consumption [according to the parameter μ in the numerator of the first term of Eq. (3.30)]. Depending on the value of the relative net price (q/p) , it may also cost money that is not available for the purchase of consumption goods [second term of Eq. (3.30)]. Finally, Eq. (3.30) corresponds closely to Eq. (3.22), and therefore the parameters influencing both location and shape of the transformation curve considered in Subsection 3.4.3.2 need not be discussed again.

3.6 COMPLEMENTARITY OR SUBSTITUTABILITY IN THE PRODUCTION OF HEALTH?

3.6.1 Significance of the Issue

Important statements of the theory of production deal with the relationship between inputs. If a factor of production becomes more expensive and must therefore be used less, does this necessarily lead to an increased use of all other factors? On the one hand, given *only two production factors* and profit maximization by a firm, the answer to this question is clearly yes; that is, *substitutability* prevails. On the other hand, with three or more production factors, complementary relationships become possible. There is some evidence that suggests that the increase in the price of energy in the 1970s led to a more labor-intensive style of production, whilst at the same time slowing the

introduction of new technologies and therefore investment. As a result we can say that substitutability exists between energy and labor, but complementarity prevails between energy and capital [see Magnus (1979)].

A related issue is of considerable interest for health policy. One possibility of dampening the cost expansion in health care is to reduce the demand for medical services. Increasing the amount of cost sharing in health insurance results in an increase in the net price for medical services, serving to curtail demand. Since this measure would meet with resistance, particularly amongst the older groups of the population who constitute the majority of voters in today's industrialized countries, *substitution of medical services* by the individual's own efforts (particularly preventive ones) offers a possible solution. Reducing the labor costs of the individual is out of the question, since this would be equivalent to a reduction in the wage rate of the working population. Moreover, this measure would fail to be effective among the retired, although it is they who are the most important demanders of medical services. The use of a production factor can, however, be encouraged not only through a lowering of its price but also by an increase of its (marginal) productivity. One is therefore led to ask the question of whether due to an *improved productivity of individuals' own efforts* in favor of health, medical services could be saved.

At first sight, the model developed in Section 3.5 does not seem to give an answer to this question because (since consumption goods X do not have any influence on health status) there is *only one factor* of production in each state. In good health, this is one's own time t^H , in bad health, the utilization of medical services M [see Table 3.4, Eq. (3.13) and (3.17)]. Therefore, the two inputs appear to be entirely unrelated. However, there is an indirect relationship that takes on a different quality depending on the initial situation, to be discussed in the next two subsections.

3.6.2 Substitutability in the Healthy State

In the previous subsection it was assumed that the marginal productivity of one's health-enhancing efforts can be increased, through more education in particular. Education results in an improved productivity in market and nonmarket activities. Even if it were possible to limit the effect of education to the nonmarket domain, the time restriction (3.16) of the model highlights the fact that time allocation may not change in favor of t^H , but of t^C , that is, in favor of the production of consumption services. Having acquired marketable skills, the individual can obtain a higher wage in the labor market. This effect has caused, for example, women increasingly to enter the labor market, changing their time budget in favor of t^W , possibly to the detriment of t^H .

Should an increase in t^H occur despite these reservations, the probability of sickness π would fall, and with it the probability of future utilization of medical services. In addition, phases of good health would be lengthened and hence the demand for medical services deferred. Since, as a result of this, *life expectancy* will increase, total medical care consumed during the entire life cycle need not decrease. This is an effect reminiscent of the so-called Sisyphus syndrome in health care (see Chapter 11). Nevertheless, demand for medical services would be reduced in the short and medium term as a result of substitution.

3.6.3 Complementarity in the Sick State

If the initial state is one of sickness, then an improvement in the individual's productivity is without effect. On the other hand, the question arises as to the effects of intensified medical care in the long term. By increasing the chance of recovery, medical care provides the individual with additional opportunity to contribute his own time t^H to the maintenance of health. The expected duration of the sick period is reduced, permitting individual health-enhancing efforts to become effective sooner. In this argument, t^H and M turn out to be complements rather than substitutes. This gives rise to

CONCLUSION 3.11: The relationship between one's own efforts and medical inputs in the production of health is state-dependent. In the initial state of health, the two inputs are substitutes and become complements in the very long term. In the initial state of sickness, they are complements.

In view of the empirical research into the production of health (to be discussed in Chapter 4), this conclusion leads to very differentiated predictions. For groups of basically healthy, young people, the relationship between their own health-enhancing efforts and the demand for medical services should result in relatively unambiguous substitutability. To the extent that the sample contains also older and less healthy individuals, complementarity between the two inputs will prevail. At a regional or national level of aggregation, complementarity rather than substitutability may again be observed, because the published data reflect cases of very intensive medical treatment, which are not present in samples of individuals.

APPENDIX

This appendix is devoted to the derivation of the production possibility frontier. First, its short-term version is established both for the state of health and sickness. Then the longer run is considered as an extrapolation of the two health states, prevailing in the current period.

3.A.1. Production in the Healthy State, Short Run (Tradeoff No. 1)

The objective is to determine the sign of the $dC_h/d(1 - \pi)$. To this end, Eq. (3.13) is differentiated

$$d(1 - \pi) = -d\pi = -\frac{\partial \pi}{\partial t^H} dt^H \quad (3.A.1)$$

Total differentiation of the production function for consumption services, given by Eq. (3.14), yields

$$dC_h = \frac{\partial C_h}{\partial X} dX + \frac{\partial C_h}{\partial t^C} dt^C \quad (3.A.2)$$

The budget constraint of Eq. (3.15) implies that additional consumption goods can only be financed out of additional labor income, which requires more working time

$$dX = \frac{w}{p} dt^W \quad (3.A.3)$$

The change of time available for consumption dt^C is derived from the time constraint (3.16)

$$dt^C = -dt^H - dt^W \quad (3.A.4)$$

Substitution of the expressions (3.A.3) and (3.A.4) into Eq. (3.A.2) for dC_h yields

$$dC_h = \frac{\partial C_h}{\partial X} \left(\frac{w}{p} dt^W \right) - \frac{\partial C_h}{\partial t^C} (dt^H + dt^W) = -\frac{dC_h}{dt^C} dt^H + \left(\frac{\partial C_h}{\partial X} \frac{w}{p} - \frac{\partial C_h}{\partial t^C} \right) dt^W \quad (3.A.5)$$

If the individual uses time and goods for producing consumption services in an optimal manner, the ratio of marginal productivities $[(\partial C_h / \partial t^C) / (\partial C_h / \partial X)]$ is equal to the price ratio w/p , with w indicating the so-called shadow price of time that could be used for earning additional labor income. Given this assumption, the second term of Eq. (3.A.5) is zero, and after dividing through by Eq. (3.A.1), one obtains the expression discussed previously [Eq. (3.21)],

$$\frac{dC_h}{d(1 - \pi)} = \frac{\partial C_h}{\partial t^C} \bigg/ \frac{\partial \pi}{\partial t^H} < 0 \quad (3.A.6)$$

The curvature of the transformation curve is given by the sign of $d^2 C_h / d(1 - \pi)^2$. Treating $\partial C_h / \partial t^C$ as a constant, one has, using Eqs. (3.A.6) and (3.A.1),

$$\begin{aligned} \frac{d^2 C_h}{d(1 - \pi)^2} &= \frac{d}{dt^H} \left(\frac{dC_h}{d(1 - \pi)} \right) \frac{dt^H}{d(1 - \pi)} = \frac{d}{dt^H} \left(\frac{\partial C_h}{\partial t^C} \bigg/ \frac{\partial \pi}{\partial t^H} \right) \frac{(-1)}{\frac{\partial \pi}{\partial t^H}} \\ &\quad (+) \quad (+) \\ &= \frac{0 \frac{\partial C_h}{\partial t^C} - \frac{\partial C_h}{\partial t^C} \frac{\partial^2 \pi}{(\partial t^H)^2}}{\left(\frac{\partial \pi}{\partial t^H} \right)^2} \frac{(-1)}{\frac{\partial \pi}{\partial t^H}} = \frac{\frac{\partial C_h}{\partial t^C} \frac{\partial^2 \pi}{(\partial t^H)^2}}{\left(\frac{\partial \pi}{\partial t^H} \right)^3} < 0 \quad (3.A.7) \\ &\quad (-) \end{aligned}$$

This shows that the transformation curve must be concave from the origin.

3.A.2. Production in the Sick State, Short Run (Tradeoff No. 2)

According to assumption, the probability of being healthy in the subsequent period can only be increased by the use of medical services. Hence differentiation of Eq. (3.17) yields

$$d(1 - \pi) = -d\pi = -\frac{\partial \pi}{\partial M} dM \quad (3.A.8)$$

With regard to the production function for consumption services, there are no modifications except for the fact that marginal productivity of consumption goods as well as of time are likely to be smaller than in the healthy state

$$dC_s = \frac{\partial C_s}{\partial X} dX + \frac{\partial C_s}{\partial t^C} dt^C \quad (3.A.9)$$

Due to the budget constraint of Eq. (3.19), the purchase of additional consumption goods entails a sacrifice of medical services

$$dX = \frac{-q}{p} dM \quad (3.A.10)$$

Finally, total differentiation of the time constraint (3.20) yields

$$dt^C = -\mu dM \quad (3.A.11)$$

Now, the two expressions (3.A.10) and (3.A.11) can be substituted into Eq. (3.A.9), resulting in

$$dC_s = \frac{-\partial C_s}{\partial X} \frac{q}{p} dM - \mu \frac{\partial C_s}{\partial t^C} dM \quad (3.A.12)$$

Dividing this by Eq. (3.A.8), one obtains Eq. (3.22),

$$\frac{dC_s}{d(1 - \pi)} = \frac{\begin{matrix} (+) \\ \frac{\partial C_s}{\partial t^C} \end{matrix} \mu}{\frac{\partial \pi}{\partial M}} + \frac{\begin{matrix} (+) \\ \frac{\partial C_s}{\partial X} \end{matrix} \frac{q}{p}}{\frac{\partial \pi}{\partial M}} < 0 \quad (3.A.13)$$

$\begin{matrix} (-) & & (-) \end{matrix}$

The sign of this expression is unambiguously negative. Assuming $\partial C_s/\partial X$, q/p , and $\partial C_s/\partial t^C$ to be independent of M , it is possible to show in full analogy to Eq. (3.A.7) that

$$\frac{d^2 C_s}{d(1 - \pi)^2} < 0 \quad (3.A.14)$$

Thus the transformation curve is again concave from the origin.

3.A.3 Production in the Healthy State, Longer Run (Tradeoff No. 1')

On the longer run, the time constraint (3.16) is replaced by (3.24), which is given by

$$T_h\{\pi(t^H)\} = t_h^C + t^H + t^W \quad (3.A.15)$$

The mean duration of a phase in good health therefore can be increased by spending additional health-enhancing time t^H . This increase is given by

$$dT_h = \frac{\partial T_h}{\partial \pi} \frac{\partial \pi}{\partial t^H} dt^H \quad (3.A.16)$$

Total differentiation of Eq. (3.A.15) and solving for dt^C (dropping the subscript h for simplicity) yields, due to Eq. (3.A.16),

$$\begin{aligned} dt^C &= dT_h - dt^H - dt^W = dT_h - \frac{1}{\frac{\partial T_h}{\partial \pi} \frac{\partial \pi}{\partial t^H}} dT_h - dt^W \\ &= \left[\left(\frac{\partial T_h}{\partial \pi} \frac{\partial \pi}{\partial t^H} - 1 \right) \middle/ \frac{\partial T_h}{\partial \pi} \frac{\partial \pi}{\partial t^H} \right] dT_h - dt^W \end{aligned} \quad (3.A.17)$$

Substituting expressions (3.A.3) and (3.A.17) into Eq. (3.A.2) for dC_h , one obtains

$$\begin{aligned} dC_h &= \frac{\partial C_h}{\partial X} \left(\frac{w}{p} dt^W \right) + \frac{\partial C_h}{\partial t^C} \left\{ \left[\left(\frac{\partial T_h}{\partial \pi} \frac{\partial \pi}{\partial t^H} - 1 \right) \middle/ \frac{\partial T_h}{\partial \pi} \frac{\partial \pi}{\partial t^H} \right] dT_h - dt^W \right\} \\ &= \left[\frac{\partial C_h}{\partial t^C} \left(\frac{\partial T_h}{\partial \pi} \frac{\partial \pi}{\partial t^H} - 1 \right) \middle/ \left(\frac{\partial T_h}{\partial \pi} \frac{\partial \pi}{\partial t^H} \right) \right] dT_h + \left(\frac{\partial C_h}{\partial X} \frac{w}{p} - \frac{\partial C_h}{\partial t^C} \right) dt^W \end{aligned} \quad (3.A.18)$$

Again, in the neighborhood of an optimum, the term involving dt^W can be neglected [see the justification of Eq. (3.A.6)]. The division of Eq. (3.A.18) by (3.A.16) then gives [see Eq. (3.25)]

$$\frac{dC_h}{dT_h} = \frac{\frac{\partial C_h}{\partial t^C} \left(\frac{\partial T_h}{\partial \pi} \frac{\partial \pi}{\partial t^H} - 1 \right)}{\frac{\partial T_h}{\partial \pi} \frac{\partial \pi}{\partial t^H}} \geq 0 \quad (3.A.19)$$

(-) (-)

The curvature of this transformation curve depends on the sign of d^2C_h/dT_h^2 . Starting from Eq. (3.A.16), assuming $\partial C/\partial t$ independent of t^H , and using Eq. (3.26) to derive $\partial T_h/\partial \pi = -1/\pi^2$, one obtains

$$\begin{aligned} \frac{d^2C_h}{dT_h} &= \frac{d}{dt^H} \left(\frac{dC_h}{dT_h} \right) \frac{dt^H}{dT_h} \\ &= \frac{d}{dt^H} \left[\frac{\partial C_h}{\partial t^C} \left(\frac{-1}{\pi^2} \frac{\partial \pi}{\partial t^H} - 1 \right) \right] \left/ \left(\frac{-1}{\pi^2} \frac{\partial \pi}{\partial t^H} \right) \right] \frac{1}{\frac{\partial T_h}{\partial \pi} \frac{\partial \pi}{\partial t^H}} \\ &= \frac{\partial C_h}{\partial t^C} \frac{d}{dt^H} \left(1 + \pi^2 \left/ \frac{\partial \pi}{\partial t^H} \right. \right) \frac{1}{\frac{\partial T_h}{\partial \pi} \frac{\partial \pi}{\partial t^H}} \\ &= \frac{\partial C_h}{\partial t^C} \left(2\pi \frac{\partial \pi}{\partial t^H} - \pi^2 \frac{\partial^2 \pi}{(\partial t^H)^2} \right) \left(1 \left/ \frac{\partial \pi}{\partial t^H} \right. \right)^2 \frac{1}{\frac{-1}{\pi^2} \frac{\partial \pi}{\partial t^H}} < 0 \end{aligned} \quad (3.A.20)$$

(+) (-) (+) (+) (+)

This proves concavity from the origin for this transformation curve as well.

3.A.4 Production in the Sick State, Longer Run (Tradeoff No. 2')

With the time constraint (3.20) replaced by Eq. (3.29), its differentiation with respect to M gives

$$dT_s = \frac{\partial T_s}{\partial \pi} \frac{\partial \pi}{\partial M} dM \quad (3.A.21)$$

The second modification concerns the time available for consumption. Here, Eq. (3.A.21) is used to replace (3.A.11) by

$$dt^C = dT_s - \mu dM = \left(\frac{\partial T_s}{\partial \pi} \frac{\partial \pi}{\partial M} - \mu \right) dM \quad (3.A.22)$$

Combined with Eq. (3.A.10) for dX , the change in consumption goods follows from Eq. (3.A.9)

$$dC_s = \frac{-\partial C_s}{\partial X} \frac{q}{p} dM + \frac{\partial C_s}{\partial t^C} \left(\frac{\partial T_s}{\partial \pi} \frac{\partial \pi}{\partial M} - \mu \right) dM \quad (3.A.23)$$

Dividing this expression by Eq. (3.A.21), one obtains [see Eq. (3.30)]

$$\frac{dC_s}{-dT_s} = - \frac{\begin{matrix} (+) \\ \frac{\partial C_s}{\partial t^C} \left(\frac{\partial T_s}{\partial \pi} \frac{\partial \pi}{\partial M} - \mu \right) \end{matrix}}{\begin{matrix} (-) \\ \frac{\partial T_s}{\partial \pi} \frac{\partial \pi}{\partial M} \end{matrix}} + \frac{\begin{matrix} (+) \\ \frac{\partial C_s}{\partial X} \frac{q}{p} \end{matrix}}{\begin{matrix} \frac{\partial T_s}{\partial \pi} \frac{\partial \pi}{\partial M} \end{matrix}} < 0 \quad (3.A.24)$$

(+) (-) (+) (-)

In the denominator of this expression, the negative value of T_s appears in order to facilitate comparison with expression (3.A.13) for the case of good health.

3.7 EXERCISES

Exercise 3.1:

What is the condition that would make the indifference curves shown in Figure 3-1 run completely vertical? Is it also conceivable that they run completely horizontal? What effects would these two extreme cases have on the individual's behavior?

Exercise 3.2:

A policy-maker suggests increasing sick pay, to be financed by a deduction from the wage of employed persons. What would be the expected effects on the population's health status and health behavior?

3.8 SELECTED READINGS

Section 3.2: The Concept of Health Production

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4 EMPIRICAL STUDIES OF THE PRODUCTION OF HEALTH

4.1 SUMMARY OF THE ISSUE

The concept of a transformation curve for consumption services and health, introduced in the previous chapter, raises a number of questions that are of considerable importance for health policy. To illustrate this, take the increase in life expectancy in industrialized countries (see Table 4.1).

Between 1900 and 1930 the average life expectancy (measured at time of death) increased, for example, in the United States, by more than 0.3 years annually. However, the annual increase between 1950 and 1980 only amounted to about 0.2 years. With the exception of women in Germany, all countries shown in Table 4.1 register a considerable *slowdown of the increase in life expectancy*.

Since the per capita health care expenditure grew considerably in all OECD countries during the period 1950–1980, many observers have come to the conclusion that the marginal contribution of medicine to health has gone to zero (“Flat of the curve medicine”). Can this conclusion be upheld if the economic concept of the production of health is utilized?

The transformation curve reminds us that the observed developments may well be due to other factors, both on the input and the output side.

1. Choice of output indicator. Neither the life expectancy nor the mortality rate of the population (another frequently used indicator) provides direct information about health status, that is, the variable of interest to the individual. If one uses QALYs (as described in Section 2.4) as the output measure, then additional life years only count for full if spent in a state of perfect health. Some epidemiologists doubt that additional life years fully translate into QALYs, pointing to the prevalence of chronic conditions in the later part of the life span [see Verbrugge (1984)]. Rheumatic diseases, for example, are rarely fatal, but do limit the quality of life. Moreover, they cause a great

Table 4.1: Development of Life Expectancy at Birth in Selected Countries

		<i>Annual increase (in years)</i>		
		<i>1900–1930</i>	<i>1930–1950</i>	<i>1950–1980</i>
Germany	M	0.503	0.225	0.203
	F	0.483	0.275	0.303
France	M	0.300	0.480	0.226
	F	0.343	0.535	0.303
United Kingdom	M	0.343	0.385	0.150
	F	0.350	0.420	0.188
Japan	M	0.027	0.635	0.509
	F	0.057	0.715	0.585
Switzerland	M	0.330	0.385	0.203
	F	0.363	0.410	0.279
United States	M	0.327	0.395	0.166
	F	0.343	0.510	0.225

Source: OECD (1987).

deal of absence from work, resulting in considerable loss of output. By the same token, the state of health of a population may improve even though life expectancy no longer increases or only increases at a slower rate.

2. Consumption and health as outputs. If individuals are allowed to optimize at all, then additional consumption services must have an opportunity cost in terms of health (see Figure 3.5). This fact becomes very clear if additional consumption means more use of goods that are harmful to health, such as alcohol and tobacco.¹ If preferences should vary between different time periods or individuals, there is a risk of spurious correlation. An observed decline in the state of health may be the consequence of an intensified quest for consumption, whose negative side effects fail to be balanced by an increase in the productivity of medical services.

3. Relative productivity of inputs. Even if the slowdown in the increase of life expectancy is accompanied by a continuing rise of health care expenditure, we cannot conclude that the marginal productivity of medical services has been decreasing. The inverse relationship could be due to a reduction of individual efforts enhancing health or to other factors, in particular environmental influences. This argument reminds us of the fact that relative (not absolute) marginal productivity determines (jointly with relative marginal cost) the optimal input of factors in production. In recent years

1. In the model of Section 3.4, consumption goods have no effect on health, and a similar simplification is usually applied to the Grossman model in spite of Eq. (3.2). This assumption considerably simplifies the mathematical analysis.

particularly, the marginal productivity of improvements in road safety or in air quality in terms of health may well have increased.

4. Population structure. The larger the proportion of elderly people with a comparatively high probability of sickness, the more tradeoffs occur in the state of bad health, according to the model developed in Section 3.5. Hence, the productivity of medical services relative to its alternatives is very high while being rather limited in absolute terms. To the extent that population aging goes along with an increased concentration of medical care in the upper age groups, medical resources are spent precisely where this returns are low, at least if measured in terms of additional life expectancy.

In the following, empirical evidence bearing on these issues will be presented, starting with studies based on aggregate data. These studies raise certain issues that can be addressed by reviewing more recent research based on individual data.

4.2 STUDIES BASED ON AGGREGATE DATA

4.2.1 Mortality Rates As a Measure of Success?

Since official population statistics record deaths only, containing no information episodes of illness, there is a limited choice of indicators to describe aggregate output of the production of health. The quantities available are mortality rates (classified according to age and sex) and life expectancies, which can be calculated from the mortality table.

From the individual's point of view, the general mortality rate as an output measure makes little sense since the long-run *probability of death equals one*. Mortality rates relating to a group and a defined time period are only slightly more informative. The probability of a randomly selected inhabitant of an area being no longer alive at the end of the current year has no relevance for the majority of the population, since almost everyone, in one or several personal characteristics, differs from the statistical average. This objection applies to infants less than to older population groups, since many characteristics of infants are not yet defined, causing some degree of homogeneity. In view of this argument, the traditional use of infant mortality rates for comparing the performance of health care systems is not entirely unjustified.

Even in the case of infants, however, the mortality rate reflects an extreme state of "health," providing information only about its probability weight π . This is very limited information indeed when compared to the QALY benchmark for describing health [see Eq. (3.7)]. Yet mortality rates continue to serve as output indicators for empirical studies in the production of health, since they are derived from official statistics and can be compared across regions and countries.

As an alternative, one may consider *life expectancy*, particularly at the time of birth. This indicator is an aggregate of all possible states of health with the exception of death. It simply sums the amounts of time the individual finds himself in one of these states, thus mirroring the fact that the attainment of a particular age depends not only on the current survival probability but on the entire sequence of survival probabilities since birth. Life expectancy is therefore subject to influences on the state of health that occurred in the past, and in order to catch their impact, lags need to be introduced in cause and effect relationships.

4.2.2 Marginal Productivity of a Health Care System

For the economic classics it was obvious that economic conditions crucially determined the health status and especially the mortality of a population [see Malthus (1803)]. It was towards the end of the nineteenth century that medicine got a scientific foundation, paving the way for its successes in fighting infectious disease. However, it was only in the 1930s that drugs permitting causal rather than symptomatic treatment appeared on the market. These so-called sulfonamides essentially eradicated tuberculosis within a few years. This experience served to establish the conviction that medical care improves the health of the population at large.

4.2.2.1 Early Evidence from the United States

Auster et al. (1969) were the first to examine the effectiveness of medicine from an economic point of view. While their study is a bit dated, it does raise a crucial issue that continues to be at the center of debate in health economics, viz. the marginal productivity of medical care relative to other factors such as education and lifestyle in the production of health. For this reason, this pioneering study is discussed in some detail here.

The authors selected the states of the United States as their unit of observation. To begin with, they replaced the observed mortality rates by values standardized for age and sex, that is, the value that would obtain if the state's population had the same age and sex structure as the United States as a whole. They thus took into account item No. 4 (population structure) mentioned in Section 4.1. Without such a *standardization*, Florida, for example, with its large proportion of retirees in the population, would come off badly right from the start. In total, four groups of factors $\{Z, X, M, D\}$ were distinguished in order to explain the remaining differences standardized in mortality rates (S). These factors enter a generalized Cobb–Douglas production function, with S serving as an inverse indicator of health as its output. The authors assume that the standardized mortality rate of state i is thus given by (see Table 4.2)

$$S_i = CZ_i^\alpha X_i^\beta M_i^\gamma e^{D_i \delta} e^{u_i} \quad (4.1)$$

In this equation, C is a constant, the Z_i represent economic inputs, the X_i inputs linked to consumption, and M_i medical inputs. The D_i symbolize additional influences emanating from the organization of health care (see Table 4.2 for more details). Finally, the variable u_i is a stochastic variable reflecting all those influences upon output S_i in a particular state that were not recorded by the researchers. The seemingly complicated form of function (4.1) dissolves into a linear equation after taking logarithms,

$$\ln S_i = \ln C + \alpha \ln Z_i + \beta \ln X_i + \gamma \ln M_i + \delta D_i + u_i,$$

$$\text{with (e.g.) } \frac{\partial \ln S_i}{\partial \ln Z_i} = \frac{\partial S_i / S_i}{\partial Z_i / Z_i} = \eta(S, Z) = \alpha \quad (4.2)$$

The parameter α indicates the *elasticity of mortality* with respect to economic influences (e.g., income). The variable D_i , in contrast to the others, appears in

Table 4.2: Determinants of Mortality in 48 States of the United States, 1960^a

<i>Explanatory Variable^b</i>		<i>Ordinary least squares^c</i>		<i>Two-stage least squares^c</i>	
	Constant	-0.065	(0.157)	0.037	(0.251)
<i>Z_i</i>	1. Income per capita	0.105	(0.079)	0.183	(0.116)
	2. Average no. of years of schooling	-0.161	(0.121)	-0.288	(0.216)
	3. Share of population in urban areas	-0.001	(0.005)	-0.001	(0.005)
	4. Share of industry in total employment	0.051*	(0.023)	0.042	(0.040)
<i>X_i</i>	5. Alcohol consumption per capita	-0.002	(0.037)	0.013	(0.044)
	6. Cigarette consumption per capita	0.094	(0.053)	-0.097	(0.058)
<i>M_i</i>	7. Pharmaceutical outlay per capita ^d	-0.070	(0.040)	-0.076	(0.066)
	8. No. of physicians per capita ^d	0.143*	(0.064)	0.044	(0.111)
	9. Medical auxiliary staff per capita ^d	-0.190**	(0.076)	-0.031	(0.195)
	10. Capital stock of hospitals per capita ^d	-0.004	(0.048)	-0.109	(0.141)
<i>D_i</i>	11. Share of group practices	0.007	(0.012)	0.007	(0.021)
	12. Existence of a medical school	-0.034**	(0.012)	-0.024	(0.019)
	<i>R</i> ²	0.639		0.586	
	Elasticity with respect to medical services (variable Nos. 7–10)	-0.121		-0.172	

*(**): Coefficient different from zero with error probability of 5% (1%).
^a Dependent variable: natural logarithm of mortality rates, standardized according to age and sex.
^b Natural logarithm, except variable No. 12; the coefficients presented are therefore elasticities [see Eq. (4.2)].
^c Standard errors of the estimated coefficients in parentheses.
^d These regressors are endogenous and are replaced in the two-stage process by their estimated values (see Table 4.3b).

Source: Auster et al. (1969).

arithmetic form, since it also contains a so-called dummy variable or categorical variable, which can only take on the values 1 (characteristic present in state *i*) and 0 (characteristic not present).

The variables of Eq. (4.2) were further differentiated by Auster et al. (1969) into a total of 12 explanatory variables, also called regressors (see Table 4.2).

- **Economic factors *Z_i* (variable Nos. 1–4).** Income per capita (variable No. 1) stands for a number of influences. On the one hand, it may reflect differences in preference, that is, a marginal rate of substitution between consumption and health that depends on income. On the other hand, income may also influence the aggregate marginal rate of transformation between consumption and health (see Section 1.2.2). The average number of years of schooling (variable No. 2) plays a role because improved education presumably increases the productivity of the individual, not only in the market domain but also in the nonmarket domain, which also includes the maintenance of his health. Through the degrees

of urbanization and industrialization (variable Nos. 3 and 4), two indicators of environmental influences on the state of health enter the equation as well.

- **Factors related to consumption X_i (variable Nos. 5 and 6).** From the large number of consumable goods and services, the two whose health-damaging effects weigh most heavily at the aggregate level are selected, viz., alcohol and cigarette consumption per capita.²
- **Medical factors M_i (variable Nos. 7–10).** Because of the successes of chemotherapy, it is expected that drug outlay per person in a state should contribute to a lower mortality rate (variable No. 7). The second factor is physician density, internationally the most popular indicator of (potential) utilization of medical services (variable No. 8). The authors include medical auxiliary staff as another input in the production function, since many tasks in the practice and in the hospital are carried out by nurses and auxiliary staff (variable No. 9). The fourth factor is the per capita capital stock available to the hospitals in the state concerned. According to widespread opinion, equipping hospitals with capital goods is the prerequisite for their offering top-quality medical care.
- **Organizational factors D_i (variable Nos. 11 and 12).** Along with the four medical inputs, two organizational features of medical care are also considered. In particular, group practice holds the promise of improved information exchange between the physicians of different specialties, and also of a certain mutual quality control. Accordingly, their share in the total number of practices in the state appears as explanatory variable No. 11. The last regressor (variable No. 12) is a dummy variable, indicating whether the states has a medical school (=1) or not (=0).

An ordinary least-squares estimate (OLS) of Eq. (4.2) gave surprising results (see first column of Table 4.2; for an explanation of test statistics, see Tables 4.3a and b).

- Contrary to popular belief and findings from development economics, a *higher average income* does not contribute to a lower mortality rate but possibly to a higher one. The elasticity estimate of 0.105, while comparatively large, does not clearly differ from zero because of its large standard error of 0.079.
- Prolonged *schooling* appears to contribute to a reduction of the mortality rate, with an elasticity of -0.16 , but this estimate is again not significantly different from zero. Among the two variables of the set that can be counted as environmental influences (degree of urbanization and industrialization), only industrialization has a discernible effect on mortality.
- Surprisingly enough, the claim could be made that in the “fight against death” only the medical auxiliary staff have a recognizable success, with an elasticity of -0.19 . By way of contrast, increased *physician density* seems to result not in a lower but a higher rate of mortality in the state, *ceteris paribus*.

2. In this way, item No. 2 of Section 4.1 is taken into account. Consumption (and with it consumption goods X_i) and health are two quantities that are simultaneously set by the individual. However, this also means that the X_i strictly speaking are not exogenous variables but, like S_i , endogenous ones that contain a stochastic error. This complication will be pursued in Section 4.3.4.

Table 4.3a: A Simple Production Function of Health (Ordinary Least-Squares Estimation)

$$s_i = c + \alpha z_i + \beta x_i + \gamma m_i + \delta D_i + u_i,$$

with $s_i = \ln S_i$, $c = \ln C$, $z_i = \ln Z_i$, and $m_i = \ln M_i$

u_i : Random variable. *Assumption*: u_i is normally distributed, has an expected value of $E(u_i) = 0$ and a constant variance $\text{Var}(u_i)$ in every state i , and is independent of all regressors (z_i , x_i , m_i , D_i).

- The parameters $\hat{c}$, $\hat{\alpha}$, $\hat{\beta}$, $\hat{\gamma}$, $\hat{\delta}$ are to be determined such that the sum of squared deviations between observed (s_i) and calculated values ($\hat{s}_i$) of the dependent variable are minimized,

$$\sum_i \hat{u}_i^2 = \sum (s_i - \hat{s}_i)^2 = \sum (s_i - \hat{c} - \hat{\alpha}z_i - \hat{\beta}x_i - \hat{\gamma}m_i - \hat{\delta}D_i)^2 \rightarrow \min.$$

- As s_i contains the random variable u_i , the estimates $\hat{c}$, $\hat{\alpha}$, $\hat{\beta}$, $\hat{\gamma}$, $\hat{\delta}$ themselves are random variables and have a variance (or a standard error $\sigma_{\hat{\alpha}}$, respectively) that can be estimated.

Example in Table 4.2: $\hat{\alpha} = 0.105$, $\hat{\sigma}_{\hat{\alpha}} = 0.079$ for variable No. 1.

- The standardized estimated coefficients (e.g., $\hat{\alpha}/\hat{\sigma}_{\hat{\alpha}}$) are t distributed. This means that a deviation from zero of $|\hat{\alpha}/\hat{\sigma}_{\hat{\alpha}} - 0| \geq 1.96$ can only occur in 5 percent of all cases, provided that $\alpha = 0$. Deviations of $|\hat{\alpha}/\hat{\sigma}_{\hat{\alpha}} - 0| \geq 2.66$ should only occur in 1 percent and of $|\hat{\alpha}/\hat{\sigma}_{\hat{\alpha}} - 0| \geq 3.29$ in 0.1 percent of all cases (significance levels of 0.05, 0.01, and 0.001, respectively).

Example in Table 4.2: $\hat{\alpha}/\hat{\sigma}_{\hat{\alpha}} = 0.105/0.079 < 1.96$; the coefficient of variable No. 1 is (with an error probability of 5 percent) not different from zero.

- The goodness of fit of the observed values of the dependent variable is expressed by the *coefficient of determination* R^2 , i.e., the part of the variance of the dependent variable not attributable to the random variable u_i (= 0.639 or 64 percent in Table 4.2).

Unexpected results such as these give rise to critical objections. One such objection can be immediately gleaned from Section 4.2.1, where it was pointed out that the life expectancy (and also the mortality rate) of the current period depends on influences that are to be found way back in the lifespan of the deceased. The mortality rate of a state of the United States in 1960 may therefore be determined by circumstances that *date back many years*.

A second important objection concerns the assumed *direction of causality*. The OLS estimate of Table 4.2 suggests that additional physicians serve to increase the mortality rate, for example, through an excessive intensity of treatment due to supplier-induced demand (see Section 7.2). Causality could, however, just as well run the other way around. In places where people see themselves exposed to an increased risk of death, their demand of physician services is particularly high, which causes physicians to set up practice in high mortality states. A similar reversal of causality could also apply to the remaining three components of medical services.

For variable Nos. 7–10, additional regression equations can therefore be postulated, containing the mortality rate as one of the explanatory variables. Being dependent

Table 4.3b: A Simple Production Function of Health (Two-Stage Least-Squares Estimation)

$$m_i = c' + \alpha' z_i + \kappa r_i + \tau s_i(u_i) + u'_i$$

- This equation represents the assumed reverse influence of mortality on m_i (e.g., physician density), whereas r_i stands for other determinants of m_i and u'_i for another random variable.
- Thus the explanatory variable m_i depends, via s_i , on the *error term* u_i of Eq. (4.2) and is therefore endogenous.
- A purging of m_i of its error component can be achieved by a preliminary regression on all exogenous variables (first stage), resulting in the parameters $\hat{c}'$, $\hat{\alpha}'$, and $\hat{\kappa}$ estimated using OLS. The purged values of m_i are thus given by

$$\hat{m}_i = \hat{c}' + \hat{\alpha}' z_i + \hat{\kappa} r_i \tag{4.3}$$

- Equation (4.2) is estimated with $\hat{m}_i$ as a regressor instead of m_i , again using OLS (second stage).

Further explanations can be found in most textbooks of econometrics [see, e.g., Theil (1971), Greene (1993)].

variables, variable Nos. 7–10 are thus subject to the influence of an error term u'_i that is of the same type as the one appearing in Eq. (4.2). This means that the distinction between systematic impacts and random noise, which is so crucial for regression analysis, breaks down. With the help of a first-stage regression on all exogenous variables of Table 4.2 (and a few additional ones not shown), regressor Nos. 7–10 may be purged of their error component. Using these purged values in a second-stage regression, one obtains coefficients that are corrected for possible reversed causation (see Tables 4.3a and b for some more explanation). These corrected coefficients appear in the second column of Table 4.2, labeled “Two-Stage Least-Squares.” This two-stage estimation procedure yields the following results.

- The difference between the detrimental effect of income and the enhancing effect of schooling on health is strengthened, although both elasticities still do not quite reach conventional levels of significance.
- Both environmental variables (urbanization, industrialization) remain without a statistically proven influence.
- Those values with the strongest link with individual health behavior, alcohol and cigarette consumption, do not contribute to the explanation of the differences in mortality rates in a statistically significant way, as before.
- Three out of four components of the medical infrastructure display a negative relationship with mortality, with the *exception of physician density*. However, the elasticity of the mortality rate with respect to physician density, while positive, cannot be distinguished from zero.

- As indicated by the sum of the coefficients of variable Nos. 7–10, increasing all medical inputs by 10 percent would reduce mortality by about 1.7 percent (1.2 percent according to the OLS estimate). By increasing the length of schooling by 10 percent, the mortality rate could be reduced by as much as to 2.9 percent, however. The risk associated of this second alternative would be that improved education leads to an increase in income, which by fostering an unhealthy lifestyle may push up mortality again (see the positive coefficient of variable No. 1).

CONCLUSION 4.1: If the contribution of different inputs to the production of health is measured by their influence on the mortality rate, then there are indications suggesting that the marginal productivity of medical infrastructure in the United States may well fall short of that of additional schooling.

As a qualification it must be noted that this conclusion is based on observations from at least 30 years ago, which fail to reflect recent developments of high-tech medical care.

4.2.2.2 Evidence from the Comparison between Industrialized Countries

The work of Auster et al. (1969) led to similar studies using data from industrialized countries, in the main members of the OECD (Organization for Economic Cooperation and Development). Among these, the study by Cochrane et al. (1978) is

Table 4.4: Determinants of Specific Mortality Rates in 18 Industrialized Countries (around 1970)

Influencing factor (increase by one standard deviation)	Change in mortality rate in percent									
	Mothers	Birth	Infants	Age groups						
				1–4	5–14	15–24	25–34	35–44	45–54	55–64
1. Physicians ^a	1	8*	17*	3	1	0	–4	–3	–3	–1
2. GNP ^a	–15	–11*	–16*	–8*	1*	0	1	–5	–7	–9*
3. Cigarettes ^a	25	8*	10*	1	5	2	5	4	7	7
4. Alcohol ^a	18	0	5*	1	–1	0	0	–1	–3	–3
5. Pop. density	–3	0	–2	1	–2	–7*	–7	–9*	–4	–1
6. Pub. share ^b	2	–2	0	–6	–2	–16*	–10*	–9	–4	–3
7. Sugar ^a	–29	–8*	–4	–5	–6	–8	–11	–8	–3	–3
R ²	0.72	0.90	0.97	0.55	0.42	0.79	0.65	0.57	0.55	0.62

* Estimate is based on a regression coefficient that is statistically significant at the 5 percent level of confidence or better.

^a Per capita.

^b Share of public in total health care expenditure, “intervention index.”

Source: Cochrane et al. (1978).

particularly worth mentioning, since it uses age-specific mortality rates. This serves to multiply opportunities for statistical testing. The seven explanatory variables listed in Table 4.4 were selected by the authors from 17 regressors on the basis of simple correlations. They exhibit stable and statistically significant correlation coefficients with reference to three specific mortality rates (mothers, birth, infants) in most age groups.

The entries of Table 4.4 are thus based on regressions of the type

$$S_i = C + \alpha X_{1,i} + \beta X_{2,i} + \cdots + \varphi X_{7,i} + u_i \quad (4.4)$$

with S_i symbolizing the group-specific mortality rate in OECD country i , $X_{1,i}, \dots, X_{7,i}$ the seven selected regressors shown in Table 4.4, and $\alpha, \beta, \dots, \varphi$ parameters to be estimated.

In order to calculate the *relative importance of influences*, changes in X must be normalized. One alternative is to compare equal percentage changes. Starting from

$$\frac{\partial S_i}{\partial X_{1,i}} = \alpha \text{ and hence } \frac{\partial S_i/S_i}{\partial X_{1,i}/X_{1,i}} = \alpha \frac{X_{1,i}}{S_i} = \eta_i(S, X_1)$$

one may transform the regression coefficient α into the *elasticity* $\eta_i(S, X_1)$. It should be noted that contrary to the constant elasticity derived from the double-log formulation of Eq. (4.2), this elasticity varies across observations. In order to have an elasticity estimate $\hat{\eta}(S, X_1)$ representative of the sample as a whole, one often uses sample means $\{\bar{S}, \bar{X}_1\}$ to obtain

$$\frac{\partial S_i}{\bar{S}} = \hat{\alpha} \frac{\partial X_{1,i}}{\bar{S}} = \left(\hat{\alpha} \frac{\bar{X}_{1,i}}{\bar{S}} \right) \frac{\partial X_{1,i}}{\bar{X}_1} = \hat{\eta}(S, X_{1,i}) \frac{\partial X_{1,i}}{\bar{X}_1} \quad (4.5)$$

Thus, given, for example, an OLS estimate of α , one may derive the elasticity $\hat{\eta}(S, X_1)$, evaluated at the sample means $\{\bar{S}, \bar{X}_1\}$, which indicates by how many percent mortality is estimated to change when X_1 changes by one percent.

Another alternative, popular in psychology and sociology, is to normalize variables by their standard errors. One then obtains, dividing through by $\hat{\sigma}_S$ rather than $\bar{S}$ as in Eq. (4.5),

$$\frac{\partial S_i}{\hat{\sigma}_S} = \hat{\alpha} \frac{\partial X_{1,i}}{\hat{\sigma}_S} = \left(\hat{\alpha} \frac{\hat{\sigma}_{X_1}}{\hat{\sigma}_S} \right) \frac{\partial X_{1,i}}{\hat{\sigma}_{X_1}},$$

with $\hat{\sigma}_S = \{\text{Var}(S)\}^{1/2}$ and $\hat{\sigma}_{X_1} = \{\text{Var}(X_1)\}^{1/2}$ (4.6)

The regression coefficient, corrected for the ratio of standard errors of regressor and dependent variable (frequently called the “beta coefficient”) then indicates by how many standard deviations mortality is estimated to change when X_1 changes by one standard deviation.

As evidenced in Table 4.4, Cochrane et al. (1978) actually use a mixture of the two approaches defined by Eqs. (4.5) and (4.6). While changes of explanatory variables are normalized to reflect increases by one standard deviation [i.e., by setting $\partial X_{1,i}/\hat{\sigma}_{X_1} = 1$ in Eq. (4.6)], the induced change of the dependent variable appears in percentage form [$\partial S_i/\bar{S}$ as in Eq. (4.5)], which may be obtained by multiplying $\partial S_i/\partial \hat{\sigma}_S$ by $\hat{\sigma}_S/\bar{S}$.

Turning to the results displayed in Table 4.4, note that the explanatory variable Nos. 5 and 6 were absent from the study by Auster et al. (1969). In the case of infectious diseases, in particular, of the respiratory tract, it is traditionally assumed that a high population density (variable No. 5) fosters the spreading of epidemics (see also Section 4.3.4). The frequent negative signs in the corresponding line suggest that this expectation is not confirmed by the experience of industrialized countries around 1970. The other additional variable is the proportion of health care utilization paid for by *public means* (variable No. 6, called “intervention index” by the authors). It reflects their conviction that public spending on health is particularly likely to contribute to a reduction of mortality rather than merely adding to amenities in health care. The estimation results can be characterized as follows.

- Across all industrialized countries, the gross national product per capita (GNP) performs as originally envisaged by development economists. A rise in GNP by one standard deviation is related to a *decrease in the mortality rate* at birth (11 percent), during infancy (16 percent), in the first four years (8 percent), and in the last ten years of active life (9 percent) (see line 2 of Table 4.4).
- At the international level, there is now some evidence to the effect that the *mortality rate* depends positively on cigarette and alcohol consumption (lines 3 and 4).
- A large proportion of *public spending* in health care expenditure indeed seems to lower mortality rates, at least in the age groups 15–24 and 25–34, possibly also in the remaining age groups.
- Among the *variables relating to nutrition* (calorie intake, protein, fats and sugar, per person per day), surprisingly sugar consumption has a negative effect on the mortality rate (line 7).

However, one of the “puzzles” from the study by Auster et al. (1969) remains unresolved. Countries with a high *physician density* have a *higher* child and infant mortality rate, other things being equal (line 1 of Table 4.4). The reversed causality explanation, according to which a high rate of mortality signals income opportunities to physicians, thus increasing their density, is hardly convincing in this context. In the first place, migration of physicians between OECD countries is of little importance in comparison with migration between states of the United States (with the possible exception of British physicians migrating to the United States). Second, high mortality rates in the high-income middle age groups would be expected to attract doctors particularly strongly, but it is in these groups that the estimated partial correlations are negative rather than positive.

An alternative explanation could be based on the *substitutional relationship* between an individual’s own health-enhancing efforts and medical services. In contrast to the model as developed in Section 3.4, it is possible that a newborn child, being in

the “sick state” the first few months of his life, depends not so much on medical treatment (M) for his well-being, but on the care of his mother. In this case, a high value of M (made possible by a high physician density) can be a sign of substitution of motherly care, which is detrimental to the health of the (perhaps unwanted) infant [see the study by Grossman and Jacobowitz (1981)].³

4.2.2.3 Evidence from a Comparison between Two Neighboring States

Sometimes it is possible to learn a great deal from a direct comparison between two areas of almost identical characteristics [see Fuchs (1974, 1975)]. Utah and Nevada are suitable for such a comparison, as they are neighboring states of the United States, both sparsely populated and having the same desert climate, so that environmental influences upon health can be excluded immediately. Despite this, their mortality rates for 1970 are quite different. Infant mortality in Nevada was 42 percent (boys, denoted by M) and 35 percent (girls, denoted by F) in excess of Utah (see part A of Table 4.5). In the age group of 40–49 year olds, the relative mortality rate in Nevada is also higher, by 54 percent (men) and 69 percent (women).

For additional information, rates of mortality due to particular causes of death may be studied (part B of Table 4.5). Deaths of cirrhosis of the liver and lung cancer (so-called malignant neoplasm of the respiratory tract) occur almost 7 times more frequently in Nevada in the age group 30–39 of males.

This can hardly be the result of inadequate provision of medical care in Nevada, since Nevada and Utah have comparable densities of physicians and other health workers (variable Nos. 1 and 2, part C of Table 4.5).

Considering the economic factors, average income (variable No. 3) is indeed 16 percent higher in Nevada than in Utah. However, applying the elasticity estimates presented in Table 4.2, this could raise the mortality rate by 16 to 29 percent at the most, which is not sufficient to explain the difference between Nevada and Utah. Finally, length of schooling is the same in both states (variable No. 4), and the inhabitants of both states demonstrate the same preference for living on the countryside (variable No. 5).

Searching for an explanation, one comes across things that reflect a very different *lifestyle* in the two states. Nevada was evidently marked by much immigration right up to the 1970s, the proportion of adults born in Nevada being only 10 percent, compared to 63 percent in Utah (see variable No. 6). After five years of residence, only 36 percent of Nevada’s population still lives in the same place (variable No. 7), while almost half of the 35–64 year old men are single or no longer married to their first wife (variable No. 8). These features portray an extraordinarily *mobile, even unstable society in Nevada*. In fact, Las Vegas and Reno with their casinos are cities of Nevada, whilst the tone in neighboring Utah is set by the Mormons with their emphasis on abstinence. In all, this comparison supports the conclusion that lifestyle and habits may have a much greater influence on mortality rates than medical care.

3. This study, while inspired by the Grossman model as sketched in Section 3.2.2, does not purport to actually test it.

Table 4.5: Mortality in Nevada and Utah and Some Possible Determinants, 1970

			Age groups						
			<1	1-19	20-29	30-39	40-49	50-59	60-69
A. Mortality in Nevada (Utah=100)	M		142	116	144	137	154	138	126
	F		135	126	142	148	169	128	117
B. Mortality resulting from cirrhosis of the liver and lung cancer (Utah=100)	M					690	211	306	217
	F					543	396	305	327
C. Possible determinants of mortality							Nevada	Utah	
1. Physicians per 10,000 population							11.3	13.8	
2. Nonmedical personnel per 10,000 inhabitants							161	180	
3. Median value of income per capita (US\$)							10,942	9,356	
4. Median value of years of schooling							12.4	12.5	
5. Proportion of rural population, in percent							19.1	19.4	
6. Proportion of over 20 year olds born in the same state, in percent							10	63	
7. Proportion of over 5 year olds having the same place of residence in 1970 as in 1965, in percent							36	54	
8. Proportion of 35-64 year olds who were single, separated or remarried, in percent							47.4	25.5	

Source: Fuchs (1974).

Since lifestyle seems to be such an important factor, it is tempting to relate changes in longevity over time to changes in lifestyle. Now lifestyle is a quantity that should best be treated as a latent variable, mirrored by multiple, more or less fallible indicators such as alcohol and tobacco consumption and caloric intake. This approach, called structural covariance analysis, has proved very fruitful in psychometrics and sociometrics; a special version of it is known as the MIMIC model (see Section 4.4.2 for an application).

Wolfe (1986) and Wolfe and Gabay (1987) subjected a health production model incorporating lifestyle to a statistical test using time series data from 7 and 22 OECD countries, respectively. However, the relationships linking the latent variables (thus making up the structural core of the model) did not give rise to statistically significant parameter estimates. For this reason, the questions raised by the earlier studies concerning the nexus between economic development, income growth, and changes in lifestyle on the one hand and health status on the other still are not fully resolved.

The argument of the three preceeding subsections may be summed up by

CONCLUSION 4.2: Studies based on country as well as regional data confirm the view that differences in mortality rates can only very partially

be traced to medical infrastructure (and physician density in particular). The individual's productivity in enhancing and maintaining his health appear to be of greater importance.

4.2.3 The Marginal Productivity of Selected Medical Interventions

Health economics research has concerned itself comparatively little with the contribution of particular medical interventions to the health status of the person treated, and for two reasons. First, the internationally observed "cost explosion" gave rise to the suspicion that specific health care expenditure had exceeded its optimum value at any rate, so examining procedures looked like a moot point. Second, economists rarely had access to clinical data, and if they did, the objective of the study often was to prove an advantageous cost–benefit ratio for a particular pharmaceutical or technological innovation.

The gap between the macroeconomic view and product-specific studies has been filled primarily by epidemiologists and medical sociologists. They group mortality rates according to *cause of death*, estimate the possible contribution of individual medical measure, and aggregate these partial effects into a total effect. Proceeding in this way, McKeown (1976), for example, investigated the historical development of mortality due to infectious diseases. He found that at a time when these diseases (ranging from typhus, small pox, scarlet fever to tuberculosis and pneumonia) could not yet be treated medically, the respective mortality rates were already declining in several industrial countries. McKinlay et al. (1989) limited their study to the United States while covering a broad range of disease categories. Their main findings may be summarized as follows.

- **Infectious diseases.** Better control of infection was responsible for about 40 percent of the mortality decline between 1900 and 1973. Among the ten illnesses in this group, only three (influenza, whooping cough, and polio) showed a reduction in mortality rates of 25 percent or more that could be causally linked to a medical innovation (vaccination). If one aggregates these three categories, about 3.5 percent of the drop in total U.S. mortality since 1900 can be attributed to medical advances in fighting infectious disease.
- **Chronic illnesses.** Here we can distinguish two main groups, cardiovascular disease (which causes one-third of all U.S. death) and cancer. In the first group, mortality rates have been falling in the United States since 1978, whilst in former Western Germany, for example, they continued to increase until the mid-1980s [see Kannel and Thom (1984)]. The fact that the mortality rate has primarily decreased outside the hospital may be interpreted as a result of successful preventive efforts, for only about 9 percent of the reduction in mortality can be attributed to the treatment of high blood pressure. Therefore, the decrease in mortality from heart attack seems mainly to reflect a reduced *frequency* of circulatory disease and not so much the success of medical treatment. In the case of cancer, medical success was limited to the treatment of testicular cancer in young men and lymphatic leukaemia in children, which jointly account for no

more than about 8 percent of the incidence of these illnesses. In contrast, an increasing proportion of deaths is due to lung cancer, in which smoking of cigarettes in particular has been proved to be the decisive risk factor.

McKinlay et al. (1989) complete this picture of limited success of modern medicine by pointing out two surprising facts. First, the number of lost workdays per person as well as the frequency of long-term health-related limitations of customary activities have been increasing since the 1950s⁴ in the United States. Second, while life expectancy did grow between 1964 and 1985 by around 4.4 years for men and 4.5 years for women in the United States, *life years free of handicap* did not. This quantity involves a weighing of years lived not only by the probability of survival but also by the conditional probability of being able to live without handicap given survival. Between 1964 and 1985, life years free of handicap showed a decline of no less than 7.3 years (men) and 7.6 years (women). Gains of 3.9 and 3.2 years, respectively, resulted in the over 65-year-old age group only.

In sum, these estimates establish a correspondence between the weak or even perverse links connecting health and medicine found at the national level and the development of mortality in certain disease categories. In a predominantly healthy population, inputs other than medical services must play the decisive role in the production of health (see remarks in Sections 3.4 and 3.5).

4.2.4 Environmental and Economic Determinants of Health Status

After Auster et al. (1969), there have hardly been any attempts to detect environmental influences upon health status using aggregate data. To recall, in that study neither the degree of urbanization nor the degree of industrialization could be systematically related to the mortality rates of the states of the United States (see Subsection 4.2.2.1). The study by Lopez et al. (1992) is an exception. It establishes a connection, in the style of the transformation curve derived in Sections 3.4 and 3.5, between water quality and mortality from cancer of the digestive tract in the 50 communities of Nuevo-Léon in Mexico. Individuals have to weigh *consumption* and *water pollution* against each other while having an influence upon both:

- On the one hand, the creation of jobs in industry expands consumption possibilities, yet aggravates the problem of water pollution.
- On the other hand, the installation of running water and sewerage relieves problems of water pollution but limits consumption possibilities.

The authors find empirical evidence for the existence of a tradeoff between consumption and life expectancy. Surprisingly, a high degree of illiteracy and high population density appear to go along with a low rate of cancer deaths.

Still at the level of aggregate data, another issue has received emphasis, viz. the effects of economic instability on health. The debate was started by Brenner (1979), who, using annual data from 1936 to 1976, found a significant statistical connection

4. Part of this increase is, however, due to the improved insurance coverage in the case of illness. The moral hazard effects discussed in Sections 6.4 and 6.5 need to be mentioned here, although McKinlay et al. (1989) judge them insignificant.

between the *mortality rate* and the *unemployment rate* in England and Wales. The two other most important explanatory variables were the trend value of real income per capita and the deviation of current income from this trend. These relationships were justified theoretically by pointing out that unemployment forces the individual to reallocate both his time budget and his financial budget. In combination, these adjustments amount to a considerable burden, which may result in poor health and even increased probability of death.

In the framework of the model developed in Chapter 3, the influence of economic instability can be represented by a sudden change in several productivities. In the healthy state, the sudden income loss due to unemployment could cause a decrease in both the individual's contribution to consumption services ($\partial C_h / \partial t^C$) and to health in the guise of controlling his risk of illness ($\partial \pi / \partial t^H$) [see Section 3.5.1, Eq. (3.25)]. Both changes cause longer-run production possibilities in the (C_h, T_h) space to decrease; the second also moves the maximum point A_h of the transformation curve inwards, which *ceteris paribus* leads to a marked loss of healthy time (see Section 3.5). In the sick state, worsening economic conditions could limit the contribution of medical services to recovery [$\partial \pi / \partial M$ smaller in absolute value in Eq. (3.30)], resulting in an increased minimal duration of a spell of illness. In extreme cases, these effects may conceivably combine to *increase the rate of mortality*. In this way, Brenner's findings can be said to have a theoretical foundation.

Table 4.6 contains two more recent estimates, one of the global mortality rates of England/Wales and Scotland, the other for both of them combined [and thus based on twice as many observations; see Brenner (1983)]. The most important results are as follows.

- There is statistical evidence to the effect that the aggregate economic unemployment rate (variable No. 3) and, even more markedly, the unemployment rate of over 20-year-old men (variable No. 4), *increases the mortality rate*, with a lag of up to two years.
- Cigarette consumption and (in Scotland) spirits consumption (variable Nos. 7 and 8) make a statistically recognizable contribution to the explanation of mortality rates.
- Health care expenditure (for the National Health Service) seems to reduce mortality, as in the comprehensive cross-country study by Cochrane et al. (1978), discussed in Subsection 4.2.2.2. What is not clear is why health care expenditure should play a role only as a proportion of total public expenditure rather than its absolute amount.
- The equation shown in Table 4.6 also performs well in the analysis of mortality resulting from cardiovascular disease, heart disease following brain stroke, cirrhosis of the liver, suicide, murder, car accidents, and infant deaths, since very high coefficients of determination R^2 are obtained throughout.

The studies by Brenner have been subject to extensive criticism from Gravelle (1984), who emphasizes the following three points.

1. ***Inaccurate theoretical foundation.*** According to Table 4.6, consumption affects health through goods such as alcohol and tobacco, whose purchase must be

Table 4.6: Mortality Rates in England, Wales, and Scotland, 1954–1976

	[L] ^a	England and Wales	Scotland	Combined sample (N=50)
Constant		−3.53	32.8***	15.4***
1. Real income per capita, trend		−0.013***	−0.025***	−0.013**
2. Real income, change over previous year		−0.003*	0	0
3. Rate of unemployment ^b	[1–2]	0.355**	31.6**	0.035**
4. Rate of unemployment, 20–40-year- old men	[0–2]	5.10***	0.704*	0.986**
5. Rate of unemployment, 40+-year-old men	[0–2]	7.25***	0.840*	0.981*
6. Weekly working time, industrial		0	−0.344**	0
7. Cigarette consumption per capita	[2–5]	0.0027***	0.0030***	0.001*
8. Spirits consumption per capita		0	10.1***	0
9. Health expenditure/public exp.	[1]	−20.6***	−14.0	−4.5*
10. Average temperature in February		−	−	−0.016*
R ²		0.97	0.95	0.96
DW (after Cochran–Orcutt transformation) ^c		1.90	2.68	2.08

*(**,***): Coefficient different from zero with error probability of 5% (1%, 0.1%).

−: Not estimated.

0: Excluded from regression due to closeness of *t* value to zero.

^a [L] = Lag in years according to the estimated equation.

^b The rate of unemployment in Scotland is not known throughout; it is partially replaced by the one from the U.K.

^c The Cochran–Orcutt transformation is a method to free the estimated equation from the so-called autocorrelation in the error terms. A high value of $\hat{\rho} = \text{Corr}(\hat{u}_t, \hat{u}_{t-1})$, i.e., a high correlation of the residual terms over time is an indicator of such autocorrelation. The test measure of Durbin and Watson (DW) takes on the value 2 if $\rho = 0$. If the DW value is significantly different from 2, the estimated equation is multiplied by $(1 - \hat{\rho})$. This so-called Cochran–Orcutt transformation leads to estimated values for the regressions coefficients and their standard errors, which are freed from autocorrelation [see Pindyck and Rubinfeld (1991), Section 6.2.1].

Source: Brenner (1983).

financed out of disposable income. Thus income itself does not belong in the equation, unless it is used as an indicator of general lifestyle, in the manner of Auster et al. (1969) [see Subsection 4.2.2.1].

2. *Simultaneity problem.* Changes in health status (which lie at the basis of changes in mortality) have an impact on labor income. Therefore, income cannot be treated as predetermined but contains a stochastic component that is positively correlated with mortality. This positive correlation results in an overestimate of the influence of income fluctuations on the mortality rate. The impact of unemployment on health is also *exaggerated*, since income fluctuations and unemployment are closely linked.

3. **Functional form.** The trend value of income can but increase with time (even at an accelerated rate because an exponential trend is assumed). This means that at some point in the future, the model will predict *negative* mortality rates. Finally, the dependent variable amounts to a linear combination of two mortality rates, one characterizing the unemployed, the other, the rest of the population, while the theory revolves around the mortality rate of the unemployed. Since the weights of this linear combination are given by the rate of unemployment (UN_t) and its complement $(1-UN_t)$,⁵ the equation should contain regressors such as UN_t^2 and $UN_t\Delta Y_t$ (ΔY_t is change in real income). Since these regressors are likely to exhibit a great deal of cyclical variability, they probably would attain statistical significance. This would render the relationship between the mortality rate and the unemployment rate UN_t variable, depending on the values of UN_t and ΔY_t in the year under consideration.

This last-mentioned argument is quite damaging to the regression equations estimated by Brenner. Moreover, it explains why Gravelle et al. (1981) found the coefficients of these regressions to be unstable over time. *Lags of very different lengths* also occur, ranging for example in cigarette consumption from 2 to 5 years (see Table 4.6). In a comparable study of four industrial countries [see Brenner (1985)], lagged effects go from zero (Sweden) to 16 years (France) and 18 years (Denmark). An estimate for Germany (1955–1979) using variable Nos. 1–3 and 7 of Table 4.6 even found consistently negative estimated effects of unemployment on age-specific mortality rates, presumably because in a recession period, many (sick) foreign workers lose their jobs and have to leave the country [see John (1983)]. These statements give rise to

CONCLUSION 4.3: A negative impact of economic instability on the individual's health-enhancing productivity as well as on the productivity of medical services is theoretically plausible. However, the statistical evidence supporting such a relationship is not very strong at present, not least because it is based on overall mortality rather than mortality of those affected by unemployment.

4.3 STUDIES BASED ON INDIVIDUAL DATA

4.3.1 Measurement of Health Status

At the aggregate level, the choice of output variable in a production function of health is severely limited by the availability of statistical data. Individual measurements based on surveys, however, open up a broad range of possible definitions of output. Preferably, the analyst's choice should be guided by the theoretical guidelines developed in Chapter 2 [see Section 2.4.2 in particular], where (subjective) health status is defined by a probability distribution over various possible states. Only one of them materializes at the time of measurement. An ambiguity then arises as to whether this state is to be understood *prior to or after* the input of medical care (or the individual's own effort).

5. The rate of unemployment usually refers to the active population rather than to the total population. This difference is neglected here for the sake of simplicity.

As already established in Section 1.2 and in Conclusion 3.6, health has a *dual function*. On the one hand, it is a variable of crucial importance for explaining the utilization of medical care; on the other hand, it represents the outcome of a production process. This dual function creates no problems as long as the inputs are defined as potential ones (physician density, number of beds, etc.). As soon as these inputs turn into effective consumption of medical services, however, health status should be measured both at the beginning and at the end of the production process. Also, measurement of health should preferably refer to a particular healthy phase or sickness episode. This type of temporal disaggregation has hardly been undertaken up to now [with the pioneering exception of Keeler et al. (1987)].

Apart from the timing of the measurement, the source of measurement also is a far more important issue in the case of individual data than with aggregate data. In particular, a distinction needs to be made between

1. Information provided by the individual himself;
2. Information provided by a third party, in particular physicians and nursing personnel.

The first source of information is often subjective and therefore deemed less reliable. For some purposes, however, the answer of a hypochondriac (say) to the question, “How are you at the moment?” can be quite *informative*. It may betray a particularly high marginal willingness to pay for better health, from which a high probability of utilization of medical services can be inferred. Information belonging to the second category is indeed objective, being at least partially based on clinical measurements. Referring to the concept of latent variables and their fallible indicators, to be introduced in Section 4.4.2 (see Figure 4–1 in particular), clinical measurements may be said to give rise to indicators having a small share of measurement error variance in total variance. On the other hand, they frequently fail to have a strong, significant relationship with the latent variable of primary interest, “health.”

However, the search for an “objective” output measure that is independent of the individual’s willingness to pay (and also ability to pay) is hardly worthwhile in this context. It hinges on the questionable premise that *health ought to be maximized regardless* of other competing objectives.⁶ As soon as we are to allow other competing objectives, an improvement in health would need to be evaluated by marginal willingness to pay (as in Section 2.3.2). Inasmuch as willingness to pay depends on the current subjective state of health, information of type (1) appears to be very suitable.

4.3.2 Marginal Productivity of Medical Infrastructure at the Individual Level

In the United States a Health Examination Survey was conducted around 1960, in the course of which over 6,000 individuals from 39 randomly selected areas were examined for various physiological aspects of their health status. These measurements

6. This does not deny that nonmonetary (medical) measures of health status may contribute to an optimal allocation in health care, e.g., when it comes to the choice between various treatment alternatives within a previously established health budget.

Table 4.7: Estimated Influences on Clinical Health Indicators, United States (around 1960)^a

<i>Clinical indicators</i>	<i>Medical infrastructure^b</i>			<i>Individual productivity^c</i>		<i>Environmental influences^d</i>	
	<i>PHYS</i>	<i>OTH</i>	<i>BEDS</i>	<i>EDU</i>	<i>INC</i>	<i>URB</i>	<i>RACE</i>
1. Hypertension ^e	−0.06	0.20	0.32	−0.25	−0.006	−0.001**	0.12*
2. Diastol. blood pressure, conditional ^f	−0.02	−0.003	0.009	0.01	−0.003	−0.007	2.48*
3. Excess lipid concentration in the blood ^e	0.39	−0.03	−0.13	−0.08	0.05	−0.000	−0.02*
4. Lipid concentration in the blood, conditional ^g	0.02	−0.02	0.03	0.02	−0.003	0.031	−1.01
5. Abnormal ECG ^e	0.03	−0.06	0.06	−0.02	−0.04	0.000	0.15**
6. Abnormal X-ray ^e	−0.04	−0.07**	0.07*	−0.09**	−0.03*	0.002**	−0.082
7. Varicose veins ^e	0.13	0.002	−0.44	−0.19	−0.04	0.003	−0.05*
8. Peridontal index	−	−	−	−0.77**	−0.21**	−0.002	0.19**
9. Aging index ^h	−0.004	0.001	0.01	−0.05**	−0.01**	0.002	0.72*

*(**): Significant with an error probability of 5 percent (1 percent or less). *N*=4,769.

^a The values shown are elasticities calculated using sample means, except for *URB* and *RACE*, where regression coefficients are shown.

^b *PHYS* = general practitioners, internists, and gynaecologists, *OTH* = other physicians, *BEDS* = hospital beds, all per 100,000 inhabitants. Two additional regressors (beds in hospitals run by the federal government and dentists' density) are not listed. All regressions contain age, (age)², and a categorical variable for the sex.

^c *EDU* =number of years of schooling, *INC* = family income.

^d *URB* = Proportion of the examined living in communities having more than 2,500 inhabitants; *RACE* = 1 for nonwhites, = 0 for whites.

^e Logit estimate, as the dependent variable can only take on the values 0 and 1.

^f Only cases of hypertension.

^g Only cases of excess lipid concentration.

^h Based on regressions of age on variable Nos. 2 and 4–8; every individual is assigned a “clinical age” using the average of the five calculated expected values.

Source: Newhouse und Friedlander (1980).

thus belong to category (2) of the sources of information mentioned in Section 4.3.1. They were related by Newhouse and Friedlander (1980) to the available regional medical infrastructure on the one hand and to variables characterizing these individuals and their environment on the other. Apart from the explanatory variables presented in Table 4.7, categorical variables for occupation, family status, family size, self-employment, and sex entered the regression equations, while age appears in linear and squared form.

The clinical indicator Nos. 1–4 refer to blood pressure and lipid concentration in the blood, conditions that evidently contribute to cardiovascular mortality according to the findings of large-scale epidemiological studies (see Section 4.3.3). Judging from Table 4.7, the following appear particularly important.

- Neither *medical infrastructure*, nor *education and income* of the individual examined have a statistically proven influence on indicator Nos. 1–4.
- In the case of indicator Nos. 6, 8, and 9 (X-rays, periodontal index, age index), *more schooling* appears to turn individuals into better producers of health, without the increase in income becoming counterproductive. In terms of the age index, a 10 percent increase in the length of schooling makes the individual 5 percent younger physiologically than his calendar age, other things kept equal.
- With regard to the degree of *urbanization*, only the occurrence of an abnormal X-ray (indicator No. 6) is affected positively, and significantly so.
- *Race* is the variable having statistical significance most often. Nonwhites, when compared with white people, appear less healthy. While differences in blood pressure and lipid concentration may be due to genetic differences, the periodontal index (No. 8) points to differences in behavior and even preferences.

4.3.3 The Influence of Medical Interventions at the Individual Level

In the 1960s cardiovascular diseases were the most important cause of death in the United States. High blood pressure, high lipid concentration in the blood, and tobacco consumption were suspected to constitute the decisive risk factors. At that time many long-term studies were started, among which that of *Framingham* (Massachusetts) is the most famous, having an observation period of up to 18 years. The results of this study provide insights into the production of health in a very concrete, almost technical sense. Output can be defined as a reduction in the cholesterol level of the blood, which in turn showed lower blood pressure. The medical input consists of treatment with an appropriate drug, and the patient's own contribution in maintaining a particular diet. Is the maintenance of the diet alone sufficient for reducing the cholesterol level, or is the combination with a drug required?

Since hypertensive patients can continue their normal life as if healthy, the conditional production possibility frontier model of Chapter 3 [in particular Conclusion 3.10 (see Section 3.5.1)] leads to the presumption that individual efforts could *substitute* for the use of drugs. On the other hand, the participants in these long-term studies do live through periods of illness during the period of observation, and may also use drugs in these periods. Therefore, medical inputs should contribute to survival. Surprisingly, Houston (1989, p. 928) in his extensive survey comes to the conclusion that drug therapy has no definite influence on mortality in the case of cardiovascular diseases. Indeed, certain forms of therapy can even *undermine the success of a diet* (see p. 929). This conclusion provides an excellent illustration of the idea developed in Section 3.4, according to which medical services cannot be substituted by individual efforts to improve health in the event of illness, while substitution becomes possible in the longer run. Therefore, the results of the studies based on individual data justify the interim

CONCLUSION 4.4: The concept of the production of health, with the individual's own contribution and medical services as inputs, can also be applied to individual data, exhibiting considerable extent of substitutability in the case of specific diseases.

4.3.4 Environment and Health

In the studies presented so far, environmental influences either have been neglected, or else they proved statistically insignificant. The first systematic attempt to assess the effects at least of air pollution on human health was made by Lave and Seskin (1977). However, they had to work with English counties as their unit of observation, with no indicators describing individual productivity such as education or average income available. Usually [as in Lopez et al. (1992); see Section 4.2.4], population density must assume the role of both an (inverse) income indicator as well as of a factor favoring the spreading of communicable disease. This often leads to instability of the estimated coefficients or even implausible results.

4.3.4.1 Air Quality and Smoking as Exogenous Factors

The problems associated with the use of aggregate data were first overcome by Ostro (1983), who combined *individual responses* to the health survey of 1976 in the United States with measurements of the air quality in 84 metropolitan areas. *TSP* (total suspended particles) and *SULF* (sulfuric compounds) were selected from a large number of values of measurement and introduced as regressors. Both indicators had been consistently measured over a long period of time, and their use ensured comparability with earlier studies. As indicators of health status, the number of days of restricted activity due to health reasons as well as the number of workdays lost during the two weeks preceding the survey were used (see Table 4.8).

The regression coefficients entered in column A of Table 4.8 suggest that *TSP* (variable No. 1) has the expected positive influence upon the number of restricted activity days, whereas *SULF* (variable No. 2) shows no significant effect. More relevant to economics in the traditional sense is the number of lost workdays (column B). Again, the *TSP* concentration plays a statistically significant role, with an elasticity of 0.45 (calculated at the sample mean). An increased *TSP* concentration of 10 percent results in a 4.5 percent increase in the number of workdays lost—a value that according to Ostro (1983) is ten times that of previous investigations.

In analyzing the connection between air quality and health state, the concept of health production by the individual comes in handy once more. By *avoiding tobacco consumption*, the individual can improve his air quality substantially. The number of smoked cigarettes appears for this reason in the equations of columns A and B in Table 4.8 (as variable No. 14). The corresponding regression coefficient proves both times to be statistically insignificant, however. Yet the low coefficients of determination of 0.09 and 0.01 point to the possibility that many determinants of the number of days of restricted activity and the days of absence from work are still absent from the regression equation. These factors may also include differences between smokers and nonsmokers not sufficiently reflected by the number of cigarettes smoked (*CIGARETTES*). For

Table 4.8: Influence of Air Quality on Health, United States (1976)

<i>Variable^a</i>	<i>Restricted activity days (A)^b</i>	<i>Workdays lost (B)^c</i>	<i>Probability of $B > 0$ (C)^d</i>	<i>Workdays lost, given $B > 0$ (D)^d</i>
Constant	-0.83***	-0.47**	-3.66***	-0.39
1. <i>TSP</i>	0.00282**	0.00145*	0.0071*	0.002
2. <i>SULF</i>	-0.00008	-0.001	-0.051	-0.009
3. <i>CHRONIC</i>	1.25***	0.25***	0.48***	0.93*
4. <i>AGE</i>	0.0063***	0.0033**	-0.0048	0.075*
5. <i>INCOME</i>	-0.009***	-0.002	-0.004	0.012
6. <i>MARRIED</i>	-0.011	-0.011	0.227	-1.24**
7. <i>RACE</i>	0.17**	0.045	-0.04	-0.46
8. <i>TEMPERATURE</i>	0.013***	0.0065*	0.003	0.097*
9. <i>BLUE COLL. WORKER</i>	-	-0.046	0.29	-1.26**
10. <i>EMPLOYED</i>	-0.114*	-	-	-
11. <i>POP.DENSITY</i>	0.0057	0.0056	0.030	-0.050
12. <i>PRECIPITATION</i>	-0.0004	-0.0004	0.0097	-0.040
13. <i>SEXF</i>	0.093*	0.067*	-	-
14. <i>CIGARETTES</i>	0.0032	-0.0006	-	-
R^2	0.09	0.01	-	0.17
χ^2			25.8***	
N	13,230	8,294	4,473	263

*(**,***): Coefficient different from zero with an error probability of 5 percent (1 percent, 0.1 percent).

^a *TSP* = concentration of total suspended particels in the air, annual average (in microgram/m³); *SULF* = concentration of sulfuric compounds in the air, annual average (in microgram/m³); *CHRONIC* = number of chronic diseases; *AGE* = age in years; *INCOME* = family income (in thousands of USD); *MARRIED* = 1 if individual is married, = 0 otherwise; *RACE* = 1 if nonwhite, = 0 otherwise; *TEMPERATURE* = annual average temperature (in Fahrenheit); *EMPLOYED* = 1 if employed, = 0 otherwise; *BLUE COLLAR WORKER* = 1 if blue collar worker, = 0 otherwise; *POPULATION DENSITY* = population of the metropolitan area (in 1,000 per square mile); *PRECIPITATION* = annual average precipitation; *SEXF* = 1 if female, = 0 otherwise; *CIGARETTES* = number of smoked cigarettes per day.

^b Sample of 18–65-year-old persons in 84 metropolitan areas.

^c Sample only includes employed (*EMPLOYED* = 1).

^d Sample only includes male, employed nonsmokers.

Source: Qstro (1983), Tables I and III.

instance, the number of chronic health problems (*CHRONIC*) probably has a completely different (also sex-specific) significance in the case of smokers with their frequent respiratory diseases than in the case of nonsmokers. The effect would be to bias the estimate of the influence of air quality on health status.

In order to estimate the effects of air pollution on the number of lost workdays without such bias, the sample was restricted to male, employed nonsmokers (see columns C and D of Table 4.8). A full 94 percent of these had no absences from work in the two weeks before investigation, while 6 percent had lost one or more workdays. This calls for estimating the *probability* of absence from work (column C of Table 4.8)

prior to estimating the length of an absence when applicable (column D). Accordingly, the dependent variable of column C is a dummy variable, which takes on the value of 1 when at least one workday was lost for health reasons, and the value of 0 otherwise.⁷ This restriction on the range of the dependent variable can be lifted by means of the so-called Logit transformation. As a consequence, however, the values entered in column C are coefficients of a logistic regression, which cannot be interpreted as partial influences on probability [see Pindyck and Rubinfeld (1991), p. 258]. The values in column D, in contrast, are ordinary least-squares estimates.

Since columns C and D of Table 4.8 refer to nonsmokers only, they should provide information about the influence of air quality upon work capability quite independently of the behavior of the individual, viz., choice of smoking status. The following observations seem appropriate.

- Air quality, measured as TSP concentration, has a statistically significant influence on the *probability of absence from work*, but not on the *length of this period*. The influence of the other component of air quality, that of sulfuric compounds, cannot be proved.
- Other environmental factors such as average air temperature or rainfall do not seem to have an *influence* on the absence from work (probability of occurrence or duration of absence).

4.3.4.2 Smoking as an Endogenous Factor

All studies discussed so far (including those using aggregate data, in Section 4.2) have a common flaw because they analyze influences on health only, whilst both the Grossman model of Section 3.2.2 and the alternative developed in Section 3.4 simultaneously explain *consumption* and *health*. Analyzing health becomes problematic as soon as the consumption decision has an immediate impact on health, as in the case of smoking (see item 2 of Section 4.1). In an attempt to control for this impact, the number of cigarettes smoked appeared in Table 4.8 as a regressor. However, Rosenzweig and Schultz (1983) argue that this solution fails to do justice to the theoretical model, calling attention to background variables that may well have an effect both on health (particularly respiratory diseases) and on tobacco consumption. If, for example, anxious people hardly ever smoke, but nevertheless are more ill than others, an *artificial positive correlation* between health and tobacco consumption results from the data. This effect serves to neutralize the expected negative correlation (i.e., the damaging effect of tobacco on health). It could be the reason for the lack of statistical significance of the regressor *CIGARETTES* in Table 4.8.

Evidently, the explanatory variable *CIGARETTES* must be purged of an error component that is due to its endogeneity. This can be done through a preliminary regression (amounting to the first stage of two-stage least-squares; see Table 4.3b), in which the observed values of *CIGARETTES* are replaced by their expected conditional values.

7. This means that the dependent variable is extremely skewed to the left, which does not square well with the customary assumption of a normally distributed error term, the normal distribution being symmetrical around the mean (see also Table 4.3a and b in Subsection 4.2.2.1).

Mullahy and Portney (1990) applied an estimation procedure of this type, the so-called generalized method of moments (GMM) to the same sample as Ostro (1983), but using data for 1979 instead of 1976. The dependent variable is the number of restricted activity days during the two weeks prior to the investigation, occurring as a *result of respiratory diseases*. This variable should thus bear a closer relationship to air quality than the dependent variable of Table 4.8, which was the number of restricted activity days for (all) health reasons. Since 96 percent of those interviewed had no such days, the probability of having days with health-related limitations is examined in column A of Table 4.9. The parameter estimates of the equation explaining the duration of these limitations are shown in column B. To demonstrate how important it is to purge the

Table 4.9: Air Quality and Smoking in the Production of Health, United States (1979)

Variable ^a	Probability of nonzero restricted activity days ^b		Number of restricted activity days, conditional on A > 0	
	A		B	
	OLS	GMM ^c	OLS	GMM ^c
Constant	0.035	0.185**	0.0078	0.789**
1. <i>CIGARETTES</i>	-0.060	-1.91***	-0.034	-8.44**
2. (<i>CIGARETTES</i>) ²	0.25	5.81***	1.09	27.95***
3. <i>OZONE</i>	0.064	0.51	4.25*	6.64*
4. (<i>OZONE</i>) ²	-1.60	-3.97	-23.7*	-35.8*
5. <i>SULF</i>	-0.20	-0.20	-1.52	-1.77
6. (<i>SULF</i>) ²	0.65	0.56	4.87	5.07
7. <i>TEMPERATURE</i>	-0.055	-0.090	-0.44**	-0.61**
8. <i>PRECIPITATION</i>	-0.017	-0.034	0.033	-0.028
9. <i>AGE</i>	0.35	-0.16	2.14*	-0.90
10. (<i>AGE</i>) ²	-0.48	0.077	-2.82*	0.60
11. <i>DCHRONIC</i>	0.011	-0.0066	0.19	0.094
12. <i>SEXM</i>	-0.00089	-0.032**	-0.045	-0.17**
N	2,331	2,331	89	89

*(**,***): Coefficient different from zero with an error probability of 5 percent (1 percent, 0.1 percent).

^a *CIGARETTES* = number of cigarettes smoked per day (/100); (*CIGARETTES*)² = square of *CIGARETTES*; *OZONE* = ozone concentration based on daily maximum values during the 14 days prior to the investigation (in ppm, parts per million); (*OZONE*)² = square of *OZONE*; *SULF* = concentration of sulfuric compounds, mean of measurements taken during 14 days prior to the investigation; (*SULF*)² = square of *SULF*; *TEMPERATURE* = average temperature based on the maximum values of 14 days (in Fahrenheit/100); *PRECIPITATION* = mean of the measurements taken during 14 days (in inches); *AGE* = age in years; (*AGE*)² = square of *AGE*; *DCHRONIC* = 1 if individual is restricted in his activity due to chronic diseases, = 0 otherwise; *SEXM* = 1 if individual is male, = 0 otherwise.

^b Although the dependent variable can only take on the values 0 and 1, it is treated like a continuous variable here. The resulting coefficients can therefore be interpreted as estimated partial effects on the probability of the corresponding regressors.

^c GMM = generalized method of moments; generalization of the two-stage estimator of Table 4.3b.

Source: Mullahy and Portney (1990), Table 3.

variable *CIGARETTES* of its endogeneity, OLS and GMM estimates appear next to each other in Table 4.9.

The results may be summarized in the following statements.

- ***Influence of tobacco consumption.*** If tobacco consumption is treated as an exogenous variable beyond the influence of the individual (like ozone concentration), then there is no indication of any impact on health status, exactly as in the study by Ostro (1983) [but in contrast to Leu and Doppmann (1986); see Table 4.10]. This holds true even if tobacco consumption is specifically related to respiratory health (see OLS estimates of variable Nos. 1 and 2). On the other hand, once *CIGARETTES* is cleared of its endogeneity, it proves a *highly significant* determinant of health (see GMM estimates of variable Nos. 1 and 2). Actually, smoking of a few cigarettes a day seems to have a positive effect on health, but beyond about 16 cigarettes, the negative effects on the respiratory tract become predominant.
- ***Influence of air quality.*** Ozone has to be recognized as a relevant substance in air pollution. Its effects on health, in contrast to tobacco consumption, tend to decrease with increased concentration (i.e., beyond about 0.06 microgram/m³, the average value in the sample being 0.043); this follows from the negative coefficients of variable No. 4. A high ozone concentration both *increases the probability* of spending a period of restricted activity (part A of Table 4.9) and *prolongs its duration* (part B of Table 4.9).
- ***Sulfuric compound concentration as an indicator of air quality.*** Despite a narrowly defined dependent variable and improved estimation methods, no influence of sulfuric compounds (*SULF*) on health condition can be proved (see variable Nos. 5 and 6). On this point, the earlier investigation of Ostro (1983) is confirmed (see Table 4.8, variable No. 2).
- ***Comparison of magnitudes of influences.*** Using the parameter values estimated using the GMM method and the respective sample means, elasticities can be calculated. For example, with an increase in daily cigarette consumption of 10 percent, the probability of having a period of restricted activity increases by around 33 percent, in contrast to only 13 percent in the case of an increase in ozone concentration of 10 percent. The length of such a period increases by around 44 percent following increased smoking, and by about 28 percent as a result of 10 percent more ozone in the air. Therefore, the estimated relative productivities in enhancing health of the respiratory tracts shift from the exogenous influence of air quality towards *smoking* as an input in the production of health, which is under the control of the individual.

The results derived from the two studies concerning the influence of environmental factors on health status are summarized in

CONCLUSION 4.5: With the help of newly available individual data, connections between air quality and the state of health are becoming evident. However, the input “tobacco consumption” appears to play a role of at least equal importance in the simultaneous production of consumption services and health of the respiratory tract.

This conclusion supports the notion that the observed slowdown of the increase of life expectancy can be traced to a rather limited extent only to a change in relative productivity of the input factors in the production of health, caused by a deterioration of environmental quality (point 3 of Section 4.1).

4.4 DEMAND FOR HEALTH, DEMAND FOR HEALTH SERVICES

Both the Grossman model and the alternative model introduced in Section 3.4 view the individual as a demander for and a supplier of health at the same time. So far it has been assumed without any further explanation that the empirical relationships reflect the production side of individual behavior. In this section, we will deal with the question of whether a differentiation between supply and demand side is necessary at all, how this could be achieved, and whether certain puzzles emerging in the literature on the production function of health can be explained.

4.4.1 How Should Individual Observations Be Interpreted?

One might claim that investigating the demand for health separately from its production by the individual is unnecessary. Given state-independent “average” preferences and production possibilities with regard to consumption and health, the marginal rate of substitution (MRS) can be simply inferred from the observed marginal rate of transformation (MRT). Figure 1–2 of Section 1.2, for example, shows that these two values coincide at the optimum (point R^*). While the location of this optimal point does depend on the preferences of the observed individual, this does not preclude an unbiased estimation of the parameters of the individual’s production function as long as preferences differ *randomly* between individuals. In this case, they only affect the error term of the estimated equation (see the variable u_i in Tables 4.3a and b), causing a loss of statistical fit but having no further consequences. This is the “classical view” based on Grossman (1972) [see Rosenzweig and Schultz (1983)].

The model developed in Section 3.4 constitutes a major departure from this view. According to this model, the individual is not really able to choose his state of health. Rather, his options are limited to *influencing the probability* of a certain state of health occurring. Consequently, he chooses an optimal probability $(1 - \pi^*)$ to remain in good health or to be in good health again, with $0 < (1 - \pi^*) < 1$. During a given period of time, the individual is (given only two possible states) either healthy ($h = 1$) or sick ($h = 0$). This simply means that the target is missed by most of the people most of the time. In terms of the Grossman model, actual and optimal stocks of health diverge most of the time (see Subsection 3.2.2.2). These divergencies compensate one another when taking averages, but only for a large group or a large number of periods.

At the level of the single individual and during a shorter observation period, the deviations between actual health h_i and targeted chance of being healthy $(1 - \pi^*)$ give rise to an additional error term $\tilde{u}$, which does not exist in the “classical view.” However, there is a *feedback* from $\tilde{u}$ to the inputs of the production function of health.

- $h = 1 > (1 - \pi^*) \rightarrow \tilde{u} > 0$: The individual is in good health, while aiming at a chance of being healthy amounting to $(1 - \pi^*) < 1$. Accordingly, he will reduce efforts to maintain his state of health (t^H in Table 3.4). The correlation between t^H and $\tilde{u}$ is *positive* (“being too healthy goes together with neglecting prevention”).
- $h = 0 < (1 - \pi^*) \rightarrow \tilde{u} < 0$: The individual is sick, although aiming at a probability of being healthy of $(1 - \pi^*) > 0$. Accordingly, he will demand medical services. The correlation between M and $\tilde{u}$ is *negative* (“poor health goes together with a lot of medicine”).

These considerations show that not only must discrepancies between MRT and MRS be expected when observing individual behavior, but also that these discrepancies *feed back on* the production of health. Causality not only runs from M and t^H to $(1 - \pi^*)$ but also from the discrepancies $[h - (1 - \pi^*)]$ back to M and t^H , causing them to be endogenous rather than exogenous regressors. In order to avoid biases in estimating the production function of health, two possible tracks can be pursued;

1. *Purging of the regressors* M and t^H from their error components due to endogeneity, i.e., using a two-stage estimation method (see Table 4.3b).
2. *Purging the error term* of its component $\tilde{u} = [h - (1 - \pi^*)]$ that is correlated with the regressors. This procedure obviously requires a measure of “target health status” $(1 - \pi^*)$, which is independent of “actual health status” h .⁸

Fortunately, the quality of health status measurements has been improving. Therefore, the second method will be analyzed more closely here. Thus an equation of the form

$$h = c + \alpha M + \beta t^H + (u + \tilde{u}), \quad \text{with} \quad \tilde{u} = [h - (1 - \pi^*)] \quad (4.7)$$

is to be estimated, the component $\tilde{u}$ of the error term being correlated with the regressors M and t^H . If measurements of $\tilde{u}$ are available, however, $\tilde{u}$ can be separated from the error term and treated as an additional regressor, such that

$$h = c + \alpha M + \beta t^H + \gamma [h - (1 - \pi^*)] + u, \quad \text{with} \quad \gamma = 1 \quad (4.8)$$

This equation may be solved for the input factor M ,

$$M = -\frac{c}{\alpha} - \frac{\beta}{\alpha} t^H - \frac{1}{\alpha} [h - (1 - \pi^*)] + \frac{1}{\alpha} h - \frac{1}{\alpha} u \quad (4.9)$$

Estimation of this equation leads to different results, depending on what health status measurement actually means.

8. As soon as the individual only has an influence on the probability of occurrence of various states, optimal target health status is defined by this very probability; see Section 2.4.2 or Figure 3–3. Therefore, $(1 - \pi^*)$ equals target and h actual health status, respectively.

- If one succeeds in measuring *target* health status $(1 - \pi^*)$, having the term $(1/\alpha)(1 - \pi^*)$ in the regression equation is sufficient because actual health status h can be dropped from Eq. (4.5). As $\alpha > 0$, the correlation of medical services and the state of health should be *positive*, as predicted by the production function for health.
- If respondents mark their health as “good” when it was *better than expected* at the time of measurement, the estimated Eq. (4.5) contains the term $(-1/\alpha)[h - (1 - \pi^*)]$. There will thus be a *negative* estimated correlation of the measured state of health and medical services in the sample.⁹
- If the survey is limited to measuring *actual* health status, then the term $(1/\alpha)h$ enters Eq. (4.5) as an explanatory variable, causing the correlation of medical services and health to be positive. However, $(-1/\alpha)[h - (1 - \pi^*)]$ is added to the error term in this case, resulting in Eq. (4.4) again, which means that α ($1/\alpha$, respectively) cannot be estimated without bias.

These different possibilities will now be illustrated using an estimate based on Swiss data.

4.4.2 Demand for Medical Services as a Derived Demand

If it is in fact possible to record the target health status in the sense of the probability $(1 - \pi^*)$ of being healthy, then according to Eq. (4.5) demand for medical services should depend positively on $(1 - \pi^*)$. “The greater the desired output, the greater the required input.” The use of medical services may be interpreted as a derived demand in this case, as stated in Table 3.2 of Subsection 3.2.2.2.

In their study of the state of health of the Swiss population, Leu and Doppmann (1986) undertook considerable efforts to record the demand for health, defining the *demand for health as a latent variable* in a so-called MIMIC (multiple indicators multiple causes) model [see Jöreskog (1973), Goldberger (1974)]. This definition squares well with the variable $(1 - \pi^*)$, since a probability is a latent variable, too. The demand for health is related to income (see Figure 4.1), which is also theoretically well founded, as the location of the optimum point depends on income (see Figure 3–3). At the same time, improved health contributes to an increase in income, as evidenced in the long-term tradeoff between “consumption” and time spent in good health of Section 3.5.1. The individual’s own health-enhancing efforts t^H are presented by six indicators of lifestyle (ranging from tobacco consumption to physical activity). In addition, environmental impacts such as noise and dust as well as personal life events (loss of close relatives and friends, moving house) are considered.

Four health indicators, presenting the statistical aggregate of several subindicators gathered through the survey, reflect target health status $(1 - \pi^*)$. As a reference indicator [assumed to vary 1:1 with the latent variable, apart from measurement errors (symbolized by vectors $\underline{\epsilon}$ in Figure 4–1)], the *self-assessed general state of health* is used. At this point already, it should be noted that self-assessed health may well derive

9. To obtain unbiased parameter estimates, actual health status should appear as an additional regressor. If this is added to the error term u_i , the combined error term $(1/\alpha)(h + u)$ is again correlated with the explanatory variables M and t^H .

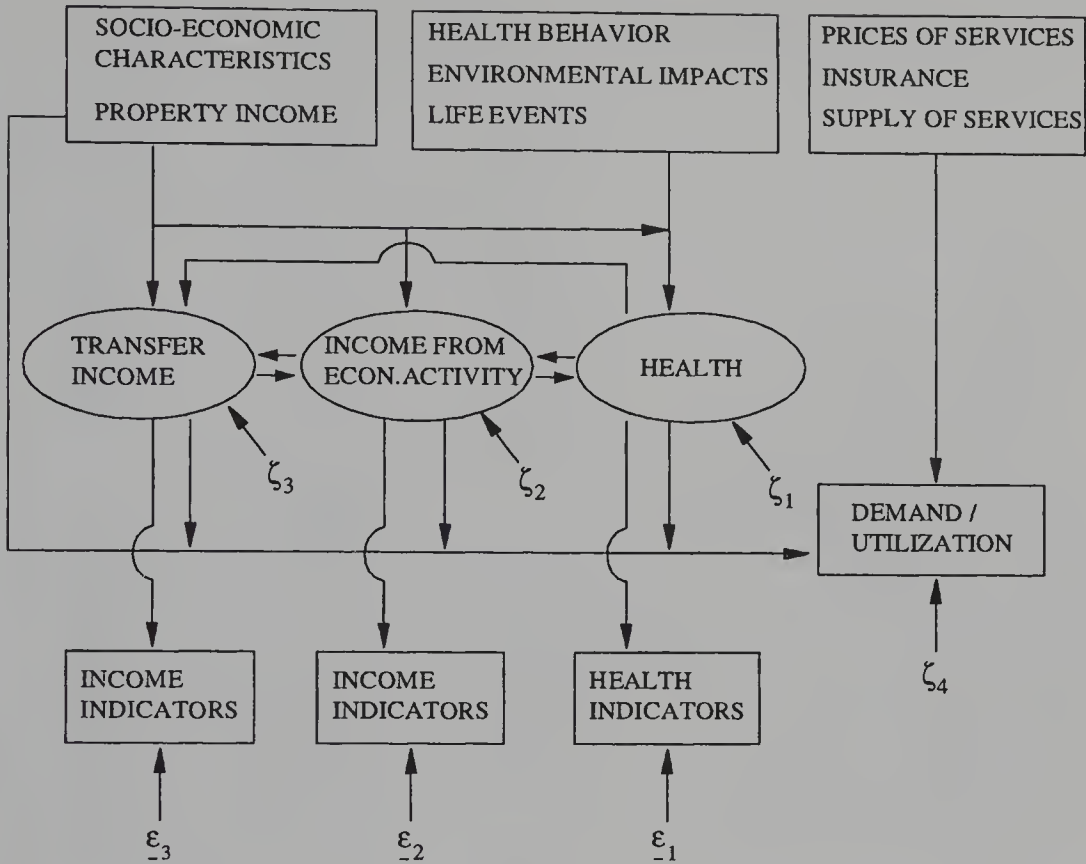


Figure 4-1. The state of health as a latent endogenous determinant of the demand for medical services.

from an implicit comparison between actual health status h and target health status $(1 - \pi^*)$. The three remaining indicators stand for the physical, psychic, and social components of health status.

The model shown in Figure 4-1 is estimated simultaneously for 3,155 adults. In this way a one-dimensional health index can be assigned to each person sampled, which is calculated from three regressions of the type of Eq. (4.3). This equation represents a production function solved for medical inputs in the guise of Eq. (4.5). Here, the inputs M are the number of consultations, number of days spent in hospital, as well as the number of days spent at a health resort in 1980 (see Table 4.10). Explanatory variables portraying individual health-enhancing efforts t^H and their productivity (age, education, family size) are complemented by others representing the relative cost of these efforts and of the utilization of medical services M (treatment time price, distance to hospital, insurance status). These additional regressors serve to recall the fact that the connection to be estimated is part of an optimization problem, at least as long as health-dependent states are not or cannot be distinguished.¹⁰

10. In the (fairly extreme) model of Section 3.4, the short-term problem of optimally combining medical services and individual efforts does not present itself, since in the sick state only the first factor and in the healthy state only the second contributes to the increase in probability $(1 - \pi)$ of spending the subsequent period in a state of good health.

Table 4.10: Health Production in Switzerland, 1980 (3,155 Adults)

	<i>Ambulatory care visits (ln)</i>	<i>Days in hospital (ln)</i>	<i>Days at a health resort (ln)</i>
1. Health ^a	-0.675***	-0.140***	-0.114***
2. Labor income ^a (ln)	0.002	0.006	-0.005
3. Transfer income ^a (ln)	0.007	0.012	-0.001
4. Property income ^b ($\times 10^4$)	0.022*	0.018*	0.002
5. Health insurance (=1 : yes)	0.525***	0.014	0.119*
6. Access time price ($\times 10^{-3}$)	-0.006**	—	—
7. Treatment time price ($\times 10^{-3}$)	0.034**	—	—
8. Distance from house to hospital	—	0.002	—
9. Shadow price of time	—	-0.000	0.000
10. Sex (=1 : male)	-0.221***	-0.047*	0.056***
11. Age	-0.009***	-0.001	0.002*
12. Education	0.007	0.003	0.003
13. Family size	-0.023*	0.004	-0.006
14. Physician density	0.183***	-0.048	-0.002
15. Bed density	-0.100	0.011*	-0.004
16. Ambulatory care visits (ln)	—	0.163***	0.017
17. Days in hospital (ln)	—	—	0.105***
Constant	1.603***	0.131	-0.110
R^2	0.27	0.11	0.08

*(**, ***) Coefficient different from zero with an error probability of 5 percent (2.5 percent, 1 percent).

^a Latent variable.

^b Externally estimated permanent income component.

Source: Leu and Doppmann (1986).

The results of estimation, shown in Table 4.10, give rise to the following comments.

- **Role of health status.** The use of all three categories of medical services is negatively correlated with latent health status (see the significant negative coefficients of variable No. 1). As can be gleaned from Table 3.2 of Subsection 3.2.2.2 (columns 3 and 4), this contradicts the Grossman model, a finding that has been replicated in a later study by Leu and Gerfin (1992). Under the alternative proposed in Section 3.4, the argument developed in the previous subsection applies, suggesting that the authors, contrary to their intention, had failed to record target health status ($1 - \pi^*$). Rather, their surveyed indicator, “self-assessed health” seems to reflect *deviations from target health status* [$h - (1 - \pi^*)$]. As Eq. (4.5) shows, actual health status has to appear in this case as an explanatory variable in the regression in order to avoid bias in the estimation of $(1/\alpha)$. A corresponding variable is, however, missing in Table 4.10,

implying that the value of α and hence of the elasticity of health with regard to M , given below $[\eta(H, M)]$, must be a biased estimate. This elasticity is related to α in the following way:

$$\begin{aligned}\eta(H, M) &:= \frac{\partial \ln H}{\partial \ln M} = \frac{\partial H}{\partial M} \frac{M}{H} = \alpha \frac{M}{H} \\ &:= \frac{\partial \ln H}{\partial \ln M} = \frac{\partial H}{\partial \ln M} \frac{\partial \ln H}{\partial H} = \frac{1}{b} \frac{1}{H}\end{aligned}\quad (4.10)$$

where b is the regression coefficient of H in an equation such as (4.5), but with $\ln M$ (instead of M) as the dependent variable, as in Table 4.10.

By equating target health status with actual health status for the sample average, assuming a mean value of $(1 - \pi^*) = 0.95$, we may retrieve the values of the elasticities $\hat{\eta}(H, M) = \{1.56; 7.52; 9.23\}^{11}$ from the three coefficients of variable No. 1. While these values probably overestimate the impact of medical care on health status, their ordering is plausible, suggesting that 10 percent more ambulatory care visits *have a smaller influence* on health than 10 percent more hospital or health resort days. A comparison with the (much lower) elasticity estimates in Table 4.2 of Subsection 4.2.2.1 is striking. Evidently it makes a difference whether health status is measured by mortality rates or by special indicators (see also item No.1 of Section 4.1).

- **Influence of age.** Contrary to widespread opinion (and the prediction implied by the Grossman model; see Table 3.2, columns 3 and 4 again), age does not necessarily go along with an increased demand for medical services. As soon as the deviation from target health status is taken into account as a *separate* explanatory variable, only the number of days at a health resort increases significantly with age among the three equations of Table 4.10. Furthermore, the elasticity with respect to age is very small. With A denoting age, the coefficient of regressor No. 11 appears in the elasticity formula as follows,

$$\eta(M, A) = \frac{\partial \ln M}{\partial \ln A} = \frac{\partial \ln M}{\partial A} \frac{\partial A}{\partial \ln A} = a \left(\frac{1}{A} \right)^{-1} = aA \quad (4.11)$$

where a is the regression coefficient of A in an equation such as (4.5), but with $\ln M$ (instead of M) as the dependent variable, as in Table 4.10.

With an average age of $\bar{A} = 45$ years, the elasticities belonging to variable No. 11 merely amount to $\hat{\eta}(M, A) = \{-0, 405; -0, 045; 0,09\}$. Similar results in Newhouse and Phelps (1976) suggest the conclusion that it is not age that influences the demand for medical services but the *deviation from target health status*. The age structure of the population may therefore have a rather limited

11. The first value is calculated from Eq. (4.6) and Table 4.10 as follows, $\hat{\eta}(H, M) = (1/0.675)(1/0.95) = 1.56$.

impact on the future development of health care expenditure (see item No. 4 of Section 4.1 and Section 11.4).

- ***Influence of opportunity cost.*** If a sick person is insured, he can purchase the means promising recovery at little opportunity cost, that is, at a small sacrifice of consumption [see Eq. (3.22)]. Insurance coverage thus is expected to encourage the use of medical services. The positive coefficients of variable No. 5 (existence of insurance cover) in the equations for ambulatory care visits and health resort days confirm this expectation. In contrast, it is *not possible to prove an influence of insurance coverage on days spent in hospital*. One explanation would be that the (countervailing) influence of the physician on utilization of medical services is particularly strong in the hospital (see Chapter 7).

Part of the opportunity cost are costs of time, where the distinction is made between access time (i.e. time spent traveling and waiting) and treatment time (variable Nos. 6 and 7 of Table 4.10). However, time does not have an immediate price in the guise of a wage rate in the sick state, at least according to the model alternative presented in Section 3.4 [see Eq (3.20)]. The lack of statistical significance of the two variables and particularly of the shadow price of time (variable No. 9) in the equations for health resort and hospital days is therefore not surprising. The time costs of treatment has an ambivalent meaning, reflecting not only cost to the patient but also the amount of care devoted by the physician and hence quality of treatment from the patient's point of view (see the positive coefficients of variable No. 7).

- ***Indication of supplier-induced demand.*** Table 4.10 displays statistically significant positive relationships between physician density (variable No. 14) and the number of visits, and between hospital beds (variable No. 15) and the number of days in hospital at a disaggregated level. Individuals living in areas with a high density of supply thus demand more medical services. Here, this effect *cannot* be explained by pointing out that an increased density of supply results in reduced traveling and waiting costs, since these *time costs* have already been taken into account by variable Nos. 6 and 7 (access time price, treatment time price). Thus it may in fact be true that Swiss physicians induce additional demand for their services in response to an increased density of supply (see Section 7.2 for this so-called supplier-induced demand).

The share of explained variance in total variance (as indicated in Table 4.10) is rather low, attaining 27 percent at a maximum. Other studies using individual data do not come off much better in this respect, however. The two following reasons may be cited in particular.

1. ***Situation-dependent possibilities in the production of health.*** This argument is based on the difference between the tradeoffs according to health status (see Section 3.5). An analysis based on annual data averaging periods of health and sickness thus suffers from a problem of aggregation. Depending on the mix of various tradeoffs occurring during the observation period, the marginal productivity of medical services varies at the individual level. Hence, the estimated relationships between the demand for medical services and their

economic determinants may be marked by considerable parameter instability, contributing to a low coefficient of determination.

2. **Delegation of decision-making power.** In the event of illness, many aspects of treatment are decided by the physician rather than by the patient. This delegation of authority goes farther in hospital care than in outpatient care. The low coefficient of determination for the number of days in hospital (see Table 4.10) may reflect the difficulty of explaining what is going on in the hospital by using only variables that characterize the patient rather than also the physician in charge.

The knowledge gained from the studies based on individual data can be summarized in

CONCLUSION 4.6: The concept of a situation-dependent production correspondence, with consumption services and health care as outputs, also appears to be valid when tested against observations at the individual level. The demand for medical services not only depends on factors associated with “need” but also on factors reflecting productivity and costs of individual health-enhancing efforts. State dependence is reflected by the finding that these factors contribute little to explaining the utilization of medical services in a clear case of sickness.

Since, in the sick state, the treatment process is governed by the individual in a few aspects only (choice of doctor, preference for drug, termination of treatment), the physician obtains leeway to pursue his own objectives.

This consideration calls for an investigation of physician decision-making, which is undertaken in Chapter 7. In a sense, these partial analyses must remain unsatisfactory, however, since they fail to fully portray the interaction between the physician and his patient. Through comparing and combining empirically tested implications of such partial models, it may still be possible approximately to predict the effects of particular health policies.

4.5 EXERCISES

Exercise 4.1:

One criticism of Brenner’s (1983) equation (see the discussion of Table 4.6) states that it should include regressors such as UN_t , ΔY_t , and UN_t^2 , the dependent variable being the mortality rate of the total (active) population rather than that of the unemployed. This would imply that the connection between the mortality rate and the rate of unemployment does not remain constant but varies over time. Prove these statements by formulating two simplified estimated equations [one for S'' (mortality rate of the unemployed) and one for S' (mortality rate of the employed)] using the regressor Nos. 1–3 of Table 4.6 for each one. Then form their linear combination, as mentioned in the text.

Exercise 4.2:

The comment on the study by Mullahy and Portney (1990) says that more than 16 cigarettes a day of additional tobacco consumption is damaging to health, whereas beyond 0.06 micrograms/m³, an increased ozone concentration has no negative effect. (a) Try to explain how these results come about, using the coefficients of Table 4.9 (part A). Draw the graph of the partial functions $P = f(\text{CIGARETTES})$ and $P = g(\text{OZONE})$, with P the probability of a phase of restricted activity. (b) Write down the estimated equation underlying Table 4.9 and determine by means of partial differentiation the algebraic value of *CIGARETTES* where additional consumption turns from a health-enhancing into a health-damaging effect. (c) Insert the GMM-estimated coefficients of Table 4.9 and calculate this extreme point algebraically, taking into account that *CIGARETTES* is the effective number divided by 100. Determine the extreme point in the case of *OZONE* in analogous manner.

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5

PECULIARITIES OF HEALTH GOODS AND THEIR ALLOCATIVE IMPLICATIONS

5.1 THE ISSUE

Considerable deviations from the principles of a market economy can be found in the allocation of health goods, particularly of medical services. This is evident even in the case of Western industrialized countries, which are otherwise organized according to market principles. In general, neither the decision to offer a medical service (e.g., an appendectomy) nor to demand it is made by sovereign individuals or firms that are confronted with the full financial consequences of their choices, nor are the resulting individual economic plans coordinated by the price mechanism. The decision process is thus completely different from that characterizing the production and purchase of refrigerators, for example.

In particular, the United Kingdom and Italy have a national health service with permanently employed physicians who provide their services to patients free of charge, being financed completely from general tax revenue. In other countries, all or at least the majority of citizens are subject to *compulsory health insurance*, with the choice of insurer often being limited. In addition, the benefits of the plans are laid down by law in some countries, while the prices of medical services are regulated by fee schedules enforced by public authorities.

These deviations from the market system are generally justified by claiming that health goods present particular features rendering them different from other goods. These features are said to cause a “*market failure*,” that is, to result in an equilibrium that does not correspond to a Pareto-optimal allocation in nonregulated markets. In addition, the violation of generally acknowledged criteria of equity is adduced as a cause for rejecting the market mechanism.

The point of departure for the theory of market failure is provided by the *First Theorem of Welfare Economics*. This proposition states that in the absence of external effects and public goods, every competitive equilibrium (i.e., an allocation in which every consumer maximizes utility and every producer maximizes profit, and the resulting individual plans are compatible) constitutes a Pareto optimum. The significance of a market failure therefore first of all demands proof that at least one of the requirements cited in this main proposition of economics is not fulfilled in the case of health goods, notably because one of the following conditions fails to be met.

- Health goods have the characteristics of *public goods*;
- The consumption of health goods gives rise to *external effects*;
- The production of health goods is characterized by *increasing returns to scale*;
- The prerequisites of a perfectly competitive market, that is, *market transparency* and *consumer sovereignty*, are not met.

In the following, we will recap the most important peculiarities of health goods cited in the literature, investigating whether they might constitute one of the four reasons for market failure listed above. Should this be the case, it is appropriate to examine the suitability of *alternative* social institutions (e.g., public provision of goods, compulsory insurance) for securing a higher degree of (Pareto) efficiency than the market would provide. In this context, however, it must always be recognized that health goods are not homogeneous in themselves, making it possible that the market presents a suitable form of allocation in one instance but may fail in another.

Another distinction between possible types of market failure will turn out to be useful:

1. Failure in the market for *medical services themselves*, providing reasons for their public provision;
2. Failure in *private insurance markets* providing coverage against the financial risks associated with illness.

We should emphasize here that our treatment will be concerned only with static efficiency, which means that we take existing health care technology as given. Thus we exclude the more difficult questions of dynamic efficiency, such as, the impact of the health care financing and delivery system on the development of new technology. Section 5.2 is devoted to a discussion of the peculiar features of health goods that may cause failure in the market for these goods. In Section 5.3, characteristics of private markets for health insurance suggesting market failure will be dealt with. Section 5.4 contains a formal statement of the risk selection problem and shows that compulsory health insurance may be Pareto-improving under certain conditions. Whilst efficiency criteria for the evaluation of market outcomes will be presented for both cases, criteria of equality and their implications for the regulation of markets both for health goods and health insurance will be reviewed in Section 5.5. In Section 5.6, these results will be used to evaluate systems of social health insurance as they exist today.

5.2 MARKET FAILURE IN MARKETS FOR HEALTH GOODS

5.2.1 Externalities, Public Goods, and Increasing Returns to Scale

If the consumption of a good h by individual i has direct effects on the utility of individual j , giving rise to an externality, then the market mechanism does not lead to a Pareto-optimal allocation in a situation of perfect competition. The reason is that in this case, consumer i , in equilibrium, buys a quantity of good h such that his own marginal utility (measured in monetary units) derived from the last unit of the good is equal to its price and hence its marginal cost of production. In comparison, if h is a public good, then the Pareto optimum is characterized by the summation of marginal utilities of individuals j and i , with this *sum* equal to marginal cost.

Positive externalities result in an increase of j 's marginal utility due to i 's consumption of h . They therefore are as a rule associated with underconsumption of the good, while negative externalities are associated with overconsumption in a market economy, relative to the Pareto-optimal solution. However, nothing has been said so far concerning the existence of institutions that help to reach such a solution.

In the case of health goods, *positive externalities* are rather more relevant, and indeed the consumption of health good h by the individual i can either

1. Directly improve the health of individual j [Culyer (1971) calls this a "physical externality"]; or
2. Simply lead to increased satisfaction for individual j ("psychological externality").

(1) *Physical externality*. Physical externalities arise through the treatment of or prevention taken against communicable diseases in individual i , serving to reduce the probability of j 's contracting this disease. In a two-person case, with j the only one, apart from i himself, benefiting from i 's consumption (e.g., a vaccination), a Pareto optimum could easily be brought about by j voluntarily subsidizing the vaccination costs of i . In reality, external advantages are of course spread over many people, giving rise to the *collective-good property* of external effects, an additional complication that can be explained as follows.

A "collective good" (also called "public good") is characterized by *nonrivalry* in consumption. If a consumer i obtains a unit of good h , then another consumer j can also use the same unit without diminishing i 's enjoyment of the good. This condition is satisfied in an ideal way in the case of the external advantage created by the vaccination (good h) of individual i , since this affects several other individuals and yet does not diminish if their number increases.

Moreover, the *exclusion principle* cannot as a rule be applied to collective goods, meaning that no one can be barred from using them, even though he has not contributed to their provision. In a perfectly competitive market, there again will be underprovision of these goods. The reason is that the individual consumer has no incentive to contribute to the financing of such a good (in the present case to participate in subsidizing the vaccination of individual i), being able to benefit from the advantages of a reduced risk of infection while others take over the financing.

These considerations suggest that mass vaccinations to prevent communicable diseases (such as polio or dangerous flu viruses) or measures designed to prevent epidemics (typhus, cholera), provided by the state and financed from general public revenue, can lead to a Pareto improvement. The appropriate level of implementation (community, state, or federal level) should be decided depending on the spatial distribution of the external benefit.

However, in view of the low proportion of total health care expenditure attributable to infectious disease, this line of argument cannot be used to justify *free provision of all medical care* by the state in present times. Apart from this, not all citizens are exposed to the danger of infectious disease to the same extent. The positive externality of fighting an epidemic vanishes, for example, when the individual—as in the case of AIDS—can effectively protect himself from infection at a sufficiently low cost. Conversely, if someone who is not himself at risk is forced to contribute to the costs of public control of an epidemic through his tax payment, then public provision no longer leads to a Pareto improvement relative to the market solution.

(2) **Psychological external advantages.** For these, the term *altruism* is also used. The utility level of the altruistic individual j not only depends on his own consumption, but also (again positively) on the consumption of individual i . As a rule, a positive externality arises only as long as the consumption of a fellow citizen is perceived as being “unbearably” low. Indeed, it causes most people worry to see others die from starvation or lack of medical care, particularly if their distress was not caused by their own actions.

The existence of altruistic attitudes in society now raises the question whether the corresponding positive externalities

1. Can be internalized through private action or if some *public intervention* is required for attaining a Pareto improvement;
2. Require subsidization or even free provision of certain goods (e.g., medical treatment), or whether they can be settled using money transfers, that is, whether “*transfers in kind*” are considered superior to “*transfers in cash*” according to welfare criteria.

Collective-good properties, as in the case of infectious diseases, again speak in favor of having the government organize relief. This is because in a society with many wealthy members, everyone profits when someone other than individual i supports the poor, and in the absence of organization by the government, this situation would result in underprovision of relief for the needy.

At first sight, free provision in the guise of transfers in kind seems to be inferior to transfers in cash because transfers in kind distort relative prices, inducing the needy to consume the subsidized (here medical) goods in excess of a Pareto optimum. This argument, however, overlooks the specific effects of recipient’s consumption on the donor. If the donor is not interested in increasing the recipient’s utility but rather his consumption of a particular good,¹ then his willingness to make the transfer increases if he can *earmark* it. The increased size of the transfer may serve to raise the utility of

1. This case is referred to as a “good externality,” distinct from a “utility externality.”

the recipient above the maximum level attainable with the (smaller) income transfer, therefore resulting in a Pareto-superior outcome.

This argument seems to apply primarily to nutrition, appropriate housing, and basic medical care, the consumption of which is deemed worthy of support by donors, calling for earmarked transfers in their interest. In the case of free supply of basic medical care for those particularly in need, it is, however, in no way implied that this supply necessarily has to be tax financed. In Section 5.6, suitable institutions for the financing of medical care will be discussed. The peculiarities of health goods as expounded here can be used even less to justify the organization of supply by the state, as in a national health service. This is because the “market failure” associated with external effects and collective goods affects the demand side exclusively, and not the supply side of the market for health goods.

Finally it should be mentioned that there is no reason to conclude that the production of health services is characterized by *increasing returns to scale* in the usual range of activity. While this may be the case for some services (such as laboratory tests) at low levels, economies of scale will not persist forever, since unlike the case of electricity or railroad services, networks do not play a major role in health service production.

5.2.2 Option-Good Property of Medical Services

Another peculiarity of most medical services is that the demand for them cannot be predicted by the individual. However, when it arises, it may often be of extreme urgency. On the other hand, capacity for providing medical services, particularly in the hospital, cannot be made available at such short notice. The existence of a certain reserve capacity therefore constitutes an option good, meaning that the *mere existence* of the good yields utility to the consumer. In order to give hospitals an incentive to hold reserve capacity, this must be paid for separately, implying that the hospital’s revenue must not depend on bed occupancy only.

Does this imply the necessity of *provision by the government*? The answer is in the negative, since option demand can be completely met within the domain of (private) health insurance. The insurer simply guarantees that a hospital bed will be available for the insured in case of need. Part of the premium revenue is then used to pay a hospital in the insured person’s area of residence for holding reserve capacity. Such a mission has actually been legislated for the sick funds in Germany, but it could also be secured through private contract.²

Only to the extent that the option in its turn has the character of a collective good would the financing of reserve hospital capacity through taxes be appropriate. In fact, the condition of nonrivalry is fulfilled, since one and the same hospital bed can provide several potential patients with the utility of secure access at the same time. However, it needs to be investigated whether or not the *exclusion principle* holds, which requires

2. It may be argued that a hospital may contract with several insurers, conjuring up the risk of insufficient capacity as soon as several insureds wish to exercise their option (e.g., in the wake of an epidemic). While this is true, this still does not justify public ownership of hospitals because the public hospital is confronted with exactly the same problem.

that, in the case of a shortage, a person who has not negotiated an appropriate insurance contract must be discriminated against. Since in an emergency this principle would presumably be violated for humanitarian reasons, it is justified to force all citizens into financing the option through taxes.

CONCLUSION 5.1: The market “fails” in its allocation of health goods inasmuch as these goods exhibit collective-good properties (vaccinations, hospital beds held in reserve) or give rise to goods externalities. In all these cases, suitable institutions, if necessary organized by the government, need to be found for financing. However, market failure in no way implies the necessity of having health services provided by the government.

5.2.3 Reasons for a Lack of Consumer Sovereignty

Apart from externalities and the collective goods properties, another frequently cited reason for the “failure” of free markets for health goods is seen in the consumer’s inability of deciding about his demand for medical care rationally, that is, in a utility-maximizing way. The most widespread arguments adduced to support this allegation are discussed in the following.

5.2.3.1 Incapability of Rational Decision Making

The state of illness presents an exceptional human situation, in which, in the most extreme case, even life is at stake. It is therefore doubted by many whether a person in this situation fits the role of the “sovereign consumer,” who, among the alternatives offered to him, seeks out the one maximizing his utility, taking into account the associated costs in a rational balancing process. Abstracting from the problem of evaluating quality for the time being (see Section 5.2.4), then at least three levels can be distinguished when examining the capability of rational decision making.

1. Perfect incapability of rational decision making. This occurs, for example, in a state of unconsciousness or mental illness. In this situation, however, medical services are not special, since the affected person cannot make a rational decision with regard to any good, which means that somebody must decide for him. Now the question arises of who could be trusted to act in the way the affected individual would if capable to decide. If anyone, then a close relative is the person most likely to meet this requirement, much sooner than a government agent [see Zweifel (1994) for the extreme case of psychic illness].

2. Limited capability of rational decision making. This occurs in illnesses that are life-threatening but do not limit mental capacity. Here, the patient is indeed in a weak position relative to the supplier of health services, since he will frequently be prepared to pay any amount of money up to his total wealth in order to recover his health. He will therefore hardly seek out less costly treatment offers as soon as they (seemingly) entail a greater risk of failure. On the other hand, the purchase of *health insurance* covering catastrophic costs of treatment can take care of such a situation. While

insurance serves to weaken the patient's interest in low-cost treatment alternatives, it does remedy his structural inferiority to the provider of care.

3. General capability of rational decision making. This occurs in illnesses that are not life-threatening, that is, in the greatest number of sickness episodes. Since survival is not at stake, the capability of a rational decision is not really limited, and the above-mentioned doubts lose their justification.

Consequently, none of the three cases seems to present sufficient reason to presume that a form of allocation other than the competitive market results in a higher degree of efficiency. Either the individual himself is perfectly capable of making a rational decision concerning his demand for medical care (levels 2 and 3), or someone close to him must make the decision for him (level 1). Public bureaucrats, who cannot judge his preferences so well, would presumably deal with this task in a less satisfactory way. The reasons mentioned provide even less support for having the government organize the supply of medical services.

5.2.3.2 *Excess Discounting of Future Wants*

By nature, illness is a stochastic event, the frequency and severity of which increases with the age of a person, causing the likelihood of treatment costs that cannot be paid out of current income to increase as well. A rational and risk-averse individual will, in the absence of public intervention, provide for this risk by saving, or better still, by buying insurance. However, since most people estimate future consumption as being less valuable than present consumption, many will only attach limited importance to this eventuality and fail to do a sufficient amount of saving. Consequently, means will be lacking in old age, resulting in—so the argument goes—a systematic underprovision with health goods.

This argument has a *paternalistic touch*, since evidently the preferences of the individual with their excessive discounting of future wants are not acknowledged as his “true” preferences. But it is not clear who should have the right to decide what these true preferences are. The argument outlined also fails to point out alternative institutional arrangements that may be expected to produce a “better” outcome.

Let us assume that politicians were allowed to decide the appropriate amount of saving (e.g., as part of compulsory capital-funded health insurance), and that they periodically have to put themselves up for election by the citizens of a democracy. From this process, a larger amount of future provision could only be expected to result if a corresponding platform was supported by the majority of voters. Younger voters, however, would have to disagree unless they behave schizophrenically in this respect. Acting as voters, they would have to discount future wants to a lower degree than when acting as consumers on the market. Older voters, on the other hand, while considering such a provision for the future necessary in view of their own (negative) experience, would not benefit from such a savings program anymore. Therefore, they would also have insufficient motivation to support it.

For these reasons, the “irrationality” of the consumer does not provide cogent justification for public intervention in the market for health goods.

5.2.4 Lack of Perfect Information in Markets for Health Goods

Perfect market transparency constitutes another requirement for the first main proposition of welfare economics (as stated in Section 5.1) to hold, which does not seem to be fully satisfied in the case of health goods. Market transparency means that potential buyers are informed about qualities offered and prices quoted by all suppliers on the market. Perfect knowledge of product quality is generally precluded in the case of services, where production and consumption occur simultaneously (“*uno actu* principle”). The various offers cannot be inspected and compared before making the decision.

Medical services are not unique in this respect, however. Incomplete information about quality also characterizes the services of hairdressers, banks (investment advice), restaurants, and performances of artists. Nevertheless three additional features can be identified which distinguish health goods from most other services.

1. Lack of opportunity for sampling. The quality of the work of a hairdresser or of an innkeeper can be experienced through trial, and, within certain limits, by relying on the judgment of others who have already tried these services. In contrast, medical services, particularly the ones crucial for survival in the case of life-threatening illness, are consumed irregularly, causing a lack of experience for evaluation. Furthermore, experiences of others cannot be transferred easily, since on the one hand health problems are never really comparable and on the other the treatment process contains a strong individual physician–patient component. On the latter point, health goods also differ from durable goods such as washing machines, which are admittedly not bought on a regular basis, but for which an objective quality evaluation is possible (e.g., by consumer test institutes).

2. Impossibility of quality evaluation. Frequently, the quality of a medical service cannot be judged correctly even after utilization, since the causality between treatment and change of health status may be blurred by other biological processes, such as the self-healing powers of the body.

3. Special properties of information. In the case of diagnostic services in particular, the good demanded is information. This makes it impossible for the patient to evaluate the quality of the service prior to its purchase because this would require him already to be in possession of the desired information. The information advantage of the supplier of course gives him a certain degree of power over the buyer. Again, however, medical services are certainly not unique in this respect. One may think, for example, of the services of a car repair shop, which in most cases begins with diagnosing the failure and defining the repair needs. Lawyers likewise have an information advantage with regard to the client’s prospects of winning a trial.

For the reasons mentioned, specific interventions in health care markets are justified if they aim at securing a *minimum quality level*, such as licensure of physicians and other health professions.³ Furthermore, a strengthening of physician liability (malpractice

3. It should be pointed out, however, that once-and-for-all licensure, granted at the start of a professional career, provides much less quality assurance than would the requirement of relicensure every few years [see the striking argument of Benham (1991)].

suits) should serve to prevent sloppiness on the part of the service supplier, which is to be expected in view of the patient's incapability of evaluating service quality.

On the other hand, none of the features of health services mentioned above justifies the belief that an organization of supply by the *government* would result in better welfare than a coordination through the market. As several examples (e.g., the case of postal services) show, bureaucratically structured institutions, which inevitably come with government organization, are not very responsive to consumer wants. Finally, a high level of quality of services may well be ensured by competition rather than government regulation.

CONCLUSION 5.2: Another reason for “market failure” is seen in the consumer's incapability of making rational decisions about his demand for health services. However, this does not provide justification for public provision, but rather for public measures to improve the information of consumers and possibly to secure product quality.

5.3 MARKET FAILURE IN HEALTH INSURANCE

While the peculiarities of health goods outlined in Section 5.2 suggest that a few targeted policy measures may be welfare-enhancing, they do not justify either the monopolization of supply through a national health service nor the financing of all services through tax revenue. Consequently, it seems that health goods should be paid for by individual consumers. Since the costs of illness occur stochastically, households can hedge their net income against inevitable uncertainty by buying health insurance. The question to be answered in the following is whether private insurance markets may be expected to deal with this demand in an optimal way, or whether there is reason to fear market failure that could be remedied by public intervention, in particular compulsory health insurance.

5.3.1 Free Rider Problem

In Section 5.2.1, the theory of goods externalities was used to show that in a wealthy society, basic medical care should be made freely available to citizens who are needy through no fault of their own. A society accepting this argument, however, must define and impose criteria for a situation of need to be a “no fault” case. Since this is very difficult, low income and lack of wealth will have to serve as substitute criteria for claiming free treatment.

This use of substitute criteria can hardly be avoided for practical reasons. If, for example, the victim of an accident or a seriously ill person is rushed to hospital, it is not acceptable, particularly to a rich society, to make the provision of medical care dependent on the patient's ability to pay. In fact, a decision maker denying medical care for this reason risks to be sued for failure to give assistance. If subsequently the patient turns out to have no health insurance or to lack the means for paying for his treatment, then the bill has to be covered by transfers.

However, there is the danger that individuals who are close to the limit in terms of income and wealth *lose all incentive* to insure against the cost of illness. By making do without insurance, they save the premium, enjoying a higher level of consumption as long as they remain in good health. As soon as sizable payments for medical care occur, however, these individuals very quickly satisfy the criteria for free treatment. This means that *ex ante*, under the veil of uncertainty regarding their future demand for health goods, they can reach a higher level of expected utility without buying insurance.

If the group of people falling back on health care financed by society becomes too large, then this system is turned away from its original purpose of helping those who cannot help themselves. In order to protect the system from exploitation by “free riders,” the principle of free treatment for the needy may be replaced by that of *compulsory insurance*. Alternatively, a sufficiently large subsidy of insurance premiums for the needy would serve the same purpose.⁴

Through both measures, individuals characterized by a strong discounting of future wants (and hence little wealth formation) are led to contribute to the future cost of health care themselves. Contrary to the justification of compulsory insurance given in Subsection 5.2.3.2, this does not occur with the objective of doing these people a favor but to protect the rest of society from being exploited.⁵ At the same time, the designated recipients of public transfers could also be helped in a system of compulsory insurance by charging them a reduced insurance contribution or no contribution at all. This leads to

CONCLUSION 5.3: Compulsory health insurance and/or subsidization of insurance premiums for the needy serve to protect charitable organizations in the health service from being exploited by “free riders,” and thus from being turned away from their actual purpose.

5.3.2 Asymmetric Information Concerning Health Risks

The introduction of a compulsory health insurance may result in an increase in welfare if the population is heterogeneous in terms of the risk to be insured and asymmetry of information is present. Here, asymmetry of information means that the individual knows his risk whereas the insurance company cannot observe it. Therefore, the company is constrained to offer insurance on the same terms to everyone.

The impact of asymmetric information on the working of markets was first expounded by Akerlof (1970) in his essay, “The Market for Lemons.” The quintessence

4. This conception was realized in Switzerland between 1911 and 1995. However, subsidies continued to be paid per enrollee, resulting in some 97 percent of the population being voluntary members of sick funds. This in turn paved the way to general compulsory insurance (with free choice of sick fund as a rule), effective from 1996.
5. This should not be taken to say that the proponents of compulsory social health insurance saw this as their primary objective. Rather, protecting society from free riders can be adduced as an economic justification for compulsory insurance.

of this famous essay is that an information advantage of the seller over the buyer of a good with nonuniform quality causes bad quality to chase good quality from the market (“*adverse selection*”). In the following it will be shown that an analogous conclusion holds for the market for insurance contracts.

Let us assume that the population is heterogeneous in terms of the risk of becoming ill π^j as well as in terms of the cost of treatment in case of illness, L^j , and that every individual j knows his pair of values. The insurer, however, *lacks discriminatory power* to assign the appropriate values of risk parameters to applicants. He will therefore only offer a standard policy, which is characterized, for example, by a partial payment of claims in case of illness, r ($0 < r \leq 1$) and by a premium P . The values of r and P must be set such that expected payments for claims of the insurance company are at least covered by the premium. Apart from costs of administration, this expected value depends on the probability of illness and the cost of treatment (with a given rate of reimbursement r) of those who actually purchase the contract offered.

Let us presume that the insurance company only writes an insurance contract with full coverage ($r = 1$), which it believes to be selling to a representative sample of the population. This means that the premium P is calculated on the basis of expected values of π and L . Such a policy will be attractive to all “bad risks,” which are those individuals with a high value of π^j and L^j , and, given risk aversion,⁶ also for average risks. In contrast, very good risks, that is, individuals with values of π^j and L^j well below average, will *avoid* buying such a policy (unless they are extremely risk-averse), since the ratio between premium and expected costs of illness is very unfavorable to them. On the other hand, because of the reasons mentioned (lack of discriminatory power on the part of the insurance company), a more favorable premium cannot be offered to this group because all other insureds would also demand these better terms.

Since the best risks drop out, the basis upon which the insurance company calculated its risk premium changes, causing the premium to rise. Thus the offered policy becomes unattractive for another group of individuals with relatively low values of π^j and L^j , who drop out in their turn, so that the premium required to cover costs rises even more. This process, called *adverse selection*, continues until in equilibrium, only the worst risks remain in the insurance pool.

As a consequence, the good risks, remaining outside the pool, have no insurance protection, unless the company offers them an alternative contract, which is characterized by a more favorable relationship between benefits and premium but also by a greatly reduced degree of coverage (r clearly below 1). This limitation of coverage is necessary to keep the bad risks from buying such an alternative policy.

It is shown that, in the situation just described, the introduction of *compulsory insurance* with uniform premiums can potentially lead to a Pareto improvement for society. However, only part of the treatment cost should be covered ($r < 1$), with the remainder left to individually contracted complementary insurance.⁷ The

6. See Subsection 2.3.2.2.2 for a precise definition of risk aversion.

7. Even the combination of social and private insurance should optimally have a global $r < 1$ in many cases when moral hazard is introduced into the picture (see Section 6.5).

advantage for the bad risks lies in that their premium is reduced because of the pooling of risks, although they obtain the same coverage as when buying insurance from a private company. In turn, the good risks benefit because they now obtain more comprehensive coverage by combining compulsory and complementary insurance. They only have to pay for the higher costs of the bad risks in the compulsory segment.

CONCLUSION 5.4: Given “adverse selection,” that is, if risks of illness are heterogeneous and not observable to the insurer, then good risks cannot obtain comprehensive insurance cover at favorable conditions in a market equilibrium. In this case, the introduction of compulsory insurance may result in a Pareto improvement.

5.4 MARKETS FOR HEALTH INSURANCE WITH ASYMMETRIC INFORMATION: A FORMAL STATEMENT

5.4.1 Model Assumptions

Consider an extremely simplified model of health risks with only two possible states of nature for every individual. The state of sickness (s for short) entails a financial loss (treatment costs and loss of earnings) of L monetary units, while the state of good health (h for short) has income Y , which is the same for everyone.⁸

In addition, there are two populations A and B with differing risk of sickness π_A and π_B ($\pi_A < \pi_B$). The share μ of the total population belongs to group A , the share $(1 - \mu)$, to group B . Let y_i^k be the *net income* of an individual of group i ($i = A, B$) in state k ($k = s, h$) and $u(y_i^k)$ maximum attainable utility, dependent on net income. Ex ante expected utility for every member of group i is thus given by

$$EU_i = \pi_i u(y_i^s) + (1 - \pi_i) u(y_i^h), \quad i = A, B \quad (5.1)$$

The question to be addressed using this model is whether an efficient allocation of the financial risk associated with sickness can be achieved by means of individual contracts struck on the market or whether public intervention is necessary. Later on, we will see that the answer to this question very much depends on whether an individual's membership in group i is known to *everyone* (“public information”) or only to *himself* (“private information”). These two cases are dealt with in the two following sections. Although the case of public information cannot be considered realistic, it still serves as a benchmark.

An overview of the model is given in Table 5.1.

8. This model is from Eckstein, Eichenbaum, and Peled (1985).

Table 5.1: Model of an Insurance Market with Heterogenous Health Risks

$$EU_i = \pi_i u(y_i^s) + (1 - \pi_i)u(y_i^h), \quad i = A, B \quad (5.1)$$

$$\mu[\pi_A y_A^s + (1 - \pi_A)y_A^h] + (1 - \mu)[\pi_B y_B^s + (1 - \pi_B)y_B^h] = Y - [\mu\pi_A + (1 - \mu)\pi_B]L \quad (5.2)$$

Pooling equilibrium

$$P_A = \sigma^v I_A, \quad P_B = \sigma^v I_B \quad (5.9)$$

$$\sigma^v = \frac{\mu\pi_A I_A + (1 - \mu)\pi_B I_B}{\mu I_A + (1 - \mu)I_B} \quad (5.11)$$

Separating equilibrium

$$y_i^s = Y - L + I_i - P_i = Y - L + (1 - \sigma_i')I_i \quad (5.14)$$

$$y_i^h = Y - P_i = Y - \sigma_i' I_i, \quad i = A, B \quad (5.15)$$

$$\frac{\pi_i u'(y_i^s)}{u'(y_i^h)} = \sigma_i' = \pi_i, \quad i = A, B \quad (5.16)$$

y^h, y^s : Net income of the individual in the state of health and sickness, respectively

Y : Gross income per capita

L : Cost of medical treatment

π_i : Health risk of a member of group i ($i = A, B$)

μ : Share of good risks (A) in the population

$u(y)$: Utility of the individual

EU : Expected utility

I : Insurance benefits

P : Insurance premium

σ^v, σ^i : Unit price of insurance in the pooling and separating equilibrium, respectively

5.4.2 Heterogeneous Health Risks and Public Information

Assuming that in accordance with the law of large numbers, probabilities at the individual level converge to relative frequencies at the societal level, the per capita budget constraint for the society considered is given by

$$\begin{aligned} & \mu[\pi_A y_A^s + (1 - \pi_A) y_A^h] + (1 - \mu)[\pi_B y_B^s + (1 - \pi_B) y_B^h] \\ & = Y - [\mu \pi_A + (1 - \mu) \pi_B] L \end{aligned} \quad (5.2)$$

where Y stands for per capita income of the economy considered.

First of all, we shall define the terms *feasible allocation* and *Pareto-optimal allocation*.

DEFINITION 5.1: An allocation $\{(y_i^s, y_i^h), i = A, B\}$ is called feasible if Eq. (5.2) is valid. It is Pareto-optimal if there are no other feasible allocations $\{(\tilde{y}_i^s, \tilde{y}_i^h), i = A, B\}$ such that

$$EU_i(\tilde{y}_i^s, \tilde{y}_i^h) \geq EU_i(y_i^s, y_i^h), \quad i = A, B \quad (5.3)$$

for at least one i .

It can be shown that every Pareto-optimal allocation follows from the maximization of an arbitrarily weighted sum of the individual expected utilities

$$EU_A (y_A^s, y_A^h) + \delta EU_B (y_B^s, y_B^h), \quad \delta > 0$$

given condition (5.2). Using Eqs. (5.1) and (5.2), one may write the maximization problem as follows,

$$\begin{aligned} z := & EU_A (y_A^s, y_A^h) + \delta EU_B (y_B^s, y_B^h) \\ & - \lambda \{ \mu [\pi_A y_A^s + (1 - \pi_A) y_A^h] + (1 - \mu) [\pi_B y_B^s + (1 - \pi_B) y_B^h] \\ & - Y + [\mu \pi_A + (1 - \mu) \pi_B] L \} \end{aligned} \quad (5.4)$$

The first-order conditions for a Pareto optimum thus read,

$$\begin{aligned} \frac{\partial z}{\partial y_i^s} &= \pi_i u' (y_i^s) - \lambda \mu \pi_i = 0 \\ \frac{\partial z}{\partial y_i^h} &= (1 - \pi_i) u' (y_i^h) - \lambda \mu (1 - \pi_i) = 0 \end{aligned} \quad (5.5)$$

Division of these two conditions yields

$$\frac{u' (y_A^s)}{u' (y_A^h)} = \frac{u' (y_B^s)}{u' (y_B^h)} = 1 \quad (5.6)$$

This means that *ex post* marginal rates of substitution between income in the state of health and in the state of sickness are equal for the two individuals. Because of $\pi_A < \pi_B$, however, it immediately follows that in a Pareto optimum the *ex ante* marginal rates of substitution between incomes in the two states of health are *not* equal between the two individuals. This can be seen by totally differentiating Eq. (5.1) to obtain the formula of the indifference curve, $dEU_i = \pi_i u' (y_i^s) dy_i^s + (1 - \pi_i) u' (y_i^h) dy_i^h = 0$.

Solving to obtain the expression for $-dy_A^s/dy_A^h$, one gets the inequality

$$-\frac{dy_A^s}{dy_A^h} = \frac{\partial EU_A / \partial y_A^h}{\partial EU_A / \partial y_A^s} = \frac{(1 - \pi_A) u' [y_A^h]}{\pi_A u' [y_A^s]} < \frac{(1 - \pi_B) u' [y_B^h]}{\pi_B u' [y_B^s]} = \frac{\partial EU_B / \partial y_B^h}{\partial EU_B / \partial y_B^s} \quad (5.7)$$

In a second step, we define the meaning of equilibrium on a private market for health insurance. Thus let every contract k be characterized by the benefits to be paid to the insured in the event of illness I_k and by the premium P to be paid by the insured regardless of the state of health realized. Furthermore, let

$$\sigma_k = P/I_k \quad (5.8)$$

be the “unit price of insurance.” The market is organized such that the suppliers of insurance determine the price σ_k , while the demanders determine I_k .

DEFINITION 5.2: An “equilibrium on the market for health insurance” is characterized by a set of contracts such that

1. Each one of them secures the insurer a nonnegative expected profit;
2. No potential contracts not belonging to the set would yield a nonnegative expected profit;
3. Either a “*separating equilibrium*” obtains (if insureds having differing probabilities of sickness buy their contracts at differing prices) or a “*pooling equilibrium*” obtains (if they all buy the same contract).

It can be easily seen that in a state of equilibrium, all contracts actually concluded result in zero expected profit, for every contract yielding strictly positive expected profit can be undercut by a contract written by another insurance company, offering the same benefits for a lower premium yet still making a nonnegative profit.

- **Possibility of a pooling equilibrium.** According to Definition 5.2, all contracts offered in a state of equilibrium must have the same price σ^v in this case. Letting I_A and I_B be the utility-maximizing benefits for all individuals of both groups, the corresponding premiums are

$$P_A = \sigma^v I_A, \quad P_B = \sigma^v I_B \quad (5.9)$$

and the zero expected profit constraint is given by

$$\mu P_A + (1 - \mu) P_B = \mu \pi_A I_A + (1 - \mu) \pi_B I_B \quad (5.10)$$

Substituting Eq. (5.9) into Eq. (5.10) and solving for σ^v yields

$$\begin{aligned} \mu \sigma^v I_A + (1 - \mu) \sigma^v I_B &= \mu \pi_A I_A + (1 - \mu) \pi_B I_B \\ \sigma^v &= \frac{\mu \pi_A I_A + (1 - \mu) \pi_B I_B}{\mu I_A + (1 - \mu) I_B} \end{aligned} \quad (5.11)$$

Letting μ first go to one and then to zero and recalling $0 < \mu < 1$ and $\pi_A < \pi_B$, one obtains from Eq. (5.11)

$$\pi_A < \sigma^v < \pi_B \quad (5.12)$$

It would thus pay for an insurer exclusively to sell contracts to members of the A group at a price σ_A^v lying between π_A and σ^v . Being cheaper than σ^v , this contract would be preferred by all potential buyers, resulting in a pooling equilibrium. But being priced in excess of π_A , this contract would generate a

positive expected profit to the insurer, which violates the definition of an equilibrium. Thus it has been shown that there exists *no pooling equilibrium* if the group membership of individuals is generally known and compulsory insurance is excluded.

- **Possibility of a separating equilibrium.** In the following, we examine the existence and the property of a separating equilibrium (superscript t). The zero profit condition of Eq. (5.10) must now hold separately for the A group and the B group. Consequently, the prices of the contracts offered to the members of the corresponding groups are

$$\sigma_A^t = \pi_A, \quad \sigma_B^t = \pi_B \quad (5.13)$$

A pooling of risks therefore occurs within each of the two risk groups, with premiums paid by the members of group A covering their benefits and likewise for group B . The premiums are thus *actuarially fair*⁹ with regard to each risk group.

The benefits optimally purchased are derived from the maximization of the utility function (5.1),

$$EU_i = \pi_i u(y_i^s) + (1 - \pi_i) u(y_i^h), \quad i = A, B$$

under the following constraints

$$y_i^s = Y - L + I_i - P_i = Y - L + (1 - \sigma_i^t) I_i \quad (5.14)$$

$$y_i^h = Y - P_i = Y - \sigma_i^t I_i, \quad i = A, B \quad (5.15)$$

In both equations, the second equality sign follows from Eq. (5.8). From the necessary conditions for an optimum with respect to I_i , one obtains, using Eq. (5.13),

$$\frac{\pi_i u'(y_i^s)}{u'(y_i^h)} = \frac{\sigma_i^t}{1 - \sigma_i^t} (1 - \pi_i) = \sigma_i^t = \pi_i, \quad i = A, B \quad (5.16)$$

By dividing both sides of Eq. (5.16) by π_i , one can see that separating insurance contracts lead to an allocation that satisfies the condition of a Pareto optimum as given by Eq. (5.4). Furthermore, the allocation meets the society's resource constraint (5.2), as one can easily check, using Eqs. (5.14) and (5.15). Consequently, the allocation is *feasible*.

In case of a strictly concave utility function in income $u(y)$, the optimum condition (5.16) holds if disposable income is *equal* in both states of health. This means that the risk associated with treatment costs is completely covered by insurance, the benefit amounting to $I_i = L$ ($i = A, B$).

9. The actuarially fair premium is a premium that is equal to the expected value of the benefits.

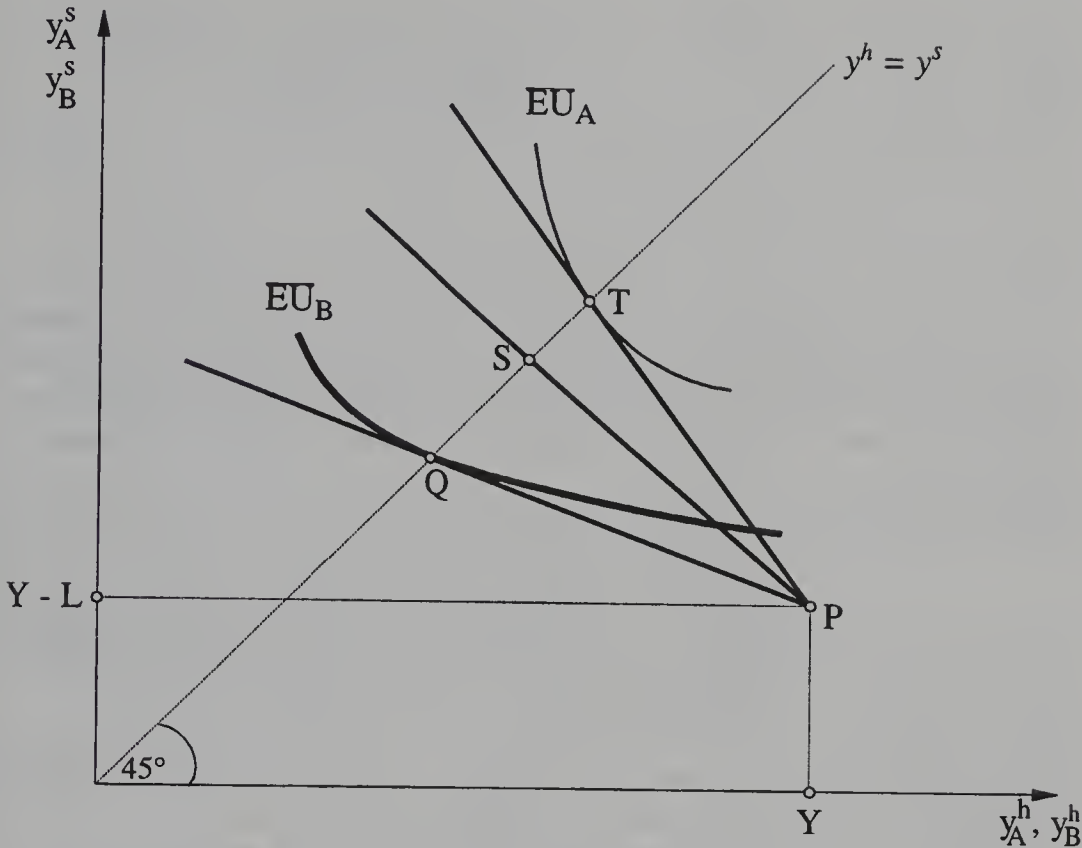


Figure 5-1. Insurance contracts in a model with contingent claims.

This result can be illustrated in the (y^h, y^s) space of a diagram (see Figure 5-1). Every single point, whose coordinates can be calculated from Eqs. (5.14) and (5.15), symbolizes a specific contract (I_j, P_j) . A given price σ_j^t shows all possible contracts and consequently all possible (y^h, y^s) combinations by solving Eq. (5.15) for $I_i = Y/\sigma_i^t - y_i^h/\sigma_i^t$ and substituting into Eq. (5.14)

$$y_i^s = \frac{1}{\sigma_j^t} Y - L - \frac{1 - \sigma_j^t}{\sigma_j^t} y_i^h, \quad i = A, B \tag{5.17}$$

One can see that all consumption vectors lie on a straight line with slope $-(1 - \sigma_j^t)/\sigma_j^t$. Furthermore, different values of σ_j^t give rise to different lines in Figure 5-1, which, however, originate from the common point $\{Y, (Y - L)\}$ indicating the initial endowment of every demander of insurance.

The lines PQ , PS , and PT of Figure 5-1 exemplify three particular consumption vectors. PT has slope $-(1 - \pi_A)/\pi_A$ and stands for all actuarially fair insurance contracts for members of the A group as given by Eq. (5.13). Line PQ , of slope $-\pi_B/(1 - \pi_B)$, contains the actuarially fair contracts for the B group. Finally, line PS has slope [since $I_A = I_B$ in Eq. (5.11)]

$$\begin{aligned}
\frac{dy_i^s}{dy_i^h} &= -\frac{1 - \sigma^v}{\sigma^v} = -\left(\frac{1}{\sigma^v} - 1\right) = -\frac{\mu I_A + (1 - \mu)I_B - \mu\pi_A I_A - (1 - \mu)\pi_B I_B}{\mu\pi_A I_A + (1 - \mu)\pi_B I_B} \\
&= -\frac{\mu(1 - \pi_A) + (1 - \mu)(1 - \pi_B)}{\mu\pi_A + (1 - \mu)\pi_B} \quad (5.18)
\end{aligned}$$

and is thus the locus of contracts that are actuarially fair for the population at large.

Given a pooling equilibrium, all contracts would have to lie on *PS*. The separating equilibrium as calculated in Eqs. (5.13) and (5.16) is shown by the pair of points *Q* and *T*. Both points not only lie on their respective fair insurance line but also on the 45° line from the origin where $y^h = y^s$. This means comprehensive coverage, as may be gleaned from Eq. (5.16).

CONCLUSION 5.5: If individuals are discriminated according to their individual health risk, resulting in different insurance contracts being offered to them, then a competitive private insurance market leads to a Pareto-optimal risk allocation. Thus compulsory insurance is not necessary from an allocative viewpoint.

If compulsory health insurance with actuarially fair contributions for the several risk groups is introduced nevertheless, the demand for private insurance is curtailed accordingly. The resulting allocation is again Pareto-optimal.

However, the compulsory insurance scheme may refrain from discriminating between risks (even though publicly available information would permit this), charging a uniform contribution instead [e.g., an actuarially fair one for the total population, as in Eq. (5.11)]. This also causes the demand for private insurance to be reduced; however, the resulting combination of compulsory and private insurance is not actuarially fair for the several risk groups and therefore does not lead to a Pareto-optimal allocation.

5.4.3 Heterogeneous Health Risks and Private Information

In the following we modify the model of the previous section by assuming that only the individual himself knows to which group of the population—*A* or *B*—he belongs. Contrary to the situation of a separating equilibrium and public information where a member of the *A* group paid a lower premium than a member of the *B* group for the same amount of benefits, he now cannot demonstrate his favorable risk status to the insurer. Conversely, the increased risk of a member of the *B* group cannot be proved at the time the insurance contract is concluded.

Therefore, *each* contract must be offered to *both* types of individuals, the equilibrium condition being non-negative profits attained from insuring those who actually buy insurance. However, the parameters (π_A , π_B , and μ) are assumed to be known to all parties, including the government, making it possible to evaluate a compulsory insurance scheme that is actuarially fair at the aggregate level.

Moreover, we presuppose that the suppliers of insurance contracts not only determine price (i.e., the ratio of benefits to premium) but also the amount of benefits.

In contrast to the model of the previous section, the demanders of insurance therefore *are not free* to decide the amount of benefits I_i ($i = A, B$) but can only choose from a limited set of price–quantity combinations offered. Without this restriction (which incidentally is quite common in insurance), a separating equilibrium with private information cannot exist, as will become clear shortly.

In the following, we examine the existence of an equilibrium on the market for private insurance. Later on, we turn to the question of whether a compulsory health insurance run by the government may bring about an improvement of social welfare.

In full accordance with the case of public information, “pooling” and “separating” equilibria are to be distinguished (see Definition 5.2 again). The following proposition, however, states that there exist at most separating equilibria.

PROPOSITION 5.1: Given private information and two risk groups with strictly concave utility functions, there exists no pooling equilibrium in the insurance market.¹⁰

PROOF: Suppose that a pooling equilibrium does exist. According to its definition, all individuals of both risk groups demand the same insurance contract (σ^v, I^v) , which secures an expected profit of zero to the supplier. Thus we have

$$I^v \sigma^v = I^v [\mu \pi_A + (1 - \mu) \pi_B] \quad (5.19)$$

Dividing this equation by I^v , one can see that σ^v must lie between the extremes given by π_A and π_B ,

$$\pi_A < \sigma^v < \pi_B \quad (5.20)$$

A pooling equilibrium can thus only be realized by a contract that is located on a line like PS in Figure 5–2 (which reproduces Figure 5–1). Suppose it is represented by point V .

Moreover it is evident that the indifference curves of two individuals belonging to different risk groups have *different slopes* at any point on PS . Consider a member of group i ($i = A, B$). Forming the total differential of Eq. (5.1) as a function of dy_i^h and dy_i^s , setting it to zero ($dEU_i = 0$), and using Eqs. (5.14) and (5.15) to express $u'[y_i^h]$ and $u'[y_i^s]$ respectively, one obtains i 's marginal rate of substitution (MRS) between the incomes in the two states of health. This MRS holds at a specific point determined by the parameters of the contract V and is given by

$$-\frac{dy_i^s}{dy_i^h} = \frac{(1 - \pi_i)u'[Y - \sigma^v I^v]}{\pi_i u'[Y - L + (1 - \sigma^v)I^v]}, \quad i = A, B \quad (5.21)$$

10. This proposition can be found in Eckstein et al. (1985) in the context of annuities.

Let us consider a contract (σ^v, I^v) represented by point V on line PS in Figure 5–2. Because of $\pi_A < \pi_B$, the right-hand side of Eq. (5.21) is larger for A members than for B members and thus, at point V , indifference curves of the A members run *steeper* than the ones of the B members. This fact is illustrated by the curves $\overline{EU}_A^1$ and $\overline{EU}_B^1$, respectively.

Therefore a point W exists, lying to the southeast of V and having the following properties. It is located

- 1. Above the indifference curve $\overline{EU}_A^1$;
- 2. Below the indifference curve $\overline{EU}_B^1$; and
- 3. Below the line PT .

If now, in addition to (σ^v, I^v) , a contract symbolized by point W of Figure 5–2 (i.e., characterized by parameters σ_W and I_W) is offered, it will be purchased by all members of group A but none of group B . Since point W lies below the line PT representing actuarially fair premiums for A members, the new contract guarantees its suppliers a *positive* expected profit. This finding, however, contradicts the definition of an equilibrium at point V . QED.

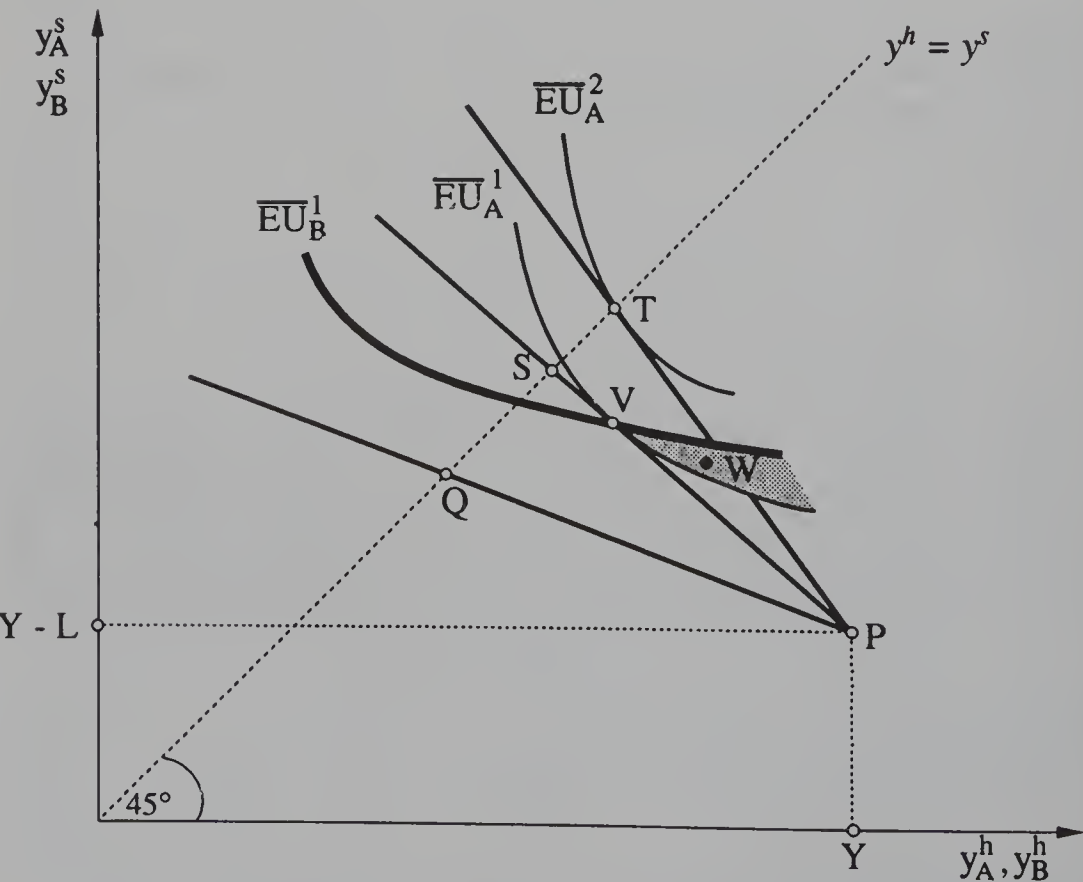


Figure 5-2. Pooling equilibrium on the insurance market.

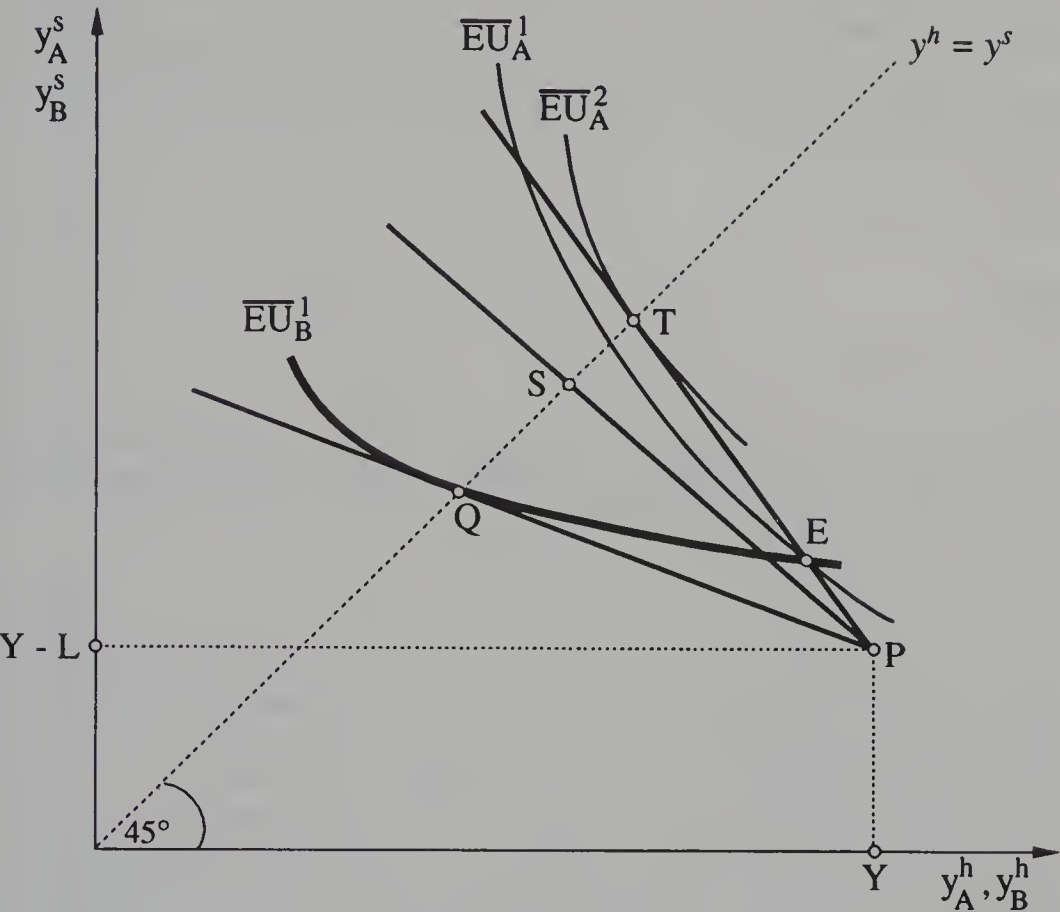


Figure 5-3. Separating equilibrium in the case of two risk groups.

This argument shows that under the conditions assumed in this section, only a *separating* equilibrium may exist on the private insurance market. With the help of Figure 5–3, we will now try to characterize its distinctive features.

As explained in the text following Definition 5.2, in a separating equilibrium *every single contract* must be associated with an expected profit of zero. Hence it follows that, in equilibrium, all contracts that are chosen by A members lie on the line *PT* of Figure 5–3. Likewise, all contracts chosen by B members lie on the line *PQ*.

Among all possible contracts available to B members, only the one represented by point Q meets the condition of expected utility maximization. This is due to condition (5.16) and the strict concavity of the utility function *u*. Only at point Q are the consumption levels y_B^s and y_B^h equal.

The contract offered to A members, however, cannot lie to the northwest of point E, where $\overline{EU}_B^1$ (i.e., the indifference curve through Q), intersects the line *PT*. Such a contract would be purchased not only by members of group A but also by B members, who cannot be discriminated given private information. Consequently, any policy offering benefits in excess of E at the favorable terms of A members would result in a negative expected profit to its supplier. Since A members have indifference curves

running more steeply than PT in the neighborhood of E , they prefer point E to all other possible points on PT (i.e., points to the southeast of E).

If a separating equilibrium is to exist, it must therefore be characterized by two different contracts. In particular, contract (σ'_B, I'_B) features *comprehensive risk coverage* (see point Q of Figure 5–3).

$$y_B^s = y_B^h = Y - \pi_B L \quad (5.22)$$

and thus

$$I'_B = L; \quad \sigma'_B = \pi_B \quad (5.23)$$

Let I'_A and $\sigma'_A (= \pi_A)$ designate the parameters of the contract preferred by A members. At point E of Figure 5–3, B members are *indifferent* between this contract and the one characterized by Eq. (5.23). Therefore, we have

$$\pi_B u[Y - L + (1 - \sigma'_A)I'_A] + (1 - \pi_B)u[Y - \sigma'_A I'_A] = u[Y - \pi_B L] \quad (5.24)$$

Using Figure 5–3 once more, we can state a condition for a separating equilibrium to hold. Specifically, the indifference curve $\overline{EU}_A^1$ for A members, running through point E , must lie *completely above* the line PS . Otherwise, there would be a point on PS such that the corresponding contract is preferred or equivalent to the contract in the potential separating equilibrium (points Q and E , respectively) by both A and B types, while an expected profit of zero is achieved. Therefore, the combination $\{Q, E\}$ cannot be an equilibrium in this case. The requirement for the existence of a separating equilibrium evidently has a greater chance of being fulfilled the *closer* line PS lies to line PQ , that is, the smaller the share of A members μ in the population [see also Eq. (5.18), with $\mu \rightarrow 0$].

CONCLUSION 5.6: If individuals cannot be discriminated according to their individual health risk, a separating equilibrium may exist at most on a private insurance market, provided that the share of “good risks” is not too large.

Figure 5–3 also shows that the absence of public information about group membership entails an externality for the members of the A group, that is, the “good risks.” Whereas in a separating equilibrium with public information, they attain point T and thus a utility level symbolized by the indifference curve $\overline{EU}_A^2$, in a separating equilibrium with private information, they only reach point E and indifference curve $\overline{EU}_A^1$. The “bad risks” of group B , however, obtain the same allocation in the two equilibria (point Q , indifference curve $\overline{EU}_B^1$). According to the Pareto criterion, the separating equilibrium with private information is therefore *inferior* to the separating equilibrium with public information.

5.4.4 Pareto-Improving Compulsory Insurance

As shown at the end of the preceding subsection, the separating equilibrium with private information is Pareto-inferior to the separating equilibrium with public

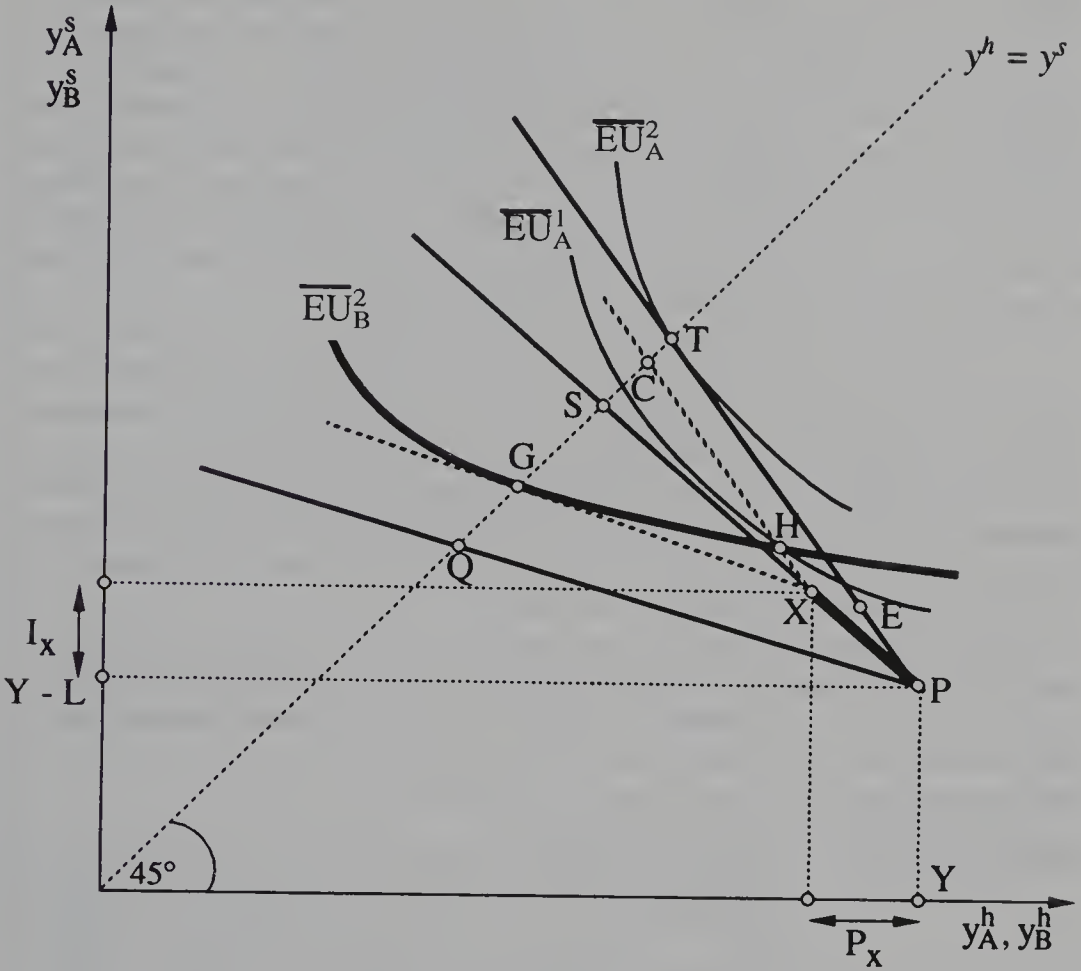


Figure 5-4. Pareto improvement through compulsory insurance and separating contracts.

information. The question now arises as to whether this welfare loss can at least be partially compensated by implementing a compulsory insurance scheme, of course on the premise that the government cannot recognize the group membership of citizens, knowing the values of the parameters π_A , π_B , and μ only.

Figure 5–4 shows that compared to the separating equilibrium with private information, a Pareto improvement can indeed be achieved by implementing a *compulsory insurance* scheme. Due to its compulsory character, it is sufficient for a nonprofit national health insurance to be actuarially fair at the level of the population as a whole [see Eq. (5.11)]. This means that in Figure 5–4 only points on line PS need to be considered. Let us assume that X is such a point; that is, the contribution is P_X and the benefits for all the sick are I_X .

If the purchase of private insurance contracts is still possible, then X represents the new “endowment point” from which the budget lines XC and XG for actuarially fair insurance contracts originate, with slopes $-(1 - \pi_A)/\pi_A$ and $-(1 - \pi_B)/\pi_B$, respectively. By purchasing *complementary private insurance*, B members may reach point G , where disposable income is equal in both states of health. Indeed, G lies on a higher

indifference curve than Q because the part of coverage provided by the government offers more favorable terms to this group than the part provided by the private one.

As to the A members, the line of argument developed above (referring to the separating equilibrium in the absence of a compulsory insurance) can be applied again. Thus their complementary policy is represented by a point on the line XC below point H , which is on the same indifference curve as point G for B members. If point H , as shown in Figure 5–4, is on a higher indifference curve for A members than point E , where the latter symbolizes a corresponding contract for the separating equilibrium without public intervention, a *Pareto improvement* obtains.

This Pareto improvement can be explained in economic terms as follows. “Bad risks” (group B) profit from the fact that a uniform premium is charged for the compulsory part of the insurance (shown by the line PX). Here, they are subsidized by the “good risks.” While the “good risks” pay more than the fair premium for the first part of their insurance coverage (PX), they can buy the amount XH privately but on terms that are not contaminated by the “bad risks.” Thus the extent of their insurance coverage rises from PE to PH . Given a sufficient degree of risk aversion, this effect may compensate them for the higher total price.

It is interesting to see that in a separating equilibrium, compulsory and private health insurance contracts are both viable (for both risk groups). With heterogeneous health risks and private information, compulsory and private health insurance are not entirely substitutes but partly complements. This also means that compulsory insurance featuring comprehensive coverage cannot be Pareto-improving.

CONCLUSION 5.7: Under the conditions stated in Conclusion 5.6 (existence of a separating equilibrium), a compulsory insurance covering but a part of the costs and charging uniform contribution leads to a Pareto improvement. Compulsory and private insurance complement one another in this case.

5.5 JUSTICE AS AN ARGUMENT IN FAVOR OF GOVERNMENT INTERVENTION IN HEALTH CARE

5.5.1 The “Right” to Medical Care

Whilst, up to this point government intervention in health care has been discussed with regard to allocative efficiency, considerations of justice often are also invoked to justify such interventions. Particularly in Anglo-Saxon countries, “*equal access*” to health care constitutes a goal of health policy besides the efficiency goal.¹¹ This means that willingness to pay (and a fortiori ability to pay) of an individual or a group (e.g., inhabitants of a region of a country) should not constitute a criterion for the allocation of health services, since everyone has a right to them.

11. In other countries, equal access could be construed to follow from constitutional principles, such as equality (*égalité*) in France or comparable living conditions in the member states in the case of Germany.

Now we may ask whether such an ideal situation can be reached at all. Indeed, an institution like the National Health Service in the United Kingdom provides free medical care, but in quantities that are limited by its share of general tax revenue. Consequently, a certain *rationing* in terms of waiting lists for elective surgery is practiced. Service quality, measured, for example, by time spent per patient in ambulatory medical care, is also limited. As is the case in the United Kingdom, a private market for health services will therefore develop (if it is allowed to), offering better quality or reduced waiting times to those able and willing to pay. If one does not want to have recourse to totalitarian means, which do not go together with a free country, then this market can hardly be suppressed.

A further problem with the entitlement theory is that it transforms the particular importance of health for a person's existence and realization of his or her potential into an exclusive emphasis on medical services. As already expounded in Chapters 3 and 4, medical services are certainly not the only factor influencing the health of a person, and they may not even be the decisive one. Other goods, such as sufficient and balanced nutrition, as well as housing of sufficient quantity and quality, are of comparable importance and would have to be distributed free of charge on the same grounds. Abolishing the market mechanism over such a wide range of goods would, however, create strong disincentives to work, thus considerably impairing the global efficiency of an economy.

Finally, influences upon health related to behavior, such as nutrition, consumption of alcohol and tobacco, and extent of physical activity, should not be negated. If they are neglected, the positive effect on health of free provision of medical care risks to be counteracted. This calls for exerting control over individual health behavior as an accompanying measure, giving rise to an insurmountable conflict with the basic principles of a free society.

These considerations show that all attempts to abolish completely ability to pay as a criterion for the allocation of health services are doomed to failure. The equality ideal behind it can probably be reached more easily and cheaply by making the *ability to pay itself* the target of social policy through payment of transfers to the lowest income groups.

5.5.2 Equality of Opportunities and Compulsory Insurance

Principles of justice are relevant in one more aspect, however, viz. when considering the equalization of starting opportunities determined by nature. In a market for private health insurance, every insured will have to pay a premium that in competitive equilibrium is equal to the expected value of his future health care costs, given the extent of coverage. People who have a higher innate susceptibility to illness (e.g., those born with a disease or handicap) would thus have to pay a higher premium than those less susceptible.

This market solution could be perceived as being "unfair" since the inequality described is not the fault of the individual affected but only reflects unequal starting opportunities. A suitable measure for correcting this would be the introduction of compulsory health insurance, subject to the provision that the premium must not be graded according to the health status of the person taking out insurance. In addition,

compulsory open enrolment by insurance companies prevents those disadvantaged in terms of health from being discriminated against, while the compulsory nature of insurance guarantees redistribution in favor of this group through the equality of premiums.

CONCLUSION 5.8: Equality of opportunities with regard to health cannot be accomplished by abolishing the market mechanism and through public provision of health services. It can be achieved, however, through compulsory insurance in combination with open enrolment and requiring insurance premiums to be independent of the state of health.

5.6 IMPLICATIONS FOR SOCIAL HEALTH INSURANCE

The discussion in this chapter has brought to light several peculiarities of the good, “medical care.” While they do not justify government intervention in the production of these services, they do justify intervention in the market for health insurance. The *free rider problem* can only be solved through compulsory insurance, and reasons of justice call for “social” terms and conditions of this insurance. Now the social health insurance systems of industrialized countries need to be examined in order to find out whether they satisfy the criteria formulated above. Depending on the outcome, other possible alternatives may have to be discussed.

As one such example, consider the statutory health insurance scheme of Germany. In 1988 (i.e., prior to reunification), almost 88 percent of the population of the Federal Republic of Germany belonged to statutory health insurance, about a tenth of which voluntarily. Thus about 20 percent are exempt from compulsory insurance, those earning high incomes.¹² This exemption is noteworthy because the rules for calculating the premiums amount to a *double redistribution*.

- Since contributions are not graded according to individual health risk, there is an implicit redistribution in favor of the *less healthy*.
- Since the contribution is raised as a fixed percentage of income (up to an upper limit), *those having low incomes* are subsidized by those having high ones.

The latter property, however, obtains only to the extent that those with high incomes are members of statutory health insurance, which, as shown above, is not the case for those with really high earnings. In addition, this redistribution is limited to members of the same sick fund. Compensating payments between the 1,100 funds occur only for the varying proportions of retired people among the insured.¹³ As an unavoidable effect,

12. Out of an active population of some 27 million, 3 million had private health insurance and 3.7 million were voluntary members of statutory health insurance schemes. Only a very small part of those 6.7 million people exempt from compulsory insurance were low income earners, whereas most of them had an income in excess of the limit below which insurance is compulsory.
13. This feature will be changed in 1996 when risk-compensating payments among funds will be introduced.

sick funds having an “unfavorable” insured population (low average income, high proportion of chronically ill) have to charge higher rates of contribution than others. The objective of “social solidarity” referred to above is therefore not fully achieved.

These facts also have an implication for competition between the carriers of social health insurance, which according to the opinion of some experts (e.g., the Council of Experts on the Health Care System in the case of Germany) should be fostered in order to improve the global efficiency in the production of health services. There is a danger that competing sick funds will concentrate their efforts on achieving a favorable composition of risks insured rather than on exerting pressure on suppliers of services to attain the best possible performance–cost ratio.

Even if open enrolment is secured (i.e., if an insurer is obliged to contract with every interested buyer on legally mandated terms), several opportunities for cream skimming among potential buyers of insurance still exist. In particular, the young, healthy, and high income earners can be attracted by offering extra benefits, whereas chronically sick people can be pushed to leave the plan through poor service. This “cream skimming,” which is undesirable from a normative point of view, is the less profitable to the insurer the more accurately the premium paid by the insured reflects the expected value of claims. If risk factors such as age, sex, and pre-existing conditions are accounted for in the calculation of premiums, then the insurer is indifferent as to whether his insured clientele represents a rather more “expensive” or “cheap” risk pool because his premiums globally are in accordance with the risks covered.

At this point the observant reader will be startled, since it has just been stated above that, for reasons of justice, the insured’s contribution should not be linked to risk to the extent that health risks are innate and therefore not the individual’s responsibility. The apparent contradiction is resolved if we distinguish between the “contribution” paid by the insured and the “premium” received by the insurer. Specifically, imagine that an *equalization fund* is built up between the two partners to the contract. Such a fund was created, for example, in the Netherlands in the course of a comprehensive reform of health insurance during the 1980s [see Van de Ven and Van Vliet (1992)]. It collects income-dependent contributions from the insured and distributes the proceeds to insurers according to the composition of their enrolled populations. In this way, cream skimming from undermining the redistribution desired for the purpose of “justice” may be prevented even if sick funds are exposed to competition. This redistribution is designed to favor individuals with a low income and suffering from innate health-related handicaps.

5.7 EXERCISES

Exercise 5.1:

- (a) Explain the possible reasons for market failure on the markets for health goods and how the government may intervene appropriately.
- (b) What arguments serve to justify compulsory health insurance?
- (c) What arguments may justify its organization by government?
- (d) Discuss alternative institutions and regulations which bar “cream skimming” in a mixed market economy.

Exercise 5.2:

Consider the following table.

Table 5.2: Existence of an Equilibrium on the Insurance Market

	Public information	Private information
Pooling equilibrium	I	II
Separating equilibrium	III	IV

(a) Comment on the existence of an equilibrium in situations I to IV in the market for health insurance. (b) Describe the principal characteristics of the existing equilibria. Compare them in terms of efficiency. (c) What are the benefits of implementing a compulsory health insurance scheme in these cases?

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Section 5.6: Implications for Social Health Insurance

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6

OPTIMAL DESIGN OF HEALTH INSURANCE CONTRACTS

6.1 THE ISSUE

In Chapter 5, several reasons were presented and discussed as to why a developed society may opt in favor of social health insurance with compulsory membership. This means that individuals are not completely free in deciding the amount of their insurance coverage against the cost of illness, since they are not permitted to have less than a minimum level of protection.

Such compulsion can only be justified if the insurance cover prescribed by the law has certain *optimal properties* with regard to its type and extent. Specifically, it should be designed such that it would be voluntarily selected by a representative individual¹ who behaves rationally while abstaining from the free-rider behavior described in Section 5.3.1 of the previous chapter. This consideration raises the issue of optimal design of health insurance contracts from the point of view of the person to be insured. In particular, the discussion will focus on the conditions under which a complete shifting of risk by means of a comprehensive insurance policy is optimal, and the conditions under which cost-sharing clauses are optimal for the insured.

In dealing with this issue, the general findings of the economic theory of insurance well as the peculiarities of the risk “illness” must be considered. Two of these peculiarities are mentioned here.

1. Dual loss in the event of illness. The onset of an illness causes two types of loss. On the one hand, the individual suffers a financial loss in terms of the cost of treatment occasioned by the illness and forsaken income from work. On the other hand, he suffers the nonfinancial loss of health itself, in particular if the initial health status cannot be

1. The concept of “representative individual” may be understood in different ways. It may mean an individual with “average” properties in terms of his or her preferences and endowments, or a member of a group that constitutes the majority of the population.

recovered. Illness therefore simultaneously entails an insurable and a noninsurable risk, and we shall see that this fact has implications for the optimal amount of cover against the insurable risk.

2. Dual moral hazard. The phenomenon of “moral hazard,” well known in insurance economics, occurs in two different forms in the case of a health risk. The term *moral hazard* states that “the insurance policy might itself change incentives and therefore the probabilities upon which the insurance company has relied.”² The crucial point is that the insurer typically cannot observe the actions of the insured and their modification in response to insurance (“hidden action”). The two types of moral hazard relevant to health insurance are:

a. The individual may influence the *probability* of falling ill through prevention, which may amount to a general change in lifestyle;

b. In the event of illness, the associated *financial loss* (cost of treatment) is not necessarily fixed, since a choice between cheaper and more expensive treatment alternatives frequently is available.

In the case of (a), the individual acts in a sense prior to “nature,” the embodiment of chance, and therefore “ex ante.” In the case of (b), the individual acts after nature (“ex post”). Whilst an ex ante influence upon risk (through prevention) is possible in almost all lines of insurance, ranging from theft insurance to life insurance, the ex post influence is a *special feature of the risk “sickness.”* It reflects the fact that when writing the contract the benefit of health insurance cannot simply be expressed by a fixed amount of money (e.g., the current value of a stolen object). Rather, it must be circumscribed, for example, as “the cost of sufficient, appropriate and economical treatment.” Here the insured and the provider of services have ample room to define what they consider sufficient, appropriate, and economical. Incentives influencing the choice of definition by the provider are ignored at this point because they will be treated in detail in the following chapter. This approach admittedly neglects possible interdependencies between patient and provider behavior.

One of the key questions addressed in this chapter will be to determine the conditions under which complete shifting of the costs of treatment from the individual to the health insurer becomes optimal, rather than a direct contribution to these costs (copayment, cost sharing).

This chapter is structured accordingly. First, an overview of possible and effective types of cost sharing is given in Section 6.2, complemented by a description of some other characteristics of health insurance policies. A few preliminary considerations regarding the implied incentives governing the demand for health services will also be offered. In the three following sections, the optimal design of health insurance will be investigated (see Figure 6–1). Section 6.3 deals with the case of no moral hazard, that is, the case in which the distribution of losses cannot be influenced by the insured. In Sections 6.4 and 6.5, the occurrence of moral hazard in its two versions (a) and (b) defined above is discussed. Section 6.6 contains some conclusions concerning the design of compulsory health insurance.

2. See Arrow (1970), p. 142.

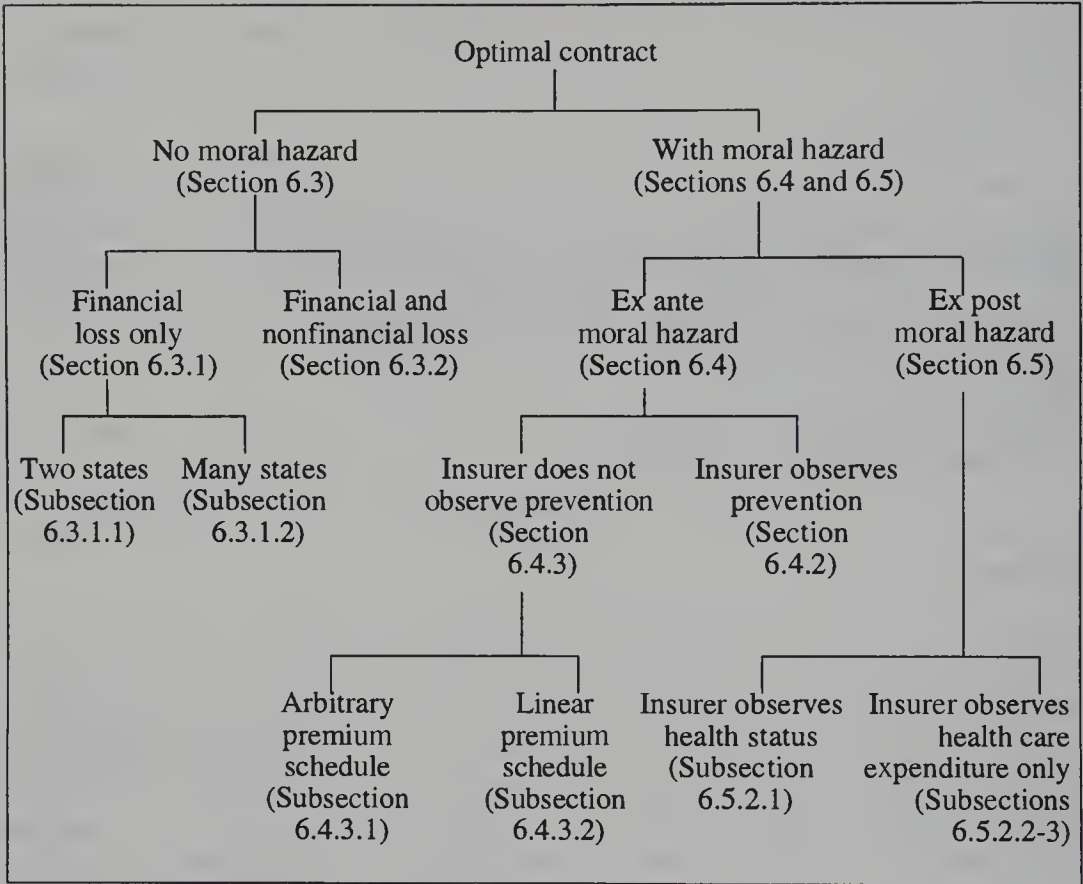


Figure 6-1. Overview of health insurance contracts.

The formal models are—in accordance with the introductory character of this text—kept as simple as possible. They are therefore all *static* in character; that is, the conclusion of the insurance contract, possible prevention, and the stochastic event “illness” all take place within the same period. Furthermore, it is presumed that only one case of illness may occur during the period considered. Additional uncertainties, for example, regarding the effectiveness of prevention and medical treatment, are also ignored. Nevertheless, the models provide many general and basic insights into the structure of optimal health insurance.

Finally, a fundamental assumption is that health insurance is not used as a vehicle for income redistribution. The insurance premium of each insured thus bears a well-defined relationship to the expected value of his claims.

6.2 TYPES OF HEALTH INSURANCE CONTRACTS AND THEIR INCENTIVE EFFECTS

The simplest type of insurance is that of *comprehensive coverage*. In this case, the insurer reimburses the full cost of medical care to the insured. Health goods therefore have an effective price of zero to the insured, who has an incentive to extend his

demand to the point where their marginal utility is zero ("satiation quantity"). Deviations from comprehensive coverage relate to (1) the type of health goods included in the plan, (2) the type of providers allowed to deliver certain services, and (3) the extent of compensation paid by the insurer.

1. Differentiation according to health goods included. This may take on the following forms. On the one hand, *entire categories of services* (in-patient, out-patient, or dental care) may not be covered. In private health insurance, separate plans for different service categories are common (ambulatory, hospital, and dental care, e.g.). In the United States, many plans do not cover expenses for drugs. Such provisions create incentives to substitute insured services for noninsured ones wherever possible. On the other hand, there are *selective exclusions*. This means that, for example, so-called "convenience drugs" and nonmedical treatments are excluded from insurance coverage. Again, this creates an incentive, for example, to replace drugs that are not reimbursed by those that are while having comparable therapeutic effects.

2. Differentiation according to type of provider. Ambulatory care is usually covered only if provided by a licensed physician in private practice or the outpatient ward of a hospital. In addition, the physician may need to be registered with social insurers, as in Germany, where sick funds control the number of residences.

3. Differentiation according to the extent of compensation. This may alternatively bear on (a) the quantity of services consumed, (b) the price of the service, or (c) the product of quantity and price, that is, total expenditure.

a. Limitations of the *quantity* of a particular medical service provided per unit of time occur, for example, in the case of psychotherapeutic treatment or of eyeglass frames.

b. Limitations of *price* may consist in the use of fee schedules. However, the plan may simply fix the compensation per unit of medical service, permitting service providers to "overbill" (so-called indemnity plans). One instance of this is provided by the statutory health insurance system in Germany, where fixed indemnities for drugs of a given therapeutic class were introduced in 1989 (see also Section 9.4). Since the difference between the actual price and the indemnity has to be paid by the patient out-of-pocket, limitations of this type give service providers an incentive to keep within the price limit. However, they have no effect on utilization in terms of quantity. Imposing an absolute *fee per service unit* (e.g., a fixed copayment per prescription) similarly results in a positive effective price to the patient. Its incentive effect, however, differs since the patient pays the more the larger the quantity consumed.

c. Limitations in terms of *expenditure* present the most general case. Caps on expenditure are usually meant when people refer to "cost sharing by the insured." Several forms may be distinguished.

- i. Proportional cost sharing.* In this case the insurer generally bears a fraction k of all treatment costs ($0 < k < 1$). The fraction $(1 - k) = c$, the so-called "rate of coinsurance," is borne by the patient. The effective price of the treatment to the insured thus equals the proportion c of the price set by the service provider. Accordingly, his interest in the cost effectiveness of care increases with an increase in the rate of coinsurance.
- ii. Per-period deductible.* In this case the insured person pays the first x monetary units (MU) of his total health expenditure per period (quarter or year) out of his own pocket. The effective price therefore amounts to the supply price as long as total expenditure is relatively low (up to x MU). Above this threshold, however,

it drops to zero. If the insured expects to exceed this level in the course of the period anyway, then his incentive for cost effectiveness in consumption vanishes from the beginning. Nevertheless, this form of copayment is widespread because it relieves the insurer from the processing of petty claims, as claims with total value below x will never be submitted.

- iii. *Per-period caps on benefits provided* have exactly the opposite incentive effect, but are not very common because they limit insurance in exactly those situations where it is most needed, namely, when the cost of treatment is jeopardizing economic survival. However, caps on lifetime benefits, generally between 250,000 and 1 million US\$, are quite common in employer-based insurance plans in the United States.
- iv. So-called *indemnity insurance* represents a borderline case of category (c) above. Here, benefits are not tied to the costs of treatment at all. Rather, they consist of a lump-sum payment in the event of illness, differentiated according to type of illness. This means that the price incentives for the insured remain intact, since his marginal share in the cost of treatment amounts to 100 percent as soon as the cost exceeds the indemnity payment.

Of course, variations of these “pure” forms of cost sharing are conceivable, for example, a copayment graded according to the amount of expenditure (i.e., more or less than proportional coinsurance), or mixed forms, for example, a deductible combined with coinsurance on expenditures in excess of the deductible.

Apart from these differentiations on the benefit side, incentives can also be created on the premium side. The simplest example of this is the *fixed bonus for no claims* amounting to y MU per year. However, this does not need to be considered further, since in this simple model covering a single time period and assuming perfectly rational behavior, the incentive effects of such a bonus do not differ from those of a fixed deductible of equal amount.³ The premium can, however, be differentiated according to particular features relevant to health behavior, such as body weight or smoking habits, creating incentives to give up patterns of behavior damaging to health, thus to fight the origins of illness.

6.3 OPTIMAL INSURANCE PROTECTION IN THE ABSENCE OF MORAL HAZARD

In this section, the phenomenon of moral hazard will still be left out of consideration. Thus the financial (and nonfinancial if applicable) loss resulting from sickness is assumed to be a pure random variable, whose distribution is not influenced by the insured, neither by lowering the probability of illness nor by limiting the cost of treatment in the event of illness through preventive effort. As a result of these

3. The actual experience of the insurers suggests that saving one's bonus represents a value in itself to many insureds, causing the physician to take into account an increased resistance against excessive bills on the part of his patient. Apart from this, bonus options permit the insured to shift the incidence of the financial consequences of illness out of the period of incidence of the illness itself. These differences as well as the effects of dynamic (experience-rated) bonus options that make premium rebates increase with the number of claim-free years, will not be discussed here any further. Interested readers are referred to Zweifel (1992).

restrictive assumptions, the conclusions to be drawn from the models dealt with here can only provide preliminary insights, serving as a point of reference to models incorporating moral hazard effects (see Sections 6.4 and 6.5).

6.3.1 Financial Consequences of Illness Only

As yet another restrictive assumption, let medical treatment completely eliminate the nonfinancial consequences of illness so that the remaining loss is of exclusively financial nature, amounting to the cost of treatment M . This is a random variable because its value is not known at the beginning of the period. Let Y symbolize the (exogenously given) gross income of the individual and y the *income disposable for consumption*. In the absence of health insurance, this is itself a random variable, given by the equation

$$y = Y - M \quad (6.1)$$

An *insurance contract* is generally characterized by the *benefit schedule* $I(M)$ depending on the size of loss M on the one hand and the *premium* P on the other. Given that health insurance is indeed purchased, Eq. (6.1) for disposable income becomes

$$y = Y - P - M + I(M) \quad (6.2)$$

In the case considered here, where nonfinancial consequences of illness are absent, the utility of the individual depends only on his consumption of goods but not on his health status. If we assume constant prices of consumption goods and utility-maximizing behavior, then the maximum utility u achievable (in the sense of the so-called indirect utility function) may be expressed as a function of disposable income y . As usual, the “marginal utility of money” is positive but decreasing; that is, the function u is assumed to have the following properties:

$$u = u(y), \quad u'(y) > 0, \quad u''(y) < 0 \quad (6.3)$$

The last-mentioned property (“strict concavity”) means that the individual is *risk-averse* (see Subsection 2.3.2.2.2 for a definition of risk aversion).

Finally the individual is assumed to pursue the goal of maximizing his *expected utility* $EU := E[u(y)]$. The symbol E represents the expected value that results from multiplying each utility value u (associated with the value y_j taken on by the random variable y in the j th state of nature) with the accompanying probabilities π_j and summing,

$$EU = \sum_{j=1}^J \pi_j u(y_j) \quad (6.4)$$

Table 6.1 gives an overview of the model developed in the sequel.

Table 6.1: Basic Model of Optimal Health Insurance in the Absence of Moral Hazard

$y = Y - P - M + I(M)$	(6.2)
$EU(I) = EU\{y(I)\}$	(6.6)
$= \pi u[Y - P(I) - L + I] + (1 - \pi)u[Y - P(I)]$	
$\frac{dEU}{dI} = \pi\{1 - P'[I^\circ]\}u'[Y - P(I^\circ)] - P[I^\circ] - L + I^\circ - (1 - \pi)P'[I^\circ]u'[Y - P(I^\circ)]$	(6.7)
$\left\{ \begin{array}{ll} \leq 0, & \text{if } I^\circ = 0 \\ = 0, & \text{if } 0 < I^\circ < L \\ \geq 0, & \text{if } I^\circ = L \end{array} \right.$	
$P(I) = P^n(I) + C(I) = C_0 + \mu\pi + (1 + \lambda)\pi I$	(6.10)

y :	Disposable income of the individual	P :	Insurance premium
Y :	Gross income	P^n :	Net premium, expected value of claims
M :	Cost of treatment (general)	$C(I)$:	Benefits-related costs of administration
L :	Cost of treatment in the state of illness	C_0 :	Fixed costs of administration
π :	Probability of sickness	λ :	Surcharge ("loading") on net premium, proportional to P^n
$u(y)$:	Individual's utility	μ :	Loading for settlement of claims, proportional to π
EU :	Expected utility		
I :	Insurance benefits		

6.3.1.1 A Model with Two States of Health

In the simplest case, there are only two states of health. With probability $\pi(0 \leq \pi \leq 1)$, the individual falls ill, with the cost of treatment M amounting to L . With probability $1 - \pi$, the individual remains in good health and thus $M = 0$. Consequently, all possible insurance contracts can be characterized by two parameters only, the premium P (to be paid in any event) and insurance benefits I received in the event of sickness ($0 \leq I \leq L$).

Benefits I may be expressed relative to loss L by defining the ratio $k = I/L$, which equals the rate of reimbursement written into the policy. Conversely, the contract's *rate of coinsurance* is given by $c = 1 - k = (L - I)/L$.

The premium P of course depends on the amount of benefits I ,

$$P = P(I) \quad (6.5)$$

Using Eq. (6.2) in both states, "sickness" ($L > 0, I > 0$) and "health" ($M = 0, L = 0, I = 0$), expected utility EU can be expressed as a function of I ,

$$EU(I) = EU\{y(I)\} = \pi u[Y - P(I) - L + I] + (1 - \pi)u[Y - P(I)] \quad (6.6)$$

In this simple case, the choice of optimal health insurance boils down to the choice of the value of I that maximizes expected utility EU as given in Eq. (6.6), which is denoted by I° (optimum in the *absence of moral hazard*). One obtains the necessary

condition for an optimum by differentiating EU and applying the theorem by Kuhn and Tucker,⁴

$$\frac{dEU}{dI} = \pi \{1 - P'[I^\circ]\} u'[Y - P[I^\circ] - L + I^\circ] - (1 - \pi) P'[I^\circ] u'[Y - P[I^\circ]] \begin{cases} \leq 0, & \text{if } I^\circ = 0 \\ = 0, & \text{if } 0 < I^\circ < L \\ \geq 0, & \text{if } I^\circ = L \end{cases} \quad (6.7)$$

Figure 6–2 illustrates the necessary first-order conditions for an optimum. In panel (a), an interior optimum (with I° strictly between 0 and L) is shown. In panel (b) an optimum at the lower boundary ($I^\circ = 0$) is characterized, while panel (c) shows an optimum at the upper boundary ($I^\circ = L$) of the admissible domain of I .

One recognizes that the solution I° to Eq. (6.7) depends both on the nature of the utility function and the shape of the premium function. Therefore, a plausible form of the premium function is proposed in order to obtain additional insights. In the case of a nonprofit organization typical of compulsory health insurance, the premium must cover *expected claims* plus all administrative expenses. The first component, often called *net premium*, is then given by

$$P^n(I) = E\{I(M)\} = \pi I \quad (6.8)$$

With regard to administrative expenses C , they may be assumed to consist of three components. The first (C_0) is a fixed amount (cost of negotiating an insurance contract). The second is proportional (with factor μ) to the individual's probability of filing a claim (costs of claims processing), and the third is proportional (with factor λ) to the net premium (e.g., risk surcharge, loading, and commission),

$$C(I) = C_0 + \mu\pi + \lambda\pi I, \quad C_0, \mu, \lambda \geq 0 \quad (6.9)$$

The gross premium therefore amounts to

$$P(I) = P^n(I) + C(I) = C_0 + \mu\pi + (1 + \lambda)\pi I \quad (6.10)$$

Note that it increases with benefits,

$$P'(I) = (1 + \lambda)\pi > 0 \quad (6.11)$$

An insurance contract is called “*fair*” if the premium is equal to the net premium, that is, if there are no administrative expenses. It is called “*marginally fair*” if the surcharge does not increase with the expected value of claims, that is, if $\lambda = 0$.

4. See Intriligator (1971), Chap. 4.

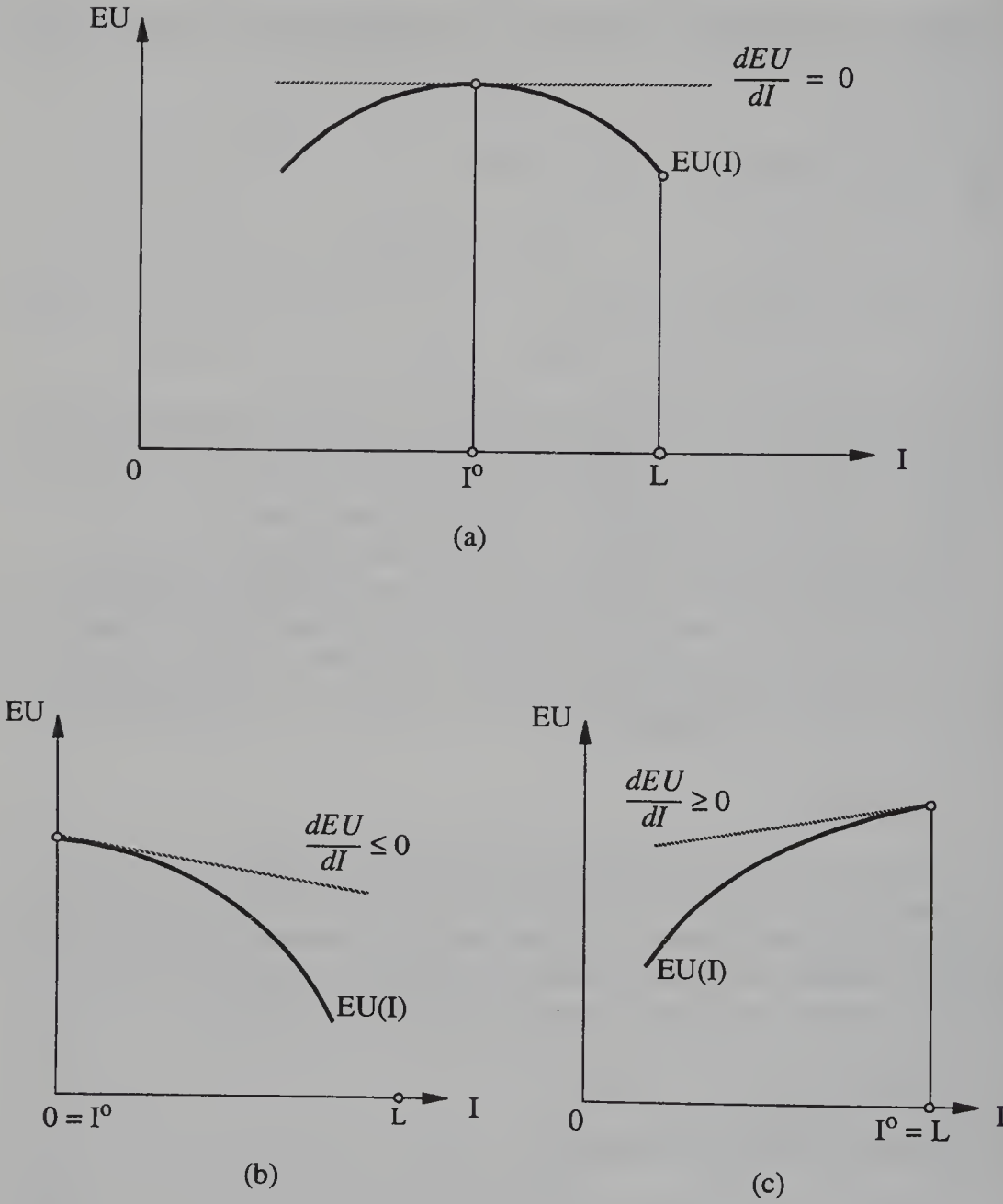


Figure 6-2. Types of optima under the constraint $0 \leq I^0 \leq L$ with (a) interior optimum: $0 < I^0 < L$, (b) boundary optimum: $I^0 = 0$, and (c) boundary optimum: $I^0 = L$.

In the following we shall concentrate on the conditions calling for *full reimbursement* of health care costs, that is, rendering the choice $I^0 = L$ optimal. Inserting this value into the optimality condition (6.7) and assembling terms in $P'(L)$ gives the following inequality:

$$(\pi - P'[L])u'[Y - P[L]] \geq 0 \tag{6.12}$$

Because of Eq. (6.3), u' is always positive; so the inequality becomes, using Eq. (6.11),

$$\pi - (1 + \lambda)\pi \geq 0 \quad (6.13)$$

This condition is satisfied if $\lambda \leq 0$. Therefore, the condition (6.7) for comprehensive coverage being optimal is fulfilled if there are *no proportional costs* related to the expected value of claims, that is, if insurance is “marginally fair.”

However, strictly speaking the fulfillment of the marginal condition (6.7) only states that the individual cannot increase his expected utility by reducing his insurance cover I by a small amount. The option, “full coverage” therefore needs to be compared in terms of expected utility with the option, “no insurance.” Full coverage is preferred if

$$EU[I^o = L] \geq EU[I^o = 0], \quad \text{or if} \quad u[Y - P[L]] = u[Y - (C_0 + \mu\pi + \pi L)] \geq \pi u[Y - L] + (1 - \pi)u[Y] \quad (6.14)$$

From the definition of risk aversion it follows that the individual always prefers fair insurance to the alternative of no insurance, which implies

$$u[Y - \pi L] > \pi u[Y - L] + (1 - \pi)u[Y] \quad (6.15)$$

A comparison of Eqs. (6.15) and (6.14) shows that an individual’s likelihood of opting for no insurance (rather than full coverage) is greater

- The larger the fixed cost C_0 ;
- The greater the probability of loss π (whenever $\mu > 0$).

The second result may at first seem somewhat surprising. But one should bear in mind that for a given expected value of claims, coverage of a small but highly probable loss is more expensive for the policy holder than coverage of a large but rather unlikely loss as soon as the insurer charges a loading for claims processing.

For this simplified model, the findings may be summarized in

CONCLUSION 6.1: In the absence of moral hazard,

- The necessary condition for full insurance coverage requires that the contract offered be *marginally fair*, that is, that the premium does not include a proportional loading on the expected value of claims;
- The sufficient condition for a risk-averse individual to opt for comprehensive insurance coverage is that the *insurance be “fair”*, that is, that its premium is equal to the net premium;
- The decision to *buy no insurance rather than full coverage* against the financial risk of illness is more likely to be optimal the greater the cost of insurance (the fixed fee and the loading for claims processing) and the greater the individual’s probability of illness.

6.3.1.2 A Model with an Arbitrary Number of States of Health

We now turn to the more realistic case where many different states of health exist. Therefore, the cost of treatment M may take on any nonnegative value. This case can be dealt with in two ways:

- M is a continuously distributed random variable with known density function $f(M)$.
- M takes on a finite number of values M_j , including the value zero ($j = 1, \dots, J$), with fixed positive probabilities π_j .

Case (a) is a borderline case of (b) if the number of possible states J (and hence of possible values of loss) goes to infinity. However, the amounts of money involved cannot be divided up indefinitely, and even more important, there will always be an upper limit to the individual's health expenditure (the country's GNP). This makes model (b) the more realistic of the two. Moreover, model (b) permits use of the Lagrange method instead of more complicated control theory without affecting results substantively.

Thus let the insurance company offer the individual to choose a benefit I_j for every value of the loss M_j , imposing only the restriction

$$0 \leq I_j \leq M_j, \quad j = 1, \dots, J \quad (6.16)$$

A premium P is associated with each vector $(I_1, \dots, I_J)$. Assuming for simplicity that P is proportional to the net premium and hence to the expected value of the claim, one has

$$P = (1 + \lambda)E(I_j) = (1 + \lambda) \sum_{j=1}^J \pi_j I_j, \quad \lambda \geq 0 \quad (6.17)$$

Again it is true that the contract is fair if $\lambda = 0$. Moreover, $P = 0$ if *no insurance* is bought, that is, if all $I_j = 0$.

The individual's expected utility function to be maximized may be formulated in analogy to Eqs. (6.2) and (6.4):

$$EU = E[u(y)] = \sum_{j=1}^J \pi_j u[Y - M_j + I_j - P] \quad (6.18)$$

The Lagrange function for this maximization problem, involving the decision variables $I_1, \dots, I_J$, P , and the Lagrange multiplier μ for the constraint (6.17), reads as follows:

$$Z = \sum_{j=1}^J \pi_j u[Y - M_j + I_j - P] + \mu \left(P - (1 + \lambda) \sum_{j=1}^J \pi_j I_j \right) \quad (6.19)$$

Due to the nonnegativity condition (6.16) on I_j , the Kuhn–Tucker theorem must be applied, yielding the following necessary conditions,

$$\frac{\partial Z}{\partial I_j} = \pi_j u'[Y - M_j + I_j - P] - \mu(1 + \lambda)\pi_j$$

$$\begin{cases} \leq 0, & \text{if } I_j = 0 \\ = 0, & \text{if } I_j > 0 \end{cases} \quad (j = 1, \dots, J) \quad (6.20)$$

$$\frac{\partial Z}{\partial P} = - \sum_{j=1}^J \pi_j u'[Y - M_j + I_j - P] + \mu \begin{cases} \leq 0, & \text{if } P = 0 \\ = 0, & \text{if } P > 0 \end{cases} \quad (6.21)$$

Let s and t be two states with positive payments ($I_s > 0$, $I_t > 0$). In view of the assumption $\pi_s > 0$, $\pi_t > 0$, one obtains from Eq. (6.20),

$$u'[Y - M_s + I_s - P] = \mu(1 + \lambda) = u'[Y - M_t + I_t - P] \quad (6.22)$$

Given strict concavity of the utility function ($u'' < 0$), the marginal utility of income in the two states s and t can only be the same if *disposable incomes* y_s , y_t themselves are equal [see Eq. (6.1)]

$$y_s = Y - P - M_s + I_s = Y - P - M_t + I_t = y_t \quad (6.23)$$

and therefore

$$M_s - I_s = M_t - I_t := D \quad (6.24)$$

Let D be the difference between the cost of treatment and the insurance benefit; it is the same across all states with positive benefits. Because of Eq. (6.16), it must be the case that $D \geq 0$. From Eq. (6.24), the vector I° of optimal benefits can be characterized as follows. There is a value $D \geq 0$ such that

$$I_j^\circ = \begin{cases} M_j - D, & \text{if } M_j \geq D \\ 0, & \text{otherwise} \end{cases} \quad (6.25)$$

Thus D may be interpreted as a *deductible*. The insurer agrees to pay for all the health care expenditure exceeding the sum D . In all cases j where expenditure M_j amounts to at least D , insurance benefits I_j° permit the individual to reach the same disposable income (y^*) whereas for $M_j < D$, benefits are zero, resulting in a value of y_j greater than y^* .

$$y_j = \begin{cases} Y - P - D = y^*, & \text{if } M_j \geq D \\ Y - P - M_j > y^*, & \text{if } M_j < D \end{cases} \quad (6.26)$$

Figure 6-3 shows insurance benefits I and disposable income y_j as a function of the realization of the random variable M_j , given a contract with deductible.

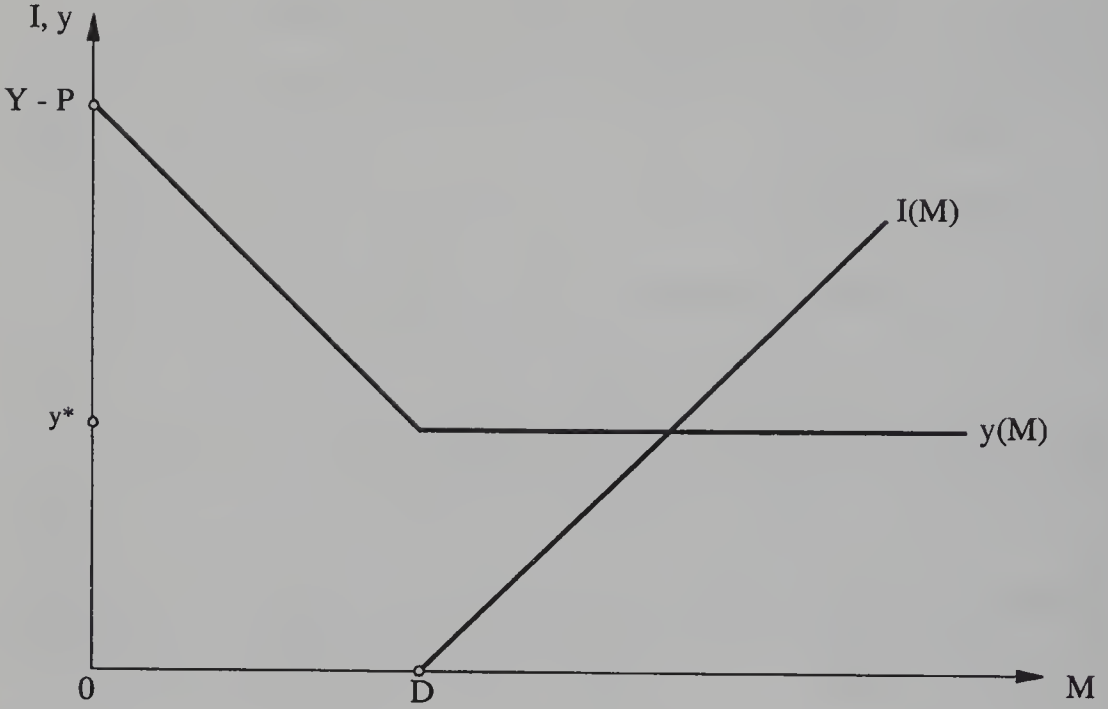


Figure 6-3. Insurance benefits (I) and disposable income (y) in case of a deductible (D).

Thanks to such a deductible, disposable income cannot fall below y^* while utility cannot fall below $u[y^*]$. On the other hand, marginal utility has an upper limit given by $u'[y^*] = \mu(1 + \lambda)$ [see Eq. (6.22)], since Eqs. (6.20) and (6.23) imply

$$u'(y_j) \leq u'[y^*] = \mu(1 + \lambda) \quad (6.27)$$

However, it follows from condition (6.21) that if an insurance contract is bought at all,

$$E[u'(y_j)] = \mu \quad (6.28)$$

The marginal utility of income therefore attains μ at most (in the case of marginally fair insurance). This means that income cannot fall below a certain limit in any of the states considered—which should be the effect of insurance after all.

Let us now examine how the *optimal deductible* D is determined for a positive loading λ . Designate with $T(D) \subset \{1, \dots, J\}$ the set of states in which health care expenditures M_j exceed a given deductible D , resulting in positive insurance benefits, and with $T^c(D)$ all the remaining states. Expected utility EU can then be expressed as a function of the deductible as follows

$$EU(D, \lambda) = \sum_{j \in T(D)} \pi_j u[y^*] + \sum_{j \in T^c(D)} \pi_j u[Y - M_j - P(D, \lambda)] \quad (6.29)$$

In order to maximize EU with respect to D , the first derivative of Eq. (6.29) is set to zero to obtain, using Eq. (6.26),

$$\begin{aligned}
\frac{\partial EU(D, \lambda)}{\partial D} &= \sum_{j \in T(D)} \pi_j u'[y^*] \left(-1 - \frac{\partial P}{\partial D} \right) + \sum_{j \in T^c(D)} \pi_j u'[Y - M_j - P(D)] \left(-\frac{\partial P}{\partial D} \right) \\
&= -\frac{\partial P}{\partial D} \sum_{j=1}^J \pi_j u'[y_j] - \pi(T) u'[y^*] = 0
\end{aligned} \tag{6.30}$$

Moreover, let $\pi(T)$ denote the total probability of all states with a positive benefit,

$$\pi(T) = \sum_{j \in T(D)} \pi_j \tag{6.31}$$

At first sight, Eq. (6.30) appears to be rather unwieldy. However, its economic interpretation is intuitively plausible.⁵ If the insured increases his deductible by a (marginal) monetary unit, then $-\partial P/\partial D$ indicates the concomitant reduction in his premium. The first expression on the right-hand side of Eq. (6.30) shows the increment of utility thus obtained. Conversely, there is a loss of utility given by $u'[Y - M_j - P]$ in all those cases (whose total probability appears in the second term) where M_j is greater than D , causing the insured to pay one monetary unit more in the way of copayment. If the two effects exactly balance, the optimal value of D is found.

This result can be shown more precisely if we insert Eq. (6.25), the equation for the optimal benefit structure, into Eq. (6.17), the equation determining the premium,

$$P(D, \lambda) = (1 + \lambda) \sum_{j \in T(D)} \pi_j (M_j - D) \tag{6.32}$$

Through partial differentiation of this equation with respect to D we can see how the premium changes if there is a marginal increase in the deductible,

$$\frac{\partial P}{\partial D} = -(1 + \lambda) \pi(T) \tag{6.33}$$

Inserting Eq. (6.33) into Eq. (6.30) and dividing by $\pi(T)$, the *condition for the optimal deductible* D° can be stated,

$$\frac{\partial EU(D^\circ, \lambda)}{\partial D} = (1 + \lambda) \sum_{j=1}^J \pi_j u'[y_j] - u'[y^*] = (1 + \lambda) E[u'(y_j)] - u'[y^*] = 0 \tag{6.34}$$

Is it possible that the optimal deductible amounts to $D^\circ = 0$? Due to Eq. (6.25), one would have $I_j = M_j$ in this case, implying $y_j = Y - P[0]$ and $u'[y_j] = u'[Y - P[0]]$ for every j , and hence $u'[y^*] = u'[Y \geq P[0]]$. Since the expected value of a constant is the

5. An analogous interpretation of the level of the optimal deductible in the case of a continuous distribution of health expenditure is given by Gould (1969, p. 145ff).

constant, the expectation operator may be dropped from Eq. (6.34), resulting in the condition

$$\{(1 + \lambda) - 1\}u'[Y - P[0]] = 0 \quad (6.35)$$

Because of $u' > 0$, this condition is *only* satisfied if $\lambda = 0$. Given positive transaction costs, a positive deductible is thus *always* chosen; conversely, a policy featuring full coverage is never taken out.

CONCLUSION 6.2: In the case of a finite number of possible health states, proportional loading, and no moral hazard, the optimal insurance policy provides for a fixed deductible. This deductible is zero only if the loading is zero, that is, if the insurance premium is “fair.”

To complete the analysis, let us examine how the optimal deductible, implicitly given by condition (6.34), is affected when there is an increase in the loading λ . To this end, we form the total differential of the left-hand side of Eq. (6.34), supposing that λ increases by $d\lambda$ and that the optimal value D° changes by dD° in response. Denoting the left-hand side of Eq. (6.30) by EU_D for short, we obtain

$$dEU_D = \frac{\partial^2 EU[\lambda, D^\circ]}{\partial D^2} dD^\circ + \frac{\partial^2 EU[\lambda, D^\circ]}{\partial D \partial \lambda} d\lambda = 0 \quad (6.36)$$

because the optimality condition (6.34) must still hold after λ has increased. Solving for $dD^\circ/d\lambda$ gives

$$\frac{dD^\circ}{d\lambda} = - \frac{\partial^2 EU[\lambda, D^\circ]/\partial D \partial \lambda}{\partial^2 EU[\lambda, D^\circ]/\partial D^2} = - \frac{\partial EU_D/\partial \lambda}{\partial EU_D/\partial D} \quad (6.37)$$

Since D° maximizes EU , the second-order condition requires the first derivative on the right-hand side of Eq. (6.36) and thus the denominator on the right-hand side of Eq. (6.37) to be negative. Therefore, the sign of the expression $dD^\circ/d\lambda$ is identical to that of $\partial EU_D/\partial \lambda$. The sign of this expression can be determined by differentiating Eq. (6.34) with respect to λ , yielding,

$$\frac{\partial EU_D}{\partial \lambda} = E[u'(y_j)] + \{u''[y^*] - (1 + \lambda)E[u''(y_j)]\} \frac{\partial P}{\partial \lambda} \quad (6.38)$$

The first term of this expression is positive because $u' > 0$. Likewise $\partial P/\partial \lambda$ is positive because of Eq. (6.32). This leaves the sign of the term in the curly bracket to be determined. Since $u'' < 0$, it is positive if the absolute value of $E[u''(y_j)]$ is at least as great as $1/(1 + \lambda)$ times the absolute value of $u''[y^*]$. This requirement is satisfied, for instance, if u'' is constant, that is, if the marginal utility of income decreases at a

constant rate in the relevant domain, or if the optimal deductible D° is small because there is little difference between the largest possible values of y_j and y^* .

CONCLUSION 6.3: Under fairly general conditions the optimal deductible increases if insurance coverage becomes more expensive.

6.3.2 Direct Health Effects on Utility

In the following, we will drop the assumption that the occurrence of illness has no effects on the individual apart from the costs of treatment. Rather, health status may now appear directly as an argument of the utility function. Utility therefore depends on disposable income y and health status H ,

$$u = u(y, H) \quad (6.39)$$

To simplify, we limit ourselves to the case discussed in Subsection 6.3.1.1, where health status can assume only two values, $H = s$ ("sick") and $H = h$ ("healthy"). Furthermore, $u(y, s)$ and $u(y, h)$ are replaced by $u_s(y)$ and $u_h(y)$. In this way, different utility functions are made to apply to the two realizations of H , with each one depending only on the individual's disposable income, however.

With the probability of illness again represented by π , expected utility is given by

$$EU = E[u(y)] = \pi u_s(y) + (1 - \pi) u_h(y) \quad (6.40)$$

exactly as in the model of Subsection 6.3.1.1. Let Y be gross income and L the cost of treatment in state s . Furthermore, let I be the benefit written into the policy and $P(I)$ the associated premium. However, it now must be possible that I exceeds the costs of treatment because illness also entails a nonfinancial loss. Payment in excess of L may be interpreted as damages, that is, as compensation for suffering.

The individual is confronted with the problem of finding the value of I that maximizes his expected utility,

$$EU(I) = \pi u_s[Y - P(I) - L + I] + (1 - \pi) u_h[Y - P(I)] \quad (6.41)$$

The necessary condition for a maximum reads

$$\begin{aligned} \frac{dEU}{dI} &= \pi \{1 - P'[I^\circ]\} u'_s[Y - P[I^\circ] - L + I^\circ] \\ &\quad - (1 - \pi) P'[I^\circ] u'_h[Y - P[I^\circ]] \begin{cases} \leq 0, & \text{if } I^\circ = 0 \\ = 0, & \text{if } I^\circ > 0 \end{cases} \end{aligned} \quad (6.42)$$

Focusing on the equality case, this condition can be simplified by writing the arguments of u'_s and u'_h as disposable incomes y_s and y_h , respectively,

$$\pi u'_s[y_s] = P'[I^o] \{ \pi u'_s[y_s] + (1 - \pi) u'_h[y_h] \} = P'[I^o] EU'(y) \quad (6.43)$$

The left-hand side of Eq. (6.43) shows the expected utility gain from increasing benefits I by a (marginal) monetary unit, while the right-hand side shows the expected utility loss due to the increase in premium. The two effects must balance at an interior optimum.

If the administrative costs of claims settlement is abstracted from ($\mu = 0$), a simplified version of the premium function (6.10) results,

$$P(I) = C_0 + (1 + \lambda)\pi I, \quad C_0, \lambda \geq 0 \quad (6.44)$$

The premium still increases with benefits, exactly as in Eq. (6.11),

$$P'(I) = (1 + \lambda)\pi > 0 \quad (6.45)$$

Given these specifications, optimality condition (6.43) reduces to

$$u'_s[y_s] = (1 + \lambda) \{ \pi u'_s[y_s] + (1 - \pi) u'_h[y_h] \} \quad (6.46)$$

This can be rewritten to read

$$u'_s[y_s] \{ 1 - \pi (1 + \lambda) \} = (1 - \pi) (1 + \lambda) u'_h[y_h] \quad (6.47)$$

The following conclusions can be drawn from this condition for optimum health insurance in the case where the utility function depends on health status.

I. If insurance is *marginally fair* (i.e., $\lambda = 0$), then Eq. (6.47) becomes

$$u'_s[y_s] = u'_h[y_h] \quad (6.48)$$

that is, benefits will be chosen such that the marginal utility of income is *equal* in both health states. In contrast to the model of Subsection 6.3.1.1, however, this does *not* mean that disposable incomes are equalized (by setting $I = L$), since u'_s and u'_h derive from different utility functions. Thus we can make the following distinction.

a. If illness means that an individual's *ability to enjoy consumption* is reduced, that is, if $u'_s[y] < u'_h[y]$ holds at a given disposable income y in both states, then Eq. (6.48) is satisfied if $y_s < y_h$ because of the concavity of u_s and u_h . This calls for $I^o < L$, implying *partial insurance coverage*, that is, *cost sharing* as the optimal solution. For example, one may think of an enthusiastic tennis player who is prepared to spend a lot of money for his hobby as long as he is in good health. If unable to play because of illness, he has no use for his money, yielding him comparable utility; therefore, he is not interested in completely equalizing his disposable income across states of health.

b. On the other hand, it is conceivable that, in the case of sickness, the marginal utility of income even *increases*, that is, that $u'_s[y] > u'_h[y]$. Then Eq. (6.48) requires $y_s > y_h$ and thus $I^o > L$: Being *overinsured* and contracting for *damages* (i.e., compensation of nonfinancial loss) is optimal for such an individual. An economic explanation of this possibility is that in the event of illness, an additional need arises for certain consumer goods that are not medical services in a strict sense, such as an apartment that is suitable for the handicapped.

2. If, however, the policy is *not marginally fair* (i.e., if $\lambda > 0$), then Eq. (6.47) is only satisfied for $u'_s[y_s] > u'_h[y_h]$; that is, the marginal utility of income must be greater when the individual is sick than when he is healthy. This in turn means that because of the strict concavity of the utility functions ($u''_s, u''_h < 0$), *less coverage* I° is demanded than in the case of marginally fair insurance. In combination with the above-mentioned case (b), this may coincide with full coverage.

3. The preceding points relate to the necessary condition for an *interior* optimum. This condition is also sufficient if the insurance premium does not contain a fixed component for administrative costs. However, if $C_0 > 0$, then the solution resulting from Eq. (6.47) and calling for $I = I^\circ$ has to be compared with the decision to have no insurance at all ($I = 0$). Having no insurance at all is preferred if

$$\begin{aligned} \pi u_s[Y - L] + (1 - \pi)u_h[Y] &> \pi u_s[Y - L - (1 + \lambda)\pi I^\circ - C_0 + I^\circ] \\ &+ (1 - \pi)u_h[Y - (1 + \lambda)\pi I^\circ - C_0] \end{aligned} \quad (6.49)$$

This condition is more likely to be satisfied the lower the costs of treatment L and the greater the administrative costs C_0 .

CONCLUSION 6.4: If an illness gives rise to an immaterial loss in addition to the financial loss due to treatment costs, then the marginal utility of income is equalized across health states provided the premium is “marginally fair.” However, optimal insurance benefits may be higher or lower than the cost of treatment.

6.3.3 Summary

Even ignoring the possible incentive effects due to the existence of an insurance contract (“moral hazard”), there are two separate reasons for the optimal health insurance policy not to simply cover the cost of treatment, namely (a) the existence of *transaction costs*, and (b) the fact that illness entails a *nonfinancial* besides a financial loss.

Transaction costs may be constant or dependent on the expected value of payments. Constant components of these costs may cause the individual not to take out any insurance, but they do not influence the extent of coverage if insurance is taken out. If the premium includes a proportional loading on expected payments, then it is generally optimal—in the absence of immaterial losses due to illness—to choose a policy with *cost sharing*, resulting in a smaller disposable income in the state of sickness than in good health. However, in all states of health in which insurance benefits are paid, disposable income should be equalized.

Conversely, if illness entails immaterial as well as financial loss, that is, if it changes the marginal utility of other consumption, then the optimal policy may provide coverage falling short or in excess of the cost of treatment even though the insurance premium does not contain a loading for transaction costs.

6.4 OPTIMAL INSURANCE COVERAGE WITH EX ANTE MORAL HAZARD

The analysis of Section 6.3 rested on the assumption that the cost of treatment is a random variable having a distribution that cannot be influenced by the individual. This is a rather unrealistic assumption because one can influence both the probability of illness and the course of illnesses through one's lifestyle. This wellknown fact is taken into account in the model to be discussed below, focusing on the ex ante version of moral hazard (for a definition of ex ante moral hazard; see Section 6.1). Through prevention, the individual is now assumed to be able to influence the *probability distribution* of treatment costs. However, once the event "illness" has occurred, the cost of treatment is exogenously fixed.

Furthermore, we presuppose that the prevention of illness costs money but has no other direct impact on the individual's utility. Thus health-enhancing activities such as physical exercise, which for some may be a pleasure and for others a burden, are ignored, as is the abstinence from alcohol and tobacco, for example. In addition, expenditures on prevention are assumed to reduce the income of the insured to their full extent rather than being paid by the insurance company. For one, this is justified by the fact that such expenditures are decided upon by the individual and therefore are not a random variable and hence not an "insurable risk." On the other hand, money spent on the prevention of illness often does not go into pure health goods, for example, into expensive health food or into atomizers that help to prevent respiratory disorders. It is hardly conceivable that a health insurer would ever cover expenditures on items such as these.

In order to make this analysis as easy as possible to understand, let us go back to the model developed in Subsection 6.3.1.1, where only two health states ("sick" and "healthy") exist. In this model, the prevention of illness may affect

- a. The *probability of illness* π with costs of treatment L constant in the event of illness; or
- b. The value of *health care costs* L given a certain probability of illness π .

Type (b) is somewhat less plausible in the case of health insurance.⁶ In the following, we will therefore deal with case (a), making the probability of ill health π a function of preventive effort V with the following properties:

$$\pi = \pi(V), \quad \text{with } \pi[0] > 0, \pi'(V) < 0, \pi''(V) > 0, \text{ for all } V \geq 0 \quad (6.50)$$

While prevention always has a nonzero marginal productivity ($\pi' < 0$), the strict convexity of the prevention function (i.e., the assumption $\pi'' > 0$) reflects a declining marginal productivity of prevention. Thus health risks cannot be totally eliminated by preventive effort because of the natural restriction $0 < \pi \leq 1$ (see Figure 6–4).

6. The intensity of medical treatment is largely decided by the physician. In other areas, the insured may have more influence on the size of the loss once it occurs. The purchase of a fire extinguisher constitutes an example. The probability of a fire breaking out is in no way influenced by the purchase, but certainly the possible damage is.

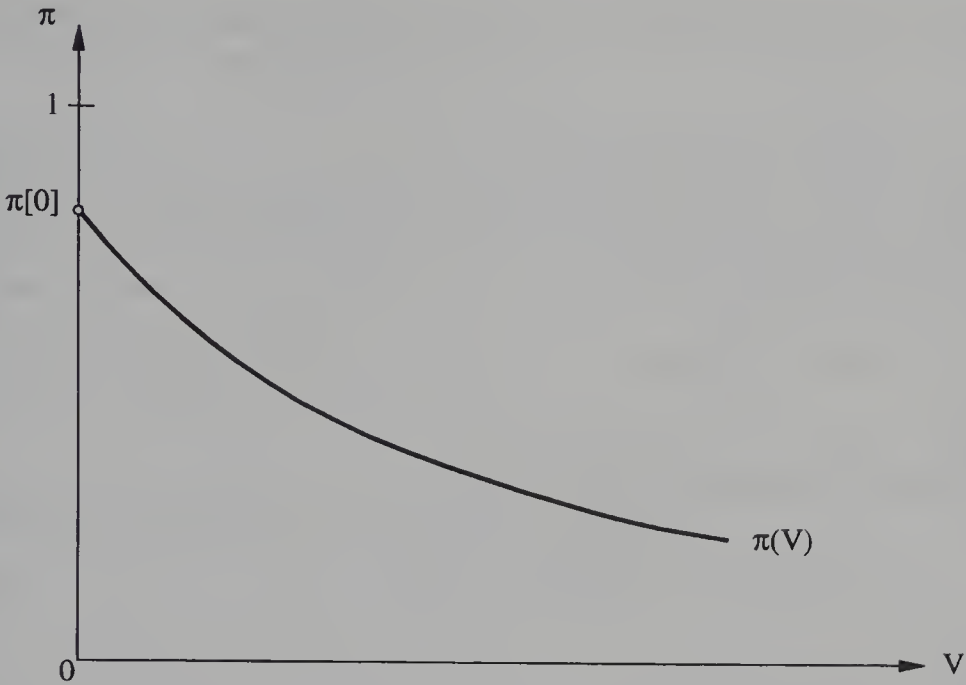


Figure 6-4. Declining marginal productivity of prevention.

An important feature distinguishing the different models is whether or not the insurer can *observe* the preventive efforts of his clients, which would enable him to reward them in setting the premium accordingly. Prior to making this distinction, however, we need to know the optimal level of prevention in the absence of health insurance (Section 6.4.1). This analysis will provide a benchmark for establishing the relationship between prevention and insurance coverage.

Table 6.2 contains an overview of the model used for simultaneously determining optimal prevention and optimal insurance coverage.

6.4.1 Optimal Prevention with No Insurance Available

Expenditure on prevention diminishes the individual’s disposable income in all states of nature because by definition they have to be made prior to the occurrence of the probabilistic event “sickness.” Referring to Eqs. (6.6) and (6.50), and equating preventive effort to expenditure on prevention, the expected utility in the absence of health insurance is given by

$$EU(V) = E\{u[y(V)]\} = \pi(V)u[Y - V - L] + \{1 - \pi(V)\}u[Y - V] \tag{6.51}$$

If this expected utility is to be maximized through the choice of a suitable value of preventive effort, the necessary condition for an interior optimum ($V^* > 0$) is given by Eq. (6.52). In this equation, y_s stands for $[Y - V - L]$ and y_h for $[Y - V]$, respectively,⁷

7. The associated second-order condition for a maximum is not examined, although its fulfilment cannot be taken for granted in the case of very strong risk aversion [see Ehrlich and Becker (1972), p. 639].

Table 6.2: A Model of Optimal Prevention and Optimal Insurance Coverage

$$EU(V) = E\{u[y(V)]\} \\ = \pi(V)u[Y - V - L] + \{1 - \pi(V)\}u[Y - V] \quad (6.51)$$

$$\frac{dEU}{dV} = \pi'[V^*]u[y_s] - \pi[V^*]u'[y_s] - \pi'[V^*]u[y_h] - \{1 - \pi[V^*]\}u'[y_h] = 0 \quad (6.52)$$

$$P(V, I) = (1 + \lambda)\pi(V)I \quad (6.54)$$

$$\frac{\partial EU}{\partial I} = \pi[V^0] \left(1 - \frac{\partial P}{\partial I}\right) u'[y_s] - \{1 - \pi[V^0]\} \frac{\partial P}{\partial I} u'[y_h] \begin{cases} \leq 0, & \text{if } I^0 = 0 & (a) \\ = 0, & \text{if } 0 < I^0 < L & (b) \\ \geq 0, & \text{if } I^0 = L & (c) \end{cases} \quad (6.56)$$

$$\frac{\partial EU}{\partial V} = \pi'[V^0]u[y_s] - \pi[V^0]u'[y_s] \left(1 + \frac{\partial P}{\partial V}\right) \\ - \pi'[V^0]u[y_h] - \{1 - \pi[V^0]\}u'[y_h] \left(1 + \frac{\partial P}{\partial V}\right) \quad (6.57)$$

$$\frac{u'[y_s]}{u'[y_h]} = 1 + \frac{\lambda}{1 - (1 + \lambda)\pi[V^0]} \quad (6.60)$$

y_h : Disposable income in the “healthy” state
 y_s : Disposable income in the “sick” state
 Y : Gross income
 L : Cost of treatment and loss of income in the “sick” state
 V : Preventive effort (equal to expenditure)
 V^* : Optimal prevention with no insurance available

V^0 : Optimal prevention given observability of preventive effort
 $\pi(V)$: Probability of “sick” state
 $u(y)$: Utility of the individual
 EU : Expected utility
 I : Insurance benefit
 P : Insurance premium
 λ : Proportional loading on net premium

$$\frac{dEU}{dV} = \pi'[V^*]u[y_s] - \pi[V^*]u'[y_s] - \pi'[V^*]u[y_h] - \{1 - \pi[V^*]\}u'[y_h] = 0 \quad (6.52)$$

Taking into account that $\pi[V^*]u'[y_s] + \{1 - \pi[V^*]\}u'[y_h] := EU'(y)$, the expected value of the marginal utility simplifies to

$$\pi'[V^*] \{u[y_h] - u[y_s]\} = EU'(y) \quad (6.53)$$

The left-hand side of Eq. (6.53) represents the increase in utility due to a reduction in the *probability of illness* π brought about by an increase of V by a (marginal) monetary unit. The right-hand side shows the expected loss of utility as a result of the *associated reduction of disposable income in both states*. In order to calculate this expected loss of utility, the marginal utility of income must be identified for each state of health and then both values added after weighing them using the associated probabilities π and $(1 - \pi)$, respectively.

6.4.2 The Insured's Optimum When Prevention Is Observable

6.4.2.1 Optimal Insurance Coverage

Let the individual be in a position to take out an insurance policy providing reimbursement I ($0 \leq I \leq L$) of his cost of treatment in the event of illness, with free choice of the value of I . Furthermore, let the insurance company be in a position to *observe* expenditure on prevention V and to adjust its premium accordingly. The premium continues to be proportional to expected payments (with loading factor λ), exactly as in Eq. (6.17), that is,

$$P(V, I) = (1 + \lambda)\pi(V)I \quad (6.54)$$

Let the individual's objective be to maximize his expected utility by a suitable choice of his decision variables V (prevention) and I (insurance),

$$\begin{aligned} EU(V, I) = E\{u(y)\} &= \pi(V)u[Y - P(V, I) - V - L + I] \\ &+ \{1 - \pi(V)\}u[Y - P(V, I) - V] \end{aligned} \quad (6.55)$$

Let us assume that an optimum with values V°, I° ($V^\circ \geq 0, 0 \leq I^\circ \leq L$) exists. Then these values must satisfy the following *first-order conditions*, in accordance with the theorem of Kuhn and Tucker (where the symbols again y_s and y_h stand for disposable income in the event of sickness and good health, respectively),

$$\begin{aligned} \frac{\partial EU}{\partial I} &= \pi[V^\circ] \left(1 - \frac{\partial P}{\partial I}\right) u'[y_s] - \{1 - \pi[V^\circ]\} \frac{\partial P}{\partial I} u'[y_h] \\ &\begin{cases} \leq 0, & \text{if } I^\circ = 0 & (a) \\ = 0, & \text{if } 0 < I^\circ < L & (b) \\ \geq 0, & \text{if } I^\circ = L & (c) \end{cases} \end{aligned} \quad (6.56)$$

$$\begin{aligned} \frac{\partial EU}{\partial V} &= \pi'[V^\circ] u[y_s] - \pi[V^\circ] u'[y_s] \left(1 + \frac{\partial P}{\partial V}\right) - \pi'[V^\circ] u[y_h] \\ &- \{1 - \pi[V^\circ]\} u'[y_h] \left(1 + \frac{\partial P}{\partial V}\right) \begin{cases} \leq 0, & \text{if } V^\circ = 0 & (a) \\ = 0, & \text{if } V^\circ > 0 & (b) \end{cases} \end{aligned} \quad (6.57)$$

In the following, we first look for solutions which satisfy conditions (6.56) and (6.57) as equalities. From the premium function (6.54), we have

$$\frac{\partial P}{\partial I} = (1 + \lambda)\pi(V) \quad (6.58)$$

$$\frac{\partial P}{\partial V} = (1 + \lambda)\pi'(V)I \quad (6.59)$$

With the help of the first of these two relationships, condition (6.56) may be converted to

$$\pi[V^\circ] \{1 - (1 + \lambda) \pi[V^\circ]\} u'[y_s] = \{1 - \pi[V^\circ]\} (1 + \lambda)\pi[V^\circ]u'[y_h]$$

yielding the ratio of marginal utilities,

$$\begin{aligned} \frac{u'[y_s]}{u'[y_h]} &= \frac{\{1 - \pi[V^\circ]\} (1 + \lambda) \pi[V^\circ]}{\{1 - (1 + \lambda) \pi[V^\circ]\} \pi[V^\circ]} = \frac{1 + \lambda - \pi[V^\circ] (1 + \lambda)}{1 - (1 + \lambda) \pi[V^\circ]} \\ &= \frac{1 - (1 + \lambda) \pi[V^\circ] + \lambda}{1 - (1 + \lambda) \pi[V^\circ]} = 1 + \frac{\lambda}{1 - (1 + \lambda) \pi[V^\circ]} \end{aligned} \quad (6.60)$$

If the insurance contract is *marginally fair*, then $\lambda = 0$, and the right-hand side of Eq. (6.60) is equal to 1, implying $u'[y_s] = u'[y_h]$. Due to the strict concavity of u , the equality of marginal utilities across states then requires

$$u[y_s] = u[y_h], \quad \text{and hence} \quad y_s = y_h \quad (6.61)$$

However, because of the definition of the two incomes, this means that $I^\circ = L$, which is compatible with the condition (6.56). Therefore, *full insurance coverage* will be chosen.

On the other hand, if λ is *positive*, the right-hand side of Eq. (6.60) exceeds 1, calling for a higher marginal utility of income in the sick state than in the healthy state. Because of the strict concavity of the utility function, this means $y_s < y_h$. Therefore, *partial insurance coverage* $I^\circ < L$ is optimal in the case of positive transaction costs [again in accord with condition (6.56)].

Finally, if λ is so high as to make the denominator of the ratio on the right-hand side of Eq. (6.60) approach 0 and hence the ratio itself ∞ , the condition (6.56) will be satisfied in its version (a). In this case optimal I° is equal to zero; that is, insurance will *not be taken out* at all.

CONCLUSION 6.5: In the presence of ex ante moral hazard but with marginally fair premiums graded to preventive effort of the insured, full insurance coverage is optimal. The greater the loading for transactions cost, the smaller the optimal degree of coverage; in the extreme, no insurance at all will be chosen.

The similarity with Conclusion 6.1 of Subsection 6.3.1.1, which refers to the case of no moral hazard, is noteworthy. In fact, as long as the insurer is able to estimate correctly the probability of illness associated with a given amount of prevention, he succeeds in neutralizing the effect of moral hazard.

6.4.2.2 Optimal Prevention

In principle, one would guess that preventive effort is reduced under the influence of insurance coverage. On the other hand, the insurance company by assumption can observe the insured's behavior, using premiums as an award or a sanction. Therefore, it is of interest to examine more closely condition (6.57), which determines the optimal amount of prevention. In analogy to the derivation of Eq. (6.53), it may be rewritten as

$$\pi'[V^\circ] \{u[y_s] - u[y_h]\} = \left(1 + \frac{\partial P}{\partial V}\right) E[u'(y)] \quad (6.62)$$

The left-hand side of Eq. (6.62) indicates the increase of utility occasioned by a decrease in the probability of illness, which in turn is brought about by an additional monetary unit of expenditure on prevention. The right-hand side reflects the associated reduction in utility, since disposable income in both states decreases by $1 + \partial P/\partial V$ (the former being the additional unit of expenditure on prevention, the latter the concomitant premium reduction). Multiplication by the expected marginal utility of income transforms this sum of money into utility units. The increase and decrease of expected utility must obviously balance at an interior optimum.

Let us now return to the special case of a *marginally fair* insurance policy ($\lambda = 0$). Because of Eq. (6.61), the left-hand side of condition (6.62) then takes on the value of zero. With $\partial P/\partial V$ given by Eq. (6.59), $I^\circ = L$, and $y_s = y_h$, one has

$$0 = \{1 + \pi'[V^\circ]L\} u'(y) \quad (6.63)$$

Because of $u' > 0$, one obtains immediately

$$-\pi'[V^\circ] L = 1 \quad (6.64)$$

On the left-hand side, we find the reduction of the (fair) premium due to an additional monetary unit spent on prevention. On the right-hand side, the additional expenditure on prevention itself appears.

It is interesting to see that Eq. (6.64) also constitutes the condition for the maximization of expected income *in the absence of a health insurance*, the expected value if income amounts to

$$E(y) = \pi(V) (Y - V - L) + \{1 - \pi(V)\} (Y - V) \quad (6.65)$$

The first derivative of $E(y)$ with respect to V is given by

$$\begin{aligned} \frac{dE(y)}{dV} &= -\pi(V) + \pi'(V) (Y - V - L) - \{1 - \pi(V)\} - \pi'(V) (Y - V) \\ &= \pi'(V) (Y - V - L) - \pi(V) - \pi'(V) (Y - V) - \{1 - \pi(V)\} \end{aligned}$$

$$= -\pi'(V)L - 1 \begin{cases} \leq 0, & \text{if } V = 0 \\ = 0, & \text{if } V > 0 \end{cases} \quad (6.66)$$

However, this condition is equivalent to Eq. (6.64) for positive values of V .

CONCLUSION 6.6: If the insurer can observe the extent of prevention and if the premium is fair, preventive measures are undertaken to the point where expected disposable income—in the absence of insurance—is maximized. The equalization of this maximum expected income across the two states of health is achieved by taking out full insurance coverage.

From Eq. (6.66) we can also deduce that preventive measures will always be undertaken unless prevention is *ineffective* from the first unit on, in the sense that it fails to increase expected income, that is, if $|\pi'[0]| \leq 1/L$. For example, this would be the case if the first dollar spent on the prevention of illness reduced the probability of illness by less than 1 percentage point, while maximum health care costs amount to $L = 100$ US\$.

6.4.3 Optimal Insurance Coverage with Unobservable Prevention

6.4.3.1 Free Choice of Premium Function: Two-Stage Optimization

In the following, we will turn to a more realistic case, where the insurance company is *not able to* observe expenditure on prevention. Consequently, the insurance premium P does not depend on the extent of prevention V but only on benefits paid I ,

$$P = P(I) \quad (6.67)$$

In contrast to the premium function in Eq. (6.54), there is no incentive for the individual to increase expenditure on prevention in return for a reduced premium. Does this imply that the individual will totally refrain from prevention as a result of lacking incentives, or is there still a way to influence V through the choice of the function (6.67)?

A “reward” for increased preventive effort could be brought about as follows. The higher the degree of coverage, the smaller the incentive for the insured to take preventive measures, the higher therefore the probability of illness π , and the more expensive fair insurance *per unit benefit* I [in accordance with Eq. (6.8)]. Provided the insurer is aware of this, he might charge a premium that increases progressively rather than linearly with the level of benefits chosen. Consequently, a contract featuring full coverage would have a premium more than twice as high as, for example, a contract featuring a copayment of 50 percent. The individual will take this into account in his decisions, thus refraining from buying full insurance coverage while keeping up his preventive effort to some extent.

Formally, these connections can be modeled using a two-level utility maximizing calculus of the individual. At the *lower level*, he determines the optimal value of his expenditure on prevention V , taking the insurance benefit I as given. If this calculation is hypothetically carried out for every I , the corresponding optimal values may be written in the form of a “reaction function”

$$V = V(I) \quad (6.68)$$

At the *upper level*, the individual chooses I^* in full knowledge of his own reaction (6.68). Here, I^* stands for the best possible insurance coverage for an arbitrary premium function given that the insurer is unable to observe the behavior of the insured.

6.4.3.2 Optimal Prevention, Terms of Insurance Given

First, let us turn to the lower level, that is, the maximization of expected utility as a function of prevention V for parametrically fixed values of I and P ,

$$EU(V) = E\{u[y(V)]\} = \pi(V)u[Y - V - P - L + I] + \{1 - \pi(V)\}u[Y - V - P] \quad (6.69)$$

The first-order condition for a maximum to this problem is formally identical to Eq. (6.52), with the symbols y_s denoting $[Y - V - P - L + I]$ and y_h denoting $[Y - V - P]$, respectively,

$$\begin{aligned} \frac{dEU}{dV} &= \pi'(V)u[y_s] - \pi(V)u'[y_s] - \pi'(V)u[y_h] \\ &\quad - \{1 - \pi(V)\}u'[y_h] \begin{cases} = 0, & \text{if } V > 0 \\ \leq 0, & \text{if } V = 0 \end{cases} \end{aligned} \quad (6.70)$$

Introducing the expression $\pi(V)u'[y_s] + \{1 - \pi(V)\}u'[y_h] := EU'(y)$ and considering those values of I which pertain to an interior solution ($V > 0$), condition (6.70) reduces to

$$\pi'(V) \{u[y_s] - u[y_h]\} = EU'(y) \quad (6.71)$$

Furthermore, let the second-order condition for a maximum, $d^2EU/dV^2 < 0$, be satisfied.

The following characteristics of the reaction function $V(I)$, defined in Eq. (6.68), can be derived from the optimality conditions (6.70) and (6.71).

1. $V[0]$ corresponds to Eq. (6.53), which implicitly determines optimal prevention *without* availability of insurance, V^* .

2. $V[L]$ has the value 0; that is, given full insurance coverage, there is *no incentive* for spending money on prevention. Inserting the condition $I = L$ or $y_s = y_h = y$,

respectively, in Eq. (6.70) yields $dEU/dV = -u'(y) < 0$, causing (6.70) to be satisfied as an inequality for $V = 0$.

3. Surprisingly, the relationship between I and V *does not have to be negative throughout*. This means that it is uncertain whether increasing insurance coverage is accompanied by increasing or decreasing prevention. In order to see this, form the total differential of the left-hand side of Eq. (6.70), letting I increase by dI and the optimal value V° change by dV in response. With EU_V representing the left-hand side of Eq. (6.70), one obtains

$$dEU_V = \frac{\partial^2 EU(V, I)}{(\partial V)^2} dV + \frac{\partial^2 EU(V, I)}{\partial V \partial I} dI = 0 \quad (6.72)$$

as the optimality condition (6.70) must continue to hold after the change. Solving for dV/dI yields

$$\frac{dV}{dI} = - \frac{\partial^2 EU(V, I)/\partial V \partial I}{\partial^2 EU(V, I)/(\partial V)^2} = - \frac{\partial EU_V/\partial I}{\partial EU_V/\partial V} \quad (6.73)$$

Because V° is a maximum of EU , the second-order condition requires the first term on the right-hand side of Eq. (6.72) and thus the denominator on the right-hand side of Eq. (6.73) to be negative. Therefore, the sign of dV/dI equals the sign of the partial derivative of Eq. (6.70) with respect to I . Recalling that $\partial y^s/\partial I = 1$, one obtains

$$\frac{\partial EU_V}{\partial I} = \pi'(V)u'[y_s] - \pi u''[y_s] \quad (6.74)$$

Due to our assumptions $\pi' < 0$, $u' > 0$, and $u'' < 0$, the sign of this expression is *indeterminate*. The economic explanation of this indeterminacy is as follows. With increasing insurance coverage, the marginal utility of prevention may decrease (as the income gap between the states “sick” and “healthy” decreases), but so do the opportunity costs of prevention (as the marginal utility of money is smaller at a higher income). Thus the net effect of an increase of I on prevention V remains ambiguous.

CONCLUSION 6.7: With full insurance coverage, zero expenditure on prevention is predicted. With decreasing coverage, optimal expenditure on prevention generally increases, but not necessarily as a monotonic function.

6.4.3.3 Optimization of Terms of Insurance

How does this behavior with regard to prevention affect the shape of the premium function that was described in general terms by Eq. (6.67)? In order to answer this question, we assume actuarially *fair* insurance for the sake of simplicity, specifying the relationship between benefits and premium to be

$$P(I) = \pi\{V(I)\}I \quad (6.75)$$

with derivative

$$P'(I) = \pi\{V(I)\} + I\pi'\{V(I)\}V'(I) \quad (6.76)$$

The shape of this function may be described more precisely using the second derivative,

$$P''(I) = 2\pi'(V)V'(I) + I\pi''(V)V''(I) + I\pi''(V)\{V'(I)\}^2 \quad (6.77)$$

Due to our assumptions $\pi' < 0$ and $\pi'' > 0$, the right-hand side of Eq. (6.77) is positive if $V'(I) < 0$ and if $V''(I)$ is negative or only weakly positive. As a rule, the premium therefore *increases progressively* as a function of the level of insurance benefits chosen.

Turning to the higher level of the decision-making problem, benefits I are treated as the decision variable while the premium function (6.67) and the reaction function of the insured (6.68) are taken as predetermined. The objective function to be maximized thus reads

$$\begin{aligned} EU(I) = E\{u[y(I)]\} &= \pi\{V(I)\}u[Y - V(I) - P(I) - L + I] \\ &+ [1 - \pi\{V(I)\}]u[Y - V(I) - P(I)] \end{aligned} \quad (6.78)$$

The first-order condition for an interior optimum ($0 < I^+ < L$) is

$$\begin{aligned} \frac{dEU}{dI} &= \pi'(V)V'(I)u[y_s] - \pi u'[y_s] \{V'(I) + P'(I) - 1\} - \pi'(V)V'(I)u[y_h] \\ &- (1 - \pi)u'[y_h] \{V'(I) + P'(I)\} \\ &= V'(I) \{ \pi'(V)(u[y_s] - u[y_h]) - E[u'(y)] \} - P'(I)E[u'(y)] + \pi u'[y_s] = 0 \end{aligned} \quad (6.79)$$

Due to condition (6.71), the expression in curly brackets is zero. So Eq. (6.79) reduces to the following *implicit equation for optimal insurance coverage* I^+ , given premium function $P(I)$ and reaction function $V(I)$. With $\pi u'[y_s]$ written out in full, it becomes

$$P'(I^+)E[u'(y)] = \pi[V(I^+)]u'[Y - V(I^+) - P(I^+) - L + I^+] \quad (6.80)$$

The right-hand side of this optimality condition states the gain of utility associated with an increase of insurance coverage by a marginal monetary unit. This utility gain is given by the product of the probability of sickness multiplied by the marginal utility of money. On the left-hand side, we see the loss of utility associated with the increase of the insurance premium, transformed into utility units by the expected value of the utility of money because the additional premium must be paid in both states of the world. At the optimum point, the two effects should balance.

However, the optimal values I^* for benefits and $V^* = V[I^*]$ for prevention, determined by the conditions (6.80) and (6.70), respectively, represent a second-best solution only. This means that they result in a lower expected utility than the first-best optimal values determined in Section 6.4.2 for the case of a variable premium, where $I^\circ (= L)$ and V° come from Eq. (6.64). This follows from the fact that given a variable premium, the values I^* and V^* were already available but not chosen. Equality of $\{I^*, V^*\}$ and $\{(I^\circ = L), V^\circ\}$ occurring by coincidence can be excluded, as the values $\{L, V^\circ\}$ are not compatible with the reaction function given in Eq. (6.68). Indeed, as shown above, the reaction function requires $V[L] = 0 < V^\circ$, since preventive effort V° is in general strictly positive, according to condition (6.64).

CONCLUSION 6.8: If the insurer cannot observe the extent of prevention while insurance is fair, the premium increases progressively with coverage as a rule, causing the insured to choose less than full coverage. This is a second-best solution because the insured's expected utility is lower *ceteris paribus* than if preventive effort was observable.

In contrast to the case of Section 6.4.2, the premium function of Eq. (6.75) does not require observability of individual expenditure on prevention. However, the insurance company still needs to know the form of the functions $\pi(V)$ and $V(I)$. Thus information requirements continue to be quite high.

As the premium tends to rise progressively rather than proportionally as a function of insurance coverage [see Eq. (6.77)], it is necessary to prevent the insured from combining several insurance contracts, each featuring small benefits, in order to achieve his target coverage, thus undermining the progression of the premium. The second-best optimum of Conclusion 6.8 therefore can be sustained only if there is a single supplier of health insurance or if there is a general *obligation to register contracts*, permitting the premium to be made dependent on total insurance coverage taken out.

6.4.3.4 The Case of a Linear Premium Function

If the condition mentioned at the end of the previous paragraph is not satisfied, viz. if the insurer fails to know total coverage taken out by the insured, then a progressive premium function cannot be enforced. To recall, the insured may avoid progression by signing several contracts of small coverage (thus staying in the flat of the premium function), causing the premium function to be essentially linear in I . Thus one has

$$P(I) = \sigma I \quad (6.81)$$

where σ is the premium per unit benefit. Function (6.69), which shows the relationship between expected utility, prevention, and insurance, therefore changes to read

$$EU(V, I) = \pi(V)u[Y - V - \sigma I - L + I] + \{1 - \pi(V)\}u[Y - V - \sigma I] \quad (6.82)$$

Individual behavior is described by the following first-order conditions for an optimum, the first being formally equivalent to Eqs. (6.70) and (6.71), respectively,

$$\frac{\partial EU}{\partial V} = \pi'(V) \{u[y_s] - u[y_h]\} - EU'(y) \begin{cases} = 0, & \text{if } V > 0 \\ \leq 0, & \text{if } V = 0 \end{cases} \quad (6.83)$$

$$\begin{aligned} \frac{\partial EU}{\partial I} &= -\pi(V) (\sigma - 1)u'[y_s] - \{1 - \pi(V)\} \sigma u'[y_h] \\ &= \pi(V)u'[y_s] - \sigma EU'(y) \begin{cases} = 0, & \text{if } I < L \\ \geq 0, & \text{if } I = L \end{cases} \end{aligned} \quad (6.84)$$

We now want to prove that in the case of actuarially fair insurance ($\sigma = \pi$), the values $I = L$ and $V = 0$ satisfy the conditions (6.83) and (6.84) for a boundary optimum. Indeed, if $I = L$ is inserted in the definition of disposable income, it follows that $y_s = y_h$ and thus $u[y_s] - u[y_h] = 0$ in Eq. (6.83). Due to $u' > 0$, condition (6.83) then must hold in case of the $<$ sign. Analogously, for $I = L$ it follows that $u'[y_s] = EU'(y)$, so that inequality (6.84) reduces to

$$\pi[0] - \sigma \geq 0, \quad \text{or } \sigma \leq \pi[0] \quad (6.85)$$

The right-hand side of Eq. (6.85), however, equals the fair premium in the case where *no prevention* is undertaken. Assuming that the insured do without prevention, the insurance company will set the premium per unit of insurance coverage, σ , to $\pi[0]$ in the absence of administrative expenditures, causing Eq. (6.84) to be satisfied.

On the other hand, conditions (6.83) and (6.84) show that an individual who maximizes his expected utility, regarding the per unit insurance premium as exogenous, does in fact go without prevention while taking out *full coverage* if “fair” terms are offered. In turn, this behavior corresponds to the insurer’s premium calculation, which is based on the expectation that no preventive effort is undertaken. The solution therefore represents an equilibrium.

Finally, we would like to point out that the combination $\{I = L, V = 0\}$ is a third-best solution. This becomes evident when recalling that this alternative was already available in the choice set for a second-best optimum but failed to be opted for.

CONCLUSION 6.9: If the premium is a linear function of the level of benefits selected, there is no incentive for the insured, given fair insurance, to undertake preventive efforts. At the same time, he will opt for comprehensive coverage, resulting in a value of expected utility that is still lower than in the second-best solution described in Conclusion 6.8.

6.4.4 Summary

Doing without full insurance coverage is optimal not only if insurance is unavailable at marginally fair terms. It may also be justified in terms of the insured’s own incentives for loss prevention. As soon as one has to pay for part of one’s cost of treatment, one is inclined to spend time and money on prevention. However, for such an incentive to be effective, the insurance premium must increase *more than*

proportionally with benefits. With a merely proportional relationship, the incentive vanishes, resulting in a solution that is worse than the one inducing preventive effort, judged by the expected utility criterion.

Full insurance coverage is only compatible with an effective incentive for prevention in the (unrealistic) case that the insurance company is able directly to observe preventive effort and *adjust its premium* accordingly. If these information requirements were satisfied, a higher level of welfare could be reached.

6.5 OPTIMAL INSURANCE COVERAGE WITH EX POST MORAL HAZARD

In contrast to Section 6.4, this section is devoted to the effect of moral hazard on optimal health insurance coverage in its ex post form. In the model to be presented below, there is no possibility to reduce the risk of illness, but there is a choice between different treatment alternatives once an illness has occurred.

6.5.1 Model Assumptions

Consider a single-period situation with only two goods, a homogeneous health good (“medical treatment”) and other consumption. The relative price of medical treatment is supposed to be constant and can thus be normalized to 1. Accordingly, the respective *quantities* of the goods consumed by the individual may be replaced by the *expenditure* on medical treatment M and on other consumption y . Given health insurance coverage, I denoting benefits, P the premium, and Y the individual’s gross income, his budget constraint reads

$$y = Y - P - M + I \quad (6.86)$$

Suppose there are two possible states the individual can be in. State “sick” occurs with the (known) probability π and state “healthy” with probability $1 - \pi$. Moreover, suppose the individual does not know his future health status at the beginning of the period. Let the two health states be characterized by different utility functions (see Section 6.3.2), since medical services naturally are valuable to a sick person but useless (at best) to a healthy one. Therefore, the sick person’s utility function is given by $u^s(M, y)$ and the healthy person’s, by $u^h(M, y)$. The following assumptions are made concerning these utility functions.

ASSUMPTION 6.1: The utility of a healthy person depends on other consumption only; that is, we have $u_M^h := \partial u^h(M, y) / \partial M = 0$.

ASSUMPTION 6.2: Both conditional utility functions are strictly increasing and strictly concave with respect to the consumption good; that is, for $j = h, s$, we have

$$u_y^j := \frac{\partial u^j(M, y)}{\partial y} > 0, \quad u_{yy}^j := \frac{\partial^2 u^j(M, y)}{\partial y^2} < 0 \quad (6.87)$$

ASSUMPTION 6.3: Holding consumption constant, utility in the sick state is a strictly concave function of medical care M ,

$$u_{MM}^s := \frac{\partial^2 u^s(M, y)}{\partial M^2} < 0 \quad (6.88)$$

The quantity of treatment M^* at which marginal utility is precisely zero will be called “satiation quantity”

$$u_M^s[M^*, y] = 0 \quad (6.89)$$

ASSUMPTION 6.4: At a given level of consumption, marginal utility of consumption is always higher in the healthy than in the sick state,

$$u_y^h[0, y] > u_y^s[M, y], \quad \text{for all values of } M$$

ASSUMPTION 6.5: Finally, medical care and consumption are *weakly complementary goods* in the “sick” state; that is, the marginal utility of consumption does not fall with increasing utilization of the health good as long as the latter is below the satiation level,

$$u_{My}^s := \frac{\partial^2 u^s(M, y)}{\partial M \partial y} \geq 0, \quad \text{if } M < M^* \quad (6.90)$$

Assumptions 6.1 and 6.2 do not require justification. In Assumptions 6.3, M^* is seen as the “medically indicated” quantity of medical services. Assumptions 6.3 and 6.4 state that with improvements in health the capacity to enjoy consumption increases, but no amount of medical services is sufficient to restore health as to permit the patient fully to enjoy consumption y .

From Assumption 6.1 it follows that in the healthy state, the optimality condition requires $M = 0$ always, that is, zero demand for medical care. The insured’s expected utility, evaluated prior to the random occurrence of illness, is then given by

$$\begin{aligned} EU(M, I) &= \pi u^s[M, y_s] + (1 - \pi) u^h[0, y_h] \\ &= \pi u^s[M, Y - P(I) - M + I] + (1 - \pi) u^h[0, Y - P(I)] \end{aligned} \quad (6.91)$$

where $P(I)$ denotes the premium associated with insurance benefits I . The elements of this model appear in Table 6.3.

Table 6.3: Optimal Insurance Coverage and Optimal Demand for Medical Services

$$\frac{\partial EU}{\partial M} = \pi \{u_M^s[M^\circ, y_s] - u_y^s[M^\circ, y_s]\} = 0 \quad (6.92)$$

$$\begin{aligned} \frac{\partial EU}{\partial I} &= \pi u_y^s \{1 - P'(I)\} - (1 - \pi) u_y^h P'(I) \\ &= \pi u_y^s - P'(I) \{ \pi u_y^s + (1 - \pi) u_y^h \} = 0 \end{aligned} \quad (6.93)$$

$$\frac{du^s}{dM} = u_M^s - u_y^s \{1 - I'(M)\} = 0 \quad (6.97)$$

$$I(M) = (1 - c)M, \quad 0 \leq c \leq 1 \quad (6.98)$$

$$\frac{du^s}{dM} = u_M^s[M, Y - P - cM] - cu_y^s[M, Y - P - cM] = 0 \quad (6.99)$$

$$EU(c) = \pi u^s[M(c), Y - P(c) - cM(c)] + (1 - \pi) u^h[0, Y - P(c)] \quad (6.105)$$

$$\frac{1 - c^\circ}{c^\circ} e(M, c) = 1 - \frac{u_y^s}{(1 + \lambda) \{ \pi u_y^s + (1 - \pi) u_y^h \}} \quad (6.113)$$

y :	Disposable income of the individual	I :	Insurance benefits
Y :	Gross income	P :	Insurance premium
M :	Expenditure on medical care	c :	Rate of coinsurance
π :	Probability of illness	$e(M, c)$:	Elasticity of demand for medical care with respect to the rate of coinsurance
$u(M, y)$:	Utility of the individual		
EU :	Expected utility		

6.5.2 Optimal Demand for Medical Services and Optimal Insurance Coverage

6.5.2.1 Global Optimization

Suppose that before the start of the period considered, that is, still under the “veil of uncertainty” about his future health status, the individual chooses his demand for medical care M in the event of illness as well as insurance benefits I in such a way as to maximize his expected utility (6.91). The necessary first-order conditions are obtained by setting the first partial derivatives with respect to M and I equal to zero,

$$\frac{\partial EU}{\partial M} = \pi \{u_M^s[M^\circ, y_s] - u_y^s[M^\circ, y_s]\} = 0 \quad (6.92)$$

$$\frac{\partial EU}{\partial I} = \pi u_y^s \{1 - P'(I)\} - (1 - \pi) u_y^h P'(I) = \pi u_y^s - P'(I) \{ \pi u_y^s + (1 - \pi) u_y^h \} = 0 \quad (6.93)$$

Equation (6.92) states that due to $\pi > 0$, in the state of illness the individual purchases the quantity of medical services that equalizes the marginal utilities of the two goods.

Due to Assumptions 6.2 ($u_y^s > 0$) and 6.3 (decreasing marginal utility of medical services), the optimal quantity must be below the satiation level such that $M < M^*$.

In order to interpret Eq. (6.92), we consider the premium function [in analogy to Eq. (6.17)],

$$P(I) = (1 + \lambda)\pi I \quad (6.94)$$

Its first derivative is given by

$$P'(I) = (1 + \lambda)\pi \quad (6.95)$$

From this it follows that, given fair insurance [i.e., $\lambda = 0$ and hence $P'(I) = \pi$], condition (6.92) is satisfied if and only if

$$u_y^s[M^\circ, y_s] = u_y^h[0, y_h] \quad (6.96)$$

that is, when the marginal utility of consumption is equal in both states of health.

By Assumption 6.4, $M^\circ < M^*$, and Assumption 6.5 (complementarity), equality of marginal utility of income in both states of nature is possible only if $y_s > y_h$, such that [from Eq. (6.86)] the optimal insurance benefit I° must *exceed* expenditure associated with the optimal level of treatment M° .

Does this mean that if actuarially fair insurance is available, the insurance benefit I° is determined as to exceed medical expenditure by a certain percentage?

To answer this question, let us consider the utility-maximizing behavior of the individual in the ex post situation, that is, after the occurrence of illness, with insurance benefits and premium already determined. Moreover, suppose that benefits depend on health care expenditure one way or another; that is, $I = I(M)$. Then the sick individual maximizes his utility $u^s\{M, Y - P - M + I(M)\}$ by choosing the quantity of M that satisfies the necessary first-order condition,

$$\frac{du^s}{dM} = u_M^s - u_y^s \{1 - I'(M)\} = 0 \quad (6.97)$$

However, Eq. (6.97) is consistent with the global optimizing condition (6.92) only if $I'(M) = 0$, that is, if *benefits do not depend* at all on the individual's expenditure on medical treatment.

Such a lack of dependence in turn requires the insurer to be able to observe not only the individual's health expenditure but also his health status. In this case, benefits take the form of a lump-sum payment ("indemnity") amounting to I° in the event of illness. Having this type of policy, the insured will choose just the optimal quantity M° when ill [see Eq. (6.96)]. Therefore, the indemnity I° , while exceeding the amount of optimal medical expenditure M° , does not depend on it. The insurance contract thus provides for a *lump-sum payment in the event of illness* that is independent of *actual* medical expenditure.

CONCLUSION 6.10: If the insurer is able to observe the insured's health status, then the optimal form of the contract provides for a lump-sum benefit that depends on health status only. This so-called indemnity exceeds the utility-maximizing amount of medical expenditure in the sick state.

6.5.2.2 Optimal Demand for Medical Care

In reality, the globally optimal solution characterized in the previous subsection cannot be attained because of the insurer's inability unambiguously to determine the health status of the insured. Rather, health care expenditure itself serves as an indicator of illness, causing benefits to depend on expenditure through a function $I(M)$. Then the question arises of the optimal form of this function I (as a second-best solution). In the following, we shall limit ourselves to the case of a single parameter, the rate of coinsurance. Thus the benefit function is of the form

$$I(M) = (1 - c)M, \quad \text{with } I'(M) = 1 - c, \quad 0 \leq c \leq 1 \quad (6.98)$$

In this formulation, c symbolizes a constant rate of coinsurance.⁸ For example, $c = 0.2$ calls for a 20 percent copayment by the insured.

The individual thus faces a *two-step optimization problem*. In the event of illness, the utility-maximizing bundle of goods $\{M, y\}$ has to be found, the rate of coinsurance c being exogenous. Ex ante, the value of c that maximizes expected utility as given by Eq. (6.91) has to be determined, taking the previously established ex post behavior into account.

Let us first turn to the *optimal demand for medical care* in the event of illness (ex post optimization), which has already been considered in Subsection 6.5.2.1 for the case of a general benefit function. Inserting the specific form (6.97) into the corresponding optimality condition (6.96), noting that $1 - I'(M) = c$, and writing out in full the relevant values of disposable income, one has

$$\frac{du^s}{dM} = u_M^s[M, Y - P - cM] - cu_y^s[M, Y - P - cM] = 0 \quad (6.99)$$

Due to Assumptions 6.2, 6.3, and 6.5, the second-order condition for a maximum is satisfied,

$$\frac{d^2u^s}{dM^2} = u_{MM}^s - 2cu_{My}^s + c^2u_{yy}^s < 0 \quad (6.100)$$

Now the *demand function for medical care*, with the rate of coinsurance c as its argument, may be derived from Eq. (6.99),⁹

$$M = M(c), \quad \text{with } M[0] = M^* \quad (6.101)$$

8. Generally, the optimal rate of coinsurance is a nonlinear function of M [see Spence and Zeckhauser (1971), p. 387].

9. If $c = 0$, then Eq. (6.99) is satisfied at $M = M^*$, due to condition (6.89).

Suppose this function is continuously differentiable, and let us determine the sign of its first derivative. In doing this, we will find the effect of a change dc in the rate of coinsurance on the demand for medical care dM . Since the first-order condition (6.99) must continue to hold after this change, the total derivative of Eq. (6.99) is given by

$$d \left(\frac{du^s}{dM} \right) = \frac{\partial^2 u^s}{\partial M^2} dM + \frac{\partial^2 u^s}{\partial M \partial c} dc = 0 \quad (6.102)$$

Solving this equation for dM/dc , one obtains

$$\frac{dM}{dc} = - \frac{\partial^2 u^s / \partial M \partial c}{\partial^2 u^s / \partial M^2} \quad (6.103)$$

Due to the second-order condition (6.100), the denominator on the right-hand side of this equation is negative. Thus, the whole expression has the same sign as [see Eq. (6.99)]

$$\frac{\partial^2 u^s}{\partial M \partial c} = -u_{My}^s M + cu_{yy}^s M - u_y^s < 0 \quad (6.104)$$

This expression is negative because of Assumptions 6.2 and 6.5. We thus have shown that dM/dc is negative.

CONCLUSION 6.11: The lower the rate of coinsurance, the larger is the utility-maximizing consumer's demand for medical care in the state of illness ("moral hazard of the ex post form").

6.5.2.3 Optimal Proportional Cost Sharing

Next we determine the ex ante optimal insurance coverage. To do this, let us assume that the individual optimizes the expected utility function represented in Eq. (6.91). It is rewritten below to show explicitly that both the demand for medical care M and the premium P depend on the rate of coinsurance c selected,

$$EU(c) = \pi u^s[M(c), Y - P(c) - cM(c)] + (1 - \pi) u^h[0, Y - P(c)] \quad (6.105)$$

The premium function $P(c)$ is a special case of the function (6.94). Thus, using Eq. (6.98), and introducing $M'(c) := dM/dc$, one has

$$P(c) = \pi(1 + \lambda) (1 - c)M(c), \quad \text{with} \quad (6.106)$$

$$\begin{aligned} P'(c) &= \pi(1 + \lambda) \{ (1 - c)M'(c) - M(c) \} \\ &= \pi(1 + \lambda) \{ (1 - c)M'(c) \} - \pi(1 + \lambda)M(c) \end{aligned} \quad (6.107)$$

Forming the first derivative of Eq. (6.105) with respect to c and equating it to zero, we obtain the following implicit condition for the optimal rate of coinsurance c° ,

$$0 = EU'[c^\circ] = \pi u_M^s M'[c^\circ] - \pi u_y^s \{P'[c^\circ] + M[c^\circ] + c^\circ M'[c^\circ]\} - (1 - \pi) u_y^h P'[c^\circ] \quad (6.108)$$

This expression can be transformed in the following way. Inserting Eq. (6.98) into the optimality condition (6.97), one obtains

$$u_M^s = u_y^s \{1 - (1 - c^\circ)\} = c^\circ u_y^s \quad (6.109)$$

Condition (6.108) may thus be written, collecting terms in $P'(c^\circ)$,

$$\begin{aligned} 0 &= M'[c^\circ] \{ \pi c^\circ u_y^s \} - \pi u_y^s M[c^\circ] \\ &\quad - \{ \pi u_y^s + (1 - \pi) u_y^h \} P'[c^\circ] - \pi c^\circ u_y^s M'[c^\circ], \quad \text{or} \\ \{ \pi u_y^s + (1 - \pi) u_y^h \} P'[c^\circ] &= -\pi u_y^s M[c^\circ] \end{aligned} \quad (6.110)$$

Using the shorthand notation $EU'(y) := (\pi u_y^s + (1 - \pi) u_y^h)$ and Eq. (6.107), Eq. (6.110) becomes

$$EU'(y) \pi (1 + \lambda) \{ (1 - c^\circ) M'[c^\circ] - M[c^\circ] \} = -\pi u_y^s M[c^\circ] \quad (6.111)$$

After dividing this expression by $EU'(y) \pi (1 + \lambda)$, one obtains the following implicit equation for c° :

$$(1 - c^\circ) M'[c^\circ] = M[c^\circ] \left(1 - \frac{u_y^s}{(1 + \lambda) EU'(y)} \right) \quad (6.112)$$

As a final step, Eq. (6.112) may be converted into a condition involving the elasticity of the demand for medical care with respect to the rate of coinsurance. This elasticity is given by $e(M, c) := (dM/dc) (c/M) = M'[c^\circ] (c^\circ/M)$ if evaluated in the neighborhood of c° . Dividing Eq. (6.112) by $M[c^\circ]$ and multiplying the left-hand side by $c^\circ/c^\circ = 1$, one has

$$\frac{1 - c^\circ}{c^\circ} e(M, c) = 1 - \frac{u_y^s}{(1 + \lambda) EU'(y)} = 1 - \frac{u_y^s}{(1 + \lambda) \{ \pi u_y^s + (1 - \pi) u_y^h \}} \quad (6.113)$$

This condition determines the *optimal rate of coinsurance*. Perhaps the easiest way to find the solution is using a graph (see Figure 6–5). First, the graph of the left-hand side (LHS) of Eq. (6.113) is constructed, assuming $e(M, c)$ to be independent of c . For $c = 0.5$, the left-hand side coincides with $e(M, c)$. According to the evidence presented in Section 6.6, $e(M, c) = -0.2$ may be considered realistic. The graph thus must run through point A of Figure 6–5 (solid line), starting from the value $-\infty$ when $c \rightarrow 0$ [due to the factor $(1 - c)/c$] and approaching 0 when $c \rightarrow 1$ [again due to the factor $(1 - c)/c$]. It must be in the negative orthant since $e(M, c) < 0$.

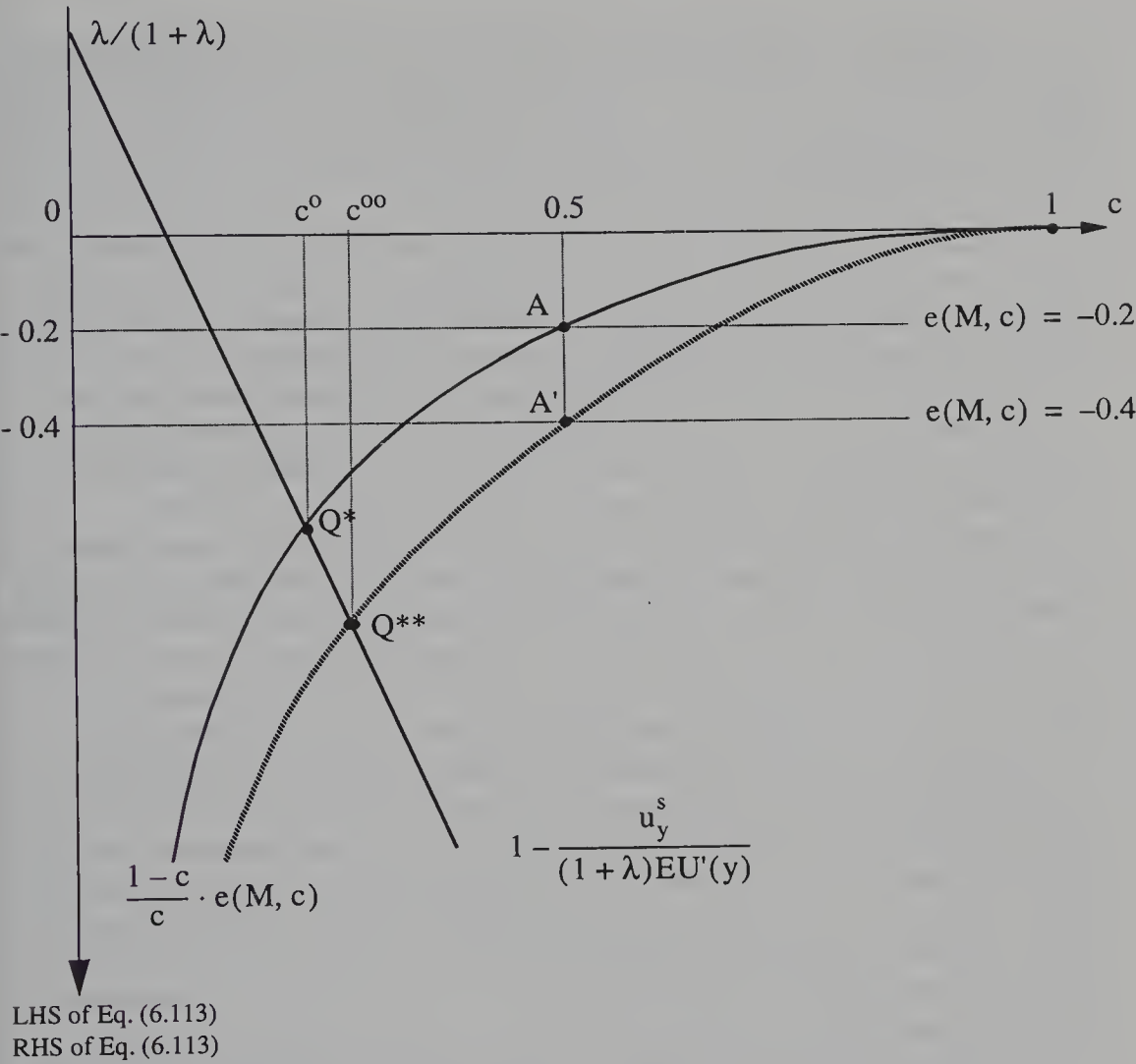


Figure 6-5. Optimal rate of coinsurance with ex post moral hazard.

The right-hand side (RHS) of Eq. (6.113) depends on the rate of coinsurance c in a complicated way because a change in c affects income in both states [and hence u_y^s as well as $EU'(y)$]. However, its impact on $EU'(y)$, resulting from two opposing effects involving u_{yy}^s and u_{yy}^h , is likely to be small. Treating $EU'(y)$ as a constant, one obtains a negative relationship between the rate of coinsurance and the RHS of Eq. (6.113), since the marginal utility of income in the sick state u_y^s will be greater the higher the rate of coinsurance selected. The limit value of the RHS for $c = 0$ is $1 - 1/(1 + \lambda) = \lambda/(1 + \lambda)$ because without coinsurance, the marginal utilities of income are equal across states of health ($u_y^s = u_y^h$). Altogether, the right-hand side of Eq. (6.113) may be represented by a negatively sloped line in the negative orthant of Figure 6-5. The intersection of the two loci at point Q^* then determines the optimal amount of coinsurance, c° .

The argument based on Eqs. (6.105)–(6.113) gives rise to the following three observations:

1. Suboptimality of full insurance coverage. It can be shown that full coverage is suboptimal. Heuristically, this follows from the left-hand side of condition (6.113), which goes to minus infinity for a constant elasticity $e(M, c)$, as can also be gleaned from Figure 6–5, whereas the right-hand side is finite because by Assumption 6.2, the denominator of the fraction is always strictly positive.

2. Effect of an increase in the health risk π . When the health risk π increases, the denominator of the fraction on the right-hand side of Eq. (6.113) becomes larger (because of $u_y^s > u_y^h$), causing the absolute value of the right-hand side (which must be negative) to decrease. Assuming again a constant elasticity $e(M, c)$, the term $(1 - c^\circ)/c^\circ$ on the left-hand side of Eq. (6.113) must decrease and the optimal *rate of coinsurance must increase*. In Figure 6–5, the straight line representing the right-hand side of Eq. (6.113) would have to rotate upward (not shown), causing the point of intersection to be shifted to the right.

At first sight, the response $dc^\circ/d\pi > 0$ does not seem plausible. However, it may be explained as follows. Provided the premium is loaded with transaction costs, then the greater the probability of occurrence, the less it pays to be insured against it (see Conclusion 6.1). Here, the transaction costs are represented by the incentive to inflate the demand for medical care if insurance coverage is ample (see Conclusion 6.11). Since these transaction costs increase with π , the insured will reduce his coverage.

3. Effect of an increased elasticity of demand. Finally, let us examine the relationship between the optimal rate of coinsurance and the elasticity of the demand for medical care with respect to coinsurance. This can be done quite easily with the help of Figure 6–5. Recall that when c is set at 0.5, the graph representing the left-hand side of Eq. (6.112) shows the elasticity $e(M, c)$ itself. If $e(M, c) = -0.4$ (say) rather than -0.2 as assumed previously, the reference point A thus moves downward to A' . The entire graph thus shifts downward, causing the point of intersection to move away from the origin from Q^* to Q^{**} .

The optimal rate of coinsurance c° thus *increases* if the elasticity of demand for medical care with respect to coinsurance (which is in fact a price elasticity) increases. This result is plausible since a higher price elasticity means that the insured responds more strongly to price incentives, and in order to let these incentives take effect, the insured has to choose a higher rate of coinsurance.

We may summarize the results of this subsection in

CONCLUSION 6.12: The optimal rate of coinsurance must be strictly positive since the demand for medical care increases with decreasing coinsurance. In addition, it must *ceteris paribus* increase with a higher health risk as well as with a higher price elasticity of the demand for medical care.

6.5.3 Summary

We have learned that apart from the motives given in Sections 6.3 and 6.4, there is still another reason for refraining from purchasing full insurance coverage. Full coverage is not optimal if the insured, in his attempt to maximize utility in the “sick” state, is able

to influence the amount of medical care expenditure. In this case, it is to his advantage to choose an insurance contract with coinsurance, creating an incentive to *use health goods in a cost-conscious manner*. If he had to pay a zero effective price, he would consume medical care up to the point of satiation, that is, the quantity that has a marginal utility of zero.

Yet, full optimality in the sense of maximum expected utility is attained only if the marginal utility of one unit of expenditure is equal in all possible types of use. This, however, requires that the insured bears the full marginal costs of medical treatment; that is, benefits must not depend on the insured's health care expenditure but be linked to his health status ("indemnity" insurance). But this would require the insurer to be able to observe the insured's health status.

These considerations show clearly that the above-mentioned choice of coinsurance in case of nonobservable health status represents only a *second-best solution*. In the event of illness, disposable income is still smaller and hence marginal utility of income larger than in good health. Thus the global maximum of expected utility described in Subsection 6.5.2.1, calling for equal marginal utility of consumption in all states, is not attained.

This failure stems from the insured being forced to choose between the lesser of two evils. Opting for a low rate of coinsurance, he induces himself to consume a lot of costly medical services, causing him to face the "Scylla" of an *expensive insurance premium*. Deciding in favor of a high rate of coinsurance, he runs into the "Charybdis" of an *incomplete shifting* of risk onto the insurer. The course of the golden mean between Scylla and Charybdis depends on the price elasticity of the insured's demand. For instance, if his demand is totally inelastic, incentives are ineffective anyway, permitting him to opt for a complete shifting of risk through a contract without coinsurance.

These considerations highlight the great importance of empirical research into the effects of cost sharing on the demand for medical care. The following section is dedicated to this issue.

6.6 THE EMPIRICAL RELATIONSHIP BETWEEN INSURANCE COVERAGE AND THE DEMAND FOR HEALTH CARE SERVICES

In the 1970s numerous empirical studies dealing with the impact of health insurance on the utilization of medical services were published in the United States. In contrast to, for example, Germany, where members of a sick fund all have the same coverage, the United States with its enormous variety of insurance contracts is perfectly suitable for this type of investigation.

Since the rate of coinsurance changes the (effective) price, the insured has to pay for his medical care, the primary goal of these surveys was to estimate the price elasticity of demand for medical services. Of course, as many influencing factors as possible had to be considered, such as age, sex, income, and state of health, calling for multiple regression analysis in most cases. The *values of the price elasticity* found in these studies [see, e.g., Phelps (1975), Newhouse (1981)] mainly lie between 0 and

−0.2. They thus had the expected sign, although they were relatively low. However, some studies [see Newhouse, Phelps, and Marquis (1980), p. 378] found considerably higher values, up to −2.1.

In the 1980s some studies from the German-speaking countries were presented. Using data from Swiss social health insurance and German private health insurance, respectively, Schulenburg (1987) and Zweifel (1988) examined the reaction of demand to fixed annual deductibles and rebates for no claims. Dynamic bonus options, linking the rebate to the time elapsed since the last claim presented, turned out to be associated with a significant reduction in the utilization of medical care [Zweifel (1992)].

The methodology used in these studies was criticized, however, on the grounds that the negative correlation between the deductible and the utilization of medical services might also be the consequence of the consumer's *choice of the insurance contract*. Individuals who foresee only little utilization of medical services for themselves (due to their robust health or their aversion against orthodox medicine) may well have selected a higher deductible than others. These others in turn would be more likely to opt for full insurance coverage. Causality then runs in reverse, from low (expected) demand to cost sharing rather than from cost sharing to reduced demand [see Newhouse et al. (1980, 1993)].

In order to exclude this effect of self-selection of risks when estimating the price elasticity of demand for medical care, the Rand Corporation initiated a large-scale "health insurance experiment" at the end of the 1970s, costing over 80 million US\$. At six different locations, approximately representative of the entire United States, around 1,000 persons each were enrolled for 3 to 5 years in an insurance company especially created for the purpose of this study. Participants were assigned *randomly* to one of sixteen health insurance contracts, with the rate of coinsurance varying between 0 and 95 percent, while some policies also featured fixed deductibles.

The statistical analysis of this experiment [see Manning et al. (1987)] yielded estimates of the price elasticity of demand for medical services lying approximately between −0.1 and −0.2, that is, within the range of earlier studies but towards the lower end. In particular, people confronted with a high deductible turned out to have a high probability of not seeing a doctor at all within a given period of time. But once a doctor is consulted, the quantity of services demanded does not seem to vary much with insurance coverage [see Keeler and Rolph (1983)].

Furthermore, the experimental character of the study allowed the gathering of information about participants' health status at the beginning and at the end of the experiment. In this way, it was possible to test the widespread opinion that the financial barrier created by cost sharing would lead to an underconsumption of medical services and indirectly to a *deterioration of health status*.¹⁰

Therefore six different indicators of health were measured [see Brook et al. (1983)]. Three of these indicators (high blood pressure, myopia, and caries) actually showed a slight improvement among insureds with low incomes when there was full insurance coverage. The authors argue, however, that the same outcome could have been

10. Although this claim is particularly popular in Germany, no empirical evidence supporting it was found in connection with bonus options for no claims in (private) German health insurance [see Zweifel (1992), Chapter 8].

achieved through comparably cheap targeted mass screening programs rather than costly “100 percent-insurance.” Nonetheless, it has to be conceded that long-term health consequences of this experiment are not known yet, and it is doubtful whether they can ever be fully known.

CONCLUSION 6.13: The demand for medical services, especially for initial physician visits, reacts in a statistically significant way to the extent of insurance coverage, although the estimated price elasticity is a low -0.1 to -0.2 . The additional utilization induced by more extensive insurance coverage appears to result in a merely marginal improvement of the state of health.

6.7 IMPLICATIONS FOR THE DESIGN OF SOCIAL HEALTH INSURANCE

In Chapter 5, reasons for the existence of a compulsory health insurance scheme were presented. Specifically, compulsory insurance serves to tackle the “free-rider problem” and to redistribute resources in favor of those who are disadvantaged in terms of their innate health. For its implementation, however, a minimum package must be defined to prevent the scheme from being circumvented by plans providing but nominal coverage.

In Germany’s statutory health insurance, for example, this problem is solved by providing coverage without cost sharing for almost all types of medical treatment. The only exceptions are

- Dentures, where copayment by patients may attain 50 percent;
- Pharmaceuticals, for which a limit price has been set (see Section 10.3). If the patient chooses a drug of the same substance category whose price exceeds the limit, he has to pay the full difference. While this regulation may encourage price competition among suppliers of a homogenous product, it fails to create an incentive for the insured to limit his *quantitative* consumption as long as the price of the drug does not exceed the limit price, a feature distinguishing it from coinsurance.

In view of the theoretical analysis of this chapter, full insurance coverage of this type cannot be considered optimal. The following points may be cited in support of this claim.

1. Administrative expenditure does increase with benefits (see Section 6.3, Conclusion 6.1). This problem has been solved in German statutory health insurance by providing benefits in kind. Thus the insured do not submit their bills to the sick fund. Rather, the fund receives a quarterly billing for all its members directly from the medical providers. Although cost-saving, this practice causes a lack of transparency in that not even the insured himself can keep track of services consumed, let alone the expenditure associated with them.

2. There is too little incentive to invest in prevention, since sick funds do not make any efforts at observing preventive effort. Moreover, they fail to honor them through a premium reduction to combat “ex ante moral hazard” (see Section 6.4, Conclusion 6.7).

3. The price elasticity of demand for health care services, although not very high, is different from zero, giving rise to “ex post moral hazard” (see Sections 6.5 and 6.6, Conclusion 6.13).

Finally, the formal analysis showed that the *optimal* deductible depends on individual health risk and individual preferences and thus may be *very different among individuals* [see Eq. (6.47), Conclusion 6.8, and Eq. (6.112)]. Therefore, mandated uniform insurance coverage cannot be welfare-maximizing. It seems to be a better idea to let lawmakers determine a minimal amount of insurance coverage while leaving it up to the individual to extend his coverage by buying complementary health insurance.

The more extensive the compulsory part of insurance coverage, the more redistribution can be achieved between the different risk groups, but the smaller are the incentives for prevention and efficient utilization of health care services. Thus a tradeoff between efficiency and social solidarity seems to impose itself. However, as shown in Breyer (1991), it is not optimal to do without any efficiency incentives, mandating full coverage for everyone. This is because those benefitting from the redistribution mentioned above gain from the continuing existence of incentives that encourage insureds to utilize medical care services in a cost-effective way.

6.8 EXERCISES

Exercise 6.1:

(a) What are the conditions rendering full insurance coverage optimal in the case of non-financial consequences of illness and no moral hazard? Under what conditions does the optimal insurance contract have a positive deductible? (b) Explain why in the basic model of health insurance it may be optimal for those having a high health risk to go without insurance coverage. (c) What characterizes an optimal health insurance policy, offered at a fair premium, if illness entails a nonfinancial loss? Explain in economic terms.

Exercise 6.2:

(a) What is “moral hazard” and in which forms does it show in the context of health? (b) Discuss and compare, for the models presented here, (expected) utility maximizing behavior of an individual with regard to preventive effort and coinsurance (given fair premiums) if the insurer (i) is able to observe the extent of preventive effort; (ii) is unable to observe preventive effort. (c) What can be said about the optimal rate of coinsurance, given moral hazard in its ex post form and using the framework introduced in this chapter?

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7

THE PHYSICIAN AS A SUPPLIER OF MEDICAL SERVICES

7.1 INTRODUCTION

The private-practice physician plays a key role in the production and distribution of medical services (see Section 1.4, Figure 1-4). Most people see him first when they have a health problem they are unable to deal with according to their beliefs. He is also the first agent to decide upon diagnosis, treatment, prescription of drugs, and referral to other suppliers of medical services (specialists, hospitals, pharmacists, etc.). Consequently, he is regarded by many as the gatekeeper to the health system, a role that will be examined more closely in the next chapter.

In this chapter, we will look only at those services that a practicing physician supplies himself or in collaboration with his employees, using his own medical supplies and technical devices.¹ We want to find out whether one can expect the production and allocation of ambulatory care services to be performed according to basic principles of efficiency, ensuring that a given treatment outcome is achieved using the smallest possible amount of scarce resources. The achievement of efficiency is doubtful because of the dual role of the physician in relation to his patients, acting both as a supplier of medical services and as an advisor concerning the services to be demanded by the patient.

This peculiarity (which does not exclusively apply to physicians but to lawyers and a few other professions as well) has caused a fierce debate among economists as to whether or not physicians use their power to secure a high volume of business even in circumstances where one would expect to see idle capacity because of high physician density. This hypothesis, known as “supplier-induced demand” in the market for

1. Services initiated by a physician in other areas will be the subject of later chapters (Chapters 9 and 10).

ambulatory medical care, is discussed in detail in Section 7.2. The focus will be on the question of whether there is empirical evidence suggesting that physicians abuse their advisory role in their own interest. We will show that the answer to this question importantly hinges on the way they are paid for their services. Accordingly, the incentives emanating from different systems of payment are examined in Section 7.3.

7.2 THE THEORY OF SUPPLIER-INDUCED DEMAND FOR AMBULATORY CARE

7.2.1 Physician Density and Utilization of Medical Services

In most industrialized countries, the medical profession enjoys a steadily increasing popularity as an occupational choice, perhaps more so than any other academic profession. It combines two features that attract many young people, viz. the fulfilment of the desire to help other people and the expectation to earn an above-average income. Due to the rush to the medical schools, the number of physicians has greatly increased during the past few decades (not only in absolute terms but also relative to population) and continues to rise. During the period 1970–1990, the total number of practicing physicians in the United States increased from 15 per 10,000 inhabitants to 23 per 10,000.²

Surprisingly, the continuous growth in the number of practices *did not cause underemployment* among physicians because the demand for ambulatory care services kept growing as well. Cross-section studies indicate that a greater supply of physicians goes hand-in-hand with an increased utilization of medical services per capita, as measured by health insurers' expenditures on ambulatory care for their beneficiaries. Far-reaching conclusions were drawn from this observation, viz. that the market for medical services "does not function" and that health care will remain affordable only if the number of physicians is limited through licensing restrictions.

The mere fact that utilization of medical services increases with the number of suppliers is not at all surprising, however. If the market for ambulatory care were competitive, then an increase in the number of suppliers should lead to an outward shift of the market supply function. Combined with a downward-sloping demand curve, the quantity transacted M is predicted to increase, accompanied by a fall in price p (see Figure 7–1). The effect on total expenditure (pM) is a priori uncertain and depends on the price elasticity of demand.

However, in many countries the utilization of medical services per capita rises with an increase in physician density even though *fees do not drop*. Using data covering the years 1963 and 1970 for different regions of the United States, Fuchs (1978) found that a 10 percent increase in surgeon density *ceteris paribus* led to 3 percent more operations per capita while causing surgical fees to increase. A positive correlation between physician density and per capita utilization of medical services also exists in countries like Germany, where fees are fixed at a uniform rate across regions so that an

2. See *OECD Health Data File* (1991).

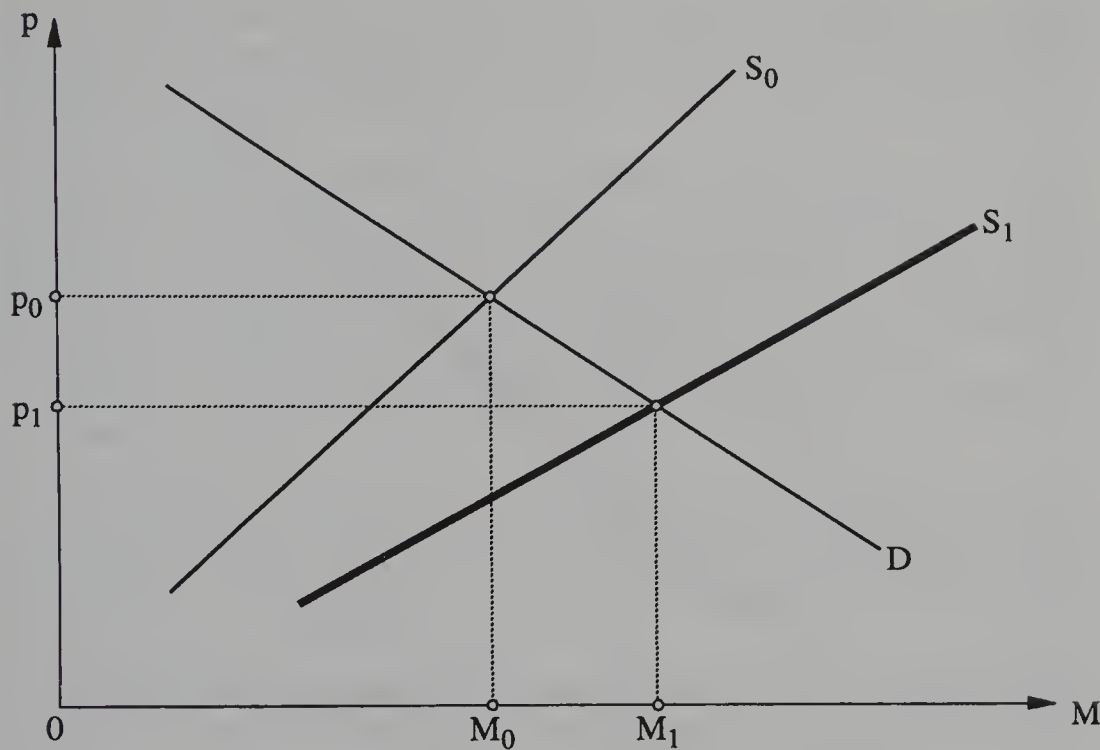


Figure 7-1. Impact of an increase in physician supply, “Standard Case.”

increase in supply cannot reduce prices. Of course, in these countries total medical expenditure rises in step with an increase in physician density.

7.2.2 Peculiarities of the Physician–Patient Relationship

The relationship between supplier (physician) and demander (patient) that is peculiar to health care (see Chapters 5 and 8) may play a key role in the existence of supplier-induced demand.

The most striking feature of the physician–patient relationship is the demander’s *incomplete information* about his true needs. A patient who consults his doctor knows only that he requires some medical treatment (of the diagnostic or therapeutic type) because he is not satisfied with his health status. Thus he delegates the choice of treatment to the better-informed doctor while deciding merely whether or not to follow the doctor’s advice and comply with his prescriptions. Not even in this decision is the patient completely sovereign, as there is a consensus among physicians and patients that the patient’s confidence in the doctor’s competence is an important precondition of successful treatment. This means that the demand curve, which depicts the volume of planned utilization of medical treatment at alternative prices, predominantly mirrors the decisions of suppliers, not of demanders of medical care. In the limit, the demand for medical services is thus supplier *determined*.

This fact remains without consequences as long as the doctor acts as a perfect agent for the patient, choosing what the patient would have chosen had he possessed the necessary technological knowledge. It does become a problem, however, as soon as the

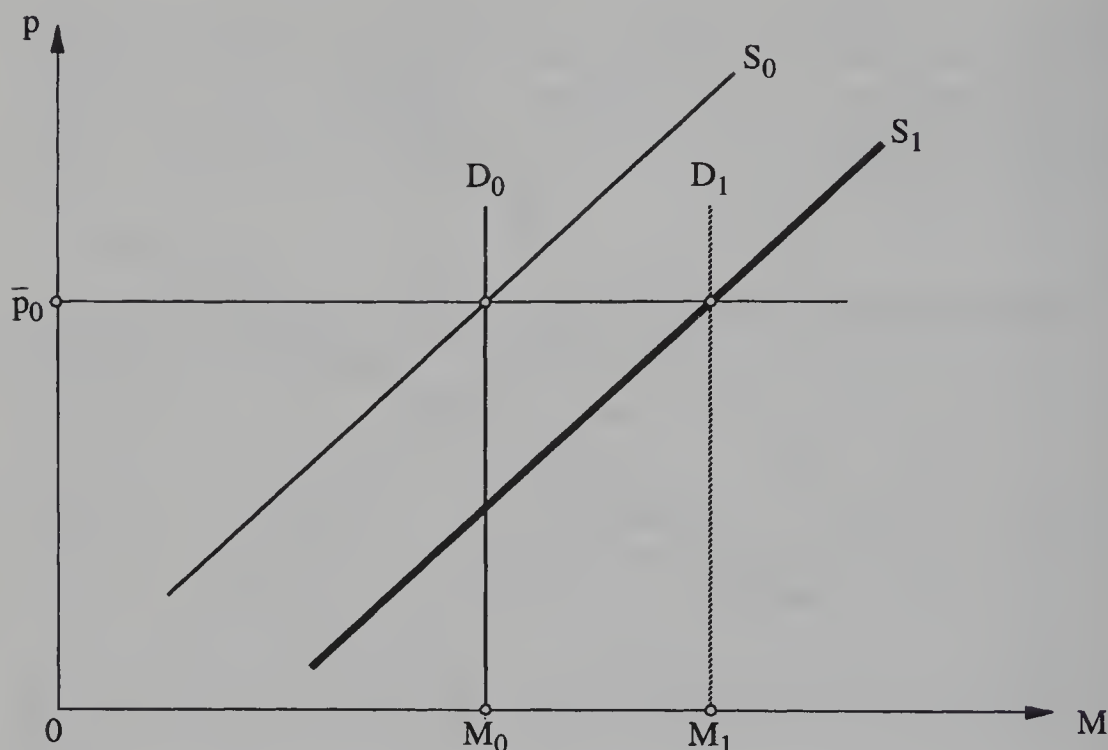


Figure 7-2. Supplier-induced demand as a response to an increase of physician density.

doctor's decisions on behalf of the patient are influenced by his own interests. If doctors vary the information provided to their patients systematically with an increase in physician density so as to secure their own full employment, then supplier-determined demand becomes *supplier-induced demand*.

The case of supplier-induced demand is illustrated in Figure 7-2, assuming full insurance coverage of patients. Therefore, demand does not depend on price. Let D_0 denote "primary demand," that is, the volume of services that meets the ideal standard of medical care given the number of patients in the catchment area, prevalence of health conditions, and the decisions of patients to contact the doctor. Let governmental regulation fix the unit price at $\bar{p}_0$, assumed to equate primary demand and supply of medical services given initial physician density (the S_0 curve).

We now assume that due to an increase in the number of physicians the supply curve shifts outward (to S_1). If providers had no direct influence on demand or if they acted as perfect agents for the patients, the demand curve would remain stable. At the regulated price, the quantity of services actually transacted would remain constant at $D_0 = M_0$. Evidently, providers produce fewer services than they would like to at this price.

Assume, however, that physicians do not act as perfect agents, but in advising their patients pursue their own interests. In this case they would recommend their patients medically unnecessary, marginally effective, or totally ineffective services such that the

planned quantities of supply $M_1 = S_1[\bar{p}_0]$ will actually be “demanded.” In terms of Figure 7–2, the demand curve would shift outward to the point where the quantity demanded (at any price, here at the regulated price level $\bar{p}_0$) equals supply S_1 planned at price $\bar{p}_0$.

If it is possible for doctors to boost the demand for their services at their own discretion, then the distinction between supply and demand loses its relevance. According to the “supplier-induced demand” hypothesis, the amount of services transacted on the market is exclusively determined by supply as desired by the physicians. The “demand” required to match supply is created artificially through the advice given to patients, whether medically appropriate or not.

Evidently, there are conditions facilitating the artificial creation of demand. The two most important are

1. Comprehensive health insurance. If patients are fully insured for the cost of medical treatment (in Germany, e.g., medical services are insured 100 percent), their own willingness to pay, which would put a limit to the consumption of services, is no longer relevant. The only reason to resist artificial demand creation is the time cost associated with the consumption of medical care.

2. Riskless medical technology. If doctors have a choice of services that, though providing only minimal benefits, do no harm (e.g., diagnostic procedures that serve to exclude the existence of a concomitant disease causing similar symptoms), they have little reason to refrain from creating demand.

CONCLUSION 7.1: Due to his superior information, the physician exerts considerable influence on patient demand. “Supplier-induced demand,” however, prevails only if physicians act not as perfect agents but according to their own interests, in particular by systematically modifying the information provided to patients in response to an increase in physician density in the aim of securing their own employment.

The occurrence of such a marked deviation from perfect agency would have important implications for health policy, especially in a system providing full insurance coverage while striving to contain health care expenditure. In particular, it may create pressure to limit access to the medical profession (see Section 8.5.2). This calls for an empirical examination of the supplier-induced demand hypothesis under conditions of full insurance. We develop a formal model of physician behavior in the following Section 7.2.3. This model implies empirically testable hypotheses concerning the relationship between physician density and per capita utilization of medical services. Other explanations for this relationship are examined in Section 7.2.4. In Section 7.2.5 we will present the results of empirical tests of the supplier-induced demand hypothesis.

7.2.3 Utility Maximization of the Physician

7.2.3.1 A Model of Physician Behavior

In this model, a utility-maximizing physician decides on the amount of services to be provided to his patients (see Table 7.1). Medical treatment is considered a

Table 7.1: A Model of Physician Behavior

$M = M(s),$	with $M'(s) > 0$	(7.1)
$h(s) = RM(s),$	with $h'(s) = RM'(s)$	(7.3)
$R = n/a$		(7.2)
$Y = Y(t),$	with $Y'(t) > 0$	(7.4)
$u = u(Y, t, s),$	with $u_Y > 0, u_t < 0, u_s < 0$	(7.5)
$t \leq h(s) = RM(s)$		(7.6)
$\frac{du}{ds} = Y'(t)RM'(s)u_Y + RM'(s)u_t + u_s = 0$		(7.8)

n : Number of inhabitants	t : Physician's working time (relative to total time available, $0 < t \leq 1$)
a : Number of physicians	s : Artificial demand creation by physician
R : Inhabitants per physician ($1/R$: physician density)	u : Utility of physician
M : Medical services per inhabitant	h : Medical services per physician relative to total time available ($0 < h \leq 1$)
Y : Income (consumption) of physician	

homogeneous good. Let there be one additional consumer good, with the units of both goods scaled such that their prices are unity.³ In this way, quantity and expenditure on goods coincide. Saving and borrowing are excluded. Consumption of medical services per patient, M , is assumed to be a monotonically increasing function of artificial demand creation by the physician, $s \geq 0$, such that

$M = M(s),$

with $M'(s) > 0$

(7.1)

where $M[0]$ represents primary demand. M is measured in units of physician working hours.

Supply of these hours is denoted by t , expressed as a fraction of his total available time ($0 < t < 1$). There are a identical physicians in an area counting n inhabitants. Thus

$R := (n/a)$

(7.2)

denotes the number of inhabitants per physician, while the reciprocal $1/R$ symbolizes *physician density*.

Thus the demand for an individual physician's working time is given by

$h(s) = RM(s),$

with $h'(s) = RM'(s)$

(7.3)

3. This simplifying assumption would raise problems only if the effects of a variation in relative prices were to be analyzed. This is not the objective of the present analysis, however.

The disposable income Y of the physician is derived from the difference between his revenue and his practice costs and taxes. It is described as a monotonically increasing function of his working hours t (with decreasing marginal returns),

$$Y = Y(t), \quad \text{with } Y'(t) > 0 \text{ and } Y''(t) < 0 \quad (7.4)$$

Because the price of the only consumer good is normalized to unity, Y also denotes the physician's consumption.

His utility depends positively on the consumption Y but negatively on his working hours t and on artificial demand creation s . The latter is valued negatively because provision of marginally effective (or even superfluous) services is *incompatible* both with professional ethics and a possible legal obligation to provide services in a rational way, stipulated in many countries having a social health insurance scheme. One therefore obtains⁴

$$u = u(Y, t, s), \quad u_Y > 0, u_t < 0, u_s < 0 \quad (7.5)$$

The physician maximizes his utility as given by Eq. (7.5) by choosing his consumption level, his working hours, and the amount of artificially created demand. Besides Eqs. (7.3) and (7.4), he has to observe the restriction

$$t \leq h(s) = RM(s) \quad (7.6)$$

which states that his actual working hours cannot exceed the quantity demanded, although excess demand may exist. However, positive excess demand ($RM > t$) may only exist if capacity is exhausted ($t=1$), since doctors have an obligation to supply medical services to their patients.

Ignoring for a moment this disequilibrium case and inserting Eq. (7.6) as an equality into the utility function, the latter can be expressed as function of a single parameter s ,

$$u = u\{Y[h(s)], h(s), s\} \quad (7.7)$$

The necessary first-order condition for an interior optimum of $u \{ \cdot \}$ is thus⁵

$$\frac{du}{ds} = Y'(t)RM'(s)u_Y + RM'(s)u_t + u_s = 0 \quad (7.8)$$

Condition (7.8) is easy to interpret. The factors appearing in front of the marginal utilities u_Y and u_t are strictly positive due to assumptions. Artificial demand is created to the point where the marginal benefit of additional consumption equals the marginal utility loss of the additional working hours and the "bad conscience" resulting from demand creation.

4. We write u_Y instead of $\partial u / \partial Y$, etc.

5. An interior maximum is a maximum with $s > 0$ and $0 < t < 1$.

7.2.3.2 Comparative Statics of the Model

We now would like to know how an increase in physician density, that is, a decrease of the exogenous parameter R , affects the amount of services provided per patient, M in the neighborhood of an interior optimum. In order to evaluate the expression

$$\frac{dM}{dR} = M'(s) \frac{ds}{dR} \quad (7.9)$$

we have to find out how the physician's choice variable s varies if R increases by dR . As the necessary first-order condition (7.8) has to be satisfied before and after the change, the following equation must hold:

$$\frac{\partial}{\partial s} \left(\frac{du}{ds} \right) ds + \frac{\partial}{\partial R} \left(\frac{du}{ds} \right) dR = 0, \quad \text{with } \frac{\partial}{\partial s} \left(\frac{du}{ds} \right) := \frac{d^2u}{ds^2} \quad (7.10)$$

Solving for ds/dR yields

$$\frac{ds}{dR} = - \frac{\partial(du/ds)/\partial R}{d^2u/ds^2} \quad (7.11)$$

The denominator in Eq. (7.11) is negative because of the second-order condition of utility maximization. The term ds/dR therefore has the same sign as the partial derivative of du/ds with respect to R . To obtain this derivative, write Eq. (7.8) explicitly as function of R ,

$$\begin{aligned} \frac{du}{ds} = & Y' \{h(R, s)\} RM'(s) u_Y \{Y(h(R, s)), h(R, s), s\} \\ & + RM'(s) u_t \{Y(h(R, s)), h(R, s), s\} + u_s \{Y(h(R, s)), h(R, s), s\} \end{aligned} \quad (7.12)$$

where $h(R, s)$ stands for $RM[s]$ and therefore $\partial h/\partial R = M$. Partial differentiation of Eq. (7.12) with respect to R then yields

$$\begin{aligned} \frac{\partial(du/ds)}{\partial R} = & Y''(t) MRM'(s) u_Y + Y'(t) M'(s) u_Y + \{Y'(t)\}^2 MRM'(s) u_{YY} \\ & + 2Y'(t) MRM'(s) u_{Yt} + M'(s) u_t + RM'(s) M u_{tt} + Y'(t) M u_{Ys} + M u_{st} \\ = & Y''(t) MRM'(s) u_Y - \frac{u_s}{R} + \{Y'(t)\}^2 MRM'(s) u_{YY} + 2Y'(t) MRM'(s) u_{Yt} \\ & + MRM'(s) u_{tt} + Y'(t) M u_{Ys} + M u_{st} \end{aligned} \quad (7.13)$$

The sign of expression (7.13) cannot be determined without additional assumptions since its first, third, and fifth terms are negative, whereas the second term is positive and the sign of the remaining terms depend upon the mixed second derivatives of the utility function. However, in two limiting cases, the sign of Eq. (7.13) is determinate. This may be seen after transforming a few terms into elasticities. Moreover, the concept of “income relative to share of time devoted to work” (Y/t) will prove useful. Thus, Eq. (7.13) is divided through by $u_Y > 0$ and multiplied by $1 = s/s$ and $1 = t/t$ to obtain, noting that $u_{Yt} = \partial u_Y / \partial t$,

$$\begin{aligned} \frac{\partial(du/ds)}{\partial R} \doteq & Y''(t)MRM'(s) - \frac{u_s/u_Y}{R} + \left(Y'(t) \frac{t}{Y}\right)^2 \left(\frac{Y}{t^2}\right) MRM'(s) \frac{u_{YY}Y}{u_Y} \\ & + 2Y'(t) \frac{t}{Y} \frac{Y}{t} MRM'(s) \frac{\partial u_Y / u_Y}{\partial t/t} \frac{1}{t} + MRM'(s) \frac{u_{tt}}{u_Y} \\ & + Y'(t) \frac{t}{Y} \frac{Y}{t} M \frac{\partial u_Y / \partial s}{u_Y/s} \frac{1}{s} + M \frac{u_{st}}{u_Y} \end{aligned} \quad (7.14)$$

Here, the “ $\doteq$ ” means “proportional to”. Now introduce the following elasticities,

- $e(Y, t) := (\partial Y / \partial t) (t/Y) = Y'(t) (t/Y) > 0$;
- $e(Y', t) := (\partial Y' / \partial t) (t/Y')$ such that $Y''(t) = e(Y', t) (Y'/t)$
 $= e(Y', t) (\partial Y / \partial t) (t/Y) (Y/t) (1/t) = e(Y', t) e(Y, t) (Y/t^2) < 0$;
- $e(u_Y, t) := (\partial u_Y / \partial t) (t/u_Y) = u_{Yt} (t/u_Y) < 0$ [and analogously $e(u_Y, s) < 0$].

Finally the coefficient of relative risk aversion with respect to variations in income is defined as $RRA := -(u_{YY}/u_Y)Y > 0$. Substituting into Eq. (7.14), one obtains,

$$\begin{aligned} \frac{\partial(du/ds)}{\partial R} \doteq & e(Y', t) e(Y, t) \frac{Y}{t^2} \underset{(-)}{MRM'(s)} - \frac{u_s/u_Y}{\underset{(-)}{R}} - \{e(Y, t)\}^2 \frac{Y}{t^2} \underset{(+)}{MRM'(s)} RRA \\ & + e(Y, t) M(Y/t) \{ \underset{(+/-)}{2RM'(s)} (1/t) e(u_Y, t) + e(u_Y, s) (1/s) \} \\ & + \underset{(-)}{MRM'(s)} \underset{(-)}{(u_{tt}/u_Y)} + \underset{(+/-)}{Mu_{st}/u_Y} \end{aligned} \quad (7.15)$$

Definite predictions can only be had at the price of additional assumptions. Specifically, the following two appear to be rather plausible.

ASSUMPTION 7.1: The elasticity of income with respect to working time is about one and greater than the elasticity of maginal income with respect to working time; that is, $|e(Y', t)| < e(Y, t) \approx 1$.

Assumption 7.1 may be justified by noting that if a physician wants to work additional time, he moves down his individual demand curve, catering to patients that present less urgent cases. In most conceivable modes of physician payment, this

means that 1 percent more working time will generate less than 1 percent additional income, however, that is, $e(Y, t) \approx 1$. In addition, this elasticity must be far greater than the absolute value of the elasticity $e(Y', t)$, which reflects how income reacts to additional working time at the margin. For otherwise, $e(Y, t)$ would quickly fall to zero or even become negative with increasing t , which is counter factual.

ASSUMPTION 7.2: With increasing intensity of demand induction, the marginal utility of income decreases; that is, $u_{Ys} < 0$.

Assumption 7.2 is realistic in view of the following thought experiment. Imagine first that the physician has additional income available for consumptions, having indulged in little or no demand induction. Quite likely he will be able to enjoy it greatly since he is free of any feelings of guilt. Next imagine the same additional income accruing to the same physician, who, however, has exerted a great deal of pressure to increase the demand for his services among his patients. Bad conscience seems likely to reduce his marginal utility from consumption in this situation.

On the basis of Assumptions 7.1 and 7.2, it is possible to examine

CASE 1. R large, that is, physician density $1/R$ is low. Neglecting the second term of Eq. (7.15) for simplicity, one has

$$\begin{aligned} \lim_{R \rightarrow \infty} \frac{\partial(\partial u / \partial s)}{\partial R} &\doteq e(Y', t) e(Y, t) \underset{(-)}{(Y/t^2) M R M'(s)} \\ &\quad - \{e(Y, t)\}^2 \underset{(+)}{\{(Y/t^2) M R M'(s) R R A\}} \\ &\quad + e(Y, t) (Y/t) M \{ \underset{(+/-)}{2 R M'(s) (1/t) e(u_{Ys}, t) + e(u_{Ys}, s) (1/s)} \} \\ &\quad + \underset{(-)}{M R M'(s) (u_{tt}/u_Y)} + \underset{(+/-)}{M u_{st}/u_Y} \\ &< 0 \end{aligned} \quad (7.16)$$

because the third term turns negative as well [regardless of the sign of $e(u_{Ys}, s)$], when R is large. Thus $ds/dR < 0$, or $ds/d(1/R) > 0$: *demand inducement* in response to an increase in physician density.

CASE 2. $R \rightarrow 0$, that is, physician density $1/R$ is high. In view of Eq. (7.6), this implies $t \rightarrow 0$.

$$\begin{aligned} \lim_{\substack{R \rightarrow 0 \\ t \rightarrow 0}} \frac{\partial(\partial u / \partial s)}{\partial R} &\doteq e(Y', t) e(Y, t) Y \underset{(-)}{(1/2t) M M'(s)} - \frac{u_s/u_Y}{\underset{(+)}{R}} \\ &\quad - \{e(Y, t)\}^2 Y \underset{(+)}{(1/2t) M M'(s) R R A} \\ &\quad + e(Y, t) M Y (1/t) \left[\underset{(-)}{M'(s) e(u_{Ys}, t) + e(u_{Ys}, s) (1/s)} \right] \\ &\quad + \underset{(+/-)}{M (u_{st}/u_Y)} \\ &< 0 \end{aligned} \quad (7.16a)$$

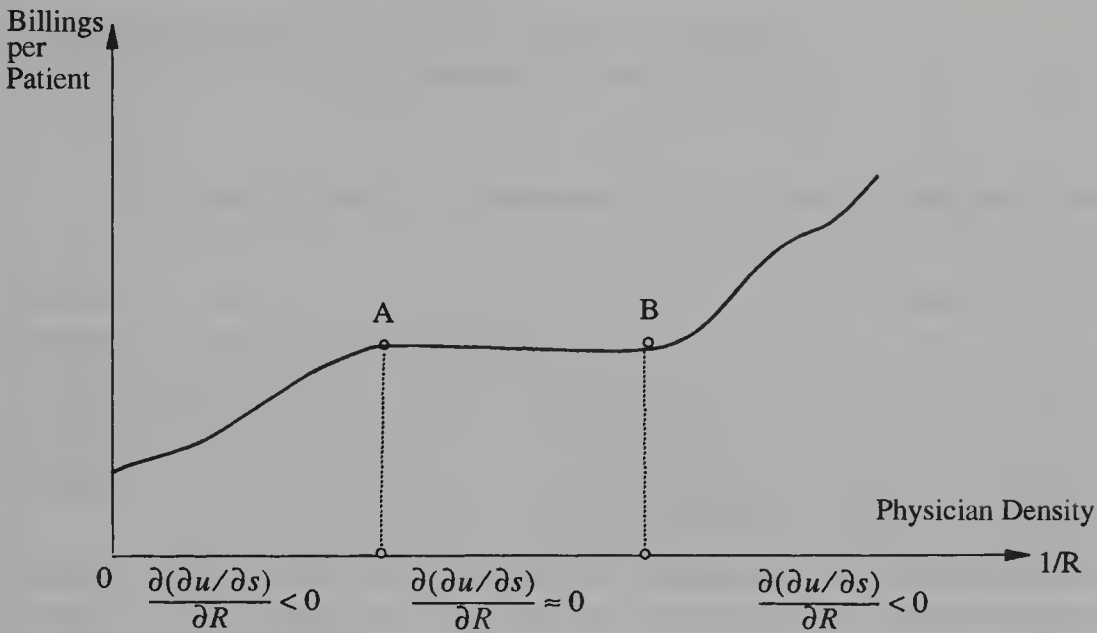


Figure 7-3. Physician density and predicted physician billings per patient.

applying L'Hôpital's rule to the terms containing R in the numerator and t in the denominator.⁶ The first three terms all increase without bounds when $R \rightarrow 0, t \rightarrow 0$, as do the two terms in the bracket of the fourth term. The sum of these terms goes to $-\infty$; therefore, $ds/dR < 0$, or $ds/d(1/R) > 0$: demand inducement in response to an increase in physician density.

In sum, there are circumstances under which an economic model of physician behavior, augmented by not too unreasonable auxiliary assumptions, generates the prediction of demand inducement—provided that patients are fully insured, of course. Specifically, demand inducement may occur at low and high values of physician density ($1/R$). At the same time, there must be a set of intermediate values $\{1/R, t\}$ that in combination produce a value of (approximately) zero for Eq. (7.15). In a country where patients have comprehensive insurance coverage (such as Germany), the relationship between billings per patient and physician density might look somewhat like Figure 7–3. At low as well as at high levels of physician density, billings per patient are predicted to increase, while at medium values, workload is adjusted so as to neutralize the relationship.

CONCLUSION 7.2: The assumption of rational physicians whose utility depends on income, working hours, and demand creation is sufficient to derive a positive relationship between physician density and per capita utilization of medical services in extreme cases only. However, demand inducement becomes a rather general

6. L'Hôpital's rule calls for evaluating $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\partial A/\partial x}{\partial B/\partial y}$ if $A \rightarrow 0$ and $B \rightarrow 0$ in the ratio A/B .

implication of such a model as soon as some auxiliary assumptions are introduced—provided always that patients are fully insured.

7.2.4 Physician Density and Utilization of Medical Services: Alternative Explanations

“Supplier-induced demand” for medical services constitutes only one explanation of the observation that the utilization of medical services per capita grows with increasing physician density, in spite of constant prices of medical services. This explanation is based on the hypothesis that physicians exploit the delegation of decision-making powers by the patient to their advantage by prescribing medically useless (ineffective) or economically inappropriate (inefficient) services. However, alternative explanations of the same empirical phenomenon exist. They include the following three.

1. Permanent excess demand. The observed correlation between physician density and utilization of medical services may result from permanent excess demand on the market for medical services, due to price regulation. In this situation, physicians work to capacity but still must turn away some patients. In the price-quantity diagram of Figure 7-4, demand is rationed. This causes points on the (shifting) supply curve instead of on the (stable) demand curve to be observed. When supply increases, the observed quantity of services provided increases as well, from M_0 to M_1 .

2. Decreasing indirect costs, improved quality of treatment. The growth of service volume associated with increasing physician density can also be the result of demand decisions by rational patients. The opening of new practices, particularly in rural areas, reduces the average time and transport costs of consultations. In addition,

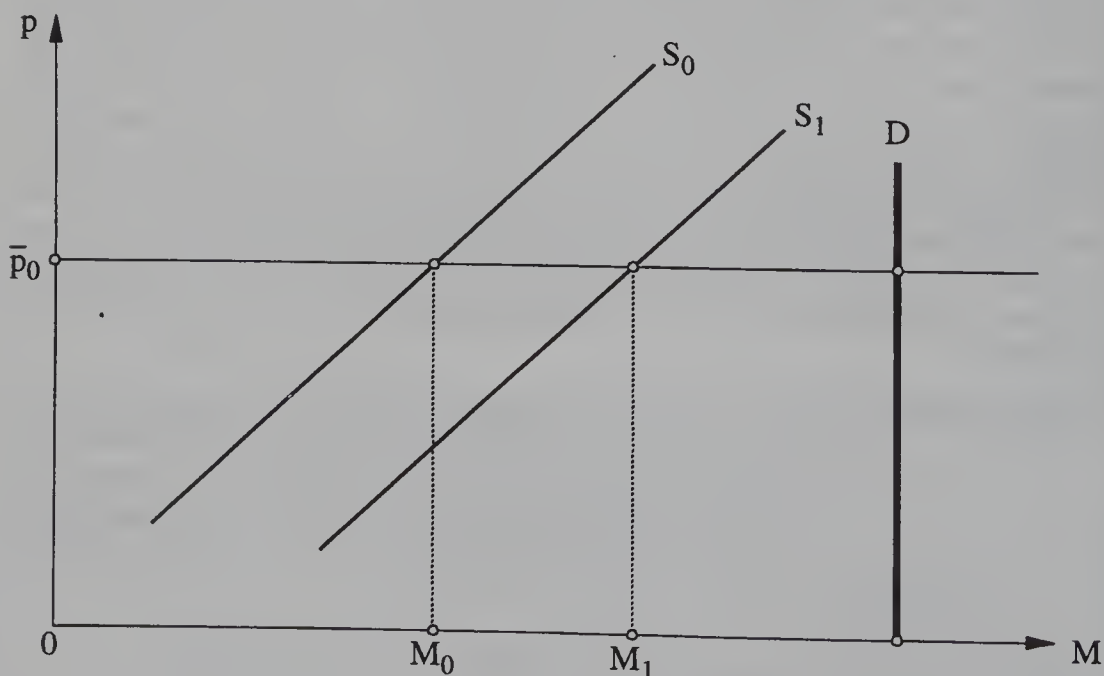


Figure 7-4. Effects of a supply extension at regulated prices and excess demand.

the average time spent in the waiting room falls when physicians are less busy. The time the patient has to wait for an appointment will also drop on average. As many diseases heal without medical intervention, an early appointment causes the probability of a condition receiving medical treatment to increase. Finally, the average amount of time devoted to a visit will be longer when the physician is less busy. As long as the duration of a visit is deemed an essential quality characteristic by patients, this serves to increase their demand for visits. All these changes cause patients' demand for medical care to shift outward at a given price (especially so at a zero money price) in response to an increased physician density.

3. Reversed causality. The third explanation for the observed positive statistical correlation of physician density and per capita utilization refers to a reversed direction of causality. When choosing a location for his practice, a young doctor will try to determine whether demand will be strong enough to generate a sufficient amount of revenue. Regions with high demand for medical services (perhaps due to their age or mortality structure) should thus attract more physicians than those with low demand. Consequently a high per capita utilization of medical services leads to a high physician density on average.⁷

CONCLUSION 7.3: The joint occurrence of rising per capita utilization of medical services with rising physician density does not prove that demand is created artificially by physicians. There are other explanations of the same phenomenon: pre-existing excess demand may be reduced, indirect costs to patients may have fallen, and variations in the physician density itself could be the result of variations in the demand for medical care.

7.2.5 Empirical Tests of the Supplier-Induced Demand Hypothesis

A careful empirical study should ideally provide discriminating evidence with regard to the three explanations given in the preceding subsection by taking into account the following implications.

- a. The excess demand explanation requires a positive (and proportional) correlation between physician density and service volume *at low physician densities* that *vanishes* once a certain density is reached.
- b. The explanation hinging on artificial demand creation predicts that this correlation is absent at *intermediate physician densities*.
- c. Locational choice by physicians should be *controlled for* to rule out the "reversed causality" explanation.
- d. The patients' *time costs* should be controlled for to rule out the "rational demand shift" explanation.

In his cross-section study of the United States between 1963 and 1970, Fuchs (1978) takes into account point (c) in a two-equation model, designed to explain the

7. In contrast to the first explanation, each region is in an equilibrium situation here.

supply of surgeons as well as the frequency of surgery. He also shows that surgeons' degrees of capacity utilization are relatively low in his sample, which rules out reason (a). As operations are usually scheduled well in advance, differences in time costs are low, ruling out explanation (d). Thus, Fuchs's finding that an increase of 10% in surgeon density leads *ceteris paribus* to a 3% increase in the frequency of surgery strongly supports the *artificial demand creation* hypothesis.

Adam (1983), Breyer (1984a), and Breyer, Mühlenkamp, and Adam (1986) analyzed data provided by the German local sickness funds, aggregated to the level of the *Länder*, between 1977 and 1982 using multiequation models in which physician density and per capita expenditure on medical services are explained simultaneously. Holding constant other determinants such as the population's age structure and the difference between urban and rural areas, statistically significant elasticities of per capita expenditures with respect to physician density were estimated in the 0.1~0.4 range. Thus, while a positive influence of physician density on observed service volume may be ascertained, reasons (a), (b), and (d) could be responsible for it.

Finally Breyer (1984b) analyzed data from local sickness funds⁸ in Baden-Württemberg, distinguishing areas of lower and higher physician density as of 1979. A simple regression demonstrates that billings per patient exhibit a relationship with physician density that depends on physician density, in a way similar to Figure 7-3. This observation supports the hypothesis of a reduction of a permanent excess demand but may also be compatible with the pattern shown in Figure 7-3, which was derived from an economic model of physician behavior allowing for the possibility of demand inducement.

In systems where patients have cost-sharing clauses in their insurance contracts, demand inducement should meet with some resistance. Yet physicians may want to compensate for the implied demand reduction by creating artificial demand. To test for this possibility, Kraft and Schulenburg (1986) analyzed data from Switzerland, the unit of observation being a medical practice in the Canton of Berne. They introduced an equation explaining physician density and used a Hausman test to check whether the latter was an endogenous variable. The test revealed that physician density can be considered exogenous. The authors discover a positive correlation between physician density and expenditure per "case", that is, per medical insurance record card submitted to the sick fund considered. From this they conclude that there was demand creation despite cost sharing by the insured.

However, they fail to see that in Switzerland in the 1980s the usual copayment clause for ambulatory care was, "10 percent per medical insurance record card submitted, with a minimum of 30 Swiss francs" (about 15 US\$ at the time). For medical expenditures between 30 and 300 Swiss francs, this means that copayment is zero at the margin. This is relevant because more than half of all billings fell into this interval. Thus the study cannot really predict physicians' possible strategic reactions to increased cost sharing.

8. That is the type of fund with the largest number of members (almost 45 percent of all persons insured in the statutory health insurance system).

CONCLUSION 7.4: Empirical studies in the United States and Germany show a statistically significant correlation between physician density and per capita expenditure on medical services. In the case of Germany during the 1970s and 1980s, the interpretation of a gradual elimination of excess demand cannot be refuted.

We would like to stress once more the fact that there is one crucial condition for artificial demand creation: an ambulatory medical care to prevail, resulting in a squandering of scarce resources. Medical revenues must rise in proportion with the amount of services supplied. However, only fee-for-service payment of medical services has this property. This indicates that supplier-induced demand is not really the key issue for efficient resource use, but rather the *choice of payment system*. The problem of economic efficiency would be solved if the doctor were paid in a way that he attains his individual optimum when using the resources under his control efficiently. The following section will discuss the influence of payment systems on physician behavior, highlighting their implications for rational resource use in ambulatory medical care.

7.3 SYSTEMS OF PAYMENT FOR MEDICAL SERVICES AND THEIR ALLOCATIVE EFFECTS

The term *payment system* is used here to denote the body of price regulations existing in the market for medical services. Under a payment system, the individual doctor is not free to define his services and the prices he charges his patients or their insurance companies.

7.3.1 Payment Procedure, Payment Basis, Payment Schedule

7.3.1.1 Payment Procedure

By payment procedure we mean the institutional mechanism of reimbursement. Thus one needs to define (a) from what sources and (b) through which intermediaries the remuneration is paid. Sources of payments may be

- a1. The patient himself;
- a2. His health insurance.

If the source is the patient, he may be insured (fully or partially) for medical expenses. Since he is reimbursed by insurance later on, we use the term, *indemnity principle* in this case. This system is typical for private health insurance. If the source is the health insurer, the patient receives medical treatment without being involved in payment procedures at all. The term *benefits in kind* is used in this case.

With regard to possible intermediaries in the payment process, one may distinguish

- b1. A procedure where patients or insurers pay the doctor who supplied the service directly;
- b2. A procedure involving intermediation, where a clearing house (in Germany, the statutory medical association) distributes the funds among physicians according to criteria that differ from those applied by health insurers.

7.3.1.2 Payment Basis

Indicators that determine the level of the payment in a certain period of time define a “payment basis”. Possible indicators are

1. The *factors of production* employed (working hours of the physician and his ancillary staff, the use of equipment, prescription drugs, and other medical goods, rent, heating and electricity used in the private practice). The basic idea underlying this choice of payment basis is to reimburse the costs incurred.
2. The *number and kind of services provided* (consultation, injections, radiation treatment, ultrasound examinations). This “fee-for-service” model is usually based on a fee schedule that values each service with an absolute or relative price.
3. The number of *treatment episodes*. This amounts to a flat rate per treatment.
4. The number of *treated patients*. The fee only depends on the number of medical insurance record cards submitted in a certain period, the quantity and intensity of the treatment being irrelevant.
5. The number of *potential patients*. At the beginning of the accounting period, every insured person has to register with the doctor he wants to consult in the case of illness. The physician’s fee depends on the number of registrations.⁹
6. A *fixed amount per period* (similar to a salary), which is independent of all the above- mentioned variables.¹⁰

7.3.1.3 Payment Schedule

A “payment schedule” is the function that connects revenue E with the components making up the payment basis, $Z_1, \dots, Z_n$. In its most general form, it can be written as

$$E = E(Z_1, \dots, Z_n) \quad (7.17)$$

The form of the function determines whether gross earnings are proportional, progressive, or regressive with respect to a change in a given factor. It also determines the average revenue per factor quantity. In the ensuing analysis it is important to know whether this unit revenue is greater or smaller than the unit price of the respective factor of production.

As we shall see, a physician’s incentives depend on the combination of payment basis and payment schedule, the payment procedure playing only a minor role.

7.3.2 A Simple Model of the Provision of Medical Services

For a formal analysis of the allocative effects of payment systems we make the following assumptions.

9. This payment basis is similar to a system practiced in Old China, where a doctor was only paid as long as his patients were healthy.
10. This differs from a salary in that the doctor recovers his practice costs from this fixed amount per period.

ASSUMPTION 7.3: The physician offers only two different kinds of services (e.g., advice and home calls) whose quantities are M_1 and M_2 .

ASSUMPTION 7.4: Each of these services is produced using the physician's working time and one other factor of production. The production function for the i th service is given by

$$M_i = f_i(t_i, x_i), \quad \text{with } \frac{\partial f_i}{\partial t_i} > 0, \frac{\partial f_i}{\partial x_i} > 0, i = 1, 2 \quad (7.18)$$

where t_i denotes working time and x_i the quantity of the other factor. The physician buys the x_i at an exogenously given price w in a competitive market.

ASSUMPTION 7.5: Therapeutic success H is a function of the quantities of both medical services,

$$H = H(M_1, M_2) \quad (7.19)$$

At small values of M_1 and M_2 , H increases with both these quantities. Beyond a certain level of services $(\tilde{M}_1, \tilde{M}_2)$, additional medical care becomes harmful, causing H to decrease.

ASSUMPTION 7.6: The physician's utility is an increasing function of his revenue Y and the achieved therapeutic success H and a decreasing function of his working hours t .¹¹

$$u = u(Y, t, H), \quad u_Y > 0, u_t < 0, u_H > 0 \quad (7.20)$$

When maximizing his utility the physician must take three restrictions into account. His total working hours are the sum of the working hours used in the production of the two services

$$t = t_1 + t_2 \quad (7.21)$$

His income is given by his nonlabor income Y_0 plus revenue $E(\cdot)$ less costs from his practice,¹²

$$Y = Y_0 + E(\cdot) - w(x_1 + x_2) \quad (7.22)$$

Finally we may combine Eqs. (7.21) and (7.22) to obtain a third relationship, linking treatment success to input quantities employed,

$$H = H\{f_1(t_1, x_1), f_2(t_2, x_2)\} \quad (7.23)$$

11. Inclusion of H represents a slight modification of the utility function used in Eq. (7.5). The difference is not fundamental, however, as soon as Assumption 7.5 is taken into consideration, which implies that the physician's marginal utility of providing services turns negative beyond a certain level.

12. The point in parentheses following E stands for variables to be inserted according to the choice of payment system.

By inserting Eqs. (7.21)–(7.23) into the utility function (7.20), we obtain the objective function,

$$u = u(Y_0 + E(\cdot) - w(x_1 + x_2); t_1 + t_2; H\{f_1(t_1, x_1), f_2(t_2, x_2)\}) \tag{7.24}$$

which is maximized by the physician’s choice of quantities of inputs (t_1, t_2) and (x_1, x_2) . Naturally, this maximum depends on the type of the revenue function $E(\cdot)$.

7.3.3 Incentive Effects of the Payment Basis

7.3.3.1 Questions To Be Answered

The basic aim of our analysis is to see what incentives for efficient behavior are inherent in the different payment systems, where we distinguish the following aspects of efficiency.

1. *Productive efficiency*, indicating whether a given bundle of services is produced using the smallest possible input quantities;¹³
2. *Treatment efficiency*, indicating whether a given treatment success is achieved using the smallest possible quantities of services;
3. “*Optimal*” *treatment success*, indicating whether the consumption of resources is justified by the marginal improvement of health status.

Production efficiency is achieved when the marginal rates of substitution between both factors of production equal their relative factor prices. Treatment efficiency requires that the marginal rate of substitution between both services in the production of treatment success equal their relative marginal costs. Determining “optimal” treatment success would call for a social evaluation of therapeutic success (e.g., according to the concept of marginal willingness to pay; see Chapter 2), which involves some unsettled issues. Nevertheless, an attempt will be made to include this third aspect into our assessment of the different payment systems.

Table 7.2 gives an overview of the model, which will be used to analyze four different payment systems: payment by input, fee-for-service, fixed amount per period (“salary”), and payment according to number of registered patients.

7.3.3.2 Payment by Input

In this case, only factor quantities appear in the revenue formula (7.17). With fees h^t and h^x fixed, the formula now reads

$$E = h^t(t_1 + t_2) + h^x(x_1 + x_2) \tag{7.25}$$

13. Production efficiency requires in particular that neither factors of production nor services are wasted; that is, the production occurs on the production frontier (7.18).

Table 7.2: Incentive Effects of Payment Systems for Physicians' Services

$$M_i = f_i(t_i, x_i), \quad i = 1, 2 \quad (7.18)$$

$$H = H(M_1, M_2) \quad (7.19)$$

$$u = u(Y, t, H), \quad u_Y > 0, u_t < 0, u_H > 0 \quad (7.20)$$

$$t = t_1 + t_2 \quad (7.21)$$

$$Y = Y_0 + E(\cdot) - w(x_1 + x_2) \quad (7.22)$$

$$u = u\{Y_0 + E(\cdot) - w(x_1 + x_2); t_1 + t_2; H[f_1(t_1, x_1), f_2(t_2, x_2)]\} \quad (7.24)$$

Payment according to factor input:

$$E = h'(t_1 + t_2) + h^x(x_1 + x_2) \quad (7.25)$$

$$\frac{\partial u}{\partial t_i} = u_Y h' + u_t + u_H \frac{\partial H}{\partial M_i} \frac{\partial f_i}{\partial t_i} = 0, \quad i = 1, 2 \quad (7.26)$$

$$\frac{\partial u}{\partial x_i} = u_Y (h^x - w) + u_H \frac{\partial H}{\partial M_i} \frac{\partial f_i}{\partial x_i} = 0, \quad i = 1, 2 \quad (7.27)$$

Fee-for-service:

$$E = p_1 M_1 + p_2 M_2 = p_1 f_1(t_1, x_1) + p_2 f_2(t_2, x_2) \quad (7.29)$$

$$\frac{\partial u}{\partial t_i} = \left(u_Y p_i + u_H \frac{\partial H}{\partial M_i} \right) \frac{\partial f_i}{\partial t_i} + u_t = 0, \quad i = 1, 2 \quad (7.30)$$

$$\frac{\partial u}{\partial x_i} = \left(u_Y p_i + u_H \frac{\partial H}{\partial M_i} \right) \frac{\partial f_i}{\partial x_i} - u_Y w = 0, \quad i = 1, 2 \quad (7.31)$$

Fixed payment per period:

$$\frac{\partial u}{\partial t_i} = u_t + u_H \frac{\partial H}{\partial M_i} \frac{\partial f_i}{\partial t_i} = 0, \quad i = 1, 2 \quad (7.36)$$

$$\frac{\partial u}{\partial x_i} = -w u_Y + u_H \frac{\partial H}{\partial M_i} \frac{\partial f_i}{\partial x_i} = 0, \quad i = 1, 2 \quad (7.37)$$

Payment per registered patient:

$$\frac{du}{dt} = \{u_Y q P'(H) + u_H\} H'(t) + u_t = 0 \quad (7.44)$$

$E(\cdot)$:	Gross revenue from the practice	q :	Honorarium per registered patient
E^* :	Periodic prepayment	t :	Total physician working time
f_i :	Production function of the i th medical service	t_i :	Physician worktime for the production of the i th service
H :	Therapeutic success; contribution to patients' health	u :	Physician's utility
h', h^x :	Input prices	w :	Factor price
M_i :	Amount of the i th medical service	x_i :	Other factor input in the production of the i th service
P :	Number of registered patients	Y :	Physician's income
p_1, p_2 :	Service prices	Y_0 :	Physician's nonlabor income

The necessary first-order conditions for a maximum of the physician's utility (interior optimum) are thus

$$\frac{\partial u}{\partial t_i} = u_Y h^i + u_t + u_H \frac{\partial H}{\partial M_i} \frac{\partial f_i}{\partial t_i} = 0, \quad i = 1, 2 \quad (7.26)$$

$$\frac{\partial u}{\partial x_i} = u_Y (h^x - w) + u_H \frac{\partial H}{\partial M_i} \frac{\partial f_i}{\partial x_i} = 0, \quad i = 1, 2 \quad (7.27)$$

From Eq. (7.27) we may draw the following conclusions. Since $u_Y > 0$, $u_H > 0$, and $\partial f_i / \partial x_i > 0$, while the unit fee h^x typically exceeds the factor price w , this condition can only be met at a level of services such that $\partial H / \partial M_i < 0$, that is, where additional medical services are detrimental to the health of patients. Too many services are provided in this case.

If, however, the fee h^x falls short of w , the physician suffers a loss each time he uses this factor. Consequently a utility-maximizing physician would cease practicing unless he draws very much utility at the margin from treatment success (u_H high). Since such "Samaritans" are rare, an insufficient fee level will result in excess demand for medical services.

Next, Eqs. (7.26) and (7.27) may be solved for marginal productivities $\partial f_i / \partial t_i$ and $\partial f_i / \partial x_i$, respectively. Division of the two expressions then yields

$$\frac{\partial f_i / \partial t_i}{\partial f_i / \partial x_i} = \frac{-(u_t / u_Y) - h^i}{w - h^x} = \frac{h^i + (u_t / u_Y)}{h^x - w} \quad (7.28)$$

Equation (7.28) may be evaluated in the light of requirements for productive efficiency. Productive efficiency in the provision of medical services of type i requires that the marginal rate of substitution between both factors [the left-hand side of Eq. (7.28)] equals the factor price ratio. While no market price for the physician's labor exists, his marginal utility of leisure relative to income, $-u_t / u_Y$, can be interpreted as the shadow price of his labor. Therefore, the factor price ratio is given by $-(u_t / u_Y) / w$. The condition (7.28) indicates that at the physician's utility maximum, the marginal rate of substitution between the two factors of production (physician time and auxiliary services) is not equal to the factor price ratio, but to the ratio of their unit profits. The latter is only equal to the factor price ratio if the proportion of the two fees h^i / h^x coincides with the factor price ratio.

With payment according to factor inputs, not only are too many services produced but they are also produced inefficiently. On top of this the physician has an incentive to waste factors of production directly, that is, to produce below his production frontier (7.18).

CONCLUSION 7.5: If the physician is paid according to factor inputs, he is led to produce his services inefficiently. Moreover, he will supply too many services if the fees exceed factor prices and not enough services if they fall short of them.

7.3.3.3 Fee-for-Service

Usually, fee-for-service systems are based on a schedule fixing a price p_i ($i = 1, 2$) for each unit of the i th service performed. If these rates are constant, the revenue function (7.17) becomes

$$E = p_1 M_1 + p_2 M_2 = p_1 f_1(t_1, x_1) + p_2 f_2(t_2, x_2) \quad (7.29)$$

Inserting Eq. (7.29) into the utility function (7.24), we obtain the following necessary first-order conditions for an (interior) utility maximum

$$\frac{\partial u}{\partial t_i} = \left(u_Y p_i + u_H \frac{\partial H}{\partial M_i} \right) \frac{\partial f_i}{\partial t_i} + u_t = 0, \quad i = 1, 2 \quad (7.30)$$

$$\frac{\partial u}{\partial x_i} = \left(u_Y p_i + u_H \frac{\partial H}{\partial M_i} \right) \frac{\partial f_i}{\partial x_i} - u_Y w = 0, \quad i = 1, 2 \quad (7.31)$$

Dividing these equations by each other, one obtains

$$\frac{\partial f_i / \partial t_i}{\partial f_i / \partial x_i} = \frac{-u_t / u_Y}{w} \quad (7.32)$$

The marginal rate of substitution between the factors of production equals their factor price ratio, which implies an efficient production of services.

If a given therapeutic success is to be achieved using the minimal possible amount of services, then the marginal therapeutic success brought about by an increase in the quantity of any one factor x_i must be the same for both kinds of services, M_1 , and M_2 . Therefore, we must have

$$\frac{(\partial H / \partial M_1)(\partial f_1 / \partial x_1)}{(\partial H / \partial M_2)(\partial f_2 / \partial x_2)} = 1 \quad (7.33)$$

However, from the optimality conditions (7.30) and (7.31) we obtain

$$\frac{(\partial H / \partial M_1)(\partial f_1 / \partial x_1)}{(\partial H / \partial M_2)(\partial f_2 / \partial x_2)} = \frac{(p_2 u_Y + u_H \partial H / \partial M_2)(\partial H / \partial M_1)}{(p_1 u_Y + u_H \partial H / \partial M_1)(\partial H / \partial M_2)} \quad (7.34)$$

The right-hand side of Eq. (7.34) equals one only if the fees are set such that

$$\frac{p_1}{p_2} = \frac{\partial H / \partial M_1}{\partial H / \partial M_2} \quad (7.35)$$

To achieve an optimal allocation of services from the economic point of view, fees have to reflect exactly their marginal productivities in terms of therapeutic success. As this condition is not met in general, physician's services are not produced at minimal cost.

CONCLUSION 7.6: Under fee-for-service payment, the physician may well choose a combination of services that is not cost-minimizing.

7.3.3.4 Fixed Payment per Period ("Salary")

Let us now consider the case where revenue is independent of all other variables, being fixed at a level E^* per period. Maximization of Eq. (7.24) then leads to the following first-order conditions for an interior optimum

$$\frac{\partial u}{\partial t_i} = u_t + u_H \frac{\partial H}{\partial M_i} \frac{\partial f_i}{\partial t_i} = 0, \quad i = 1, 2 \quad (7.36)$$

$$\frac{\partial u}{\partial x_i} = -w u_Y + u_H \frac{\partial H}{\partial M_i} \frac{\partial f_i}{\partial x_i} = 0, \quad i = 1, 2 \quad (7.37)$$

Division of the two equations results in Eq. (7.32), which is repeated here for convenience,

$$\frac{\partial f_i / \partial t_i}{\partial f_i / \partial x_i} = \frac{-u_t / u_Y}{w}$$

The condition for an efficient production of services is therefore satisfied. Payment by salary also induces the choice of a cost-minimizing combination of services. This can be seen by using Eq. (7.36) for $i = 1$ and 2 and dividing. The result conforms to the efficiency condition (7.33).

The disadvantage of this payment system lies in the fact that the physician's personal interest in the success of the treatment is his only incentive to provide any services at all. If $u_H = 0$, we see that neither Eq. (7.36) nor (7.37) is fulfilled, indicating that there is no interior optimum. The only solution is a *corner solution* with $M_1 = M_2 = 0$. Given a positive but weak interest of the physician in therapeutic success (i.e., a small value of u_H relative to u_Y and $-u_t$), the optimality conditions (7.36) and (7.37) are only met if the marginal factor productivities as well as the marginal productivities of both services in achieving treatment success are large. As the production function has decreasing marginal returns, this indicates that factors are employed but in small quantities and only few services are produced. Thus it is likely that therapeutic success falls short of the level desired by society.

CONCLUSION 7.7: While a fixed payment per period ("salary") induces the physician both to produce his services efficiently and to achieve the desired therapeutic success in an efficient way, there is no mechanism (apart from a possible ethical motivation) to ensure optimal therapeutic success.

7.3.3.5 Payment According to the Number of Registered Patients

Analyzing the three modes of payment in the preceding subsections, we found that none of them meets all three requirements (productive efficiency, treatment efficiency, and optimal intensity of treatment). Neither payment according to factor input nor fee-for-service provides any incentives for treatment efficiency, being input-oriented. In the case of a fixed salary, we expect physicians to provide an *insufficient amount of treatment*. However, this payment system does foster productive as well as treatment efficiency.

The failure of the fixed salary to reward output sufficiently could be corrected by choosing a payment that does not depend on input variables, but rather on variables positively related to output. For example, consider linking the salary to the number of registered patients, which would presumably reflect the degree of success of the physician's treatment. Let P denote the number of registered insured [dependent on the treatment success H , where $P'(H) = dP/dH > 0$] and q the fee per insured and per period. Thus the revenue function becomes

$$E = qP(H) \quad (7.38)$$

How does this mode of payment affect the degree of success aimed at by the physician? To answer this question, we simplify our model of Section 7.3.3 to include only one type of service, which is produced using only one factor of production, physician time.¹⁴

The production function then becomes

$$M = M(t) \quad (7.39)$$

where M represents the total amount of services to be distributed among P patients. The average degree of success in treatment is hypothesized to depend on the quantity of services per registered patient M/P ,

$$H = H\left(\frac{M(t)}{P(H)}\right) \quad (7.40)$$

The physician's objective function (7.24) now reads

$$u = u\{Y_0 + qP(H), t, H[M(t)/P(H)]\} \quad (7.41)$$

Before solving the utility maximization problem, let us determine the effects of an extension of physician working hours, t , on the average degree of success in treatment. To this end, differentiate Eq. (7.40) to obtain

$$H'(t) = \frac{\partial H}{\partial (M/P)} \frac{P(H)M'(t) - M(t)P'(H)H'(t)}{P^2} \quad (7.42)$$

14. This simplification is legitimate, since we have shown in the previous subsection that the choice of efficient combinations of services and factors of production is guaranteed if the mode of payment is independent of inputs.

This expression has to be solved for $H'(t)$. Denoting $\partial H/\partial(M/P)$ by $H'(M/P)$, one has

$$H'(t) = \frac{P(H)H'(M/P)M'(t)}{P^2 + H'(M/P)M(t)P'(H)} > 0 \quad (7.43)$$

With this result in hand, we may write the first-order condition for utility-maximizing working hours as

$$\frac{du}{dt} = \{u_Y q P'(H) + u_H\} H'(t) + u_t = 0 \quad (7.44)$$

Because of $H'(t) > 0$, a *positive degree of therapeutic success* H is strived for even when $u_H = 0$, and the physician is not interested in the success of his treatment per se. At the optimum, the loss of utility due to leisure time sacrificed must be equal to the marginal utility of additional income due to additional registrations, which in turn are the result of an increased degree of success in treatment brought about by increased time devoted to the practice.

Furthermore, any level of treatment success H^* desired by society may be achieved through an appropriate choice of the payment parameter q [see Eq. (7.38)]. To see this, solve the optimality condition (7.44) for q and insert H^* (and the corresponding quantities t^* and Y^*),

$$q = \frac{-u_t[Y^*, t^*, H^*] - u_H[Y^*, t^*, H^*]H'(t^*)}{u_Y[Y^*, t^*, H^*]P'[H^*]H'[t^*]} \quad (7.45)$$

CONCLUSION 7.8: Payment according to the number of registered patients can meet all the above-mentioned efficiency requirements. While bringing about productive and treatment efficiency, it induces physicians to choose the society's desired degree of therapeutic success.

We will return to the issue of choice of payment system in Chapter 8 in a more general context.

7.4 EXERCISES

Exercise 7.1:

Please explain what a payment schedule is without looking it up. (a) A freely negotiated price for medical services between physician and patient. (b) A market price for medical services to which the physician adjusts the services he provides. (c) A state-regulated system of functional correlations between revenue and payment indicators.

Exercise 7.2:

What are the consequences if the unit price were $h^x = w$?

Exercise 7.3:

Show that Eq. (7.34) follows from Eqs. (7.31) and (7.32).

Exercise 7.4:

Explain in terms of economics how the choice of the parameter of payment q determines the level of treatment success.

7.5 SELECTED READINGS**Section 7.2: The Theory of Supplier-Induced Demand for Ambulatory Care**

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8

FORMS OF DELIVERY OF MEDICAL CARE

8.1 THE ISSUE

Forms of delivery of medical care differ considerably between countries. These differences ultimately reflect basic differences of philosophy. Where the state grants the citizen a right to health, government tends to be directly responsible for the provision of health care services. Examples include the former Eastern Bloc countries as well as Great Britain and Italy with their National Health Services. Conversely, in the United States it is the individual himself who in principle assumes sole responsibility for his or her health. Nonetheless, the public purse finances 40 percent of total U.S. health care expenditure, supports public hospitals, and acts directly as a health insurer through the programs Medicare (for pensioners) and Medicaid (for the poor). However, even in countries that seem to be of a similar philosophical orientation with regard to health, forms of delivery of medical care may vary widely. In Sweden, for example, hospitals play a key role in the provision of medical care, whereas in Germany hospitals (with the exception of those of former Eastern Germany) are prohibited to run outpatient departments.

This variety in the organization of medical care is in a sense surprising, since the particular relationship between the physician and his patient forms the cornerstone of all health care systems. The physician is expected always to put his knowledge and capabilities to the advantage, and never to the detriment, of his patient's health. In actual practice, however, he will pursue his own personal objectives (the most important of which are discussed in the previous chapter) that run counter his patient's interests if only his income increases with medical care expenditure whereas the patient would prefer to pay less rather than more for treatment of a given quality level. The so-called *principal-agent relationship* between the patient (the principal) and the physician (the agent) is therefore not without conflict, and for reasons to be discussed in Section 8.2, both parties may choose to enlist the help of a complementary agent to resolve it. In this chapter, however, only those complementary principal-agent

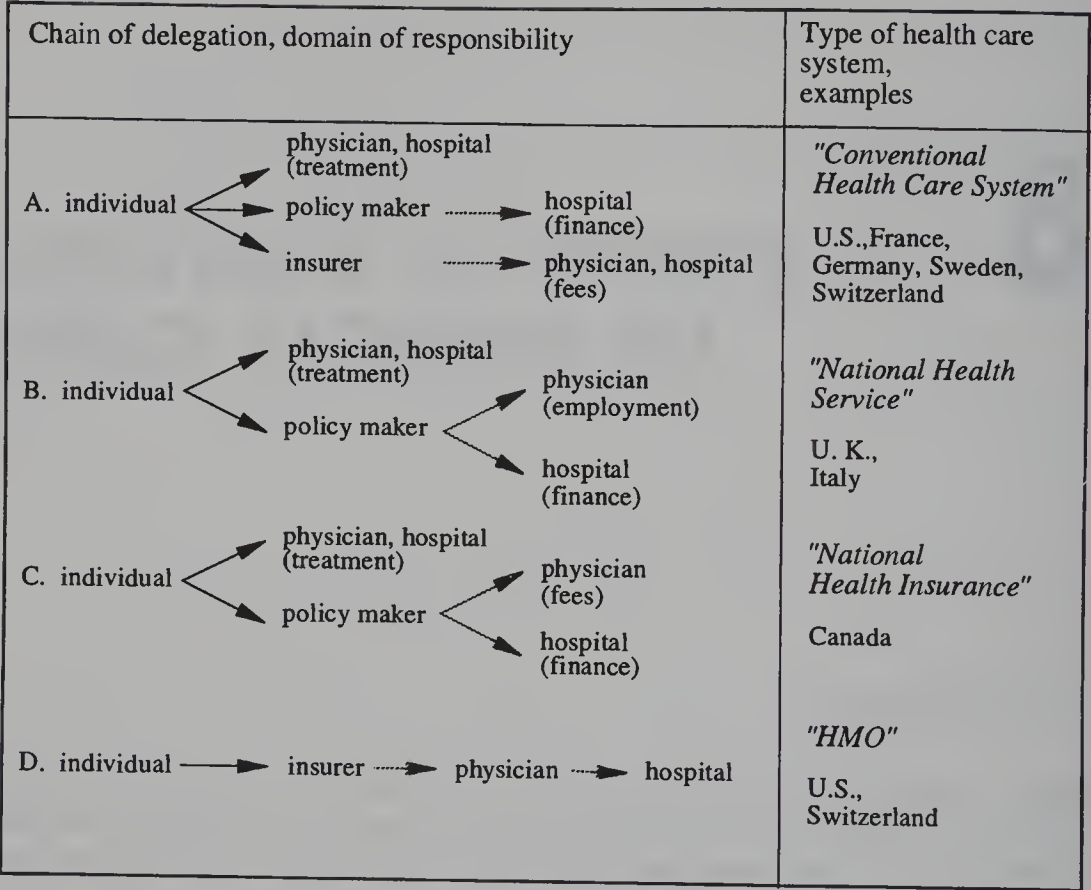


Figure 8-1. Principal-agent relationships as a characteristic of health care systems.

relationships will be examined that are of importance to the (potential) patient. In fact, it seems that the choice of the leading complementary relationship accounts for major differences in the delivery of medical care across countries.

Figure 8-1 gives an overview of possible arrangements of *complementary principal-agent relationships in health*. As far as conventional medical care is concerned (case A), which still constitutes the rule in western European countries and in the United States, the individual selects a physician to act as his agent who does not necessarily, however, represent his interests in the hospital. Indeed, politicians enter as complementary agents who attempt to safeguard patient interests by, for example, financing hospitals through their budget allocations. In general, interventions in the health care system are attractive to politicians, who want to win votes. As a result of the key role played by health insurance, redistribution in favor of certain voter groups is easily concealed. Redistribution is the main task of insurance, namely from those who have been without damage to those who have suffered a loss. It is difficult for a single insured voter to differentiate between this random, unsystematic redistribution and a hidden systematic redistribution. It is for this reason that *politicians acting as agents* have room for manipulation when it comes to structuring the complementary

principal-agent relationships in health care, resulting in a variety of health care systems.

In Sweden, for example, where there are 26 health districts, politicians have a comparatively large amount of power since they are able to raise taxes for public health in addition to overseeing hospitals. It is in these hospitals that the majority of medical services are provided, while physicians in private practice play only a marginal role in the health care system [see Ståhl (1990)]. Conversely, in Germany the *ban on establishing outpatient departments* has led to a marked division between ambulatory and hospital care, which serves to limit the political agent's influence mainly to hospitals. In France, a significant share of medical care is provided in private hospitals, which grant community physicians access to their equipment [see Rosa and Launois (1990)].

The close relationship between random and systematic *redistribution through insurance* is also exploited in various ways in different countries. In Germany, for example, statutory health insurance is forced to subsidize pensioners, whose contribution (paid by the government) falls short of the cost of their medical care. Redistribution also comes into play when low income earners are not allowed to choose between statutory and private health insurance or are automatically assigned to a certain health insurance scheme because of their occupation or employment in a particular company. In France everybody, even the self-employed, is a member of social health insurance, which is directly integrated into the national budget [see Rosa and Launois (1990)]. Conversely, in Switzerland federalism has traditionally gone so far as to leave it up to the individual canton to make insurance compulsory for low income earners. In their turn, most cantons leave it up to the individual to decide whether or not to take out health insurance.

Political agents not only in Great Britain and Italy, but also in Canada, are in possession of much more extensive powers (case B of Figure 8-1). Social health insurance no longer exists in Great Britain and Italy, having been replaced by a national health service that is directly financed out of public purse. In Canada, politicians act as the voter's agent by overseeing national health insurance, which in its turn is responsible for physician payment and hospital finance (case C).

Finally, Figure 8-1 (case D) displays still another mode of organization, in which the private health insurer becomes the most important agent of the individual in matters of health care. This is the concept of the *Health Maintenance Organization (HMO)*, by now quite popular in the United States. Since the HMO selects participating physicians, the physician-patient relationship is the result of its intermediation. At the same time, however, the HMO physician also safeguards the interests of his insureds in the hospital to an extent hardly known in conventional health care. HMOs provide an example of downstream integration (the retailer taking over control over the producer) in health care. In fact, existing boundaries between firms in health care may reflect the interests of professional associations to a greater extent than economics of scope and hence may be expected to change when health policy changes.

Evidently, the form of delivery of medical care has much to do with the choice of principal-agent relationship designed to complement the basic physician-patient relationship. In order to explain the demand for such complementary agents, the basic theory of the principal-agent relationship needs to be expounded and its application to

the physician–patient relationship examined. This is done in Section 8.2, providing at least partial explanation of why individuals do not rely on a physician as their unique agent in health care but may prefer to also enlist the help of politicians (“the government”), insurers, or even employers. Section 8.3 deals with the pros and cons of these complementary agents. The possible role of the health insurer as a complementary agent in health care is further examined in a separate Section 8.4, which deals with Health Maintenance Organizations and other organizational innovations in health care. Finally, professional associations in health care, above all the medical associations, have considerable influence on the delivery of health care services. Their role will be analyzed in the last section of this chapter.

8.2 THE PRINCIPAL–AGENT RELATIONSHIP AS THE KEY PROBLEM

In this section, a general formulation of the principal–agent relationship is presented, with the physician–patient relationship serving as the main example. Suppose a principal secures himself the services of an agent, whose efforts he is not able to assess. Moreover, let the outcome of these efforts importantly depend on factors that are beyond the agent’s control. In this situation, the principal’s problem is to devise a *payment function* giving the agent sufficient incentive to exert effort in his favor. An optimal payment function, while not ensuring a satisfactory outcome in every case, should still permit the principal to attain the greatest net benefit on average.

In health care, the physician–patient relationship constitutes the archetype of such a relationship. Most often the patient is not able to judge the appropriateness of the

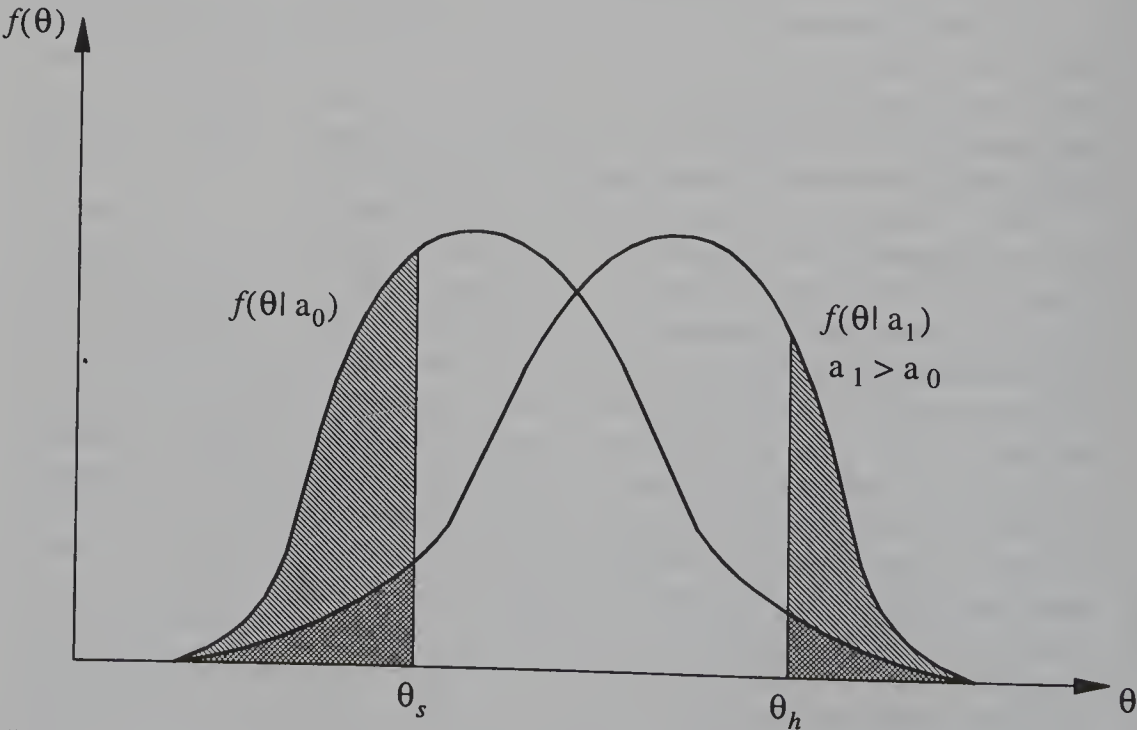


Figure 8-2. Outcome due to agent’s effort as a random variable.

measures taken by the physician, whose true effort thus amounts to an unobservable quantity. In addition, even intensive medical efforts do not guarantee an improvement of health status, while spontaneous healing occurs without medical treatment. These facts speak in favor of viewing the outcome of treatment in terms of health status as a random variable θ whose density function $f(\theta)$ depends on the physician's effort.¹ Figure 8-2 represents θ as a continuous variable whose expected value increases with physician effort a but whose range remains unchanged. According to these assumptions, intensified effort ($a_1 > a_0$) serves to lower the probability of an unfavorable result $\theta \leq \theta_s$, and to increase the probability of a favorable result $\theta > \theta_h$, for all values of θ . At the same time, even an arbitrarily large effort cannot rule out an unfavorable outcome, $\theta \leq \theta_s$.

Suppose the resulting health status lies in the neighborhood of θ_s . Unless the payment function is fixed in advance, this outcome likely leads to *conflict between the physician and his patient*. Based on his efforts, the physician bills for effort a_1 , while the patient, observing θ_s , expects a billing for a_0 . If, however, the physician's work actually amounted to a_1 while he is only paid for a_0 , he will reduce his effort level to a_0 next time. In view of these difficulties, the patient has an interest in negotiating a contract prior to undergoing treatment, settling on terms that induce the right amount of physician effort on average. Given that true effort is unobservable, payment may be based on outcome θ .

In terms of the general formulation of the principal-agent problem, the design of the contract is now reduced to a rule $p(\theta)$ that shares out the benefits of the transaction between the two parties. The *principal's decision problem* is to find the rule $p^*(\theta)$ maximizing his expected utility [see Holmström (1979) and Levinthal (1988) for the ensuing development]. The elements of this decision problem are summarized in Table 8.1.

Since the optimization refers to the choice of a function, Eq. (8.1) is a so-called objective functional. The functions $p(\theta)$ and $\{\theta - p(\theta)\}$ determine the shares of the benefit from the transaction going to the agent and the principal, respectively. For this formulation to be applicable to the physician-patient relationship, it must be assumed that health outcomes can be expressed *in terms of money*—a condition that may be considered satisfied in view of the willingness-to-pay approach to the valuation of changes in health status developed in Chapter 2 (see Subsection 2.3.2.2). The probability density function of possible outcomes $f(\theta|a^*)$ is conditional on the effort level of the agent, who will select the value a^* that is optimal to him. For simplicity, objectives other than income are neglected here. Thus the principal's *objective functional* reads

$$\text{Max}_{p(\theta)} \int_{\theta} u^P \{ \theta - p(\theta) \} f(\theta|a^*) d\theta \quad (8.1)$$

The agent will not strike the contract unless he benefits from it, his second-ranking alternative yielding some utility level $\bar{u}^A$. He too is supposed to maximize expected

1. This view was already espoused in Chapter 3, where the production of health was defined as exerting influence on the probability of occurrence of different states of health.

Table 8.1: General Model of the Principal-Agent Relationship

$\text{Max}_{p(\theta)} \int_{\theta} u^P \{ \theta - p(\theta) \} f(\theta a^*) d\theta$	Principal's objective functional	(8.1)
$\int_{\theta} u^A \{ p(\theta) \} f(\theta a^*) d\theta - v^A[a^*] \geq \bar{u}^A$	Agent's participation constraint (λ)	(8.2)
$\int_{\theta} u^A \{ p(\theta) \} \frac{\partial f(\theta a^*)}{\partial a} d\theta - v'^A[a^*] = 0$	Incentive compatibility constraint as first-order optimum condition (μ)	(8.4)
$L = \int_{\theta} u^P \{ \theta - p(\theta) \} f(\theta a^*) d\theta + \lambda \left(\int_{\theta} u^A \{ p(\theta) \} f(\theta a^*) d\theta - v^A[a^*] - \bar{u}^A \right) + \mu \left(\int_{\theta} u^A \{ p(\theta) \} \frac{\partial f(\theta a^*)}{\partial a} d\theta - v'^A[a^*] \right)$	Lagrange functional based on Eqs. (8.1), (8.2), and (8.4)	(8.5)
$\frac{u'^P[\theta - p^*(\theta)]}{u'^A[p^*(\theta)]} = \lambda + \mu \frac{\frac{\partial f(\theta a^*)}{\partial a}}{f(\theta a^*)}$	Condition determining optimal payment function	(8.7)

- $u^P\{\cdot\}$: Principal's utility function
 $f(\theta|a)$: Density function of possible outcomes; depends on agent's effort
 θ : Outcome
 $p(\theta)$: Payment of agent
 a^* : Optimal effort level from the agent's point of view
 $u^A\{\cdot\}$: Agent's utility function
 $\bar{u}^A$: Utility level attained by the agent failing conclusion of contract
 $v^A[a^*]$: Cost associated with effort a^* , in utility terms
 $\frac{\partial f(\theta|a^*)}{\partial a}$: Change of the density function due to an increase in effort

λ, μ : Lagrange multipliers, pertaining to constraints (8.2) and (8.4)

Source: Levinthal (1988).

utility and to be *risk averse* [i.e., the second derivative of his utility function with respect to payment $p(\theta)$ is negative]. With $v^A[a^*]$ denoting the cost of effort expressed in terms of utility,² the so-called *participation constraint* imposed by the agent reads

$$\int_{\theta} u^A \{ p(\theta) \} f(\theta|a^*) d\theta - v^A[a^*] \geq \bar{u}^A \tag{8.2}$$

2. The brackets indicate that the function has to be evaluated at the point indicated, i.e., $v^A[a^*]$ at the point $a = a^*$.

Moreover, given the nonobservability of his effort, the agent is free to choose a such that his own expected utility less cost is maximized. This gives rise to the so-called *incentive compatibility constraint*, with $\text{Argmax}\{\cdot\}$ rendering the value of a which maximizes the expression in braces,

$$a^* = \text{Argmax}_a \left\{ \int_{\bar{\theta}} u^A \{p(\theta)\} f(\theta|a) d\theta - v^A(a) \right\} \quad (8.3)$$

The agent's choice a^* must satisfy the first-order condition

$$\int_{\bar{\theta}} u^A \{p(\theta)\} \frac{\partial f(\theta|a^*)}{\partial a} d\theta - v'^A[a^*] = 0 \quad (8.4)$$

Optimal effort a^* is determined uniquely by condition (8.4) only if the probability of the outcome θ falling short of a certain threshold value $\bar{\theta}$ monotonously decreases with increasing a [see Rogerson (1985)]. This is the case in Figure 8-2. Since the density function $f(\theta|a_1)$ is obtained by moving probability mass to the right while increasing effort from a_0 to a_1 , the probability of an outcome $\theta < \bar{\theta}$ decreases for $\bar{\theta} = \theta_s$. It also decreases for the threshold value $\bar{\theta} = \theta_h$, which may be verified by noting that the area below the density function $f(\theta|a_0)$ to the left of any outcome θ is larger than below the density function $f(\theta|a_1)$.

The objective functional (8.1) and the constraints (8.2) and (8.4) may be combined to form a Lagrange functional, with θ below the integral sign indicating integration over all possible values of θ and hence calculation of the expected value,

$$\begin{aligned} L = & \int_{\bar{\theta}} u^P \{ \theta - p(\theta) \} f(\theta|a^*) d\theta \\ & + \lambda \left(\int_{\bar{\theta}} u^A \{ p(\theta) \} f(\theta|a^*) d\theta - v^A[a^*] - \bar{u}^A \right) \\ & + \mu \left(\int_{\bar{\theta}} u^A \{ p(\theta) \} \frac{\partial f(\theta|a^*)}{\partial a} d\theta - v'^A[a^*] \right), \quad \text{with } \lambda, \mu \geq 0 \end{aligned} \quad (8.5)$$

This expression is differentiated pointwise with respect to the value that the function p takes on at point θ [Gâteaux method; see Blanchard and Bruening (1982), Chapter II.3]. This means that at the optimum, each term under the integral must satisfy the first-order condition

$$-u'^P[\theta - p^*(\theta)]f(\theta|a^*) + \lambda u'^A[p^*(\theta)]f(\theta|a^*) + \mu u'^A[p^*(\theta)] \frac{\partial f(\theta|a^*)}{\partial a} = 0 \quad (8.6)$$

with, for example, $u'^A[p^*(\theta)]$ symbolizing the agent's marginal utility if his payment assumes the value $p^*(\theta)$. Dividing by $u'^A[p^*(\theta)]f(\theta|a^*)$, one obtains

$$\frac{u'^P[\theta - p^*(\theta)]}{u'^A[p^*(\theta)]} = \lambda + \mu \frac{\frac{\partial f(\theta|a^*)}{\partial a}}{f(\theta|a^*)} = \lambda + \mu E,$$

$$\text{with } E := \frac{\frac{\partial f(\theta|a^*)}{\partial a}}{f(\theta|a^*)} \quad (8.7)$$

This condition holds for all possible values of θ and may be interpreted as follows. Its left-hand side represents the *optimal sharing of the benefit* accruing from the transaction between the principal P and the agent A , expressed in terms of subjective marginal utility units. The higher the party's marginal utility at the optimum, the smaller his share of the benefit, because of decreasing marginal utility owing to risk aversion. Thus, the greater the value of the fraction on the left-hand side of Eq. (8.7), the smaller is the principal's "net profit" compared to the agent's fee, and the greater is the agent's share of the benefit created by the transaction.

The *objective determinants* of the optimal sharing of the benefits appear on the right-hand side of condition (8.7). The first term λ represents the principal's opportunity cost caused by failure to conclude the contract because the agent's participation constraint is not satisfied (see Table 8.1). The multiplier of the second term μ indicates the consequences a violation of incentive compatibility would have for the principal. The greater the value of this multiplier, the greater is the importance of the incentive component in the payment of the agent. The incentive component E reflects a quantity that might be dubbed "marginal relative effectiveness" (in stochastic terms) of the agent. The sign of E determines whether the optimal payment function contains a bonus for excellent service or a penalty for poor service (see Figure 8-2 again).

- If the outcome θ lies in the neighborhood of θ_s , then the probability of such an unfavorable result decreases with increasing efforts on the part of the agent: $f[\theta_s|a_1]$ has a smaller value than $f[\theta_s|a_0]$, with $a_1 > a_0$, or in infinitesimal terms, $\partial f(\theta|a)/\partial a < 0$. This serves to lower the value of the right-hand side of Eq. (8.7), implying a reduced share for the agent. This amounts to a *penalty for poor service*.
- If the outcome lies in the neighborhood of the high value θ_h , then the probability of achieving such a favorable result increases with an increase in the agent's efforts— $f[\theta_h|a_1]$ has a greater value than $f[\theta_h|a_0]$, and thus $\partial f(\theta|a)/\partial a > 0$. This serves to increase the value of the right-hand side of Eq. (8.7), implying an increased share for the agent. This amounts to an *incentive payment for excellent service*.

The determination of the optimal sharing rule may be illustrated as follows. In Figure 8-3, the value of the realized outcome θ is given by the distance from O^P to θ (if measured from the principal's origin) or the distance from O^A to θ (if measured from the agent's origin). Due to risk aversion ($u''^P < 0$), the line

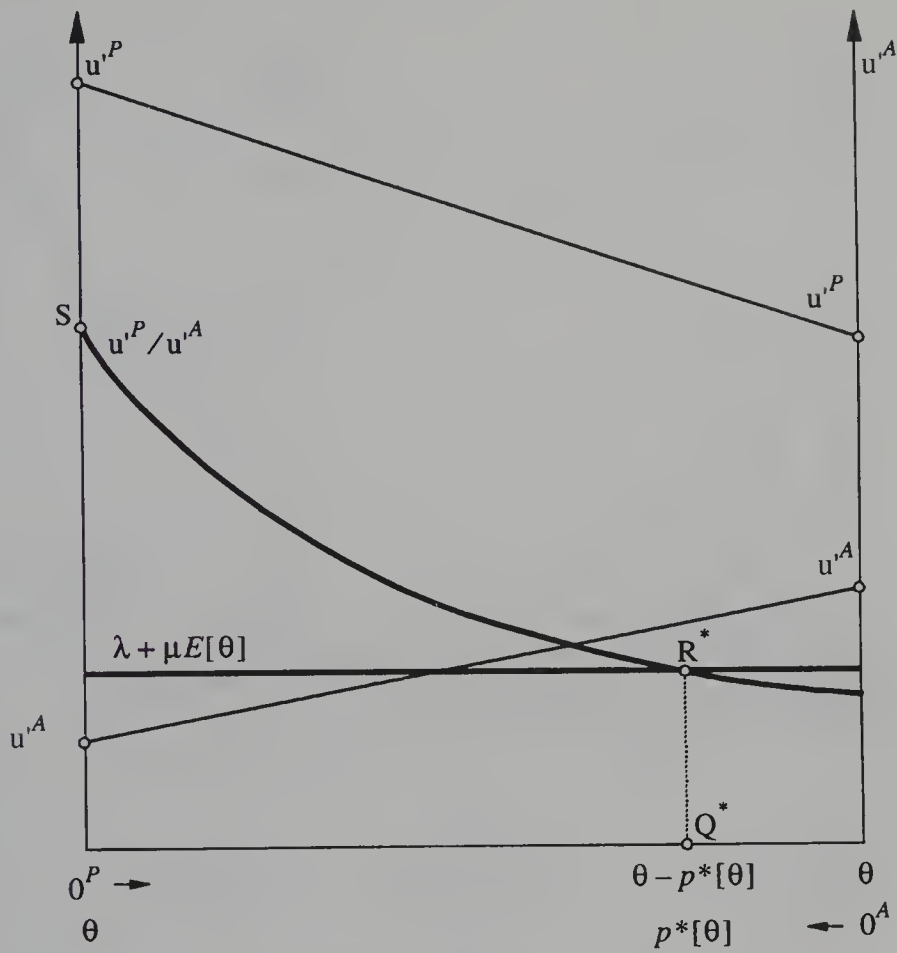


Figure 8-3. Derivation of the optimal payment for a particular outcome θ .

representing the principal’s marginal utility u^P slopes down with $p(\theta)$ and hence increasing distance from O^P . For simplicity, this relationship appears as a linear function in Figure 8–3. Similarly, u^A decreases with $\theta - p(\theta)$ and hence increasing distance from O^A ($u^A < 0$). The ratio u^P/u^A , which corresponds to the left-hand side of condition (8.7), is shown by the negatively sloped curve originating from point S of Figure 8–3.

The right-hand side of condition (8.7) is a horizontal line indicating the value $\lambda + \mu E$ for a given outcome θ , with E denoting the physician’s marginal effectiveness in improving the distribution of outcomes. The intersection of the two loci occurs at R^* ; there, the optimality condition (8.7) is satisfied. Optimal shares $\theta - p^*[\theta]$ and $p^*[\theta]$, respectively, may be read off from point Q^* on the horizontal axis.

In order actually to trace out the *optimal functions* $\theta - p^*[\theta]$ and $p^*[\theta]$, respectively, this procedure has to be repeated for different outcomes θ , as shown in Figure 8–4. In the case of an outcome $\tilde{\theta}$ smaller than the one shown in Figure 8–3, the base line of the figure must be shortened, causing the new agent’s origin $\tilde{O}^A$ to move to the left. With the function u^A also shifted to the left (see $\tilde{u}^A$), the

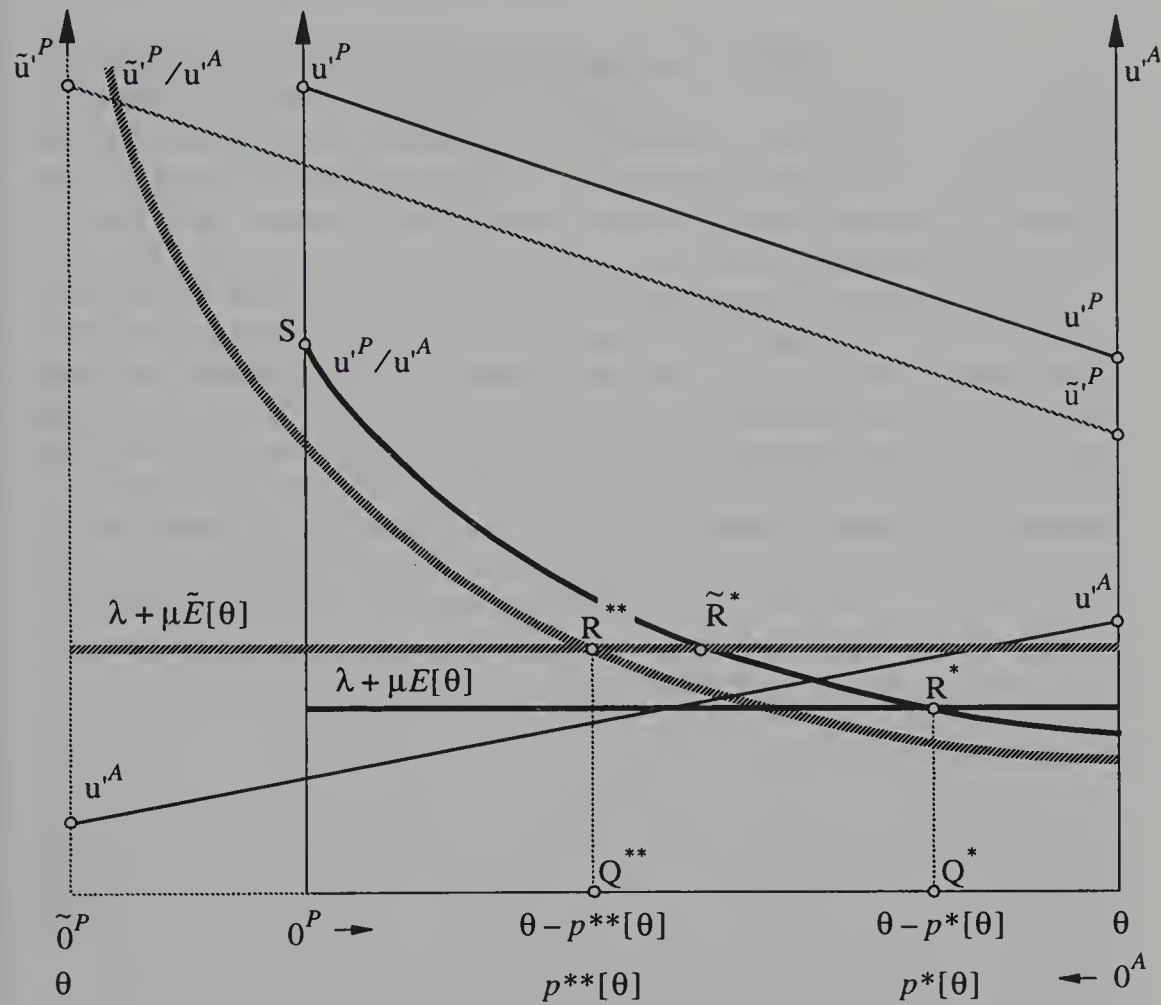


Figure 8-5. Dependence of contract design on urgency and initial health.

large value of λ , however, means that the patient depends very much on the physician. The greater λ , the greater the decrease in the value of the objective functional (8.1) if the participation constraint (8.2) were to be violated, for example, due to an exogenous increase of the agent's reservation utility $\tilde{u}^A$. A very high value of λ obtains, for example, if the *survival of the patient* depends on the conclusion of the contract. In the extreme, the line labeled $\lambda + \mu E$ would intersect the u^P/u^A curve at point S of Figure 8-5, indicating that the utility gain to the patient (given by the area below the u^A function) approaches zero. Thus the patient would be willing to offer the physician a fee amounting to the total benefit from his survival (i.e., his entire future income).

2. Incentive component of payment. The more important the physician's effort in determining the probability of a favorable outcome [i.e., the greater the absolute value of $\partial f(\theta|a)/\partial a$], the greater must be the incentive component of his fee. This plausible implication of condition (8.7), however, reveals the *core problem* of the principal-agent relationship in health care. Since the patient generally does not dispose of a sufficient number of observations, he is hardly able to assess the value of $\partial f(\theta|a)/\partial a$. For this reason, clauses of contracts providing for a penalty for poor service (i.e., for results in the neighborhood of θ_s in Figure 8-2) are rare in actual practice. Such clauses

could in principle be replaced by *physician liability*, but the small number of liability lawsuits won by patients indicates that the courts also have difficulties in estimating $\partial f(\theta|a)/\partial a$. If this parameter cannot be proved to differ significantly from zero, the right-hand side of condition (8.7) reduces to λ , which means that the patient will have to settle for a contract that fails to make the physician participate in the (lack of) success of his treatment.

3. Basic problems of an incentive payment. In principle, there is no compelling reason for $p^*(\theta)$ to increase monotonously with an increase in θ . It might be advantageous for the patient to provide for an extra reward for excellent performance only while remunerating average outcomes, which can also be attained through low values of a , by a fixed payment. Only if both utility functions are concave (i.e., reflect risk aversion), and if the fraction on the right-hand side of Eq. (8.7) represents a monotonously increasing function of θ , does $p^*(\theta)$ monotonously increase in θ [see Holmström (1979)]. With risk aversion likely to be present in both physician and patient, the choice between the two types of payment function hinges on the sign of $\partial E/\partial \theta$. However, to the extent that the patient is unable to assess either the distribution of possible health states as a function of physician's effort $f(\theta|a)$ or the effect of this effort on the probability of reaching θ , $\partial f(\theta|a)/\partial a$, lack of knowledge about E may well obstruct an optimal choice of payment function at this basic level.

4. Dependence of contract design on health status. If the patient suffers from bad health initially, a given outcome θ means a greater improvement of health status. In Figure 8-5, worse initial health may be depicted by moving the patient's origin from O^P to a new virtual origin $\tilde{O}^P$ while keeping θ constant. Accordingly, the function u'^P shifts horizontally to the left to become $\tilde{u}'^P$, as does the curve $\tilde{u}'^P/u'^A$. With a more unfavorable initial health status, the values of λ and μ increase because violations of the participation and incentive compatibility constraints have a greater impact on the patient in this situation. Therefore, assuming that the outcome is favorable ($E > 0$), the line $\lambda + \mu E$ shifts upward, giving rise to a modified point of intersection R^{**} . Comparing points Q^* and Q^{**} (which now indicate different optimal sharing rules), it will be noticed that the patient's share (measured from O^P as before because θ is constant) has unambiguously decreased. Thus an *already ill person* may consent to a contract that allocates the lion's share of the benefit to the physician.

These considerations may be summarized in

CONCLUSION 8.1: The principal having an information deficit relative to his agent will seek to strike a contract that fixes the sharing of the benefits prior to the occurrence of illness. The terms and conditions of this contract will, however, favor the agent the more strongly the principal depends on its conclusion. The optimal payment function contains an incentive component, which is the more important the greater the agent's influence on the probability of a favorable outcome. In health care, however, the patient's information gap will most often prevent him from identifying the optimal contract with a physician.

8.3 COMPLEMENTARY PRINCIPAL-AGENT RELATIONSHIPS IN HEALTH CARE

As the explanations of the optimality condition (8.7) have shown, parameters such as $\bar{u}^A$ (agent's reservation utility), u'^A (agent's marginal utility of wealth), and $\partial f(\theta|a)/\partial a$ (the agent's influence on possible outcomes) play an important role in the optimal design of the contract. It is precisely these parameters that a *patient is unable to assess* under most circumstances. Rather than trying to determine and enforce an optimal payment function $p^*(\theta)$ himself, he will therefore tend to call on a complementary agent to negotiate a favorable payment function on his behalf. In view of the high transaction costs associated with the writing of an agency contract, these complementary agents are likely to be partners of already existing contracts. This could be the employer, the health insurer, but also a political pressure group, or even a politician. However, the relationship between a politician and his constituency may at best be considered as an implicit, unwritten agreement.

8.3.1 The Employer as a Complementary Agent

Employers have a lot of experience with the principal-agent problem described in Section 8.2. From their perspective, employees are their agents whose activities contribute to the company's value added but cannot be monitored in detail. Since a favorable outcome may well be the result of chance (weather, exit of a competitor, regulatory measures) rather than effort, a payment system has to be found that is *incentive compatible*. This means that employers have considerable experience in the design of labor contracts that take moral hazard effects on the part of the employee into account. In view of Eq. (8.7), the solution to this problem requires in particular a reliable estimate of $\partial f(\theta|a)/\partial a$, the employee's influence on the economic success of the firm (with θ symbolizing the company's value added for now). Therefore, the employer might use his experience with employee assessment for contracting with physicians and other service providers in health care. It is in this way that the first so-called *Health Maintenance Organization* was set up in the United States (see Section 8.4).

The employees of a company will, however, have reservations with regard to having their employer select physicians and hospitals for them. When negotiating wages and employment, risk-averse employees may want to conceal fluctuations of their marginal productivity in order not to jeopardize their jobs. Because such fluctuations are closely related to state of health, medical information provided by participating physicians constitutes an important potential source of information. Usually, professional ethics will safeguard the confidentiality of medical records. However, the more physicians and hospitals depend on the employer, the more likely they will have to yield to employer pressure to provide such information.

Such an outcome may be predicted using condition (8.7) once more. This time, let condition (8.7) describe the rule of dividing up the benefits of transaction between *the employer and the providers of health care*, with the principal (P) and the agent (A) denoting the employer and the participating physician, respectively. The employer benefits from the transaction through an increased productivity of his employees due to both improved health on the one hand and more accurate knowledge of marginal

productivity based on medical information on the other. The right-hand side of condition (8.7), $\lambda + \mu E$, has a low value here in comparison to the situation where the individual is the principal because failure of a single physician to conclude the contract is of less importance to the employer, resulting in a low value of λ . This means that the optimum will be reached at a point to the right of R^* in Figure 8–3, indicating a sharing rule more in favor of the principal, with part of the benefit from the transaction accruing to the employer in the guise of valuable medical information about his workforce.

CONCLUSION 8.2: Employers can act as complementary agents shaping the contractual relationship with health care providers in the interest of their employees. Since they would be likely to obtain an information advantage relevant to wage setting and employment, however, they are accepted as complementary agents by employees only in exceptional cases.

This conclusion is the reason why employers are not listed in Figure 8–1 as viable complementary agents in health care. Only in exceptional situations (remote construction sites, extended work areas, shift work) may the savings on transaction costs be so large as to cause the employer to take over the provision of medical services by hiring physicians. As a rule, however, the employer's role is limited to providing health insurance, as, for example, in the United States, where large employers are legally obliged to offer a choice of plans. Employers typically have abstained from engaging in negotiations with providers of health care, with the exception of the so-called *Preferred Provider Organization (PPO)* in the United States. In a PPO, firms strike preferential deals with groups of physicians and hospitals in order to obtain a rebate on local fees. In terms of condition (8.7), this means that employers do not bother so much about estimating $\partial f(\theta|a)/\partial a$ but try to form a regional purchasing cartel designed to lower the physician's reservation utility $\bar{u}^A$. This relaxation of the participation constraint (8.2) serves to lower the value of λ , resulting in an increased employer's share of benefits from the contract.

8.3.2 The Government as a Complementary Agent

In democracies, voters have the option of replacing market and bargaining allocation mechanisms by government-controlled ones, whether it be directly by popular initiative as in some federal states of the United States and in Switzerland, or indirectly by electing a government that promises, for example, "health for all." This can take place in two ways. (1) *The government organizes the provision of medical services* directly as was the case, for example, in the former Eastern Bloc countries and as is the case in the British and Italian National Health Services, or (2) the government assumes the role of a monopolistic *national health insurer*, as is the case in Canada.

8.3.2.1 Public Provision of Health Care

Here, public authorities, while acting as a complementary agent of voters, take over the task of contracting with physicians and hospitals. In these negotiations, they assume the

role of the principal. The variables of the optimality condition (8.7) may then be interpreted in two ways.

a. Public authorities have a *budget reserved for health purposes* at their disposal. This amount ultimately must reflect the monetary equivalent of the health improvement brought about by health care services. For simplicity, it is denoted by θ , the symbol of health improvement used in Section 8.2. The amount of θ is fixed. Through a payment system $p(\theta)$, they remunerate physicians and hospitals for their services. If there is anything left over from the budget, it may be used for other purposes, to the benefit of the government or public administration.

b. The government regards θ as equivalent to the *total public budget*. It seeks to create incentives for the providers of health care services to contribute both to the nation's health and wealth through their efforts. A healthier workforce not only produces a higher National Product but also more revenue from taxation, resulting in a surplus after payment of health care providers (this interpretation corresponds to the model developed in Section 1.2).

According to both interpretations, the government, acting as a complementary agent in health care, achieves a surplus it may pass on to the electorate in the guise of reduced taxes or additional public goods. Whether this surplus is actually passed on to the electorate depends, as in the case of markets, on the *intensity of competition*, that is, on the competition between political parties for votes. Thus the fact that for years spending on health care expenditure in the UK has been making up a mere 6 percent of the GDP, a percentage figure far below that of other industrialized countries (see Table 1.1), is not a sufficient condition for the net burden on taxpayers being low; the money saved in health care may well be channelled into other domains of public administration.

In spite of these reservations, the government may be deemed a complementary agent meriting favorable consideration because of its power to shape the design of contracts in health care at will.

- The agent's participation constraint (8.2) may be manipulated by the government by reducing the agent's *reservation utility level* $\bar{u}^A$. Through regulating private health care, the government can make alternatives to public health service *unattractive* to physicians. In the extreme, $\lambda = 0$ would apply in Eq. (8.7), resulting in a lower optimal value of u'^P/u'^A and hence to a greater share of benefits going to the principal (the government in this case).
- Insofar as members of the health care profession depend on the government for their livelihood, they can be *forced* to maintain a certain effort level a . In terms of the model presented in the preceding section, this serves to render the incentive compatibility constraint (8.4) less effective, thus reducing the value of the Lagrange multiplier μ . As long as the effort level is high enough, causing $\partial f(\theta/a)/\partial a$ to be positive, the value of the right-hand side of the optimality condition (8.7) decreases if $\mu \rightarrow 0$, and with it the optimal value of the ratio of the marginal utilities u'^P/u'^A on the left-hand side. This again means that the government gets a bigger share of the benefits from the contract.

Yet, such compulsory measures may prove counterproductive in the longer term. In the national economy, health care competes with other industries for resources, such as

workers. If wages diverge too strongly from wages paid in other industries, the health care sector cannot keep its workers. In addition, international competition plays an important role because physicians in particular can easily find employment in another country.

For these reasons, public authorities may stop short of using their monopsony power, finding themselves in a situation comparable to that of employers. From the individual's point of view, however, there are two additional *disadvantages* to having the government rather than an employer as a complementary agent.

- As opposed to private employers, public authorities are protected from competitive pressures that would force them to adjust payment to the agent's "marginal productivity" $\partial f(\theta|a)/\partial a$ and thus to create incentives to make *special efforts* in favor of patients. Thus the individual's main problem of not being able to estimate this parameter is not solved by the government acting as his complementary agent.
- Competition between political parties is, as a rule, not as strong as competition between employers on the labor market. Therefore, there is *less pressure* on the government to pass on to its electorate any savings achieved in health care bargaining than on an employer to pass them on to his employees.

8.3.2.2 National Health Insurance

A national health insurance scheme constitutes an intermediate solution between public health care and the use of private health insurers as complementary agents, an alternative to be examined in Subsection 8.3.3. Since the representatives of a national health insurance scheme can always fall back on the power of the government if necessary, they will likewise try to draw up contracts in their favor, for example, by obstructing the practice of medicine outside the scheme. This would serve to lower the value of λ (importance of the participation constraint) in the optimality condition (8.7). Nevertheless, physicians in the Canadian provinces of Alberta and Ontario have kept the right to "overbill" their patients rather than having to accept the fees negotiated with national health insurance as full payment. As a result, an insured who wants to have free choice of physician must pay part of the cost, in a way reminiscent of the indemnity insurance described in Chapter 6 [see Evans (1984), pp. 333–35]. Apart from this, a national health insurance agency will act in a similar fashion as a private insurer inasmuch as its revenue must cover costs, but with the important *difference* that there is no competition forcing it to pass any savings achieved in contracting on to the insured.

8.3.3 The Private Health Insurer as a Complementary Agent

The possibility of relying on the private health insurer as a complementary agent dealing with physicians (or providers of health care services in general) seems to be an obvious alternative. In contrast to the employer, the health insurer usually does not obtain additional information about the insured that he could use to his advantage. As opposed to public authorities or a national health insurance scheme, a private health

insurer is also under competitive pressure to pass on savings achieved to the insured.

8.3.3.1 Derivation of the Optimal Payment Function

In the following, a model developed by Selden (1990) is presented in order to examine the possibility of an insurer, acting as a complementary agent, to develop and implement a payment system. Competition between insurers would then permit individuals indirectly to choose between payment systems through their choice of health insurer. Let the objective functional of the insured be additively separable, with component $B\{\cdot\}$ symbolizing utility flowing from health, and component $V\{\cdot\}$ symbolizing utility from income or consumption (see also Table 8.2),

$$\text{Max}_{p(M)} \int_h [B\{H(M^*, h)\} + V\{\bar{y} - b(M^*)\}] f(h) dh$$

(8.8)

The analogy with the objective functional (8.1) is obvious. The nonobservable effort level a of the agent corresponds with medical services M here, fixed at M^* by the physician in view of his own optimality conditions. The insured has to weigh the

Table 8.2: Applying the Principal-Agent Model to the Private Health Insurer

$\text{Max}_{p(M)} \int_h [B\{H(M^*, h)\} + V\{\bar{y} - b(M^*)\}] f(h) dh$	Objective functional of the insured	(8.8)
$M^* = \underset{M}{\text{Argmax}} [\Pi(M) + \alpha B\{H(M, h)\} + \delta V\{\bar{y} - b(M)\}]$	Condition for incentive compatibility on the part of the physician	(8.9)
$\Pi(M) = C + p(M) - wM$	Net income function of physician	(8.10)
$\int_h \Pi\{M^*(h)\} f(h) dh = 0$	Participation constraint on the part of the physician	(8.11)
$\int_h [C + p\{M^*(h)\} - b(M^*)] f(h) dh = 0$	Participation constraint on the part of the insurer	(8.12)

- B: Utility of health
- H: Health status achieved, dependent on medical services
- M: Medical services, fixed at M^* by the physician
- h: Health prior to treatment, with probability density function $f(h)$
- V: Utility of net income available for consumer goods
- $\bar{y}$: Income of the insured, exogenously determined
- $b(M)$: Insurance premium function
- $\Pi(M)$: Net income of the physician
- α, δ : Relative weights of patient utility, $0 \leq (\alpha, \delta) \leq 1$
- C: Capitation paid to physician, per insured
- $p(M)$: Fee schedule, function to be optimized
- w: Unit cost of service for physician

Source: Selden (1990).

benefits of M^* in terms of health achieved H against the fact that a high value of M^* , resulting in high premiums $b(M^*)$, curtails his income available for consumption. In contradistinction to Eq. (8.1), however, optimizing occurs over the possible states of health h prior to the medical intervention rather than over the states θ realized as an outcome of intervention. For an insurance contract to be viable, the premium function $b(M)$ has to be determined before the event and to hold regardless of the state of health realized in the future.

The first constraint (8.9) secures *incentive compatibility*. As the insured cannot observe physician effort M , he has to leave it up to the physician to determine M^* according to his preferences. In contrast to the constraint (8.3), the physician is assumed not only to consider his own net income $\Pi(M)$ but also the utility of the patient in his objective function,

$$M^*(h) = \underset{M}{\text{Argmax}} [\Pi(M) + \alpha B\{H(M, h)\} + \delta V\{\bar{y} - b(M)\}] \quad (8.9)$$

A physician characterized by $\alpha > 0$ is influenced by professional ethics because he makes the success of treatment (as evaluated by the patient) an argument of his own utility function. If $\delta > 0$ as well, then the physician even considers the financial consequences for the patient in his choice of treatment.

The *net income of the physician* is given by a possible fixed amount per patient C (a so-called capitation) and a fee according to the payment function $p(M)$, from which the costs occasioned by the efforts to provide the service, wM , are subtracted,

$$\Pi(M) = C + p(M) - wM, \quad \text{with } p[0] = 0 \quad (8.10)$$

Generally, M is a vector that includes not only the services of the physician but those of his aides and technical equipment as well. For the sake of simplicity, however, this vector is assumed to consist of one element only, valued at price w .

The model is completed by two conditions that guarantee participation in the system. First, the physician's net income (including a wage for his own working time) must have nonnegative expected value, which equals zero under competitive conditions,

$$\int_h \Pi\{M^*(h)\} f(h) dh = 0 \quad (8.11)$$

Second, the *insurer* must break even on average,

$$\int_h [C + p\{M^*(h)\} - b(M^*)] f(h) dh = 0 \quad (8.12)$$

Given competition, no net profits are possible, with premiums $b(M^*)$ just equal to costs, that is, the capitation C plus payment of medical services according to the function $p\{M^*(h)\}$. The notation used in Eq. (8.12) makes clear that the physician chooses his level of treatment M^* in view of h , the initial health status of the

patient. This fact would show in the payment function $p(M^*, h)$ as well as in the premium function $b(M^*, h)$ of the insured, were it not for the health insurer who converts them into a *payment function* $p(M)$ and a *premium function* $b(M)$, respectively, neither of which depends on the initial health status of the insured [see Eqs. (8.8) and (8.10)].

Based on these considerations, the building blocks of an optimal arrangement, with the insurer acting as the agent of the patient, may be determined rather easily. Since the insurer is deemed to charge his premium b independently of the insured's health status h , his participation constraint (8.12) requires

$$b^*(M) = \int_h [C^* + p^*\{M^*(h)\}]f(h)dh \quad (8.13)$$

Whatever the optimal payment function, it has to be designed in a way that the capitation plus the expected value of resulting payments to physicians is recovered by premium income $b^*(M)$. The *optimal capitation* is derived by substituting Eq. (8.10) into the participation constraint (8.11) of the physician and solving for C^* ,³

$$C^* = \int_h [wM^*(h) - p^*\{M^*(h)\}]f(h)dh \quad (8.14)$$

The optimal fixed payment thus has to bridge the gap between the cost of services provided $wM^*(h)$ and the payment for those services $p^*\{M^*(h)\}$, a gap that may occur when the expectation over an insured's possible states of health is formed (among which there may be some causing very high values M^*). Thus the participating physician must bear the risk of incurring costs in excess of revenues in some states of health, while he is relieved from this risk in fee-for-service payment.

Next, constraint (8.9) is transformed into a first-order condition for an *optimum on the part of the physician* in analogy to Eq. (8.4), who is assumed to consider $b^*(M)$ as predetermined. The differentiation of Eq. (8.9) then yields, using Eq. (8.10),

$$\frac{\partial \Pi}{\partial M} + \alpha \frac{\partial B}{\partial H} \frac{\partial H}{\partial M} = \frac{\partial p^*(M)}{\partial M} - w + \alpha \frac{\partial B}{\partial H} \frac{\partial H}{\partial M} = 0 \quad (8.15)$$

For reasons explained in the Appendix to Section 8.3, the derivative of B with respect to M may be determined from another, first-best optimum involving the Lagrangean multiplier $\tilde{\lambda}$ such that

$$\frac{\partial B}{\partial H} \frac{\partial H}{\partial M} = \tilde{\lambda} w, \quad \text{with } \tilde{\lambda} \geq 0 \quad (8.16)$$

3. Note that $\int_h hC^*f(h)dh = C^*$, the capitation being a constant.

Combined with Eq. (8.16), Eq. (8.15) yields

$$\frac{\partial p^*(M)}{\partial M} - w + \alpha \tilde{\lambda} w = 0 \quad (8.17)$$

Therefore, *optimal payment for the services provided by the physician M* is governed by the condition

$$\frac{\partial p^*(M)}{\partial M} = w(1 - \alpha \tilde{\lambda}) \quad (8.18)$$

Integration over M results in the optimal payment function

$$p^*(M) = wM(1 - \alpha \tilde{\lambda}) + C \quad (8.19)$$

In view of the participation constraint (8.11), the constant of integration amounts to the capitation C of Eq. (8.10), with optimal value C^* given by Eq. (8.14).

8.3.3.2 Properties of the Optimal Payment Function

The following properties of the payment function (8.19) deserve special mention.

1. Cost sharing on the part of the service provider. As soon as $\alpha > 0$, there should not be cost sharing on the part of the insured only [to the extent that his premium reflects expected cost; see Eq. (8.13)] but also on the part of the service provider. The insurer should, in the interest of the patient, take advantage of the fact that physicians and hospitals have an interest in the success of their treatment ($\alpha > 0$) and not only in their net income. The providers motivated by professional ethics need, however, to be identified as such.

2. Extent of optimal cost sharing. Eq. (8.19) indicates that the insurer is faced with considerable problems in this regard. First, he has to estimate α , the importance of professional ethics in the treatment decisions of the physician (or the hospital, respectively). Furthermore, an estimate of the Lagrangean multiplier λ is needed, that is, knowledge of the decision situation *that would obtain* if the insured himself were able to determine the intensity of treatment in full consumer sovereignty (see Appendix to Section 8.3). According to Eq. (8.16), this would imply knowledge of the subjective valuation of health improvement $\partial B/\partial H$ and of the marginal productivity of medical services $\partial H/\partial M$, the nonstochastic counterpart of the parameter $\partial f(\theta|a)/\partial a$ of Eq. (8.7). As the majority of the service providers have no incentive to reveal this parameter (see Section 8.3.1), the health insurer is confronted with the task of *finding* physicians and hospitals that (possibly thanks to their professional ethics) are prepared to have the productivity of their activities measured and the results transmitted to the insurer.

3. Opposition from medical associations. The payment function (8.19) has been derived under the assumption that service providers exactly recover their costs on average [see Eq. (8.11)]. They would thus fail to earn any economic rent due to, for example, restricted admission to medical schools and medical practice. Yet the goal of medical associations presumably is to *secure rents* for their members, that is, to keep net income above the level that would result from unrestricted access to the medical profession (see Section 8.5.2). Medical associations therefore should constitute difficult negotiating partners for an insurer seeking to work out a fee schedule that provides for cost sharing on the part of service providers.

4. Passing on of savings to the insured. The savings resulting from cost sharing by service providers should, according to Eq. (8.13), show in the premium function $b^*(M)$ of the insured. However, Eq. (8.13) is based on the participation constraint (8.12), which states that the insurer only recovers his expenses. Therefore, one may suspect that a possible premium reduction is fully passed on by the insurer only if there is *competition* in the market for health insurance.

CONCLUSION 8.3: A payment system that is designed by a health insurer acting as the complementary agent of the insured should include cost sharing by service providers having a professional ethics orientation. The health insurer will meet with opposition from the medical associations while passing on the resulting savings in the guise of premium reductions to the insured only under the pressure of competition.

A form of health care provision that seems to implement the optimal payment function just derived is the Health Maintenance Organization (HMO). A separate section is devoted to the HMO, as it represents an important innovation in the health care system of the United States.

APPENDIX

This appendix is devoted to the derivation of Eq. (8.16), repeated here for convenience,

$$\frac{\partial B}{\partial H} \frac{\partial H}{\partial M} = \tilde{\lambda}_w, \quad \text{with } \tilde{\lambda} \geq 0 \quad (8.A.1)$$

The objective is to prove that this constraint follows from solving a so-called first-best problem. In our context, this means that the optimizing insured determines himself the extent of medical treatment rather than delegating this decision to a physician. In such a world of *full consumer sovereignty*, the individual would choose M according to the objective function,

$$\max_M \int_h [B\{H(M, h)\} + V\{\bar{y} - b(h)\}] f(h) dh \quad (8.A.2)$$

In contrast to Eq. (8.8), disposable income depends on the state of health h in this formulation. The reason is that the premium function no longer has M but h as its argument. As the decision maker determines M^* himself in this world, a premium function $b(M)$ would make little sense. For an insurer to participate in the system, one must have

$$\int_h [b(h) - wM(h)]f(h)dh = 0 \quad (8.A.3)$$

If this constraint is added to Eq. (8.A.2) using a Lagrangean multiplier $\tilde{\lambda}$, differentiation with respect to M below the integral yields the first-order condition for an optimum,

$$\frac{\partial B}{\partial H} \frac{\partial H}{\partial M} = \tilde{\lambda}w, \quad \tilde{\lambda} \geq 0 \quad (8.A.4)$$

This result is used as Eq. (8.16) in order to derive Eq. (8.17) from Eq. (8.15) in the text.

8.4 THE HEALTH MAINTENANCE ORGANIZATION AS AN ALTERNATIVE FORM OF PROVISION OF CARE

The Health Maintenance Organization is an alternative form of provision of medical care services, based on the following principles [see Enthoven (1980)].

- The insured pays a monthly or yearly *fixed premium* in advance to the HMO.
- In return, he is guaranteed comprehensive *treatment as specified by the HMO physician* in the event of illness, with no copayment as a rule.
- The HMO physician is *not paid* according to the quantity of services provided; he may have a stake in the financial success of the HMO, however.
- The patient *must accept* being treated by participating physicians of the HMO; that is, he must give up his free choice of provider to some extent.

The first HMO was founded by Henry J. Kaiser, a building contractor in the 1930s, who had to organize medical care for his employees on remote construction sites. Therefore it was an *employer* who was to become the complementary agent to patients under the pressure of circumstances, and whose idea was well received by employees despite the possible conflict of interest pointed out in Section 8.3.1. The Kaiser Foundation, which devotes itself exclusively to health insurance, is even today the largest single HMO with an enrolment of more than 5 million in 16 federal states of the United States.

In the United States, the number of HMOs has risen from 26 in 1970 with 3 million members to more than 500 in 1995 with 50 million members, representing a *market share of 15 percent* [see InterStudy (1988, 1995)]. This rapid development was fostered

by the HMO laws of 1973, which required employers also to include an HMO among the health insurance options offered. Furthermore, nonprofit HMOs received government subsidies, which were abolished in 1981, however. While the ensuing analysis will focus on the HMO, there are other forms of “managed care” today that are enjoying rapid growth. For example, the insured may consent to utilization review or even actual case management by the insurer (both mainly designed to curtail hospitalization). Or he may sign up for so-called point of service plans that permit him access to nonparticipating providers in return for a certain amount of copayment. In a traditional HMO, copayment would be 100 percent in this case.

8.4.1 The HMO as a Complementary Agent in Health Care

In the following, the HMO is described as a complementary agent in health care, with the properties of the payment function (8.19) discussed in Subsection 8.3.3.2 serving as the guiding principle.

1. Cost sharing on the part of the service provider. HMO physicians usually receive part of their income as a bonus, whose size depends on the financial status of the organization. Because they may lose this bonus in the event of an unfavorable cost development, there is in fact some cost sharing. The incentive to be a prudent purchaser of externally provided services is reinforced by the fact that the HMO physician (a general practitioner as a rule) receives a *budget for the insureds that are on his list* from which he must pay the cost of these farmed-out services. These external purchases make up more than 70 percent of the HMO physician’s total disposable budget, as illustrated in Figure 8–6. Apparently, internal control (facilitated, e.g., by a group practice setting) is effective in preventing a diversion of funds into the general practitioner’s pockets.

Due to their incentives and decision-making authorities, HMO physicians are expected to achieve savings especially through reduced referrals to specialists and hospitals. This expectation has been confirmed in an extensive study carried out by Luft (1981). However, low rates of hospitalization may also be due to risk selection because HMOs presumably attract healthy individuals (see Section 6.6). This objection has been invalidated by the Health Insurance Study of the Rand Corporation [see Manning et al. (1984)]. Participants in that experiment gave up the right to choose a plan in return to a lump sum payment. They were randomly assigned to plans, making risk selection impossible. From Table 8.3 it can be seen that people who entered an HMO in the context of the experiment exhibit a low number of days spent in the hospital quite similar to permanent HMO members. Their figure amounts to 49 per 100 enrolment per year compared to 83 for conventional insurance. This difference can be traced back to the 40 percent lower rate of hospitalization in HMOs.

2. Extent of optimal cost sharing. In the discussion of Eq. (8.19) of Subsection 8.3.3.2, the difficulties of fixing the service provider’s optimal share in cost were pointed out. Little wonder that there are three types of HMOs that differ considerably in this respect.

- **Company type.** The HMO itself is the owner of one or more group practices, employing physicians whose income typically contains a profit-related component. Thus physician income depends (negatively) on the amount of cost generated.

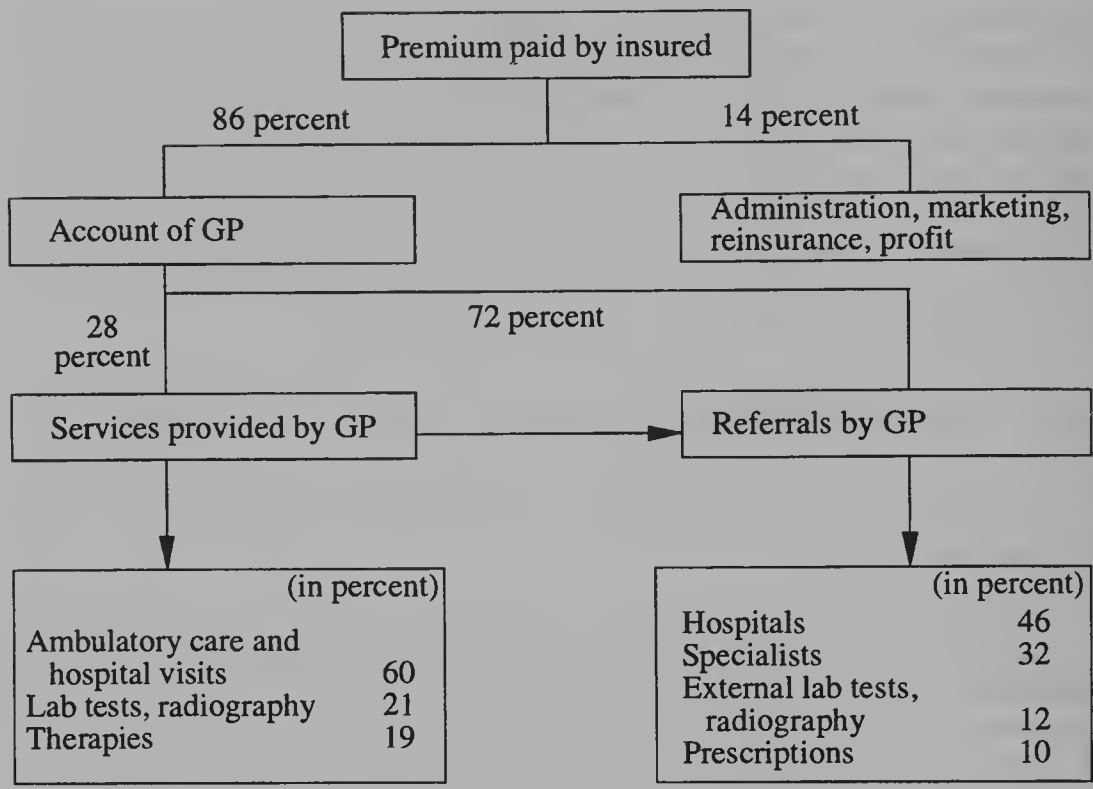


Figure 8-6. Use of premium income of an HMO (network type). Note: GP = general practitioner. Source: Moore (1979).

Table 8.3: Comparison of Utilization of Medical Care Services and Expenditure, 1983

Type of organization	Hospital admis- sions per 100 enrolment	Days spent in hospital per 100 enrolment	Physician visits per enrollee	Expenditure in US\$ per enrollee
HMO				
● Participants in the experiment	8.4	49	4.3	439
● Control group	8.3	38	4.7	469
Conventional insurance				
Without copayment	13.8	83	4.2	609

Source: Manning et al. (1984).

- **Network type.** Participating physicians are themselves owners of one or several group practices that contract with the HMO for the treatment of patients. Physicians retain the right to treat conventionally insured patients in addition to members of the HMO. Thus there may be cost shifting to these other patients in an attempt partly to balance cost sharing within the HMO (see Section 8.4.2).
- **Independent Practice Association (IPA) type.** An HMO may contract not only with individual physicians but also with groups of physicians. IPA enrollees must select their

family physician among participating doctors, who will provide primary medical care and also retain control after hospitalization. His bonus and hence degree of cost sharing is minimal, however.

This graduation of cost sharing on the part of service providers across the three different types of HMO may well reflect differences in ethical orientation (α) but also differences of opinion regarding the importance of financial constraints in an alternative world with complete consumer sovereignty of the patient [$\tilde{\lambda}$; see Eq. (8.A.4)].

3. Opposition from medical associations. The history of the HMO has been one of tension with the American Medical Association (AMA), revolving around the charge that incentives set by the HMOs will cause participating physicians to practice low-quality medicine, jeopardizing the health of the patient. This allegation has been examined in dozens of studies. On balance HMOs seem to offer *better* care than conventional fee-for-service [see Cunningham and Williamson (1980)]. However, a comparison of health status reached at the end of the Health Insurance Study performed by the Rand Corporation suggests that low-income individuals with poor initial health made smaller gains in an HMO than the comparison group in conventional health care [see Ware et al. (1986)].

The true reason for AMA opposition to HMOs seems to lie in the mode of payment, as predicted in Section 8.3.3. The value of the capitation C [see Eq. (8.10)] is fixed for the life of the contract, which makes *price discrimination according to the insured's willingness to pay* difficult. HMOs make it easy for the buyers of health insurance to shop around, seeking out the package with the most favorable cost-benefit ratio. Ex ante information costs of the buyer of a conventional fee-for-service plan with copayment are much higher, who will tend to gather information regarding the cost of treatment only in case of a serious illness. For a well-to-do individual, however, it will be too late to avoid price discrimination according to income in this case. Such price discrimination provides physicians with opportunities to increase their net incomes, which may serve to explain AMA's consistent stance in favor of fee-for-service [see the classic by Kessel (1958)].

4. Passing on of savings to the insured. In strongly contested markets, HMOs seem to pass on much more of any cost advantages to their insureds than in less competitive ones. Conversely, where the HMO is the only alternative to conventional health care in a market, its premium was found to be higher, *ceteris paribus* [see Schlesinger et al. (1986), Feldman et al. (1990)].

This short characterization of HMOs shows enough parallels to the terms of contract elaborated in Section 8.3.3 for drawing the following conclusion.

CONCLUSION 8.4: In their terms of contract, relationship to medical associations, and market behavior toward the insured, Health Maintenance Organizations correspond to a considerable extent to the predictions of Conclusion 8.3, which were obtained from a model of a private health insurer acting as a complementary agent in health care.

8.4.2 Cost Effects of HMOs at the Systems Level

If a supplier succeeds in producing his services at reduced cost due to an innovation, the usual presumption is that the cost of production decreases also at the level of the industry and even of the economy as a whole. However, this rule does not seem to hold in the case of health care. Critical observers point to the possibility of “cost ballooning,” that is, cost shifting between insurers and alternative modes of provision of medical care.

- The *government as a purchaser* of medical care services may push through lower fees for itself while causing providers to charge other purchasers more in compensation. The introduction of hospital payment based on diagnosis-related groups (DRG, see Section 9.2.3) by the U.S. Medicare administration has presumably resulted in higher charges for other groups, such as higher fees for the privately insured and a curtailment of free services for the uninsured poor. Likewise, the German government has been saving billions of dollars by not fully reimbursing the health care costs of pensioners to statutory health insurance. This shortfall has been compensated through higher contributions from the active population.
- Private innovators in health care are also suspected of *shifting cost rather than reducing them* by putting pressure on prices of services purchased from external sources. The preceding section mentions HMOs of the network type allowing participating physicians to compensate fee reductions negotiated with the HMO by treating other patient groups. Similarly, a hospital might settle for low fees with the important purchaser HMO while charging those having traditional insurance higher fees in compensation.

Apparently, the scope for cost shifting is particularly large in health care, making assessment of the economic benefits flowing from new forms of payment for and delivery of medical care difficult. Therefore, this section is devoted to an analysis of the determinants and likely extent of *cost shifting caused by HMOs*, using a model developed by Dranove (1988). To exemplify the situation, think of a hospital treating HMO patients as well as privately insured patients. Will a typical hospital actually charge private patients higher prices for comparable services if the HMO is capable of negotiating lower prices for their insured?

In order to answer this question, an objective function for the hospital, containing two groups of arguments, is postulated. On the one hand, management seeks to make a profit (Π) consisting of a contribution from selling services to patients having traditional insurance (Π_T) and a contribution from treating HMO patients (Π_H ; see Table 8.4). On the other hand, it pursues the objective of providing service to the community, which may be thought to reflect professional ethics. Accordingly, the amounts of services provided to the respective patient groups (Y_T and Y_H , respectively) also enter the utility function.⁴

4. Similar objective functions for hospitals are presented in Chapter 9. The function shown here does not include the quality of treatment for the sake of simplicity and thus fails to do justice to the influence of physicians on the goals pursued by the hospital.

Table 8.4: Cost Shifting by the Hospital

$u = u\{\Pi_T(p_T, C_T) + \Pi_H(p_H, C_H); Y_T(p_T), Y_H(p_H)\},$ with $\frac{\partial u}{\partial \Pi_T} = \frac{\partial u}{\partial \Pi_H} = \frac{\partial u}{\partial \Pi} > 0,$ $\frac{\partial u}{\partial Y_T} > 0, \frac{\partial u}{\partial Y_H} > 0,$ $\frac{\partial^2 u}{\partial \Pi_T^2} = \frac{\partial^2 u}{\partial \Pi_H \partial \Pi_T} = \frac{\partial^2 u}{\partial \Pi^2} < 0$	Objective function of the hospital	(8.20)
$\frac{\partial \Pi_T}{\partial C_T} < 0, \frac{\partial^2 \Pi_T}{\partial C_T \partial p_T} \geq 0; \frac{\partial \Pi_T}{\partial p_T} > 0, \frac{\partial^2 \Pi_T}{\partial p_T^2} < 0$	Characteristics of the market for patients having traditional health insurance	(8.21)
$C_T = C_H, \quad p_H = \bar{p}_H, \quad Y_H = \bar{Y}_H$	Characteristics of the market for HMO patients	(8.22)

Π : Total profit, $= \Pi_T + \Pi_H$
 Π_T : Profit from treatment of patients having traditional insurance
 Π_H : Profit from treatment of HMO patients
 p_T, p_H : Prices to be charged to members of either group
 C_T, C_H : Marginal cost (= average cost) of treating a member of the respective patient groups
 Y_T, Y_H : Number of billed patient days, negatively dependent on the respective price

Source: Dranove (1988).

$u = u\{\Pi_T(p_T, C_T) + \Pi_H(p_H, C_H); Y_T(p_T), Y_H(p_H)\},$

(8.20)

with $\frac{\partial u}{\partial \Pi_T} = \frac{\partial u}{\partial \Pi_H} = \frac{\partial u}{\partial \Pi} > 0,$ $\frac{\partial u}{\partial Y_T} > 0, \quad \frac{\partial u}{\partial Y_H} > 0,$

$\frac{\partial^2 u}{\partial \Pi_T^2} = \frac{\partial^2 u}{\partial \Pi_H \partial \Pi_T} = \frac{\partial^2 u}{\partial \Pi^2} < 0$

The first equality in the second line of Eq. (8.20) derives from the assumption that the first argument of the objective function is the *sum* of profits. The management of the hospital is therefore viewed as being indifferent with regard to the source of profit. Partial differentiation of the equalities of the second line with respect to Π_T yields the equalities in the third line of Eq. (8.20). Finally, the utility function is concave from below in profit, which may be justified by the assumption of risk aversion as soon as the model were to be generalized to include uncertainty (see Subsection 2.3.2.2.2).

Billed patient days $\{Y_T, Y_H\}$ should generally depend negatively on *price charged* [apart from the assumptions (8.22) to be introduced below]. For patients having traditional health insurance, this certainly is a plausible assumption, given the important degree of cost sharing for hospital care in the United States. But also in the

case of HMO patients, physicians as prudent purchasers of hospital services are likely to respond to an increased price with a reduction of patient days ordered.

The following sign restrictions characterize the market for patients having traditional insurance,

$$\frac{\partial \Pi_T}{\partial C_T} < 0, \quad \frac{\partial^2 \Pi_T}{\partial C_T \partial p_T} \geq 0; \quad \frac{\partial \Pi_T}{\partial p_T} > 0, \quad \frac{\partial^2 \Pi_T}{\partial p_T^2} < 0 \quad (8.21)$$

According to these assumptions, the contribution to profit from this market segment is reduced by cost increases ($\partial \Pi_T / \partial C_T < 0$). This effect is mitigated, however, if price can be raised at the same time ($\partial^2 \Pi / \partial C_T \partial p_T \geq 0$). Since objectives as stipulated in Eq. (8.20) cause management to deviate from profit maximization, there is scope for an increase of the service price to raise profits ($\partial \Pi_T / \partial p_T > 0$), albeit at a decreasing rate ($\partial \Pi_T^2 / \partial p_T^2 < 0$).

Additional assumptions serve to characterize the HMO market segment from the point of view of the hospital,

$$C_T = C_H, \quad p_H = \bar{p}_H, \quad Y_H = \bar{Y}_H \quad (8.22)$$

The first assumption indicates that the treatment of an additional HMO patient generates the same cost as that of a patient having traditional insurance. Consequently, *cost differences are excluded* as a reason for any ballooning of cost between the two patient groups.

For the sake of simplicity, the price for HMO patients $\bar{p}_H$ is considered exogenous, although—in contrast to the fees for Medicaid and Medicare patients, which are set by the government—this price is the result of a negotiating process. At this negotiated price, the HMO commits to paying for a *fixed number of patient days* ($\bar{Y}_H$).

These additional assumptions leave the hospital with just p_T as its decision variable. The first-order condition for an (interior) optimum may be found by differentiating Eq. (8.20),

$$\frac{\partial u}{\partial p_T} = \frac{\partial u}{\partial \Pi_T} \frac{\partial \Pi_T}{\partial p_T} + \frac{\partial u}{\partial Y_T} \frac{\partial Y_T}{\partial p_T} = 0 \quad (8.23)$$

(+ +) (+) (−)

Increasing the price for those having traditional insurance has a positive effect on profit but a negative effect on the professional objective of providing service, since patients are deterred by the higher price.

Now, if the optimum is disturbed by, for example, a *reduction of the fee paid for HMO patients* ($d\bar{p}_H < 0$), both sides of the first-order condition (8.23) change. However, its right-hand side must again be zero after the change, indicating conservation of optimality. On its left-hand side, the decision variable adjusts by dp_T , yielding

$$\frac{\partial^2 u}{\partial p_T^2} dp_T + \frac{\partial^2 u}{\partial p_T \partial \bar{p}_H} d\bar{p}_H = 0 \quad (8.24)$$

Solving for $dp_T/d\bar{p}_H$, one obtains, using Eq. (8.23),

$$\begin{aligned} \frac{dp_T}{d\bar{p}_H} &= \frac{(-1)}{\frac{\partial^2 u}{\partial p_T^2}} \frac{\partial^2 u}{\partial p_T \partial \bar{p}_H} \\ &= \frac{(-1)}{\frac{\partial^2 u}{\partial p_T^2}} \left(\frac{\partial^2 u}{\partial \Pi_T \partial \Pi_H} \frac{\partial \Pi_H}{\partial \bar{p}_H} \frac{\partial \Pi_T}{\partial p_T} + \frac{\partial^2 u}{\partial Y_T \partial \Pi_H} \frac{\partial \Pi_H}{\partial \bar{p}_H} \frac{\partial Y_T}{\partial p_T} \right) \quad (8.25) \\ &\quad \begin{matrix} (+) & (-) & (+) & (+) & (+/-) & (+) & (-) \end{matrix} \end{aligned}$$

Due to the second-order condition for a maximum, $\partial^2 u/\partial p_T^2 < 0$, which is assumed to hold without further proof. Eq. (8.25) shows that three cases may be distinguished.

1. Parallel movement of prices, $dp_T/d\bar{p}_H > 0$. If the HMO has to be granted a rebate, those having traditional insurance profit from a reduced fee as well. This case may occur only if $\partial^2 u/\partial Y_T \partial \Pi_H < 0$; that is, the more services are provided to patients having traditional insurance, the smaller the management's marginal utility of profit from the treatment of HMO patients.

2. Independent movement of prices, no cost shifting, $dp_T/d\bar{p}_H = 0$. A sufficient condition for this case is

$$\frac{\partial u}{\partial Y_T} = 0 \quad (8.26)$$

for due to Eq. (8.23), one obtains $\partial \Pi_T/\partial p_T = 0$ [so the first term in the large parentheses of Eq. (8.25) may be dropped], while Eq. (8.26) implies $\partial^2 u/\partial Y_T \partial \Pi_H = 0$ (so the second term may be dropped). In other words, as soon as the hospital pursues pure profit maximization, disregarding the objective of service provision, ballooning of cost can be ruled out.

3. Compensating price movement, cost shifting, $dp_T/d\bar{p}_H < 0$. A rebate in favor of HMO patients is compensated by a higher price charged to patients having traditional insurance. According to Eq. (8.25), it suffices for this case to occur that

- The marginal utility of additional profit from admitting HMO patients $\partial u/\partial \Pi_H$ is independent of the quantity of services provided to patients having traditional insurance [$\partial^2 u/\partial \Pi_H \partial Y_T = 0$, so the second term in the large parentheses of Eq. (8.25) may be dropped], or
- The marginal utility of additional profit from admitting HMO patients $\partial u/\partial \Pi_H$ increases with the quantity of services provided to patients having traditional insurance

$[\partial^2 u / \partial \Pi_H \partial Y_T > 0$, causing the second term in the large parentheses of Eq. (8.25) to be negative]. This condition corresponds to the usual notion that a hospital that predominantly serves patients having traditional insurance (in particular the privately insured) is more likely to be *profit-oriented*.

These results may be summarized and generalized in

CONCLUSION 8.5: If service providers in health care were pure profit maximizers, there would be no reason to expect cost shifting (e.g., from HMO patients to patients having traditional insurance). Cost shifting may, however, occur if their profit orientation is unaffected or becomes more important as they increasingly turn to patients having traditional insurance.

The empirical relevance of cost shifting appears to be a largely open question. Dranove (1988) does present empirical evidence from hospitals in Illinois that suggests that there is a compensating increase in the price of services charged to patients having traditional insurance, amounting to 0.5 US\$ for every US\$ lost due to reduced charges for Medicaid and Medicare patients; that is, that $dp_T/d\bar{p} = -0.5$.

8.4.3 Final Assessment of HMOs

Relying on an HMO as a complementary agent may be advantageous for many individuals, who obtain comprehensive health care at a lower cost than the conventionally insured, as a rule without a loss of health status. Therefore, one may conclude that HMOs result in an efficiency gain in health care. Efficiency gains are generally believed to be possible in health care, particularly in the hospital industry. Accordingly, so-called slack variables reflecting inefficiency are introduced in the cost function of the hospital to be presented in Section 9.2. This is justified with the hospital's lack of interest in cost minimization, which in turn goes back to its *rejection of pure profit maximization*. However, this rejection of pure profit maximization proves to be a necessary condition (but not a sufficient one) for health service providers to agree to a contract with a health insurer that provides for cost sharing on their part. The sufficient condition actually amounts to an *ethical objective* [as defined by Eq. (8.9); see Table 8.2] complementing the profit motive. Conversely, however, Conclusion 8.5 states that the rejection of pure profit maximization is responsible for the tendency *towards costs shifting*. Thus the same condition that permits an HMO to be a viable complementary agent prevents a final assessment of its efficiency because savings achieved by the HMO may result in cost increases falling on those having traditional insurance.

Even though HMOs may result in some ballooning of health care costs, they increase consumer welfare by adding to the choices among complementary agents available to consumers. This advantage is not to the detriment of physicians as a group as long as some of them view their participation in an HMO as a way of living up to their professional ethics [as formulated in Eq. (8.9), e.g.]. To the extent that hospitals also voluntarily agree to the terms of contract offered by the HMO, they as a whole are not on the losing side either.

Despite these advantages, it is questionable whether HMOs will acquire a significant market share outside the United States. In countries whose population is assigned to a national health service or to social health insurance without the freedom of choice, the insurer lacks the incentive to offer his services as a supplementary health care agent. However, even in countries such as France, Germany, the Netherlands, or Switzerland, where some competition exists between health insurers, it may be difficult for HMOs to gain a large market share. As long as *hospitals* in these countries are *subsidized by the government*, permitting them to charge fees below actual cost to socially insured patients, HMOs have a much smaller cost advantage than in the United States, where they achieve their savings mainly through less hospitalizations and patient days.⁵ But even if HMOs will only attract a minority of physicians and hospitals of a health care system, their viability proves that financial incentives may make health care providers use resources economically without necessarily indulging in low-quality medicine. A similar line of reasoning holds for other, more recent forms of managed care.

8.5 THE ROLE OF ASSOCIATIONS IN HEALTH CARE

While the issues surrounding the principal–agent relationship expounded in Section 8.1 are of particular relevance for the physician–patient relationship, they are also of concern to the service providers themselves. The individual physician, dentist, or pharmacist entrusts a professional association with the representation of his interests. He therefore faces the problem of how to tie an agent to his interests, in the same way as any other principal. Using the *medical association* as an example, it will be shown, however, that this problem may be considered solved by and large. At the same time, the demand for services provided by an association turns out to be especially marked in the case of the medical profession. The second part of this section is devoted to the consequences of the lobbying efforts exerted by these associations. Finally, the possibility of competition, if not between individual physicians but between their professional associations, will also be explored.

8.5.1 Why Are Professional Associations So Important in Health Care?

Professional associations and especially the medical associations play a decisive role in the health care systems of all industrialized countries. This observation may be interpreted as the result of a particularly large demand for association activity on the part of the service providers on the one hand and a particularly large supply on the part of the “entrepreneurs in matters of association” on the other. When considering the *demand side* there are three important reasons for the individual physician to join and support an association [see Zweifel and Eichenberger (1992); for an insightful discussion of the U.S. case, see Feldstein (1988), Chap. 3].

5. This difference is reflected by the first European HMO, which started business in Zurich (Switzerland) at the beginning of 1990. Its premiums are only 15 percent below those of conventional sick funds, compared to up to 30 percent in the United States.

1. Protection against competition. According to the economic theory of cartels, associations have the task to limit price and quality competition between their members in order to keep their incomes high. Professional associations in health care have a much better chance of achieving this aim than associations in other industries because consumers, being covered by insurance, have little incentive to seek out a lower-priced outsider. Even if they should go in search for one, it is difficult for them to compare costs as soon as there exists a detailed fee schedule permitting physicians a choice between a finely itemized and a global invoice.

2. Access to public funds. Medical associations serve as a partner in negotiations not only with the health insurers but also with public authorities. Since health care accounts for eight percent or more of the Gross National Product (see Table 1.1), two-thirds or more of which is financed by public funds in most industrialized countries [see OECD (1990), p. 10], medical associations provide access to a much larger volume of public purchases than do, for example, bar associations of lawyers.

3. Associations as guarantors of redistribution. The individual physician is involved in a formidable process of redistribution that shifts the cost of medical care occasioned by the older to the younger members of social health insurance, who typically pay the same contribution. The extent of this redistribution will continue to increase in the future due to the aging of the population (see Section 11.2). By defending "medical need" as the sole guideline of a physician's decisions, medical associations secure the continuity of this redistribution while consolidating the physician's key role in the process.

On the *supply side* too, a number of factors may be discerned that serve to lower the cost of forming and sustaining a medical association.

1. Easy price discrimination. It is comparatively easy to maintain discriminatory price collusion as medical services are always delivered to persons. In particular, a rich patient is not able to circumvent price discrimination according to income by having a poor person purchase the service at a lower price and sell it again to himself. So, in contrast to goods, there is no secondary market.

2. Easy control of market access. An association is able to restrict access to the market for medical services at little cost. The supply of medical services being tied to the medically trained individual, a firm from another industry wanting to gain access to the market would have to employ a great number of physicians. And as soon as the public authorities have been convinced of the need of limiting admissions to medical schools, entry into the profession is also highly restricted.

3. Homogeneous interests. Physicians are not very numerous, at least within a given area and of the same speciality. In addition, their investments in medical technology and human capital are very specific. These factors ensure considerable homogeneity of interests. Also, most medical practices are so small as to leave sufficient scope for medical associations to provide their members with exclusive services such as counselling on locational choice, accountancy, tax filing, and liability insurance. Exclusive services, however, create the basis for the continued existence of an association [see Olson (1965)].

These observations also apply to the other important providers of medical services, namely *dentists* and *pharmacists*. In contrast to them, nursing associations and other nonmedical associations are disadvantaged since they are not negotiating party to social health insurers. The following conclusion therefore holds true only to a limited extent for these nonmedical associations.

CONCLUSION 8.6: The demand for the services of professional associations is particularly strong in health care. At the same time, the cost of creating and sustaining an association is comparatively low. In combination, these facts explain the important role of professional associations in health care.

Once an association has been set up, it acts as an *agent of its members*. Although each individual member cannot monitor everything the association does, the principal–agent problem is definitely *not as severe* as in the relationship between the patient and his physician. The chairman of the association is usually a fellow colleague who after a limited term of service joins the rank and file again. In this way, a divergence of interest between the board of the association and its members can be avoided.

8.5.2 Tasks of Associations in Health Care

Professional associations in health care mainly have to deal with three groups, the insured and patients, politicians, and their own members. They should perform one main function with respect to each one of these three groups.

- Ensuring quality of treatment for patients;
- Performing tasks of coordination in the interest of politicians;
- Protecting the incomes of their members.

8.5.2.1 Quality Assurance

As shown in Section 8.2, one of the reasons for a demand for a complementary agent by patients to exist is patients' inability to judge the quality of treatment, sample sizes often being too small. Associations of health care therefore like to justify their existence by pointing out their contribution to the maintenance of professional ethics serving to guarantee a high standard of treatment.

This claim does not appear credible. In their analysis of Canadian physicians, Roos et al. (1977) found important differences in the style of practice within very small areas that cannot be easily reconciled with the notion of uniform quality standards being enforced by the local medical association. Benham (1991) quotes several cases in which American colleges of medicine shied away from repealing a licence to practice although the physician was patently incompetent. The reason for this failure may stem from the strong ties between the board of the association and its members mentioned in the preceding section. When a member of the board resigns from his position after a certain number of years to become a “simple” physician again, he must be prepared to face the consequences of any unpopular decision he has made during his time of service on the board. Finally, a proven way to guarantee a good quality of treatment

would be a *re-examination at regular intervals*. Pilots, who too are entrusted with the lives of many people, are administered regular tests to prove their skills. Medical associations, however, have never introduced this for their members.

8.5.2.2 Performing Tasks of Coordination in the Interest of Politicians

Besides patients, politicians form another group for whom the existence of associations is important. In a democratic society, politicians need to win votes for election, and professional associations in health care may be helpful in this respect. Medical associations again play the most prominent part among these professional associations because their voting recommendation may be spread through tens of thousands of practices. In the longer run, medical associations can also contribute to vote-getting by guaranteeing the *continued existence of redistribution* in the health care system. This redistribution not only occurs through uniform, age-independent contributions to social health insurance, but also through a differentiation of fees between members of social and private health insurance. Such a fee structure is in the interest of the majority of physicians, as it basically corresponds to the price differentiation of a profit-maximizing monopolist [see Kessel (1958)].

Medical associations may also support governments whose aim is to *stabilize the share of health* in the GDP (see Section 1.2 for a critical review of this policy). To start with, such a policy requires information that in many cases can only be provided by professional associations. This especially applies to limiting admissions to medical schools, which serves to keep health care expenditure low, at least in the short run. For administering the rationing process, medical associations have many experts who “perfectly know” how to choose the “right” candidates for a medical career. They can furthermore agree to exercise restraint in fee negotiations for a certain time, preventing the stabilization of the quantity of services from being offset by an increase in prices. In the longer run, however, they will want to use their increased market power to satisfy the demands for securing and raising the incomes of their members, as described in the following subsection.

8.5.2.3 Securing the Incomes of Members

Starting from the presumption that professional associations in health care exist primarily to protect their members from competition, one would predict that they seek to avoid an increase in the density of supply. In an international comparison, this should be easily achievable in a country like Germany with its particularly *high degree of organization*, which stems from the fact that membership in the medical association is a precondition for being able to contract with statutory health insurance. Thanks to its predominance, the medical association successfully lobbied in favor of limiting admissions to medical schools. In a country like Belgium, where medical associations are separated by language (Flemish versus French) and religious denomination (Catholic versus Protestant), they have no influence on the number of students admitted to medical schools. Physician density should therefore have increased rapidly over a number of years in Belgium, contrary to Germany.

Figure 8–7 shows the *development of physician density* (defined as a proportion of total employment) in four countries (along with the development of relative incomes; see below). During the 1970s the smallest increase of the density of supply took place in the United States, where the American Medical Association is known to control access to the market rather successfully through its influence on the eligibility of

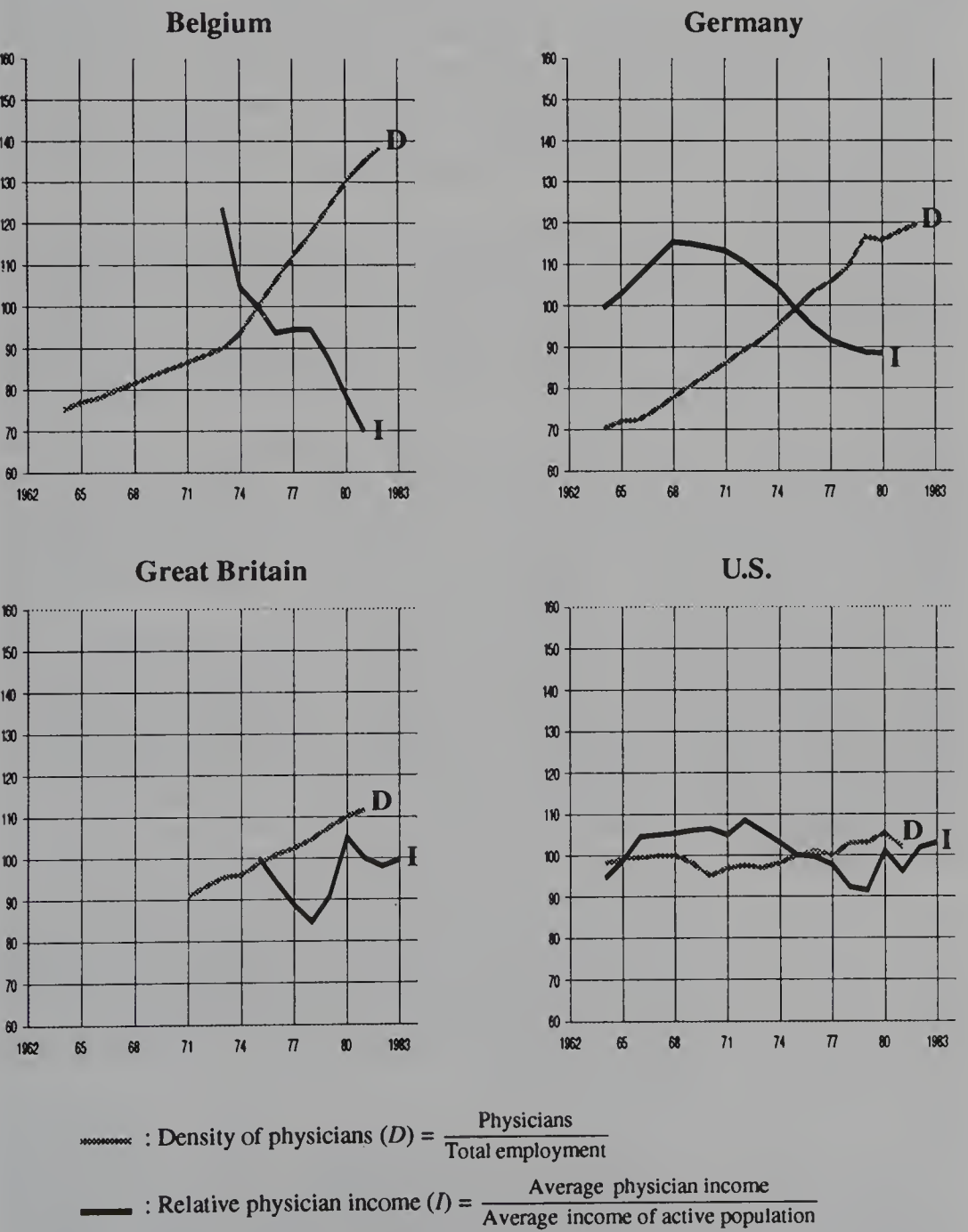


Figure 8-7. Development of the density of physicians and their relative income, 1975 = 100.
Source: Zweifel and Eichenberger (1992).

teaching hospitals for subsidization by the government. Density of supply increased very moderately also in Great Britain, where the government, being directly responsible for funding the NHS (National Health Service), plans and controls the number of physicians. Belgium is found at the opposite end of the spectrum as expected, with Germany occupying an intermediate position.

If an increase in the density of supply occurs despite its efforts, the medical association may still temper the adverse effects on its members. For example, it may negotiate to extend the benefits of social health insurance to the detriment of nonmedical suppliers ("counseling on prescription rather than by appointment with a psychologist"), or to allow general practitioners to administer and charge for therapies that were previously administered on an inpatient basis by hospitals. This leads to the prediction that in countries where medical associations play an important role, an increase in physician density does not necessarily bring about a marked drop of relative medical incomes.⁶

Using Figure 8-7, one can roughly estimate the *strength of the relationship between the density of supply and physician income* (relative to the average income of the active population). Great Britain is a special case because there seems to be no systematic relationship between the quantities. The sudden drop in income after 1975 was a result of the Labour government's attempt to stabilize spending on the NHS by cutting physicians' fees. This was reversed three years later by the new Conservative government headed by Mrs. Thatcher. The comparison between *Belgium* on the one side and the *United States* and *Germany* on the other is instructive, however. In Belgium, the density of physicians increased slowly during the 1960s. Nonetheless, a marked drop in their relative income occurred already in the early 1970s, whereas the decrease was very moderate in Germany, where regional medical associations are important negotiating partners of social health insurers.

These observations imply that in some countries medical associations have managed to maintain a career in medicine financially attractive using market closure. This presumption is supported by estimates of the so-called internal rate of return of a medical career, which will be discussed in Section 11.5. As long as graduating from medical schools remains an investment of above-average profitability, the threat of a physician glut will continue to exist, however.

CONCLUSION 8.7: Medical associations serve the interests of patients, politicians, and their own membership. Their behavior, however, gives rise to the suspicion that the interests of the second and third group [contributing to the (re-)election of politicians and securing of medical incomes] prevail, whereas there is little evidence in favor of a commitment to the interests of the first group (quality assurance).

6. Causality could also run the other way, from relative incomes to density of supply. A high relative income of physicians may encourage many school leavers to study medicine, giving rise to a positive correlation between the density of supply and income (allowing for a lag of a few years). Judging from Figure 8-7, this direction of causality seemed to be relevant in Germany until the end of the 1960s, but to have played a much less important role since then.

8.5.3 Competition between Service Providers, Competition between Associations

At first glance, unfettered competition between individual service providers attracting patients with low spot prices and a high standard of care would seem to be in the best interest of the insured. However, as demonstrated in Section 8.2, uncertainty surrounding future health status and inability to assess the physician's performance cause patients to prefer to have fees fixed *ex ante*. When negotiating such fees, the individual will usually enlist the help of the *health insurer as a complementary agent*. However, as soon as this health insurer, alone or in combination with others, begins to represent a large portion of a physician's clientele, he may try to exert pressure on the negotiated fee. It is against such a threat that the professional association protects its members. Given that it has successfully performed in fee negotiations, there is an especially strong incentive for the individual physician to join the association so as to avoid the conflict over the payment of nonobservable efforts, as described in Section 8.2.

Thus price competition between individual service providers is unlikely. What then are the chances of having professional associations compete in their stead, and what are the consequences of such competition? This question will again be examined from the point of view of three different groups.

1. Insured and patients. Competition between professional associations would be to a considerable advantage for this group. For one, it provides health insurers with the opportunity of negotiating lower fees. More important still, competition especially between medical associations also facilitates the establishment of entirely *new forms of health care delivery*. For example, successful creation of an HMO requires physicians to be able to break away from an existing association that clings to fee-for-service payment. In addition, a professional ethics orientation seems to be a necessary condition for the viability of plans that make physicians participate in the cost of health care [see Eqs. (8.9) and (8.19) of Section 8.3]. Physicians having such a professional ethics orientation could then join one of these *competing associations* catering to different physician preferences (e.g., with respect to the tradeoff between income and regular hours of work, besides professional ethics) while serving as negotiators for payment.

2. Politicians. Competition between professional associations in health care is not very advantageous to politicians because of the asymmetry of information between them and the associations [see Schulenburg (1987)]. In the long run, only those associations will survive that fight most for the interests of their members. Government and administration thus have increasingly to deal with associations that *filter and even distort information in their own interest*. In contrast to this, a professional association with compulsory membership may permit itself to paint a true picture, not being forced to compete for members.

3. Physicians. From the point of view of physicians, freedom of choice of professional association is a double-edged sword. On the one hand, medical associations forge a collective monopoly for their members in negotiations with

health insurers, resulting in fees for their services that are likely to be higher than in the case of competition between associations. This advantage would be lost if medical associations were to compete for contracts. On the other hand, competition between associations also allows for *competition between systems of health care provision* in the same country, which could in particular help newly licensed physicians to find job opportunities.

CONCLUSION 8.8: Competition between professional associations in health care may pave the way to new designs of health care provision, serving the interests of the patients and insured. For the majority of physician members as well as for politicians, however, competition brings more disadvantages than advantages.

In the longer run, associations in health care with compulsory membership will be able to survive with great difficulty only (at least in Europe), as they go against the intended opening up of labor markets within the European Union (EU). Efforts to achieve this opening have already resulted in the mutual recognition of diplomas of EU member countries. They will fail to achieve their objective, however, as long as an immigrant physician, dentist, or pharmacist sees himself forced to join a monopoly national association if he wants to become partner of contract with the social security system of his new country of residence.

8.6 EXERCISES

Exercise 8.1:

For the payment function (8.7) to create appropriate incentives, increased effort on the part of the agent must lower the probability of an unfavorable result. (a) Explain why this condition is important. (b) Rather than increasing the expected value of the result as pictured in Figure 8–2, increased efforts on the part of the agent could reduce its variance (without changing the expected value). Would this still meet the stated condition? Can you imagine instances of such a case? (c) If one distinguished simply between the states “healthy” θ_h and “sick” θ_s (as was the case in Section 3.4) and supposing that the physician’s efforts serve to reduce the probability of θ_s , would one face the same problems?

Exercise 8.2:

“Ballooning of cost” could also describe a situation in which an increase in the cost of treatment of HMO patients results in higher prices charged to patients covered by conventional insurance. What does the occurrence of ballooning of cost of this type depend on? [Hint: Delete the cost equality $C_T = C_H$ in Eq. (8.22) and use $d\tilde{C}_H$ instead of $d\bar{p}_H$ in Eq. (8.24)].

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9

HOSPITAL SERVICES AND THEIR PAYMENT

9.1 THE ISSUE

In debates about the economic problems of health care, the hospital plays a key role. This is due to the quantitative importance of the hospital industry. In most industrialized countries, hospital services account for the largest single block of health care expenditure. Moreover, this share has been rising considerably during the last two decades in developed countries (see Table 9.1).

However, in the course of economic development there are always industries that grow at an above-average rate. Furthermore, the hospital industry is part of the services sector whose increase in relative importance is generally regarded as reflecting a certain saturation of the demand for material consumer goods. So why should one be

Table 9.1: Hospital Expenditure as a Share of Total Health Care Expenditure (in percent)

<i>Year</i>	<i>1960</i>	<i>1970</i>	<i>1980</i>	<i>1990^a</i>
Canada	43.6	52.2	52.6	49.6
Federal Republic of Germany	n.a.	35.7	36.1	35.4
France	34.7	38.0	48.1	44.1
Italy	43.2	47.6	54.0	48.3
Japan	34.1	26.4	30.7	32.0
Netherlands	n.a.	55.1	57.3	53.9
U.S.	37.8	44.1	48.9	46.4

^a For Canada: 1987, for Federal Republic of Germany and Japan: 1988, for Italy: 1989.

Source: OECD (1991), Health Data File.

worried about this development and why should economists be expected to deal with it?

In answering this question, one should refer back to the general conclusion stated in Chapter 1, viz. that the *development* of health care expenditure is *not* really of interest to economists but their *amount and structure*. Efficient use of resources has to do with the issue of whether the “right” quantity of health care goods is supplied in the “right” composition to the “right” consumer and produced at the lowest social cost possible. Therefore, a strongly expanding part of the health care system may function in perfect efficiency, reflecting growth of technological change in medicine, whereas a part in decline may just as well be inefficient.

These considerations apart, there are two main reasons motivating a close examination of expenditure in the hospital sector. The first reason has to do with the *structure of the hospital market*. Patients do not have to bear the full cost of the inpatient services they consume; as a rule they face a nominal copayment only and hence an effective price of almost zero. This is due to health insurance, which spreads the financial risk of sickness, thereby increasing the welfare of risk-averse consumers. The beneficial effects of health insurance are even more important in the case of hospital care than other areas of health care, because the financial risk of a stay in hospital looms particularly large.

This benefit of insurance coverage has to be traded off for a price mechanism that cannot bring supply and demand into line in the same way as in a competitive market. For as soon as the consumer does not have to pay for a product, he no longer demands a quantity reflecting his relative valuation in comparison to its scarcity. Therefore, the price mechanism can no longer bring about the “right” amount of resources for use in an industry, that is, an allocation in accordance with the preferences of consumers.

In view of this “market failure,” legislation in most industrialized countries stipulates that the financial means to be made available to the hospital industry shall be negotiated *between hospitals and social health insurers*, following defined rules. With such negotiations, the danger arises that hospitals provide a quantity of services that is too large or too small compared to the social optimum, the outcome depending on the relative strength of negotiating parties.

The second reason for the economist’s interest has to do with *the structure of firms* in this market. Hospitals are in their vast majority not for profit and consequently have—at least *prima facie*—no intrinsic motivation to produce at minimum cost. This indicates that the quantity of services negotiated will not as a rule be supplied using the minimal amount of valuable resources.

For economists, it is appropriate first of all to give the term *efficient use of resources* a more precise meaning with reference to a specific area of application, the hospital industry in this case, and then to examine the effect of incentives on the efficient use of resources emanating from alternative systems of control in this industry.

The *principle of efficient use of resources* generally calls for the minimization of the cost of an economic activity for a given return. While “costs” result from the consumption of productive resources (manpower, energy, raw materials) that have alternative uses, the definition of “return” in health care and especially in the hospital is not as obvious. Difficulties arise in defining hospital output as well as in making the

definition operational for measurement. These problems will be discussed in detail in Section 9.2.

The most important element of the system of control is the choice of procedure for allocating resources to hospitals, in short, the “payment system.” The analysis of payment systems not only provides insights into their incentive effects but also leads to recommendations for hospital policy, since the choice of payment system is determined by public regulation and legislation. The detailed analysis of alternative payment systems for hospital services will be discussed in Section 9.4. Since the effect of a specific incentive system also depends on the objectives of the actors involved, these objectives will be first dealt with in Section 9.3, using a simplified economic model of the hospital.

9.2 THE HOSPITAL AS A PRODUCTIVE UNIT

9.2.1 Hospital Output: Health as a Latent Quantity

In order to measure hospital “output,” it is not enough to describe the tasks that are carried out (surgery, radiotherapy, medication, wound dressing, and accommodation, to name a few) or bundles of tasks such as “medical services,” “nursing services,” and “hotel services.” All these tasks are only a means to an end. One gets closer to true output by asking the question of what the patients (or referring physicians acting in their interest) want, what they expect from their hospitalization, and what the taxpayers expect to obtain in return for their contribution to the financing of the hospital.

In the majority of cases, expectations are in terms of a positive *influence on the patient's state of health*, that is, the curing of disease (or control of its development) and the alleviation of pain. Even though there is little disagreement with regard to these objectives, the degree of their realization nevertheless may hardly serve as a basis for payment of hospital services. The difficulties lie with the measurement as well as the imputation of effects to services performed.

In order to measure the extent of recovery of patients, their health status would have to be evaluated at the beginning and at the end of hospital treatment, using objective criteria. This is—except for obvious indicators such as the survival and complication rates of operations—a fairly hopeless undertaking, because health is not only multidimensional but also contains a considerable subjective component. The latter is of growing importance as medical care gets more sophisticated.

Even if this difficulty is surmounted, one should not simply tie payment for hospital services to the measured change in the state of health achieved during the period of hospitalization. For the relevant comparison for assessing hospital performance is not with the patient's actual state before admission, but the (hypothetical) state that *would have been realized without hospital treatment* at the end of the observation period. The importance of this distinction becomes very clear in cases where hospital treatment can only slow down the progressive course of an incurable disease. The extent of “deterioration averted” defies measurement, however.¹

1. Prognoses from “comparable” cases are not very reliable, because no two cases are ever the same.

Patients hold expectations not only with regard to the final state of their health after hospital treatment but also with regard to their physical and mental well-being *during hospitalization* (to the extent that the disease will permit), as life goes on in the hospital. This aspect takes on special significance if the disease itself can no longer be fought and only suffering alleviated, that is, when dealing with incurable and terminally ill patients. However, objective and reliable measurement of subjective well-being during hospitalization is just as difficult as measuring the influence of hospitalization on the patient's health status.²

Finally, the patient group does not just cover people who will actually be treated as patients in a hospital, but the whole population of its catchment area. The mere existence of hospitals provides people with the security that in case of accident or serious illness inpatient treatment will be available. This so-called "option demand" is satisfied by hospital beds held in reserve, along with the necessary staff and equipment.

CONCLUSION 9.1: Hospital output consists of improving or maintaining the patient's state of health on the one hand and the capacity to satisfy an option demand on the other hand. The former part of output is particularly difficult to operationalize and can only partially be attributed to the hospital.

9.2.2 The Multistage Character of Production in the Hospital

As the final outcome of hospital activity (particularly the improvement of a patient's health status) may only be measured imperfectly, observable quantities that are suitable to serve as *output indicators* have to be identified for obtaining an operational definition of the term *efficient use of resources*. In the present context, it seems appropriate to list various indicators of hospital activity and to classify them according to the stage of production, using a scheme that may help to describe hospital activity from an economic perspective.

The indicators most commonly used are [Breyer (1987)]:

- Quantities of *factors of production* (hours worked by physicians, by the nursing staff, and by other employees, drugs and dressings, electricity, fuel, etc.);
- Quantities of *individual medical and nursing services* performed (examinations, operations, medications, injections, physical therapies, temperature measurements, meals, etc.);
- The *number of patient days*, possibly differentiated according to intensity of care;
- The *number of patients* (or *cases*) treated, possibly differentiated according to various types of diseases.

2. This argument will, however, tend to be somewhat less important in the future, as instruments for measuring the (marginal) willingness to pay as presented in Chapter 2 are used more and more [for example, see Johannesson, Jönsson, and Borquist (1991)].

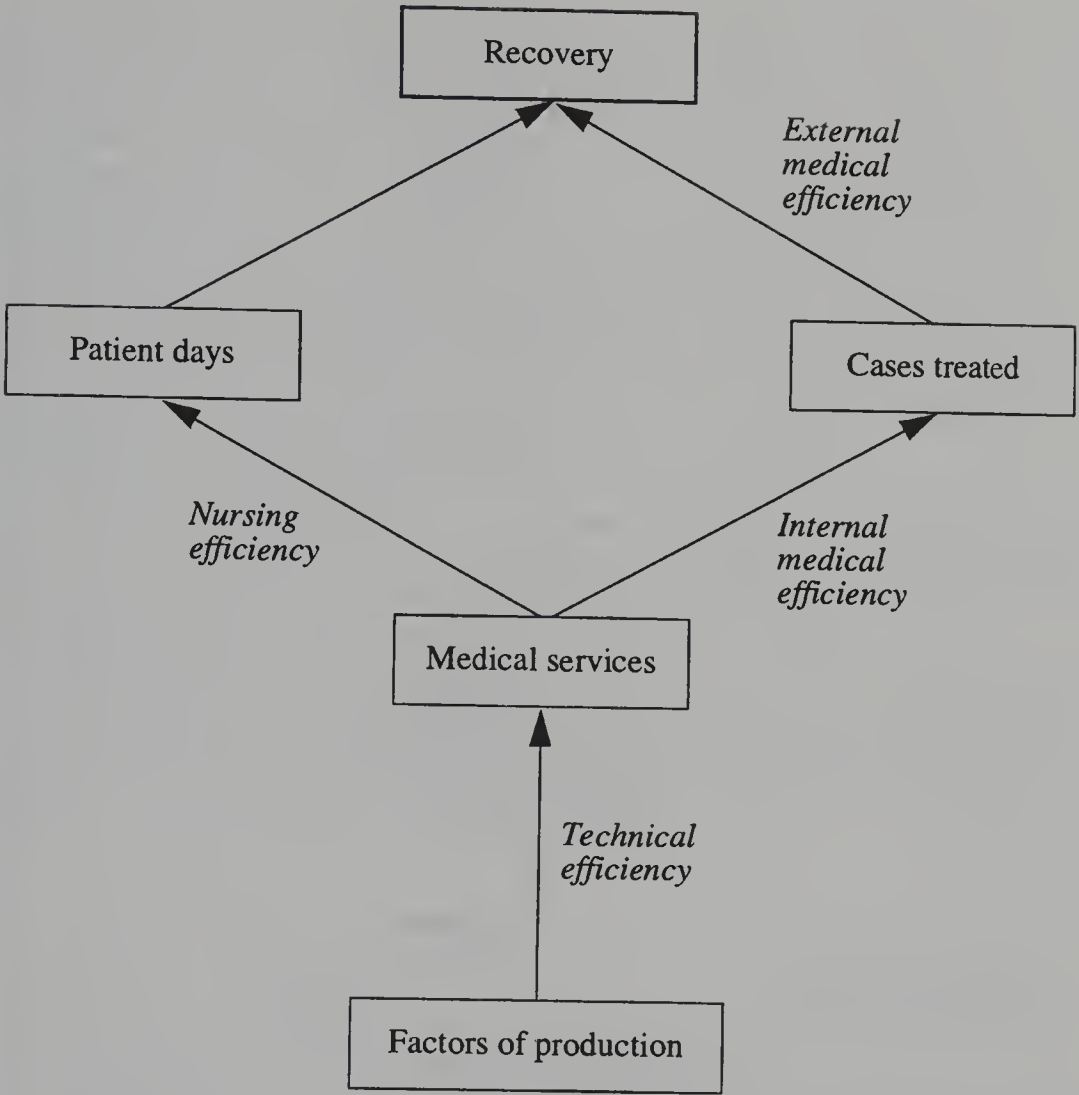


Figure 9-1. Patient days and cases treated as intermediate hospital products.

Figures 9-1 and 9-2 show two alternative views of the different stages of hospital production involving these quantities. In both of them, the factors of production, which can be interpreted as primary inputs, appear at the bottom level. They serve to produce various medical services, which are therefore located at the second level (secondary inputs). Minimizing factor inputs for producing a given bundle of services may be called “technical efficiency.”

With regard to the two remaining indicators, cases treated and patient days, two different views can be justified.

I. Cases treated and days in hospital may be seen as indicators of different *intermediate products* at the same stage (Figure 9-1), which thus enter immediately below the true output, “recovery.” The number of cases treated reflects the medical component and the number of patient days, the nursing component of hospital services. Different types of services in the form of (direct) inputs may be assigned to the two

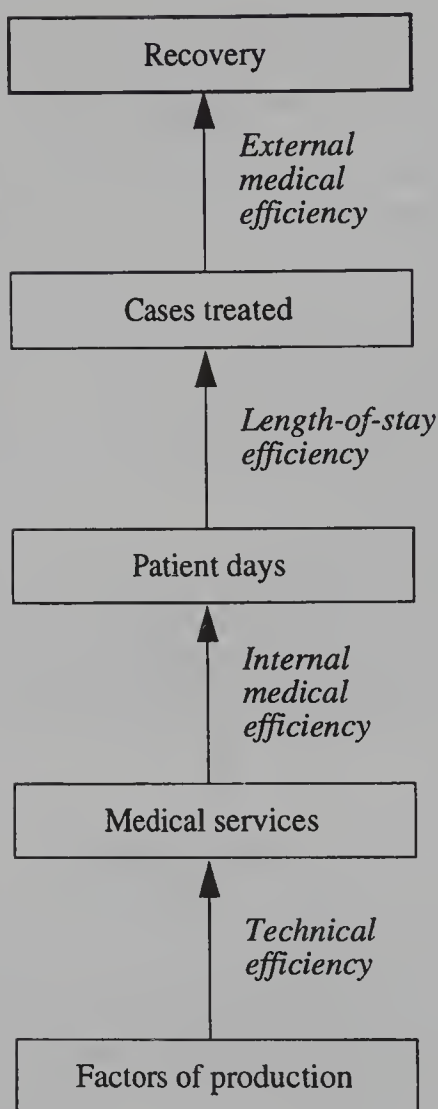


Figure 9-2. Patient days as an input to the treatment process.

components. This results in two different types of efficiency, viz., the use of the least amount of nursing services possible per day (“nursing efficiency”) and the least amount of medical services possible per case (“internal medical efficiency”) for a given improvement of health status.³

2. Alternatively, one may argue that hospitalization per se neither gives rise to utility (because the patient is taken away from his normal environment) nor improves the patient’s health status as such. In contradistinction to case (1), *days spent in hospital are themselves considered as an input* into the treatment process (Figure 9–2). In this view, minimizing the length of stay for a given spectrum of diseases constitutes a type of efficiency of its own (“length-of-stay efficiency”).

3. In addition, the economic objective of the health system as a whole is to achieve a given degree of health improvement at the lowest cost possible (for hospitalization as well as other services). This enters as “external” medical efficiency in Figure 9–1.

This second approach obviously presupposes the existence of binding norms for measuring health status at discharge because otherwise it would be possible to increase length-of-stay efficiency by simply sending patients home “quicker but sicker.”

In this context, the question arises as how to characterize a “bed.” At first sight, the bed seems to be an input to the production of hospital services simply because a hospital cannot exist without beds. However, with a given (maximum) number of cases simultaneously treated in the hospital, the required input of the factor “beds” is uniquely determined, causing additional beds to have a marginal productivity of zero. Thus one may conclude that a fixed coefficients type of technology applies to beds with respect to the output, “cases treated.” A further problem with this approach is that no (running) costs are associated with the factor “beds.” One could possibly regard the depreciation on the purchase price of beds as the user cost of beds.

Conversely, in accord with the notion of option demand discussed in Section 9.2.1, an *empty bed*, like excess capacity in other industries such as electricity generation, *also represents an output* of the hospital in question. This holds true as long as the number of empty beds or beds occupied by patients ready for discharge remains within the bounds defined by requirements in case of a catastrophic event.

Following this latter approach, one may assign that part of a hospital’s factor endowment that must be allocated independently of the number of patients to the service capacity held on reserve, making the number of beds another indicator of a hospital’s output. Completing the scheme of Figure 9–1 in this way yields Figure 9–3.

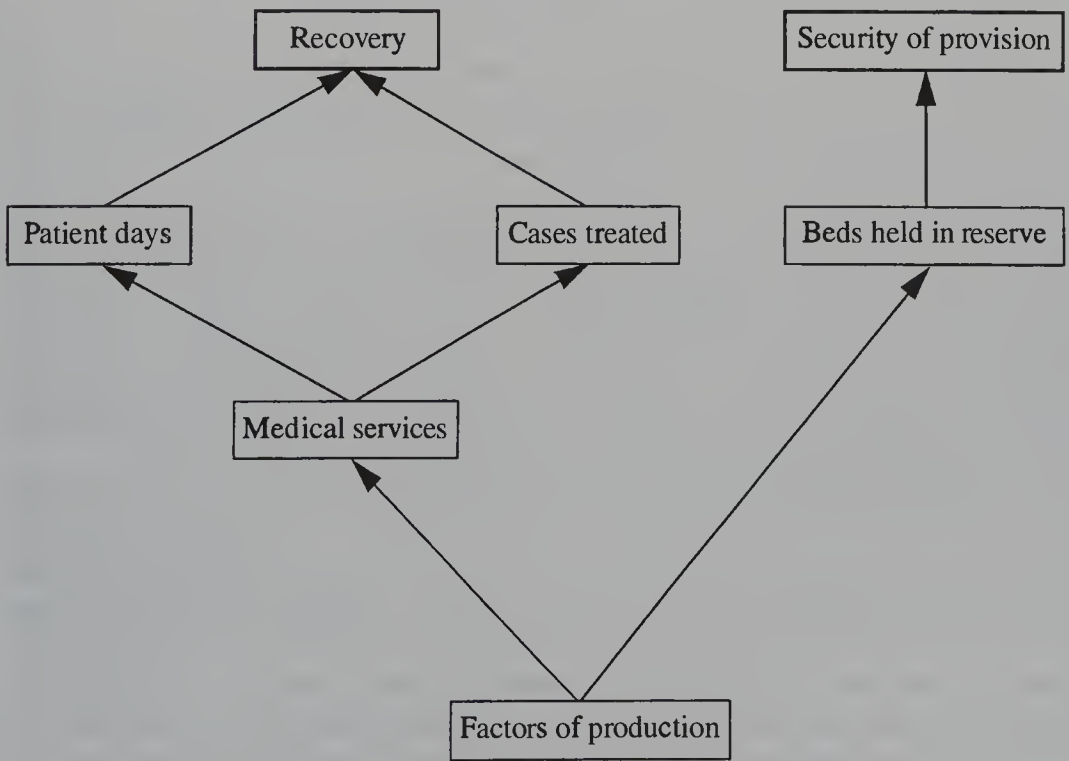


Figure 9-3. Beds held in reserve as an additional hospital output.

CONCLUSION 9.2: A hospital's "output" can be described as the outcome of a multistage process, with each stage being assigned its specific concept of efficiency.

9.2.3 The Heterogeneity of Hospital Output

Another problem with defining and measuring the output of a hospital stems from the fact that there exists a considerable amount of heterogeneity, be it at the level of output indicators or of intermediate products. Consider the number of cases treated in a hospital in the course of a year. Is it possible to describe adequately a hospital's output simply by counting the number of cases? Is it fair to say that a hospital that treats 1,000 cases achieves more than the one that treats 995? Obviously, it would be foolish to agree because the 1,000 cases of the first hospital might consist of 500 minor fractures and 500 uncomplicated tonsillectomies whereas the second hospital might be a heart unit specializing in transplantations.

One must therefore take into consideration that the "case treated" does not constitute a homogeneous quantity but a mental construct that needs to be specified through its characteristics. This means that a treatment case must be differentiated along various dimensions, for example, according to

- The *type of illness* that has called for hospital treatment (principal diagnosis);
- The *severity* of the illness and complications arising during treatment;
- The *stage* of the disease (e.g., in the case of cancer);
- *Concomitant diseases* (secondary diagnosis);
- *Patient characteristics* reflecting his or her contribution to the "production of recovery," such as age and possibly sex.

In view of these distinctions, to which still more could be added, a purist must come to the conclusion that the heterogeneity of the case mix can only adequately be taken into account by considering each patient as an output category of his own. However, if one were to follow this principle, one would forego the possibility of comparing the output vectors of two or more hospitals. This would put an end to the economic analysis of the hospital in the quest of, for example, measuring the degree of efficiency or determining a performance-based payment system.

A reasonable compromise between the rigorous approach just outlined and the total abandonment of case differentiation may be achieved by dividing patients into a manageable number of groups, using the distinguishing characteristics mentioned in the preceding paragraph. This division is known as "*patient classification*" and purports to form a manageable number of patient groups that are as homogeneous as possible. Moreover, assignment to a group should be unique and reproducible using objective criteria. Obviously, there is a conflict between the criteria "manageable number" and "greatest possible homogeneity within each group" that can only be resolved by weighing the relative disadvantages caused by their violation.

The three most common patient classification systems are:

1. The “*international classification of diseases*” (ICD), which was originally developed as a basis for mortality statistics and thus solely refers to (principal) diagnoses. In its three-digit version, the ICD consists of more than 900 groups, while aggregation to the level of 110 main groups is already very coarse. For instance, all benign neoplasms constitute one single group in this categorization.

2. The “*diagnosis-related groups*” (DRG), developed at Yale University in the 1970s, with the explicit objective to create relatively cost-homogeneous groups. Besides the principal diagnosis, the DRG take into account the existence of concomitant disease and complications, the age of the patient, and the type of treatment (surgical or conservative), getting by with 467 groups [see Fetter et al. (1980); Health Care Financing Administration (1983)].

3. The “*patient management categories*” (PMC), also developed in the United States (Pittsburgh), consists of a total of 840 groups. Compared to DRGs, the PMC put greater emphasis on concomitant diseases as well as the treatment strategies chosen by the hospital [see, e.g., Young (1991)].

CONCLUSION 9.3: A patient classification system represents the attempt to do justice to the heterogeneity of the hospital output while making comparisons between hospitals possible. All systems share the idea of describing hospital output in some detail, if not with regard to treatment outcomes, that is, the improvement of health status, at least with regard to the difficulty of the task (see Section 9.2.1).

9.2.4 Hospital Cost Functions

Considering the hospital as a productive unit suggests the application of elementary concepts of production theory to hospitals. One of the core concepts of production theory, also of utmost empirical relevance, is the cost function. Unlike the production function, the cost function is defined for multiproduct firms as well, *assigning minimal costs of production to each output bundle*. The cost function, while containing the same information as the production function,⁴ is more easily amenable to econometric estimation. The quantities on the right-hand side of the production function are input quantities. Being chosen by the firm, they cannot be considered exogenous. In contrast, the quantities on the right-hand side of the cost function are factor *prices* and output quantities. These variables—and in the case of a short-term cost function, the amount of fixed factors—may be considered *exogenous* provided perfect competition prevails on the factor market.

The empirical estimation of cost functions is helpful for answering a number of important economic questions.

1. From the shape of the cost function, one may recognize *economies* or *diseconomies of scale*, which are important for determining the optimal size of the unit.

4. This is a key result of duality theory [see, e.g., Diewert (1974)].

Optimal size is of relevance to policy because in some countries the hospital industry is subject to public regulation prohibiting hospitals freely to choose their number of beds.

2. Differentiation of the cost function with respect to the number of patients of a given category yields the *marginal costs of treatment* of this patient type. This information is useful for calculating prices in the context of a performance-based type of payment (see Subsection 9.4.2.4).

3. From the size of the residuals of an estimated cost function, that is, the difference between actual and estimated cost of hospitals, statements about their *relative efficiency* can be derived. This may facilitate the monitoring of hospital performance under a payment system based on cost reimbursement.

As mentioned above, the microeconomic cost function may contain explanatory variables of three types: output quantities, factor prices, and (in their short-term variant) the amounts of fixed factors of production. Furthermore, their estimation proceeds on the assumption that the units included in the sample all aim to minimize costs and that deviations from this target are random.

This assumption is hardly sustainable in the case of hospitals due to the predominance of government and community ownership, since the absence of the profit motive diminishes the pressure to minimize costs. This led Evans (1971) to coin the term *behavioral cost function*, that is, a function designed simply to explain the costs resulting from actually observed behavior. This function may also include determinants such as the type of ownership and organization of a hospital or the system of payment for hospital services, which should have no influence on minimum cost but that help to explain systematical deviations from minimum cost.

Therefore, a typical *behavioral cost function* takes on the form

$$C = C(X_1, \dots, X_m; Y, Z; W; T, D_1, \dots, D_n) \quad (9.1)$$

with C denoting the hospital's total cost, $X_1, \dots, X_m$ a vector containing the number of cases in the m patient groups distinguished, Y the number of hospital days, Z the number of beds, W a measure of the factor price level, T the type of ownership, and $D_1, \dots, D_n$ additional hospital characteristics. By including the activity variables $X (= \sum X_i)$, Y , and Z in the linear as well as the quadratic form, increasing and decreasing returns to scale may be modeled. For econometric reasons, both sides of Eq. (9.1) are divided by the number of cases X prior to estimation,⁵ making *cost per case* the dependent variable.

From the wealth of estimations of hospital cost functions presented in the literature,⁶ a study by Breyer, Paffrath, Preuss, and Schmidt (1988) shall be described in some detail here. It is based on data from 614 hospitals in the Federal Republic of Germany, referring to 1983. As an indication of case mix, the shares of patients falling

5. Dividing through by the number of cases serves to avoid heteroskedasticity (meaning that the variance of the residuals varies systematically with one of the explanatory variables). Indeed, the variance of total cost usually increases with the size of the hospital.

6. For surveys of the literature, see Cowing, Holtmann, and Powers (1983).

Table 9.2: Hospital Costs in Germany (1983, *n* = 614)

Dependent variable: Cost per case (<i>C/X</i>)			
Independent variables		Coefficient (OLS) ^a	Standard error
Constant		600.64**	(187.57)
1/ <i>X</i>	(1/number of cases)	−559,412**	(71,244)
<i>X</i>	(number of cases)	−0.031	(0.059)
<i>Z/X</i>	(beds per case)	20,552**	(6,919)
<i>Z</i> ² / <i>X</i>	(beds squared per case)	42.65	(95.03)
<i>Y/X</i>	(patient days per case)	119.56**	(25.41)
<i>Z</i>	(number of beds)	−0.789	(4.15)
<i>Y</i>	(number of hospital days)	12.09	(15.56)
<i>ZY/X</i>	(interaction term)	−0.246	(0.353)
Public ownership (= 1 if applicable)		219.74*	(103.94)
Ownership by a voluntary institution (= 1 if applicable)		−113.86	(101.03)
Number of inhabitants of community of residence		20.93*	(8.22)
Teaching hospital (= 1 if applicable)		−54.87	(68.47)
Emergencies per admission		40.51	(236.75)
Referrals received per admission		710.15	(903.22)
Patient days in single room accommodation		−183.94*	(77.13)
Patient days in double room accommodation		−28.67	(35.10)
(+ 30 factors for case mix, of which			
6 significant at the 0.01 level of significance			
5 significant at the 0.05 level of significance)			
$\overline{R}^2 = 0.747$			

^a ** = estimated coefficient significant at the 0.01 level of significance.

* = estimated coefficient significant at the 0.05 level of significance.

Source: Breyer et al. (1988), 84–86.

into 110 main ICD categories mentioned in the preceding section were recorded. The results of an ordinary least-squares (OLS) estimation of the equation for cost per case are summarized in Table 9.2. In order to simplify the interpretation, both sides of the estimated equation are multiplied by the number of cases *X* again. In this way, the following equation for total cost is obtained (with asterisks indicating statistical significance; see footnote to Table 9.2),

$$C = - 559,412^{**} + 600.64^{**}X - 0.031X^2 + 20,552^{**}Z + 42.65Z^2 + 119.56^{**}Y - 0.789XZ + 12.09XY - 0.246YZ + \hat{\alpha}(K + D)X \quad (9.2)$$

where *X* is the number of cases, *Y* the patient days, and *Z* the number of beds.

All hospital characteristics and diagnostic variables listed in the lower part of Table 9.2 are lumped into *K* and *D*, respectively, with $\hat{\alpha}$ representing the regression coefficients pertaining to them.

The results may be interpreted as follows.

1. Economies of scale. Judged by the insignificant coefficients of X^2 and Z^2 , there is no evidence of deviations from constant returns to scale, which seems to indicate that German hospitals do not have a definite optimal size of firm.

2. Marginal cost. The marginal cost of holding an additional bed in reserve can be calculated by taking the partial derivative of Eq. (9.2) with respect to Z ,

$$\frac{\partial C}{\partial Z} = 20,552 + 85.3Z - 0.789X - 0.246Y$$

Evaluated at the sample means ($\bar{Z} = 299$, $\bar{X} = 6,399$, $\bar{Y} = 90,700$), the marginal cost of a bed is DM 18,696 or roughly 6,400 US\$ per year at 1983 prices and exchange rates. With 21.4 cases per bed and year on average, these capacity costs amount to DM 874 (US\$ 300) per case. Analogously, the marginal cost of an additional patient day, $\partial C/\partial Y$, amounts to DM 123.43 evaluated at the sample means. With an average length of stay of 14.2 days, this gives DM 1,753 per case, whereas the pure marginal cost of an additional case $\partial C/\partial X$, with number of beds and the number of patient days held constant, amounts to DM 1,053 (again calculated at sample means). Therefore, the total marginal cost per case is DM 3,680 ($= 874 + 1,753 + 1,053$), of which 24 percent is accounted for by the provision of beds, 47 percent by the cost of care, and 29 percent by costs caused by an additional admission. The high value of the capacity cost share of 24 percent is at least in part due to the fact that total cost comprises but operating expenditures, thus neglecting depreciation and interest on capital.

3. Impact of disease type. A considerable fraction of the variation in costs per case treated can be traced to differences in the diagnostic mix of hospitals. The corresponding regression coefficients allow one to identify "expensive" and "cheap" diseases. The estimated costs per case treated (not shown in Table 9.2) in the 53 most frequent ICD categories vary between DM 689 (for skin and subcutaneous cellular tissue infection) and DM 9,340 (for other diseases of the respiratory system), again at prices of 1983.

4. Influence of property rights. Public hospitals clearly display higher costs per case than for-profit private hospitals (the benchmark category not entered in Table 9.2), which corroborates the traditional microeconomic theory of property rights. Hospitals owned by voluntary institutions appear to operate at even lower cost than for-profit ones, the difference not being statistically significant, however.

5. Relative efficiency. A hospital's predicted costs (i.e., the cost calculated based on its characteristics, especially its case mix, using estimated coefficients) may be compared with its observed expenditure to obtain a clue for assessing its relative efficiency. In 15 of the 614 hospitals ($= 2.4$ percent), actual cost differs by more than 30 percent from predicted cost, with maximum deviations amounting to -41 percent and +91 percent, respectively.

9.3 AN ECONOMIC MODEL OF THE HOSPITAL

In the following, a simplified economic model of the hospital is presented with the aim of

- Applying elementary relationships of production theory to the treatment process in the hospital;
- Describing “the hospital” as an economic actor with its own objectives by first studying individual groups of actors in the hospital and their respective preferences;
- Creating the framework for a formal analysis of the effects of alternative payment systems for hospital services, to be performed in Section 9.4.

9.3.1 Notation and Assumptions

Ignoring the fact that hospitals are multiproduct firms, we assume that there is only one type of patient (case treated). As in the preceding section, the number of cases per period is X , the number of beds is Z , and the quantity of medical and nursing services performed on patients is Y , which also includes the number of patient days. Let the quality of treatment Q depend on the amount of services provided per patient (see Table 9.3 of Section 9.4 for an overview of the model),

$$Q = g\left(\frac{Y}{X}\right), \quad \text{with } g'' < 0 \text{ and } g'\left(\frac{Y}{X}\right) \leq 0 \text{ for } \frac{Y}{X} \leq \gamma \quad (9.3)$$

This means that the function g is concave from below, increasing in Y/X up to a *level of saturation* $Y/X = \gamma$ and decreasing thereafter. This assumption may be justified with reference to a level of medical and especially technical services beyond which the stress imposed on the patient is not compensated by the improved outcome of treatment.

For the sake of simplicity, the numerous inputs are aggregated into two groups only. The first includes (medical and nonmedical) labor, their quantity and unit price being L and w , respectively. The second group comprises all other factors (capital, energy, medical supplies, etc.), their quantity being symbolized by K and their price by r . Thus actual operating expenditure of the hospital is given by

$$A = wL + rK \quad (9.4)$$

These factors are used for holding beds on reserve and for the production of medical services. In order to keep the analysis simple, assume constant factor prices. This means that for a given quantity of beds and services, the cost-minimizing factor quantities are uniquely determined, at $L^*(Y, Z)$ and $K^*(Y, Z)$, respectively. The hospital will, however, not necessarily use the factors in a technically efficient, let alone cost-minimizing, manner, resulting in *observed factor quantities* that are given by

$$L = L^*(Y, Z) + L^S \quad (9.5)$$

$$K = K^*(Y, Z) + K^S \quad (9.6)$$

The superscript S ("slack") indicates that L^S and K^S are wasted inputs, such as working at a slow pace or careless handling of materials.

The cost function is defined in the usual microeconomic terms as the *minimal* expenditure for providing a given output,

$$C = C(Y, Z) = wL^*(Y, Z) + rK^*(Y, Z) \quad (9.7)$$

In this function, the number of beds Z is considered exogenous, giving rise to fixed costs that are equated to the minimal cost of providing this same number of beds Z while the number of patient days is held at zero,

$$C^f = C^f(Z) = C[0, Z] = wL^*[0, Z] + rK^*[0, Z] \quad (9.8)$$

Turning to the revenue side, a general revenue function is postulated that comprises all possible payment systems as special cases,

$$R = R(Z, X, Y, A) \quad (9.9)$$

On the one hand, the *intermediate outputs*, "patient days" or equivalently "medical services," symbolized by Y , and actual expenditure, symbolized by A , are determined by the hospital. On the other hand, the *number of beds* Z is *exogenous* (set by the regulatory agency). The number of cases treated X constitutes a decision variable that however may be varied by the hospital within bounds only, being partially controlled by referring physicians, referring hospitals, and patients themselves (e.g., in emergencies).⁷ Even in the case of so-called spontaneous admissions, the patient has to check in at the outpatient department first, so again the number of cases cannot be increased in arbitrary fashion.

However, it is quite realistic to assume that the hospital competes for patients and thus *indirectly influences the number of admissions* by earning a reputation for high-quality treatment among community physicians and for good care with the general public (see Figure 9-3). We therefore assume the number of patients to depend on the quality of treatment within certain exogenous bounds $[\underline{X}, \bar{X}]$,

$$X = \varphi(Q) = \varphi \left[g \left(\frac{Y}{X} \right) \right], \quad \text{with } \varphi' > 0 \text{ and } \underline{X} \leq X \leq \bar{X} \quad (9.10)$$

7. The number of patients may be influenced to a greater degree by a hospital that has attending physicians on its staff, as is common in the United States. This fact results in the bounds $[\underline{X}, \bar{X}]$ covering a comparatively wide range [see Eq. (9.10)].

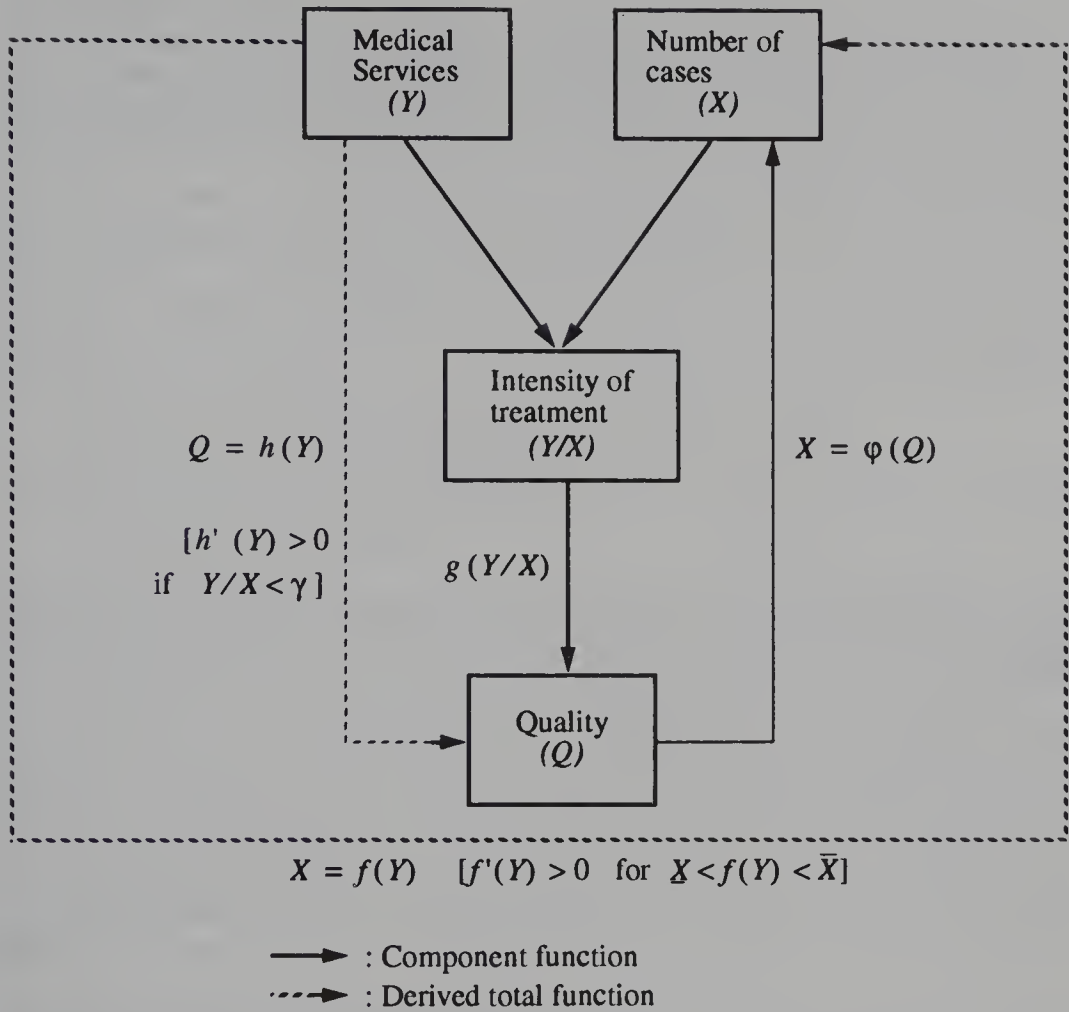


Figure 9-4. Medical services (Y) as the ultimate decision variable of the hospital.

This implicit equation in X may be solved to show explicitly the dependence of X on the hospital’s ultimate decision variable Y,

$$X = \begin{cases} \underline{X}, & \text{if } f(Y) < \underline{X} \\ f(Y), & \text{if } \underline{X} \leq f(Y) \leq \bar{X} \\ \bar{X}, & \text{if } f(Y) > \bar{X} \end{cases} \tag{9.11}$$

with $0 < f'(Y) < X/Y$ as long as $Y/X < \gamma$.⁸ This “demand function” states that given a low level of medical services Y, the hospital attracts some minimum number of patients

8. Total differentiation of $X = \varphi[g(Y/X)]$ yields $dX = \varphi'(Q)g'(Y/X)\{dYX - YdX\}/X^2$, and after ordering terms,

$$f'(Y) = \frac{dX}{dY} = \frac{\varphi'(Q)g'(Y/X)X}{\varphi'(Q)g'(Y/X)Y + X^2} = \frac{X}{Y + B}, \text{ with } B = X^2/\{\varphi'(Q)g'(Y/X)\}$$

As long as $Y/X < \gamma$, $g' > 0$ according to Eq. (9.3), $B > 0$, and thus $f'(Y) < X/Y$.

$\underline{X}$. By increasing Y , the hospital attracts additional patients, at least as long as service intensity $Y/\underline{X}$ remains below the critical value γ . However, the hospital can never attract more than $\bar{X}$ cases per period. Figure 9-4 illustrates the "total" relationship $X = f(Y)$ between medical services and number of cases, which takes into account the feedback from quality Q on the number of cases X .

Alternatively, quality can be expressed as a function of the quantity of medical services Y as follows (see Figure 9-4 again),⁹

$$Q = g \left[\frac{Y}{X} \right] = h(Y),$$

$$\text{with } h'(Y) > 0, \text{ if } \begin{cases} X = \underline{X} \\ X = f(Y), \\ X = \bar{X} \end{cases} \quad \text{as long as } Y/X < \gamma \quad (9.12)$$

Profits, or equivalently, *excess revenue* of the hospital Π are given by the well-known identity

$$\begin{aligned} \Pi &= R - A = R(Z, X, Y, A) - wL - rK \\ &= R(Z, X, Y, A) - C(Y, Z) - wL^S - rK^S \end{aligned} \quad (9.13)$$

Substituting Eq. (9.11) into Eq. (9.13), one obtains

$$\Pi = R\{Z, f(Y), Y, A\} - C(Y, Z) - wL^S - rK^S \quad (9.14)$$

For a given revenue function $R\{\cdot\}$, demand function $f(\cdot)$, and an exogenously determined number of beds Z , this equation shows the relationship between the quantity of medical services Y on the one hand and achievable excess revenue Π and the extent of slack on the other. In fact, Eqs. (9.12) and (9.14) form the constraints subject to which the actors of the hospital pursue their objectives, to be described in the following subsection.

Finally, revenue may be split into an *endogenous* and an *exogenous* part, a distinction that will prove useful in the subsequent analysis. The endogenous part depends only on variables the hospital is free to set according to its preferences, whereas the exogenous part depends on variables that cannot be influenced by the hospital,

$$R = R^{\text{end}}(Y, A, X) + R^{\text{ex}}(Z, \underline{X}, \bar{X}) \quad (9.15)$$

9. Total differentiation of Eq. (9.12) yields $dQ = g'(Y/X)\{dYX - YdX\}/X^2$, and after ordering terms,

$$h'(Y) = \frac{dQ}{dY} = g' \left(\frac{Y}{X} \right) \frac{X - f'(Y)Y}{X^2} > 0 \text{ if } g' > 0, \text{ due to } f'(Y) < X/Y \text{ and hence } X > f'(Y)Y$$

[see Eq. (9.11)].

Assigning the number of cases to one of the categories creates a certain problem, however, because this quantity may by assumption only be varied within a possibly narrow margin. Eq. (9.15) solves this problem by treating the number of cases (X) itself as a determinant of endogenous revenue while including the upper and lower bounds ($\underline{X}$, $\bar{X}$) among the determinants of exogenous revenue.

9.3.2 Hospital Agents and Their Objectives

In order to identify the objectives of “the hospital,” it seems useful to start by discussing the groups of agents whose interests enter the joint decision-making process in the hospital. These are

1. The medical directors and other hospital-based physicians;
2. The nurses and the rest of the staff;
3. The hospital manager;
4. The owners of the hospital.¹⁰

The *satisfaction of hospital-based physicians* depends first on the quality of medical treatment they are able to provide (an objective reflecting their professional ethics), second on their effective income, and third on their position in the power structure of the hospital. As the salary of hospital-based physicians is fixed by contract, their effective income is the higher the less hours of effective work per physician.¹¹ In other words, they value the “leisure content of work,” a quantity that corresponds to slack variable L^S of the preceding section. Finally, their *position of power* depends on the extent to which they contribute to the revenue of the hospital through their work. We therefore postulate the following utility function for the group of hospital-based physicians (denoted by the superscript P)

$$u^P = u^P(Q, L^S, \alpha), \quad \frac{\partial u^P}{\partial Q} > 0, \frac{\partial u^P}{\partial L^S} > 0, \frac{\partial u^P}{\partial \alpha} > 0 \quad (9.16)$$

where $\alpha = R^{\text{end}}/R$ stands for the share of endogenous revenue in total revenue.

Nurses and the rest of the staff presumably value a slower pace of work, too, symbolized again by L^S . Furthermore, this group is mainly responsible for the proper handling of equipment, drugs, and other inputs, which is a less demanding task if there is some slack K^S . The utility function of this group (superscript N) may therefore read,

$$u^N = u^N(L^S, K^S), \quad \frac{\partial u^N}{\partial K^S} > 0, \frac{\partial u^N}{\partial L^S} > 0 \quad (9.17)$$

10. In this model, no distinction according to ownership is made, although the relative position of power of an agent may well vary with type of ownership.

11. Medical directors at least may profit from a reduced number of working hours to earn additional income, treating private patients both inside and outside the hospital.

With regard to *ownership*, one must distinguish between for-profit and nonprofit (i.e., public or voluntary) hospitals in order to reflect the owners' difference of views with regard to making profits. As the for-profit group is quantitatively not very significant in most European countries,¹² we shall focus on the second group, which appears to be reasonably homogeneous concerning the profit issue. Profits are not welcome (to public ownership) or even prohibited by law (in the case of voluntary ownership), whereas deficits are not desirable either because they are in conflict with other objectives, whose attainment calls for expenditures. The owners' utility function thus attains a maximum at the value of zero profits. In mathematical terms, the marginal utility of profits $\partial u^O/\partial \Pi$ must have the opposite sign of profits Π themselves; that is, the product of the two, $\Pi(\partial u^O/\partial \Pi)$, is negative.

Besides this, owners are interested in the prestige of their hospital, which relates to the quality of output Q , whereas slack, which is input without output, cannot be in their interest. The owner's utility function (superscript O) is thus given by

$$u^O = u^O(Q, \Pi), \quad \frac{\partial u^O}{\partial Q} > 0, \quad \Pi \frac{\partial u^O}{\partial \Pi} < 0 \quad (9.18)$$

With regard to the objectives of *hospital managers*, there is the considerable divergence of opinions in the literature. On the one hand, some say that the performance of a manager of a nonprofit firm is not measured in terms of profits, which therefore cannot possibly be his objective. In this way, the utility function of the manager is tied to the reputation of the hospital, suggesting that factor inputs (e.g., costly equipment) are among its arguments. This would justify entering slack quantities into the manager's utility function.

On the other hand, however, considering that the administrative director of a hospital usually has an MBA degree, it is likely that he will at least for himself define success on the job in terms of profits achieved. In the case of hospitals that have to be restructured because they are in permanent deficit, this objective does agree with that of the owner. The manager's income may increase with profits in this case, for example, in the form of premiums.

Besides success defined in this way, the manager's position in the power structure of the hospital should also be of interest to him. The stronger the *economic forces* impinging on the hospital from the outside (i.e., the financiers), the stronger the manager's own position. Put more pointedly, if the existence of the hospital is at stake, the manager, being an economic expert, becomes a dictator. If, however, there is no need for a rational use of resources (e.g., because all costs are reimbursed after the fact), the manager is reduced to the position of a purchasing agent by the physicians.

A good indicator for measuring the strength (or, more precisely, the absence) of external economic pressure is the share α of endogenous revenue R^{end} in total revenue, keeping in mind that α may be *influenced by the hospital*. Since R^{end} is attributed to

12. In 1983, only 3.8 percent of all acute-care beds were in private, profit-seeking hospitals in Germany [see Daten des Gesundheitswesens (1985), p. 250].

the physicians, a high value of α implies a weak position of the manager. Thus managers (M) are assumed to have the utility function,

$$u^M = u^M(\Pi, \alpha), \quad \frac{\partial u^M}{\partial \Pi} > 0, \quad \frac{\partial u^M}{\partial \alpha} < 0 \quad (9.19)$$

If one is to derive predictions about the behavior of the hospital as a whole from the objectives of the groups involved, one must make assumptions regarding the *decision-making process*. In the following, we assume that the different parties' efforts to influence α , the share of endogenous revenue, neutralize each other. On one side, there is the administrative director who aims at minimizing this share, and on the other, the physicians wanting to maximize it. Given neutralization of these opposing forces, α may be deleted from the aggregate objective function (9.20).

With regard to the remaining variables, the interests of "the hospital" will again reflect a compromise struck between the intentions of the groups involved. This means that all variables desired by at least one group appear with a positive sign in the objective function of the hospital.

As far as the variable "profit" is concerned, all parties involved presumably are interested in avoiding deficits, provided the owner makes it clear that he is not willing or able to cover these deficits for a longer period of time. Avoiding deficits amounts to safeguarding the existence of the hospital and hence one's own job in this situation. Things are less clear if the hospital already is beyond the break-even level, making a profit. Will the parties involved try to increase profits still more, or will they tend to increase costs, causing profits to disappear? The answer to this question depends very much on whether the hospital is free or subject to public regulation with regard to its *use of profits*.

Free use of profits permits, for example, paying out premiums to employees or spending money on their on-the-job training. This may enhance motivation and promote a more economical use of resources generally. However, if there are public regulations concerning the use of profits, they might not be in accordance with the interests of those working in the hospital, making it unattractive to be in the black.

Assuming that there are opportunities to use profit freely, we formulate the objective function

$$u = u(Q, \Pi, L^S, K^S), \quad \frac{\partial u}{\partial Q} > 0, \frac{\partial u}{\partial \Pi} > 0, \frac{\partial u}{\partial K^S} > 0, \frac{\partial u}{\partial L^S} > 0 \quad (9.20)$$

Given a revenue function, the hospital is supposed to maximize the objective function (9.20) subject to the profit constraint given by Eq. (9.14).

CONCLUSION 9.4: Since the hospital is a social institution comprising different groups of agents with different interests, its objective function is a compromise between the objectives of these groups. Quality of treatment, excess revenue, as well as slack inputs are expected to be positively valued.

9.3.3 Issues to Be Addressed in Subsequent Analysis

The questions relevant to health policy to be dealt with in Section 9.4 are:

1. Will the hospital produce in a *technically efficient manner*; that is, will the slacks L^S and K^S be zero?
2. If the answer to question No. 1 is no, will at least the ratio of inputs L/K selected correspond to the *efficient combination of inputs* $L^*(Y, Z)/K^*(Y, Z)$ at a given output?
3. Will the quantity of medical services per case be so high as to achieve the *highest possible quality of treatment* $Q = g(\gamma)$, causing the marginal impact of services on quality to equal zero, or will Q be lower?

The answers to these questions will presumably depend on the payment system. This dependence will permit identification of the incentives for efficient behavior inherent in the different systems of payment for hospital services.

9.4 PAYMENT SYSTEMS AND THEIR INCENTIVE EFFECTS

Hospital services may be financed in a variety of ways that differ considerably in terms of their basic design. At one extreme, there is the principle of cost recovery, calling for reimbursement of hospital costs by the payers (social and private health insurers as well as uninsured individuals). At the opposite end, there is the payment of a prospectively set price for a service that likewise has been defined in advance. To the extent that the payment in the second case does not correspond to actual costs incurred, the hospital and its owner may face a profit or a loss.

9.4.1 The Principle of Cost Recovery: The Pros and Cons

9.4.1.1 Health for Profit

As the principle of cost recovery prevents the providers of health care from making either a profit or a loss, many people consider it as an ethically sound payment system for health care services. They believe that "compensation of expense" guarantees that providers simply are reimbursed for the money they had to spend themselves, whereas the realization of a profit in health care means that one may make money on people's hardship. This simplified view of ethics is open to several criticisms.

Most people support the *ethical norm* that one should not make a profit on a fellowman's hardship. It is likewise universally accepted that illness creates hardship. In exactly the same way, however, there is little doubt that hunger, thirst, homelessness, and freezing also create hardship. The term *hardship* may be defined to mean that there is always hardship when the satisfaction of basic human needs is in jeopardy.¹³ Consistent application of this maxim would therefore make it unethical to draw a profit from supplying food, drink, accommodation, and fuel for heating.

13. Immaterial needs may also be included to the extent that their nonsatisfaction results in psychic hardship.

Admittedly the need for health care services differs from other basic human needs because its occurrence as a rule is *not predictable*, constituting an event whose financial consequences easily *exceed* the individual's ability to pay. This problem can, however, be solved by introducing (possibly compulsory) health insurance, which serves to separate the risk of needing health care services from the risk of having to pay for them (see Section 5.3 and Chapter 6). Therefore the moral argument advanced above *applies to health care services* even to a lesser degree than to other "basic needs," at least in contemporary industrialized countries.

However, a ban on making a profit from producing and distributing food, housing, and fuel is only seldom advocated, and with good reason. In a free economy, the prospect of making a profit is an essential incentive for potential suppliers to produce the goods and put them on the market for sale. The greater the opportunities for making money, the smaller the risk of some consumers being unable to obtain the desired quantity of the good.

This theoretical reasoning is confirmed by experience. For decades, a housing shortage has continued to exist in Germany despite massive government subsidies. A plausible explanation for this is continued rent control, which caused both the upkeep of existing housing and the construction of new housing to be poor investments compared to other alternatives.

Making a profit can be morally questionable if the scarcity of the good is due to either a *contrived shortage*, as in the case of legally enforced restrictions on market entry, or to unforeseeable and uninsurable events like *natural disasters*. Only here is "making money on a hardship" really based on the exploitation of a shortage that cannot be corrected. In the absence of competition, the supplier may dictate price, while due to existing (government-imposed or natural) barriers, profits fail to create the incentives that in the long term serve to put an end to the shortage.

In addition, the distinction between profit and other forms of income according to moral criteria is highly questionable. In the case of self-employed individuals (physicians, for example) and not-for-profit associations, income *cannot be divided into wages and profit without arbitrariness*, because the wage is not defined and cannot easily be inferred through comparison with other professions. The common distinction between for-profit and nonprofit firms has no ethical relevance either, since the latter might primarily aim to increase the incomes of their workers. So it cannot be claimed in any way that only for-profit firms are "making money on illness."

In view of these arguments, the seemingly innocent request of charitable and other nonprofit enterprises to have their own costs reimbursed appears in a different light. Their own costs include employees' wages and salaries, which may be set by the enterprise itself. Even if wage rates are fixed at the industry level, the enterprise still has an amount of discretion with the grading of employees and the granting of fringe benefits that may be exploited to *increase the effective income* per hour worked or to create comfortable working conditions.

9.4.1.2 The Ambiguity of "Cost"

The argument advanced in the preceding paragraph points to misapprehension concerning the principle of cost recovery. The term *cost* suggests something objective

and exogenous—which it is not at all in reality. The interpretation of this term is not as clearcut as one may think at first.

For one, the term *cost* may mean “minimum expenditure to produce a given bundle of output,” which would agree with the terminology of microeconomic theory. This cost (represented by C in the formal model of Section 9.3) indeed *cannot be influenced* by the firm in question, given its technological knowledge and factor prices.

However, in a concrete case it can be very difficult to determine whether a hospital has been operating efficiently, or, if this was not the case, what its costs would have been if it had been managed efficiently. Thus recovery of cost in practice relates to actual expenditure (represented by the symbol A in the formal model). But actual expenditure quite obviously *depends on the behavior* of the hospital in question, in particular its management, rendering it neither objective nor exogenous. Thus a key argument for the ethical acceptance of cost reimbursement loses its relevance.

9.4.1.3 Incentive Effects of Cost Reimbursement

It goes without saying that reimbursing a hospital for its actual expenditure utterly fails to create incentives for an economical use of scarce resources. It inevitably induces hospital management to try to *prove and justify* costs rather than to cut them. This may be shown as follows, using the formal model developed above.

If total revenue R is always equal to expenditure A , then profit Π is zero by definition [see Eq. (9.13)]. The hospital's objective function (9.20) then becomes

$$u(Q, 0, L^S, K^S) \rightarrow \max! \quad (9.21)$$

As long as there are no additional constraints, utility may be increased without bounds by improving the quality of treatment or by squandering resources. As for quality, it is assumed to have a maximum value that cannot be exceeded by increasing the quantity of medical services per patient, according to Eq. (9.3). In view of Eqs. (9.4) and (9.5), no such limit on the use of resources exists, and the model indeed leaves open the question of why in such a regime *costs do not simply rise to infinity*. The answer seems to lie in the existence of the public supervisory agencies that impose regulations directly limiting hospital expenditures, such as benchmark figures for the use of personnel per bed in hospital planning.

For the case of cost reimbursement, the answers to the questions raised in Section 9.3.3 are thus as follows.

1. Production will be *technically inefficient*.
2. The relative use of factors depends on the relative rigorousness of regulation. A *bias in favor of material inputs* may be expected because the use of personnel is easier to control.
3. The *quality of treatment* is likely to be maximum.

Payment according to prospective costs rather than actual costs has quite similar effects. There may be a certain incentive to avoid costs during the stage of budget implementation, but it is in the interest of the hospital to establish the highest possible

value of costs at the stage of budget planning. The incentive to cut costs is further weakened by the fact that hospital management must be afraid that these cuts will be taken into account when planning the budget for the following period, causing one-time successes of cost containment to translate into a long-term reduction of revenue.

It follows from these arguments that the principle of cost recovery is *diametrically opposed* to the aim of creating incentives for a rational use of resources. Attempts to achieve efficiency by administrative fiat, for example, by establishing limits to specific factor inputs, are counterproductive because they restrict the freedom of decision-making that hospitals need to produce efficiently. In particular, cost-saving substitution of factors of production is inhibited by rigid norms of this kind.

In addition, simple reimbursement of costs incurred fails to reward good performance because neither patient mix (defined with reference to type of illness, age, or some other criterion) nor any other indicators are used to assess the performance of the hospital. Once again, the transition to reimbursement of planned costs within a prospectively determined budget does not affect this finding since future budgets usually amount to projections of previous budgets. Historical differences in payment levels among hospitals that often have nothing to do with performance are perpetuated in this way.

CONCLUSION 9.5: Payment of hospital services according to the principle of cost recovery cannot be derived from recognized ethical norms in a stringent manner. Moreover, it runs counter to the aim of creating incentives for efficiency as well as rewarding good performance.

9.4.2 Output-Based Payment Systems

Payment systems that are based on the payment of a certain price per unit of defined service form the counterpole to the principle of cost recovery. The separation of revenue from expenditure provides the hospital with the opportunity to make profits and losses that are not merely due to errors in the calculation of the budget.

Payment systems based on performance are possible in a multitude of forms that differ from each other in

1. Their choice of the output indicator;
2. Their method of price determination;
3. Their domain of application of prices;
4. Their choice of standard of reference for price.

For each of these distinctive characteristics, the most important alternatives that have been tried in actual practice or suggested in the literature will be presented and their effects analyzed, using the formal model developed in Section 9.3.

The most important elements of that model are listed in Table 9.3.

9.4.2.1 The Indicator of Output

In Section 9.2.1 it was found that the “true” output of a hospital, viz. improving the state of health and the well-being of patients as well as guaranteeing the security of

Table 9.3: An Economic Model of the Hospital

$$q = g\left(\frac{Y}{X}\right), \quad \text{with } g'' < 0 \text{ and } g'\left(\frac{Y}{X}\right) \leq 0 \text{ for } \frac{Y}{X} \leq \gamma \quad (9.3)$$

$$A = wL + rK \quad (9.4)$$

$$L = L^*(Y, Z) + L^S \quad (9.5)$$

$$K = K^*(Y, Z) + K^S \quad (9.6)$$

$$C = C(Y, Z) = wL^*(Y, Z) + rK^*(Y, Z) \quad (9.7)$$

$$C^f = C^f(Z) = C[0, Z] = wL^*[0, Z] + rK^*[0, Z] \quad (9.8)$$

$$X = \varphi(Q) = \varphi\left[g\left(\frac{Y}{X}\right)\right], \quad \text{with } \varphi' > 0 \text{ for } \underline{X} \leq X \leq \bar{X} \quad (9.10)$$

$$\Pi = R\{Z, f(Y), Y, A\} - C\{Y, Z\} - wL^S - rK^S \quad (9.14)$$

$$R = R^{\text{end}}(Y, A, X) + R^{\text{ex}}(Z, \underline{X}, \bar{X}) \quad (9.15)$$

$$u = u(Q, \Pi, L^S, K^S), \quad \frac{\partial u}{\partial Q} > 0, \frac{\partial u}{\partial \Pi} > 0, \frac{\partial u}{\partial K^S} > 0, \frac{\partial u}{\partial L^S} > 0 \quad (9.20)$$

Prospective budget

$$\frac{\partial u}{\partial \Pi} \frac{\partial C}{\partial Y} = \frac{\partial u}{\partial Q} h'(Y) \quad (9.24)$$

Fixed payment per case

$$\frac{\partial u}{\partial \Pi} \left(\frac{\partial C}{\partial Y} - p_X f'(Y) \right) = \frac{\partial u}{\partial Q} h'(Y) \quad (9.28)$$

Fee-for-service and per diem

$$\frac{\partial u}{\partial \Pi} \left(p_Y - \frac{\partial u}{\partial Y} \right) = - \frac{\partial u}{\partial Q} h'(Y) \quad (9.31)$$

Z : Number of beds

X : Number of cases

$\underline{X}, \bar{X}$: Lower and upper bounds on the number of cases

Y : Patient days, or equivalently, quantity of medical services provided

Q : Quality of treatment

$\varphi(Q)$: Demand function relating cases to quality of treatment

L : Labor input

K : Inputs other than labor

w, r : Factor prices

L^*, K^* : Cost-minimizing input quantities

L^S, K^S : Slack inputs

A : Actual expenditure of the hospital

C : Minimal costs of the hospital

C^f : Fixed costs of the hospital

R : Hospital revenue

R^{end} : Endogenous component of revenue

R^{ex} : Exogenous component of revenue

Π : Excess revenue of the hospital

$u(\cdot)$: Objective function of the hospital

R^f : Fixed budget

p_X : DRG payment per case

p_Y : Per diem, or equivalently, unit price of medical services

provision, largely defies measurement based on objective criteria. In the context of a payment system based on performance, a number of proxies are in use. While not reflecting the performance of a hospital in the sense stated above, they constitute “*intermediate products*” in the process resulting in improved health and well-being. They are

- The number of *beds*;
- The quantity of *medical or nursing services* provided (operations, injections, medications, etc.);
- The number of *patient days* provided (“days”);
- The number of *patients* treated (“cases”).

Payment systems that are based on these indicators will now be formally examined as to their incentive effects.¹⁴

9.4.2.1.1 Prospective Budget

The first payment system is the prospectively fixed budget R^f that might be linked to the number of beds of a hospital. This does not make any difference for the analysis as long as the number of beds cannot be chosen by the hospital. Replacing Q by $h(Y)$ according to Eq. (9.12) and combining the objective function (9.20) with the constraint (9.14), one obtains the Lagrangean

$$\Lambda(Y, \Pi, L^S, K^S) = u\{h(Y), \Pi, L^S, K^S\} + \mu\{R^f - C(Y, Z) - wL^S - rK^S - \Pi\} \quad (9.22)$$

The first-order conditions for a maximum are

$$\frac{\partial \Lambda}{\partial \Pi} = \frac{\partial u}{\partial \Pi} - \mu = 0 \quad (9.22a)$$

$$\frac{\partial \Lambda}{\partial Y} = \frac{\partial u}{\partial Q} h'(Y) - \mu \frac{\partial C}{\partial Y} = 0 \quad (9.22b)$$

$$\frac{\partial \Lambda}{\partial K^S} = \frac{\partial u}{\partial K^S} - \mu r \leq 0; \quad = 0, \text{ if } K^S > 0 \quad (9.22c)$$

$$\frac{\partial \Lambda}{\partial L^S} = \frac{\partial u}{\partial L^S} - \mu w \leq 0; \quad = 0, \text{ if } L^S > 0 \quad (9.22d)$$

Starting with the conditions referring to optimal slack, Eqs. (9.22c, d) may be combined to yield, for the case of an interior optimum (with $L^S, K^S > 0$),

14. Of course, one might also consider a payment consisting of a combination of different output indicators. Capacities held on reserve could be compensated by a flat rate per bed, medical services by diagnostic related groups (DRG), and nursing by a fixed fee per patient day.

$$\frac{\partial u / \partial K^S}{\partial u / \partial L^S} = \frac{r}{w} \quad (9.23)$$

Thus the marginal rate of indifferent substitution between the quantities of slack inputs has to correspond to the ratio of their input prices. Condition (9.23) determines the optimal ratio of slack inputs from the hospital's point of view. This ratio may equal the ratio of efficient inputs only by coincidence.

Focusing on the first two conditions now, one obtains from Eqs. (9.22a, b)

$$\frac{\partial u}{\partial \Pi} \frac{\partial C}{\partial Y} = \frac{\partial u}{\partial Q} h'(Y) \quad (9.24)$$

Since $\partial u / \partial \Pi$, $\partial u / \partial Q$, $\partial C / \partial Y > 0$, it follows that $h'(Y)$ must be positive at the optimum point. This means that the hospital chooses a smaller quantity of services than the one that maximizes the quality of treatment. With given marginal costs $\partial C / \partial Y$, the level of quality selected is higher (and h' lower), the stronger the preference for quality (symbolized by $\partial u / \partial Q$) in relation to the preference for profits (symbolized by $\partial u / \partial P$).

Does a fixed budget allow a *waste of resources* (i.e., positive values for K^S and L^S), as assumed in the derivation of condition (9.23)? In order to answer this question, let $V(R^f)$ denote the set of vectors (Y, Π, L^S, K^S) —or alternatively, since Y and Q are related to each other, the set of vectors (Q, Π, L^S, K^S) —that solve Eq. (9.24) for a given budget R^f . Next, consider the combination of a fixed budget R^f and a quality regulation that imposes a minimum quality Q^* on the hospital. Now if all vectors contained in $V(R^f)$ with $Q = Q^*$ have $L^S = K^S = 0$, then we know that the described policy prevents the hospital from wasting resources. We therefore draw the

CONCLUSION 9.6: If the hospital obtains a prospective budget per period, it will provide a smaller quantity of services than would be necessary to maximize quality of treatment. While positive quantities of slack inputs are possible in general, the hospital can be induced to produce technically efficiently by a suitable fixing of the budget combined with a regulation of (minimum) quality.

9.4.2.1.2 Fixed Payment Per Case

One instance of a payment system that uses “cases” as a basis is the DRG (diagnosis-related group) system adopted by the U.S. Medicare administration starting in 1983. Neglecting heterogeneity among patients for the moment and assuming that the hospital receives the same flat rate p_X for every case, constraint (9.14) may be written,

$$\begin{aligned} \Pi &= p_X X - C(Y, Z) - wL^S - rK^S \\ &= p_X f(Y) - C(Y, Z) - wL^S - rK^S \end{aligned} \quad (9.25)$$

The Lagrangean function (9.22) for utility maximization now becomes

$$\Lambda(Y, \Pi, L^S, K^S) = \{h(Y), \Pi, L^S, K^S\} + \mu \{p_X f(Y) - C(Y, Z) - wL^S - rK^S - \Pi\} \quad (9.26)$$

While the first-order optimality conditions (9.22a), (9.22c), and (9.22d) remain the same, Eq. (9.22b) changes to become,

$$\frac{\partial \Lambda}{\partial Y} = \frac{\partial u}{\partial Q} h'(Y) + \mu \left(p_X f'(Y) - \frac{\partial C}{\partial Y} \right) = 0 \quad (9.27)$$

Recalling that $\mu = \partial u / \partial \Pi$ [by Eq. (9.22a)], this can be rewritten to read

$$\frac{\partial u}{\partial \Pi} \left(\frac{\partial C}{\partial Y} - p_X f'(Y) \right) = \frac{\partial u}{\partial Q} h'(Y) \quad (9.28)$$

In Section 9.3.1 it was found that f' and h' have the same sign [see Eqs. (9.11) and (9.12)]. Due to $\partial u / \partial \Pi, \partial u / \partial Q > 0$, condition (9.28) can only be satisfied if f' and h' are positive, while the expression in large parentheses must be greater than zero. This means that output is increased to the point where payment for an additional patient *does not cover marginal cost*. The point of maximum quality is not exceeded, however. For in contradistinction to the optimality condition for a fixed budget (9.24), a negative term, $-p_X f'(Y)$, enters on the left-hand side of Eq. (9.28). This implies that Eq. (9.28) is satisfied at a value of $\partial C / \partial Y$ that is larger and/or a value $h'(Y)$ that is smaller than its counterpart in Eq. (9.24), pertaining to the case of a fixed budget. Due to the convexity of the cost function and the concavity of the production function for quality, $h(Y)$, it immediately follows that a fixed payment per case induces a higher (but not maximum) quality by use of a *larger quantity of medical services* than a fixed budget.

With regard to the possibility of positive slack in inputs, the same result applies as in the case of the fixed budget. A combination of a low payment rate p_X and a high minimum quality can force the hospital to produce in a technically efficient way.

CONCLUSION 9.7: The incentive to skimp on medical services is weaker in the case of a fixed payment per case (of the DRG type) than under a fixed budget. Concerning the possibility of slack, the same can be said for the payment per case as for the fixed budget: it depends on the size of the payment and the imposed level of quality.

9.4.2.1.3 Fee-for-Service and Per Diem

The two performance indicators, medical services and patient days, are not distinguished analytically in this model. They are both symbolized by Y , to be freely determined by the decision makers of the hospital. It is assumed that the payment

consists of a fixed fee p_Y per unit of this type of service. The constraint (9.14) in the hospital's maximizing problem now reads

$$\Pi = p_Y Y - C(Y, Z) - wL^S - rK^S \quad (9.29)$$

The Lagrangean function (9.22) and the optimality conditions (9.22a)–(9.22d) change to read

$$\Lambda(Y, \Pi, L^S, K^S) = u\{h(Y), \Pi, L^S, K^S\} + \mu\{p_Y Y - C(Y, Z) - wL^S - rK^S - \Pi\} \quad (9.30)$$

$$\frac{\partial \Lambda}{\partial \Pi} = \frac{\partial u}{\partial \Pi} - \mu = 0 \quad (9.30a)$$

$$\frac{\partial \Lambda}{\partial Y} = \frac{\partial u}{\partial Q} h'(Y) + \mu \left(p_Y - \frac{\partial C}{\partial Y} \right) = 0 \quad (9.30b)$$

$$\frac{\partial \Lambda}{\partial K^S} = \frac{\partial u}{\partial K^S} - \mu r \leq 0; \quad = 0, \text{ if } K^S > 0 \quad (9.30c)$$

$$\frac{\partial \Lambda}{\partial L^S} = \frac{\partial u}{\partial L^S} - \mu w \leq 0; \quad = 0, \text{ if } L^S > 0 \quad (9.30d)$$

From Eq. (9.30a, b), one obtains

$$\frac{\partial u}{\partial \Pi} \left(p_Y - \frac{\partial C}{\partial Y} \right) = - \frac{\partial u}{\partial Q} h'(Y) \quad (9.31)$$

To interpret this optimality condition, one first has to determine the sign of the expression in large parentheses. We know from Section 9.2.4 that hospitals typically have constant marginal costs with respect to services, while Eq. (9.8) points to positive fixed costs. The cost function may thus be approximated by

$$C(Y, Z) \approx C^f + cY, \quad \text{with } c := \frac{\partial C}{\partial Y} \quad (9.32)$$

Introducing the plausible assumption that the rate p_Y is calculated so as to permit the hospital to achieve a positive contribution to its fixed costs, one has $p_Y > c$, and the expression in large parentheses is positive, making the left-hand side of Eq. (9.31) positive as well. Consequently, $h'(Y)$ has to be negative in the optimum; that is, *more services are provided than would be necessary to maximize quality*. From a social perspective, this result reveals another type of inefficiency, in addition to the direct waste of resources in the form of positive values of L^S and K^S [see Eq. (9.23)].

This finding applies to both “patient days” and “medical services” as the unit of settlement. In the first case, a medically unjustified prolongation of the length of stay

of patients occurs, while in the second case, services that are not medically indicated will be provided, in particular of the diagnostic type.

CONCLUSION 9.8: Fee-for-service and per diem payment will cause hospitals to provide services and patient days in excess of maximum quality of treatment. Furthermore, there will be positive slack in inputs, with the actual factor ratio generally deviating from the efficient ratio.

9.4.2.2 The Method of Price Determination

With regard to the process of price determination, the three leading alternatives are:

1. Market prices;
2. Negotiated prices;
3. Administered prices.

1. Market prices. Here legislators merely fix the output indicator to be used for payment of hospital services, leaving the determination of the level of remuneration to market forces. However, due to the reasons mentioned in Section 9.1, the resulting prices do not ensure an optimal allocation of resources in the hospital industry because consumers, being protected by insurance, have no incentive to seek out the supplier with the most favorable cost–performance ratio.

2. Negotiated prices. The market mechanism is replaced by negotiations between some agent representing consumers (e.g., an association of health insurers; for other alternatives, see Section 8.3) and the service provider. The outcome of such negotiations depends crucially on the bargaining skills and the relative power of the parties involved. The power of a party in turn depends on whether or not his negotiating partner has alternatives at his disposal.

If, for example, the supply side is represented by a single representative of the regional hospital association (in the guise of “collective bargaining”), it has better prospects for pushing through hospital interests than if insurers negotiate with each hospital separately. In the latter case, a hospital enjoying a regional monopoly has a better negotiating position than one among several hospitals located in the same urban area and offering about the same range of services.

Under these auspices, prices resulting from negotiations in different regions with different competitive situations are unlikely to reflect differences in performance. In addition, incentives for efficiency are stifled where suppliers are able to dictate prices, due to a regional monopoly. In these cases, hospitals will achieve payment levels that allow them to operate at a low effort.

3. Administered prices. The third possibility is the determination of prices by decree of public authorities or some institution charged with this task. Uniformity of payment for identical performance is best achieved by this choice of method. The term *identical performance* points to the need for a high degree of *homogeneity in the output indicator* used for payment. Thus “identical performance” lends itself to operationalization more easily for payment according to a patient classification system than for per diem payment.

Furthermore, to the extent that administrative determination of prices amounts to prospective budgeting or fixed payment per case, it may force hospitals to be run efficiently, as has been shown above (see Subsections 9.4.2.1 and 9.4.2.2). However, to ensure efficiency an adjustment process is required guaranteeing that prices reach the level at which hospitals choose to eliminate slack in order to reach their aspired level of profits and quality of care [see the RHS of inequality (9.28)]. This issue will be examined more closely in Subsection 9.4.2.4.

9.4.2.3 *The Domain of Application of Prices*

Prices may be individually determined for each hospital, they may apply to all hospitals of a given area (or type of institution, respectively), or they may apply uniformly to all hospitals. In the first case, the arguments are the same as those stated in connection with negotiated prices. The principle of performance-based payment is violated if hospitals providing the same service are paid differently. In addition, authorities are tempted to base their decision on the costs of the individual hospital, causing incentives for efficiency to get lost in the long term.

The principle of performance-based payment also requires payment per case not to be the same across patients but to be differentiated according to the type of illness and possibly additional criteria (e.g., the patient's age) that relate to the difficulty of the task faced by the hospital. This requirement is largely met by the DRG (diagnosis-related groups) payment system.

9.4.2.4 *Standards of Reference for Price*

Even if a payment system or a combination of payment systems is found that induces a minimum amount of inefficiency while securing the desired quality of treatment, the problem of *determining an adequate level of payment* remains to be solved, especially in a system of administered prices.

In the preceding subsection, it was argued that the principle of performance-based payment does not go along with a differentiation of price among hospitals providing the same service. Therefore, the formula used for price determination must amount to a uniform norm, applicable to all hospitals.

This norm should in principle be derived from the utility of inpatient treatment conferred on patients or on society as a whole. Even if one abstracts from the problem of measuring the success of treatment at the level of the individual patient (see Section 9.2.1) by assuming a typical course of recovery in every case category, the problem of *putting a monetary value* on the improvement of health status would have to be solved. Indeed, theoretical concepts for this exist, such as the "willingness-to-pay approach" to valuing the increase in life expectancy due to treatment. However, their practical implementation is fraught with great problems of measurement and valuation. In particular, no convincing procedure is available for getting individuals to reveal their true willingness to pay (see Chapter 2).

The desired norm for determining price must therefore be based on cost. One possible solution is provided by having price equal the expected value of resources used to produce the service in question. In the case of DRG payment, say, the expected

cost associated with patients of a given diagnosis-related group would serve as the determinant of price (prospective cost-based pricing). However, this norm has to be detailed in *two aspects*. On the one hand, it has to be indicated whether one is dealing with

- Fictitious; or
- Historically observed

costs. Under the first option, one must be able to list the necessary resource quantities and to assign prices to them in order to add them up to obtain total resource cost. In view of the considerable differences in opinion between physicians with regard to the adequate treatment of many diseases and necessary factors of production, this possibility has to be discarded.

On the other hand, if the price-fixing rule is based on *historically observed cost*, the problem still remains whether

- Minimum; or
- Average

production costs of a comparable service, averaged over all hospitals, should constitute the standard of reference for payment.

From a theoretical point of view, reliance on *minimum costs* appears preferable. However, rigorously limiting payment to minimum costs would mean a considerable loss of revenue for the majority of hospitals. Failing a rapid reduction of expenditure, their very existence would be at stake, and this in turn would endanger the security of provision enjoyed by the general population.

The main objection against the use of *average costs* as the standard of reference for payment is that it “compares inefficiency with inefficiency” and confusates the normative character of the concept of costs. For only a hospital having minimal unit cost fulfills the necessary condition for technically and economically efficient production.

However, this does not mean that efficiency would be totally sacrificed if payment were to be based on costs observed in the average hospital. On the one hand, this would force hospitals with above-average costs to achieve savings in order to avoid losses. On the other hand, the lump-sum character of the payment leaves an incentive for efficient production even for hospitals with below-average expenditure. The *key prerequisite* for this is that they are interested in making a profit, a condition more likely to be satisfied the less the use of profit is constrained by public regulation. If the hospital is allowed to distribute profits to employees in the form of premiums, this incentive is stronger than if it is obliged to invest them internally.

Finally, under payment systems using administered prices, one may *ceteris paribus* expect a decrease of average unit cost over time, because hospitals with high costs will be forced to adjust to the average or to withdraw from the market (or specialize in activities where they do not exceed the average). Through periodic recalculation of the cost average, the reference standard given by “norm costs” may be brought closer to the ideal of minimal costs.

CONCLUSION 9.9: In the absence of a functioning market mechanism in the hospital industry, the criteria of performance-based payment and incentives for efficiency seem to be satisfied to a comparatively high degree by a system of administered prices. These prices may be based on unit costs averaged over all hospitals; however, there must be an adjustment process ensuring that in the course of time this average is brought in line with minimum cost. In contrast, negotiated prices, reflecting regional differences in competitive conditions, will fail to reflect performance and to convey incentives for efficiency, at least in cases of regional monopolies.

9.5 EXERCISES

Exercise 9.1:

Why is a fixed budget an output-based payment system?

Exercise 9.2:

What is the difference between a fixed payment per case and a fixed prospective budget (with "output" given by the number of beds) in terms of quality, supposing a common utility function and equality of marginal cost?

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10

THE MARKET FOR PHARMACEUTICALS

10.1 ISSUES INVOLVED

One cannot possibly imagine present-day health care without pharmaceuticals for at least the following three reasons:

1. Pharmaceuticals represent a form of therapy that does not involve *injuring or removing organs* while permitting causal treatment (not only aimed at alleviating symptoms) in several instances. Chemotherapy, employed in the treatment of tuberculosis, is a historic example of such causal treatment; vaccination against the AIDS virus may become a future example.

2. Pharmaceuticals are in general *prescribed in a dosed manner* and may be replaced by other drugs in the event of adverse effects. Apart from organ transplants, surgical mistakes cannot be put right. Once the womb has been removed in the course of a hysterectomy, for example, it can never again be reinserted.

3. In contrast to nursing services, pharmaceuticals are industrially manufactured and therefore have a particularly large *scope for increased productivity* in the provision of health care services, permitting considerable savings of time not only for the physician and auxiliary staff, but equally for the patient himself.

At the same time, pharmaceuticals account on average for only approximately 10 percent of total public health care expenditure in OECD countries, compared with 12 percent in 1970 [see OECD (1987)]. With this background, it seems surprising at first that drug manufacturers of almost every Western country face so much criticism, which comprises four main points.

1. Many drugs are branded *useless or even harmful*. For example, out of 14 active substances newly introduced in the German market in 1988, only 3 were classified by the federal association of sick funds as new and therapeutically significant [see Schwabe and Paffrath (1989), p. 6]. In other words, almost 80 percent of the small

number of preparations newly introduced are considered to be marginal advances to existing preparations commonly called “me too” drugs.

2. *Expenditure on advertising* is largely seen as excessive. In 1991, the total of “general and administrative costs, advertising, and promotion” amounted to no less than 35 percent of revenues of leading U.S. pharmaceutical companies [see Comanor and Schweitzer (1995)]. Conversely, expenditure on advertising by car manufacturers in Germany in 1980 varied between 0.2 percent (Daimler-Benz) and 4.6 percent (Toyota) of turnover [see Oberender (1984, pp. 207, 283)].

3. The production processes of the pharmaceutical industry also have been under fire time and time again, in particular, for their *testing on animals* when searching for new active substances. Similarly—the productivity of pharmaceuticals being part of the chemical industry—the impact of *toxic waste* on the environment is another issue.

4. As a result of what are considered *exorbitant prices*, revenues and profits are condemned as being excessive. During 1985, the seven German research-based companies witnessed a return on turnover of almost 3 percent, in 1986 of 1.6 percent [see MPS (n.d.), p. 8]. The automobile industry has also achieved returns of around 3 percent over several years, however, with comparable volatility.

Economic theory may contribute to clarifying the majority of these points of criticism, while not always resulting in conclusive suggestions for policy. This is mainly because the traditional criterion of “price equal to marginal costs” cannot be applied directly. Suppliers are not so much in competition with one another with a given product. Rather they attempt to safeguard their economic survival through *innovation*—up until that hardly determinable point where the expected marginal revenue of an additional innovation exactly covers their expected marginal costs.

As a result, therefore, the development of a new pharmaceutical—from the design of its desired therapeutic effects to its market launch and on to the end of its useful economic life—should be clearly understood. The pharmacological properties of a new substance constitute the starting point of Section 10.2, since within the prevailing system of public regulation of market access, pharmacological requirements largely determine whether or not development of the innovation will be pursued. Only if there is substantial evidence to the effect that market entry will be granted does the innovation become a worthy investment.

The profits of a pharmaceutical innovation will be examined in Section 10.3. They depend very much upon the length of patent protection granted by public authorities. Therefore, Section 10.4 is dedicated to the question of optimal patent life. After expiration of patent life at the latest, price competition will be possible between drugs of the same diagnostic category. Whether and how this takes place shall finally be examined in Section 10.5.

10.2 THE DEVELOPMENT OF A NEW DRUG

Traditional microeconomic theory of demand takes an existing good with established characteristics as the unit of analysis. Consequently, this theory is badly suited to

describe so-called product innovations. In contrast to process innovations, which aim at lowering the cost of production of an already existing product, product innovations create products with modified—and even completely new—characteristics. In the context of the development of a new pharmaceutical, it is therefore appropriate to make its characteristics the basic element of analysis, as is the case in the so-called *new theory of demand* [see Lancaster (1966, 1971), Becker (1965)].

10.2.1 The Consumption Technology of a Pharmaceutical

Pharmaceuticals may be described along several quality dimensions, among them

- Beneficial principal effect (per daily dose);
- Desired drug reaction;
- Adverse drug reaction;
- Rate of absorption in the organism;
- Convenience of handling (e.g., tablets instead of injection);
- Storability.

For concreteness, take a drug for the treatment of rheumatism. It is assumed that existing preparations definitely act as pain relievers (featuring characteristic c_1), without, however, being able to stop the joints from degenerating and therefore being unable to restore the mobility of joints (characteristic c_2). Let product A be one of the preparations already on the market, with a daily dose of A relieving pain during a period of four hours. Its so-called *consumption technology* is thus described by point 1A of Figure 10–1. Let a competing company develop a new drug, a dose of which is effective during seven hours (point 1N of Figure 10–1). On a market with perfectly informed and fully insured patients, the producer of N would be able to conquer the market, since for fully insured patients (and their physicians), only the quality features of a drug play a role, not its price.

10.2.2 Regulation of Market Access by Public Authorities

The hitherto considered innovation N was marginal in that it offered only more of the same quality feature c_1 , “pain relief.” Now let the patient suffering from rheumatism be interested not only in pain relief, but also in the restoration of mobility of the joints (c_2). In technical terms, both c_1 and c_2 are arguments of his utility function. In the following, an innovation that provides a good or service *featuring additional characteristics* of relevance to demand will be called a *breakthrough innovation*. The consumption technology of such a breakthrough innovation is represented by point 1B of Figure 10–1.

Although B is pioneering, its access to the market is not certain, since in Western industrialized countries, public authorities (the Food and Drug Administration in the United States, the Ministry of Health in other countries) decide whether a new chemical entity (NCE) is admitted to the market. In the example of Figure 10–1 the agency

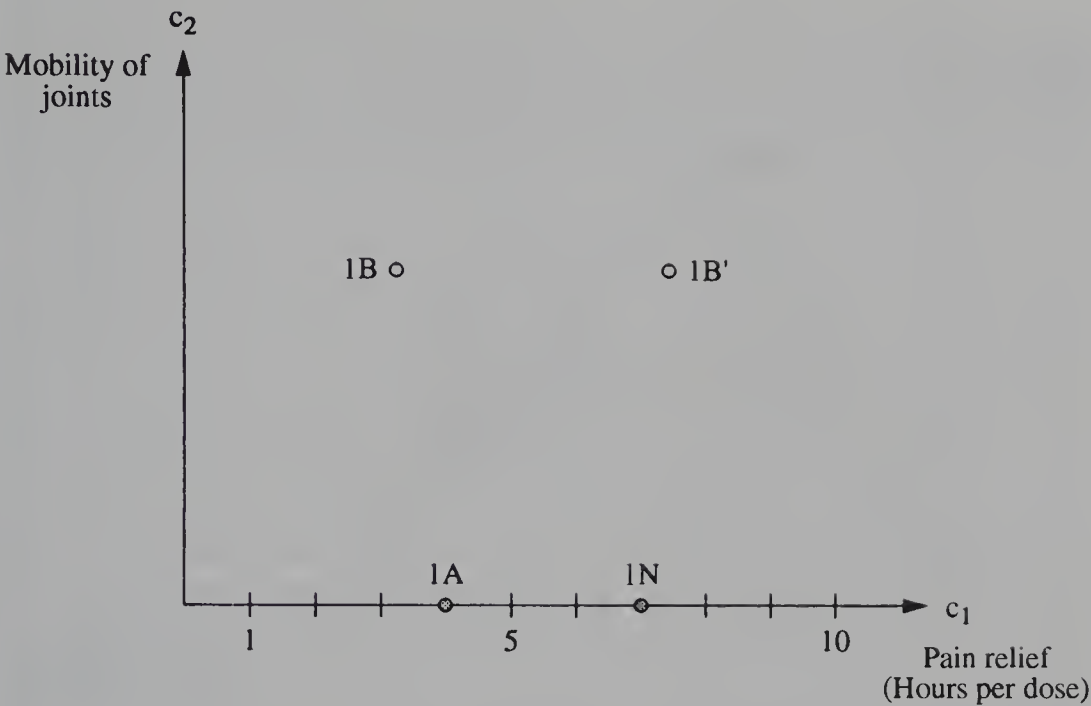


Figure 10-1. Consumption technology of three preparations for the treatment of rheumatism.

regulating market access could assert that product *B* will not even guarantee the same pain relief as does *A*, let alone that of *N*. In fact, it must weigh restoring mobility of joints against pain relief, and it is an open question whether it correctly reflects patients' preferences in this process.

The innovator's access to the market is only guaranteed if he succeeds in attaining *technological dominance* (e.g., point 1*B'* in Figure 10–1). A product is technologically dominant if it contains more of at least one and no less of the remaining characteristics per unit than its competitors. Since the price of a drug plays no role whatsoever for fully insured patients, such a breakthrough innovation will have guaranteed market success as well.

CONCLUSION 10.1: Apart from marginal (“me too”) innovations, which offer more of the same characteristics, only those breakthrough innovations that achieve technological dominance are certain of access to the market under traditional regulation. Given comprehensive health insurance, both types of innovation will have a guaranteed market success, however, once access to the market is granted.

10.2.3 Effects of Cost Sharing by the Patient

As soon as the patient does not have full insurance coverage and consequently shares in the cost of pharmaceutical treatment, product prices in addition to quality features become important. In the following, price is the net amount the insured has to pay per daily defined dosis.

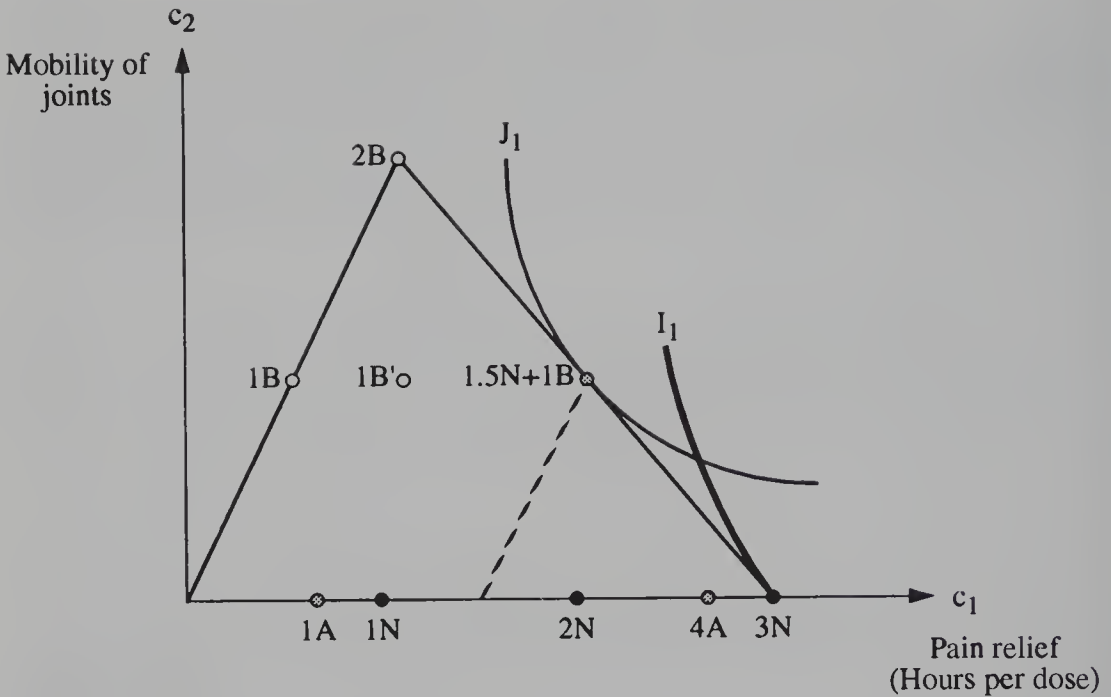


Figure 10-2. Pharmacological and economic aspects of a pharmaceutical innovation.

The decision situation of the patient is presented in Figure 10-2 under the assumption that his budget for pharmaceutical treatment is a fixed amount of 12 monetary units (MU). If the traditional antirheumatic drug A costs 3 MU per dose, his budget buys 4 doses. Given linear consumption technology, these 4 doses are four times as effective as a single dose. Accordingly, point 1A of the consumption technology becomes the economically relevant point 4A of Figure 10-2. However, this linearity assumption has its problems in many cases. On the one hand, a drug may fully take effect after several applications only, whereas on the other hand some substances may be addictive, implying that their dose must be successively increased to reach a given effect [see Stigler and Becker (1977)]. In the first case, one may speak of a consumption technology exhibiting increasing returns to scale, while in the second case, it has decreasing returns to scale (with respect to the quantity of desired characteristics).

By assumption product N is more effective than A. If its price were not higher than that of A, its superiority in terms of pharmacological effectiveness would transform into economic superiority. In general, however, the producer will skim off at least part of the higher willingness to pay for the innovative product by charging a higher price (see Section 10.5.1 for some empirical evidence). Let this price be 4 MU per dose, so that the budget of 12 MU buys 3 doses of N only. At this price, product N still dominates product A on the market (see points 3N and 4A of Figure 10-2).

Now let the breakthrough innovation B be even more expensive, costing 6 MU per dose. Assuming as before that a doubling of the dose results in a doubling of output in terms of characteristics, the patient using this preparation reaches point 2B. Whether

the patient or his physician actually chooses this point depends on *subjective preferences*, in keeping with the traditional theory of demand. The indifference curve I_1 of Figure 10–2 represents a patient who considers pain relief so important as to make him opt for the marginal innovation (point $3N$). Comparing points $3N$ and $4A$ of Figure 10–2, one finds that innovation N (qualified as “me too” above) confers a benefit upon consumers.

Other patients may be more interested in regaining mobility of joints (c_2). Accordingly, their indifference curves run flatter (like J_1), causing a combination of preparations B and N to be optimal. However, the optimum ($1.5N + 1B$) indicated cannot be realized if an integer number of doses is required (e.g., if the medicine is provided in the form of ampoules) or if the combined use of substances gives rise to adverse effects of interaction. If patients of type J are numerous enough, it may be worthwhile for a producer to *combine* characteristics c_1 and c_2 in a new way and to set a price such that his product lies on or beyond the existing efficiency boundary $2B3N$. Even if this means that “only” a marginal product differentiation is performed, this producer still contributes to a better provision of rheumatic patients with drugs. This advantage though has to be weighed against the cost of *product differentiation*, since smaller lot sizes in the production of a given product specification serve to keep average cost high. The problem of “excessive” product differentiation as mentioned in Section 10.1 can then be attributed to the fact that the insured consumer only has to bear part of the cost of such product differentiation while fully reaping its benefits.

The effect of a price change on demand can be shown to *differ according to type of innovation*. First, consider a doubling of price of the breakthrough innovation. In Figure 10–3, this causes the efficiency boundary to rotate from $2B3N$ to $1B3N$. A patient with indifference curve K_1 , who badly wants to see the mobility of his joints improved, will shift from point $2B$ to a new optimum Q^{**} . Rather than use product B exclusively as before, he now combines it with N to some extent. At the market level, this implies a loss of market share of product B . Thus a breakthrough innovation has a downward-sloping demand curve as soon as the buyer has to pay part of the price through cost sharing.

Yet, as long as patients value the additional characteristic provided by the breakthrough innovation, the new product is unlikely to be substituted completely, *regardless of price*. In Figure 10–3, with indifference curves like K_1 and K_0 , the efficiency boundary would have to run almost horizontal for the extreme point $3N$, indicating complete substitution of product B , to become the optimal solution. This implies that the price elasticity of demand for a breakthrough innovation must be small (in absolute value), since complete substitution in general does not occur.

By way of contrast, consider a tripling of the price of the “me too” innovation, causing the efficiency boundary to rotate from $2B3N$ to $2B1N$. Since the new boundary has positive slope in characteristics space, even an extreme type I patient (with indifference curves I_0 and I_1) will now opt for point $2B$ of Figure 10–3. Although exclusively interested in pain relief, this patient *switches to the breakthrough innovation* (which also provides pain relief to some extent) if the pure pain killer becomes sufficiently expensive. Thus, the price elasticity of demand for the incumbent product must be large (in absolute value) somewhere along its demand curve, in particular at high prices relative to the breakthrough innovation.

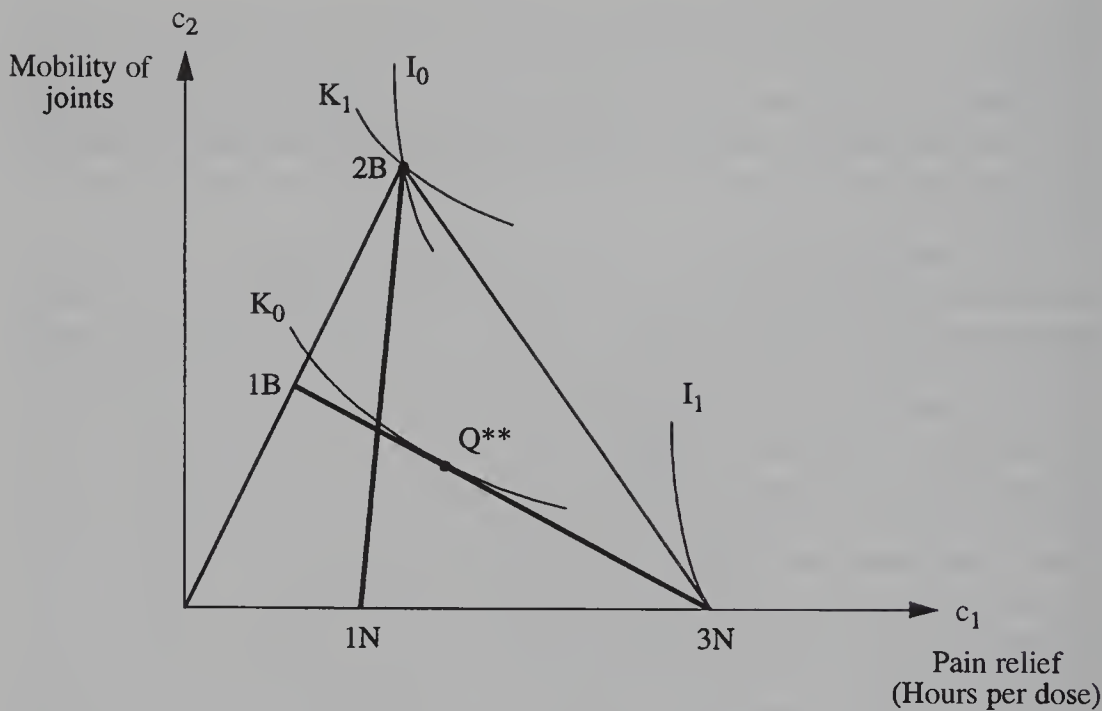


Figure 10-3. Price elasticity of demand for a breakthrough and a “me too” innovation.

CONCLUSION 10.2: In a market with less than complete insurance coverage, suppliers of a breakthrough innovation are faced with a falling demand curve for their product, whose price elasticity is, however, closer to zero than that of the competing traditional product.

10.3 INNOVATION AS AN INVESTMENT

At first glance, one would be inclined to say that the development of a new drug has little to do with investment in property, buildings, or machinery. Decisions in both areas have something very important in common, however, because they entail expenditures that are certain and occur in the present, whereas profit is uncertain and lies in the future. In this section, this similarity will be brought into play in order to characterize drug innovation from an economic point of view.

10.3.1 The Chronology of a Drug Innovation

At present the development of a new drug spans on average a period of more than over ten years, calculated from the point in time when, after thousands of synthesizing experiments, there at last emerges a promising substance. The subsequent preclinical phase lasts about two years, during which, based on weeks or months of testing on animals, evidence accumulates as to whether the use of the substance in humans comes into consideration at all. This is followed by a comparatively short phase, by the end

of which the properties of the substance are established so that a provisional registration with the Ministry of Health can be made (New Drug Application, NDA, with the Food and Drug Administration, FDA, in the case of the United States). Next, there are several years of clinical development that serve on the one hand to prove the drug's efficacy, and on the other hand its safety not only in the short but also in the long term. Finally, documentation must be prepared for submission to the authority deciding about market access (typically the Ministry of Health; the FDA in the United States), which lasts approximately two years. Parallel testing of the drug in several hospitals and countries is used to discover any adverse drug reaction that may arise in long-term application.

During the 1970s, this whole process lengthened considerably [see Walker and Parrish (1988)]. At present, it takes about *eleven years*, compared to less than four years at the beginning of the 1960s. This extension is due to a tightening of requirements for registration, with the United States playing a leading role. It represented the reaction to the Thalidomide tragedy when hundreds of babies with malformations were born because their mothers had used this sleeping drug. It is perhaps paradoxical that the added requirements for market access probably *cost more lives than they saved* because effective drugs too became available with delay only, causing many people to die from diseases that could have been treated in the meantime [see Peltzman (1973)].

10.3.2 Probability of Success and the Cost of Innovation

The making of a pharmaceutical not only takes many years, but is also accompanied by many setbacks. According to estimates prepared by Wardell, DiRaddo, and Trimble (1980) for the United States, the synthesis of no fewer than 10,000 substances is necessary in order to develop one marketable drug. Thus, out of every 10,000 synthesized substances, only 15 to 40 will be tested for toxicity, of which only 2 to 3 will survive a long phase of clinical trials. In the course of investigations for long-term toxicity, this number boils down to one single new chemical entity (NCE). Previous estimates arrived at a probability of success of 1:3,000 during the 1960s [see Vane (1964)].

The simultaneous decrease in the probability of success and lengthening of the development phase has caused a surge in the average cost of a new marketable drug. Discounted at 8 percent, the development of a new product cost 7.5 million US\$ during the 1950s [see Baily (1972)]. This figure rose to 54 million for products introduced during the 1970s [see Hansen (1979)] and on to 125 million during the 1980s [see Wiggins (1987)].¹ The most recent estimate amounts to roughly 200 million US\$ [see DiMasi et al. (1991)]; it refers to new substances that were tested during the period 1970–1982 and whose sales prices were compounded at 8 percent until 1987. Since sales during the same period did not increase as rapidly, the *number of market introductions worldwide fell*, as Figure 10–4 illustrates.

Whereas during the early 1960s, almost twenty NCEs were launched every year in the United States alone, this number dropped to about 15 during the 1980s in the United

1. In this calculation, the accumulated costs at the time of market launch are discounted to the present value at the time of decision to develop, that is, 10 years earlier.

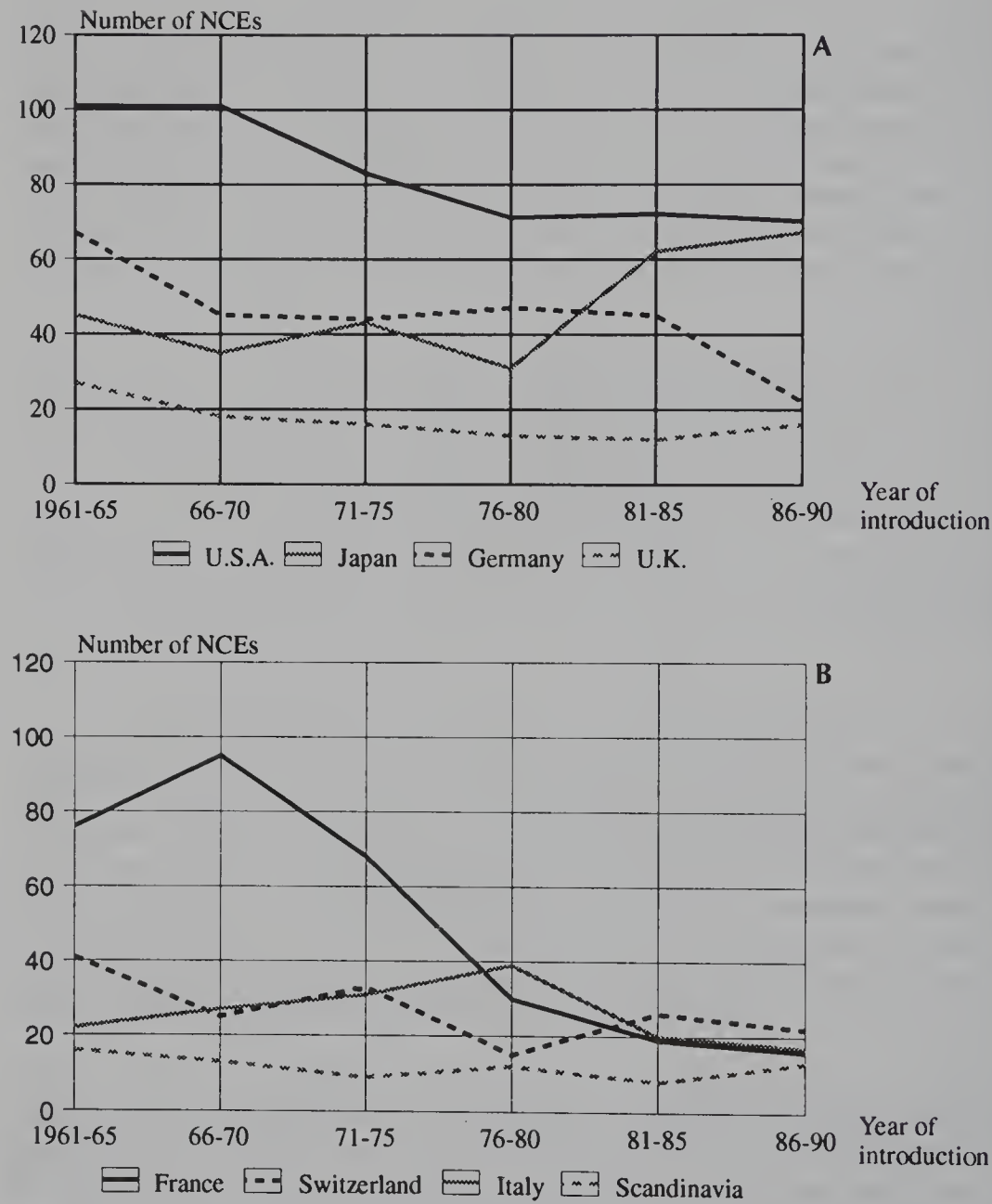


Figure 10-4. New chemical entities (NCEs) and country of introduction: (a) United States, Japan, Germany, and United Kingdom and (b) France, Switzerland, Italy, and Scandinavia. Source: Reis-Arndt (1993).

States, with comparable declines in other major markets such as Germany and the United Kingdom. France, where the number of new introductions almost reached the level of the United States in the later 1960s, continues on a falling trend, the likely reason being its rather strict regime of price control for drugs that are on the formulary of social health insurance. The one country registering a rising number of NCE introductions is Japan, where social health insurers accept high initial prices for

innovations but request “rebates” later on very much like the country’s physicians do, who prescribe drugs on their own account.

CONCLUSION 10.3: Drug innovation can be considered an investment. Since the early 1960s, the cost of drug innovation has rapidly increased, while the number of newly introduced active substances tends to decrease worldwide, with the notable exception of Japan.

10.3.3 Great Returns to Pharmaceutical Innovation?

Since innovation represents an investment, the present value criterion (*PV*) is suitable for gauging profitability. In its most simple form, neglecting the several stages of development with their specific survival probabilities, the *PV* criterion is given by

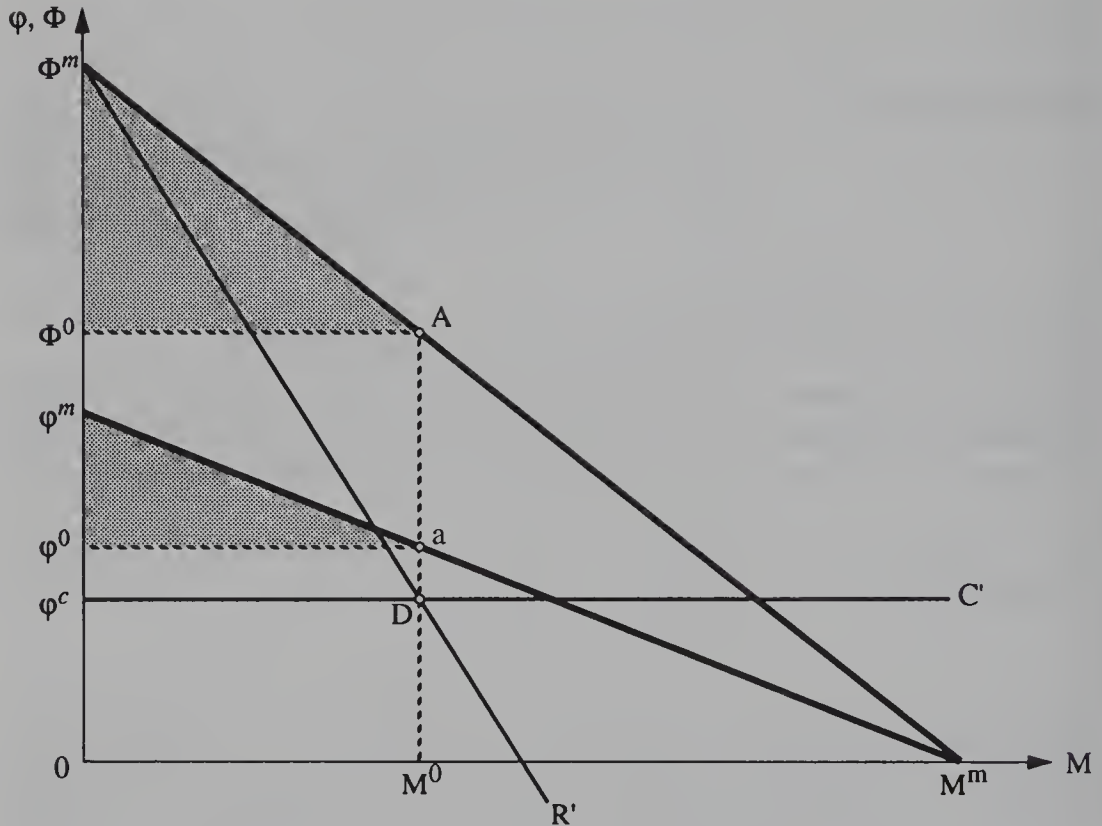
$$PV = \sum_{t=0}^T \frac{R_t}{(1+r)^t} - \sum_{t=0}^T \frac{C_t}{(1+r)^t} \quad (10.1)$$

In this formula, R_t symbolizes a flow of revenues that takes on positive values no sooner than after twelve years on average. The expenditure flow, C_t , however, assumes high values at the beginning of the research and development phase but drops to the variable costs of production towards the end of the life of the product. If *PV* is positive, then the project contributes to the investor’s assets. The flows C_t and R_t as well as the interest rate r used for discounting can reliably be estimated only for products that are approaching the end of their economic life. Therefore, the estimates to be cited below reflect the situation in the mid-1970s.

Whereas the present value criterion represents the theoretically preferred measure of profitability because of its unambiguity, the *internal rate of return* is the more popular. The internal rate of return is equal to the interest rate r^* that yields a present value $PV = 0$ in Eq. (10.1). It indicates the extent to which the investment project may be “charged” with capital user cost while remaining profitable.

Joglekar and Paterson (1986) calculated expected internal rates of return of pharmaceutical innovations in the United States that were initiated in 1976 or earlier and were registered in 1988 at the latest. Based on trend extrapolations of the probable development of returns, they arrived at a real internal rate of return of 6.1 percent after taxes for the average NCE. In addition, they give an impression of the *uncertainty* that faced an investor in the 1970s. Since the arithmetic mean is strongly influenced by outliers (i.e., the very few successful drugs in this case), the median value of the internal rate of return is far below the average value of 6.1 percent, amounting to -5.5 percent. According to the study, one-half of the 218 innovations that were introduced in the United States during the period 1962–1977 will not have paid back their development costs even after 36 years.

For an assessment of the social value of pharmaceutical innovation, its *total economic rate of return* is relevant. This may be positive although the NCE turns into a loss to the investor. For in order to calculate total rate of return, *consumer surplus* must be added, constituting the part of willingness to pay for the innovation not



- | | |
|---|---|
| Φ : Observed price of the drug | M : Quantity demanded |
| ϕ : Net price faced by the insured, $= c \Phi$ ($c = 0.5$) | M^m : Satiation quantity |
| Φ^0 : Profit-maximizing price | $\Phi^0 A \Phi^m$: Consumer surplus inferred from observed market data |
| ϕ^0 : Net price corresponding to profit-maximizing price | $\phi^0 a \phi^m$: True consumer surplus $= c \Phi^0 A \Phi^m$ |
| ϕ^m : True maximum willingness to pay (observed: Φ^m) | $\phi^c D A \Phi^0$: Producer surplus (profit) |
| R' : Marginal revenue | |
| C' : Marginal cost, constant (and equal average cost) | |

Figure 10-5. True and observed willingness to pay (50 percent coinsurance).

appropriated by its producer. For three particular pharmaceutical innovations, Wu (1984) calculated total internal rates of return in excess of 25 percent. However, this certainly was an overestimation because drug prices paid on the market exaggerate patient's willingness to pay in the presence of health insurance.

This is demonstrated in Figure 10-5. Let the patient's true marginal willingness to pay during some period t of the life of the drug be represented by the demand function $\phi^m M^m$. With a rate of coinsurance of 50 percent ($c = 0.5$), the observed market demand function $\Phi^m M^m$ runs twice as steeply as $\phi^m M^m$. If, for example, the patient's true maximum willingness to pay is $\phi^m = 5$ MU, then the maximum price of the drug may be set at $\Phi^m = 10$ MU. Because the innovator, assumed to act as a monopolist, equates marginal revenue R' to marginal cost C' , the price actually paid is Φ^0 .²

2. For the sake of simplicity, we abstract from the possibility of price negotiations between health insurers and drug producers.

Based on price–quantity observations, an econometrician will estimate the demand function $\Phi^m M^m$, obtaining a consumer surplus given by the area $\Phi^0 A \Phi^m$. The sum of revenue and consumer surplus per period (i.e., $OM^0 A \Phi^m$) corresponds to a typical element in the calculation of the total internal rate of return analogous to R_i of Eq. (10.1). The term C_i is symbolized by the area $OM^0 D \varphi^c$ below the marginal cost function.

The *true aggregate willingness to pay* for M^0 units of the drug can be read off from the area below the line $\varphi^m M^m$, symbolizing true marginal willingness to pay. It amounts to the area given by $OM^0 a \varphi^m$. Due to proportionality, this is obviously one-half of observed consumer surplus if the rate of coinsurance is 50 percent. So when Wu (1984) estimated the total rate of return of a pharmaceutical innovation to be around 200 percent of the private rate of return, the surcharge of 100 percent for consumer surplus must be cut to 50 percent in case of $c = 0.5$ and even to 25 percent if $c = 0.25$. Widespread scepticism with regard to the social usefulness of pharmaceutical innovations may well reflect this discrepancy between willingness to pay observed on the insured market and true willingness to pay on the part of the patient [see Zweifel (1984)].

CONCLUSION 10.4: Low real internal rates of return on average at about 6 percent for the innovator contrast with higher total rates of return of pharmaceutical innovations for society, whose consumer surplus component is however overestimated due to the effect of health insurance.

10.4 THE ROLE OF PATENT PROTECTION

10.4.1 Why Patent Protection?

Patent protection grants the inventor of a new product or process a monopoly for the commercial use of his innovation. While this monopoly is limited in time—to 20 years from the date of filing a patent application in most Western industrialized countries at present—it does not agree with the principles of perfect competition. Perfect competition ensures that relative prices of commodities simultaneously reflect their relative marginal utilities on the demand side and their relative marginal cost of production on the supply side; this means that the economy is in a Pareto-optimum. This equality is disrupted by the existence of monopolies, which cause relative commodity prices to differ from their relative marginal costs. Therefore, patent protection is responsible for long-term deviations from the optimal state of the economy.

This point of view proves too narrow, however, as soon as the commodity set contains not only the already existing products but also the ones *still to be invented* in the future. Absent patent protection, imitating someone else’s innovation would be much more profitable than investing in innovation because the cost stream C_i in Eq. (10.1) is much smaller in the case of imitation. Patent protection thus is a way to lower the profitability of imitation and enhancing that of innovation. It serves to curtail the consumption of currently available goods (in the way of deadweight losses due to monopolization) in the interest of the prospect of consuming newly developed goods

in the future. In an ultimate sense, patent protection brings to the fore the latent willingness to pay for innovations, which would have to be reflected by a price surcharge over and above the marginal cost of current output.

Through patent protection, the innovator becomes a *monopolist for a limited period of time* during which he is capable of enforcing such a surcharge on the market. It is easy to show that the surcharge brought about by the patent usually is not socially optimal. The profit-maximizing price φ^* is given by the equality of marginal revenue and marginal cost,

$$\varphi^*(1 + 1/\eta) = C', \quad \text{with } \eta := \frac{\partial M}{\partial \varphi} \frac{\varphi}{M} < 0$$

which determines the optimal relative surcharge

$$\frac{\varphi^*}{C'} = \frac{1}{1 + 1/\eta} \quad (10.2)$$

Here η represents the price elasticity of demand, which generally varies with output. Evidently the surcharge over marginal cost φ^*/C' as set by the innovator depends on η and hence output, whereas the marginal cost of an innovation (to be recovered by the patent) is a fixed quantity independent of output.

Therefore, patent protection is a *second-best solution* to the problem of innovation. The problem stems from the fact that innovation is based mainly on information. The pharmaceutical industry is concerned with the search for knowledge that permits the creation or identification of an NCE with desirable characteristics. Once this knowledge is available in the form of a drug, its chemical structure can be found out at comparably low cost. Hence, the knowledge that makes the innovation possible becomes a *public good*.

The imitator does not have to bear the cost of innovation, enabling him to recover average cost at a lower price than the original manufacturer. He can therefore increase his market share at the innovator's expense through a reduction of price. The innovator will hesitate to lower his price, for example, for fear of having difficulties with justifying the price of his next innovation in negotiations with health insurers and the government. Clinging to a high price, however, will sooner or later make him lose the market for this particular drug. To avoid this, the innovator has to lower the price to the level of the imitator, that is, to marginal production cost. Thus, without patent protection, he is not able to recover his research expense. Whatever his pricing policy, the innovator will come to the conclusion that innovation does not pay. Therefore it is the key function of patent protection to safeguard the incentives for innovation.

10.4.2 The Decision Situation of the Innovator

Given that patent protection constitutes at least a feasible solution to the problem of keeping technological change going, the question of its extent and duration remains.

With regard to extent, novel drugs enjoy protection of substance rather than merely process. This means that the innovator may reap the additional returns arising from new uses of an NCE. Since the scope of patent protection is so extensive in the case of drugs, the issue of its *duration* becomes all the more crucial, motivating the study of optimal length of patent protection.

In the early literature [see Arrow (1962); Nordhaus (1969)], the innovator is assumed to have already made his invention, his expenditure on research and development constituting sunk costs. He makes his invention available to a firm, enabling it to produce at lower cost. The question to be answered was how long the firm should have to pay for the use of the invention. The length of this period defines the optimal life of the patent. In these models, patent protection is not granted prior to the final development of the new product. But at least in the pharmaceutical industry, patents are applied for early in the process for securing the protection of substance. At this time, the *amount of investment in innovation is not determined yet*, implying that protection afforded by the patent will exert its influence on the incentive to invest in innovation to its full extent.

These facts are taken into account in a model developed by DeBrock (1985), which is presented in the following. From the present value criterion of Eq. (10.1), it follows that lengthening the useful life of an NCE (an increase of T) makes the innovation more profitable. In the following, T is equated with the effective length of patent protection. Denoting innovative effort by I , let the innovator maximize the present value of profits Π

$$\Pi = R(I, T) - C(I) \quad (10.3)$$

Here R symbolizes the present value of revenues and C the present value of cost. The following assumptions are made.

ASSUMPTION 10.1: Marginal returns to innovation are always positive and may even increase at first, reflecting the notion that effort must exceed some threshold really to pay off,

$$R_I := \frac{\partial R}{\partial I} > 0$$

$$R_{II} := \frac{\partial^2 R}{\partial I^2} \begin{cases} > 0 \text{ for small values of } I \\ < \text{ for large values of } I \end{cases}$$

ASSUMPTION 10.2: Additional innovative effort always costs more, and since it requires highly specialized resources, it progressively drives up cost at the margin, that is,

$$C_I := \frac{\partial C}{\partial I} > 0, \quad C_{II} := \frac{\partial^2 C}{\partial I^2} > 0, \quad C_{III} := \frac{\partial^3 C}{\partial I^3} > 0$$

ASSUMPTION 10.3: The present value of revenues is greater the longer the patent protection. Moreover, additional effort pays off with longer

Table 10.1: Simultaneous Determination of Innovation Effort and Length of Patent Protection

$\Pi = R_{(+)(+)}(I, T) - C_{(+)}(I) \rightarrow \max$	Innovator's objective function	(10.3)
$W = S_{(+)(-)}(I, T) - P_{(+/-)(+)}(I, T) \rightarrow \max$	Patent Office's objective function	(10.6)

Π : Present value of profits from innovation	S : Present value of consumers' (social) rent after market launch
I : Innovative efforts; decision variable of the innovator	T : Length of patent protection; decision variable of the Patent Office
C : Present value of cost of innovation	W : Social welfare due to the innovation
R : Present value of returns from innovation	
P : Present value of producer's (private) rent after market launch	

Note: The signs in parentheses indicate signs of partial derivatives with respect to the relevant argument.

patent protection, or equivalently, patent protection makes a difference especially when the marginal return on innovative effort is high (as in the case of a breakthrough innovation),

$$R_T := \frac{\partial R}{\partial T} > 0, \quad R_{IT} := \frac{\partial^2 R}{\partial I \partial T} > 0$$

Altogether an increase of research effort (I) has an ambiguous effect on the present value of profits Π . While additional research I promises a greater success after market launch and thus an increased present value of returns, the costs not only of development but also of production will generally increase with I . The innovator's *decision variable* is innovative effort I , whereas the Patent Office determines the length of patent protection T (see also Table 10.1).

The first step consists in deriving a reaction function for the innovator, displaying his optimal adjustment of I to changes in T . For this purpose, consider the first-order condition for a maximum, noting that R_I continues to depend on T ,

$$\Pi_I = R_I(T) - C_I = 0 \tag{10.4}$$

Let this optimum be disturbed by a change in effective patent life dT . In order to restore the first-order optimum condition, the innovator's adjustment must satisfy the following condition:

$$R_{II} dI + R_{IT} dT - C_{II} dI = 0 \tag{10.5}$$

Solving for dI/dT , one obtains the slope of the *innovator's reaction function*,

$$\frac{dI}{dT} = - \frac{R_{IT}}{R_{II} - C_{II}} > 0 \tag{10.6}$$

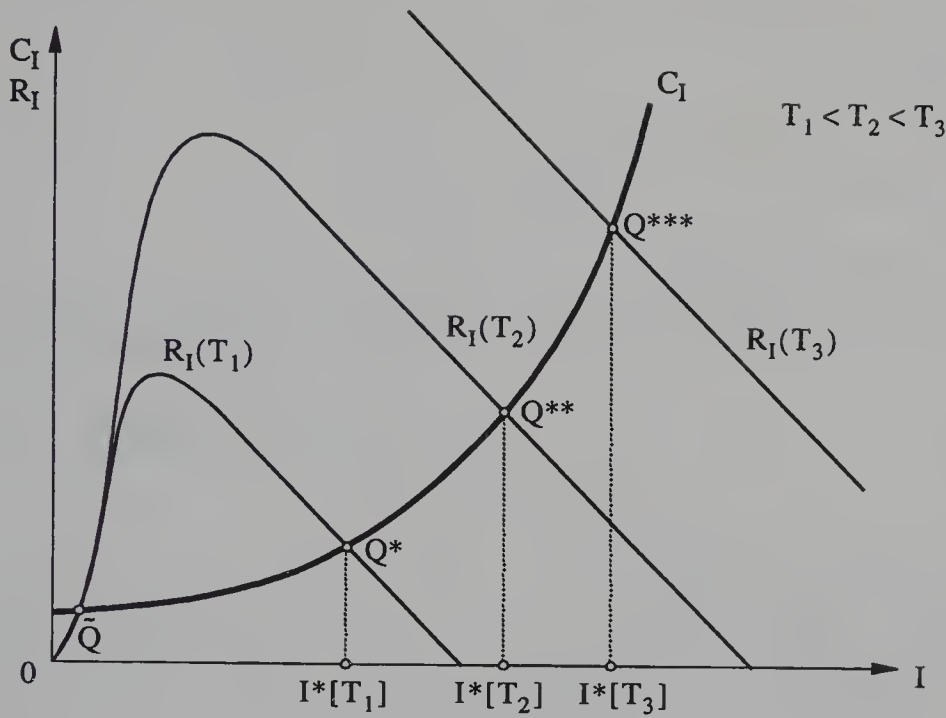


Figure 10-6. Iso-profit curves, marginal cost of innovation, and reaction function of innovator.

The positive sign of Eq. (10.6) derives from Assumption 10.3 and the sufficient condition for maximum profit, $\Pi_{II} = R_{II} - C_{II} < 0$. The innovator's reaction function (with second-order derivatives R_{II} and R_{IT} independent of I , at least for large values of I) is illustrated in Figure 10-6. Note that $\bar{Q}$ does not qualify for an optimum because the slope of the R_I function exceeds that of the C_I function, implying $R_{II} - C_{II} > 0$, which violates the sufficient condition for a maximum. Due to the convexity of the marginal cost function ($C_{III} > 0$; see Assumption 10.2), the sequence of optima $\{Q^*, Q^{**}, Q^{***}\}$ defines a sequence of optimal effort levels $\{I^*[T_1], I^*[T_2], I^*[T_3]\}$ that increases in T as required by Eq. (10.6), but at a decreasing rate. This expansion path appears as the $g(T)$ locus of Figure 10-7.

10.4.3 The Length of Patent Protection from the Regulator's Point of View

In this section, the viewpoint of the regulatory agency is adopted, assuming that it does not pursue its own objectives but has its eyes on aggregate welfare. The agency's own interests would probably be served best by a long period of patent protection, causing applicants to follow the instructions of the Patent Office with diligence because large stakes are involved (see Assumption 10.3).

In the following, this institutional bias will be disregarded in favor of the assumption that the public authority seeks to draw the greatest possible benefit from the innovation. If the economy as a whole profits, so does the government through tax revenue. The gain in welfare (W) from an innovation that has made it to the market has

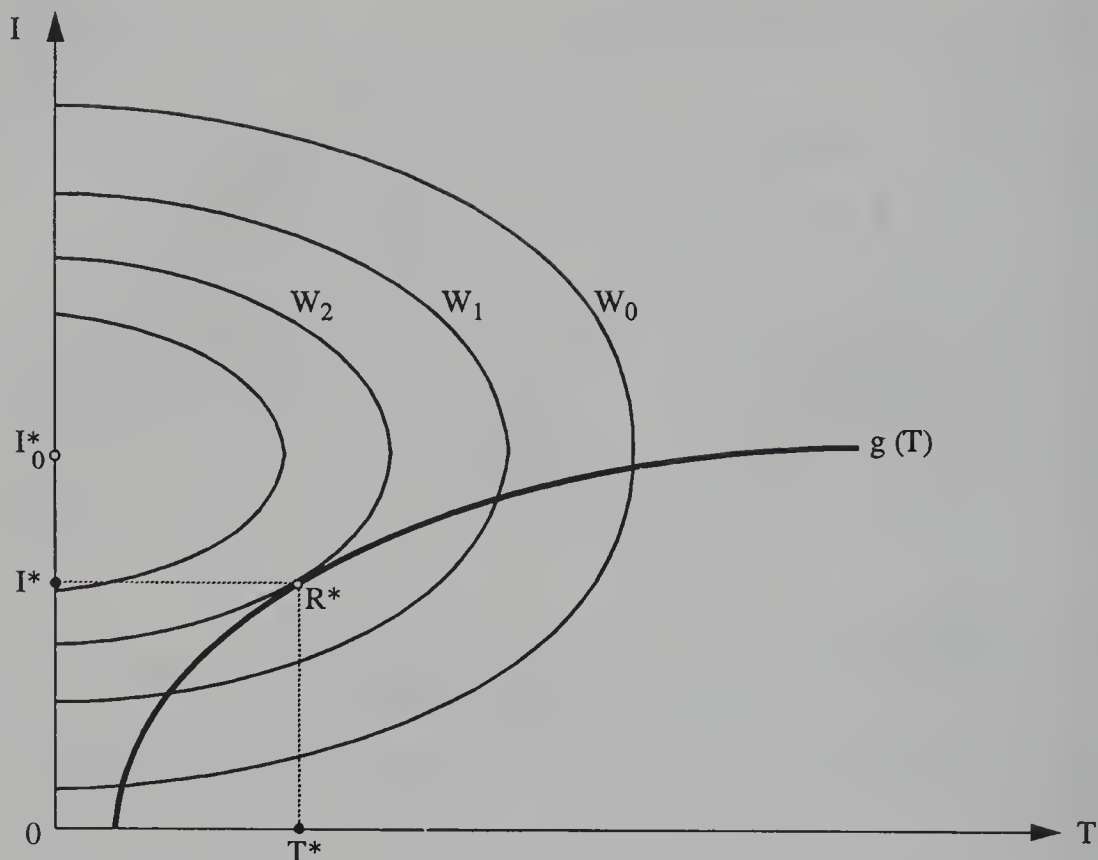


Figure 10-7. Optimal patent life in equilibrium. *Source:* DeBrock (1985).

two components. First, there is the *social component*, consisting of consumer surplus and any nonappropriated positive externalities in the rest of the economy, whose present value is denoted by S . Second, there is the *private component*, given by the present value of producer's rent P . Since profits Π are an important part of producer's rent, P depends on innovative effort and effective patent life in the same way as does Π [see Eq. (10.3) and Assumptions 10.1–10.3]. The function to be maximized by the Patent Office is thus (see also Table 10.1)

$$W = S(I, T) + P(I, T) \quad (10.7)$$

In the following, some additional assumptions are introduced.

ASSUMPTION 10.4: The present value of the social component of the welfare gain increases with innovative effort but decreases with effective patent life. Thus

$$S_I > 0, \quad S_T < 0$$

ASSUMPTION 10.5: The present value of the private component of the welfare gain increases for low values of innovative effort I but decreases

for high values of I . However, it increases with effective patent life T throughout.

$$P_I \begin{cases} > 0 \text{ for small values of } I \\ < \text{ for large values of } I \end{cases}$$
$$P_T > 0$$

An iso-welfare curve may be derived from the objective function (10.7) by total differentiation and setting dW to zero,

$dW = S_I dI + S_T dT + P_I dI + P_T dT = 0,$ yielding

$$\frac{dI}{dT} = - \frac{S_T + P_T}{S_I + P_I}, \quad \text{with } S_T + P_T < 0 \tag{10.8}$$

The condition $S_T + P_T < 0$ follows from the argument that an increase in T raises the so-called *deadweight loss*, because the loss in consumer surplus caused by monopolization cannot be fully transformed into additional producer surplus. The slope of the iso-welfare curve thus depends on the denominator of Eq. (10.8), which in its turn hinges on P_I . For small values of innovative effort, the marginal effect of I is positive on consumer as well as producer surplus, causing the iso-welfare curves to be positively sloped (see Figure 10–7). They bend back at the point where the loss of producer surplus ($P_I < 0$) is precisely compensated by the gain in consumer surplus ($S_I > 0$), resulting in a value of zero of the denominator in Eq. (10.8).

If the Patent Office could choose T according to its preferences without having to consider the reaction of the innovators, it would aim at reaching satiation point I_0^* as an *absolute optimum* where the deadweight loss is reduced to zero. However, positive innovation efforts require a positive amount of patent protection, as argued in Section 10.4.1. Consequently, I_0^* , not being on the reaction function of the innovator, cannot be reached by the Patent Office.

The point of equilibrium R^* marked in Figure 10–7 is the outcome of a noncooperative game between the Patent Office and the innovator, with the Patent Office taking the reaction function of the innovator as given and seeking its optimum on that function.³ The agency thus will strive for the highest attainable iso-welfare curve $W_2 (> W_1)$ by granting patent protection of length T^* .

CONCLUSION 10.5: The optimal length of patent protection can be regarded as the result of a noncooperative game between the Patent Office and the innovator.

10.4.4 Adjustment of Optimal Patent Life to Exogenous Change

Since new substances are submitted for patenting right after synthesis, that is, very early in the research and development process, the extension of the clinical phase since

3. This equilibrium concept corresponds to the duopoly solution of Stackelberg with a leader and a follower. The Patent Office here assumes the role of the independent decision-maker.

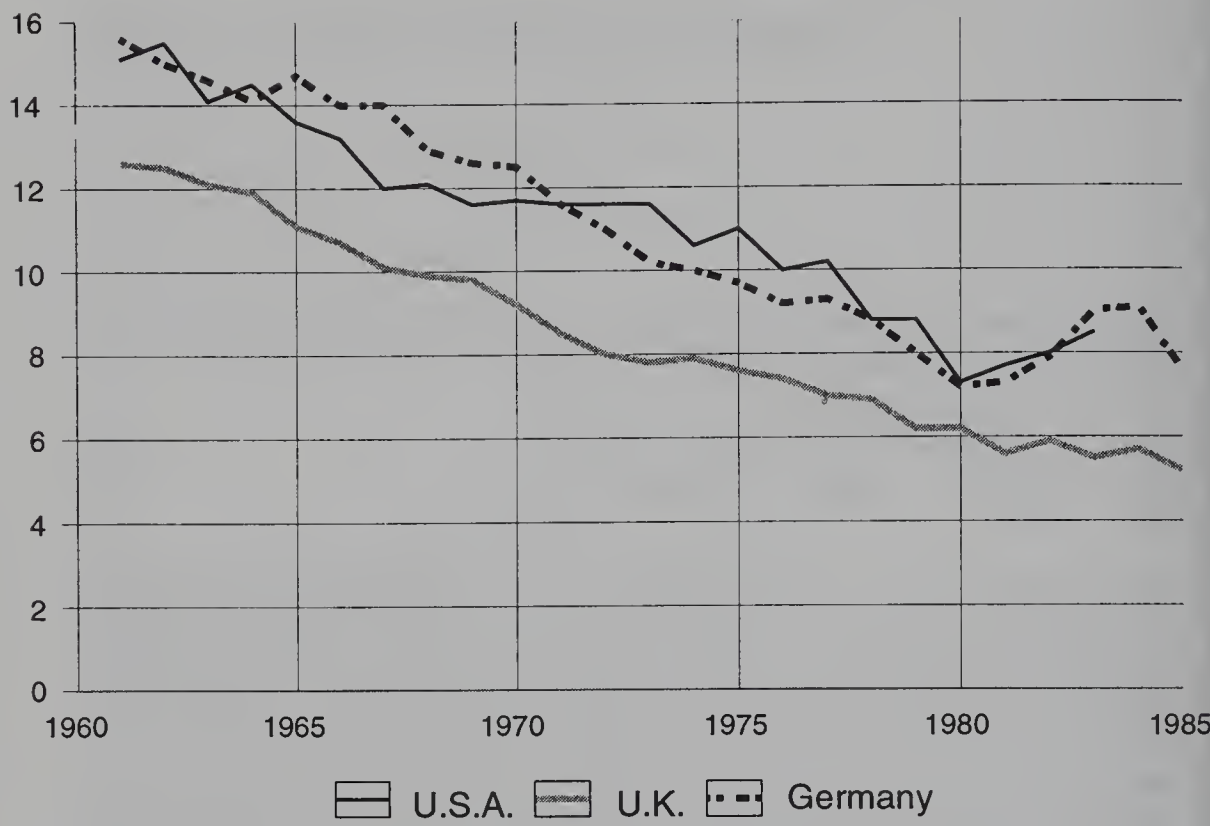


Figure 10-8. Effective patent life in the United States, the United Kingdom, and Germany (in years). Source: Walker and Parrish (1988).

1960 has led to a decline of so-called effective patent life. This is the period of time during which the innovation is on the market while being legally protected from imitation. This reduction of effective patent life is documented by Walker and Parrish (1988) for some important pharmaceutical markets (see Figure 10–8). According to an unpublished update, the downward trend was halted in the second half of the 1980s, with effective patent life converging to eight years in the United States, United Kingdom, and Germany.

When discussing these changes, the claim is usually made that the *nominal length of patent protection should be raised*, restoring effective patent life to its former value. The validity of this claim can be tested using the model just developed. Let T now represent the effective rather than the nominal length of protection. Although patent authorities in fact fix nominal patent life, they indirectly determine its effective value as well.

The *lengthening of the registration process* for new drugs causes additional innovative effort to be even more than before associated with costs of clinical testing and documentation. This may be modeled by another exogeneous shock representing administrative hurdles, $dA > 0$ (say), impinging on the marginal cost side but not on the marginal revenue side of condition (10.4), which means $C_{IA} := \partial^2 C / \partial I \partial A > 0$. Conservation of first-order condition (10.4) requires

$$R_{II} \, dI - C_{II} \, dI - C_{IA} \, dA = 0 \tag{10.9}$$

Solving for dI/dA , one has

$$\frac{dI}{dA} = \frac{C_{IA}}{R_{II} - C_{II}} < 0 \tag{10.10}$$

Thus for a given length of patent protection T , more stringent requirements in terms of clinical testing and documentation have caused innovative effort by pharmaceutical firms to be lower than otherwise. This is shown as a shift of the reaction function from $g(T)$ down to $g'(T)$ of Figure 10–9. For the sake of the ensuing argument, two additional assumptions are introduced at this point.

ASSUMPTION 10.6: The amount of shift from $g(T)$ to $g'(T)$ in Figure 10–9 is independent of effective patent life T . This requires

$$\frac{\partial}{\partial T} \left(\frac{dI}{dA} \right) = 0$$

ASSUMPTION 10.7: The objective function W of the public authority is homothetic.

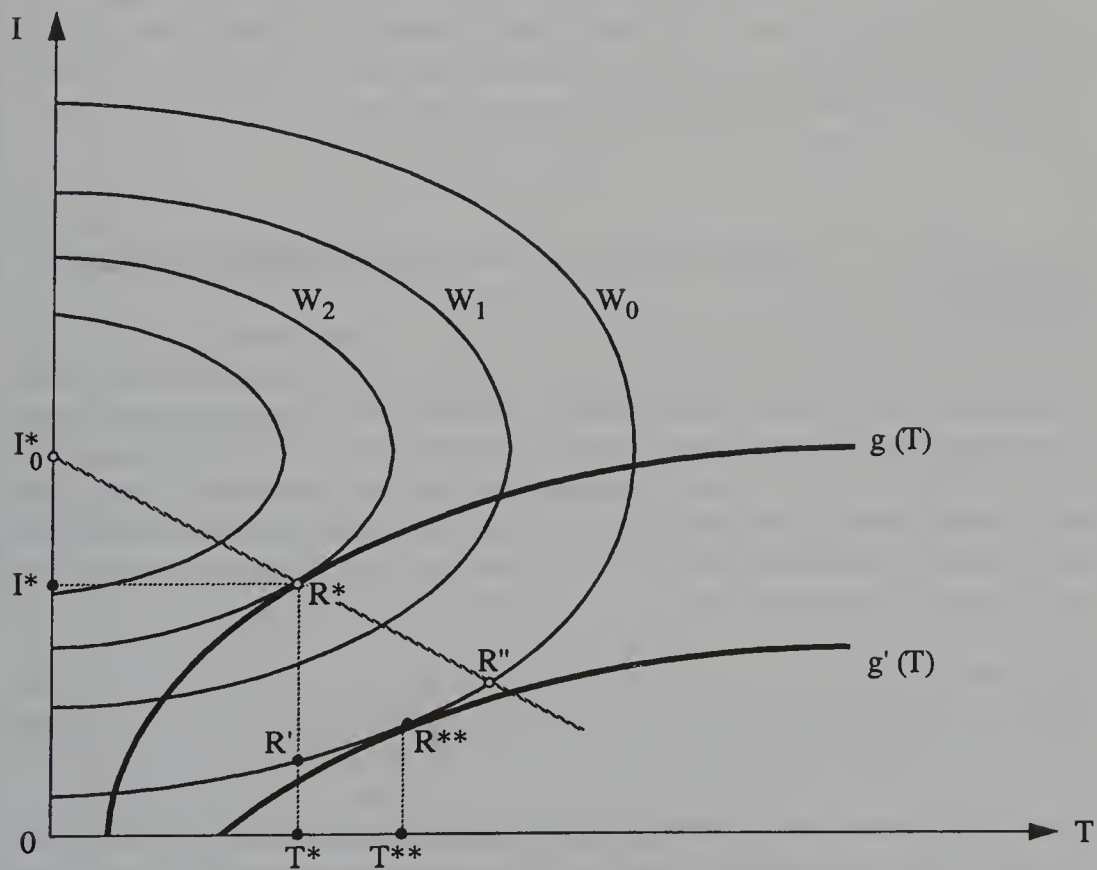


Figure 10-9. Extension of patent life in response to increased marginal cost of innovation.

Assumption 10.7 says that the iso-welfare curve W_0 of Figure 10-9 has the same slope at point R'' as the iso-welfare curve W_2 at point R^* . We now intend to show that the optimal length of effective patent life and hence patent protection increases in response to added administrative hurdles ($dA > 0$). In Figure 10-9, consider point R' lying on W_0 , which corresponds to an unchanged length of patent protection T^* despite the shift to $g'(T)$. In view of Assumption 10.7, W_0 must have a flatter slope at point R' than W_2 at the initial optimum R^* . However, at $T = T^*$, the slope of the new reaction function $g'(T)$ is the same as in R^* by Assumption 10.6, $g'(T)$ being the result of a parallel downward shift of $g(T)$. At $T = T^*$, the slope of $g'(T)$ thus exceeds that of W_0 , implying that T^* cannot be the optimum value of T in the new situation. Indeed, equality of the slopes of the iso-welfare curve and the reaction function $g'(T)$ is attained at a *higher value* of T such as T^{**} , corresponding to the new equilibrium point R^{**} .

However, this line of argument takes as given an *unchanged valuation of the social benefits* of pharmaceutical innovation, whereas policy makers may be of the opinion that consumer welfare is no longer augmented to the same extent as before through pharmaceutical innovation. Such a change in belief could be modeled by assigning a smaller value to the parameter S_I of Eq. (10.8). As a result of this, the rising branch of the iso-welfare curve of Figure 10-7 would run steeper and bend back earlier, which must lead to a reduced optimal length of patent protection. The observed fall in effective patent life might thus also be interpreted as the expression of a more sceptical social valuation of the innovation efforts by the pharmaceutical industry.

This alternative interpretation does not seem to apply to the U.S. government and the EU Commission, however. In 1984, the U.S. Drug Price Competition and Term Patent Restoration Act extended patent protection for drugs from 17 to a maximum of 22 years, while similar attempts are underway in the EU.

10.5 PRICE COMPETITION IN THE PHARMACEUTICAL MARKET

In the wake of the Thalidomide catastrophe, the U.S. pharmaceutical industry was examined by the Kefauver Committee of the Senate. The Committee found that the production costs in many cases do not exceed 15 percent of the selling price. If one equates these production costs to marginal cost, Eq. (10.2) indicates a considerable degree of monopoly, as measured by a so-called *Lerner index* [see Steele (1962)]. The Lerner index is derived from the relative deviation of price from its benchmark value under competition, the marginal cost C' . From Eq. (10.2), one obtains

$$\frac{\Phi^* - C'}{C'} = \frac{\Phi^*}{C'} - 1 = \frac{1}{1 + 1/\eta} - 1 = \frac{-1/\eta}{1 + 1/\eta} = \frac{1/\eta}{-1 - 1/\eta} \quad (10.11)$$

Since a high relative excess of price over marginal cost is associated with a high value of $1/\eta$, the index amounts to the inverse of the price elasticity of demand, expressed in absolute terms (Lerner index = $1/|\eta|$).

However, there are problems with such a simplified view of competition in pharmaceutical markets. The Lerner index is only suitable for an industry characterized by a stable technology. As soon as the increase in output comes about as a result of product innovation involving a new technology, the expenditure on the development and introduction of this new technology becomes part of the marginal cost. In this context one must also remember the fact that the successful innovation (and only successes can be observed on the market) is not typical of a company's product spectrum. One thus runs the risk of singling out the comparatively rare winners of a lottery [see Comanor (1986)]. Therefore, the issue of price competition can only be clarified in the context of product innovation.

10.5.1 Competitive Pricing Despite Insurance Coverage and Market Closure

As was demonstrated in connection with Figure 10–1, only differences in the quality of treatment but not differences in prices are important for patients who have full insurance coverage. In addition, many national pharmaceutical markets are shut off from each other to a large extent. For example, German producers frequently sell their products at much lower prices in countries where there is price regulation (mainly France, Italy, and Spain), creating opportunities for arbitrage. However, this arbitrage is prevented by the prohibition of so-called *parallel imports*.

The high expenditure on research and development of new drugs as described in Section 10.3.3 also contributes to market closure. This barrier to entry could be overcome by buying marketing licenses from innovating pharmaceutical companies. However, licensing hardly occurs, possibly because innovators want to keep the right to imitate their own invention, enabling them to exploit the goodwill of their trademark.

From the late 1970s there are some remarkable examples of *price differentials between drugs sold by the innovator and imitators* that continued to exist for a number of years on the German market. When the patent protection for the tranquilizer Valium produced by Roche expired in 1978, various imitators were allowed to enter this market, but for a series of years failed to attain considerable share of the market—although their prices were about 70 percent below that of the original product. From 1976 sales of Valium decreased, but sales of Lexotanil, a successor with improvements in several dimensions of quality (also produced by Roche) increased at the same time, suggesting successful price maintenance in the face of competition in a closed market [see Oberender (1984), p. 276].

In his study of prices of NCEs launched in the United States between 1958 and 1975, Reekie (1978) paints quite a different picture. In the first year after launch, NCEs cost an average 62 percent more than the drugs of the same therapeutic category that had already been on the market for some time (see Table 10.2, Panel A). Four years after launch, the difference had dropped to 29 percent. Of particular importance is the fact that the variance of relative prices charged for NCEs decreased significantly over time. Within a few years, the innovation obviously must fit into the price band established by existing substances.

Table 10.2: Prices of NCEs Relative to Drugs on the Market

	1st year	2nd year after market launch	3rd year	4th year
A. United States, 1958–1975				
Maximum	15.516	11.969	5.128	5.449
Mean	1.618	1.519	1.345	1.287
Coefficient of variation	1.169	0.936	0.626	0.549
Number of observations	105	176	163	146
Ratio of variances				
2nd year/1st year		0.565		
3rd year/1st year			0.198	
4th year/1st year				0.140
B. Netherlands, 1970–1977				
Maximum	24.873	27.190	11.732	12.559
Mean	2.084	1.785	1.544	1.61
Coefficient of variation	1.764	1.641	0.994	1.082
Number of observations	94	100	98	87
Ratio of variances				
2nd year/1st year		0.797		
3rd year/1st year			0.174	
4th year/1st year				0.224

Source: Reekie (1978, 1981).

In a second study, Reekie (1981) studied the development of relative prices charged for NCEs in the Dutch market between 1970 and 1977. In the first year after launch, NCEs cost on average 108 percent more than existing competing products, and still 61 percent after four years (see Table 10.2, panel B). Therefore, there was again a drop in relative price. But the initial premium on NCEs was higher, and the process of *price adjustment took much longer* than in the United States. Finally the variance of relative prices dropped as well, even though after four years the most expensive NCE was still twelve times more expensive than the already existing preparations. A likely explanation of the differences between the United States and the Netherlands can be found in the respective design of health insurance. In the United States, the insured either pay for pharmaceuticals themselves or at least face a substantial amount of *cost sharing*, whereas in the Netherlands pharmaceuticals are part of the benefit package for the roughly 70 percent of the population who have statutory health insurance.

In the case of the U.S. market, the effect of competition on the pricing of NCEs is known in more detail, thanks to a recent study by Lu and Comanor (1994). They first seek to explain the introductory prices of NCEs launched between 1978 and 1987 and then trace the development of real NCE prices over a period of eight years. In view of

the distinction between breakthrough and “me too” innovations introduced in Section 10.2, an NCE’s rating by the U.S. Food and Drug Administration (FDA) is of particular interest. New drugs providing important therapeutic gains (rated 1A) should be more likely to be of the breakthrough type than those providing a modest but real advantage over existing substitutes (rated 1B), while those providing little or no therapeutic gains (rated 1C) may correspond to the “me too” category.

Table 10.3: Determinants of Relative Introductory Prices in the United States, 1978–1987

Explanatory variables	Dependent variables ^a		
	lnRELP ^b	lnRATIO (t) ^c	
	t = 0 (1)	t = 4 (2)	t = 8 (3)
Constant	0.250 (1.94)	0.262** (8.56)	0.558** (9.63)
RATE1A (= 1 if NCE receives FDA rating 1A)	0.439* (2.03)	−0.303** (4.99)	−0.464** (−3.88)
RATE1B (= 1 if NCE receives FDA rating 1B)	0.228 (1.80)	−0.072 (1.81)	−0.141* (−1.96)
ACUTE (= 1 if NCE for acute illness)	0.365** (2.97)	−0.053 (−1.34)	−0.162* (−2.16)
lnNUM (log of number of existing substitutes)	−0.288** (−3.34)	−	−
NEW4 (number of competing NCEs during the following 4 years)	−	−0.055** (−3.85)	−0.053** (−2.93)
RATE1B* lnRELP (interaction variable; RATE1B = 1 if NCE receives FDA rating 1B)	−	−0.096* (−2.55)	−
RATE1C* lnRELP (interaction variable; RATE1C = 1 if NCE receives FDA rating 1C)	−	−0.132** (−2.68)	−0.195* (−2.16)
R ²	0.23	0.26	0.25
N	131	115	86

Notes:
^a OLS estimation; *t* ratios in parentheses; * (**): statistically significant at the 0.05 (0.01) level (two-tailed test).
^b RELP = Price of NCE relative to existing substitutes in the year of introduction.
^c RATIO (*t*) = Real price of NCE (deflated by Consumer Price Index) in year *t* relative to *t* = 0.
Source: Lu and Comanor (1994).

Turning first to the introductory price of NCEs relative to existing substitutes (in natural logarithms, see dependent variable $\ln\text{RELP}$ in column 1 of Table 10.3), the following results merit attention.

- ***Breakthrough innovations are launched at a higher price.*** According to the coefficient of RATE1A , NCEs qualified as important therapeutic gains by the FDA were priced about 55 percent higher ($e^{0.439} - e^0 = 0.55$) than those pertaining to the “me too” (1C) category, *ceteris paribus*. NCEs rated 1B by the FDA were priced about 26 percent higher than those rated 1C, although the coefficient of the RATE1B variable does not quite reach statistical significance, judged by conventional criteria. This evidence is in accord with Conclusion 10.2, which states that a breakthrough innovation has a low price elasticity of demand at least locally, a fact that will permit a monopolistic supplier to set a high price [see Eq. (10.2) again].
- ***Competition from existing substitutes is effective.*** The coefficient of $\ln\text{NUM}$, the logarithm of the number of existing substitutes at the time of market launch, suggests that a doubling of this number (i.e., a 100 percent increase) causes the price of the NCE to be 29 percent lower than otherwise.
- ***Fear of adverse drug reactions matters.*** New drugs for the treatment of acute illness ($\text{ACUTE} = 1$ in Table 10.3) command a particularly high price. This may well reflect physician concern about adverse drug reactions (which are part of a drug’s consumption technology; see Section 10.2). Adverse drug reactions weigh heavily in the evaluation of a drug used for treatment of a chronic condition. In the case of an acute illness, however, the drug’s beneficial principal effect is of overriding importance, causing the physician to switch easily to a new alternative promising more effective therapy.

Evidence with regard to the *development of NCE prices over time* is presented in columns 2 and 3 of Table 10.3. Here, the dependent variable is $\ln\text{RATIO}(t)$, the log of the price of the NCE (relative to all other goods and services represented in the Consumer Price Index), measured t years after market launch and compared to its initial price. For example, $\text{RATIO}(4) = 1.30$ indicates that the real price of an NCE has increased by 30 percent over the course of four years from its introduction. Judging from the constant, this would indeed have been the average price increase ($e^{0.262} = 1.30$) had there not been the following mitigating factors.

- ***Lower price increase for breakthrough NCEs.*** The negative coefficients of RATE1A indicate that prices of major innovations rose at a slower pace than those of “me too” drugs. To the extent that prices of “me too” preparations developed similarly to all existing substitutes, this implies that the price of a breakthrough innovation must sooner or later fit into a common price band as well, corroborating the findings of Reekie (1978, 1981). Price convergence is enhanced by the fact that “me too” products that were priced especially low at market launch tend to catch up later on. To see this, note that the interaction variable $\text{RATE1C} \cdot \ln\text{RELP}$ takes on a negative value if the NCE was rated 1C and if its relative price was below one; this negative value is then transformed into a positive impact through its negative regression coefficient.

- *Skimming versus penetration price strategies.* Breakthrough innovations and drugs for acute treatment share the same sign pattern: positive in column 1 and negative in columns 2 and 3 of Table 10.3. There is a common explanation in that gains due to substitution from existing drugs are likely to be large at the time of introduction, resulting in strong instantaneous demand but weak demand growth in subsequent periods. A monopolistic supplier of an NCE of this type will therefore set a high initial price, skimming off early willingness to pay. Conversely, the supplier of a “me too” product must boost initial demand through a low price at first, giving physicians and (chronic) patients time to make favorable experiences, in particular with regard to adverse drug reactions. When demand picks up, he may increase price later on.
- *Increasing effect of competition on price.* While the number of existing substitutes is relevant for pricing at the time of market launch, the number of competing innovations subsequently appearing on the market should be decisive when it comes to price adjustment over time. This number (NEW4) is measured with reference to the first four years after market launch. Being zero in many cases, NEW4 is not in logarithmic form. Its regression coefficient suggests that a doubling of the number of rivals causes the price after four years to be some 6 percent lower than otherwise.⁴

CONCLUSION 10.6: Pharmaceutical innovations (and breakthrough innovations in particular) are launched at prices which may be many times higher than the average price of competing drugs already on the market. In the course of a few years, however, competitive pressure causes their relative price to decrease.

10.5.2 The Effect of Maximum Allowable Prices for Drugs: A Case Study

In the Federal Republic of Germany, members of statutory health insurance (i.e., around 90 percent of all patients) enjoyed virtually complete insurance coverage until 1989. They paid a charge of 3 DM (some 2 US\$ at 1988 exchange rates) per prescription, regardless of the price of the drug. Beginning in 1989, maximum allowable prices (MAP) were imposed on categories of drugs deemed “therapeutically equivalent,” some differences of consumption technology as defined in Section 10.2.1 notwithstanding. In the following, the effects of this regulatory change are studied building on the theory of Section 10.2. By assumption, the patient is fully informed, with no influence exerted by his physician.

4. To calculate the elasticity of RATIO with respect to NEW4, note that

$$\frac{\partial \ln \text{RATIO}}{\partial \ln \text{NEW4}} = \frac{\partial \ln \text{RATIO}}{\partial \text{NEW4}} \frac{\partial \text{NEW4}}{\partial \ln \text{NEW4}} = \frac{\partial \ln \text{RATIO}}{\partial \text{NEW4}} \text{NEW4}$$

Since $\partial \ln \text{RATIO} / \partial \text{NEW4} = -0.055$ from column 2 of Table 10.3, while NEW4 has a mean value of approximately 1.1, the elasticity (evaluated at the mean of NEW4) amounts to about -0.06 , the figure used in the text.

Thus consider the market diagram of Figure 10–10 for a brand-name drug. Prior to 1989, observed market demand coincides with true demand $\varphi^m M^m$ only for prices below the deductible of 3 DM, which is shown by OD . For prices in excess of D , observed demand was totally price inelastic due to full coverage, resulting in the kinked demand function $M_m AB$. In this situation, a monopolist charging an ever higher price would not meet a sanction from the market in the guise of a reduction of quantity demanded.

Therefore, price could theoretically be set at an *infinite value*. This, however, would serve to destroy the existing system of statutory health insurance, causing it to be replaced by a more strongly regulated one incorporating price controls for drugs. To fix ideas, let π symbolize the probability of a maximum allowable price Φ^a being introduced, with π depending positively on the price Φ of the drug currently charged. Moreover, assume marginal cost C' of production to be constant and equal to average cost. Finally, in view of the low deductible on drugs, neglect the difference between observed quantities demanded along $M^m AB$ and the fixed satiation quantity M^m (the difference being drawn large in Figure 10–10 for purposes of exposition). Expected profit is then given by

$$E\Pi = \pi(\Phi)(\Phi^a - C') M^m + \{1 - \pi(\Phi)\} (\Phi - C') M^m, \quad \text{with } \partial\pi/\partial\Phi > 0 \quad (10.12)$$

The necessary condition for optimal price Φ^* is

$$\frac{\partial E\Pi}{\partial\Phi} = \frac{\partial\pi}{\partial\Phi} (\Phi^a - C') M^m - \frac{\partial\pi}{\partial\Phi} (\Phi^* - C') M^m + \{1 - \pi\} M^m = 0 \quad (10.13)$$

Dividing through by M^m and solving for Φ^* yields

$$\Phi^* = C' + (\Phi^a - C') + \frac{1 - \pi}{\partial\pi/\partial\Phi} = \Phi^a + \frac{1 - \pi}{\partial\pi/\partial\Phi} \quad (10.14)$$

This can be rewritten by introducing $e(\pi, \Phi) := (\partial\pi/\partial\Phi)(\Phi/\pi) > 0$, an elasticity that indicates how much an increase of price by 1 percent *raises the likelihood of price regulation being imposed* (in percent). Use of this elasticity results in

$$\Phi^* = \Phi^a + \frac{1}{e(\pi, \Phi)} \Phi^* \left(\frac{1 - \pi}{\pi} \right)$$

which may once more be solved for Φ^* ,

$$\Phi^* = \frac{\Phi^a}{1 - \frac{1}{e(\pi, \Phi)} \frac{1 - \pi}{\pi}} \quad (10.15)$$

Thus prior to 1989 the price of a brand-name drug was likely to reflect the following factors⁵

- The *maximum allowable price* Φ^a that could reasonably be expected under an alternative system of drug prices fixed by administrative fiat;
- The *security of the status quo*. The more the odds are in favor of the status quo ($\pi \rightarrow 0$), the smaller the denominator of Eq. (10.15), and hence the higher the optimal price Φ^* . Conversely, Φ^* approaches the expected allowable price Φ^a if $\pi \rightarrow 1$, that is, if the change of regulatory regime is almost certain;
- The *importance of the firm's pricing policy*. The smaller $e(\pi, \Phi)$, the smaller the value of the denominator of Eq. (10.15), resulting in a high value of Φ^* . Therefore, if the supplier believes it to have a negligible impact on policy (e.g., because his share in the country's drug bill is small), he will set the price of his product very high.

These considerations help to explain why the level of pharmaceutical prices was high (but finite) prior to the reform of 1989. In addition, the several rounds of "voluntary price restraint" by the German pharmaceutical industry during the 1980s may be interpreted as a reaction to a perceived increase in π , the probability of a system change.

The same type of argument can be used to explain why *generics* did make some inroads into the German market in spite of the lack of price elasticity on the part of fully insured patients. Physicians face a certain risk of being audited by their regional association, with the possible sanction of a cap on the volume of services remunerated. Denote this probability by $\pi^P(\Phi^P M)$, with $\partial \pi / \partial (\Phi^P M) > 0$ to reflect the fact that such a sanction becomes more likely when the value of drug prescriptions ($\Phi^P M$) is high. By shifting to generics, physicians are able to lower price Φ^P while keeping total drug consumption per case M constant, to the satisfaction of their patients. In this way, they may avoid being audited.

On 1 January 1989, the law on health care reform was put into force in Germany, authorizing sick funds and medical associations jointly to fix a maximum allowable price (MAP) for each category of "therapeutically equivalent" drugs to be reimbursed by statutory health insurers. If the price of the drug exceeds the MAP, the insured has to cover the gap himself. By 1990, 43 drugs with a total turnover of 4.3 billion DM had been subjected to the MAP scheme, with the remainder to follow in the years to come.

The effect of this regulation on price and consumption may be predicted in the case of a product having a *small volume of sales*, using Figure 10-10. Let the maximum allowable price be Φ^a , causing the true demand function $\varphi^m M^m$ to be replaced by the observed function $M^m E \Phi^m$ as true willingness to pay is augmented by the amount of the MAP; that is, the amount the sick fund contributes to the cost of the drug. Since marginal revenue R' coincides with the portion $M^m E$ of the observed demand function,

5. The argument proceeds on the assumption that the denominator in Eq. (10.15) is positive, which requires $e(\pi, \Phi) > (1 - \pi)/\pi$. Hence, the pricing policy of the pharmaceutical firm considered must have a nonnegligible impact on the probability π of price regulation being introduced.

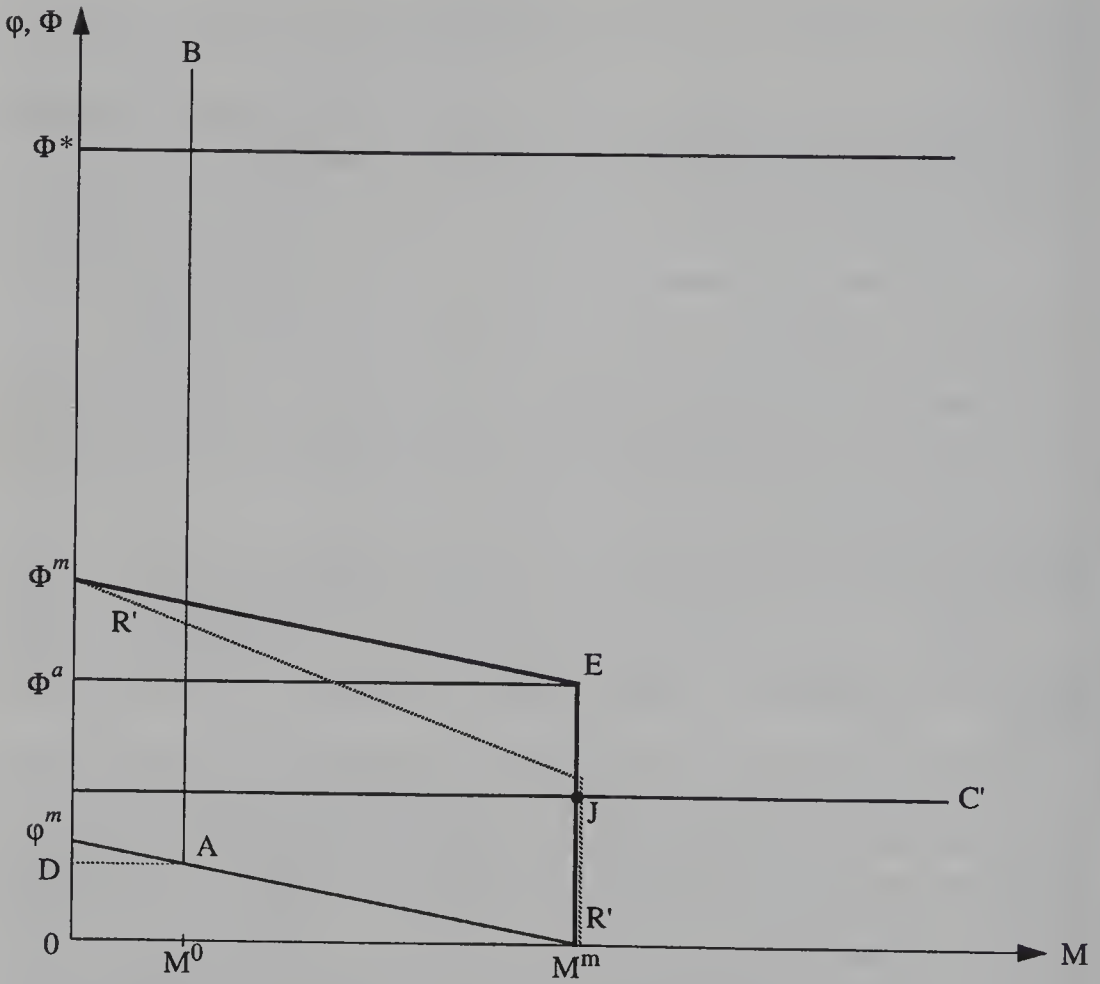


Figure 10-10. Effect of introducing a maximum allowable price (small sales).

profit maximization calls for equalizing marginal cost and marginal revenue at point *J*, which implies realization of point *E* on the observed demand function. Therefore, it is optimal for the supplier to accept the maximum allowable price Φ^a in return for an increase of sales from M^0 to M^m in this case.

However, lowering the current price of the drug to the limit price may not be optimal for a product achieving a large volume of sales. In Figure 10–11, let true maximum willingness to pay φ^m be large relative to MAP and let true willingness to pay decrease slowly with quantity along $\varphi^m M^m$, say. With marginal cost C' as low as before, the drug has large sales prior to the introduction of the new scheme, such as M^0 . Compared with Figure 10–10, the observed demand function $M^m E \Phi^m$ is much less price elastic here, specifically at prices in excess of the MAP. Therefore, marginal revenue drops faster with increasing quantity, resulting in equality with marginal cost C' at point *K*. Now it is optimal to charge price Φ^{**} , making consumers pay the difference $(\Phi^{**} - \Phi^a)$ out of their pocket. This time, drug consumption is predicted to decline, from M^0 to M^{**} .

As an example of the first case (small sales), consider Trental by Roussel Hoechst, a preparation to enhance circulation. Due to generics, the market share of this drug had

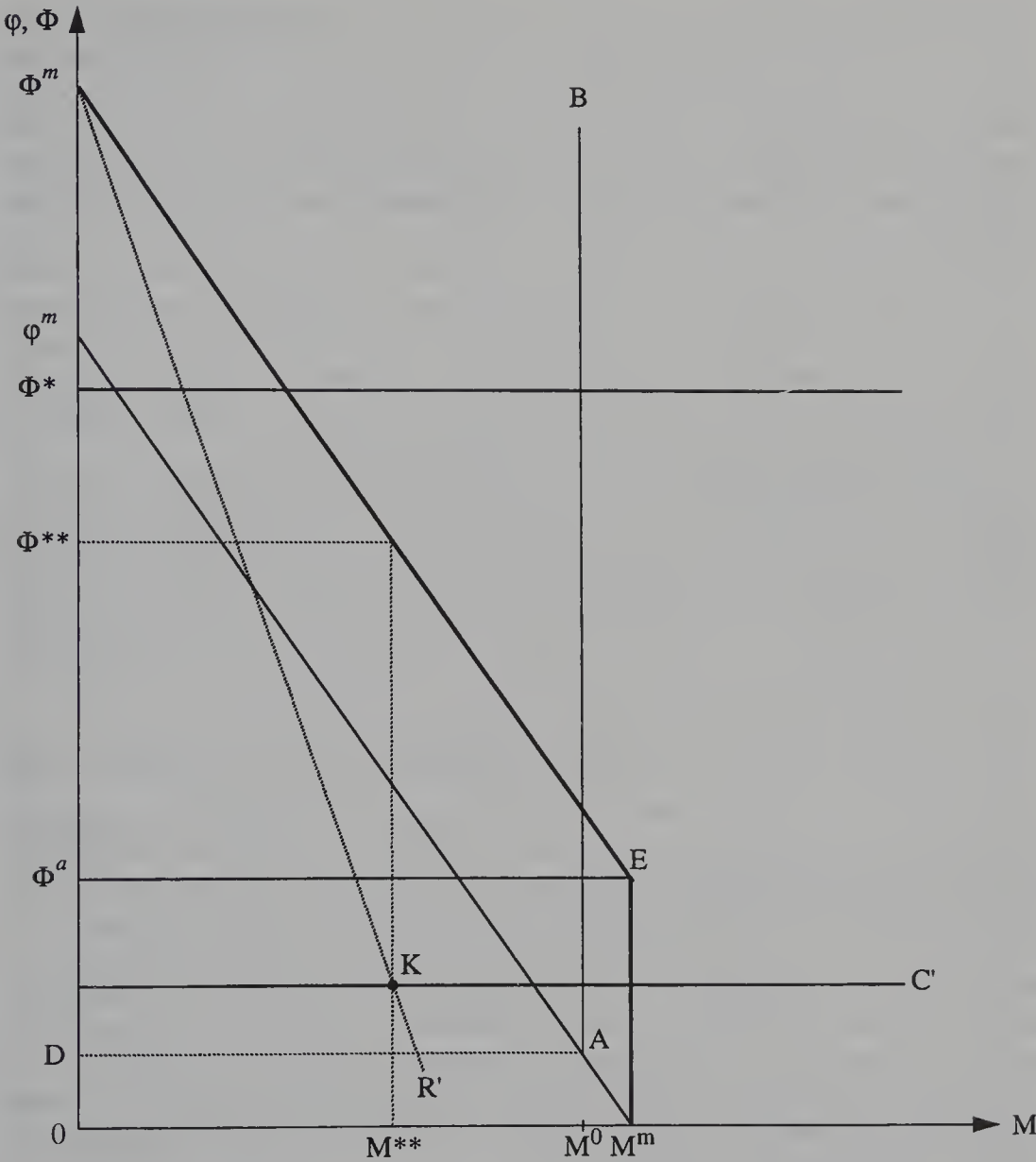


Figure 10-11. Effect of introducing a maximum allowable price (large sales).

already dropped to 40 percent when the MAP was set at 63.69 DM (some US\$ 35 at 1990 exchange rates). Thus Trental could not be considered to have large sales any longer; moreover, price elasticity of true demand clearly was nonnegligible. As predicted on the basis of Figure 10–10, the *price of Trental was lowered* by Roussel Hoechst to coincide with the MAP. The market share of the original product rose back to 45 percent during the following months, a development that corresponds to the increase from M^0 to M^m in Figure 10–10 [see Litsch, Reichelt, and Selke (1990), pp. 206ff].

The fate of Adalat may serve as an example of the *second case* (large sales). Bayer is the original manufacturer of this calcium antagonist, which is used for the treatment

of cardiovascular disease. Up to September 1989, it had been sold for 110.55 DM (about US\$ 60 at 1989 exchange rates) and held a market share of 25 percent, although there were 14 generics on the market, costing up to 60 percent less than the original drug [see Litsch, Reichelt, and Selke (1990), pp. 126ff], evidence of a rather low price elasticity of true demand. When the maximum allowable price was set at 66.95 DM, Bayer lowered the price of Adalat to 75.75 DM. Thus patients had to pay 8.80 DM out of pocket.

While in September 1989, the market share volume of Adalat still amounted to 14 percent, it fell to 10 percent in the following months, corresponding to the movement from M^o to M^{**} in Figure 10–11. It is interesting to note that the suppliers of generics failed to pick up market share, although their prices constantly remained below the maximum allowable price. The main beneficiary was a seller of parallel imports of Adalat, whose price still made consumers pay an extra 5 DM.

Another example for the second pricing policy is Lexotanil (already mentioned in Section 10.5.1), a drug for anxiety and stress relief. As with many other psychotropic drugs, it can lead to addiction, which should result in a *decreased price elasticity of demand* (in absolute value). Before the introduction of the MAP of 19 DM (about US\$ 10 at 1989 exchange rates), Lexotanil (supplied by Roche) had a market share between 67 and 78 percent; thus it had large sales. When the MAP was introduced in September 1989 for all drugs that were chemically equivalent to Lexotanil, Roche kept its price at the previous level, implying a copayment of 7.55 DM by patients. This was probably a major reason why the market share of Lexotanil dropped to 40 percent before the end of 1989 [see Litsch, Reichelt, and Selke (1990), pp. 52ff]. The main beneficiary was supplier No. 2 in the market, who increased his share from 15 percent to 32 percent. The price of this substitute was 1 DM below the MAP, whereas other competitors fixed their prices even 4 DM below MAP. Nevertheless, the reaction on the demand side, with approximately –45 percent (from an average of 72 percent to 40 percent of a roughly constant market volume), was moderate, since the net price could actually be reduced to zero by switching over to No. 2 and other competitors, a reduction of 100 percent. The estimated low implied value of the price elasticity ($\eta = -0.45$) is in accordance with the steep slope of the demand curve shown in Figure 10–11. Thus Roche's policy of maintaining a price in excess of MAP for Lexotanil is in accordance with the particular properties of this drug.

CONCLUSION 10.7: As soon as price differences between drugs result in cost differences to the insured, a considerable price elasticity of demand is observed even on the insured German market. Suppliers reacted to the introduction of a maximum allowable price (MAP) scheme in a predictable fashion, accepting the MAP for drugs with large (absolute) price elasticity of demand and small sales while requiring copayment for drugs having small price elasticity and large sales.

The simple analysis performed in this section cannot explain, however, why a few suppliers sell their products *below the limit price*. Probably, they cater to physician interests. As used to be the case with generics, physicians can now substantiate their quest for cost effectiveness by prescribing drugs priced below the MAP.

10.6 EXERCISES

Exercise 10.1:

It has been demonstrated that the consumer surplus of a pharmaceutical innovation is overestimated due to insurance coverage. (a) Explain the logic of the argument without referring to Figure 10–4. (b) Referring to Figure 10–4 now, are there additional elements that may be affected by insurance coverage, and what is their impact on social benefit?

Exercise 10.2:

In contrast to what has been assumed in Section 10.4.3, the Patent Office could remain passive, like the innovator, reacting to innovative efforts by altering the length of patent protection. Try to derive a reaction function $h(I)$ for the regulatory agency. Note that $P_T + S_T < 0$ for all values of T . Determine the so-called Cournot equilibrium between the two reaction functions $g(T)$ and $h(I)$. What are the effects of an increase in the marginal cost of the innovation in this model?

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11 FUTURE CHALLENGES TO HEALTH CARE SYSTEMS

11.1 THE ISSUE

Actors on markets are under continuous pressure to adjust. Consumers' changes in taste lead to changes in demand, new technologies provide rivals with competitive advantages, and public authorities step in to regulate business. In a market economy, this pressure to adjust is transmitted by price signals, indicating to firms the need to adapt the marketing and production of their goods and services. In health care, however, fluctuating market prices for medical services are out of the question because they are incompatible with the key principal-agent relationship between the patient and the service provider (see Chapter 8). One possibility of avoiding fluctuating prices is bargaining over fee schedules, which paves the way for the important role of professional associations and public authorities in health care.

This departure from allocation through price has the adverse effect of reducing the system's *speed of adjustment*, however. For example, structural adjustment of a fee schedule usually takes years. On the one hand, this sluggishness prevents physicians, dentists, and hospitals from initiating and shortly thereafter terminating contractual agreements with insurers because of temporary advantages, which would be to the detriment of many patients. On the other hand, though, it does result in considerable delays in adjustment to exogenous shocks. The ensuing disequilibria are perceived as challenges to health care systems. Towards the end of this century, such challenges have emerged in four areas. In the following, they appear in the order of their popularity in political debates rather than their degree of long-term relevance.

1. The demographic challenge. At first sight, this is simply to say that more and more people are getting older. In 1980, the share of people aged 75 or over in the United States was about 5 percent and is predicted to increase to 10 percent by the year 2025, a figure that will be even exceeded by Japan with a share of 10.5 percent [see

Heller et al. (1986)]. Old age is associated with an increased demand for medical and in particular nursing services, and the issue is how to meet this demand.

On closer inspection, however, another demographic change of considerable significance emerges. Within just twenty years, the number of *single person units* has greatly increased, for example, from 13 to 23 percent in the United States [see Roussel (1986)]. Persons living alone are much less able, in the case of illness, to fall back on support and care from relatives, causing them to demand more health care services.

2. The technological challenge. In the health care systems of industrial countries “. . . an explosion of the feasible is taking place. In 1980 alone the magazine Newsweek presented the following medical innovations. A new piece of equipment of a significance comparable to the CT scanner which makes brain-waves visible, revolutionary surgical methods for eliminating shortsightedness and infertility in women, new drugs for jaundice, sexually transmitted diseases and gout, various new cancer treatments, an operation for safe implantation of artificial breast following breast removal in females, new life-saving techniques in child heart surgery, and a new type of electric shock treatment to regenerate muscle and nerve tissue. Through this, one day even paraplegics will be able to walk.” [Krämer (1982), p. 37/38].

Almost all these innovations are *product innovations*, that is, they save lives or contribute to an improved quality of life, although at (much) higher cost. “When Christiaan Barnard transplanted the first human heart on 3rd December 1967, at that very moment the cost of such a treatment rose from zero to DM 200,000” [Krämer (1982), p. 92]. Conversely, *process innovations* that enable a particular service to be produced at lower cost are rare. There seem to be even fewer *organizational innovations* in health care, which promise cost savings through a rebundling of production processes resulting in economies of scope. Thus technological change in medicine threatens to become the driving force of future “cost explosions.”

3. The challenge of the so-called Sisyphus syndrome. The successes of modern medicine remind one of Sisyphus, the hero of Greek mythology who was condemned to roll a lump of rock up a mountain, only to see it slip out of his grasp just before reaching the summit, forcing him to start all over again. Inasmuch as technological change in medicine lengthens human life, it may also increase the number of those who make more than average demands on the health care system. Because of their increased political clout, the elderly might exert their influence in public spending, especially in favor of costly medical innovation in health care. As a result of this process, the successes of medicine may turn into a growing burden on policy makers who would like to retain their freedom of choice in allocating the public budget.

4. The challenge of international competition. This often overlooked challenge to health care has its origin in the increasing economic integration of nations. In the future, when labor will be able to move about freely within the European Union (and some day within the North American Free Trade Area), workers will be attracted not only by attainable income but also by things such as the relative performance of a country's health care system. Physicians and medical staff will be able to move about more freely, too, and finally, international *direct investment* into private health insurance and hospitals will become more prominent. Altogether, national health care systems will change from being a highly protected to being an only partially protected sector.

These four challenges will be discussed in the following sections. Confronting them in ways that enhance efficiency, while granting access to health care to everyone, constitutes a metachallenge that is the topic of the concluding Chapter 12.

11.2 THE DEMOGRAPHIC CHALLENGE

11.2.1 Aging of the Population

11.2.1.1 Improved Control over the State of Health

As expounded in Chapter 4, life expectancy in industrialized countries is still increasing, albeit at a markedly slower rate during the past few years (see Table 4.1). This slowdown is often interpreted as evidence proving that modern medicine has to fight against *decreasing marginal returns*. *Ceteris paribus*, marginal willingness to pay for an input factor having decreasing marginal returns should fall off particularly quickly with more intensive use. However, in countries where individual willingness to pay for medical services tends to prevail over political considerations of cost control (for instance, the U.S., the Netherlands, or Switzerland), the share of medical expenditure in the national product shows no tendency to fall. In order to resolve this contradiction, many observers take refuge in the hypothesis of supplier-induced demand (see Section 7.2).

An alternative explanation for medical services being consistently in high demand might be *people's aversion to risk in relation to their health*. For a risk-averse individual, the achievements of medicine possibly consist not so much in an increase in life expectancy or mean number of QALYs (see Section 2.4) as in a reduced volatility of these quantities. From this point of view, medicine has the advantage of directly lowering the risk associated with health status rather than lowering the wealth risk emanating from bad health, as does health insurance. This risk reduction is worth a premium to individuals who in the extreme may strive to attain complete control of their health status. Indeed, the ideal of Western man seems to be to remain perfectly healthy far into old age and then to drop dead, as it were.

If a whole population were successfully to gain control over its health, then a *rectangularization of the survival curve* would have to be observed at the aggregate level. This term conveys the notion that in the course of time, successive cohorts lose fewer and fewer of their members through premature death. Rather, deaths cluster around a biologically determined life span [see Fries (1980), Fries and Crapo (1981)]. Thus the curve indicating the number of survivors of a cohort as a function of their age (the survival curve for short) should take on a more *rectangular shape* over time, due to the increasing degree of control over health status. This also holds true of the aggregate survival curve, derived from observed deaths during a period of ten years, say. Although composed of several cohorts, population should move towards the rectangular ideal, increasingly reflecting the more recent cohorts.

In actual fact, survival curves of Western countries have successively come closer to the rectangular ideal (see Figure 11-1). While deaths during the 1890s in England and Wales seem to characterize a cohort that lost one-half of its members by the age of 55, deaths of the years 1965-1977 call to mind a cohort that hardly suffers losses up to the age of 50.

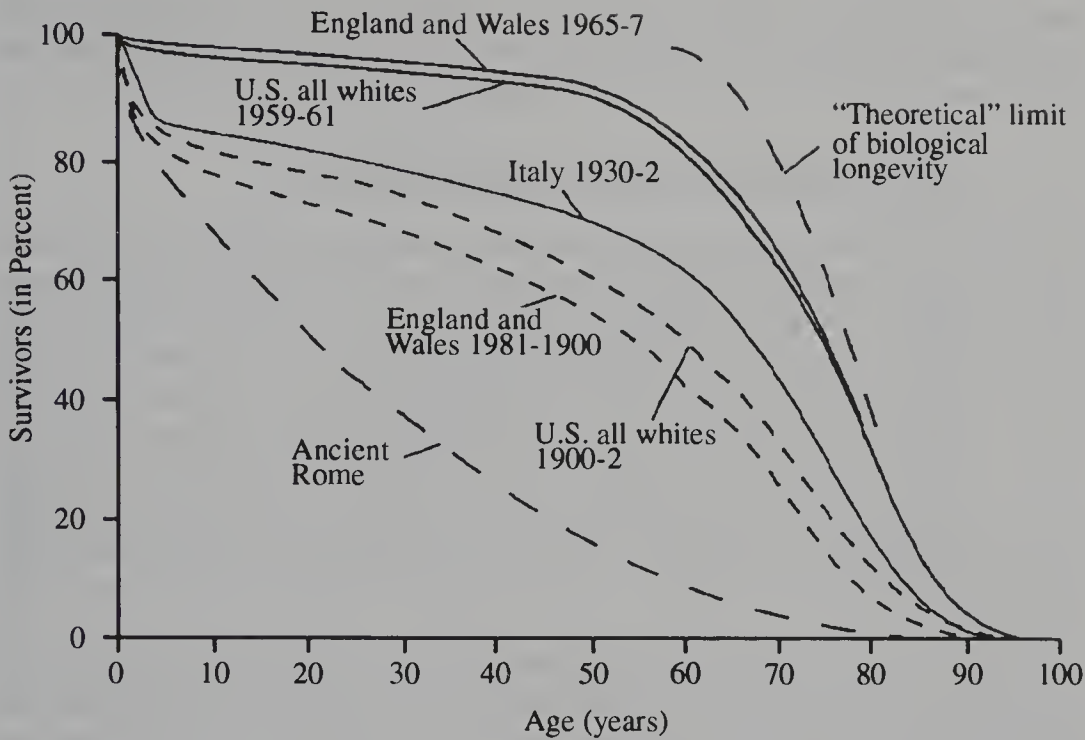


Figure 11-1. Rectangularization of survival curves. *Source:* Walford (1981).

CONCLUSION 11.1: The progressive rectangularization of the survival curve may be interpreted as the result of efforts to gain better control over health status. To the extent that this control is due to modern medicine, the unabated willingness to pay for medical services can be explained with reference to people’s risk aversion.

The question of whether modern medicine actually has contributed to the rectangularization of the survival curve is examined more closely in Section 11.4. At this point, however, a phenomenon related to Conclusion 11.1 merits mention. In fact, an important portion of medical expenditure seems to benefit individuals whose remaining life expectancy amounts to one year or less. Lubitz and Prihoda (1984) discovered that in the United States approximately one-half of a year’s Medicare expenditure was spent on persons who, one year later, were no longer alive. However, this figure overstates the importance of the “cost of dying” in the health care system as a whole.¹

Apparently, a great many people are reaping the benefits of health care, although their remaining life expectancy can be counted in months rather than years. If one were to measure the marginal productivity of this expenditure exclusively in terms of time spent in “good” or “better” health, respectively (as suggested, e.g., in the simple model of Section 3.5), one would be tempted to speak of a *waste of resources*, benefitting at best hospital physicians but not patients themselves [see Zweifel (1990a)]. On the other

1. It should be noted that the population covered by Medicare is 65 and older. In this population, the share of individuals being in their last year of life is far higher than average.

hand, Conclusion 11.1 reminds us that even rather aged patients might display considerable willingness to pay for medical services that, while failing to improve their health status, still enhance to their control over it.

11.2.1.2 The Redistributive Effects of Aging

Every health insurance system whose premiums are independent of age in spite of age-dependent health costs acquires features of a pay-as-you-go system of provision for old age and thereby “inherits” its problems as well [see Breyer (1990), especially Chapter 1]. As in provision for old age, the working generation pays contributions in excess of benefits received. This excess is used in social health insurance to balance the deficit resulting from the provision of care to children and pensioners (see Figure 11-2).

As soon as the number of insured pensioners increases, this deficit multiplies. Therefore, contributions paid by workers must be increased in order to safeguard the financial equilibrium of the scheme. Safeguarding the financial equilibrium in response to population aging thus implies an *intensified redistribution* from workers to pensioners.

Therefore, during a period of population aging, membership in a social health insurance scheme may turn into an unprofitable investment for an entire generation of workers who must finance the high cost of treatment of several pensioners but in their turn can consume the benefits of the scheme only once when in retirement age. The present value of their contributions exceeds the present value of the benefits they stand to receive in the course of their lifetime. Such a cohort of workers pays

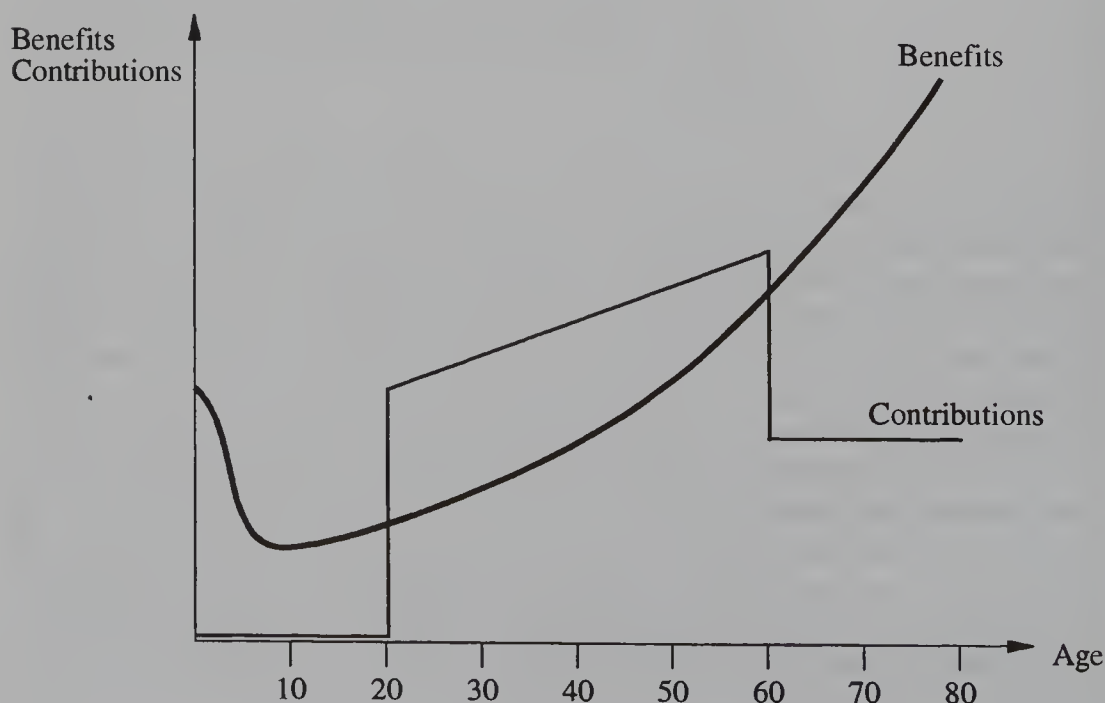


Figure 11-2. Children, workers, and pensioners in social health insurance. Source: Schulenburg (1989), p. 286.

Table 11.1: Average Net Lifetime Transfers to German Statutory Health Insurance, in Constant US\$ of 1980

<i>Cohort</i>	<i>Variant I (pure demographic effect)</i>	<i>Variant II (demographic effect and effect of cost inflation)</i>
1900	-5,886	119
1920	-6,012	-707
1940	-3,180	-1,463
1960	-2,201	199
1980	8,088	8,082
2000	8,350	18,857
2020	10,673	54,200
2040	7,323	83,760
2060	2,965	76,653
2080	299	18,711
2100	0	0

Note: In variant I, constant health care expenditure is assumed for the whole period of observation. In variant II it is assumed that the health care expenditure rises 1 percent p.a. until 1960, 10 percent p.a. from 1960 to 1990, 5 percent p.a. from 1990 on (due to increased efforts at cost control). Discounting to present value occurs at 2 percent p.a. until 1960 and at the rate of cost increase from 1960 on. Net transfers on a life cycle approach zero due to constant assumed rate of birth from the year 2030 on.

Source: Schulenburg (1989, p. 296).

a net lifetime transfer to survivors of former cohorts. Table 11.1 gives an impression of the estimated magnitude of such intergenerational transfers in German statutory health insurance. In variant I, which takes into account demographic effects only, the cohort born in 1980 will be the first one since 1900 to pay a net transfer into the scheme, amounting to some 8,000 US\$ in present value (positive entry in Table 11.1). If the effect of past and future cost inflation in health care is taken into account as well (variant II of Table 11.1), the cohort of 1960 is already about to pay a net lifetime transfer. These involuntary transfers will increase considerably in the future, reaching a point where acceptance of social health insurance in its present-day form could be seriously undermined.

CONCLUSION 11.2: The aging of the population jeopardizes the financial equilibrium of social health insurance. The adjustment of contributions for safeguarding equilibrium may turn membership into an unprofitable investment for present and future generations of workers.

11.2.2 Changes in Household Composition

In industrialized countries, household composition has been changing at least as much as the population’s age structure is about to. In the early 1960s, single person units were

Table 11.2: Share of Single Person Households in Selected Industrialized Countries (in Percent)

	<i>Around 1960</i>	<i>Around 1980</i>	<i>Around 1990</i>
Germany	20	30	33 ¹
France	20	24	28 ¹
United Kingdom	11	22	27 ¹
Netherlands	12	22	29 ¹
Sweden	20	33	36 ²
United States	13	23	25 ³

Sources: Roussel (1986); for 1990: (1) Eurostat, 1993, (2) Statistical Abstract of Sweden, 1991, (3) Demographic Yearbook 1990, UN, New York, 1992.

the exception, while they may make up a third of all households today in some Western countries. With a share of 25 percent, the United States is lagging behind in this development towards a “singles society” (see Table 11.2). The growing share of single person units can be traced to two trends. On the one hand, marriage rates have tended to decrease, combined with an increase in the divorce rate that was not balanced by an increase of remarriages. On the other hand, life expectancy in industrialized countries has been developing differently between the sexes. Around 1950, an American woman aged 40 years could expect to live another 35 years, her husband of the same age, another 32 years. However, around 1980, remaining life expectancy at age 40 amounted to 39 years for a woman and 33 years for a man [see OECD (1987), p. 39]. The probability of having to spend the last years of one’s life alone has obviously increased considerably for women.

The relevance of single person units to the health care system may be illustrated by a simple probability argument. Let individual *A* fall ill with probability $\pi(A)$. If *A* lives alone, $\pi(A)$ also denotes his probability of demanding health care services. On the other hand, if *A* lives together with another person *B*, the healthy partner usually takes care of the sick one. In a two-person unit, claims on health care services are therefore only made if *both members* of the household are sick. Using the formula for conditional probabilities, one may write the probability of a two-person unit exerting a demand for health care services as follows:

$$\pi(A, B) = \pi(A|B)\pi(B) = \pi(B|A)\pi(A) \tag{11.1}$$

where $\pi(A, B)$ is the probability of *A* and *B* being ill simultaneously (and hence demanding health care services), and $\pi(A|B)$ is the probability of *A* being ill, given that *B* is ill. From this equation it immediately follows that $\pi(A, B) \leq \pi(A)$ and $\pi(A, B) \leq \pi(B)$, because the conditional probabilities $\pi(A|B)$ and $\pi(B|A)$ can be one at most. The probability of both members of a two-person household being ill at the same time therefore is *less than the risk of illness of a single person* in general. This means that living together in a family tends to relieve the health care system from some demands.

The extent of this relief effect is, according to Eq. (11.1), dependent on the conditional probabilities $\pi(A|B)$ and $\pi(B|A)$, respectively, which can be interpreted as the risk of infection. If these probabilities approach one, the sickness of one partner inevitably conditions the sickness of the other.

In a country afflicted by epidemics, living together in households is hardly an advantage for health care. However, since communicable diseases no longer pose a threat in today's industrialized countries, these effects are no longer decisive.

CONCLUSION 11.3: The spreading of single-person units increases the likelihood of a health loss translating into a demand for health care services while its contribution to the control of epidemics is marginal in today's industrialized countries.

11.3 THE TECHNOLOGICAL CHALLENGE

11.3.1 Three Types of Innovation

In the economic literature, the following types of innovation are distinguished.

- **Process innovation.** Process innovation permits the production of a given good or service at lower cost. As an example in health care, twice as many parameters of a blood test can be examined in the same time by means of automated serological analysis.
- **Product innovation.** Product innovation provides a good or the recipient of a service with new qualities or with a new combination of existing qualities. It usually entails increased production costs compared to existing alternatives. New drugs and therapies for hitherto incurable diseases are examples of product innovations in health care.
- **Organizational innovation.** The cost of producing a good or service can also be lowered by reorganizing production processes, including the restructuring of entire firms. The separation of geriatric care from internal medicine in hospital care, the creation of Health Maintenance Organizations and of group practices exemplify organizational innovations within the health care system.

The optimal allocation of these three types of innovations in health care from the individual's point of view may be modeled as follows. Through a sacrifice in terms of consumption during the current period, he has the opportunity to invest in innovations, whose payoff will be in terms of improved health status during the following period.

In order to keep the analysis as simple as possible, uncertainty about health status is neglected. Thus the individual is assumed to evaluate innovations based on a utility function that averages over the different states of health, some of which may entail very high marginal utilities for (specific) innovations in health care. Moreover, and contrary to Chapter 3, no distinction is made between consumption and consumer goods; thus the utility function has consumption expenditures (X_1 , X_2) of the two periods as its arguments. For the sake of simplicity, utilities (u_1 , u_2) will be additive and not

discounted to present value [see Lyttkens (1990) for an analogous macroeconomic formulation],

$$u = u_1(X_1, H_1) + u_2(X_2, H_2) \quad (11.2)$$

with (H_1, H_2) denoting health status, in period 1 and 2, respectively.

In the following section, the three types of innovations will be included into the production function for health step by step.

- **Process innovation only.** The expenditure on process innovation occurs in the first period; it is simply symbolized by R^C without a subscript for the period. In the second period, process innovation acts as if the quantity of medical services M_2 were augmented by a factor g^C . This multiplier takes on the value of one without investment in process innovation [$g^C(0) = 1$] and increases with R^C . Thus the production function of health is given by

$$H_2 = h_2(\tilde{m}_2), \quad \text{with } \tilde{m}_2 = g^C(R^C)M_2, \quad g^C[0] = 1, \quad \frac{\partial g^C}{\partial R^C} > 0 \quad (11.3)$$

Thanks to process innovation, a given health status can be achieved at a lower cost in the second period. It causes medical services to have the same effect as $g^C M_2$ units; so M_2 itself may take on a lower value. This means less expenditure on health in the second period, the price of M_2 being normalized to one [see the budget constraint (11.6)].

- **Process and product innovation combined.** A successful product innovation in health care enables individuals to improve their health status beyond the value $h_2(\tilde{m}_2)$ attained using existing medical technology. This improvement is again indicated by a multiplier $g^D \geq 1$ whose value depends on the expenditure on product innovation R^D during the first period. Thus Eq. (11.3) is extended to become

$$H_2 = g^D(R^D)h_2(\tilde{m}_2), \quad \text{with } g^D[0] = 1, \quad \frac{\partial g^D}{\partial R^D} > 0 \quad (11.4)$$

- **Process, product, and organizational innovation combined.** Economies of scope are typical for organizational innovations, which means that a reallocation of activities between two productive units increases their joint efficiency. Here, we assume economies of scope to exist *between the two periods* in order to keep the model as simple as possible.² According to this definition, organizational innovation enhances the productivity of medical services in both periods rather than only in the second period. Consequently, the production functions of both periods are modified by organizational innovation R^O as follows,

$$\begin{aligned} H_1 &= h_1(m_1), & \text{with } m_1 &= g^O(R^O)M_1 \\ H_2 &= g^D(R^D)h_2(m_2), & \text{with } m_2 &= g^O(R^O)\tilde{m}_2 = g^O(R^O)g^C(R^C)M_2 \end{aligned}$$

2. Thus, rather than having synergies result in an improved division of labor between "productive unit No. 1" (ambulatory care, say) and "productive unit No. 2" (hospital care) they are assumed to increase effectiveness of "period No. 1 medicine" as well as "period No. 2 medicine."

$$\text{and } g^O [0] = 1, \frac{\partial g^O}{\partial R^O} > 0 \quad (11.5)$$

Let the total income of both periods be $\bar{Y}$, to be spent on the purchase of consumer goods (X_1, X_2), medical services (M_1, M_2), and on three types of innovation (R^C, R^D, R^O). Therefore, the *budget constraint* reads

$$X_1 + M_1 + R^C + R^D + R^O + X_2 + M_2 = \bar{Y} \quad (11.6)$$

Eqs. (11.2)–(11.6) describe an optimization problem that can be examined using the Lagrange function below (see also Table 11.3),

$$\begin{aligned} L = & u_1\{X_1; h_1[g^O(R^O)M_1]\} \\ & + u_2\{X_2; g^D(R^D)h_2[g^O(R^O)g^C(R^C)M_2]\} \\ & - \lambda\{X_1 + M_1 + R^C + R^D + R^O + X_2 + M_2 - \bar{Y}\} \end{aligned} \quad (11.7)$$

In period 1, the individual faces the task of maximizing this function by his choice of X_1, M_1 , and above all his choice of R^C, R^D , and R^O .

11.3.2 Criteria for an Optimal Allocation of Innovative Effort

This section is devoted to the formulation of operational criteria for the optimal allocation of innovative effort from the point of view of an individual. These criteria amount to required marginal improvements of health status during the second period. First, by differentiating function (11.7) with respect to consumption expenditure X_1 , the value of the Lagrange multiplier λ can be determined,

$$\frac{\partial L}{\partial X_1} = \frac{\partial u_1}{\partial X_1} - \lambda = 0 \rightarrow \lambda = \frac{\partial u_1}{\partial X_1} \quad (11.8)$$

Next the optimal amount of expenditure on *process innovation* R^C has to satisfy the following condition,

$$\begin{aligned} \frac{\partial L}{\partial R^C} &= \frac{\partial u_2}{\partial H_2} \frac{\partial H_2}{\partial h_2} \frac{\partial h_2}{\partial m_2} \frac{\partial m_2}{\partial g^C} \frac{\partial g^C}{\partial R^C} - \lambda \\ &= \frac{\partial u_2}{\partial H_2} g^D \frac{\partial h_2}{\partial m_2} g^O M_2 \frac{\partial g^C}{\partial R^C} - \frac{\partial u_1}{\partial X_1} = 0 \end{aligned} \quad (11.9)$$

[see Eq. (11.5)].

Obviously, the marginal health improvement that can be achieved in the second period has to balance with the opportunity cost of process innovation. This opportunity cost in turn is determined by the utility, which could be derived from additional

Table 11.3: The Three Types of Innovation in a Two-Period Model

$u = u_1(X_1, H_1) + u_2(X_2, H_2)$	Utility function	(11.2)
$H_2 = h_2(\tilde{m}_2)$	Production of health with process innovation only	(11.3)
$\tilde{m}_2 = g^C(R^C)M_2, g^C[0] = 1, \frac{\partial g^C}{\partial R^C} > 0$		
$H_2 = g^D(R^D)h_2(\tilde{m}_2)$	Production of health with process and product innovation only	(11.4)
$g^D[0] = 1, \frac{\partial g^D}{\partial R^D} > 0$		
$H_1 = h_1(m_1), \text{ with } m_1 = g^O(R^O)M_1$	Production of health with process, product, and organizational innovation combined	(11.5)
$H_2 = g^D(R^D)h_2(m_2), \text{ with}$		
$m_2 = g^O(R^O)\tilde{m}_2 = g^O(R^O)g^C(R^C)M_2$		
$g^O[0] = 1, \partial g^O / \partial R^O > 0$		
$X_1 + M_1 + R^C + R^D + R^O + X_2 + M_2 = \bar{Y}$	Budget constraint	(11.6)
$L = u_1\{X_1; h_1[g^O(R^O)M_1]\}$	Lagrange function, to be maximized	(11.7)
$+ u_2\{X_2; g^D(R^D)h_2[g^O(R^O)g^C(R^C)M_2]\}$		
$- \lambda\{X_1 + M_1 + R^C + R^D + R^O + X_2 + M_2 - \bar{Y}\}$		
X_1 : Consumption in the first period (X_2 : in the second period)	R^C : Expenditure on process innovation, in first period only	
H_1 : Health status in the first period (H_2 : in the second period)	R^D : Expenditure on product innovation, in first period only	
λ : Lagrange multiplier pertaining to the budget constraint	R^O : Expenditure on organizational innovation, in first period only	
M_1 : Expenditure on medical services in the first period (M_2 : in the second period)	$\bar{Y}$: Total income of periods 1 and 2	

consumption during the first period available if less investment in innovation were made.

Based on the Eqs. (11.7) and (11.8), the necessary optimal condition for *product innovation* R^D is given by

$$\frac{\partial L}{\partial R^D} = \frac{\partial u_2}{\partial H_2} h_2 \frac{\partial g^D}{\partial R^D} - \frac{\partial u_1}{\partial X_1} = 0 \quad (11.10)$$

By spending more on product innovation, the health status h_2 that would have been achieved by the use of medical services and, possibly, process and organizational

innovation only can be exceeded. In an optimum, this benefit again must equal its opportunity cost in terms of consumption sacrificed.

Finally, the following condition with respect to *organizational innovation* can be derived from Eq. (11.7), using Eq. (11.8),

$$\begin{aligned}\frac{\partial L}{\partial R^O} &= \frac{\partial u_1}{\partial H_1} \frac{\partial H_1}{\partial h_1} \frac{\partial h_1}{\partial m_1} \frac{\partial m_1}{\partial g^O} \frac{\partial g^O}{\partial R^O} + \frac{\partial u_2}{\partial H_2} \frac{\partial H_2}{\partial h_2} \frac{\partial h_2}{\partial m_2} \frac{\partial m_2}{\partial g^O} \frac{\partial g^O}{\partial R^O} - \frac{\partial u_1}{\partial X_1} \\ &= \frac{\partial u_1}{\partial H_1} \frac{\partial h_1}{\partial m_1} M_1 \frac{\partial g^O}{\partial R^O} + \frac{\partial u_2}{\partial H_2} g^D \frac{\partial h_2}{\partial m_2} g^C M_2 \frac{\partial g^O}{\partial R^O} - \frac{\partial u_1}{\partial X_1} \\ &= 0\end{aligned}\quad (11.11)$$

Now conditions (11.9)–(11.11) may be compared by examining the required effectiveness of the respective expenditure on innovation, effectiveness being defined as the marginal contribution to health status in the second period. Rewriting Eqs. (11.9)–(11.11), one obtains

$$\frac{\partial h_2}{\partial m_2} \frac{\partial g^C}{\partial R^C} g^D g^O M_2 = \frac{\partial u_1 / \partial X_1}{\partial u_2 / \partial H_2}, \quad \text{process innovation} \quad (11.12a)$$

$$\frac{\partial g^D}{\partial R^D} h_2 = \frac{\partial u_1 / \partial X_1}{\partial u_2 / \partial H_2}, \quad \text{product innovation} \quad (11.12b)$$

$$\frac{\partial h_2}{\partial m_2} \frac{\partial g^O}{\partial R^O} g^C g^D M_2 = \frac{\partial u_1 / \partial X_1}{\partial u_2 / \partial H_2} - \frac{\partial u_1 / \partial H_1}{\partial u_2 / \partial H_2} \frac{\partial h_1}{\partial m_1} \frac{\partial g^O}{\partial R^O} M_1, \quad (11.12c)$$

organizational innovation

These three conditions may be interpreted as follows.

- The three criteria share a common element $[(\partial u_1 / \partial X_1) / (\partial u_2 / \partial H_2)]$. It says that, regardless of the type of innovation, the required improvement of health in the second period has to be higher the more important the *loss of utility* occasioned by the sacrifice in terms of consumption during the first period (large value of $\partial u_1 / \partial X_1$). However, the required improvement is relatively small in a situation where health is of overriding importance (large value of $\partial u_2 / \partial H_2$).
- For every monetary unit additionally spent, both process and product innovation should have the *same effect* on second period health status [LHS of Eq. (11.12a) equals LHS of Eq. (11.12b)].
- The marginal improvement of health status to be achieved in the second period is *always lower* for organizational innovation than for the other two types of innovation. This difference is larger, the greater medical expenditure during the first period (M_1). Clearly, this argument hinges on organizational innovation indeed having an effect already in the first period $[(\partial h_1 / \partial m_1) (\partial g^O / \partial R^O) > 0$ in Eq. (11.12c)].

- The benchmark for process innovation in Eq. (11.12a) depends, inter alia, on the (optimal) extent of product innovation g^D . Since $\partial g^D / \partial R^D$ appears in Eq. (11.12b), the value of g^D is only known if h_2 , the health status attained as a result of process and organizational innovation, is known. This highlights the *interdependence* of the three types of innovation from the ex ante optimizing point of view of an individual.

CONCLUSION 11.4: The requirements for the three types of innovation in health care can be expressed as required improvements of health status. From the ex ante point of view of an individual, these requirements are always equal for both process and product innovation, but lower for organizational innovation.

It should be noted that Conclusion 11.4 does not contain statements about the optimal values of R^C , R^D , and R^O . Indeed, it is not possible to infer from the marginal productivities, determined in Eqs. (11.12), the respective values of expenditures on innovation without additional assumptions. For instance, a high required value of the marginal product of product innovation does not necessarily mean that the individual wishes to reduce R^D below the amount of process innovation R^C . This would only be the case if the functional relationships between the expressions on the left-hand side of Eqs. (11.12) and R^C , R^D , and R^O were the same. However, no equality conditions of the kind $\partial g^C / \partial R^C = \partial g^D / \partial R^D = \partial g^O / \partial R^O$ were introduced as assumptions.

11.3.3 Distortions of Criteria at the Aggregate Level

In the preceding section, benchmarks for innovation were derived that an “average” individual would apply as, for example, a potential purchaser of an innovative drug or a taxpayer who had to pay for innovation in public health. According to Conclusion 11.4, such an individual would demand the same improvement of health per additional monetary unit (MU) for process and product innovation, with a discount for synergy effects of organizational innovation.

This expectation does not seem to conform with experience, however, according to evidence from Great Britain. If health effects of innovation are measured in terms of quality adjusted life years (QALYs; see Section 2.4), estimates for the British National Health Service (NHS) range from 5.88 QALYs per £1,000 to 0.068 QALYs per £1,000 (see Table 2.1, which is reproduced here for convenience as Table 11.4). The decisions taken by the NHS with regard to product innovation are based on a benchmark of only 0.068 QALYs per £1,000 for a dialysis service in hospitals, but of as much as 5.88 QALYs per £1,000 for a stop-smoking campaign (which, while not originating within the health service itself, nonetheless saves medical inputs and thus appears as a process innovation in Table 11.4).

More generally, the entries of Table 11.4 suggest that *process innovations* have to satisfy *more stringent criteria* than do product innovations. Apparently, individual ex ante preferences concerning health care innovation may be distorted at the aggregate level. There seem to be three possible reasons for such a distortion; the influence of

Table 11.4: League Table of Medical Interventions: QALYs Gained per £1,000, Various Innovations

<i>Therapy/Innovation</i>	<i>Type of Innovation^a</i>	<i>QALYs per 1,000 £^b</i>
Stop-smoking campaign	C	5.88
Implantation of pacemaker	C	1.43
Hip replacement	D	1.33
Bypass operation in case of severe angina pectoris	D	0.96
Control of serum cholesterol	D	0.59
Kidney transplant	C	0.33
Breast cancer mass screening program	C	0.29
Heart transplant	D	0.20
Bypass operation in case of light angina pectoris	D	0.079
Hospital dialysis	D	0.068

^a D: Product innovation, C: Process innovation. Product innovations include in particular life-saving therapies to which at the time of their introduction no alternative existed.

^b Present value of all future costs, in prices of 1983/84.

Source: Williams (1985).

health insurance (moral hazard), a medical imperative at the level of objectives, and an imperative at the level of means.³

- **Effects of Health Insurance.** If covered by (social) health insurance, an individual does not have to pay separately for innovations in health care. A stay in hospital usually costs just as much (frequently nothing at all), regardless of whether the hospital has just had a new operating room installed or not, or if use is made of a newly released drug that is twice as expensive as its conventional alternatives.

As soon as the individual decides as a patient, he thus does not have to pay the full opportunity cost of the innovation. His sacrifice of consumer goods X_1 (and with it his loss of utility $\partial u_1/\partial X_1$ in exchange for improved health status) are reduced by health insurance. According to Eqs. (11.12), this serves to lower the benchmark for all three types of innovation because the common term $(\partial u_1/\partial X_1) / (\partial u_2/\partial H_2)$ on the right-hand side decreases. The effect of “moral hazard” is therefore not limited to a boosting of demand for medical care in the current period (see Section 6.5), but also to speed up technological change in medicine [see also Newhouse (1981)].

In addition, health insurance distorts the balance between process innovation and product innovation by reducing the cost of medical services to the individual. Accordingly, the value of M_2 (medical expenditure) decreases in Eq.

3. The term *technological imperative in medicine* has been introduced by Fuchs (1968). It denotes the rapid diffusion of product innovation not only in U.S. health care but in all industrialized countries, in accordance with medical norms espoused by the rest of society.

(11.12a), resulting in a higher value of $\partial g^C/\partial R^C$ ceteris paribus.⁴ Decreasing marginal returns to innovation then call for a lower value of R^C . In sum, health insurance does not foster process innovation to the same extent as it does *product innovation*.

- **Medical Imperative at the Level of Objectives.** This imperative means that under the influence of physicians it becomes the norm in a society to fight disease as much as possible. Thus everything must be done to ensure that the health status of an individual in need of treatment attains a certain level.⁵ In terms of the present model, this imperative may be represented by the condition

$$\frac{\partial u_2}{\partial H_2} \leq \bar{u}'_2 \tag{11.13}$$

stating that the marginal utility of health must be low, that is, below some limit $\bar{u}'_2$. Judging from Eqs. (11.12), this condition at first seems to push up standards of performance for all three types of innovation, since the denominator of the common term, $\partial u_2/\partial H_2$, is reduced. However, one must take into account that a low value of $\partial u_2/\partial H_2$ is not exogenously given but will be the result of a high value of H_2 , which by Eq. (11.5) calls for *increases* in g^D , g^O , g^C , and M_2 , and hence in R^D , R^O , and R^C . Therefore, the medical imperative at the level of objectives turns out to have an *accelerating effect* on the rate of innovation in health care.

- **Medical Imperative at the Level of Means.** This imperative calls for availability of “sufficient” amounts of medical services. This means that medical expenditure M_2 must be pushed to a point where their marginal productivity is already low. Denote this limit on marginal productivity by $\bar{H}'_2$, differentiate Eq. (11.5) with respect to M_2 , and use this condition to obtain

$$\frac{\partial H_2}{\partial M_2} = \frac{\partial h_2}{\partial m_2} g^D g^C g^O \leq \bar{H}'_2 \tag{11.14}$$

The implications of such an imperative are disturbing. The more medical services M_2 are used, the lower the marginal productivity $\partial h_2/\partial m_2$, at a given value of g^C , and the higher the values of g^D , g^C , and g^O (and thus R^D , R^C , and R^O) that are compatible with the constraint (11.14). The medical imperative at the level of means thus not only encourages additional provision of medical services during the current period but also expenditure on all three types of innovation [whereas it is organizational innovation that should be fostered according to Eqs. (11.12)].

4. The ceteris paribus clause is violated as a rule because health insurance also causes a decrease of $\partial g^D/\partial R^D$ in Eq. (11.12b) and consequently an increase of g^D in Eq. (11.12a). Full effects would call for total differentiation of the equation system (11.12) with respect to M_2 , which shall not be undertaken here.

5. Defining quantified health objectives (e.g., a reduction of 50 percent of mortality caused by cardiovascular diseases by the year 2000), as is common in the Annual Health Reports of the U.S. government, may be interpreted as the outflow of an imperative at the level of objectives.

CONCLUSION 11.5: In the transition from the individual to the social level, the required standards of performance for all three types innovation in health care are lowered. Insurance-induced moral hazard and medical imperatives at the level both of objectives and means constitute likely reasons for this, with health insurance distorting the balance in favor of product innovation.

11.4 THE CHALLENGE OF THE SISYPHUS SYNDROME

The Sisyphus syndrome points to the possibility of “failure of success” in health care at least from a politician’s point of view. The ingredients of such a failure have already been identified. On the one hand, the rectangularization of survival curve was presented in Section 11.2, which means that life expectancy gains are concentrated among the higher age groups. In Section 11.3, on the other hand, the predominance of product innovation in health care was noted, a type of innovation that usually goes along with increased costs of treatment. Taken together, these two facts may well result in a process reminiscent of a Sisyphus syndrome in health care. Let technological progress in medicine in the guise of product innovation improve the chances of survival particularly at middle and advanced ages. Through their increased numbers, aged voters exert more influence in the political process, pushing through their preference in favor of public health care and medical innovation, causing resources to be channeled into health care, especially product innovation. This in turn lays the basis for even more successes of modern medicine, which serve to boost the ranks of the old. The result is an ever-increasing share of the public budget devoted to health care, serving progressively to curtail politicians’ freedom of maneuver.

11.4.1 A Simple Dynamic Model

The objective of this section is to give the Sisyphus syndrome a formal representation. Let time t be measured in periods of approximately 30 years, corresponding both to the span of a generation and the maximum remaining life expectancy of a retiree. The size of the aged population A_t at the middle of period t is given by the working population in the previous period W_{t-1} times the rate of survival $(1 - \pi_t)$. In accordance with the hypothesis underlying the Sisyphus syndrome, let the rate of mortality π_t depend on the input of medical services in the previous period M_{t-1} (see also Table 11.5),

$$A_t = [1 - \pi_t(M_{t-1})]W_{t-1}, \quad \frac{\partial \pi_t}{\partial M_{t-1}} < 0 \quad (11.15)$$

The second equation of the system serves to represent the argument that the quantity of medical services made available in a society depends on the population’s age structure. Thus let M_t depend positively on the *age coefficient* α_t , that is, on the ratio of the aged to the working population in the same period. For one, this reflects the

Table 11.5: Model of the Sisyphus Syndrome in the Health Care System

$$A_t = [1 - \pi_t(M_{t-1})]W_{t-1}, \quad \frac{\partial \pi_t}{\partial M_{t-1}} < 0 \quad (11.15)$$

$$M_t = M_t(\alpha_t), \quad \text{with } \frac{\partial M_t}{\partial \alpha_t} > 0 \text{ and } \alpha_t := \frac{A_t}{W_t} \quad (11.16)$$

$$dA_t = \left(\frac{-\partial \pi_t}{\partial M_{t-1}} \right) W_{t-1} dM_{t-1} \quad (11.17)$$

$$dM_t = \frac{\partial M_t}{\partial \alpha_t} \frac{\partial \alpha_t}{\partial A_t} dA_t = \frac{\partial M_t}{\partial \alpha_t} \frac{1}{W_t} dA_t \quad (11.18)$$

$$dM_t = \underbrace{\left(-\frac{\partial \pi_t}{\partial M_{t-1}} \right)}_{(+)} \underbrace{\left(\frac{\partial M_t}{\partial \alpha_t} \right)}_{(+)} \underbrace{\left(\frac{W_{t-1}}{W_t} \right)}_{(+)} dM_{t-1} \quad (11.19)$$

A_t : Aged population in the current period
 W_t : Working population in the current period, exogenously determined

α_t : Age coefficient, ratio of aged to working population
 M_t : Expenditure on medical services
 π_t : Mortality rate

fact that the demand for private medical care is likely to be strong in a relatively old (and hence less healthy) population. Second, however, the age coefficient represents the proportion of aged voters and thus their power in the political process. The larger the age coefficient, the more tax money flows into medical research and into the infrastructure of the health care system in general. Thus

$$M_t = M_t(\alpha_t), \quad \text{with } \frac{\partial M_t}{\partial \alpha_t} > 0 \text{ and } \alpha_t := \frac{A_t}{W_t} : \text{age coefficient} \quad (11.16)$$

To examine the *dynamic properties* of this interdependent model, consider its reaction to an exogenous shock. For some reason, let the quantity of medical services increase by the amount $dM_{t-1} > 0$ in the past. By Eq. (11.15), this results in an increase of the aged population in period t ,

$$dA_t = \left(\frac{-\partial \pi_t}{\partial M_{t-1}} \right) W_{t-1} dM_{t-1} \quad (11.17)$$

This increase in the number of the aged induces, following Eq. (11.16), an increase of total expenditure on medical services. Because the working population is assumed to be exogenously given, differentiation of Eq. (11.16) yields

$$dM_t = \frac{\partial M_t}{\partial \alpha_t} \frac{\partial \alpha_t}{\partial A_t} dA_t = \frac{\partial M_t}{\partial \alpha_t} \frac{1}{W_t} dA_t \quad (11.18)$$

Now expression (11.17) can be substituted into Eq. (11.18), which after slight rearrangement gives the following *equation of motion* for medical care expenditure

$$dM_t = \underbrace{\left(-\frac{\partial \pi_t}{\partial M_{t-1}} \right)}_{(+)} \underbrace{\left(\frac{\partial M_t}{\partial \alpha_t} \right)}_{(+)} \underbrace{\left(\frac{W_{t-1}}{W_t} \right)}_{(+)} dM_{t-1} \quad (11.19)$$

Since all three expressions in parentheses have a positive sign, a past increase of expenditure on medical services is predicted to entail an increase in the present, which in turn is transmitted into the future, as claimed by the Sisyphus syndrome. Each term may be interpreted as one of three factors determining the degree of transmission.

1. Success of medicine. The more the rate of mortality π_t can be reduced due to additional medical expenditure in the previous period M_{t-1} , the more acute the Sisyphus syndrome, *ceteris paribus*.

2. Influence of the aged population on health care expenditure. This is the second factor of Eq. (11.19). The more the political institutions of a country permit an increase of the age coefficient α_t to result in higher health care expenditure, the stronger the Sisyphus syndrome.

3. Development of the working population. The third factor of Eq. (11.19) shows the size of the working population in the previous period compared to the size of the working population in the current period. The less the working population grows (the more it diminishes, respectively), the higher is this factor and the more will an increase of health care expenditure carry forward into the future, enhancing the Sisyphus syndrome.

CONCLUSION 11.6: In democratically organized countries a Sisyphus syndrome might exist, which means that present successes of medicine will be burdening the health care system with additional expenditure on medical care in the future. The acuteness of the alleged syndrome depends on the extent to which medical services contribute to longevity, the influence of the aged population on the public budget, and inversely on the growth of the working population.

11.4.2 Empirical Relevance of the Sisyphus Syndrome

The equation of motion (11.19) of Table 11.5 contains three quantities, which must all have a positive sign for the Sisyphus syndrome in health care to occur. The third term, re-presenting the development of the working population over time, can be set to one and hence be neglected. The reason is that at least in OECD countries, the work force increases very slowly [see OECD (1985), p. 31]. This leaves the first two quantities to be examined. In the light of the evidence presented in Chapter 4, it is doubtful whether

$\partial \pi_t / \partial M_{t-1} < 0$ in fact holds, that is, whether medical services are capable of significantly reducing the rate of mortality. Thus, this issue has to be settled first.

The model presented in Table 11.5 cannot be tested directly, since a period of fixed length t does not quite fit the facts. In the model, t denotes the maximum remaining life expectancy of those just retired. In actual fact, the remaining average life expectancy at age 65 (an indicator of maximum remaining life expectancy) increased during the last decades. This increase differs between countries, reflecting differences in the postretirement mortality rate. Thus, in the empirical analysis, π_t (which would have to relate to a period of variable length) is discarded in favor of remaining life expectancy itself.

In order to gain additional insight into the shape of the survival curve discussed in Subsection 11.2.1.1, remaining life expectancy not only at the age of 65 years ($L65F$, $L65M$) but also at the age of 40 years ($L40F$, $L40M$) was included in the analysis, differentiated according to sex. Data about these four variables were available for 19 OECD countries, relating to the year 1980 [see OECD (1987)]. The four equations were estimated jointly under the assumption that both Gross Domestic Product per capita ($GDP70$) and health care expenditure ($HCE70$) a decade back influence remaining life expectancy equally for both ages and sexes. By way of contrast, dummy variables $AGE65$ and $SEXF$ are entered to reflect the fact that individuals aged 65 presumably differ from those aged 40 in terms of remaining life expectancy, which is also true of females compared to males. The results of a SURE estimation⁶ are shown below [see Zweifel and Ferrari (1992)].

$$\begin{aligned}
 \begin{bmatrix} L40F \\ L65F \\ L40M \\ L65M \end{bmatrix}_{1980} &= 34.9*** \\
 &- 20.7*** (AGE65, = 1 \text{ if life expectancy at age 65}) \\
 &+ 5.0*** (SEXF, = 1 \text{ if life expectancy of females}) \\
 &- 0.97* (GDP70, \text{ per capita GDP in 000 USD of 1970;} \\
 &\quad e = -0.13) \\
 &+ 13.2** (HCE70, \text{ per capita HCE in 000 USD of 1970;} \\
 &\quad e = 0.105) \\
 &+ \hat{u}_1
 \end{aligned} \tag{11.20}$$

where $*$ ($**$, $***$) is the significance level 5% (1%, 0.1%); SURE estimation; and e is the elasticity, calculated at arithmetic means.

6. Seemingly unrelated regression estimation (SURE) takes into account the fact that the error term of Eq. (11.20) may be correlated with the error term of another equation. This other equation is Eq. (11.21), postulating a feedback from remaining life expectancy to health care expenditure. This reverse causation should ideally be controlled for, e.g., by 2SLS estimation (see Subsection 4.2.2.1). However, suitable exogenous variables for purging endogenous regressors from their errors were not available. If these errors were important, they would have to result in considerable correlation between errors u_1 and u_2 and hence SURE estimates that differ from OLS ones, which is, however, not the case. For details, see Zweifel and Ferrari (1992).

Equation (11.20) may be interpreted as follows.

- **Extent of rectangularization.** According to this estimate, the remaining life expectancy of individuals aged 65 is some 21 years less on average than that of individuals aged 40, *ceteris paribus*. This follows from the coefficient (-20.7) of the explanatory variable AGE_{65} , which takes on the value of one if remaining life expectancy is measured at age 65, and zero otherwise. If the survival curve between years 40 and 65 ran horizontal, all 40-year-old individuals would survive until the age of 65. This would amount to a difference of exactly 25 years (instead of the estimated 21 years) in remaining life expectancy. Since the gap is small, the *survival curve must run almost horizontal* between ages 40 and 65 in today's industrialized countries.
- **Difference between the sexes.** The remaining life expectancy of women exceeds that of men by 5 years, *ceteris paribus*. This is shown by the coefficient ($+5.0$) of the dummy variable $SEXF$, which takes on the value one if it represents the life expectancy of females and is zero otherwise.
- **Influence of income.** The remaining life expectancy appears to react with a considerable lag to income (measured by per capita GDP of comparable purchasing power). According to this estimate, an income difference of 10 percent in 1970 translates into a differential of remaining life expectancy of 1.3 percent *ten years later* (this amounts to about 2 months for a 65 year old person). Interestingly, the estimated partial relationship turns out to be negative, suggesting that lifestyles associated with (still) higher income detract from health and longevity.
- **Influence of health care expenditure.** This result is most noteworthy in that remaining life expectancy seems to depend positively on health care expenditure (HCE) incurred about a decade earlier. Thus a 10 percent increase in per capita health care expenditure in 1970 is estimated to result in a lengthening of remaining life expectancy by 1 percent (1.5 months for a 65 year old person) ten years later. In view of the difficulty of establishing the relationship between medical care and health in the production of health (see Chapter 4), this comes as a surprise. Note, however, that the literature reviewed there does not specifically focus on *remaining life expectancy* at middle and advanced ages, a focus motivated by the process of rectangularization that seems to continue.

Thus one of the conditions ($\partial\pi_t/\partial M_{t-1} < 0$) for the existence of the Sisyphus syndrome is fulfilled. The other empirically relevant factor states that an increased proportion of the aged population leads to more health care expenditure [see Eq. (11.17)]. This may take two forms. For one, the aged themselves spend more on medical care, resulting in an increased per capita value of private health care expenditure ($HCEP$). Or, by participating in elections and votes, the aged might influence public spending in favor of health care, causing public health care expenditure ($HCEG$) to increase.

Both components of per capita health care expenditure are included as dependent variables in the following regression equation. In order to allow for a different impact of income on public and private health care expenditure, per capita GDP is multiplied by a dummy variable that takes on the value of one if the data refer to public health care

($PUBL = 1$) and the value of zero if referring to private health care expenditure ($PUBL = 0$). The results of SURE estimation are given below.

$$\begin{aligned} \begin{bmatrix} HCEG \\ HCEP \end{bmatrix}_{1984} &= 8.80 \\ &+ 0.078 (GDP84, \text{ per capita GDP, 1984; } e = 2.08) \\ &+ 0.073 (GDP84 \text{ } PUBL; PUBL = 1 \text{ if public HCE;} \\ &\quad e = 2.14) \\ &- 0.19 (L65F, \text{ life expectancy of 65-year-old women}) \\ &- 0.45 (L65M, \text{ life expectancy of 65-year-old men}) \\ &+ \hat{u}_2 \end{aligned} \quad (11.21)$$

SURE estimation; e is the elasticity, calculated at arithmetic means.

The following remarks seem appropriate.

- ***Influence of income.*** Contrary to OLS estimation results not shown here, income (GDP per capita) does not appear to be a significant determinant of health care expenditure. Thus, as soon as the correlation between the errors u_1 and u_2 of Eqs. (11.20) and (11.21) is taken into account, the relationship between income and health care expenditure becomes less clear than commonly assumed. Still, the implied elasticity estimates are suggestive, pointing to health care (and especially public health care with its redistributive character) being a luxury good.
- ***Remaining life expectancy as an explanatory variable.*** According to Eq. (11.16), the ratio of the aged to the working population α_t should appear as an explanatory variable. In order to emphasize the feedback relation inherent in the Sisyphus syndrome, remaining life expectancies $L65F$ and $L65M$ are used instead. This may be justified by noting that these life expectancies determine the future size of the aged population and hence the value of α_t . Moreover, most 40 year olds may count on living to age 65 [as shown below Eq. (11.20)], possibly causing them to cast their ballot in favor of public health care in the hope of reaping its future benefits, which importantly depend on the remaining life expectancy at retirement age, $L65F$ and $L65M$.
- ***Lack of significance of remaining life expectancy.*** Surprisingly, Eq. (11.21) fails to produce statistical evidence to the effect that remaining life expectancy at age 65 exerts an influence on the amount of a country's health care expenditure. This finding was already obtained in OLS regression (not shown) and is robust to the choice of indicator for the importance of the aged population (e.g., to using the population ratio with and without weighing with remaining life expectancy). In these alternative estimations, too, negative regression coefficients occur frequently, which is incompatible with the maintained hypothesis $\partial M_t / \partial \alpha_t > 0$ of Eq. (11.19). Therefore, the empirical evidence speaks in favor of

setting the second term of the equation of motion (11.19) to zero, implying that impulse dM_{t-1} is not transmitted to dM_t .

CONCLUSION 11.7: An empirical analysis using OECD country data suggests that while additional health care expenditure may contribute to higher remaining life expectancy at middle and higher ages, higher life expectancy cannot be claimed to bias spending in favor of (public) health care expenditure. Thus the Sisyphus syndrome is found to lack empirical support for the time being.

11.5 INTERNATIONAL CHALLENGES

11.5.1 Integration of Insurance Markets

Important challenges for European health insurers will originate from the integration of insurance markets with the European Union (EU); similar developments may become relevant for insurers in NAFTA countries in the more distant future. In 1987 the European Court of Justice decided that insurance is subject to the anticartel provisions of Art. 85 of the Treaty of Rome like any other branch of industry and that the EU has the authority to regulate insurance. At the same time the Court's decision accords EU member states a long transition period for liberalizing social insurance. Thus, while social health insurers are exempt from the Court's decision in the foreseeable future, they will have to redefine their role—if only because they are increasingly becoming attractive partners for profit-seeking multinational insurance companies:

- Health insurance is characterized by *frequent, comparatively small losses*, occasioning frequent client contacts and hence opportunities for additional sales at comparatively low cost.
- Social health insurance covers *the majority of the population* in most countries, with the exception of the United States, its market share ranging from 70 percent in the Netherlands to 100 percent in Canada and France, for example.
- A social health insurer may provide the partner insurance company with *information* on the health status of enrollees. This constitutes a valuable input for risk rating in other products such as life insurance or homeowner's insurance.
- The risk of ill health tends to be *negatively correlated* with risks covered under life insurance and automobile insurance. Costly medical treatment is concentrated among those aged 60 or over, when most life insurance policies have already expired. Likewise, someone who is sick in bed cannot cause a traffic accident. Negatively correlated losses serve to reduce the variance of total loss in a decisive way [see Fama (1976)]. This means that a portfolio of insurance policies containing combinations of health and other lines of insurance requires less reserves per unit premium, conferring obvious cost advantage on the insurance company.

For these reasons, multiple line insurers have considerable interest in cooperating with a social health insurer. They already have profited from this possibility in the

Netherlands, for example. Whereas only domestic insurance companies have been pursuing this strategy up to now [see Schut et al. (1991)], foreign insurers may well also adopt it in the future, appealing to new national treatment provisions under EU and GATT rules. This also means that solidarity *within* a carrier of social health insurance can hardly be sustained. The profit-seeking partner of this cooperation will press for replacing pooling contracts by stable separating contracts (see Section 5.4).

A further challenge for social health insurance will come from increased mobility of workers in common NAFTA and EU labor markets. As to the EU, ordinance No. 1408/71 stipulates portability of health insurance coverage between member countries. A guest worker from Portugal is thus free to claim German health care services after a few years of residence in Germany, even if he has moved back to Portugal in the meantime.

The Dutch special insurance scheme AWBZ for exceptionally high costs of treatment constitutes an interesting case in point [Van de Ven (1991)]. The benefits of this tax-financed scheme are comparatively generous. They are suspected to induce workers from other EU countries to seek employment in the Netherlands in order to obtain costly medical treatment free of charge. Evidently, the *problem of risk selection* takes on a new dimension here. On a national scale, it could be solved by getting (nearly) all inhabitants of the country to join social health insurance, either through subsidization or compulsory membership. However, compulsory Europe-wide health insurance cannot be counted upon to come into force very soon for citizens of EU countries. In the meantime, national health insurance schemes offering high-quality treatment are running the danger of attracting bad risks from other EU countries, causing them to lose their financial equilibrium.

11.5.2 Migration of Health Care Personnel

For some years now, national diplomas of physicians and dentists have been mutually recognized within the EU. As shown in Figure 11–3, there is a definitive incentive to migrate especially for physicians. At the end of the 1980s, a general practitioner in former Western Germany earned approximately 50,000 US\$ at prices and purchasing parities of 1980. His colleague in France had to put up with approximately 30,000 US\$; thus, by moving to the neighboring country, he would have been able to increase his income significantly. Yet language barriers in particular will continue to block migration on a large scale between EU countries in the future. Even within a common language area, income differences of several 10,000 US\$ may be insufficient to motivate physicians to migrate, judging from Canadian experience. As may be gleaned from Figure 11–3, a general practitioner's income in the United States was 50 percent higher during many years without causing much migration of Canadian physicians to the United States [see Evans (1984), p. 301].

However, the decision to migrate will hardly be based on one year's income but rather on the income stream associated with an entire career. For example, a general practitioner in Great Britain's National Health Service, according to Figure 11–3, earns a comparatively low yearly income. In return, he starts earning this income at a relatively young age. Evidently, the quantity of interest is not so much the annual income but the rate of return calculated on an entire working life.

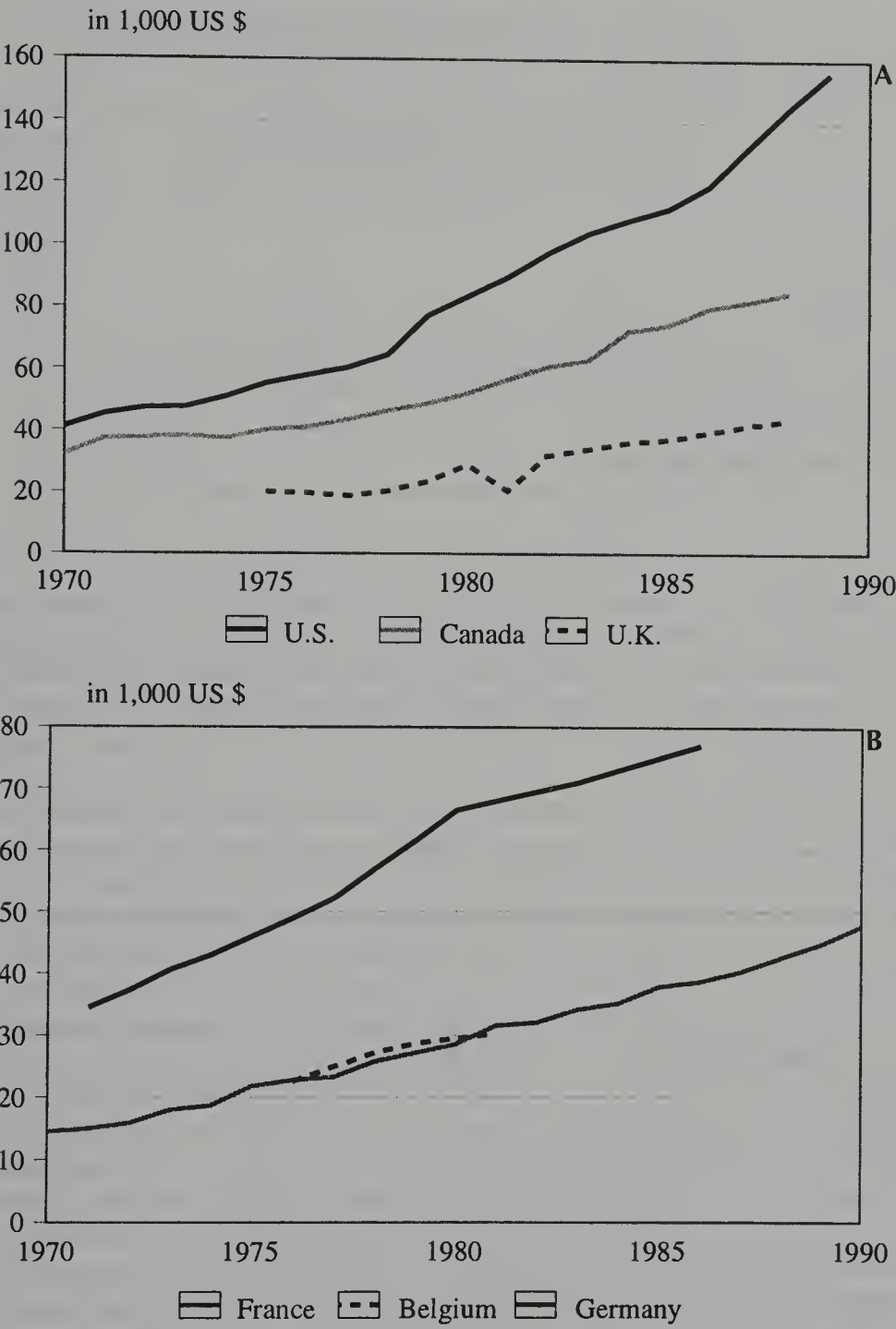


Figure 11-3. Average physician incomes for selected countries and years with 1980 prices and purchasing parities: (a) United States, Canada, and United Kingdom and (b) France, Belgium, and Germany. Source: OECD (1992), Sandier (1990).

The costs and returns of a medical career must be discounted, yielding a net present value that may be positive or negative, depending on the rate of interest used in the calculation. The rate of interest resulting in a net present value of zero is called the *internal rate of return* of a medical career. The costs of a medical

Table 11.6: Internal Rates of Return on a Medical Career, Not Corrected for Working Hours (around 1983)

	<i>All physicians</i>	<i>General practitioners</i>	<i>Specialists</i>
Belgium	< 0	< 0	1.0
Germany	8.8	--	—
Great Britain	10.5	10.7	10.3
United States ^a	14.0	16.7	11.3 ^b

^a Comparison with all high school graduates.

^b Only internists and surgeons.

Source: Zweifel and Eichenberger (1992).

career basically are opportunity costs, that is, the income the medical student would have earned in an alternative occupation. In Table 11.6, this alternative occupation is assumed to be a career in banking. While constituting a credible alternative in view of the abilities of a typical medical student, this choice has the additional advantage of data availability, as incomes by age happen to be known for banking in several OECD countries.

Considerable differences with regard to the rate of return on a physician's career emerge from Table 11.6. In Belgium, a country with almost free access to the medical market, only the rate of return on a career as a specialist is slightly positive. By way of contrast, German physicians achieve a rate of return close to 9 percent and their colleagues in the British National Health Service even one of above 10 percent. Physicians in the United States are way out in front, but for reasons of data availability, a career of an average high school graduate and not that of a bank manager had to be used in the comparison, which accounts for part of the difference.

The rates of return displayed in Table 11.6 show the relative financial attractiveness of the medical profession from the perspective of a person about to enter college, while the decision to migrate typically is made from the perspective of a person having completed his medical education. The fact that a medical career amounts to a profitable investment relative to other careers in the same country does not necessarily imply that the medical career in that country is attractive relative to a medical career in another country. Nevertheless, in view of the high level of U.S. wages in general, the United States presumably is an attractive place to work for physicians from many other countries.

In the case of *nursing staff*, the stakes associated with migration are smaller due to their generally lower income. This means that migration between countries of roughly comparable income levels is less likely for auxiliary health personnel, while the present value of the wage differential in comparison with most developing countries may be large enough to induce migration to industrialized countries [see Gray and Phillips (1993)].

11.5.3 Direct International Investment in Hospitals

A third international challenge in the future may well come from multinational hospital chains. For example, such a chain may introduce new types of surgery not offered by public hospitals of a country. The potential for multinational operations exists, as can be seen in the United States, with the biggest U.S. hospital chain—the Hospital Corporation of America—administering 54,000 beds in 364 hospitals [see *Health Week* of 1 May 1991].

In order to assess the implications of this possible challenge, it is appropriate to recall the *economic theory of the multinational enterprise* for a moment.

The existence of enterprises as such—which are founded on hierarchy rather than exchange relations between equals on a market—is explained with reference to the *role of transaction costs*. A multilateral network of contracts between members of a productive group can be very costly to implement and monitor. Thus it is replaced by a set of bilateral contracts concluded between the manager and the respective subordinate. Multinational enterprises owe their existence and economic success to the fact that transaction costs loom particularly large in exchanges of goods and services between two countries, where great savings may be achieved by carrying out internal transactions using a standardized legal framework for contracts within the firm. International transfer of know-how in particular becomes less costly under a common set of norms. Indeed, multinational enterprises are specialized in introducing an innovation into many markets within a short period of time [see Casson (1985); Williamson (1981)].

In the case of hospital treatment, transaction costs are considerable in view of the efforts a patient and his physician must undertake for estimating a hospital's chance of success in a particular field of activity and its quality of treatment in general. A multinational enterprise may lower these costs by *guaranteeing a certain quality standard* worldwide (analogous to Holiday Inn, Best Western in the case of hotel accommodation). Guaranteeing such a standard amounts to an organizational innovation in the sense of Section 11.3.1, serving to lower cost (not so much of production but rather of transaction) in several markets. This organizational innovation may be applied to hospitals, and hospital chains were indeed gaining ground in the United States during the 1980s, to the detriment of the traditional community hospital. This development has stagnated in the meantime, with multinational hospital chains suffering setbacks as well [see *Health Week* of 1 May 1991].

In the long run, however, multinational enterprises should be able to extend their market share in national hospital sectors due to their ability to transfer product innovations quickly from one country to another. Even social health insurers will be trying to do business with these new suppliers, provided they offer a favorable price–performance ratio to their members. The challenge will consist in rewriting the terms of contract between insurers and service providers in health care in a way such that the insured may benefit from the advantages of this internationalization. The advantages of a multinational for-profit enterprise in terms of efficiency will have to be weighed against the benefits of a professional ethics orientation (described in Section 8.4).

CONCLUSION 11.8: The integration of health insurance as well as the hospital sector in international markets will call for a rewriting of

contracts in the health care sectors of countries participating in economic integration. In comparison, migration of physicians and nursing staff represents a less pressing challenge in the foreseeable future.

11.6 EXERCISES

Exercise 11.1:

In the model of Table 11.3, the assumption was made that the sum of incomes Y is exogenous. This appears restrictive. (a) Which quantities of the model could Y depend on, and how? (b) Consider the possibility $Y = Y(R^D)$, $\partial Y/\partial R^D > 0$. How could such an assumption be justified, for instance in view of foreign trade? (c) What are the consequences of (b) for the optimal allocation of innovative effort? [Hint: Modify Eq. (11.7) accordingly and start from there.]

Exercise 11.2:

The dynamic stability of the law of motion (11.19) is very important for health policy. Will a one-time increase of health care expenditure taper off as time goes on, or must an explosive process be reckoned with? (a) In order to clarify the problem, transform the terms of Eq. (11.19) into elasticities as far as possible. Divide by M_t and introduce the following elasticities,

$$(\pi_t, M_{t-1}) := \frac{-\partial \pi_t}{\partial M_{t-1}} \frac{M_{t-1}}{\pi_t} \text{ and } e(M_t, \alpha_t) := \frac{\partial M_t}{\partial \alpha_t} \frac{\alpha_t}{M_t} \quad (11.19)$$

(b) Interpret the resulting equation. (c) The resulting equation can be written as

$$\dot{M}_t = c \dot{M}_{t-1} \quad \text{with } \dot{M}_t := dM_t/M_t \quad (11.19')$$

(c1) What does c stand for? (c2) Use this formula to determine $\dot{M}_{t+1}, \dot{M}_{t+2}, \dots, \dot{M}_{t+k}$. Why is it important to know whether $c < 1$ or $c > 1$? (c3) Estimate the value of c using your knowledge from Chapter 4 and about the present structure of population. Is the case $c > 1$ a possibility?

11.7 SELECTED READINGS

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12

SUGGESTIONS FOR POLICY

12.1 HEALTH AS AN ECONOMIC GOOD

In the first chapter of this book, the difference between the “economics of health” and the “economics of health care” was emphasized. Here, this distinction is taken up again because the first of our policy conclusions applies to health itself, independently of issues relating to the organization and financing in health care.

In Chapter 2 and 3 we argued that health and even life itself are not detached from mundane economic activities but can be regarded as *economic goods* and thus be analyzed using the instruments of economic theory. This applies to positive economics, which explains how health is produced with the help of inputs such as one’s own time and medical services (see Chapter 3). It also holds for normative economics, which is taking first steps toward measuring health and even human life itself on the most economic yardstick, that is, money (see Chapter 2).

The policy recommendation that derives from these considerations is to apply the instruments developed there in political decision-making. Public measures affecting the health or the mortality risk of citizens should no longer be taken before a *cost–benefit analysis* has been made, serving at least as an aid to decision-making. Such analysis should not shy away from the valuation of survival and health in monetary terms rather than just mentioning health effects in passing as “intangible quantities.”

Although many persons consider the valuing of life in monetary terms as unethical or even heartless, we must warn of the *consequences of refraining* from such a valuation:

- a. Either the health effects are *ultimately neglected* when considering the costs and benefits of a project, which is thus implicitly decided upon as though human survival and health were worth little to nothing; or
- b. The project with better health effects is preferred, *no matter its cost*; or
- c. Decisions are made on a *casewise basis*, applying rules (a) and (b) in haphazard manner.

All three alternatives have severe adverse implications. Both (a) and (b) evidently result in inconsistencies between public measures and the behavior of individuals in their domain of private decision-making, and between the actions of the government and the preferences of its citizens. Principle (c) causes a large waste of resources, barring solutions that could bring more consumption as well as more health to the members of the society through explicit inclusion of cost-benefit analysis in decisions with regard to health policy.

12.2 COMPETITION OR REGULATION IN HEALTH CARE?

During the past years, the policy debate about health care has been revolving around the question of whether it is possible and promising to introduce more market elements in this sector or whether the market is entirely unsuitable and hence should be replaced by governmental control. In terms of popular slogans, the debate was about the alternatives, "competition" and "regulation" [for a survey of reform projects in Europe, see *Social Science & Medicine* (1994), special issue; for cost containment policies in EU countries, see Abel-Smith and Mossialos (1994); for an account of the reform of the British National Health Service, see *Health Policy* (1993), special issue; for Japan, see Powell and Anesaki (1990), Chap. 9].

When choice is limited to these two alternatives, one comes to the conclusion that the market mechanism in its pure form will fail to result in an optimal allocation of resources for reasons that are discussed in Chapter 5. The most cogent of these reasons probably is the *uncertainty about the event* "sickness," causing individuals to purchase an insurance contract, usually with comprehensive coverage. However, comprehensive coverage destroys the incentives to search for reasonably priced alternatives, incentives that are vital to the functioning of the price mechanism.

However, a pure system of governmental allocation and control as incorporated, for example, in a public health service need not perform better. Since suppliers of medical services are salaried employees in such a system, they lack the incentives to provide services (see, e.g., Chapters 7 and 9), resulting in artificial shortages and queues on the demand side. More important still, actors in such a system lose interest in *innovation* of any type. The experience with the National Health Service in Great Britain tends to confirm this theoretical expectation.

When correctly interpreting the assertions of welfare theory, one finds that for an optimal allocation of resources there need not be a competitive market at every level; it is sufficient that economic subjects behave as if they were acting in such a system. In this view, competition and public regulation do not constitute polar cases; rather, public regulation may be used to create the conditions for the actors to behave as if they were in competition.

The key word here is *incentives*.¹ Therefore, the objective of public regulation should be *to set the incentives* such that actors use resources efficiently even where the market mechanism cannot work. The analysis of alternative payment systems for the

1. The motto of the First European Congress on Health Economics, held 1989 in Barcelona, was "Incentives in Health Care Systems."

services of physicians and hospitals (Chapters 7 and 9) is capable of providing some guidelines. Specifically, incentives are stronger the less payment is geared to input quantities (factor inputs, actual cost) or activity levels (services provided) and the more it depends on output quantities (treatment cases, patients to be treated). It is the task of political decision-makers (usually the legislature) to devise such payment systems.

Another important point is the *role of competition*. Competition necessarily goes along with attempts of suppliers to call consumers' attention more or less ostentatiously to their products. Most people reject the idea of attracting patients through billboard advertising, TV spots, or special offers. To them, these are attempts to manipulate patient preferences. However, advertizing also disseminates *information* that helps consumers find the good or service having the best price–performance ratio.²

Competition may thus contribute to efficiency in health care as elsewhere, provided that the demanders of health services can make their choices consciously and without the pressure of an emergency. These conditions are obviously not fulfilled for an already ill patient, for whom often “the best is just about good enough” in his quest to restore his health. They are met much more for a healthy person *choosing among different insurance packages* (see Chapter 6) or *entire systems of provision* (see Chapter 8).

By all means, there is a role for competition at the level of entire systems of provision [see Enthoven (1980)]. Again it is the task of the legislative body and government to create the necessary conditions for competition. Insofar as legal constraints (such as compulsory membership in a particular sick fund) are repealed, thus increasing individuals' freedom of choice, one can truly speak of a step in the direction of “deregulation.”

12.3 SPECIFIC RECOMMENDATIONS CONCERNING HEALTH POLICY

A few specific recommendations merit spelling out, using the principles of Section 12.2 as a backdrop. These recommendations derive from the microeconomic theory developed for specific areas of health care, and they are presented in the order suggested by the systems analysis of Chapter 1. The possibility of a “total redesign” of health care will be discussed later (Section 12.4).

12.3.1 Insureds and Patients

Many critics of today's health care systems believe that progress can only be made if individual health behavior is changed. Motivated by this belief, proponents of preventive medicine launched large-scale campaigns such as the MRFIT program in the United States [see Multiple Risk Factor Intervention Trial Research Group (1982)], designed to prevent cardiovascular disease. This study proved that mortality due to cardiovascular disease can be considerably lowered through a reduction of excess weight and giving up smoking.

2. Some of the states of the United States require a license for opticians while some do not. In the first group, professional associations restrict advertising. The result is that prices for glasses and optometric services are higher, with no recognizable advantage in terms of quality [see Benham and Benham (1975)].

From results of research such as these, it is a small step to claim that *preventive medicine should be fostered*, possibly at the expense of curative medicine. However, such a claim should be met with reservations in view of the analysis performed in Chapter 5. In fact, a market failure that would justify public intervention does not appear to be present on closer examination (Conclusions 5.1 and 5.2).

Excessive eating, drinking, and smoking are an important and integral part of quality of life to many people, and giving up these things would mean a considerable sacrifice to them. The cost of this sacrifice is to be balanced against an uncertain return in terms of prolonged healthy lifetime. This return is lost if instead of the illness to be avoided another (physical or mental) illness develops or if an accident puts a premature end to life. Presumably it is the person affected who knows best about the many risks jeopardizing his health and survival. Therefore, an apparent lack of interest in prevention relating to health may well be compatible with rational behavior. As a rule, *no negative external effects* emanate from failure to invest in prevention. Public intervention therefore cannot be justified economically, apart from certain well-known exceptions (communicable disease, particularly AIDS in recent times).

12.3.2 The Design of Health Insurance

According to the theory developed in Chapters 5 and 6, every insurer faces the problems of adverse selection and moral hazard. With regard to the problem of moral hazard, the individual insured knows best what preventive measures to apply. The results of Chapter 6 suggest that a certain degree of *cost sharing on the part of the insured* would be in his own interest (Conclusion 6.11). If, however, the intensity of treatment is no longer determined by the patient but by a physician (having a professional ethics orientation), a degree of *cost sharing on the part of the service provider* is also advisable, broadly along the lines of a Health Maintenance Organization (see Chapter 8, particularly Conclusion 8.4).

If deregulation of insurance will put an end to the separation of lines of business, private insurance companies may cover the risk arising from the cost of medical treatment in combination with any other risk. This would permit the insured to opt for an “umbrella” policy providing for a maximum amount of overall cost sharing. For example, his share of the total annual amount of damages under the titles of auto liability, homeowner’s, and health insurance would be restricted to a certain percentage.

In this context, the question arises of whether insurers should be allowed to offer policies—in isolation or in combination with another policy (e.g., written by a sick fund and stipulating a coinsurance)—providing for full coverage. There seem to be two problems with this.

1. The *moral hazard* effect caused by this type of combination amounts to a negative externality, and for two reasons. First, there is the danger that the insured without cost sharing biases choice of treatment in favor of the more costly alternative at a given point in time. To the extent that only a single standard is provided by a given physician or hospital, other insureds with positive coinsurance may run into financial problems (Conclusion 6.10).

Second, insurance also affects the speed and structure of innovation in health care, as shown in Chapters 10 and 11 (Conclusions 10.1 and 11.5). Full coverage in particular fosters product innovation, while insureds having a policy with cost sharing are more interested in process and organizational innovations. Both of these external effects speak in favor of *stipulating* a minimal rate of coinsurance unless the contract causes service providers to check moral hazard effects (as in Health Maintenance Organizations).

2. The second problem is *adverse selection*. Adverse selection exemplifies the statement made in the previous section, viz. that public regulation may be needed to ensure competition. As shown in Chapter 5, profit-seeking insurers as well as sick funds must avoid enrolling bad risks simply in order to safeguard their financial equilibrium. The more insurers are exposed to competition, the greater will be their efforts to attract good risks and avoid bad ones. Unless insurers are permitted fully to differentiate premiums according to risk, there may be bad risks that are denied insurance cover. A public insurance scheme of last resort must be set up for these risks. However, this would deprive precisely the handicapped and chronically ill of the freedom of choice and the benefits from innovation brought about by competition between insurers.

In this context, the task of the government could be seen in organizing *competition for the enrolment of such risk groups*. One way is to provide subsidies covering the expected extra costs associated with enrolling a bad risk. In order to safeguard budget neutrality, insurers having an above-average share of good risks could be mandated to pay an equalization payment in favor of competitors having an excessive share of bad risks. This would amount to the introduction of an internalizing tax, the negative externality being the rejection of bad risks [on some of the problems of implementation, see Van de Ven and Van Vliet (1992)].

Competition for bad risks can in principle be organized to cover all “problematic” groups of a population. For example, if an unemployed or low-income individual turns out to be a bad health risk, his enrolment may be fostered by attaching a subsidy to him. If the subsidy is sufficiently high, even profit-seeking insurers will strive to enrol these “problematic” individuals. The *granting of the freedom of choice* of health insurance would thus cease to be limited to high-income individuals only; conversely, private and social health insurers could compete on equal terms. In order to have something to compete on, however, social insurers must be relieved from the legal obligation to conclude contracts with any physician or hospital belonging to their respective associations, a stipulation characterizing most industrialized countries (with the notable exception of the United States). In this way, social health insurers would be able to also face international competition, one of the challenges discussed in Chapter 11 (Conclusion 11.8).

12.3.3 The Market for Medical Services

Today, many industrialized countries are confronted with a paradoxical situation regarding their supply of physicians. On the one hand, they have spent considerable amounts of tax money to build up the capacities of their (public) medical schools on the promise to improve future access to medical care. On the other hand, one

generation later there are complaints about a physician glut fueling the “cost explosion” [see Rosa (1990) for an international survey]. In fact, however, preventing this glut amounts to depriving tax payers from access to medical services they had paid for in advance on the grounds that these services are too expensive for social health insurers. However, these medical services would become affordable if it were possible to intensify *price competition in the market for medical services*, then the cost of medical services would be contained at least in the long term.

As explained in Chapters 7 and 8, prices of medical services are fixed by negotiations between medical associations and health insurance associations. Therefore, the individual physician can reduce the price of his services only indirectly by spending more time with the patient. This causes his price–performance ratio to improve without, however, bringing financial relief to the insurer. Even in the long term, health insurers have thus limited opportunity to take advantage of their increased market power under present circumstances.

Again, regulation may be changed to foster competition. In European countries in particular, social insurers should be relieved of the legal obligation to negotiate payment schedules with a single medical association or hospital association. Rather, they should have the right to choose amongst the services offered by different physician groups, at prices negotiated with their respective associations (see Conclusion 8.3).

This implies that physicians’ compulsory membership in statutory medical associations must be abolished, giving the individual physician a *choice of agents representing his interests*. In particular, he is free to join a group that guarantees medical care to inhabitants of an area or employees of a company, for example, in the guise of a Health Maintenance Organization (see Chapter 8). Young physicians who want to build their practice may now attract patients by joining a medical group offering the insurer a reduction on their fee schedule or offering flat rates for certain treatments. Three things would be achieved in this way.

1. To anyone *considering a medical career*, it is made clear that social health insurance is not there to guarantee an income to physicians. This serves to make a medical career less attractive as an investment (see Chapter 11), causing demand for admissions to medical schools to drop. The issue of open or restricted access to medical schools would lose some of its importance.

2. *Health insurers* will have to offer their members a variety of packages, the lower priced of which would typically exclude physicians and hospitals having a costly style of treatment.

3. *The insured* will have more freedom of choice not least within social health insurance. Those who want to get out of the “cost escalator” may sign up with an insurer who in turn enlists physicians and hospitals that are prepared to respect recommendations emerging from cost–benefit analysis (see Section 12.1).

Finally, lifetime licensure does not seem appropriate for guaranteeing a high *quality* of medical services (see Conclusion 8.7). Periodical reexaminations should take its place. They should focus on the physician’s contribution to the health of his patients, a requirement that can be increasingly satisfied in view of the progress made in the evaluation of health status (see Chapters 2 and 4).

12.3.4 The Market for Hospital Services

With the exception of the United States, most hospitals are either owned by the public or subsidized by the public purse. An important result of Chapter 9 states that hospitals should be paid for those services that they are expected to deliver (Conclusions 9.6 and 9.8). Therefore, payments from the public budget should be *payments for positive externalities* produced by hospitals, such as training provided to medical students and utility provided through the mere existence of a hospital in the area in the guise of an option good. It may be noted, however, that the training of medical students does not really represent an externality because students unwilling to pay could be kept out of the hospital at little cost.

The utility of an option good cannot be internalized as easily. While inhabitants of a catchment area may value having access to the hospital, they can be made to contribute no sooner than when they utilize the hospital's services. Therefore, taxing potential users of the hospital for its option utility appears to be justified. Yet, given that a vast majority of the population has health insurance of some type, this argument loses its force. After all, the premium of the insured may be calculated as to include a *supplementary charge for the option* of using the services of a certain hospital. Evidently, private hospitals already succeed in internalizing positive external effects in their negotiations with insurers.

Therefore, hospital subsidization needs to be *reduced*, first and foremost because it artificially lowers the price of inpatient care relative to outpatient care. This bias serves to stifle incentives of patients and their health insurers to avoid or shorten costly hospital stays in favor of treatment in ambulatory medical care.

12.3.5 The Market for Drugs

Of all markets in the health care system, the pharmaceutical market probably is the most competitive one. The main reason for this conjecture is that drugs can easily be traded internationally, while for the services of physicians and hospitals this is possible only at comparably high cost. However, the considerations advanced in Chapters 10 and 11 suggest that international pharmaceutical competition is distorted in favor of *quality competition* (see Conclusions 10.1 and 11.5). A foreign preparation is, independent of its price, banned from the market by the government of a typical industrialized country unless its quality is at least equal to those of the domestic competitors in each relevant dimension.

Conversely, international *price competition* is eliminated beyond the life of the patent in several European countries due to the prohibition of parallel imports. Parallel imports undermine international price differentiation by manufacturers, which is in part forced upon them by countries like France and Italy, where the same preparation must be sold at a lower price than in its country of origin (or not at all). By subjecting drugs to a strict price control, these countries in fact behave as free riders who profit from the innovative efforts financed by consumers in other countries without paying for it.

One possibility to make free riders bear more of the costs of innovation consists in an extension of international patent protection (see Chapter 10). As soon as effective patent life has reached its optimal value, however, maintaining the prohibition of parallel imports is not justified anymore.

12.4 COMPREHENSIVE OR PIECEMEAL REFORM?

The reform proposals that have been discussed up to this point mainly derive from one principle, that of allocative efficiency [for a characterization of some industrialized countries in terms of equity in the financing and delivery of health care, see Van Doorslaer and Wagstaff (1992), Wagstaff and Van Doorslaer (1992)]. Moreover, these proposals refer only to one sector of health care at a time. Accordingly, the realization of any one of them would by itself bring limited welfare gains. A comprehensive reform strategy could be expected to yield better results than any one or even the sum of these partial proposals for change. However, the successful outcome of a comprehensive reform strategy seems improbable for at least three reasons.

1. The reform proposals are inevitably addressed to several actors because none of them has the authority or the power to push them through on his own. The mere existence of a *division of authority* between government, member states, and local authorities in many countries implies such a multitude of actors.

2. Every change in the status quo has its losers as well as its winners. Because there are scarcely any institutionalized mechanisms for *compensating the losers* by the winners that could make Pareto-optimal reforms succeed, "logrolling" between the lobbying groups involved remains the main solution both within and outside of parliament. During the course of such logrolling, reform proposals must be packaged anew and also modified so that majorities may be found in a whole sequence of votes.

3. Comprehensive reform proposals are incompatible with the basic views espoused in this text, viz. that developments in health care should be increasingly *determined by market forces*, which are expected to bring forth even more organizational innovations, some of them different from anything conceivable at present, in the aim of adapting products more closely to the preferences of consumers.

These considerations support the notion that a piecemeal health policy, in spite of its lack of consistency, should not be interpreted as expressing failure, but rather as reflecting a sequence of steps in search of an improvement for all parties involved in health care. Hence, the central task of health economics cannot consist of developing comprehensive reform strategies, but identifying institutions and specific measures that are characterized by a particularly favorable relationship between benefits and costs.

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Translated from the German and adapted with examples for the U.S. and other English-speaking audiences, *Health Economics* is the most complete text available on the economics of health and health care delivery. Appropriate for both advanced undergraduate and beginning graduate students of economics, this text develops the theory necessary to examine issues analytically, presenting empirical evidence on each issue and summarizing results for easy comprehension. Issues discussed include the "cost explosion" in health care, the role of physicians and the power of medical associations, and the control of technological change in medicine. Without neglecting ethical concerns, modern microeconomic theory is applied to formulate theoretical implications and predictions, and key arguments are summarized to facilitate follow-up and review. The text explains the institutional differences of health care sectors in many industrialized countries, reflecting these differences by using microeconomic modelling. Rather than simplifying the issues facing today's health care systems, *Health Economics* models existing complexities as they are, showing students how economics can be adapted to reflect the views of the average person. Due to the rapidly expanding level of sophistication in health economics, the text also provides key analytic tools required to understand research that is currently in progress.

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